

# Hruday gauge-theory workbench: spectral bounds, local corrections, Wilson readouts, and controlled evolution

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Integrated research checkpoint: Round30

## Abstract

This integrated research compendium carries the complete Draft01 body through Round26, the Round27 and Round28 addenda, and the reviewed Round29 continuation. The following Round30 extension is admitted only through its source-bound review records. It follows regulated benchmarks, finite and full-representation  $SU(2)$  calculations, canonical readouts and a distinct homogeneous fixed-lattice model. Results include controlled spectral and heat approximations, a critical growing-time endpoint counterexample, complete source budgets and conditional nonzero physical Wilson fluctuations. A four-site creation certificate gives a unique ground and physical gap at least  $\alpha/16$  for every nonempty finite complete-factor volume at both signs of  $|\tau| \leq 10^{-8}$ , with representation cutoffs removed. A separate construction at that numerical cap supplies a locally normal full-lattice subsequential state and strongly continuous GNS evolution. In that same representation the complete original-endpoint physical sector has gap at least  $\alpha/16$ , a simple vacuum and original Wilson variance greater than  $1/5$ . The full-GNS strengthening uses the separately reviewed full-Hilbert finite-volume premise. This does not identify the earlier conditional orthant state or establish uniqueness of all thermodynamic states. For the earlier orthant state, the Round29 moment theorem retains its original restrictions. Round30 separately proves the Wilson operator domain and exact first-energy identity in the numerical-cap AQ state, with second moment at most  $\alpha^2(36 + 98|\tau|)$ . This new proof does not identify the states; equality of finite and limiting second moments remains unproved. The same-state moments now give a nonzero finite-window spectral-mass bound and an inverse-energy interval. A conditional 4097-node Euclidean protocol then achieves dimensionless interval width below  $1/500$  under its stated data and arithmetic contracts, and separates two abstract equal-moment controls. No actual AQ Euclidean samples or inverse response have been computed. Contemporary primary-source comparison materially limits distinct mathematical novelty. Hruday/HNM labels identify project records; established mechanisms retain their attribution. These fixed-model results do not construct four-dimensional continuum Yang–Mills theory or prove its continuum physical mass gap.

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# 1 Human authorship, integrated coverage and naming policy

**Human author: Hruday N M (BUNZEEY).** This is an integrated manuscript prepared with AI assistance in research, derivation checking, code, source comparison, and editing. Model-agent reviews are disclosed in the historical chapters. No external human peer review or formal proof-assistant verification is claimed; there is no asserted arXiv submission or acceptance.

The complete Draft01 research body and appendices through Round26 and the complete Round27 and Round28 research addenda are carried forward here, including failures, conditions and retrospective roadmaps. The original source packages remain unchanged. A dated roadmap is a record of the question selected then, not a claim that subsequently completed work remains unexecuted. The Round29 chapter retains its reviewed investigations and dated roadmap. The Round30 chapter contains the current three-investigation cycle and its closing roadmap. Equation numbers and cross-references are unified; original identifiers remain recoverable in the registry.

## 1.1 What Hruday and HNM do and do not name

Hruday names and the HNM namespace identify this project’s records. They do not rename Yang–Mills theory, Haar measure, Wilson observables, Jensen’s inequality, the GNS construction, Schur complements, Duhamel’s formula or any established external theorem. Mathematical notation and physical units stay intact. In particular, adding HNM to a result does not convert a conditional estimate into a theorem about another model, and does not establish historical priority. The number of newly asserted axioms is zero. HNM-T and HNM-P statement aliases identify admitted scoped theorem statements and propositions by mathematical content, including source-theorem applications. These aliases add no premises or proof steps. External axioms retain their names and attribution.

Every contribution has a stable HNM-C identifier with its legacy ID. Labelled equations have stable HNM-E keys derived from their original source labels; the printed HNM equation numbers are document locators. HNM-Q quantity entries give traceable names while retaining their conventional symbols. Shared symbols with different meanings remain local to the stated model. The exact machine-readable registry is `registry/hnm-registry.json`; the appendix prints the contribution aliases and equation concordance.

The double-star marker continues to mean derived, specialized, checked or implemented within this research record. It is not evidence that a method or equation is new to the literature. The priority table ranks usefulness to this route qualitatively, rather than assigning discovery scores or a percentage of the Millennium problem.

## 1.2 The requested primary-source comparison changes the novelty boundary

The September 2026 version of Gauvin’s preprint [1] gives a regulated SU(3) analysis in Sections II.3–II.4, with supplemental proofs: physical gap estimates, local Wilson survival, a finite energy moment, and physical GNS correlation bounds. This overlaps the general strategy of our AJ/AK branch. The Wilson variance–moment–window–Jensen mechanism therefore must not be presented as a Hruday invention. Our stated SU(2) model, constants, domains, boundary convention and source-bound computations remain individually inspectable; their distinct scientific novelty is unverified. The preprint’s constants cannot be transferred by relabeling its group or energy scales. Its abstract also leaves the interacting continuum construction open. This is a scoped comparison, not an endorsement or complete proof audit of the external work.

The numerical-cap AQ branch has a further explicit prior-method boundary: Gauvin’s Supplement A.10 already presents the thermodynamic compactness, dynamics, GNS and Fourier-test strategy. The actual unbounded-onsite dynamics placement uses Nachtergaele–Sims [2], with the source

panel also checking the modern Nachtergaele–Sims–Young framework [3]. The contribution is a separately checked SU(2) selected-strip model dictionary and numerical-cap application. Neither renaming nor a bounded-spin theorem supplies the unbounded-rotor hypotheses. Volume-norm convergence of a fixed local evolution remains distinct from norm continuity in time on every bounded local operator.

The official problem description [4] remains the governing target: a nontrivial four-dimensional quantum theory of the requisite axiomatic strength with a positive physical mass gap for each compact simple gauge group. Finite graphs, a fixed-spacing homogeneous limit, a selected boundary state and a conditional spectral estimate do not jointly supply that construction. The pending model map, scale trajectory, uniform estimates, nontrivial observables and limiting reconstruction are explicit proof obligations.

### 1.3 How to read the enlarged manuscript

The historical first part establishes the model vocabulary and source-bound calculations. Round27 studies identifiability and one nonlinear correction. Round28 restores complete sources, improves finite-graph heat control and establishes conditional physical Wilson estimates. Round29 tests the next missing implications against the reviewed external sources. Use the ranked appendix to locate a useful result, then read its full assumptions and surviving objection. A negative or insufficient gate is a completed investigation with a limited conclusion, not a solved target.

### 1.4 Current extension to the numerical-cap state

The later AT1 gate closes the Wilson operator-domain and first-moment identity obligations directly in the AQ numerical-cap state, with second moment bounded by  $\alpha^2(36 + 98|\tau|)$ . This is a fresh same-state proof, using seven bulk stars, rather than a transfer from the earlier orthant state. Earlier passages saying that AQ has no moment theorem describe their original admissions; the current registry attaches the separate AT1 extension. The full forward and reverse derivations appear in the Round30 part. AO/AQ identification, second-moment equality, boundary-independent uniqueness and the continuum construction remain open. This distinction applies throughout the carried historical chapters and appendices.

## 2 Starting point, question, and evidence boundary

The initial motivation was to turn an exploratory Einstein–QED workbench into a program addressing precise nonabelian gauge questions. The surviving early calculations concern a regulated homogeneous Maxwell–Dirac mean field and reduced gravitational consistency, rather than a construction of either quantum gravity or continuum QED. Subsequent work asks what a finite SU(2) gauge system can establish exactly, which approximations can be certified, and what additional premises would be needed to connect such results to the four-dimensional Yang–Mills existence and mass-gap problem. The latter requires a nontrivial continuum quantum theory satisfying appropriate axioms and a positive physical mass gap; a finite lattice gap alone is insufficient [4].

This paper is a integrated research compendium, not a claim that the final target has been reached. It preserves failed claims because they determine the next mathematical question. The transition from a matrix with a positive gap to an infinite representation space requires a tail bound. The transition from a finite graph to an infinite lattice requires uniform estimates. The transition from a static probability measure to a Hamiltonian requires a justified state-space and generator map. Each of these transitions appears explicitly in the history.

## 2.1 Coverage and reproducible checkpoint

The historical Round26 source checkpoint is repository commit 40960f39a3dcaa6b2735d47adaa0e3b40e6a0f02, with Git tree 169b2db4e8017f5c4e783ba5f42e4f96522ad2b1. The repository is [https://github.com/occult-kranti/yang\\_mills\\_workbench](https://github.com/occult-kranti/yang_mills_workbench). The historical ledger records 111 entries through Round25, comprising 35 named study entries and 76 explicitly numbered physics loops. Round26 adds ten loops. This gives 121 entries and 86 explicit loops across 24 recorded rounds, Rounds3–26. An entry is an audit unit, not necessarily a distinct discovery. There are no independently identifiable Round1/2 packages in this checkpoint; their contents cannot be reconstructed from a round number.

The paper groups repeated refinements into mathematical contributions while Appendix D retains every recorded entry. The exact contracts, complete longer proofs, source files, archived outputs, and historical reviews remain in the pinned repository. A concise derivation here is not a replacement for a missing historical proof. Where only a numerical study or conditional argument is available, it is described at that level.

## 2.2 What the two stars mean

The marker \*\* accompanies a workbench contribution and its central newly derived or specialized equation. It means *newly derived, computed, repaired, or implemented in this research record*. It does not mean a new law of nature, a historically first theorem, or a completed literature-priority search. Established ingredients—Haar integration, Schur complements, spectral projection, Duhamel formulas, Rayleigh–Ritz and residual bounds, and Lieb–Robinson estimates—remain established mathematics. Their use in a new calculation does not make the underlying method an invention.

The appendix classifies contributions as exact identities, conditional bounds, certified arithmetic, numerical diagnostics, obstructions, and workflow artifacts. Proposed future goals are not starred results. Literature comparison identifies the nearest checked sources and remaining differences; scientific priority is unverified throughout this manuscript.

## 2.3 Research design and attribution

The workflow pairs a forward calculation from frozen premises with a reverse reconstruction of the premises sufficient for a desired conclusion. An advisor chooses the next experiment from the first result and skeptical objection. Separate model agents wrote many forward, reverse, and skeptical records; these share the same project, inherited premises, and model infrastructure. This is not independent experimental replication. Round23 T2/U1/U2 and Round25 AA1/AA2 retain their same-author review attribution. Replays, software tests, admission repairs, and manuscript checks are not counted as new physics loops.

Newton’s analysis/synthesis motivates the question “which assumptions are necessary to reconstruct this observable?” Tesla’s mechanism and loading approach motivates “which omitted channel changes the proposed device or resonance?” These are modern methodological adaptations of selected historical sources; the inspected Opticks discussion and Tesla patent are historical context, not mathematical authority [5, 6]. They supply no premise in an operator proof. The double Fibonacci spiral was a way to organize investigation nodes; no tested physical significance or golden-ratio coupling has been established. No historical or esoteric analogy is inserted as a term in the gauge Hamiltonian.

### 3 Models, notation, and nontransfer rules

Keep a positive lattice spacing  $a$ , physical energy reference  $E_\star$ ,  $\alpha/E_\star$ , and  $\hbar$  fixed unless a stated construction changes them. A change of notation cannot serve as physical scale matching. Dimensionless eigenvalue inequalities become physical energy inequalities only after multiplication by their own energy unit.

Table 1: Distinct model families. Symbols reused in historical reports are local to the stated model.

| Family                      | Generator or object                                                             | Scope and scale                                                                                                          |
|-----------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Early mean field            | Finite-regulator Maxwell–Dirac variables and declared scalar/gravity reductions | Classical mean-field evolution; cutoff and closure assumptions remain.                                                   |
| Static compact integrals    | Normalized Haar density with specified Euclidean action coefficients            | Coupling derivatives are static responses, not physical time evolution.                                                  |
| Finite physical gauge graph | $H = \alpha L$ , with $L = K + \lambda(20 - S)$ in the latest graph             | 18 vertices, 33 links, 20 faces; all vertex Gauss constraints; dimensionless heat coordinate $\sigma = \alpha t/\hbar$ . |
| Homogeneous complete blocks | Initial interacting operator $G$ ; $M = 7 \tau $                                | Energy unit $\delta = \alpha/8$ ; original $ \tau  \leq 5/1664$ ; stronger AE2 result uses $ \tau  \leq 1/7000$ .        |
| Canonical summable pro-file | Declared $q$ -dependent full-link family                                        | Fixed spacing; $q \uparrow 1$ ; physical observation time $T_q = (z/\eta)(\hbar/\alpha)(1 - q)^{-3}$ .                   |

In finite-link gauge systems the ambient Hilbert space is a product of  $L^2(\text{SU}(2))$  spaces and the physical space is the invariant subspace under the independent vertex gauge actions. A finite number of links does not make this representation space finite-dimensional. A retained projector  $P$  must be built from actual physical vectors;  $Q = I - P$  denotes its full complement. On the latest graph,  $P_{21} = \mathbf{1}_{[0,9/2)}(K)$  is the complete strict low-electric-energy sector. Its 293-state residual enrichment reduces  $K$  but is not a complete ambient energy cutoff. The 293-coordinate representation has a nontrivial Gram matrix, so coordinate Euclidean norms cannot replace physical norms.

For positive self-adjoint  $L$  with ground energy  $\epsilon$  we use centered heat  $e^{-\sigma(L-\epsilon)}$ . For  $A = PLP$  with its own lowest eigenvalue  $\mu$ , the correct retained comparison is  $e^{-\sigma(A-\mu)}P$ . These two centers are generally different. Neither heat semigroup is a real-time unitary  $e^{-itH/\hbar}$ ; an estimate for one is not silently transferred to the other.

An observable is a full operator, together with its state and readout. Equality  $B\Omega = C\Omega$  identifies a vacuum-created vector but does not imply  $B = C$ , equal operator norms, or transferable finite-parameter error constants. An exact identity, a norm estimate, and a numerical evaluation are separate results. These distinctions are repeatedly used below.

### 4 Relation to established results

The earlier QED models draw on established, regime-specific methods: constant-field electric-probe response with a stated finite subtraction [7], homogeneous mean-field pair production with adiabatic current renormalization [8], and locally covariant current and stress renormalization with explicit background restrictions [9]. These sources do not themselves provide the workbench’s evaluated finite-mode formulas or a general strong-field Bianchi-I closure.

The lattice and Hamiltonian frameworks originate with Wilson and Kogut–Susskind [10, 11]. For the particular reductions studied here, the physical single-plaquette problem already has a

Mathieu-function formulation [12, Appendix B], and the open two-plaquette electric spectrum has explicit low-level tabulations [13, Appendix C]. Maximal-tree gauge fixing also has a developed operator treatment: its electric generators require transported derivatives, beyond the kinematic identification of gauge orbits [14, Appendix C]. Thus these reductions and low levels are comparison targets, not claims of discovery. The possible increment is the explicit graph arithmetic, Gram identities, complete residual accounting, and inspectable bounds recorded below. The Gegenbauer polynomials and variational tools used in those calculations are classical [15, 16].

Product-vacuum stability already permits infinite-dimensional onsite spaces under appropriate gap and bounded-interaction hypotheses [17]. Local stability of ground-state expectations is a related statement whose local perturbation need not preserve the perturbed global gap [18, Theorem 3]. Explicit qubit thresholds [19] and finite-dimensional local block diagonalization [20] have narrower hypotheses than some models here. However, a broader construction does cover separable onsite spaces and relatively form-bounded interactions [21]. Its relevance prevents any claim that unbounded-interaction stability is absent from the literature. Applying it quantitatively still requires matching domains, interaction geometry, weighted norms, and smallness constants. This compendium does not supply an evaluated general homogeneous threshold by citing that theorem.

Locality and approximation likewise have established frameworks. Nachtergaele–Sims treat unbounded onsite terms with controlled bounded intersite interactions [2]; their operator topologies must be retained when comparing time continuity and spatial limits. Exponential-in-volume tails offer a further lead under a finite-qudit setup [22], not an admitted extension to these rotor factors. Rigorous quantum-number truncation estimates [23] require their specified initial-state and Hamiltonian assumptions. They do not automatically certify an arbitrary retained block or its numerically centered heat evolution.

Finally, strong-coupling Wilson-measure and stochastic-generator results [24] concern explicitly specified measures and clocks. They do not identify an auxiliary Markov gap with the canonical physical-time gap. Continuum reconstruction requires a full compatible Schwinger-function hierarchy with the corrected regularity and other axioms [25]; the four-dimensional existence and physical mass-gap target remains distinct [4]. This is a selected literature comparison, not an exhaustive priority search. Scientific novelty of the model-specific certificates has not been established.

## 5 From semiclassical backreaction to a controlled finite gauge theory

The early program asked how a quantum current, electromagnetic backreaction, and gravitational constraints could be calculated consistently. The surviving record begins with Round 3. It contains imported prescribed-background results and descriptions of earlier research fronts, but no independently identifiable Round 1 or Round 2 package. We therefore do not reconstruct their missing contracts or assign them new results. The present section accounts for all 31 named studies in Rounds 3–12, using the Round 25 history inventory and the underlying mathematical reports. These studies precede the explicit paired loop contracts introduced later. Here \*\* marks a contribution to this workbench, including an implementation or an independently derived obstruction; it does not establish priority over the mathematical literature.

## 5.1 The original closure problem and its finite electromagnetic realization

**Initial scope.** The organizing Einstein–QED action contained Einstein gravity, Maxwell and Dirac terms, possible scalar-dependent coefficients, and curvature counterterms. After integrating out the electron, the intended causal object was a closed-time-path functional  $\Gamma[g_+, A_+; g_-, A_-; \rho_0]$ . Varying one branch before identifying the branches defines state-dependent current and stress. This is a specification of a problem, not its solution. A local vacuum-persistence rate is not a causal current; an Euler–Heisenberg expansion cannot be added independently to the same full electron determinant. The four contextual fronts—charged de Sitter discharge, magnetized pair creation, Cauchy-horizon dynamics, and field mixing—have different geometries and boundary data. No theorem identifies their solutions. Sources: `evidence/qeg-research/round3/chapters/01_master_action.md` and `evidence/qeg-research/round3/chapters/02_fronts.md`.

**Matched finite Maxwell–Dirac closure** \*\*. Self-consistent field/mode initial-value formulations and adiabatic regularization have established precedents [8]. The executable reduction fixes a flat homogeneous geometry, a static magnetic field, finitely many Landau levels, and a finite positive momentum quadrature. This early reduction uses Heaviside–Lorentz natural units,  $c = \hbar = 1$ ; later gauge-graph comparisons restore their explicitly stated physical scales. With  $s = mt$ ,  $a = eA_z/m$ ,  $x = eE/m^2$ , and  $b = |eB|/m^2$ , set  $p_i = k_i - a$ ,  $M_i^2 = 1 + 2bn_i$ ,  $\omega_i^2 = M_i^2 + p_i^2$ , and  $h_i = (M_i, 0, p_i)$ . The positive weights are  $w_i = b(2 - \delta_{n_i 0})w_{k_i}/(4\pi^2)$ . A prime in this subsection means  $d/ds$ . Bloch evolution and three finite sums are

$$\begin{aligned} a' &= -x, & r_i' &= 2h_i \times r_i, \\ S &= \sum_i w_i(r_{iz} + p_i/\omega_i), & C &= \sum_i \frac{w_i M_i^2}{4\omega_i^5}, & D &= \sum_i \frac{5w_i M_i^2 p_i}{8\omega_i^7}. \end{aligned} \quad (\text{HNM-1})$$

The fixed magnetic matching coefficient and divided field equation are

$$\begin{aligned} \chi_b &= \frac{e^2}{12\pi^2} \left[ b - \log(2b) - \psi \left( 1 + \frac{1}{2b} \right) \right], \\ Z &= 1 + \chi_b - e^2 C, & Zx' &= F - e^2(S + Dx^2). \end{aligned} \quad ** \quad (\text{HNM-2})$$

The susceptibility comparison uses the constant-field, second-order electric-probe and all-orders magnetic-field setting of [7]; the evolving current is obtained from the retained modes. Here  $F$  is the prescribed source. The mode subtraction is  $c_i = M_i^2 x'/(4\omega_i^5) - 5M_i^2 p_i x^2/(8\omega_i^7)$ , with matching restored once. It follows by solving  $2h_i \times r_i^{(2)} = (r_i^{(1)})'$ , with  $r_i^{(1)} = (0, -M_i x/(2\omega_i^3), 0)$ , and fixing the longitudinal component by  $|r_i| = 1$ ; the companion energy subtraction  $M_i^2 x^2/(8\omega_i^5)$  differentiates to  $xc_i$ . The finite matching is the difference between the full-momentum magnetic Landau sum and the zero-field coefficient  $e^2 \log(1 + 2bN)/(12\pi^2)$ , followed by  $N \rightarrow \infty$ . In particular  $S + Dx^2$  is a numerator, whereas the full renormalized matter current in these conventions is  $J = S - Cx' + Dx^2 + (\chi_b/e^2)x'$ . Confusing the two changes the response.

Define  $U = \sum_i w_i(h_i \cdot r_i + \omega_i)$ . Direct differentiation uses  $p_i' = x$ ,  $h_i \cdot (h_i \times r_i) = 0$ , and  $C' = -2Dx$  to give  $U' = xS$ ,  $Z' = 2e^2 Dx$ , and therefore

$$W = \frac{1}{2}Zx^2 + e^2 U, \quad W' = xF. \quad ** \quad (\text{HNM-3})$$

This is an exact finite-model identity, not a continuum error estimate. For physical  $|r_i| \leq 1$ ,  $U \geq 0$ . If  $Z(a) \geq z_* > 0$  for all  $a$ , then  $W \geq z_* x^2/2$  and, by applying the differential inequality first to  $\sqrt{W} + \varepsilon$  and then letting  $\varepsilon \downarrow 0$ ,

$$\sqrt{W(s)} \leq \sqrt{W(0)} + \frac{1}{\sqrt{2z_*}} \int_0^s |F(u)| du. \quad (\text{HNM-4})$$

The spins remain in compact balls,  $x$  is bounded on each finite interval, and  $a = a_0 - \int x$  cannot diverge there. Local Lipschitz continuity of the finite vector field proves global classical continuation for continuous drives, as assumed in the historical theorem. An initially mixed mode satisfying  $|r_j| \leq 1 - \delta$  also gives  $U \geq w_j \delta |k_j - a|$ , hence bounds  $a$  for an integrable drive. That additional premise does not hold for every pure-vacuum production run.

The historical source-off calculation reached  $x(50) \simeq -0.03757$  with 4096 momentum nodes, Landau levels 0–8, and window  $K = 40$ . The recorded 2048-to-4096 history difference  $3.26 \times 10^{-9}$  and separate Landau endpoint difference  $5.06 \times 10^{-8}$  are refinement diagnostics. The advisor's  $10^{-18}$  absolute benchmark failed at approximately  $1.82 \times 10^{-18}$ ; it remains failed. Independent spinor/Bloch and matching controls established a further methodological point: a consistently wrong prescription may conserve its own energy. The separate massless lowest-Landau-level comparator obeys  $x'' + \Omega_B^2 x = F'$ ,  $\Omega_B^2 = e^2 b / (2\pi^2)$ , in its own declared scaling; it is not obtained by dividing the massive variables by zero. Sources: `evidence/qeg-research/round3/theory.md`, `evidence/qeg-research/round3/README.md`, and `evidence/qeg-research/round5/theorem_advisor.md`.

For the weak probe about a stationary vacuum, eliminating the transverse components of the linearized Bloch equation supplies the oscillator denominator  $4\omega_i^2 - (\nu + i0)^2$ . The same zero-frequency subtraction gives

$$\chi(b, \nu) = \chi_b + e^2 \sum_i \frac{w_i M_i^2}{\omega_i^3} \left[ \frac{1}{4\omega_i^2 - (\nu + i0)^2} - \frac{1}{4\omega_i^2} \right]. \quad ** \quad (\text{HNM-5})$$

This tests matching independently of the nonlinear work identity. In an instantaneous eigenbasis, the mode current also contains a coherence:  $r_{iz} + p_i/\omega_i = 2p_i f_i/\omega_i - M_i v_i^{\text{coh}}/\omega_i$ . Retaining the occupation  $f_i$  alone cannot reproduce that current.

**Causal tangent and the quantity a stability theorem controls \*\*.** Let  $v = \delta a$ ,  $u = \delta x$ ,  $q_i = -v$ ,  $\eta_i = \delta r_i$ , and  $f = \delta F$ , with grid and matching held fixed. Differentiation gives

$$\begin{aligned} \delta S &= \sum_i w_i (\eta_{iz} + M_i^2 q_i / \omega_i^3), \\ \delta C &= - \sum_i \frac{5w_i M_i^2 p_i q_i}{4\omega_i^7}, & \delta D &= \sum_i \frac{5w_i M_i^2 (M_i^2 - 6p_i^2) q_i}{8\omega_i^9}, \\ v' &= -u, & u' &= \frac{f - e^2 (\delta S + \delta D x^2 + 2Dxu) + e^2 \delta C x'}{Z}, \\ \eta'_i &= 2h_i \times \eta_i + 2q_i e_z \times r_i. \end{aligned} \quad ** \quad (\text{HNM-6})$$

The fundamental-matrix integral gives causal support and finite-time Gronwall bounds. The first variation  $\delta W$ , satisfying  $(\delta W)' = uF + xf$ , is signed and cannot be used as a positive norm. At a stationary pure vacuum, restrict to variations preserving the Bloch norm, so  $h_i \cdot \eta_i = 0$ , and define  $t_i = \partial_{p_i}(h_i/\omega_i)$  and  $\xi_i = \eta_i + q_i t_i$ . Then  $Z_0 u' = f - e^2 \sum_i w_i \xi_{iz}$ ,  $\xi'_i = 2h_i \times \xi_i + u t_i$ , and

$$E_2 = Z_0 u^2 + e^2 \sum_i w_i \omega_i |\xi_i|^2, \quad E'_2 = 2uf. \quad ** \quad (\text{HNM-7})$$

Indeed  $h_i \cdot \xi_i = 0$  and  $\omega_i \xi_i \cdot t_i = \xi_{iz}$  cancel the exchange terms. This controls  $u, \xi$ , not every raw tangent. For one mode with  $M = w = e^2 = 1$ ,  $p_0 = 0$ , and  $Z_0 = 3/4$ ,  $u = 1$ ,  $v = -s$ ,  $\xi = (0, -1/2, 0)$ ,  $\eta = (0, -1/2, -s)$  solve the unforced tangent while  $E_2 = 1$ . The secular raw tangent defeats a stronger stability claim. Rounds 4 and 6 also retained source-center and acceptance repairs rather than overwriting earlier failures. Sources: `evidence/qeg-research/round4/response_contract.md` and `evidence/qeg-research/round6/extensions.md`.

**Finite-window positivity and its cofinal obstruction** \*\*. For continuous momentum in  $[-K, K]$ , write

$$C_{N,K}(a) = \frac{b}{16\pi^2} \sum_{n=0}^N d_n [P_{M_n}(K-a) + P_{M_n}(K+a)],$$

$$P_M(y) = \frac{u - u^3/3}{M^2}, \quad u = \frac{y}{\sqrt{M^2 + y^2}}. \quad ** \quad (\text{HNM-8})$$

Since  $P'_M = M^2/(M^2 + y^2)^{5/2}$  is even and decreases with  $|y|$ ,  $C$  is even and strictly decreases for  $a > 0$ . Its maximum is at zero. This statement is about an integral: a two-node Gauss rule with nodes  $\pm K/\sqrt{3}$  has, at a node, a lowest-level contribution  $e^2 C_{\text{disc}} \geq \alpha_{\text{em}} bK/(4\pi)$ , which can exceed the continuous bound  $\alpha_{\text{em}} b/(3\pi)$ . Independently, at  $b = 10, K = 20$ , positivity of every quadrature weight and exactness on constants, together with  $\alpha_{\text{em}} = e^2/(4\pi) < 1/137$ , suffice for the rational estimate

$$e^2 C_{\text{disc}} < \frac{67743204500}{274510827927} < \frac{1}{4}, \quad Z > \frac{3}{4}, \quad ** \quad (\text{HNM-9})$$

uniformly in every finite Landau cutoff. The estimate uses mass bounds for the first levels and the integral majorant for the decreasing tail. It verifies a premise of (HNM-4), without certifying a trajectory.

At infinite momentum window the integral is elementary:

$$C_{N,\infty} = \frac{b}{12\pi^2} \left( 1 + 2 \sum_{n=1}^N \frac{1}{1 + 2bn} \right), \quad Z_{N,\infty} = 1 - \frac{e^2}{12\pi^2} \left[ \log(2b) + \psi \left( N + 1 + \frac{1}{2b} \right) \right]. \quad ** \quad (\text{HNM-10})$$

The latter tends to  $-\infty$ . Positivity makes  $C_{N,K}(0)$  increase in both cutoffs: choose a finite  $N$  exceeding any prescribed threshold, then a finite  $K$  approximating that integral. Every cofinal cutoff path eventually exceeds both. Thus unchanged matching cannot retain a positive uniform coefficient margin. This obstructs this divided-coefficient proof route, not continuum QED or every renormalization prescription. A per-run reference choice  $\chi_{N,K} = z_{\text{ref}} - 1 + e^2 C_{N,K}(0)$  yields  $Z = z_{\text{ref}} + e^2 [C(0) - C(a)] \geq z_{\text{ref}}$  for the continuous window; it changes the matching and does not inherit that inequality for arbitrary quadrature. A scalar contribution uniformly bounded over the permitted field domain cannot cancel the cofinal divergence at fixed  $a$ , or uniformly bounded  $a$ . Sources: `evidence/qeg-research/round6/extensions.md` and `evidence/qeg-research/round7/renormalization_map.md`.

The same estimates give the coefficient-only tail  $e^2 \sup_a |C_{\infty,K} - C_{N,K}| \leq \alpha_{\text{em}} K/[\pi\sqrt{1 + 2bN}]$ , provided previously retained quadratures are unchanged. This does not bound omitted dynamical modes. Whenever a finite cutoff has  $Z(0) < 0$ , continuity and  $Z(a) \rightarrow 1 + \chi_b > 0$  produce singular coefficient surfaces at opposite values of  $a$ ; no argument here proves that a physical trajectory reaches one.

## 5.2 What gravitational and scalar extensions actually add

**An anisotropic current–stress interface** \*\*. For  $ds^2 = -dt^2 + a_{\perp}^2(dx^2 + dy^2) + a_{\parallel}^2 dz^2$ , define  $h = \dot{a}_{\perp}/a_{\perp}$ ,  $k = \dot{a}_{\parallel}/a_{\parallel}$ , and  $\theta = 2h + k$ . Maxwell evolution gives  $\dot{E} + 2hE = -j$ ,  $\dot{B} + 2kB = 0$ , with field stresses  $(\rho, p_{\perp}, p_{\parallel})_{\text{EM}} = (u, u, -u)$ ,  $u = (E^2 + B^2)/2$ . The spatial Einstein equations  $\dot{h} = (\Lambda - \kappa p_{\parallel} - 3h^2)/2$  and  $\dot{k} = \Lambda - \kappa p_{\perp} - h^2 - k^2 - hk - \dot{h}$  imply, by differentiation,

$$C_g = h^2 + 2hk - \kappa\rho - \Lambda, \quad Q = \dot{\rho} + 2h(\rho + p_{\perp}) + k(\rho + p_{\parallel}),$$

$$\dot{C}_g = -\theta C_g - \kappa Q, \quad C_g(t) = \frac{V(0)C_g(0) - \kappa \int_0^t V(s)Q(s) ds}{V(t)}, \quad V = a_{\perp}^2 a_{\parallel}. \quad ** \quad (\text{HNM-11})$$

The coefficient is  $-\theta$  for these off-constraint equations, not the  $-2\theta$  initially suggested in review. Running  $\kappa, \Lambda$  adds  $-\dot{\kappa}\rho - \dot{\Lambda}$ . Moreover  $(k-h) + \theta(k-h) = \kappa(p_{\parallel} - p_{\perp})$ . The replacement  $(\delta p_{\perp}, \delta p_{\parallel}) = (kf, -2hf)$  leaves the energy equation unchanged but changes the shear source by  $-\kappa\theta f$ . Exact mean-energy conservation therefore cannot identify both pressures.

At a fixed comoving regulator the rescaled Dirac modes have  $H_n = m\beta + P\alpha_z + c_n\alpha_x$ ,  $P = (k_z - qA_z)/a_{\parallel}$ ,  $c_n = \sqrt{2n|qB_0|}/a_{\perp}$ . Writing  $\mathcal{D} = |qB_0|/(4\pi^2V)$  and  $\mathcal{S}[X]$  for the occupied-mode trace and sum, variations of this same Hamiltonian give

$$(\rho_m, j, p_{\parallel}, p_{\perp}) = \mathcal{D}(\mathcal{S}[H], q\mathcal{S}[\alpha_z], \mathcal{S}[P\alpha_z], \tfrac{1}{2}\mathcal{S}[c_n\alpha_x]). \quad ** \quad (\text{HNM-12})$$

Using  $\dot{\mathcal{D}} = -\theta\mathcal{D}$ ,  $\dot{P} = qE - kP$ , and  $\dot{c}_n = -hc_n$  proves  $\dot{\rho}_m + 2h(\rho_m + p_{\perp}) + k(\rho_m + p_{\parallel}) = Ej$ . The two occupied columns at  $n \geq 1$  already include the spin multiplicity. A rotation of  $P\sigma_3 + m\sigma_1 - sc_n\sigma_2$  to real transverse mass adds the connection  $s(m\dot{c}_n - c_n\dot{m})/[2(m^2 + c_n^2)]$  to  $P$ ; omitting it explains why the naive effective-mass replacement fails in the imported expanding background comparator. These are bare regulated quantities, not renormalized Einstein sources. A finite counterterm  $-c \int \sqrt{-g}F^2/4$  changes the current by  $c(\dot{E} + 2hE)$  and the stress by  $c(u, u, -u)$ ; its Ward exchange cancels only when both are changed. The classical Ohmic and later scalar-gravity fixtures test this interface with their own matter. They do not compute a common covariant quantum current and directional stress. The archived physical electron scale estimate  $\kappa\rho_{\text{EM}}/m_e^2 \simeq 2.40 \times 10^{-39}$  at  $E/E_c = 1, B/B_c = 100$  also rules out interpreting the fixture's visibly large gravitational response as an electron-coupling prediction. Sources: [evidence/qeg-research/round3/gravity.md](#), [evidence/qeg-research/round3/evidence/prior\\_curved\\_sector\\_results.json](#), and [evidence/qeg-research/round5/research\\_bridge.md](#). The locally covariant Dirac analysis [9] provides a renormalization framework with explicit background qualifications; its direct free stress-conservation discussion does not supply the strong-field Bianchi-I mode sums required here.

Two retained logical controls distinguish conservation from validity. A random energy equal to  $\bar{E}/p$  with probability  $p$ , and zero otherwise, has the same mean as the eigenvalue  $\bar{E}$  but variance  $\bar{E}^2(p^{-1} - 1)$ . Mean stress alone therefore does not determine stress noise. Also exact local evolution  $e^{-i\eta\sigma_x} \otimes e^{-i\eta\sigma_x}|00\rangle$  stays separable, whereas normalizing its first-order truncation gives

$$\text{Var}(E) = \bar{E}^2(p^{-1} - 1), \quad \mathcal{C}_{\text{truncated}} = \frac{2\eta^2}{1 + 2\eta^2}, \quad \mathcal{C}_{\text{exact}} = 0. \quad ** \quad (\text{HNM-13})$$

The latter apparent witness is at the discarded order. These project counterexamples <sup>\*\*</sup> do not diagnose gravitational breakdown or a physical entanglement experiment. For the inherited Cauchy-horizon profile  $\phi \sim |V|^{\beta}$ , integrating  $|\partial_V \phi|^2 \sim |V|^{2\beta-2}$  gives the familiar threshold  $\beta > 1/2$ , with logarithmic divergence at equality; it is an energy regularity test, not a solved backreacted horizon.

**Action-derived scalar completion <sup>\*\*</sup>.** A dynamical coefficient requires its own equation and energy. Round 7 chose  $f(\phi) = 1 + g^2\phi^2$ , scalar kinetic energy  $y^2/2$ , and  $V(\phi) = \nu^2\phi^2/2$ , with  $y = \phi'$ . Variation of  $-f(\phi)F^2/4$ , with the finite-mode current unchanged, gives

$$\begin{aligned} Zx' &= F - e^2(S + Dx^2) - f_{\phi}yx, & y' &= -\nu^2\phi + \tfrac{1}{2}f_{\phi}(x^2 - b^2), \\ \widetilde{W} &= \tfrac{1}{2}Zx^2 + e^2U + \tfrac{1}{2}y^2 + \tfrac{1}{2}(\nu^2 + g^2b^2)\phi^2, & \widetilde{W}' &= xF. \end{aligned} \quad ** \quad (\text{HNM-14})$$

The scalar and field exchange terms cancel directly. Since  $f \geq 1$ , the finite  $Z > 3/4$  certificate is retained, and the same continuation argument applies; the uncoupled zero-frequency scalar is handled as an affine solution. By contrast, merely prescribing  $\chi(s)$  in the old equations produces the unaccounted power  $\chi'x^2/2$ . The separate classical Bianchi-I model with  $f = e^{2c\varphi}$

closes its own scalar/Maxwell exchanges. It supplies no identification with quantum stress. Renormalization-scale running, cutoff changes, and a dynamical scalar are different operations. Sources: `evidence/qeg-research/round7/research_report.md` and `evidence/qeg-research/round7/renormalization_map.md`.

### 5.3 Spectral logic before a nonabelian calculation

**False plateaux and conditional spectral bridges \*\*.** The move to pure Yang–Mills changes the theory and its target [4]. Its useful inheritance is explicit control of assumptions, not an abelian mass-gap result. For a nonzero finite positive spectral measure supported on  $[0, \infty)$ ,  $C(t) = \int e^{-Et} d\mu(E)$ , let  $E_* = \inf \text{supp } \mu$ . Then  $C(t+h) \leq e^{-hE_*} C(t)$ ; logarithmic convexity follows from Hölder. Consequently the effective mass  $m_h(t) = h^{-1} \log[C(t)/C(t+h)]$  decreases to  $E_*$  from above. For  $t > 0$ , differentiating gives

$$-\partial_t \log C = \langle E \rangle_t, \quad \partial_t \langle E \rangle_t = -\text{Var}_t(E). \quad (\text{HNM-15})$$

Positive measure in every  $[E_*, E_* + \varepsilon]$  fixes the asymptotic exponential rate; convexity then fixes the secant limit. Thus an early plateau near 1 in the executed  $e^{-t} + 10^{-12}e^{-0.1t}$  example is not a lower bound on the threshold 0.1. A lighter orthogonal channel may be invisible even at infinite time. The periodic massless free-field diagnostic explicitly excluded its zero mode; its lowest nonzero momentum is not the full gap.

Two standard spectral-theorem arguments were made explicit. First, if  $H_n \rightarrow H$  strongly in resolvent on a common Hilbert space and all  $\text{spec}(H_n) \cap (0, m) = \emptyset$  for the same  $m > 0$ , then compactly supported continuous  $f$  in that interval satisfies  $f(H) = \lim f(H_n) = 0$ . Positive but nonuniform gaps are insufficient:  $H_n e_k = \max(1/n, 1/k) e_k$ ,  $k \geq 1$ , and  $H_n e_0 = 0$ , converge in norm to a gapless diagonal operator. Second, if a reconstructed positive Hamiltonian already exists and centered states  $\psi_O$  span densely in  $\Omega^\perp$ , a common bound  $\langle \psi_O, e^{-tH} \psi_O \rangle \leq A_O e^{-mt}$  at all sufficiently large times implies

$$e^{-(m-\varepsilon)t} \|P_H([0, m-\varepsilon])\psi_O\|^2 \leq A_O e^{-mt} \implies H|_{\Omega^\perp} \geq m. \quad (\text{HNM-16})$$

Density removes every lower channel. These known implications do not construct the continuum Hilbert space or supply the common physical rate. The Round 5 and Round 8 planners replay finite reviewed implication libraries; they are not proof-assistant kernels. The Round 8 delayed-source refinement met its declared adjacent-grid working scale, while 21 symbolic checks were blocked by an unavailable dependency and remain unexecuted in that audit. Sources: `research/round8/spectral-proof.md`, `research/round8/advisor-dossier.md`, and `research/round8/README.md`.

**Finite Haar hierarchy, and what a single identity cannot certify \*\*.** For the finite Wilson measure  $d\mu_W = Z^{-1} e^{-S_W} dH$  with  $S_W = \beta_W \sum_p (1 - \text{Tr } U_p/2)$ , left Haar differentiation yields  $\langle L_a f - f L_a S_W \rangle = 0$ . Conversely, on a finite product of connected compact groups, if every smooth insertion satisfies every such identity under a probability measure  $\nu$ , substitute  $f = e^S \varphi$ . Then  $\int L_a \varphi e^S d\nu = 0$ . Integrating the generated flows shows  $e^S \nu$  is translation invariant; uniqueness of Haar measure gives  $\nu = Z^{-1} e^{-S} H$ , without assuming absolute continuity in advance.

For an independent link and an unnormalized staple  $A$ , write  $UA = a_0 I + i\mathbf{a} \cdot \boldsymbol{\sigma}$ . The generator  $\exp(it\sigma_a/2)$  gives  $L_a S_W = \beta_W a_a/2$  and  $L_a^2 S_W = \beta_W a_0/4$ , hence

$$\langle \beta_W^2 |\mathbf{a}|^2 - 3\beta_W a_0 \rangle = 0. \quad (\text{HNM-17})$$

The staple is not a unit quaternion. The identity assumes the varied link occurs once in each incident plaquette. On one  $\text{SU}(2)$  variable the wrong law  $(\delta_I + \delta_{-I})/2$  passes this residual

exactly but gives  $\langle L_a(u_0 u_a) - u_0 u_a L_a S \rangle = 1/2$ . The complete hierarchy characterizes the law; one measured residual does not. The six finite four-dimensional lattice chains at  $\beta_W = 0, 0.5, 2.2$  tested the declared nonabelian sampler and local identities. The  $\beta_W = 2.2$  autocorrelation evidence was insufficient. No chain supplied a continuum gap. Sources: `research/round9/advisor/advisor.md` and `research/round9/lattice/RESULTS.md`.

**An exact auxiliary transfer operator and the missing norm \*\*.** For central convolution by  $e^{\beta \cos \theta} / Z_\beta$  on full  $L^2(SU(2))$ , the class Haar measure is  $(2/\pi) \sin^2 \theta d\theta$ . With  $n = 2j + 1$ , character integration gives  $c_j = I_{n-1} - I_{n+1} = 2nI_n/\beta$ ; Schur orthogonality divides by  $n$ , yielding the complete spectrum

$$r_j = \frac{I_{2j+1}(\beta)}{I_1(\beta)}, \quad \text{mult}(r_j) = (2j+1)^2, \quad \Delta = -a_t^{-1} \log \frac{I_2(\beta)}{I_1(\beta)}. \quad (\text{HNM-18})$$

For  $\beta > 0$ , the positive integral representation identifies  $I_{n+1}/I_n$  with a mean of  $t \in (-1, 1)$  tilted by  $e^{\beta t}$ ; pairing  $t, -t$  proves this mean is strictly between zero and one. The constant is the unique stationary state. At fixed spin,  $-\log r_j = 2j(j+1)/\beta + O(\beta^{-2})$ . If  $\beta a_t \rightarrow c > 0$ , dominated convergence of square-summable Peter-Weyl coefficients gives  $T_\beta^{[t/a_t]} \rightarrow e^{-2tc/c}$  strongly, with rotor gap  $3/(2c)$ . Other choices of the clock produce zero or infinite limiting gaps. Independent endpoint gauge invariance of an open link forces constants, so this rotor is not the interacting gauge transfer operator. Negative  $\beta$  supplies a pointwise positive kernel with a negative fundamental eigenvalue, refuting positivity by inspection alone.

An additional exact counterexample isolates an operator-norm error. On  $N$  uniformly weighted atoms let  $B$  average the first two coordinates and fix the rest,  $\Pi$  average all coordinates, and  $v = (1, -1, 0, \dots)/\sqrt{2}$ . Set  $P_0 = (1-c)B + c\Pi$ ,  $P_1 = P_0 + \delta v v^T$ , with  $0 < \delta < 1-c < 1$ . Both are symmetric positive Markov matrices with strictly positive entries, yet for their density kernels  $K_i = NP_i$ ,

$$\|P_1 - P_0\|_{2 \rightarrow 2} = \delta, \quad \|K_1 - K_0\|_{L^1(H \otimes H)} = \frac{2\delta}{N}. \quad ** \quad (\text{HNM-19})$$

Thus product-space  $L^1$  control cannot replace uniform row and column control in Schur's test. For actual positive contractions sharing a vacuum,  $\|T|_{\Omega^\perp}\| \leq e^{-ma}$  and  $\|S - T\| \leq \varepsilon_a$  imply the conditional gap  $-a^{-1} \log(e^{-ma} + \varepsilon_a)$ , when its argument is below one. Telescoping gives  $\|S^n - T^n\| \leq n\varepsilon_a$ ; fixed-time convergence needs  $\varepsilon_a = o(a)$  by this route. Sources: `research/round9/transfer/PROOF.md` and `research/round9/advisor/advisor.md`.

## 5.4 Gauge reduction, infinite representation tails, and certified dynamics

**One-square certification \*\*.** For an open four-link square, all vertex Gauss laws reduce the physical space to class functions on  $SU(2)$ . Assume  $\alpha > 0$ ,  $\lambda \geq 0$ . The known character and radial reductions give

$$H_{nm} = [\alpha n(n+2) + \lambda] \delta_{nm} - \frac{\lambda}{2} (\delta_{n,m+1} + \delta_{n,m-1}), \quad n, m \geq 0, \quad (\text{HNM-20})$$

The four link Casimirs contribute  $4j(j+1) = n(n+2)$ , while  $2x\chi_{n/2} = \chi_{(n+1)/2} + \chi_{(n-1)/2}$ , with the negative-index character absent, fixes the off-diagonal factor. Equivalently, the Hamiltonian is  $-\alpha \partial_\theta^2 - \alpha + \lambda(1 - \cos \theta)$  on Dirichlet functions  $u = \sqrt{2/\pi} \sin \theta f$  on  $(0, \pi)$ . The Mathieu representation is established prior work [12]; the workbench's contribution is the explicit certified error contract. Compact resolvent and uniqueness for the second-order Dirichlet equation imply a simple ground state and a positive gap for fixed  $\alpha > 0$ . Retain  $n < N$ , let  $A = P_N H P_N$ ,  $\tau = \alpha N(N+2)$ , and choose a certified  $\mu_1(A) < R < \tau$ . The sole cross-boundary entry is  $-\lambda/2$ . Young's inequality and tail positivity give

$$B = A - \frac{(\lambda/2)^2}{\tau - R} |N-1\rangle \langle N-1|, \quad \mu_k(B) \leq E_k(H) \leq \mu_k(A), \quad k = 0, 1. \quad ** \quad (\text{HNM-21})$$

Exact rational Sturm inertia, including zero and endpoint policies, encloses these eigenvalues. After dropping the scalar  $\lambda I$ , bounded multiplication proves  $|\delta(\kappa) - \delta(\kappa')| \leq 2|\kappa - \kappa'|$  for  $\delta = \Delta/\alpha$ ,  $\kappa = \lambda/\alpha$ . A certified interval cover then gave  $\Delta \geq 0.999999\alpha$  on  $0 \leq \kappa \leq 10$ . Point samples alone would not prove this. A two-dimensional trial space localized in  $0 < \theta < w$  gives  $\Delta \leq 4\pi^2\alpha/w^2 + \lambda w^2/2 - \alpha$ , hence  $\Delta \leq 2\sqrt{2}\pi\sqrt{\alpha\lambda} - \alpha \rightarrow 0$  at fixed  $\lambda > 0$  as  $\alpha \downarrow 0$ , with the chosen  $w = (8\pi^2\alpha/\lambda)^{1/4} \leq \pi$  once  $\alpha/\lambda \leq \pi^2/8$ . This path is not a matched continuum trajectory. Source: `research/round10/advisor/advisor.md`.

**The interacting two-square quotient and its complete electric spectrum \*\*.** The canonical Hamiltonian framework is established [11]. The seven-link, six-vertex open graph has physical space  $L^2(SU(2)^2)^{\text{Ad,diag}}$ . Tree gauge leaves loops  $U, V$  based at the shared vertex. This planar spatial patch with continuous time is a finite  $2 + 1$ -dimensional regulator; increasing its polynomial cutoff neither adds spatial cells nor produces the four-dimensional target. The complete orbit coordinates are  $x = \text{Tr } U/2$ ,  $y = \text{Tr } V/2$ ,  $z = \text{Tr}(UV)/2$ , with

$$\Omega = \{|x|, |y| \leq 1 : 1 - x^2 - y^2 - z^2 + 2xyz \geq 0\}, \quad d\mu = \frac{2}{\pi^2} dx dy dz. \quad ** \quad (\text{HNM-22})$$

To derive it, write the  $SU(2)$  quaternions as  $(x, \mathbf{a})$  and  $(y, \mathbf{b})$ . The three traces fix their Gram matrix, and two vector pairs with that Gram matrix differ by a proper rotation. The scalar marginals are semicircular and the relative angle has uniform cosine; the Jacobian to  $z = xy - \mathbf{a} \cdot \mathbf{b}$  cancels both square roots. This proves the measure as well as orbit classification, including central pairs. The pair  $(x, y)$  alone loses the relative orientation.

The shared derivative acts as  $L_U - R_V$ , so the kinetic operator is  $K = 3(C_U + C_V) - \sum_a (L_U^a - R_V^a)^2$ , rather than two independent rotors. In trace coordinates,

$$K = -\text{div}(A\nabla), \quad A = \begin{pmatrix} 1 - x^2 & (z - xy)/4 & 3(y - xz)/4 \\ (z - xy)/4 & 1 - y^2 & 3(x - yz)/4 \\ 3(y - xz)/4 & 3(x - yz)/4 & 3(1 - z^2)/2 \end{pmatrix}. \quad ** \quad (\text{HNM-23})$$

Its domain is inherited from invariant  $H^2(SU(2)^2)$ , equivalently the closure of the polynomial core, not Dirichlet data on the quotient boundary. The checks  $Kz = 9z/2$  and  $K(xy) = 13xy/2 - z/2$  detect the shared link. For a monomial  $x^a y^b z^c$ , its degree-preserving coefficient is

$$\epsilon_{abc} = a^2 + b^2 + \frac{3}{2}c^2 + \frac{1}{2}ab + \frac{3}{2}c(a + b) + 2a + 2b + 3c, \quad m_d = \frac{5}{8}d^2 + 2d + \frac{3}{8}(d \bmod 2). \quad ** \quad (\text{HNM-24})$$

All other terms lower degree. Symmetry makes each orthogonal polynomial shell reduce  $K$ ; the diagonal quotient coefficients are its eigenvalues. Density of polynomials proves completeness. Writing  $j = (a + c)/2$ ,  $k = (b + c)/2$ ,  $\ell = (a + b)/2$  gives  $\epsilon = 3j(j + 1) + 3k(k + 1) + \ell(\ell + 1)$ , independently identifying the spin-network spectrum. Its low electric levels agree with the independent two-plaquette construction in [13]; this is a comparison of those levels, not an identification of all intermediate gauge-fixed formulas. At fixed  $d = a + b + c$ , the minimum has  $c = 0$  and  $a, b$  as equal as possible, giving  $m_d$ . Thus a degree cutoff  $D$  has dimension  $\binom{D+3}{3}$  and exact omitted threshold  $m_{D+1}$ . Sources: `research/round10/skeptic/coordinate-note.md` and `research/round11/advisor/advisor.md`.

**Interacting gap enclosures and the cost of a global bound \*\*.** For  $H = \alpha K + \lambda_1(1 - x) + \lambda_2(1 - y)$ , with  $\alpha > 0$ ,  $\lambda_1, \lambda_2 \geq 0$ , bounded perturbation and positivity improvement on the connected compact group give compact resolvent and a unique positive physical ground state. With  $P = P_D$ ,  $M = -\lambda_1 x - \lambda_2 y$ , and  $\tau = \alpha m_{D+1}$ , the exact boundary form is finite:

$$CC^* = PM(P_{D+1} - P_D)MP, \quad B = PHP - \frac{CC^*}{\tau - R}, \quad \mu_k(B) \leq E_k(H) \leq \mu_k(PHP), \quad k = 0, 1. \quad ** \quad (\text{HNM-25})$$

Here  $\mu_1(PHP) < R < \tau$ . Multiplication raises degree by at most one; Young's inequality then proves the same form comparison as (HNM-21). In raw monomials, inertia must be computed for  $H_{\text{form}} - rG$ , using the Haar Gram matrix  $G$ . Its entries are exact: if  $\mu_{2r} = \text{Catalan}(r)/4^r$ ,  $\mu_{2r+1} = 0$ , and  $J(n, h) = \sum_{q=0}^h (-1)^q \binom{h}{q} \mu_{n+2q}$ , then

$$\langle x^a y^b z^c \rangle = \sum_{h=0}^{\lfloor c/2 \rfloor} \frac{\binom{c}{2h}}{2h+1} J(a+c-2h, h) J(b+c-2h, h). \quad (\text{HNM-26})$$

This follows from the same relative-angle integration. The gap is 2-Lipschitz in the  $\ell^1$  distance of  $(\lambda_1/\alpha, \lambda_2/\alpha)$ . Exact cover certificates gave a floor exceeding  $0.96\alpha$  on  $[0, 2]^2$ , with recorded lower endpoint  $0.963936567779819\alpha$ . At  $(\alpha, \lambda_1, \lambda_2, \rho) = (1, 2, 3, 1)$ , the recorded full-space gap interval was  $[3.362386061099721, 3.362408390681595]$ .

An analytic estimate exposed the volume issue before larger computation. The  $\text{SU}(2)$  heat kernel at Casimir time 4 lies between  $3/4$  and  $5/4$ : its nonconstant character sum is bounded by  $4e^{-3} + \sum_{n \geq 2} (2/25)^n < 1/4$ . For the two-square kinetic operator, time  $4/3$  therefore gives kernel bounds  $9/16, 25/16$ . Feynman–Kac bounds a positive ground state's ratio by  $(25/9)e^{8(\lambda_1+\lambda_2)/(3\alpha)}$ . The ground-state transform and free physical Poincaré constant 3 give

$$\Delta \geq \frac{243}{625} \alpha e^{-16(\lambda_1+\lambda_2)/(3\alpha)}. \quad ** \quad (\text{HNM-27})$$

On any finite connected open hypercubic graph containing a square, the free physical gap is exactly  $3\alpha$ : a nonzero spin support has no degree-one vertex, hence contains at least four edges, each costing at least  $3/4$ ; a fundamental square attains the bound. Repeating the global heat comparison for  $E$  links gives  $\Delta_\Gamma \geq 3\alpha(3/5)^{2E} e^{-16 \sum_p \lambda_p/\alpha}$ . Its deterioration is not a proof of gap closure. For  $q$  disconnected copies, tensor eigenvectors give the exact same first gap as one copy while the global density-ratio bound decays exponentially in  $q$ .

There is also a direct measure obstruction. For the one-square Hamiltonian,

$$\frac{[4C + \kappa(1-x)]e^{\theta x}}{e^{\theta x}} = \kappa - \theta^2 + (3\theta - \kappa)x + \theta^2 x^2. \quad ** \quad (\text{HNM-28})$$

It is constant only for  $\theta = \kappa = 0$ . The square root of a nontrivial Wilson Gibbs density is thus not this Hamiltonian's ground state. A Langevin Poincaré theorem for the former cannot silently bound the latter; the fixed-spacing strong-coupling results of [24] retain their own measure, normalization, and generator hypotheses. For a declared Hamiltonian convention,  $\alpha = g_H^2/(2a)$ ,  $\lambda = 2/(g_H^2 a)$ , so  $\lambda/\alpha = 4/g_H^4$ ; neither a fixed graph nor a compact coupling square follows a growing-volume weak-coupling continuum limit. Sources: `research/round11/advisor/advisor.md` and `research/round12/volume/volume-bridge.md`.

A possible replacement must control the actual ground state. The conditional Bakry–Émery bound  $\text{Ric} - 2 \text{Hess} \log \phi_\Gamma \geq c_* g$ , uniformly in volume, would give  $\Delta_\Gamma \geq \alpha c_*$  through the ground-state generator  $\Delta + 2 \nabla \log \phi_\Gamma \cdot \nabla$ . No such bound was established. Similarly, transferring  $A_j \geq m_*(I - P_j)$  to a continuum operator needs a common realization, strong convergence of rank-one vacuum projections, and a limiting semigroup strongly continuous at zero. The example  $A_j = j(I - P)$  has  $e^{-tA_j} \rightarrow P$  for each  $t > 0$ , but no strongly continuous limit on the full space. Lower-gap control alone can leave only a vacuum; nontriviality is a separate obligation.

**A full-Hilbert-space dynamical error bound \*\*.** The earlier driven one- and two-square histories were finite-Galerkin studies. Round 12 added the omitted representation error. Let  $H(t) = \alpha(t)K_\rho + \lambda_1(t)(1 - X) + \lambda_2(t)(1 - Y)$ ,  $K_\rho = 3(C_U + C_V) - \rho \sum (L_U - R_V)^2$ , with fixed  $\rho > 0$ . Remove the common scalar phase. The electric propagator preserves every degree shell and the remaining bounded interaction has norm at most  $w = |\lambda_1| + |\lambda_2|$ , with shell bandwidth

one. Real locally integrable coefficients suffice for mild unitary evolution. For normalized initial support in  $P_{d_0}$ , let  $\psi_D$  be exact Galerkin evolution and  $A(t) = \int_0^t w(s) ds$ . Duhamel in the finite tail block gives inductively  $\|(P_D - P_{n-1})\psi_D\| \leq A^{n-d_0}/(n-d_0)!$ . The only full-equation defect is  $r_D = \Pi_{D+1}W\Pi_D\psi_D$ . One more integral yields

$$\|\psi(t) - \psi_D(t)\| \leq \min \left\{ 2, \frac{A(t)^{D-d_0+1}}{(D-d_0+1)!} \right\}. \quad ** \quad (\text{HNM-29})$$

No exponential factor is needed because both block propagators are unitary. Equal action bounds do not identify different protocols. For a computed path, add its preparation error and the integral of  $\|i\dot{\tilde{\psi}}_D - H_D\tilde{\psi}_D\|$ . Norm conservation alone does not bound that residual. The boundary norm has the exact Gram expression  $\|r_D\|^2 = c^*[N - MG^{-1}M]c$ , where  $M, N$  are the form matrices of  $P_DWP_D, P_DW^2P_D$ .

For a finite step of length  $h$ , midpoint  $B = H_D(m)$ ,  $C = H'_D(m)$ , and  $M_2 = \sup \|H''_D\|$ , a Taylor comparison followed by two unitary Duhamel integrals gives

$$\|U_D - e^{-ihB}\| \leq h^3 M_2/24 + h^3 \| [B, C] \| /12 + h^4 \|C\|^2 /32. \quad ** \quad (\text{HNM-30})$$

The first term integrates the quadratic Taylor remainder. The centered linear integral cancels its constant interaction-picture part, leaving the commutator term; the second Duhamel remainder is half the squared integrated linear norm. Compressed  $X, Y$  need not commute. Evaluating the exponential is a separate error: for a finite selfadjoint  $B$ , its order- $q$  Taylor remainder is at most  $(h \|B\|)^{q+1}/(q+1)!$ . The archived exact rational two-step state used degree 3, Taylor order 100,  $(\lambda_1, \lambda_2) = (1/20, 1/10)$  then  $(1/10, 1/20)$ , each for duration 1, and  $\alpha = \rho = 1$ . With  $e = (114/5)^{101}/101!$ , its total bound was  $27/80000 + 2e + e^2$ , including the product error and the unnormalized computed state. This certifies that declared protocol, not all historical drives. Bounded-observable errors are at most  $2\|O\|$  times a normalized state error; energy needs its own work identity and cannot use a fictitious global norm of  $H$ . Sources: `research/round12/adv` `isor/advisor.md` and `research/round12/solver/output/exact_step_certificate.json`.

**Closure tests that retain the missing fluctuation** \*\*. For the one-square Euclidean measure proportional to  $e^{\kappa x} \sqrt{1-x^2} dx$ , integration of  $\partial_x[(1-x^2)^{3/2} e^{\kappa x} f]$  gives, with  $m_n = \langle x^n \rangle$ ,

$$nm_{n-1} - (n+3)m_{n+1} + \kappa(m_n - m_{n+2}) = 0, \quad 3m_1 = \kappa(1 - m_1^2 - v), \quad v \geq \frac{1}{4}e^{-2|\kappa|} > 0. \quad ** \quad (\text{HNM-31})$$

The last inequality follows from domination of Haar density by  $e^{-2|\kappa|}$  and  $\text{Var}(x) = \inf_c \langle (x-c)^2 \rangle$ . Discarding  $v$  while keeping the true mean leaves residual  $\kappa v$ . At  $\kappa = 0$ , the next equation, not the first, enforces  $m_2 = 1/4$ . Adding a fluctuation symbol does not compute its next moment. The same audit checked specific displayed formulas of `arXiv:2608.05415v1` [26]: directed versus unordered plaquette counting, a derivative prefactor, a missing source phase, and the rank impossibility  $\eta^T g \eta = I_{N_c^2-1}$  when  $N_c^2 - 1 > D$ . These are local, version-specific tests and do not refute all closures. Source: `research/round12/closure/closure-audit.md`.

**Change in understanding and the next obligations.** By Round 12 the reliable object had narrowed from an intended coupled Einstein–QED system to explicit finite models with named observables, domains, and error norms. The abelian branch still required a common renormalized current and anisotropic stress for a specified state, followed by fluctuation and regulator control. The nonabelian branch had genuine infinite-representation results on finite graphs, including interacting stationary and driven bounds. Its remaining tasks were connected-volume estimates for the actual physical Hamiltonian, scale matching, nontrivial continuum observables, reconstruction, and a spectral exclusion covering the whole physical sector. Neither the formal proof planner

nor the number of passed checks supplied those premises. The machine-readable early ledger preserves every historical entry, retained failure, source path, and contribution mapping; the present manuscript synthesis does not claim to rerun every historical calculation.

## 6 Finite Haar constraints and connected lattice integrals

Rounds 13–22 developed several distinct mathematical models. Throughout this section,  $x_p = \text{Tr}(U_p)/2$  denotes a normalized fundamental Wilson trace,  $\chi_p = 2x_p$  its ordinary character, and  $C_e$  the positive link Casimir with eigenvalues  $j(j+1)$ . A finite physical Hamiltonian has the convention

$$H = \alpha \sum_e C_e - \sum_p \lambda_p x_p, \quad \alpha > 0.$$

Here  $\alpha, \lambda_p$  have energy units. Euclidean coefficients  $\kappa_p$ , introduced in densities proportional to  $\exp(\sum_p \kappa_p x_p)$ , are dimensionless and do not define a physical clock. Every spectral statement below retains a fixed positive energy reference  $E_*$ . The symbol **\*\*** identifies a result derived or computed in the workbench; it does not assert priority over the literature. The audit ledger accounts for all 62 historical entries in this range, including four named Round13 studies and 58 numbered loops.

### 6.1 Compact moments, response, and complete interval coverage **\*\***

The one-holonomy model uses normalized Haar pushforward

$$d\mu_\kappa(x) = Z_\kappa^{-1} e^{\kappa x} \frac{2}{\pi} \sqrt{1-x^2} dx, \quad -1 \leq x \leq 1.$$

Writing  $m_n = \int x^n d\mu_\kappa$ , integration of the derivative of  $x^n e^{\kappa x} (1-x^2)^{3/2}$  gives the familiar identity

$$nm_{n-1} - (n+3)m_{n+1} + \kappa(m_n - m_{n+2}) = 0. \quad (\text{HNM-32})$$

The first term is absent at  $n = 0$ . For nonzero  $\kappa$ , normalization and  $m_1 = u$  express every moment affinely in  $u$ . At zero coupling one instead uses  $m_{2j} = C_j/4^j$ ,  $m_{2j+1} = 0$ , where  $C_j$  is a Catalan number. Positivity supplies

$$(m_{i+j})_{i,j=0}^r \succeq 0, \quad (m_{i+j} - m_{i+j+2})_{i,j=0}^{r-1} \succeq 0.$$

The workbench implementation retains rational vectors whose quadratic forms are affine inequalities in  $u$ , so rejected means have exact polynomial witnesses. Its 28 finite certificates include a level-six interval at  $\kappa = 1$  of width approximately  $1.40631 \cdot 10^{-13}$ ; the analogous width at  $\kappa = 5$  is approximately  $6.03663 \cdot 10^{-8}$ . These are enclosure widths, not sampling errors.

Completeness is a separate argument. All moment and support inequalities produce a probability measure on  $[-1, 1]$ , using multiplication by  $x$  as a self-adjoint contraction on the polynomial completion. All recurrence identities extend to  $C^1$  tests and imply

$$D[(1-x^2)\mu] = [\kappa(1-x^2) - 3x]\mu.$$

The interior density is proportional to  $e^{\kappa x} \sqrt{1-x^2}$ ; the equation excludes endpoint atoms. Consequently the full hierarchy identifies the desired measure. Nested finite feasible intervals shrink to a point, without a proved rate. Successful finite computations additionally require vanishing optimization slack; the delivered level-six cap does not execute this limit.

Differentiation with respect to the coupling gives  $u' = \text{Var}(x) > 0$ . Combining this with (HNM-32) yields

$$\kappa u' + 3u = \kappa(1-u^2), \quad u(0) = 0, \quad u'(0) = \frac{1}{4}.$$

This is a static response equation. The regular solution is  $u = I_2(\kappa)/I_1(\kappa)$ ; singular modified-Bessel branches violate boundedness at the origin. Replacing  $3u$  by  $2u$  changes the prior to uniform measure on  $[-1, 1]$ , rather than improving the Haar approximation. The project also derived a useful residual certificate: fix  $0 < \kappa_0 \leq \kappa$  and a differentiable approximation  $p \geq 0$  throughout  $[\kappa_0, \kappa]$ , with  $r = p' - (1 - p^2 - 3p/\kappa)$ ,

$$|p(\kappa) - u(\kappa)| \leq \left(\frac{\kappa_0}{\kappa}\right)^3 |p(\kappa_0) - u(\kappa_0)| + \kappa^{-3} \int_{\kappa_0}^{\kappa} s^3 |r(s)| ds. \quad ** \quad (\text{HNM-33})$$

It follows by an integrating factor, since the error damping contains  $3/\kappa + u + p \geq 3/\kappa$ . A certified launch error therefore does not bound a later floating trajectory unless the entire residual is enclosed. The point-mass proposal  $m_n = (1/3)^n$  at  $\kappa = 9/8$  passes positivity and the first identity but has next-identity residual  $8/9$ . This supplies an exact counterexample to a zero-variance closure.

Rounds14–15 moved to the genuine two-holonomy density  $Z^{-1} e^{x+y+\eta z} dU dV$ , with  $z = \text{Tr}(UV)/2$ . For  $C(\eta) = \text{Cov}_\eta(x, y)$ , differentiating the normalized quotient gives

$$C'(\eta) = \mathbb{E}_\eta[(x - \mathbb{E}_\eta x)(y - \mathbb{E}_\eta y)(z - \mathbb{E}_\eta z)], \quad |C'| \leq 2.$$

The last inequality uses  $|z - \mathbb{E}z| \leq 2$ , Cauchy–Schwarz, and  $\text{Var}(x), \text{Var}(y) \leq 1$ . The first loop’s small- $\eta$  estimate did not resolve the chosen point  $(1, 1, 1/4)$ . Exact total-degree Taylor integration did. For any bounded insertion and action  $|S| \leq M < N + 2$ , the standard tail

$$R_N(M) = \frac{M^{N+1}}{(N+1)!} \frac{1}{1 - M/(N+2)}$$

bounds numerator and partition separately. Jensen gives  $Z \geq 1$  when the unweighted action mean vanishes; signed interval division retains all endpoint combinations. Point intervals  $[L, U]$  extend across a cell of radius  $\rho$  as  $[L - 2\rho, U + 2\rho]$ . All eight initial midpoint intervals were positive, while all eight cell bounds failed. Refining only the failing cells produced 8, 16, then 21 cells, with a final minimum lower bound approximately 0.0003346525 over the whole closed interval  $\eta \in [1/8, 1/4]$ . Increasing already adequate point precision would not have removed the geometric transport loss.

These results support certified static response calculations and provide countermodels to incomplete closures. Their next application requires shared-link identities for larger graphs, not an identification of  $\eta$  with time or energy. Sources: `research/round13/moments/README.md`, `research/round13/response/response.md`, `research/round14/forward/loop2.md`, `research/round15/forward/A2/result.md`.

## 6.2 Closed-surface gluing and the first shared face \*\*

On the closed cube there are eight vertices, twelve links and six faces. For  $\chi_n$ , the spin- $n/2$  character of dimension  $d_n = n + 1$ , write  $f_p = \sum_n a_{p,n} \chi_n$ . Schur orthogonality forces equal labels on adjacent faces. Each edge contributes  $d_n^{-1}$ , each surviving vertex index  $d_n$ , giving the graph-specific application

$$\int \prod_{p=1}^6 f_p(U_p) dU = \sum_{n \geq 0} d_n^{-4} \prod_{p=1}^6 a_{p,n}, \quad \mathbb{E}_0 \prod_{p=1}^6 x_p = 2^{-10}. \quad ** \quad (\text{HNM-34})$$

The identity is immediate for finite character polynomials; analytic weights require convergence or the retained Taylor remainder. If any one face weight equals one, only label zero survives and the other five class observables factorize. Thus five-face factorization coexists with a nonzero six-face moment. It would be incorrect to use this as evidence for independent face holonomies.

For  $e^{\kappa x}$ ,

$$a_n(\kappa) = \sum_{j \geq 0} \frac{(n+1)\kappa^{n+2j}}{2^{n+2j} j! (n+1+j)!}.$$

An insertion of one  $x_p$  differentiates that face's coefficient before setting all couplings equal. It does not differentiate the common-coupling partition six times. At  $\kappa = 1/4$ , total degree18 first met the specified expectation precision; degree24 gave  $\mathbb{E} \prod_p x_p \simeq 0.001023116359923$  and a partition excess over the independent-face product exceeding  $2.4598180 \cdot 10^{-7}$ . The exact fractions, including the failed coarse degrees, remain in the source certificate.

Two adjacent cubes have twelve vertices, twenty links and eleven faces. Conditioning on their shared holonomy  $H$ , each five-face disk contributes  $\sum_n d_n^{-4} (\prod_{\text{disk}} a_{p,n}) \chi_n(H)$ . The actual integral is therefore

$$\sum_{n,m,k} \frac{\prod_{p \in L} a_{p,n}}{d_n^4} \frac{\prod_{p \in R} a_{p,m}}{d_m^4} a_{s,k} N_{nm}^k, \quad N_{nm}^k = \mathbf{1}_{\{|n-m| \leq k \leq n+m, n+m+k \text{ even}\}}. \quad ** \quad (\text{HNM-35})$$

The shared-edge incidence is three, so the old one-surface single-label rule is insufficient. Center parity makes the all-eleven distinct-trace Haar moment zero. With the shared trace squared, both fusion channels  $k = 0, 2$  contribute and  $\mathbb{E}_0[(\prod_{\text{outer } 10} x_p) x_s^2] = 2^{-19}$ . Removing  $k = 2$  halves the answer.

Keeping the same all-eleven-trace observable, Round16 compared outer couplings  $1/8$  and shared coupling  $1/8$  against shared coupling zero. The degree24 difference is enclosed near  $2.4303274916 \cdot 10^{-7}$ , with width about  $4.173 \cdot 10^{-22}$ . The zero shared-action baseline is nonzero: its leading numerator at common outer coupling  $t$  is  $41t^5/(81 \cdot 2^{18})$ . Deleting a coefficient is not deleting an insertion. These finite integrals expose graph dependence and supply later trial matrix elements; they provide no transfer Hamiltonian by themselves. Sources: `research/round15/forward/B1/derivation.md`, `research/round15/forward/B2/derivation.md`, `research/round16/forward/loop1/derivation.md`, `research/round16/forward/loop2/derivation.md`.

### 6.3 Recoupling, sufficient coordinates, and surrounding weights \*\*

The four-cube  $2 \times 2 \times 1$  complex has 18 vertices, 33 links and 20 faces. Integrating its central link in the adjoint representation requires the complete invariant projector on  $(\mathbb{R}^3)^{\otimes 4}$ . The three invariant tensors  $B_1 = \delta_{ab}\delta_{cd}$ ,  $B_2 = \delta_{ac}\delta_{bd}$ ,  $B_3 = \delta_{ad}\delta_{bc}$  have Gram matrix with diagonal nine and off-diagonal three. The Clebsch–Gordan decomposition  $1 \otimes 1 = 0 \oplus 1 \oplus 2$  proves completeness, and Haar averaging gives  $P = B(B^T B)^{-1} B^T$ . The inverse has diagonal  $2/15$  and off-diagonal  $-1/30$ . For an explicitly realized tetrahedral boundary, the product of four normalized adjoint traces has expectation  $-1/405$ ; diagonal-only and one-channel substitutes instead give  $2/405$  and  $1/729$ . The negative entries cannot be dropped.

Writing the central quaternion as  $q \in S^3$ , face traces as  $q \cdot a_i$ , and the action as  $q \cdot b$ , with  $b = \sum_i \kappa_i a_i$ , the partition depends only on  $\|b\|$ . The joint observable also depends on the directions  $a_i$ . Tetrahedral and commuting boundaries can have the same  $b$  but opposite-sign joint expectations. The complete Gram matrix of  $(b, a_1, \dots, a_4)$  is sufficient: equality of Grams defines an isometry between the spans, extends to an element of  $O(4)$ , and preserves Haar sphere measure. This proof includes singular Grams and  $b = 0$ , without inverting a Gram matrix. Admissibility requires positive semidefiniteness, rank at most four, unit  $a_i$ , and both action relations

$$G_{0i} = \sum_j \kappa_j G_{ji}, \quad G_{00} = \sum_{i,j} \kappa_i \kappa_j G_{ij}.$$

An exact computational realization uses

$$M(k) = \frac{\sum_j (k_j - \delta_{ij}) G_{ij} M(k - e_i - e_j)}{|k| + 2}, \quad |k| > 0 \text{ even, } k_i > 0, \quad ** \quad (\text{HNM-36})$$

with  $M(0) = 1$  and odd moments zero. A term of zero multiplicity is omitted before its reduced multi-index is formed, so no negative exponent is evaluated. Gaussian integration by parts followed by radial separation derives this recurrence; its project contribution is the complete action/observable implementation and audit, not a new sphere-moment principle.

Integrating one actual surrounding link changes the problem again. For  $x = \text{Tr } U/2$ ,  $y = \text{Tr } V/2$ , and  $w = \text{Tr}(UV^\dagger)/2$ , character convolution gives

$$\mathbb{E}[x^a y^b w^c] = 2^{-a-b-c} \sum_n \frac{m(a, n) m(b, n) m(c, n)}{n+1}, \quad m(a, n) = \binom{a}{(a-n)/2} - \binom{a}{(a-n)/2-1}.$$

In particular  $\mathbb{E}[xyw] = 1/16$  and  $\mathbb{E}[x^2 y^2 w^2] = 1/48$ . The actual six affected faces give  $S_2 = 3x + 2y + w$ , with  $O = (4x^2 - 1)^3(4w^2 - 1)/81$ . Its normalized expectation at  $\kappa = 1/64$  is enclosed near  $1.95552781 \cdot 10^{-6}$ . The surrounding marginal is the semicircle measure multiplied by  $e^{2\kappa y} Z_U(y)$ ; averaging normalized inner expectations against bare Haar discards both weights.

Unfreezing a third actual link gives  $S_3 = 3x + y + z + w + t$ , where  $t = \text{Tr}(VW^\dagger)/2$ . The same  $V$  occurs in both branches:  $\mathbb{E}[xzw t] = 1/64$ , whereas resampling it independently gives zero. Round19 integrated all required monomials through degree16, retained the full Taylor tails, and obtained a continuous result for  $F(\kappa) = \mathbb{E}[O e^{\kappa S_3}] / \mathbb{E}[e^{\kappa S_3}]$ :

$$F(0) = 0, \quad F(\kappa) \geq \kappa^2/2048 \quad (|\kappa| \leq 1/8), \quad F(\kappa) > F(-\kappa) \quad (0 < \kappa \leq 1/64). \quad ** \quad (\text{HNM-37})$$

Here  $N_2 = 1/324$ ,  $N_3 = 13/1296$  for the numerator. Subtracting absolute higher-order coefficients and its complete tail bounds  $N/\kappa^2$ ; an upper partition bound gives the displayed direction of the quotient. The sign comparison uses  $N(\kappa)Z(-\kappa) - N(-\kappa)Z(\kappa)$ , whose leading term is  $13\kappa^3/648$ . This conditional three-link result still fixes thirty links. Full surrounding integration and physical dynamical matching remain separate tasks. Sources: `research/round17/forward/c1/report.md`, `research/round18/forward/c1/report.md`, `research/round18/forward/c2/report.md`, `research/round19/forward/c1/report.md`, `research/round19/forward/c2/report.md`.

## 7 Physical spectral bounds and an explicit infinite-volume exception

### 7.1 Finite untruncated graph bounds and complete complements \*\*

On a connected finite simple graph with Gauss law at every vertex and no external charges, a nonconstant spin network has no degree-one active vertex. Its support therefore contains a cycle. Every occupied link costs at least  $3\alpha/4$ ; a fundamental character on a shortest cycle attains the resulting energy. For graph girth  $g$ ,  $\delta_G = 3\alpha g/4$  is the exact free physical gap. A forest has only the vacuum physical sector, not a numerical first excitation.

Let  $L = \sum_p |\lambda_p|$ . Bounded multiplication gives  $E_1(H) \geq \delta_G - L$ , while the vacuum trial gives  $E_0(H) \leq 0$ . For square graphs, the orthonormal trial vectors  $1, \chi_p$  have electric energies  $0, 3\alpha$  and magnetic vacuum-to-face entries  $-\lambda_p/2$ . The relevant third Haar moments vanish. Diagonalizing only this arrowhead yields a ground-energy *upper* bound. Combining it with the independent full-space excited-energy bound gives

$$\Delta(H) \geq \frac{\delta + \sqrt{\delta^2 + \sum_p \lambda_p^2}}{2} - L, \quad \delta = 3\alpha. \quad ** \quad (\text{HNM-38})$$

This is not a Ritz-gap lower bound. On the six-face cube, it expands the common-magnitude sufficient range from  $r = |\lambda|/\alpha < 1/2$  to  $r < 12/23$ , and gives  $(\sqrt{21/2} - 3)\alpha/2 > 0$  at  $r = 1/2$ . On the eleven-face two-cube graph it gives  $r < 12/43$ . At that latter boundary an actual

shared-face adjoint state  $\chi_2(U_s) = 4x_s^2 - 1$ , of electric energy  $8\alpha$ , repairs the zero margin. With  $v_0 = 1 + \sum_p \chi_p/22$ ,  $v_\eta = v_0 + \eta\chi_2(U_s)$ ,

$$\frac{\Delta}{\alpha} \geq \frac{(6/473)\eta - (347/43)\eta^2}{45/44 + \eta^2} > 0 \quad (0 < \eta < 6/3817). \quad **$$

At  $\eta = 3/3817$  the exact bound is  $1388/284767457$ . This amplitude maximizes the quadratic numerator, not the quotient.

The next improvement required a complete complement, rather than a larger trial space alone. Round18 classified every physical state below  $9\alpha/2$ : only the vacuum and eleven fundamental faces occur. For their projector  $P$ ,  $QH_0Q \geq (9\alpha/2)Q$ , attained by a six-link rectangle. The full cross Gram is

$$(QVP)^*(QVP) = PV^2P - (PVP)^2, \quad C_{00} = C_{0p} = 0, \\ C_{pp} = \frac{1}{4} \sum_q \lambda_q^2, \quad C_{pq} = \lambda_p \lambda_q / 4 \quad (p \neq q).$$

Both terms in the subtraction matter:  $V\Omega$  already lies in  $P$ . For  $|\lambda_p| \leq r\alpha$ ,  $\|QVP\|^2 \leq 21\alpha^2 r^2/4$ . On the codimension-one space  $\Omega^\perp$ , comparison with

$$\alpha \begin{pmatrix} 3 & -\sqrt{21}r/2 \\ -\sqrt{21}r/2 & 9/2 - 11r \end{pmatrix}$$

gives the full  $E_1$  lower bound. At  $r = 3/8$ , it proves  $\Delta/\alpha \geq (27 - 3\sqrt{70})/16 > 0$  for the entire signed box.

Round19 raised the strict electric projector to  $E_{el} < 6\alpha$ . Its complete dimension is 48: vacuum, eleven four-cycles and thirty-six six-cycles. At the excluded threshold  $6\alpha$ , 99 support/label assignments represent 107 physical channels, because eight assignments have intertwiner multiplicity two. Counting supports alone would miss this distinction. The full 48-channel cross Gram gives  $\|QVR\|^2 \leq 19\alpha^2 r^2/4$ , with  $R = P - P_\Omega$ , while  $RHR \geq \alpha(3 - 2r)$  and  $QH_0Q \geq \alpha(6 - 11r)$ . Hence

$$\frac{\Delta(H)}{\alpha} \geq R(r) = \frac{9 - 13r - \sqrt{100r^2 - 54r + 9}}{2} > 0, \quad 0 \leq r < \frac{30 - 2\sqrt{87}}{23}. \quad ** \quad (\text{HNM-39})$$

In particular  $R(3/8) = (33 - 6\sqrt{5})/16$  and  $R(7/16) = 19/32$ . The failed  $r = 1/2$  estimate is a failure of this row-envelope certificate, not a proved gap closure.

These calculations yield full untruncated finite-graph theorems because the omitted physical spectrum is bounded analytically. They do not give a dense volume-uniform threshold. Under the usual lattice normalization  $\alpha = g_H^2/(2a)$ ,  $\lambda = 2/(g_H^2 a)$ , one has  $r = 4/g_H^4$ , so fixed small- $r$  regimes do not follow a weak-bare-coupling path  $g_H \rightarrow 0$  [12]. Sources: `research/round15/forward/C2/derivation.md`, `research/round17/forward/b2/report.md`, `research/round18/forward/b1/report.md`, `research/round19/forward/b1/report.md`, `research/round19/forward/b2/report.md`.

## 7.2 Dressed strips, literal boundaries, and summable perturbations \*\*

A different route obtains volume-uniform results by an explicit restriction of the interaction family. Pairwise link-disjoint nonzero plaquettes factorize the *full* link Hilbert space into four-link blocks and free links. The four-link free gap before Gauss restriction is  $3\alpha/4$ , not the physical loop energy  $3\alpha$ . With  $\max |\lambda_p|/\alpha \leq \rho < 3/4$ , min-max and zero vacuum mean give  $\Delta \geq \alpha(3/4 - \rho)$ . Each unique block ground is gauge invariant: its gauge action is a continuous one-dimensional representation of a finite product of  $SU(2)$ , hence trivial. The physical restriction contains the same ground and inherits the lower bound without factorizing the physical Hilbert space.

Three consecutive square faces form a ten-link strip. Dress its two link-disjoint ends first and add the middle bridge with coefficient  $\mu$ . The bridge contains a free Haar link in this reference, so its mean is zero and its squared norm on the reference ground is  $1/4$ . Thus

$$\Delta_{\text{strip}} \geq \alpha \left( \frac{3}{4} - \max(|\lambda_L|, |\lambda_R|)/\alpha \right) - |\mu| \geq \alpha/8 \quad \begin{cases} |\lambda_L|, |\lambda_R| \leq \alpha/2, \\ |\mu| \leq \alpha/8. \end{cases} \quad **$$

The zero-mean premise saves one perturbation norm; a generic bounded perturbation would lose two.

Complete strips anchored at  $(4i, 2j, z)$  are link-disjoint. Round18's finite-box rule initially classified incomplete boundary strips as remainder faces; a fixed face could consequently change coefficient when the box grew. Round19 replaced that rule by literal restrictions of a single infinite assignment. All possible clipped components  $L, M, R, LM, MR, LMR$  retain the same bound  $\alpha/8$ , using the bare vacuum for proper clips and dressed ends for the complete strip.

In the incomplete tensor product of complete-strip grounds and free Haar links, define

$$H_{\text{ref}} = \sum_C (H_C - E_C) + \sum_{e \text{ free}} \alpha C_e, \quad H_{\text{ref}} \geq \delta(1 - P_\Omega), \quad \delta = \alpha/8.$$

This is the closed nonnegative form sum on the finite-excitation core. Give every omitted face of anchor  $(x, y, z)$  the coefficient  $\nu_f = \alpha\tau 2^{-x-y-z}/24$ . The exact geometric sums are

$$\sum_{\text{all}} w_f = 1, \quad \sum_{\text{selected}} w_f = 28/135, \quad \sum_{\text{omitted}} w_f = 107/135, \quad \|V\| \leq \beta = \alpha|\tau| 107/135. \quad **$$

(HNM-40)

Every omitted face has a free Haar link, so  $\langle \Omega, V\Omega \rangle = 0$ . On  $\Omega^\perp$ ,  $H_{\text{ref}} + V \geq \delta - \beta$ , whereas the vacuum trial energy is zero. If  $\beta < \delta$ , a spectral subspace below  $\delta - \beta$  is nonempty and at most one-dimensional. This proves an isolated simple ground without invoking infinite-volume compact resolvent. At  $|\tau| = 1/64$ ,

$$\Delta \geq 973\alpha/8640. \quad ** \quad \text{(HNM-41)}$$

All actual face coefficients can be nonzero, but the omitted ones decay with position. This is a fully supported inhomogeneous exception.

Round20 completed each finite retained perturbation to whole reference factors. If  $t_L \downarrow 0$  is its exact omitted dyadic tail and  $\epsilon_L = \alpha|\tau|t_L$ , the correct lift is  $H_L = K_L \otimes I + I \otimes H_{\text{ref,out}}$ . Standard bounded-perturbation identities [16] give

$$\left\| (H - z)^{-1} - (H_L - z)^{-1} \right\| \leq \epsilon_L / |\text{Im } z|^2, \quad |e - e_L| \leq \epsilon_L, \quad \|P - P_L\| \leq \min(1, \epsilon_L/g_L),$$

where  $g_L = \delta - \|V_L\|_{\text{budget}} > 0$ . The projector bound follows by applying the complementary inverse of  $H_L - e$  to its residual on the true ground. A zero exterior generator would leave escaping free excitations at zero and fails norm-resolvent convergence even at zero perturbation.

Aligned literal vertex boxes  $N = 4m + 3$  coincide with full factors after subtracting the explicit reference ground energy  $c_N = \sum_{C \subset \Lambda_N} E_C$ . Other upper boundary phases admit local-state convergence: complete the support of both a fixed observable and an interior perturbation, compare their identical central ground states, then bound each omitted tail. This gives

$$|\omega_{\text{box}}(A) - \omega_\infty(A)| \leq 4\|A\| \min(1, \epsilon_M/g_M)$$

once the box contains that common completion. It does not establish arbitrary-phase global vector or norm-resolvent convergence. Sources: `research/round18/forward/a1/report.md`, `research/round19/forward/a1/report.md`, `research/round19/forward/a2/report.md`, `research/round20/forward/d1/report.md`, `research/round20/forward/e2/report.md`.

### 7.3 Physical GNS space and a nonzero Wilson fluctuation \*\*

For the dyadic model the full bounded local-factor algebra is irreducible in the chosen incomplete product. Finite-factor vacuum projections converge strongly to  $P_\Omega$ ; an operator in the commutant is scalar on  $\Omega$ , then on its dense local rank-one orbit. Every nonzero ground vector  $\Psi$  is consequently cyclic for that full algebra. The GNS null quotient maps isometrically by  $[A] \mapsto A\Psi$ . Its energy generator is  $H - e$ , since this subtraction fixes the cyclic vacuum, even though scalar shifts cancel in observable conjugation.

The physical algebra requires an additional argument. Let  $\mathcal{H}_{\text{inv}}$  be the vectors invariant under every finite-support vertex gauge transformation, and  $\mathcal{A}_{\text{phys}}$  the norm closure of bounded invariant local operators. For any invariant  $\xi$ , approximate it by  $A_n\Psi$  using full-algebra cyclicity. Haar-average  $A_n$  over its finitely many endpoint groups. The weak operator average preserves local support and, on  $\Psi$ , is a strong vector integral satisfying  $\|\mathcal{E}(A_n)\Psi - \xi\| \leq \|A_n\Psi - \xi\|$ . Therefore

$$\overline{\mathcal{A}_{\text{phys}}\Psi} = \mathcal{H}_{\text{inv}}, \quad K_{\text{phys}} = (H - e)|_{\mathcal{H}_{\text{inv}}}, \quad \text{spec } K_{\text{phys}} \subset \{0\} \cup [973\alpha/8640, \infty). \quad ** \quad (\text{HNM-42})$$

The restriction is self-adjoint on  $D(H) \cap \mathcal{H}_{\text{inv}}$ ; gauge-commuting spectral cutoffs prove density of that domain.

This sector contains a nonvacuum vector. For the actual omitted origin  $xz$  Wilson multiplier  $W$ , the reference mean is zero and variance  $1/4$ . Two free Haar links per omitted face make distinct-face covariances vanish. The resulting residual and pure-state trace distance give, throughout  $|\tau| \leq 1/64$ ,

$$d^2 = \|P_\Psi - P_\Omega\|^2 \leq 2817/64377572 < 1/22500, \quad **$$

$$\text{Var}_\Psi W \geq \frac{1}{4} - 2d - 4d^2 \geq 5321/22500 > \frac{1}{5}.$$

For  $\xi = (W - \omega(W))\Psi$ , spectral calculus now gives the nonzero imaginary-time correlation

$$0 < \langle \xi, e^{-tK_{\text{phys}}/\hbar} \xi \rangle \leq \text{Var}_\Psi(W) e^{-973\alpha t/(8640\hbar)} \quad (t \geq 0).$$

A lower spectral bound supplies this upper decay bound, not a lower exponential estimate. It also supplies no real-time magnitude decay. The result concerns the specified summable representation; equality to a homogeneous or continuum representation has not been proved. Sources: `research/round20/forward/g2/report.md`, `research/round21/forward/j1/report.md`, `research/round21/forward/j2/report.md`.

## 8 Locality, profile limits, and stationary observations

The Round13 locality study applied the Lieb–Robinson interaction-picture method to full  $SU(2)$  links with unbounded onsite Casimirs. Common onsite unitaries preserve support; only bounded plaquette interactions enter the chain expansion. If  $q_{\text{inc}} = 2(d_s - 1)$ ,  $J = \hbar^{-1} \int j(t) dt$  is the dimensionless integrated common magnetic envelope, and  $R$  is the shortest overlapping-plaquette chain from the observable support  $X$  to a removed boundary plaquette, then

$$\|\tau_{\Lambda'}(A) - \tau_\Lambda(A)\| \leq \|A\| \min \left\{ 2, \frac{|X|}{4} \sum_{k \geq R} \frac{(8q_{\text{inc}}J)^k}{k!} \right\}. \quad ** \quad (\text{HNM-43})$$

There are at most  $|X|q_{\text{inc}}(4q_{\text{inc}})^{k-1}$  chains; each commutator costs two and ordered times give  $J^k/k!$ . Unitary resummation is essential: unrestricted growing-support word counting would not give this chain bound. The factorial tail tends to zero at fixed  $J$  as the boundary recedes. This proves a fixed-spacing volume comparison, with no ground-state comparison or continuum velocity. The underlying strong-operator framework permits infinite-dimensional sites [2].

The summable model admits a separate direct Duhamel construction. Completing the local observable and interior perturbation to common factors gives all-phase finite-box dynamics convergence at fixed time, with error  $4\|A\| |t|\epsilon_M/\hbar$ . Strongly continuous implementing unitaries do not imply point-norm time continuity for every bounded local operator. On a free link the character shift  $\chi_n \mapsto \chi_{n+1}$  acquires phases with arbitrarily large frequency; at times tending to zero its norm difference is exactly two. This limitation was retained when identifying the GNS generator.

### 8.1 An exact spatial budget and its endpoint \*\*

Replace only the omitted weights by  $w_f(q) = q^{x+y+z}/24$ ,  $0 < q < 1$ . Summing all orientations and subtracting selected strips gives

$$\mathcal{B}(q) = \frac{2 + 5q + 5q^2 + 6q^3 + 3q^4}{24(1-q)^3(1+q)^2(1+q^2)}, \quad \mathcal{B}(1/2) = 107/135, \quad (1-q)^3\mathcal{B}(q) \longrightarrow 7/64. \quad ** \quad (\text{HNM-44})$$

The same bounded perturbation proof gives  $\Delta \geq \alpha[1/8 - |\tau|\mathcal{B}(q)]$ . At  $\tau = 1/64$  its sufficient boundary lies between 0.764003 and 0.764004. This is a boundary of the certificate, not a measured critical parameter.

Fix  $0 < \eta < 1$  and follow the canonical constant-budget ray  $\tau_q = \eta/[8\mathcal{B}(q)]$ . Each omitted face has two free reference links, and distinct squares share at most one link. Conditional Haar integration therefore gives  $\langle x_f x_g \rangle_\Omega = \delta_{fg}/4$ , despite possible entanglement within strip factors. Hence

$$\sigma_q^2 := \|V_q \Omega\|^2 = \frac{\alpha^2 \tau_q^2}{96} \mathcal{B}(q^2), \quad \|V_q\| = \alpha\eta/8, \quad \frac{\sigma_q^2}{\alpha^2(1-q)^3} \longrightarrow \frac{\eta^2}{5376}. \quad ** \quad (\text{HNM-45})$$

The norm equality follows by localizing finitely many links near identity and then controlling the remaining absolute tail; it does not use a delta function on an infinite identity configuration. With  $\bar{g} = \alpha(1-\eta)/8$ , the excited threshold gives

$$\|P_q - P_\Omega\| \leq \sigma_q/\bar{g}, \quad -\sigma_q^2/\bar{g} \leq e_q \leq 0.$$

Thus the global ground projection returns to the selected-strip reference although the perturbation norm stays nonzero. Every fixed omitted coefficient tends to zero. This route cannot retain a nonzero homogeneous omitted coupling at fixed  $\alpha/E_*$ .

For a bounded operator  $A$  on finitely many complete factors  $F$ , only finitely many omitted faces meet all their links  $E(F)$ . Writing their budget as  $D_F(q) \leq |E(F)|/6$ ,

$$\|V_q A \Omega\| \leq \|A\| [\sigma_q + 2\alpha\tau_q D_F(q)] \longrightarrow 0.$$

The local rank-one orbit is dense and  $\|V_q\|$  is uniformly bounded, so  $V_q \rightarrow 0$  strongly. The resolvent identity and strong Duhamel formula give strong-resolvent convergence and strong unitary convergence uniformly on compact time intervals. Neither norm-resolvent convergence nor its failure follows merely from the nonvanishing perturbation norm.

### 8.2 Stationarity improves the time window \*\*

Round21 first compared connected correlations using a vector residual, obtaining sufficient windows  $T_q = (\hbar/\alpha)C(1-q)^{-\gamma}$ ,  $\gamma < 3/2$ . Round22 exploited stationarity before estimating. For bounded complex local observables,

$$C_q^{A,B}(t) = \omega_q(A^* \beta_q(t)(B)) - \omega_q(A^*) \omega_q(B), \quad \beta_q(t)(B) = e^{-itH_q/\hbar} B e^{itH_q/\hbar}.$$

Both scalar energy phases cancel in this Heisenberg expression. The bounded cocycle proves the strong integral identity

$$\beta_q(t)(B) - \beta_0(t)(B) = -\frac{i}{\hbar} \int_0^t \beta_q(t-s)([V_q, \beta_0(s)(B)]) ds.$$

No differentiation of an arbitrary bounded  $B$  on an unbounded domain is needed. Reference evolution stays on the same complete factors, so  $\|[V_q, \beta_0(s)(B)]\| \leq 2\alpha\tau_q \|B\| D_{F_B}(q)$  for every  $s$ . Changing the state costs  $2d_q$  in the first expectation and  $4d_q$  in the two mean factors. Consequently

$$|C_q^{A,B}(t) - C_0^{A,B}(t)| \leq \|A\| \|B\| \left[ 6d_q + \frac{2\alpha\tau_q|t|}{\hbar} D_{F_B}(q) \right]. \quad ** \quad (\text{HNM-46})$$

Since  $d_q = O((1-q)^{3/2})$  and  $\tau_q = O((1-q)^3)$ , this includes every  $\gamma < 3$ , with sufficient rate  $O((1-q)^{\min(3/2, 3-\gamma)})$ . The old  $\gamma = 3/2$  endpoint is covered. Failure of the new upper certificate at  $\gamma = 3$  does not establish actual nonconvergence.

For moving origin-anchored tail-block cubes of side  $L$ , all outgoing links are retained: there are  $24L^3$  links and  $21L^3$  incident omitted faces. Downward closure in the positive orthant makes incidence equivalent to having the face anchor inside that tail rectangle. Thus

$$D_L(q) = \mathcal{B}(q)(1-q^{4L})(1-q^{2L})(1-q^L). \quad ** \quad (\text{HNM-47})$$

For  $L = \max(1, \lfloor \ell(1-q)^{-\beta} \rfloor)$ ,  $\ell > 0$ ,  $\beta \geq 0$ , and time windows  $|t| \leq (\hbar/\alpha)C(1-q)^{-\gamma}$  with  $C > 0$ ,  $\gamma \geq 0$ , the nonnegative certificate in (HNM-46) vanishes exactly when  $\beta < 1$  and  $\gamma < 3(1-\beta)$ . This necessity concerns that certificate only. At  $\beta = 0$ , its constant uses  $L_0 = \max(1, \lfloor \ell \rfloor)$ , not  $\ell$ . Logarithmically shortened fixed-support  $\gamma = 3$  windows also pass. Translated rectangles need new incoming-face counts. Actual long-time endpoint behavior remains open. Sources: `research/round13/locality/locality.md`, `research/round20/forward/h2/report.md`, `research/round21/forward/m1/report.md`, `research/round21/forward/m2/report.md`, `research/round22/reverse/n1/report.md`, `research/round22/reverse/n2/report.md`.

## 9 Static information, clocks, and exact physical reduction

### 9.1 Conditional dynamics and finite-rate identifiability \*\*

On the conditional three-link space with  $\rho_\kappa \propto e^{\kappa S_3}$ , a separately specified reversible generator is

$$A_{\kappa,m} = -\rho_\kappa^{-1} \operatorname{div}(\rho_\kappa m \nabla), \quad H = cA_{\kappa,m}, \quad c > 0.$$

For smooth uniformly positive mobility  $m$ , its closed form has domain  $H^1$ , operator domain  $H^2$ , and constant ground. This standard divergence-form construction preserves the same static law for every  $c$  and  $m$ , demonstrating why static moments cannot fix a clock. The actual shared-middle gradient produces

$$\Gamma(S_3) = \frac{1}{4}\{15 + 8y + 2x + 2z + 2r - (3x + w)^2 - (y + w + t)^2 - (z + t)^2\}, \quad r = \operatorname{Tr}(UW^\dagger)/2.$$

The mixed term involving  $r - wt$  disappears under an incorrect resampling of  $V$ . Independent optimization of  $U, W$  at fixed  $y$  proves  $\min S_3 = -5$ ,  $\max S_3 = 7$ . Bounded-density comparison with the Haar Casimir gap then gives  $\operatorname{gap}(cA_{\kappa,1}) \geq (3c/4)e^{-12|\kappa|} \geq c/6$  for  $|\kappa| \leq 1/8$ . This is a theorem for the declared conditional generator, not a derived lattice Hamiltonian.

For  $m = 1 + \zeta x$ ,  $|\zeta| < 1$ , the measurable initial rate would be  $r_f = -\hbar C'_f(0+) = c \int m \Gamma(f) \rho_\kappa$ . At known  $\kappa = 0$ , the pair  $x, x + x^2$  gives

$$r_x = 3c/16, \quad r_{x+x^2} = c(5/16 + \zeta/8), \quad c = 16r_x/3, \quad \zeta = 8r_{x+x^2}/c - 5/2. \quad **$$

The pair  $x, x^2$  is parity-blind to  $\zeta$ . For known  $|\kappa| \leq 1/8$ , choosing  $g = x + x^2/4$  proves a uniform determinant lower bound  $D \geq 1/7168$ . Indeed, if  $G(x) = (1 - x^2)/4$  and  $X, X'$  are independent copies of the actual one-coordinate marginal, symmetrization gives, for  $g = x + ax^2$ ,

$$D = 2a \mathbb{E}\{G(X)G(X')(X - X')^2[1 + a(X + X')]\} > 0 \quad (0 < a < 1/2).$$

Comparison to Haar yields  $D \geq 3a(1 - 2a)e^{-24|\kappa|}/128$ ;  $a = 1/4$  and  $e^3 < 21$  give the stated rational floor.

These inverses identify only their declared finite family. At Haar,  $p_3 = x^3 - 3x/8 = C_3^{(2)}(x)/32$  is invisible to both rates. Adding  $h = x^2 + x^3$  identifies  $m = 1 + \zeta x + \xi p_3$  through

$$\begin{pmatrix} r_x \\ r_{x+x^2/4} \\ r_{x^2+x^3} \end{pmatrix} = \begin{pmatrix} 3/16 & 0 & 0 \\ 25/128 & 1/32 & 0 \\ 59/256 & 9/64 & 9/512 \end{pmatrix} \begin{pmatrix} c \\ c\zeta \\ c\xi \end{pmatrix}, \quad \det M = 27/262144. \quad ** \quad (\text{HNM-48})$$

The box  $|\zeta|, |\xi| \leq 1/4$  guarantees  $m \geq 13c/192$ , hence gap at least  $13c/192$  on the stated  $\kappa$  range. The Gegenbauer polynomial itself is classical [15]; its blind and detecting roles here are the project calculation.

More generally  $N$  initial rates impose  $N$  linear conditions on a polynomial with  $N + 1$  coefficients. A nonzero null polynomial, normalized by its coefficient absolute sum, has sup norm at most one. It gives a whole positive interval of distinct mobilities with identical rates at fixed density and  $c$ . The explicit  $p_5 = x^5 - 5x^3/6 + x/8 = C_5^{(2)}(x)/192$  preserves all three rates above, but changes the  $x$ -correlation curvature to

$$C_x''(0+) = \frac{c^2}{\hbar^2} \left( \frac{9}{64} + \frac{\epsilon^2}{1024} \right) \quad \text{for } m = 1 + \epsilon p_5. \quad **$$

Thus the obstruction concerns finitely many initial slopes, not complete time curves. No physical rates were measured; the inversions were tested on synthetic exact inputs. Sources: `research/round20/forward/f2/report.md`, `research/round21/forward/k2/report.md`, `research/round21/forward/l1/report.md`, `research/round21/forward/l2/report.md`.

## 9.2 A failed section and a valid maximal-tree map \*\*

Round22 tested physical matching on the actual 18-vertex, 33-link graph at zero magnetic coupling. Fixing thirty links to identity leaves the old  $U, V, W$  coordinates, but fixes thirteen genuine cycles in addition to tree gauge variables. It is not a Hilbert-space reduction. If  $Q_0$  is a face fixed to identity, then  $f_n = \chi_n(Q_0)/(n + 1)$  satisfies  $\|f_n\|_2 = 1/(n + 1) \rightarrow 0$ , whereas its section equals one. The restriction is nonclosable. Three edge-disjoint such faces give an analogous sequence converging even in the electric graph norm. Moreover  $H_E x_0 = 3\alpha x_0$  before restriction, but the conditional generator annihilates its constant image. A scalar clock cannot repair this core defect.

A designated slope fit for a physical square that restricts to  $x$  gives  $c = 4\alpha$ . A held-out square restricting to  $w = \text{Tr}(UV^\dagger)/2$  has physical correlation  $e^{-3\alpha t/\hbar}/4$ , while the fitted conditional prediction is  $e^{-6\alpha t/\hbar}/4$ . The disagreement is strict at every finite positive time.

The repair uses a genuine maximal tree  $T = \{0, \dots, 15, 26\}$ , root  $(0, 2, 0)$ , and all sixteen chord holonomies  $L_e = h_{s(e)} g_e h_{t(e)}^{-1}$ . At fixed tree variables, left/right Haar invariance gives product Haar on the chords. Gauge-invariant functions become simultaneous-conjugation invariants on these sixteen variables, yielding a unitary  $J_{16}$ . The three-chord isometry  $J_3$  embeds functions of  $L_{28}, L_{27}, L_{24}$ ; its adjoint integrates the other thirteen chords, rather than evaluating them at identity. These are the standard maximal-tree principles [14], here realized on the complete signed graph.

Every tree derivative must also be transported. If cutting tree edge  $q$  leaves subtree  $S_q$ , the induced vector field is

$$D_{q,a} = \epsilon_q \left( \sum_{s(e) \in S_q} \ell_{e,a} - \sum_{t(e) \in S_q} \rho_{e,a} \right).$$

Both endpoint occurrences remain when a chord lies inside  $S_q$ . The full generator is  $-\alpha \sum_a (\sum_e \ell_{e,a}^2 + \sum_q D_{q,a}^2)$ . Its quadratic form is a sum of squared gradients, bounded above and below by positive multiples of the product Casimir form. On finite Peter–Weyl blocks these comparisons give equivalence of  $H^1$  forms and  $H^2$  operator domains and identify the closed operators. On the selected three chords the exact reduction is

$$H_{\text{eff}}/\alpha = 3C_U + 3C_V + C_W + C_{VW}^\ell + C_{VW}^\rho + C_{UVW}^\ell + 2C_{UVW}^\rho. \quad ** \quad (\text{HNM-49})$$

Here  $C_A^\ell = -\sum_a (\sum_{j \in A} \ell_{j,a})^2$ , and similarly on the right. The image reduces  $H_E$  because its orthogonal projector is Haar integration over unused original chord links and commutes with each link heat semigroup. This proves exact electric semigroup intertwining, not merely a formal compression.

The new  $x, y, z$  observables complete to physical loops of lengths six, eight, six, respectively, so their electric energies are  $(9/2)\alpha, 6\alpha, (9/2)\alpha$ . They are different completions from the old section’s training square. The prescribed  $x$ -slope now gives  $c = 6\alpha$ ; its reserved curvature forces  $\zeta = 0$ . The held-out  $y$ -curve still differs: the physical exponent is  $6\alpha$ , the conditional one  $9\alpha/2$ . Thus exact Haar reduction succeeds while matching to the scalar-clock affine-mobility family fails. Sources: `research/round22/reverse/p1/report.md`, `research/round22/reverse/p2/report.md`.

### 9.3 Magnetic leakage, projected memory, and exact channels \*\*

Keep this tree and add all twenty magnetic faces:  $H_\lambda = H_E + \alpha\lambda V$ ,  $V = \sum_{p=0}^{19} (1 - x_p)$ ,  $\lambda \geq 0$ . This  $\lambda$  is dimensionless, unlike the physical  $\lambda_p$  used earlier. With  $P = J_3 J_3^*$ , only one face survives conditional averaging, so  $J_3^* V J_3 = 20 - \text{Tr}(VW^\dagger)/2$ . Three other faces share the same omitted chord. Their nonzero conditional cross moments give

$$\begin{aligned} B_\lambda^* B_\lambda &= (\alpha\lambda)^2 M_d, & B_\lambda &= Q H_\lambda J_3, \\ d &= \frac{19}{4} + \frac{1}{2}(r + x + z) = 4 + \frac{1}{4} \|u + w + e_0\|^2 \in [4, 25/4]. \end{aligned} \quad ** \quad (\text{HNM-50})$$

Here  $r = \text{Tr}(UW^\dagger)/2$ , and  $u, w$  are unit quaternions. Hence every nonzero selected vector leaks at nonzero  $\lambda$ ;  $\|B_\lambda\| = 5\alpha\lambda/2$ . Replacing  $d$  by its mean  $19/4$  loses the correlated face terms. Also  $H_\lambda 1$  is nonconstant, with centered squared residual  $5(\alpha\lambda)^2$ ; the Haar reference is not an interacting ground.

Set  $A_\lambda = J_3^* H_\lambda J_3$  and  $T_\lambda(t) = J_3^* e^{-tH_\lambda/\hbar} J_3$ . Two bounded off-diagonal Duhamel insertions prove

$$\|T_\lambda(t) - e^{-tA_\lambda/\hbar}\| \leq \min\{1, \frac{25}{8}(\alpha\lambda t/\hbar)^2\}. \quad **$$

The scaled difference tends strongly to  $M_d/2$  as  $t \downarrow 0$ ; this is not an operator-norm Taylor expansion. It already disproves autonomy of the selected compression.

The complement has the exact electric lower bound  $3\alpha$ , by the physical girth argument and an omitted square attaining it. Thus  $C_\lambda = Q H_\lambda Q \geq 3\alpha$ . Eliminating the complementary component gives the exact memory and resolvent relations

$$\begin{aligned} \hbar u' &= -A_\lambda u + \hbar^{-1} \int_0^t K_\lambda(t-s) u(s) ds, & K_\lambda(s) &= B_\lambda^* e^{-sC_\lambda/\hbar} B_\lambda, \\ J_3^* (H_\lambda + z)^{-1} J_3 &= [A_\lambda + z - \Sigma_\lambda(z)]^{-1}, & \Sigma_\lambda(z) &= B_\lambda^* (C_\lambda + z)^{-1} B_\lambda, \quad z > 0. \end{aligned}$$

The full return  $u(s)$  remains inside the memory integral. These block identities are established methods; the actual channel content and quantified errors are the project contribution.

For the two orthonormal channels  $1, 2x$ , the leading kernel  $(\alpha\lambda)^2 B_1^* e^{-sC_0/\hbar} B_1$  has seven exact electric energies. The 11 weights at energies  $3, 9/2, 6, 15/2, 8, 17/2, 9$ , in units of  $\alpha$ , are

$$(1/8, 1/8, 3/8, 9/4, 9/8, 3/8, 3/8). \quad **$$

The 00 weight is  $19/4$  at energy three, and the off-diagonal weight is  $1/4$  at that same energy. To derive the table, intersect each face with the actual six-link  $x$  loop. A shared consecutive path of length  $m > 0$  has spin-zero and spin-one components of squared norms  $1/16, 3/16$ , with energies  $3(10 - 2m)/4$  and that number plus  $2m$ . Nine faces have disjoint edge support; the other counts for  $m = 1, 2, 3$  are six, two, two. This is an exact finite spectral orbit in the full Hilbert space, not a representation cutoff.

Since  $\|V\| \leq 40$ , bounded Duhamel and the resolvent identity give full selected-space errors

$$\begin{aligned} \|K_\lambda(s) - K_\lambda^{(2)}(s)\| &\leq 250\alpha^3 \lambda^3 (s/\hbar) e^{-3\alpha s/\hbar}, \\ \|\Sigma_\lambda(z) - \Sigma_\lambda^{(2)}(z)\| &\leq \frac{250\alpha^3 \lambda^3}{(3\alpha + z)^2}. \end{aligned} \quad ** \quad (\text{HNM-51})$$

These are uniform absolute kernel bounds in delay and uniform complementary self-energy bounds down to  $z = 0$ . They do not give uniform relative late-time accuracy or a bounded full resolvent at zero. Sources: `research/round22/reverse/q1/report.md`, `research/round22/reverse/q2/report.md`.

## 10 Homogeneous stability: explicit progress and retained obstructions

### 10.1 A matched theorem dictionary and a controlled rotation \*\*

The qualitative homogeneous stability route is independent of the summable construction. Yarotsky's product-vacuum theorem permits unbounded onsite operators and infinite-dimensional site spaces, but its range-dependent constants remain unevaluated [17]. The Round13 outgoing-link blocking gives normalized perturbation  $\epsilon \leq (4/3) \binom{d}{2} \lambda_{\max}/\alpha$ . Boundary padding handles all-contained plaquettes; it does not silently identify every thermodynamic boundary limit. A sufficiently small, existential magnetic/electric ratio gives a fixed-spacing gap at least  $3\alpha/8$ . A qubit, two-site algorithmic theorem cannot supply its missing constants [19].

The dressed-strip reference admits a sharper dictionary. Each coarse tail cell contains 24 links: ten strip links and fourteen free links. There are 24 anchored faces, of which three are selected and 21 omitted. Their full grouped support is  $G = \{0, e_x, e_y, e_z\}$ . In energy unit  $\delta = \alpha/8$ ,

$$h_b \geq 1 - P_b, \quad \phi_b = -\frac{\tau}{3} \sum_{f \in O_b} x_f, \quad \|\phi_b\| = 7|\tau|. \quad **$$

All 21 omitted faces cross a coarse block boundary. Ignoring crossing faces would erase the entire homogeneous perturbation. The matched source theorem gives an existential interval

$$|\tau| < \frac{1}{7} \min\{c_1(G), [2c_2(G)]^{-1}\}, \quad \Delta > \alpha/16,$$

including literal finite rectangles through clipped reference sites and padding. No numerical value of  $\tau$ , including  $1/64$ , is thereby certified.

The full magnetic interaction is not purely form-bounded relative to the undressed vacuum energy. On an actual square,  $\psi_t = (1 + t\chi_p)/\sqrt{1 + t^2}$  has electric energy  $3\alpha t^2/(1 + t^2)$  and magnetic

expectation  $-\lambda t/(1+t^2)$ ; their absolute ratio diverges as  $t \rightarrow 0$ . The diagonal compression alone does obey the incidence bound  $|\langle W_{\text{diag}} \rangle| \leq (16\lambda_{\max}/3\alpha)\langle H_0 \rangle$ . The missing off-diagonal term cannot be discarded.

On one four-site star, conditional Haar orthogonality gives  $\|\phi\Omega\|^2 = 7\tau^2/12$ . Let  $v = \phi\Omega$ ,  $u = (H_0|_Q)^{-1}v$ , and  $S = |u\rangle\langle\Omega| - |\Omega\rangle\langle u|$ . Then  $[S, H_0] = -(|v\rangle\langle\Omega| + \text{adjoint})$ . The finite-rank range lies in  $D(H_0)$ ; the exponential preserves that domain. Integrating the bounded commutator gives the exact identity

$$e^S(H_0 + \phi)e^{-S} = H_0 + Q\phi Q + R, \quad \|R\| \leq (707/60)\tau^2. \quad **$$

This instantiates a standard unitary dressing method with the actual local variance. The remainder is part of the Hamiltonian.

For simultaneous finite-volume rotations, every nonzero ordered commutator word adds a star meeting the previous union.  $|G - G| = 13$ , so there are at most  $13^n n!$  relative words of order  $n$ , each supported on at most  $4 + 3n$  sites. Rooting at a fixed site adds a factor at most  $4 + 3n$ . In the interaction norm  $\|\Phi\|_w = \sup_x \sum_{Y \ni x} 2^{|Y|} \|\Phi_Y\|$ , this yields a convergent all-order remainder majorant. The admitted common certificate is

$$\|R\|_w \leq \frac{25460736}{25}\tau^2, \quad |\tau| \leq 5/1664, \quad |\langle D \rangle| \leq 28|\tau|\langle H_0 \rangle. \quad ** \quad (\text{HNM-52})$$

The reverse derivation separately obtained the smaller valid constant  $1970176/5$ ; it is not substituted into the subsequently frozen common recurrence. Neither constant is a homogeneous gap threshold.

## 10.2 Generated supports, interacting inverses, and the remaining source \*\*

For a residual term on an arbitrary generated support  $Y$ , split

$$R_Y = c_Y I + A_Y + Z_Y, \quad A_Y = |Q_Y R_Y \Omega_Y\rangle\langle\Omega_Y| + \text{adjoint}, \quad Z_Y = Q_Y(R_Y - c_Y I)Q_Y.$$

The subtraction inside  $Z_Y$  is necessary when the scalar is  $c_Y I$ . The bare homological inverse has norm at most one, and the resulting diagonal update is bounded by  $2\|R_Y\|$ . For exponential cardinality weight  $\mu$ , an explicit root/intersection sum gives

$$\|[\Phi, \Psi]\|_{\mu-\ell} \leq \frac{4}{e\ell} \|\Phi\|_\mu \|\Psi\|_\mu, \quad \ell > 0.$$

Spending  $\ell/n$  per nested commutator and  $n! \geq (n/e)^n$  bounds the factorial-divided order by  $(4\|S\|_\mu/\ell)^n \|K\|_\mu$ . If  $r > 0$  bounds the residual,  $b \geq 0$  bounds the retained diagonal interaction, and  $\ell > 4r$ , the reverse majorant is

$$r' \leq \mathcal{F}(r, b; \ell) := \frac{4r}{\ell - 4r}(b + 3r/2), \quad b' \leq b + 2r, \quad ** \quad (\text{HNM-53})$$

$$\mathcal{F}(r, b; \ell) < r \iff \ell > 4b + 10r.$$

The equivalence refers to the displayed scalar majorant, not the actual residual;  $r = 0$  is the separate exact zero-residual case. Any finite total loss has  $\ell_n \rightarrow 0$ , while this majorant retains  $b_n \geq 448|\tau| > 0$ . Its positive scalar trajectory eventually leaves its convergence radius. This proves failure of that certificate, not of the true rotation algorithm or the physical gap.

An equal-weight commutator counterexample also needs its exact support convention. In the reverse structured example, regrouping to the actual support gives the special homological value  $1/64$ ; retaining indexed unions instead gives  $(9N + 16)/(64(3N + 1))$ . The post-freeze gate explicitly records this distinction. It is not legitimate to improve an interaction norm by silently changing its decomposition.

Round22 then replaced the bare inverse by the actual *initial* diagonal operator. For finite  $Y$ , let  $G_Y = H_Y + \sum_{b+G \subset Y} Q_b \phi_b Q_b$ . Four-star incidence gives, with  $a = 28|\tau| < 1$ ,

$$(1-a)H_Y \leq G_Y \leq (1+a)H_Y, \quad \|(G_Y|_{Q_Y})^{-1}\| \leq (1-a)^{-1}.$$

The inverse maps into  $D(H_Y)$ , so its rank-two skew generator has a valid domain-preserving exponential. Interior cancellation is exact, but on the full finite volume

$$[S_Y, H_0 + D] + A_Y = \sum_{\substack{b+G \cap Y \neq \emptyset \\ b+G \not\subset Y}} [S_Y, D_b].$$

For the admitted actual  $SU(2)$  diagnostic source consisting of a free link character at the origin, the squared boundary residual on the product vacuum is exactly

$$\|([S_Y, H_0 + D] + A_Y)\Omega\|^2 = 7\tau^2/432. \quad ** \quad (\text{HNM-54})$$

All 21 face contributions and their vanishing distinct-face cross moments are retained. This source is admissible but is not asserted to be an actually generated term of the earlier rotation.

Finally, the infinite initial diagonal is constructed as a relative form sum on the same incomplete product. Let  $K = H_0^{-1/2} D H_0^{-1/2}$  on  $Q_\Omega$ ;  $\|K\| \leq a < 1$ . For a vacuum-exterior source  $v \perp \Omega$ ,

$$u = H_0^{-1/2} (I + K)^{-1} H_0^{-1/2} v = \sum_{n \geq 0} H_0^{-1/2} (-K)^n H_0^{-1/2} v. \quad ** \quad (\text{HNM-55})$$

The form equation proves  $Gu = v$ . Each coefficient lies in the  $n$ -collar  $Y^{(n)}$  with norm at most  $\|v\| a^n$ ; direct finite-support action also gives an  $H_0$ -graph bound  $\|H_0 u_n\| \leq \|v\| |Y| (2n-1)^3 a^n$  for  $n \geq 1$ . Polynomially weighted geometric tails prove Hilbert, half-energy,  $H_0$ -graph and  $G$ -graph convergence, including explicit finite-volume collar errors. This response regularity does not prove global equality  $D(G) = D(H_0)$ .

The exact global vacuum rank-two source is  $A_Y \otimes P_{\text{ext}}$ . Against the desired local source  $A_Y \otimes I$ , the remaining operator is  $A_Y \otimes Q_{\text{ext}}$ , of norm one on the specified exterior character test. Moreover, the prescribed collar interaction certificate  $\sum_n e^{\mu(n+1)^3} a^n$  diverges for every fixed  $\mu > 0$ ,  $0 < a < 1$ , although the unweighted response converges. Thus full local-source cancellation and a useful weighted decomposition remain missing. These are the precise next requirements for an all-stage homogeneous stability proof; the later diagonal updates are not covered by the initial inverse.

The domain and inverse comparisons use established form and Lie–Schwinger methods [16, 21]. Their complete source inductions and numerical stability thresholds were not imported. Sources: `research/round21/forward/i1/report.md`, `research/round21/forward/i2/report.md`, `research/round22/advisor/o1-gate.json`, `research/round22/advisor/o2-gate.json`, `research/round22/reverse/r1/report.md`, `research/round22/reverse/r2/report.md`.

## 11 Full sources, endpoint observables, and certified heat: Rounds 23–26

The twenty-eight investigations S1–AF2 develop three separate models. The homogeneous initial-diagonal problem uses dimensionless generators with physical energy unit  $\alpha/8$ ; the finite-graph problem uses  $L = H/\alpha$  and heat time  $\sigma = \alpha t/\hbar$ ; the canonical summable family uses its original real-time generator and a  $q$ -dependent interaction profile. Throughout, lattice spacing  $a$ ,  $E_\star > 0$ ,  $\alpha/E_\star > 0$ , and  $\hbar > 0$  are fixed. A source filter, collar depth, Galerkin projection, joining time, or input-overlap restriction is a proof or approximation parameter, not a new term in an action.

The marker **\*\*** means a contribution newly derived or computed in this workbench; scientific priority is unverified. S1/S2/T1, all ten Round 24 loops, and all ten Round 26 loops have separately authored model-agent derivations and skeptical reviews. T2/U1/U2 and AA1/AA2 instead have same-author derivation, reverse reconstruction, and critical self-review. These categories describe provenance, not external mathematical peer review or experimental independence. Historical limited verdicts remain limited even where later work answers a previously missing premise. The detailed source-to-equation mapping is recorded in `audit/recent-ledger.json` accompanying this manuscript.

## 11.1 An actual homogeneous source and its regulated inverses

### 11.1.1 **\*\*** S1: a generated cubic source and the complete crossing defect

Use the complete 24-link onsite factors and four-site stars  $Y = \{0, e_1, e_2, e_3\}$  of the homogeneous construction. On one star let  $H = H_{0,Y}$ ,  $P_0 = |\Omega_Y\rangle\langle\Omega_Y|$ ,  $Q_0 = I - P_0$ , and  $H \geq Q_0$ . With the actual 21-face potential  $\phi_0$ , set

$$v = \phi_0 \Omega_Y, \quad u = H_{Q_0}^{-1} v, \quad a_0 = \langle u, v \rangle, \quad b_0 = \|u\|^2, \quad s_0^2 = \|v\|^2 = \frac{7}{12} \tau^2. \quad (\text{HNM-56})$$

The Haar identity follows because each face has a free Haar link and distinct faces have an unmatched free link: the 21 diagonal half-trace moments are  $1/4$  and every cross term vanishes. Define  $S = |u\rangle\langle\Omega_Y| - |\Omega_Y\rangle\langle u|$  and  $A_0 = |v\rangle\langle\Omega_Y| + \text{adjoint}$ . The specified repeated-anchor term in the previous remainder is

$$C = \frac{1}{2}[S, [S, \phi_0]] - \frac{1}{6}[S, [S, A_0]], \quad w = Q_0 C \Omega_Y = -a_0 u - \frac{b_0}{3} v. \quad ** \quad (\text{HNM-57})$$

The two double commutators have the same  $Q_0$ -column  $-3a_0 u - b_0 v$ ; their coefficients give the displayed result. In particular,  $\langle u, w \rangle = -4a_0 b_0 / 3 < 0$  for  $\tau \neq 0$ , so this is an actual nonzero source, with  $r := \|w\| \leq 4s_0^3/3$ . It is one indexed cubic term; other terms in a regrouped cubic coefficient could cancel it. Its exact local split is  $C = c_0 I + A + Q_0(C - c_0 I)Q_0$ , where  $c_0 = \langle \Omega_Y, C \Omega_Y \rangle$  and  $A = |w\rangle\langle\Omega_Y| + \text{adjoint}$ . All relative ratios divided by  $r$  below require  $\tau \neq 0$ , hence  $r > 0$ . At  $\tau = 0$  the generated source vanishes; the undivided bounds still apply.

Write  $M = 7|\tau| \leq 35/1664$ ,  $D_c = Q_c \phi_c Q_c$ , and  $G_\Lambda = H_{0,\Lambda} + \sum_c D_c$ , retaining only complete stars in a containing finite cuboid. The one-star operator  $G_Y = H + D_0$  satisfies  $(1 - M)H \leq G_Y \leq (1 + M)H$  as forms and  $D(G_Y) = D(H)$  by bounded perturbation. Therefore

$$z = (G_Y|_{Q_0})^{-1} w, \quad K_Y = |z\rangle\langle\Omega_Y| - \text{adjoint}, \quad [K_Y, G_Y] = -A, \quad \|K_Y\| \leq \frac{r}{1 - M}. \quad (\text{HNM-58})$$

The inverse places  $z$  in the operator domain. Identity extension preserves  $D(H_{0,\Lambda})$  because the local commutator with  $H$  is bounded and exterior spectral projections commute with  $K_Y$ . The graph-bounded exponential and its inverse preserve that domain as well. The complete extended equation is

$$[K_Y \otimes I, G_\Lambda] = -A \otimes I + F_Y, \quad F_Y = \sum_{c \text{ crossing } Y} [K_Y, D_c], \quad \|F_Y\| \leq \frac{6Mr}{1 - M}. \quad ** \quad (\text{HNM-59})$$

At the positive-octant origin there are three possible crossings, with anchors  $e_1, e_2, e_3$ , each on a seven-site union. For the single-source decomposition,  $\sum_c e^{7\mu} \| [K_Y, D_c] \| \leq 6e^{7\mu} Mr / (1 - M)$ . A translated interior star has twelve crossings. The reverse-origin sharper rooted count is  $12 \cdot 7 = 84$ , giving a family bound  $168e^{7\mu} Mr_*/(1 - M)$ ; the forward count 96 gives the valid weaker constant 192. These are one-step interaction bounds, not an all-order inverse.

An exterior mean-zero unit Haar character  $\xi$  distinguishes the actual source from its vacuum-projected replacement:  $(A \otimes I)(\Omega_Y \otimes \xi) = w \otimes \xi$  while  $(A \otimes P_{\text{ext}})(\Omega_Y \otimes \xi) = 0$ . Also a bounded  $D(G)$ -preserving exact inverse must satisfy  $P_E A P_E = 0$  at every energy; a positive ground gap alone says nothing about all excited Bohr differences. This explains S1's limited gate and motivates a regulated full-source equation. Sources: `research/round23/forward/s1/report.md`, `research/round23/reverse/s1/report.md`, and `research/round23/skeptic/s1.md`.

### 11.1.2 \*\* S2 and W1–W2: retain the residual and name its topology

For  $\alpha_s^G(A) = e^{isG} A e^{-isG}$ , S2 fixes  $0 < \Theta \leq 1/16$  and defines

$$\begin{aligned} \mathcal{L}_\Theta(A) &= -\frac{i}{2} \int_{-\Theta}^{\Theta} \text{sgn}(s)(1 - |s|/\Theta) \alpha_s^G(A) ds, \\ \mathcal{R}_\Theta(A) &= \frac{1}{2\Theta} \int_{-\Theta}^{\Theta} \alpha_s^G(A) ds, \quad [\mathcal{L}_\Theta(A), G] = -A + \mathcal{R}_\Theta(A). \end{aligned} \quad ** \quad (\text{HNM-60})$$

These are vectorwise strong integrals. In weak pairings on  $D(G)$ , integration by parts uses the kernel derivative  $2\delta_0 - \Theta^{-1}\mathbf{1}_{[-\Theta, \Theta]}$ . The bounded weak commutator then proves  $\mathcal{L}_\Theta(A)D(G) \subset D(G)$  by self-adjointness; domain preservation of an arbitrary initial  $A$  is not assumed. Its norm and graph bounds are  $\Theta r/2$  and  $\Theta r/2 + 2r$ . At Bohr frequency  $\omega$ ,

$$\ell_\Theta(\omega) = \frac{1 - \text{sinc}(\Theta\omega)}{\omega}, \quad r_\Theta(\omega) = \text{sinc}(\Theta\omega), \quad \ell_\Theta(0) = 0, r_\Theta(0) = 1. \quad (\text{HNM-61})$$

Thus every equal-energy block survives exactly; shortening  $\Theta$  leaves more of the source.

For the rooted norm  $\|\Phi\|_2 = \sup_x \sum_{X \ni x} 2^{|X|} \|\Phi_X\|$ , a surviving ordered commutator word starts on one star and adds at most three sites at each step. The forward rooted count, before simplex integration, is

$$N_0 \leq 64r_\star, \quad N_{n+1} \leq 64M(8 + 3n)N_n, \quad N_n \leq 64r_\star(192M)^n(8/3)_n. \quad ** \quad (\text{HNM-62})$$

Here the insertion cost includes  $2M$  from a commutator and  $2^3$  from the support weight; incoming anchors and repetitions remain. The simplex contributes  $|s|^n/n!$ , and the inverse and residual filter moments are respectively  $\Theta^{n+1}/(n+2)!$  and  $\Theta^n/(n+1)!$ . Consequently

$$\|\mathcal{L}_\Theta(\{A_b\})\|_2 \leq 32\Theta r_\star(1 - 192M\Theta)^{-8/3}, \quad \|\mathcal{R}_\Theta(\{A_b\})\|_2 \leq 64r_\star(1 - 192M\Theta)^{-8/3}, \quad (\text{HNM-63})$$

with  $192M\Theta \leq 105/416 < 1$ . Independently, reverse counts at most  $13^n n!$  relative words and  $4 + 3n$  rooted translates. With  $u = 208M\Theta$ , its orbital sum is  $16r_\star(4 - u)/(1 - u)^2$ ; the inverse and residual bounds are  $8\Theta r_\star(4 - u)/(1 - u)^2$  and  $16r_\star(4 - u)/(1 - u)^2$ . At  $u = 35/128$  the rational sum is  $6784/961$ . The full tail after depth  $N$  is

$$\sum_{n > N} (4 + 3n)u^n = \frac{u^{N+1}[(3N + 7) - (3N + 4)u]}{(1 - u)^2}. \quad (\text{HNM-64})$$

These alternative bounds concern indexed interaction decompositions, not a bounded extensive infinite sum.

The forward-origin infinite-representation identification adds a separate form argument. In the admitted product-reference representation, write  $G \leftrightarrow h_0 + d$ ,  $\kappa = 4M < 1$ , and  $\sum_b \|Q_b u\|^2 \leq 4h_0[u]$ . Subtracting resolvent form equations for  $u = (G + 1)^{-1}f$  and  $u_\Lambda = (G_\Lambda + 1)^{-1}f$  gives

$$\|u_\Lambda - u\|_{\mathcal{F}} \leq \frac{\sqrt{\kappa}}{1 - \kappa} \left( M \sum_{b \text{ omitted}} \|Q_b u\|^2 \right)^{1/2} \longrightarrow 0. \quad (\text{HNM-65})$$

Strong-resolvent convergence identifies the strong unitary limit. Stabilization of every finite connected word plus the uniform summable tail improves the single-source short-time limit to operator norm. It does not prove  $D(G) = D(H_0)$  globally or a norm limit of global propagators. Sources: `research/round23/forward/s2/report.md`, `research/round23/reverse/s2/report.md`, `research/round23/skeptic/s2.md`.

W1 then evaluates the actual source's first form moment, a reverse-origin refinement separately reviewed. The ground-transform identity and the half-trace gradient identity give

$$h[F\Omega_Y] = 8 \sum_e \int |\nabla_e F|^2 |\Omega_Y|^2, \quad \sum_a |X_e^a W_f|^2 = \frac{1 - W_f^2}{4}, \quad h[v] = 14\tau^2 = 24s_0^2. \quad ** \quad (\text{HNM-66})$$

Free-link parity also cancels cross gradients. In the spectral measure of  $v$ ,  $w = -f(H)v$  with decreasing positive  $f(\lambda) = a_0/\lambda + b_0/3$ . The covariance of  $\lambda$  with  $f(\lambda)^2$  is nonpositive, so  $h[w]/r^2 \leq 24$ ; Jensen's inequalities for the two inverse moments yield  $r \geq s_0^3/432$ . Because  $G\Omega = 0$ ,  $\mathcal{R}_\Theta(A)\Omega = \text{sinc}(\Theta G)w$ . Applying  $\text{sinc } x \geq 1 - x/2$  to this positive spectral measure and  $G \leq (1 + 4M)H_0$  gives the actual lower bound

$$\|\mathcal{R}_\Theta(A)\| \geq [1 - 12\Theta(1 + 4M)]r \geq \frac{311}{1664}r > 0. \quad (\text{HNM-67})$$

This prevents deleting the short-filter cubic residual. It does not establish an actual zero-frequency obstruction to every inverse. The complementary forward result is the source-mass certificate  $\|\mathcal{R}_\Theta(A)\| \geq (5/6)r\sqrt{\nu([0, E])}$  when  $\Theta E \leq 1$ , and  $\|\mathcal{R}_\Theta(A)\|/r \rightarrow 1$  as  $\Theta \downarrow 0$ ; neither is a full-operator upper estimate. If a bounded exact inverse  $K$  were already known, integration would instead give  $\|\mathcal{R}_\Theta(A)\| \leq \|K\|/\Theta$ , a conditional criterion only. Sources: `research/round24/reverse/w1/report.md`, `research/round24/forward/w1/report.md`, `research/round24/skeptic/w1-review.md`.

W2 holds  $G$  fixed and sets  $A_n = \mathcal{R}_\Theta^n(A)$ ,  $K_n = \sum_{j < n} \mathcal{L}_\Theta(A_j)$ . Telescoping preserves

$$[K_n, G] = -A + A_n, \quad \|K_n\| \leq n\Theta r/2, \quad (A_n)_\omega = \text{sinc}(\Theta\omega)^n A_\omega. \quad (\text{HNM-68})$$

On the actual vacuum column,  $g = 1 - 4M \geq 381/416$  supplies a gap. With the reverse bound  $\rho = 1 - (\Theta g)^2/8 < 1$ , a piecewise Taylor and  $|\text{sinc } x| \leq 1/|x|$  argument yields

$$\|A_n\Omega\| \leq \rho^n r, \quad \|K_n\Omega - G_Q^{-1}w\| \leq \rho^n r/g, \quad \|G(K_n\Omega - G_Q^{-1}w)\| \leq \rho^n r. \quad ** \quad (\text{HNM-69})$$

Forward independently obtains denominator 10 in  $1 - (\Theta g)^2/10$ . These are graph-norm convergence of a column, uniform in the containing volumes and valid in the admitted infinite representation. In each fixed finite volume, compact resolvent and dominated convergence in an energy eigenbasis additionally prove

$$\mathcal{R}_\Theta^n(A) \xrightarrow{\text{strong}} \mathcal{D}_G(A) := \sum_E P_E A P_E. \quad (\text{HNM-70})$$

The value is not known to vanish. Compact-resolvent doublets with splittings  $1/(k+1)$  and unit swap sources show why strong convergence can coexist with residual norm one and an unbounded formal inverse. In the actual model the old weighted majorants cover only  $n \leq 3$  at the cap; failure at  $n = 4$  is failure of that estimate, not divergence of the dynamics. W2 therefore does not close a changed-diagonal nonlinear iteration. Sources: `research/round24/forward/w2/report.md`, `research/round24/reverse/w2/report.md`, `research/round24/skeptic/w2-review.md`.

### 11.1.3 \*\* AB1–AB2: the surviving block is controlled by its boundary

Pinching the complete identity (HNM-59) gives

$$\mathcal{D}_G(A) = \mathcal{D}_G(F_Y), \quad \|\mathcal{D}_G(A)\| \leq \frac{6Mr}{1-M}, \quad \frac{6M}{1-M} \leq \frac{70}{543}. \quad ** \quad (\text{HNM-71})$$

The diagonal map is contractive because its input and output blocks are orthogonal. As  $r = O(|\tau|^3)$ , this is an actual uniform finite-volume quartic upper bound, not an evaluation of a nonzero resonant block. On a spectral window  $P = \mathbf{1}_{[E-\varepsilon, E+\varepsilon]}(G)$ , reverse additionally obtains

$$\|PAP\| \leq \frac{(2\varepsilon + 6M)r}{1-M}, \quad (\text{HNM-72})$$

because  $\|(G - E)P\| \leq \varepsilon$ . If no crossing star is retained,  $F_Y = 0$  and the interior inverse is exact; otherwise zero versus nonzero crossed blocks remains unresolved. Pinching need not preserve locality.

For the Gaussian density  $p_s(t) = e^{-t^2/(2s^2)}/(\sqrt{2\pi}s)$  and odd tail  $h_s(t) = \text{sgn}(t) \int_{|t|}^{\infty} p_s(u) du$ , define  $\mathcal{R}_s(A) = \int p_s(t) \alpha_t(A) dt$  and  $\mathcal{J}_s(A) = -i \int h_s(t) \alpha_t(A) dt$ . Since  $h'_s = \delta_0 - p_s$ ,  $\|h_s\|_1 = s\sqrt{2/\pi}$ , and  $\|p'_s\|_1 = \sqrt{2/\pi}/s$ , weak integration by parts proves

$$\begin{aligned} [\mathcal{J}_s(A), G] &= -A + \mathcal{R}_s(A), & \mathcal{J}_s(A)D(G) &\subset D(G), \\ \mathcal{R}_s(A) &= \mathcal{R}_s(F_Y) + i \int p'_s(t) \alpha_t(K_Y) dt, \\ \frac{\|\mathcal{R}_s(A)\|}{r} &\leq \min \left\{ 1, \frac{6M + \sqrt{2/\pi}/s}{1-M} \right\}. \quad ** \end{aligned} \quad (\text{HNM-73})$$

The residual and inverse spectral multipliers are  $e^{-s^2\omega^2/2}$  and  $(1 - e^{-s^2\omega^2/2})/\omega$ , retaining the zero-frequency block. At  $s = 4$  the bound is  $626/1629 < 0.385$ ; reverse's  $s = 2$  gives  $1042/1629 < 2/3$ . This advances W2 from a column contraction to a full-operator contraction on the fixed source and initial  $G$ . The asymptotic boundary budget is an upper bound, not a positive residual lower bound. Gaussian time support is unbounded, so the old finite-radius connected series cannot establish its locality by direct integration. Sources: `research/round26/forward/ab1/report.md`, `research/round26/reverse/ab1/report.md`, `research/round26/forward/ab2/report.md`, `research/round26/skeptic/ab1.md`, `research/round26/skeptic/ab2.md`.

### 11.1.4 \*\* AE1–AE2: locality, graph error, and the original weighted norm

AE1 uses an all-time interaction-picture commutator recurrence. Each site lies in at most four stars and a four-site star meets at most sixteen possible stars. Removing internal unitaries in the commutator recurrence leaves chains, rather than the growing-union count in (HNM-62). For the forward collar  $C_R = \{x \in \mathbb{N}_0^3 : |x|_1 \leq 2R + 1\}$ , every chain of at most  $R$  steps is retained. Duhamel and the chain exponential then give

$$\begin{aligned} \|\alpha_t^G(A) - \alpha_t^{G_R}(A)\| &\leq r \sum_{m \geq R+1} \frac{(32M|t|)^m}{m!} \leq r 2^{-R-1} e^{64M|t|}, \\ \|\mathcal{R}_s^G(A) - \mathcal{R}_s^{G_R}(A)\| &\leq r 2^{-R} e^{(64Ms)^2/2}. \quad ** \end{aligned} \quad (\text{HNM-74})$$

The last step integrates the entire Gaussian using  $\mathbb{E}e^{v|T|} \leq 2e^{v^2s^2/2}$ . At the old cap and  $s = 4$ ,  $R = 64$  gives relative operator error below  $2^{-42}$  but requires 374,660 complete factors. No local finite Hilbert dimension is assumed.

The independent reverse construction uses  $C_R = \{0, \dots, R+1\}^3$  and  $b_R = 3R^2 + 9R + 7$  crossing stars. With  $v = 64M$ , it bounds the residual and inverse errors by

$$\begin{aligned}\varepsilon_R &= 4rMb_Rs(1+vs)e^{v^2s^2/2}2^{-R}, \\ \kappa_R &= 4rMb_Rs^2e^{v^2s^2/2}2^{-R}, \quad \|\mathcal{J}_s - \mathcal{J}_{s,R}, G\| \leq 2\varepsilon_R.\end{aligned}\tag{HNM-75}$$

The graph conclusion is an additional calculation of every crossing commutator; small operator-norm error by itself would not prove it. At  $s = 2, R = 40$ , the bounds are  $\varepsilon_R/r < 2 \cdot 10^{-7}$  and  $\kappa_R/r < 10^{-7}$  on 74,088 factors. Telescoping collars produces exponential diameter decay, with polynomial translation multiplicities harmless. It does not establish the old support-cardinality norm: a factor  $e^{\mu O(R^3)}$  overwhelms this  $2^{-R}$  upper majorant. That failed majorant is not a divergence theorem. Sources: `research/round26/forward/ae1/report.md`, `research/round26/reverse/ae1/report.md`, `research/round26/skeptic/ae1.md`.

AE2 obtains a simultaneous statement in the original rooted weight by explicitly restricting  $M \leq 10^{-3}$ , equivalently  $|\tau| \leq 1/7000$ . At the fixed proof duration  $T = 3$ , use

$$p_T(t) = \frac{(1 - |t|/T)_+}{T}, \quad h_T(t) = \frac{\text{sgn}(t)}{2}(1 - |t|/T)_+^2, \quad \hat{p}_T(\omega) = \text{sinc}^2(T\omega/2).\tag{HNM-76}$$

Its derivative and moments are  $h'_T = \delta_0 - p_T$ ,  $\|p'_T\|_1 = 2/T$ ,  $\|h_T\|_1 = T/3$ . For a translated interior source, all twelve crossings cost  $24Mr_b/(1-M)$ , so the boundary identity proves

$$\|\mathcal{R}_T(A_b)\| \leq \frac{24M + 2/T}{1-M}r_b \leq \frac{2072}{2997}r_b < 0.7r_b. \quad **\tag{HNM-77}$$

Origin-only three-crossing constants cannot be used for this family. Integrating the same ordered words as S2 yields, with  $c = 192M$ ,  $p = 8/3$ ,

$$\begin{aligned}\|\mathcal{R}_T(\{A_b\})\|_2 &\leq 128r_\star \sum_{n \geq 0} \frac{(p)_n(cT)^n}{(n+2)!}, \\ \|\mathcal{J}_T(\{A_b\})\|_2 &\leq 128Tr_\star \sum_{n \geq 0} \frac{(p)_n(cT)^n}{(n+3)!}.\end{aligned}\tag{HNM-78}$$

Here  $cT \leq 72/125 < 1$ , giving the common bound  $64(125/53)^3r_\star$ . Reverse's independent  $208M$  count gives the smaller displayed sufficient budget  $844000r_\star/2209 < 383r_\star$  at  $208MT = 78/125$ . These prove finiteness, not weighted contraction: their positive residual series already begins at the initial  $64r_\star$  budget.

The same certificates require simultaneously  $T > 2/(1 - 25M)$  and  $T < 1/(192M)$ , possible only if  $409M < 1$ ; the reverse radius requires  $441M < 1$ . At the old cap  $M = 35/1664$ , the first lower bound is  $3328/789$ , while the upper bounds are  $26/105$  and  $8/35$ . This is a precise failure of those joint certificates at the old cap. At their radius endpoint the inverse majorant is integrable but the residual majorant is not. The next missing premise is weighted contraction for generated sources with all scalar, diagonal, and domain updates, not another application of the original-source factor 0.7. Sources: `research/round26/forward/ae2/report.md`, `research/round26/reverse/ae2/report.md`, `research/round26/skeptic/ae2.md`.

## 11.2 A fixed physical graph: from memory to computed relative heat

The graph throughout this subsection has vertices  $\{0, 1, 2\} \times \{0, 1, 2\} \times \{0, 1\}$ , 33 links, 20 square faces, and all 18 vertex Gauss constraints. Its full physical Hilbert space is  $\mathcal{H}_{\text{phys}} = L^2(\text{SU}(2)^{33}, d\text{Haar})^{\text{SU}(2)^{18}}$ . Define

$$\begin{aligned}K &= \sum_e C_e, \quad L = K + \lambda(20 - S), \quad S = \sum_{p=1}^{20} W_p, \\ W_p &= \frac{1}{2} \text{Tr } U_{\partial p}, \quad 0 \leq \lambda \leq 10^{-2}, \quad \sigma = \alpha t_{\text{phys}}/\hbar.\end{aligned}\tag{HNM-79}$$

In heat estimates below, integration variables  $t, s$  and joining times are dimensionless coordinates in the same units as  $\sigma$ ; physical duration is  $t_{\text{phys}} = (\hbar/\alpha)\sigma$ . The multiplication  $20 - S$  lies between zero and 40. Gauge averaging commutes with the electric spectral projections; bounded perturbation preserves the self-adjoint electric domain and its smooth invariant core. The electric resolvent is compact. Its vacuum is unique and the physical electric gap is 3: a nonempty invariant active-edge support has no degree-one vertex, hence contains a cycle, whose at least four spin-half edges cost  $4(3/4)$ . A fundamental square character attains 3. These are full-space facts, not a finite-spin assumption.

### 11.2.1 \*\* T1–T2: a full-space memory model and separate ground centering

For the inherited infinite-rank isometry  $J$  from the selected physical space, write the full Hamiltonian in retained/complementary blocks

$$H_\lambda = \begin{pmatrix} A & B^* \\ B & C \end{pmatrix}, \quad \tilde{H}_\lambda = \begin{pmatrix} A & B^* \\ B & C_0 \end{pmatrix}, \quad C - C_0 = \alpha\lambda D, \quad 0 \leq D \leq 40Q, \quad \|B\| = \frac{5}{2}\alpha\lambda. \quad (\text{HNM-80})$$

The diagonals have their actual inherited electric domains,  $A \geq 19\alpha\lambda$  and  $C_0 \geq 3\alpha$ . The scalar Schur/Young inequality gives  $\tilde{H}_\lambda \geq 18\alpha\lambda I$  throughout the stated interval;  $H_\lambda \geq \tilde{H}_\lambda$  as forms gives the same separate spectral floor for  $H_\lambda$ . No order between their heat operators is inferred. Bounded perturbation establishes both on the full electric domain.

Eliminating the complementary component shows that  $\tilde{T}(t) = J^* e^{-t\tilde{H}_\lambda/\hbar} J$  satisfies

$$\begin{aligned} \tilde{T}(t) &= e^{-tA/\hbar} + \hbar^{-2} \int_{0 \leq u \leq s \leq t} e^{-(t-s)A/\hbar} K_2(s-u) \tilde{T}(u) du ds, \\ K_2(v) &= B^* e^{-vC_0/\hbar} B. \end{aligned} \quad ** \quad (\text{HNM-81})$$

The full returned  $\tilde{T}(u)$  retains repeated excursions. Iterating a difference of two solutions bounds it by  $M_T(\|B\|t/\hbar)^{2n}/(2n)!$ , proving uniqueness at each finite window. The corresponding Schur complement has minus sign  $A + z - B^*(C_0 + z)^{-1}B$ . A one-return substitution loses a fourth-order  $(B^*B)^2$  term. The two earlier diagnostic channels cannot be made autonomous: their full second-derivative defect dominates  $\alpha^2\lambda^2 \begin{pmatrix} 19/4 & 1/4 \\ 1/4 & 19/4 \end{pmatrix} > 0$ . This last actual-model clarification originated in the independent T1 review.

Let  $d = 3 - 18\lambda$ . Duhamel for the purely complementary perturbation has one leakage factor on each side, each bounded by  $(5\lambda/2)e^{-18\lambda\sigma}(1 - e^{-d\sigma})/d$ . Thus

$$\begin{aligned} \|T(t) - \tilde{T}(t)\| &\leq 250\lambda^3 e^{-18\lambda\sigma} F_d(\sigma), \\ F_d(\sigma) &= \frac{\sigma(1 + e^{-d\sigma}) - 2(1 - e^{-d\sigma})/d}{d^2} \leq \min\{\sigma^3/6, \sigma/d^2\}, \\ \sup_{t \geq 0} \|T(t) - \tilde{T}(t)\| &\leq \frac{125\lambda^2}{9e(3 - 18\lambda)^2} \leq \frac{312500}{477144} \lambda^2. \end{aligned} \quad ** \quad (\text{HNM-82})$$

This is cubic in coupling at fixed  $\sigma$ , but its uniform quadratic bound uses decay in the unshifted energy convention.

T2 answers the centering objection rather than multiplying this estimate by uncontrolled ground exponentials. Along  $L_s = \tilde{H}_\lambda/\alpha + s\lambda D$ , min-max gives a simple ground  $\epsilon_s \leq 40\lambda$  and excitation gap  $g = 3 - 85\lambda/2 \geq 103/40$ . The complementary ground inverse has denominator  $d_0 = 3 - 40\lambda \geq 13/5$ , so its leakage is  $\ell = (5\lambda/2)/d_0$ . Differentiation gives  $\epsilon'_s = \lambda\langle\psi_s, D\psi_s\rangle$  and  $\|G'_s\| \leq 40\lambda\ell/g$ . In centered Duhamel the ground-ground term cancels exactly because  $G_s(\lambda D - \epsilon'_s I)G_s = 0$ . Two mixed, magnetic excited, and scalar excited terms cost respectively

$80\lambda\ell/g$ ,  $80\lambda\ell/g$ , and  $40\lambda\ell^2/g$ . Integration in  $s$  proves

$$\sup_{t \geq 0} \left\| J^* e^{-t(H_\lambda - e_\lambda)/\hbar} J - J^* e^{-t(\tilde{H}_\lambda - \tilde{e}_\lambda)/\hbar} J \right\| \leq 60\lambda^2,$$

$$0 \leq e_\lambda - \tilde{e}_\lambda \leq \frac{6250}{169} \alpha \lambda^3, \quad \left\| G_\lambda - \tilde{G}_\lambda \right\| \leq \frac{20000}{1339} \lambda^2. \quad **$$

(HNM-83)

The exact cap coefficient before rounding is  $1042500/17407 < 60$ . The excited remainder has initial complementary component  $-QG_s J$ , which must remain in the bound  $\left\| Q(e^{-u(L_s - \epsilon_s)} - G_s) J \right\| \leq \min\{2\ell, e^{-gu}\}$ . These results concern infinite-rank selected heat, not a finite algorithm or a relative-output theorem. T2 is same-author self-review. Sources: `research/round23/forward/t1/report.md`, `research/round23/reverse/t1/report.md`, `research/round23/skeptic/t1.md`, `research/round23/forward/t2/report.md`, `research/round23/skeptic/t2.md`.

### 11.2.2 \*\* X1: two truncation theorems and an actually assembled sector

A zero-extended finite-rank heat operator has full-space error exactly one at  $\sigma = 0$ . X1 therefore separates two meaningful questions. For the kinetic cutoff  $P_R = \mathbf{1}_{[0,R)}(K)$ ,  $R > 3$ , let  $A_R = P_R L P_R$ ,  $B_R = Q_R L P_R$ ,  $b = 20\lambda$ , and  $c = 20\lambda$ . Then  $\|B_R\| \leq b$  and  $Q_R L Q_R \geq R$ . Complementary variation of constants and its return give

$$\left\| e^{-\sigma L} - e^{-\sigma A_R} P_R \right\| \leq e^{-R\sigma} + \frac{2b}{R} + \frac{b^2\sigma}{R}. \quad (\text{HNM-84})$$

For actual ground energies  $\epsilon, \mu_R$ , Schur elimination and the full excited threshold 3 imply

$$0 \leq \mu_R - \epsilon \leq \frac{b^2}{R - c}, \quad \|G - G_R\| \leq D_R := \frac{b}{R - c} \left( 1 + \frac{b}{3 - c} \right). \quad (\text{HNM-85})$$

For any  $0 < \sigma_0 \leq T$ , separate ground centering consequently gives

$$\sup_{\sigma \geq \sigma_0} \left\| e^{-\sigma(L - \epsilon)} - e^{-\sigma(A_R - \mu_R)} P_R \right\| \leq \min\{2, \max(F_R, D_R + 2e^{-(3-c)T})\},$$

$$F_R = e^{cT} \left[ e^{-R\sigma_0} + 2b/R + b^2 T/R + T b^2/(R - c) \right]. \quad **$$

(HNM-86)

Choosing  $T$  first and then  $R$  proves uniform operator-norm convergence on every positive delayed half-line. The certified example  $\lambda \leq 0.01$ ,  $\sigma_0 = 1$ ,  $T = 4$ ,  $R = 1024$  has error below 0.002; that large matrix was not assembled. A Hermitian finite coefficient perturbation of norm  $\zeta$  adds at most  $4\zeta/(g - 2\zeta)$  uniformly when both finite matrices use their own grounds and  $2\zeta < g$ . Independent rounding of a ground exponent instead produces  $e^{\sigma\delta}$  growth or loss of the limiting ground.

The independent reverse projection retains all invariant Peter–Weyl blocks with degree  $d = \sum_e 2j_e \leq N$ . A face multiplier raises degree by at most four. For initial degree  $d_0 \leq N$ , set  $r = \lfloor (N - d_0)/4 \rfloor + 1$  and  $z = 20\lambda\sigma$ . Dyson words of order below  $r$  agree exactly, including all intermediate states; therefore their uncentered difference is at most  $2e^{-z} \sum_{n \geq r} z^n/n! \leq 2z^r/r!$ . The true ground Neumann series has ratio  $a = 20\lambda/3$  and raises degree by at most four, giving omitted ground norm  $t_N = a^{\lfloor N/4 \rfloor + 1}/(1 - a)$  and energy error  $\delta_N \leq 20\lambda t_N/(1 - t_N)$ . The centered finite-window bound is

$$\left\| (e^{-\sigma(L - \epsilon)} - e^{-\sigma(A_N - \mu_N)} P_N) P_{d_0} \right\| \leq \min\{2, 2e^z z^r/r! + \sigma\delta_N\}. \quad **$$

(HNM-87)

For  $d_0 = 0$ ,  $N = 20$ ,  $0 \leq \sigma \leq 1$ , it is below  $10^{-6}$  at the cap, but that larger matrix also remained unassembled. The cutoff, input, and time sets in (HNM-86) and (HNM-87) are different.

Both constructions identify the same actual rank-21 sector at  $R = 9/2$  or  $N = 4$ :  $\Omega = 1$  and  $\phi_p = 2W_p$ . Gauge support completeness below this energy follows from the four-cycle argument; five-edge supports cannot occur on the bipartite graph. Haar normalization and the absence of a zero parity mask among triples of faces give the complete matrix

$$(A_{21})_{00} = c, \quad (A_{21})_{0p} = -\lambda/2, \quad (A_{21})_{pq} = (3+c)\delta_{pq}, \quad \mu = c - \frac{\sqrt{9+20\lambda^2}-3}{2}. \quad ** \quad (\text{HNM-88})$$

There are nineteen dark face eigenvectors at  $3+c$  and one bright excited root  $c+(3+\sqrt{9+20\lambda^2})/2$ . The spin-one character  $\chi_p = 4W_p^2 - 1$  is omitted, normalized, and has energy 8, with  $\langle \chi_p, L\phi_p \rangle = -\lambda/2$ . Thus the small matrix is a compression with genuine physical leakage. X1's parent practical-truncation gate remains limited. Sources: `research/round24/forward/x1/report.md`, `research/round24/reverse/x1/report.md`, `research/round24/skeptic/x1-review.md`.

### 11.2.3 \*\* X2: complete residual, spectral data, and evaluated late heat

Write  $D = \sqrt{9+20\lambda^2}$ ,  $w = (D-3)/2$ ,  $h = \lambda/[2(3+w)]$ ,  $Z = 1+20h^2$ , and  $f = (\Omega + h \sum_p \phi_p)/\sqrt{Z}$ . The full residual is

$$F = S^2 - 5, \quad (L - \mu)f = -\frac{2\lambda h}{\sqrt{Z}}F, \quad \rho^2 = \frac{195\lambda^4}{4(3+w)D}. \quad ** \quad (\text{HNM-89})$$

To obtain its complete norm, use  $\mathbb{E}W_p^4 = 1/8$  and  $\mathbb{E}W_p^2W_q^2 = 1/16$  for distinct faces, even when they share an edge: integrating an exclusive link first makes each squared factor constant. No four distinct faces have zero combined center parity. Hence

$$\mathbb{E}S^2 = 5, \quad \mathbb{E}S^4 = 295/4, \quad F = \frac{1}{4} \sum_p \chi_p + 2 \sum_{p < q} W_p W_q, \quad \|F\|^2 = 195/4. \quad (\text{HNM-90})$$

This accounts for 20 spin-one and 190 orthogonal pair channels, not the whole omitted Hilbert space. Keeping only spin-one channels loses a factor 39 in the squared residual. The forward-origin electric refinement resolves the 62 shared-edge products into singlet/triplet norms  $1/64, 3/64$ , whereas 128 edge-disjoint products have energy 6. Its complete weight table is

| Electric energy       | 9/2  | 6  | 13/2 | 8   |
|-----------------------|------|----|------|-----|
| Squared weight in $F$ | 31/8 | 32 | 93/8 | 5/4 |

The weights sum to  $195/4$  and their first moment is 295. The independent form identity  $\langle F, KF \rangle = 4\mathbb{E}S^4 = 295$  follows by integration by parts from  $KS = 3S$ . A link-center transformation sends every face to its negative, so the remaining odd potential moment vanishes. Thus for  $\xi = -F/\sqrt{195/4}$ ,  $\langle \xi, L\xi \rangle = \nu = 236/39 + c$ .

The true ground  $\epsilon$  is simple and every other eigenvalue is at least 3, by min-max. Taking the spectral expectation of  $(L - \epsilon)(L - 3) \geq 0$  in  $f$ , and using the actual variational plane  $\text{span}\{f, \xi\}$ , gives

$$\delta_- := \frac{2\rho^2}{\nu - \mu + \sqrt{(\nu - \mu)^2 + 4\rho^2}} \leq \mu - \epsilon \leq \delta_+ := \frac{\rho^2}{3 - \mu},$$

$$\|G - |f\rangle\langle f|\| \leq p := \frac{\rho}{3 - \mu}. \quad ** \quad (\text{HNM-91})$$

Reverse obtains the valid weaker lower displacement  $\rho^2/(8 + 40\lambda - \mu + \rho)$  from complete kinetic support at energies at most 8. At  $\lambda = 0.01$  the forward enclosure is  $0.19983332325 \leq \epsilon \leq 0.19983333365$  and  $p < 8.32 \cdot 10^{-5}$ . Neither trial plane is asserted invariant.

Decomposing each heat operator into its own ground projection and excited remainder yields

$$\left\| e^{-\sigma(L-\epsilon)} - e^{-\sigma(A_{21}-\mu)} P_{21} \right\| \leq p + e^{-(3-\mu)\sigma} + e^{-(3+D)\sigma/2}. \quad (\text{HNM-92})$$

A continuous-coupling envelope and certified scalar evaluator make this less than 0.000085 for every  $\lambda \in [0, 0.01]$ ,  $\sigma \geq 5$ . Forward's distinct cap-only example begins at  $\sigma = 4$  with bound 0.000103. The actual finite representation is  $G_R + e^{-D\sigma}(U - G_R) + e^{-(3+D)\sigma/2}(P_{21} - U)$ , where  $U$  projects onto the vacuum/symmetric-face plane. The computed rational  $\hat{G}_R$  is exactly idempotent and the ground coefficient stays exactly one. Positive gap rounding and scalar exponential errors are budgeted separately. The reverse scalar algorithm uses the degree-160 positive Taylor enclosure for  $e^x$  on  $[0, 96]$ , reciprocates, and returns zero beyond 96. Its uniform  $e^{-x}$  error is below  $3 \cdot 10^{-30}$ , since the reciprocal width is bounded by  $96(80!)^2/[161!(1 - 96/162)]$  and the omitted tail by  $80!/96^{80}$ . The combined representation allowance is below  $3 \cdot 10^{-25}$ . Sources: `research/round24/forward/x2/report.md`, `research/round24/reverse/x2/report.md`, `research/round24/skeptic/x2-review.md`.

#### 11.2.4 \*\* Z1–Z2: a relative obstruction and a preparation-based repair

Let  $E = e^{-\sigma(L-\epsilon)}$  and  $E_R = e^{-\sigma(A_{21}-\mu)} P_{21}$ . The relative quantity is  $\|(E - E_R)x\| / \|Ex\|$  for the same nonzero retained input. Heat injectivity makes the denominator positive at every finite time, but not uniformly in time. The exact vacuum row  $L\Omega = c\Omega - \lambda S$  lies in  $P_{21}$ , and (HNM-91) proves  $\delta = \mu - \epsilon > 0$  at positive coupling. For  $u = (L - \epsilon)\Omega$  one has  $Gu = 0$  but  $\langle f, u \rangle = \delta \langle f, \Omega \rangle > 0$ , whence

$$\frac{\|(E - E_R)u\|}{\|Eu\|} \geq \frac{\delta \langle f, \Omega \rangle}{\|u\|} e^{(3-\mu)\sigma} - 1 \longrightarrow \infty. \quad ** \quad (\text{HNM-93})$$

This is a fixed actual retained vector for each fixed  $\lambda > 0$ , although preparing its exact ground-orthogonal coefficient requires unknown  $\epsilon$ . The computable  $v = (A_{21} - \mu)\Omega$  is instead a bright Ritz eigenvector with true ground overlap  $-\delta \langle \psi, \Omega \rangle \neq 0$ ; its true-relative error tends to one, while its Ritz-output-relative error diverges. At the cap forward certifies the first error above 6.9 at  $\sigma = 6$ , and the second within  $10^{-4}$  of one at  $\sigma = 8$ . The mechanism is distinct ground overlap, not divergence of either correctly centered heat operator. At  $\lambda = 0$  or  $\sigma = 0$  the retained physical error is exactly zero. Sources: `research/round24/forward/z1/report.md`, `research/round24/reverse/z1/report.md`, `research/round24/skeptic/z1-review.md`.

In the remaining heat estimates, integration coordinates  $t, s$  and joining values  $T$  are dimensionless heat coordinates, equal in units to  $\sigma$ ; a duration  $t$  corresponds to physical time  $(\hbar/\alpha)t$ .

Z2 reconstructs the entire retained-to-omitted map, not just its ground column. With  $\eta_{pq} = 4W_p W_q$ , the actual orthogonal residual channels give

$$\begin{aligned} B\Omega = 0, \quad B \sum_p b_p \phi_p &= -\frac{\lambda}{2} \left[ \sum_p b_p \chi_p + \sum_{p < q} (b_p + b_q) \eta_{pq} \right], \\ (B^* B)|_{\text{faces}} &= \frac{\lambda^2}{4} (19I + J_{20}), \quad \|B\| = \frac{\sqrt{39}}{2} \lambda. \quad ** \end{aligned} \quad (\text{HNM-94})$$

This follows by expanding the squared norm on arbitrary complex coefficients. For the exact retained vacuum evolution,

$$BE_R(s)\Omega = -\frac{\lambda^2}{D}(1 - e^{-Ds})F, \quad \|BE_R(s)\Omega\| = \beta(1 - e^{-Ds}), \quad \beta = \frac{\lambda^2 \sqrt{195}}{2D}. \quad (\text{HNM-95})$$

The zero initial omitted column therefore does not remain zero. In fact  $Q(L - \epsilon)^2\Omega = \lambda^2 F$ . The exact centered Duhamel defect is  $B + \delta P$ , not  $B$  alone. Thus

$$V(t) = \beta[t - (1 - e^{-Dt})/D] + \delta_+ t \quad \text{bounds} \quad \|(E(t) - E_R(t))\Omega\|. \quad (\text{HNM-96})$$

For  $f_0 = \langle f, \Omega \rangle$  and  $q_0 = \|(I - G_R)\Omega\|$ , an independent spectral bound is  $J(t) = p + (q_0 + p)e^{-(3-\mu)t} + q_0e^{-Dt}$ , and the true denominator is at least  $f_0 - p$ . The increasing  $V$  and decreasing  $J$  can be joined:

$$\sup_{t \geq 0} \frac{\|(E(t) - E_R(t))\Omega\|}{\|E(t)\Omega\|} \leq \frac{\max\{V(T), J(T)\}}{f_0 - p}. \quad ** \quad (\text{HNM-97})$$

For normalized  $x \in P_{21}$  with  $\|x - \Omega\| \leq \eta$ , add to the early bound  $\eta[(\rho + \delta_+)t + \|B\|(1 - e^{-3t})/3]$ , and use late bound  $p + (q_0 + \eta + p)e^{-(3-\mu)t} + (q_0 + \eta)e^{-3t}$  with denominator  $f_0 - \eta - p$ . This preserves the same prepared input in both evolutions. The continuous envelopes  $\rho, \beta \leq \sqrt{195}\Lambda^2/6$ ,  $g \geq 3 - 20\Lambda$ ,  $q_0 \leq \sqrt{5}\Lambda/3$ , and  $f_0 \geq \sqrt{1 - 5\Lambda^2/9}$  cover every  $0 \leq \lambda \leq \Lambda \leq 0.01$ , rather than a parameter grid. They give

| Input class                                  | joining $\sigma$ | all-time relative bound |
|----------------------------------------------|------------------|-------------------------|
| Vacuum                                       | 3/2              | 0.00028                 |
| Normalized $P_{21}$ ball, radius 0.01        | 8/5              | 0.00044                 |
| Normalized retained Ritz overlap $\geq 0.99$ | 2                | 0.002                   |

Reverse's ball without a normalization requirement has the distinct bound 0.000720 at join 29/10; its vacuum bound is 0.000600. These are not averaged with the sharper normalized results. Ritz overlap  $a > p$  provides the checkable denominator floor  $a - p$ ; uncertainty in overlap, coefficients, or preparation must be subtracted first. Comparing to an ideal vacuum instead of the same prepared input adds the preparation bias separately. Rational projectors and positive-gap scalar evaluation add negligible uniform numerical cost while preserving an exact ground coefficient. Z1's excluded ground-null inputs remain excluded. Sources: `research/round24/forward/z2/report.md`, `research/round24/reverse/z2/report.md`, `research/round24/skeptic/z2-review.md`.

### 11.2.5 **\*\* AC1–AC2: a 293-state physical space and delayed new-channel loading**

Adjoin every old residual channel to  $P_{21}$  and close under  $K$ . A shared-edge pair must be split into its spin-zero and spin-one common-edge parts. In exact rational coordinates the basis and metric are

| Vector type                                | count | norm squared | electric energy |
|--------------------------------------------|-------|--------------|-----------------|
| $\Omega$                                   | 1     | 1            | 0               |
| $\phi_p = 2W_p$                            | 20    | 1            | 3               |
| $\chi_p = 4W_p^2 - 1$                      | 20    | 1            | 8               |
| $d_{pq} = 4W_pW_q$ , edge-disjoint         | 128   | 1            | 6               |
| $s_{pq} = 8X_{pq,s}$ , shared-edge singlet | 62    | 1            | 9/2             |
| $t_{pq} = 8X_{pq,t}$ , shared-edge triplet | 62    | 3            | 13/2            |

Thus  $P_{293}$  has rank  $1 + 20 + 20 + 128 + 124 = 293$ . Orthogonality follows from pair center parity, distinct spin-one supports, and orthogonal common-edge Casimir projections. This is a generated  $K$ -reducing subspace, not an entire electric shell; omitted intertwiners remain in  $Q_{293}$ .

The forward-origin complete magnetic matrix follows from center parity and the exact old columns

$$S\Omega = \frac{1}{2} \sum_p \phi_p, \quad S\phi_p = \frac{1}{2}\Omega + \frac{1}{2}\chi_p + \frac{1}{2} \sum_{q \text{ edge-disjoint}} d_{pq} + \frac{1}{4} \sum_{q \text{ sharing}} (s_{pq} + t_{pq}). \quad ** \quad (\text{HNM-98})$$

All new vectors are even under simultaneous face-sign reversal and all old face vectors are odd, so same-parity magnetic entries vanish. Metric self-adjointness fixes the reverse entries, notably 3/4

from a triplet to each face, not  $1/4$ . The assembled matrix has 1088 directed nonzero magnetic entries and diagonal Gram with 231 weights one and 62 weights three. Its complete omitted map satisfies

$$B_+ = Q_{293}LP_{293} = -\lambda Q_{293}SP_{293}, \quad B_+P_{21} = 0, \quad \|B_+\| \leq 20\lambda. \quad (\text{HNM-99})$$

The bound covers all 272 new columns together, including unexpanded omitted intertwiners; it is not an evaluation of their full Gram.

Because the old Ritz residual is now retained, its angle to the new Ritz ground is at most  $p_{21} = \rho_{21}/(3 - \mu_{21})$ . Since  $B_+f_{21} = 0$ ,

$$\rho_+ \leq 20\lambda p_{21}, \quad \|G - G_+\| \leq \frac{\rho_+}{3 - \mu_+}, \quad 0 \leq \mu_+ - \epsilon \leq \frac{\rho_+^2}{3 - \mu_+}. \quad (\text{HNM-100})$$

At the coupling cap, conservative exact envelopes are  $p_{21} \leq 1/12000$ ,  $\rho_+ \leq 1/60000$ ,  $p_+ \leq 1/168000$ , and  $\delta_+ \leq 1/1008000000$ . AC1 improves ground certification but its coarse all-time ball bound 0.0014 is worse than Z2's 0.00044; increasing dimension alone does not guarantee a better error certificate. Its parent accuracy gate was therefore limited. Sources: `research/round26/forward/ac1/report.md`, `research/round26/reverse/ac1/report.md`, `research/round26/skeptic/ac1.md`.

AC2 exploits the actual loading sequence. Let  $N = P_{293} - P_{21}$ ,  $P_f = P_{21} - |\Omega\rangle\langle\Omega|$ ,  $A_+ = P_{293}LP_{293}$ , and  $C = NA_+P_{21}$ . Exact matrix identities give  $NA_+N = D_N + cN$ ,  $D_N \geq 9/2$ ,  $C\Omega = 0$ , and  $\|C\| = \sqrt{39}\lambda/2$  from (HNM-94). For original inputs  $x \in P_{21}$ , the new retained component  $y(t) = Ne^{-t(A_+ - \mu_+)}x$  obeys the full equation

$$y' = -(D_N + c - \mu_+)y - CP_f e^{-t(A_+ - \mu_+)}x, \quad y(0) = 0. \quad (\text{HNM-101})$$

Its forcing uses the entire retained trajectory and therefore includes feedback. For  $\|x\| = 1$ ,  $\|x - \Omega\| \leq \eta = 0.01$  and  $0 \leq \lambda \leq \Lambda \leq 0.01$ , put

$$g = 3 - 20\Lambda, \quad p_{21} = \frac{7\Lambda^2}{3g}, \quad q_+ = \frac{3\Lambda}{4} + \frac{3p_{21}}{2}, \quad b = q_+ + \eta, \quad \bar{C} = \frac{25\Lambda}{8}, \quad d = 9/2. \quad (\text{HNM-102})$$

The new ground/excited decomposition bounds the forcing by  $q_+ + be^{-gt}$ . Convolution yields

$$\|y(t)\| \leq \bar{C} \left[ q_+ \frac{1 - e^{-dt}}{d} + b \frac{e^{-gt} - e^{-dt}}{d - g} \right]. \quad (\text{HNM-103})$$

Duhamel with the complete  $B_+$  and its separate centering defect consequently has increasing envelope  $\bar{V}(t) = \delta_+t + 20\Lambda\bar{C}[q_+t/d + b/(gd)]$ . The independent late envelope is  $J(t) = p_+ + (2b + p_+)e^{-gt}$ . The true full output is bounded below using the old full-ground result,  $D_\eta = 1 - 5\Lambda^2/9 - \eta - p_{21} > 0$ . At  $\Lambda = 0.01$ , the join  $t_0 = 13/5$  and  $e^{-182/25} < 1/1400$  give

$$D_\eta = \frac{7127}{7200}, \quad \bar{V}(t_0) = \frac{457097}{12600000000}, \quad J(t_0) \leq \frac{2441}{78400000}, \quad (\text{HNM-104})$$

$$\sup_{\sigma \geq 0} \frac{\|(E - E_+)x\|}{\|Ex\|} \leq \frac{457097}{12472250000} < 0.000037. \quad **$$

All envelopes cover the continuous coupling interval. Reverse independently uses face-block norms  $\|NSP_f\| = \sqrt{39}/2$  and  $\|P_fS(I - P_f)P_{293}\| = \sqrt{59}/2$  to obtain a delayed-leakage integral

$$\int_0^t \|B_+E_+(s)x\| ds \leq \frac{130}{9}\lambda^2 \left[ \eta \frac{1 - e^{-3t}}{3} + \frac{4\lambda}{3} \left( t - \frac{1 - e^{-3t}}{3} \right) \right], \quad (\text{HNM-105})$$

leading to 0.000048 for the ball and 0.000043 for the vacuum. The sharper result is confined to original  $P_{21}$  inputs: new retained preparations would not have  $y(0) = 0$ . At  $\sigma = 0$  and at  $\lambda = 0$  the physical retained error is exactly zero. AC2 certifies the exact retained semigroup; numerical evaluation was still a separate task. Sources: `research/round26/forward/ac2/report.md`, `research/round26/reverse/ac2/report.md`, `research/round26/skeptic/ac2.md`.

### 11.2.6 \*\* AF1–AF2: execute the heat and prevent secular centering error

AF1 fixes  $\lambda = 0.01$ ,  $\sigma = 1$ , and input  $\Omega$ , and exports all 293 coordinates. Every norm uses the actual diagonal Gram  $\Gamma$ , with  $\text{Tr } \Gamma = 417$ . A trial residual identifies the minimum only after the spectral separation  $\lambda_2(A_+) \geq 3$  and  $\mu_+ \leq 1/5$  are proved. For a rational trial vector  $v$ , define its Rayleigh quotient  $a$  and residual  $\rho_v^2 = \|A_+ v\|_\Gamma^2 / \|v\|_\Gamma^2 - a^2$ . Then

$$a - \frac{\rho_v^2}{3 - a} \leq \mu_+ \leq a, \quad \|P_v - G_+\| \leq \frac{\rho_v}{3 - a}, \quad P_v x = v \frac{\langle v, x \rangle_\Gamma}{\langle v, v \rangle_\Gamma}. \quad (\text{HNM-106})$$

The first inequality is again the full retained spectral-measure argument, not a small residual near an unidentified eigenvalue.

Forward uses an explicit one-step electric residual correction, a center radius below  $10^{-11}$ , and the degree-100 polynomial in  $A_+ - \hat{\mu}$ ; reverse uses the old Ritz energy interval, center radius below  $10^{-8}$ , and degree 80. Exact integer or rational Horner arithmetic has no accumulated floating roundoff. Since  $\|A_+ - \hat{\mu}\| \leq 9$ , their respective polynomial bounds are  $2^{14}9^{101}/101!$  and  $2 \cdot 9^{81}/81!$ . Coordinate grids  $10^{-50}$  and  $10^{-30}$  cost at most  $21/(2 \cdot 10^{50})$  and  $11 \cdot 10^{-30}$  in the physical metric. The separate center cost is  $d_\mu/(1 - d_\mu)$  at this time. Adding the independently retained physical omission and dividing by the true floor  $7199/7200$  gives full relative point bounds  $0.000015$  forward and  $0.000014$  reverse. Exact Gram comparisons put their exported-vector difference inside the sum of their independent numerical radii. Sources: `research/round26/forward/af1/report.md`, `research/round26/reverse/af1/report.md`, `research/round26/skeptic/af1.md`.

AF2 prevents the uncertain center from being exponentiated indefinitely. It fixes the same coupling and original normalized radius-0.01  $P_{21}$  class, and defines a piecewise approximation:

$$\hat{E}_+(\sigma)x = \begin{cases} P_n(\sigma(A_+ - \hat{\mu}))x, & 0 \leq \sigma \leq 8, \\ P_v x, & \sigma > 8, \end{cases} \quad P_n(H) = \sum_{j=0}^n \frac{(-H)^j}{j!}. \quad (\text{HNM-107})$$

Forward uses degree 256 with uniform Taylor error  $2 \cdot 72^{257}/257! < 10^{-25}$ , center cost  $8d_\mu/(1 - 8d_\mu)$ , and complex export allowance  $15 \cdot 10^{-40}$ . Its corrected rational trial is certified by (HNM-106), independently of the heuristic correction's convergence. Reverse constructs  $v = (I - A_+/9)^{64}\Omega$  exactly. The excited-to-ground power ratio is at most  $15/22$  and vacuum ground overlap squared is at least  $14/15$ ; the final Rayleigh residual certifies the actual resulting vector. It uses degree 240, Taylor bound  $16 \cdot 72^{241}/241!$ , and real export allowance  $11 \cdot 10^{-30}$ .

The late branch has no uncertain ground exponent. For every  $\sigma > 8$ ,

$$\|e^{-\sigma(A_+ - \mu_+)} - P_v\| \leq \frac{\rho_v}{3 - a} + e^{-(14/5)\sigma}. \quad (\text{HNM-108})$$

The full retained excited tail is below  $1/(5 \cdot 10^9)$  in forward's rational certificate and below  $10^{-9}$  in reverse's. Each complete numerical allowance  $e_{\text{num}}$  is below  $10^{-8}$ . Therefore

$$\frac{\|\hat{E}_+(\sigma)x - E(\sigma)x\|}{\|E(\sigma)x\|} \leq \frac{457097}{12472250000} + \frac{e_{\text{num}}}{7127/7200} < 0.000037 \quad (\sigma \geq 0). \quad ** \quad (\text{HNM-109})$$

The two terms are physical compression and retained computation errors. The sample exports include the exact normalized preparations  $(159999\Omega + 800\phi_0)/160001$  and, forward only,  $(159999\Omega + 800i\phi_{19})/160001$ , whose squared vacuum distance is  $4/160001 < 10^{-4}$ . Forward exports eighteen full complex vectors at  $\sigma = 0, 1, 13/5, 8, 9, 1000$ ; reverse exports twelve real vectors, replacing the last time by  $10^6$ . Continuous coverage comes from the two analytic branch bounds, not interpolation between these exports. Outputs are not renormalized. The reverse command interface accepts exact rational times and 293 real rational coordinates, enforcing original support, normalization, and radius; the operator estimate covers complex vectors, but that does not make

the real interface a complex interface. Fixed-coupling computation is not continuous-coupling execution. Time, coefficient, or preparation measurement uncertainty requires its own budget.

The remaining finite-graph tasks are growing-graph control and additional observable/input classes with complete new omission certificates. Neither the all-time heat result nor its small error implies real-time accuracy, a homogeneous numerical gap, physical scale matching, or a continuum construction. Sources: `research/round26/forward/af2/report.md`, `research/round26/reverse/af2/report.md`, `research/round26/skeptic/af2.md`.

### 11.3 Canonical endpoint observables and an exact reached spectrum

Use the canonical summable full-link model, with complete selected-strip and free-link reference factors, unique reference ground  $\Omega$ , and

$$\begin{aligned} H_q &= H_0 + V_q, & V_q &= -\frac{\alpha\tau_q}{24} \sum_{f \text{ omitted}} q^{|a_f|1} x_f, & x_f &= \frac{1}{2} \text{Tr } U_f, \\ b(q) &= \frac{p(q)}{24(1-q)^3(1+q)^2(1+q^2)}, & p(q) &= 2 + 5q + 5q^2 + 6q^3 + 3q^4, \\ \tau_q &= \frac{\eta}{8b(q)}, & 0 < q, \eta < 1. \end{aligned} \quad (\text{HNM-110})$$

The perturbation is norm convergent with  $\|V_q\| = \alpha\eta/8$  and  $D(H_q) = D(H_0)$ . The admitted stationary ground projection satisfies

$$d_q = \|P_q - P_0\| \leq \hat{d}_q := \frac{\sigma_q}{\bar{g}}, \quad \sigma_q^2 = \frac{\alpha^2 \tau_q^2 b(q^2)}{96}, \quad \bar{g} = \frac{\alpha(1-\eta)}{8}, \quad \hat{d}_q = O((1-q)^{3/2}). \quad (\text{HNM-111})$$

For the convention  $\beta_q^t(B) = e^{-itH_q/\hbar} B e^{itH_q/\hbar}$ , define  $C_q^B(t) = \omega_q(B\beta_q^t(B)) - \omega_q(B)^2$ . All endpoint conclusions in this subsection are restricted to

$$T_q = \frac{z}{\eta} \frac{\hbar}{\alpha} (1-q)^{-3}, \quad 0 < z \leq 10^{-6}, \quad s_q = \alpha\tau_q T_q / \hbar \longrightarrow \frac{8z}{7}, \quad s_q \leq \frac{3z}{2}. \quad (\text{HNM-112})$$

The variable  $s_q$  is a proof coordinate at the original physical time  $T_q$ . The trigonometric formulas below have algebraic extensions, but no full stationary endpoint theorem beyond the displayed  $z$  interval is asserted.

#### 11.3.1 \*\* U1–U2: a first-order resonance becomes a full-system witness

Let  $O = (3, 1, 0)$  and  $D = (3, 2, 1)$ , and take paths  $X : O \rightarrow (3, 2, 0) \rightarrow D$ ,  $Y : O \rightarrow (3, 1, 1) \rightarrow D$ , and  $Z : O \rightarrow (4, 1, 0) \rightarrow (4, 2, 0) \rightarrow (4, 2, 1) \rightarrow D$ . Their eight links are free singleton factors. The two real characters  $\phi_X = \text{Tr}(XZ^{-1})$ ,  $\phi_Y = \text{Tr}(YZ^{-1})$  are orthonormal and have energy  $E = 6(3\alpha/4) = 9\alpha/2$ . Set  $\psi_+ = (\phi_X + \phi_Y)/\sqrt{2}$  and  $W = (|\psi_+\rangle\langle\Omega_F| + \text{adjoint}) \otimes I_{F^c}$ . It is a fixed gauge-invariant bounded local operator, with  $W^3 = W$ ,  $\|W\| = 1$ , and reference variance one. It is not a Wilson multiplier.

The yz face at  $O$  has  $x_{f_0} = \text{Tr}(XY^{-1})/2$ . Haar character convolution gives  $\langle\phi_Y, x_{f_0}\phi_X\rangle = 1/4$ . Every other face has an unmatched free-link center parity, and all diagonal terms vanish by the same selection rule. The actual full-model matrix element is therefore

$$\langle\psi_+, V_q\psi_+\rangle = -\frac{\alpha\tau_q q^4}{96}. \quad ** \quad (\text{HNM-113})$$

It gives first-order phase-removed response  $is_q q^4/96 \rightarrow iz/84$ . The vacuum subtraction uses  $\langle\Omega, V_q\Omega\rangle = 0$ , not  $V_q\Omega = 0$ . A derivative alone did not establish an endpoint; U1's gate is limited and its two-state plane is not assumed invariant.

U2 supplies the complete remainder. Each complete reference factor owns at most ten links and meets at most forty omitted faces; one face reaches at most four factors. Starting from eight factors, the ordered connected word count at depth  $n$  is bounded by  $\prod_{k=0}^{n-1} 40(8+3k)$ . Combining coupling  $\alpha\tau_q/24$ , commutator factor two and simplex  $|t|^n/n!$  gives

$$\|\text{order } n\| \leq \frac{(8/3)^n}{n!} (10\alpha\tau_q|t|/\hbar)^n. \quad (\text{HNM-114})$$

Strong bounded-generator Dyson evolution first justifies the expansion for each  $q < 1$ ; connected counting then supplies its useful uniform bound without assuming operator-norm continuity through the unbounded  $H_0$ . For  $x = 10\alpha\tau_q|t|/\hbar < 1$ ,

$$R(x) = (1-x)^{-8/3} - 1 - \frac{8}{3}x \leq \widehat{R}(x) := \frac{44x^2}{9(1-x)^5}. \quad (\text{HNM-115})$$

Taylor's integral remainder proves this using the second derivative  $(88/9)(1-x)^{-14/3}$ ; reverse's independent algebraic bound within the same author's reconstruction is  $x^2(6-8x+3x^2)/(1-x)^3$ . State replacement in the bounded connected expression costs  $6\widehat{d}_q$ , so

$$\begin{aligned} |e^{iEt/\hbar}C_q^W(t) - 1 - is_qq^4/96| &\leq 6\widehat{d}_q + \widehat{R}(10s_q), \\ \liminf_{q \uparrow 1} |C_q^W(T_q) - C_0^W(T_q)| &\geq \frac{z}{84} - \frac{44}{9} \frac{(80z/7)^2}{(1-80z/7)^5} > \frac{z}{168} > 0. \quad ** \end{aligned} \quad (\text{HNM-116})$$

The final strict inequality follows by monotonicity of  $\widehat{R}(80z/7)/z$  and exact arithmetic at  $z = 10^{-6}$ . This makes the earlier sufficient  $\gamma < 3$  time-window exponent sharp in the existence-of-one-observable sense; it does not decide every observable or compute the limiting trajectory. Both U loops use same-author self-review. Sources: [research/round23/forward/u1/report.md](#), [research/round23/reverse/u1/report.md](#), [research/round23/forward/u2/report.md](#), [research/round23/reverse/u2/report.md](#), [research/round23/skeptic/u1.md](#), [research/round23/skeptic/u2.md](#).

### 11.3.2 \*\* V1–V2: finite outcomes are not a finite implementation

The exact spectral projections  $P_{\pm} = (W^2 \pm W)/2$ ,  $P_0 = I - W^2$  give outcomes  $-1, 0, +1$ , with an infinite-dimensional local zero eigenspace. For self-adjoint  $\|A - W\| \leq \delta$ ,  $\|A\| \leq 1 + \delta$ , two applications of the product difference inequality, including the connected means, prove

$$|C_q^A - C_q^W| \leq 4\delta + 2\delta^2, \quad |(C_q^A - C_0^A) - (C_q^W - C_0^W)| \leq 8\delta + 4\delta^2. \quad (\text{HNM-117})$$

This is uniform in time. Choosing fixed  $\delta = z/100000$  retains endpoint margin greater than  $z/200$ ; for contractions the two-model cost improves to  $8\delta$ . A binary first-moment POVM instead has outcome-square one and does not identify  $\omega(W^2)$  with its outcome variance.

Ordinary sequential projective measurements cannot generally recover the complex unordered correlation. Set the self-adjoint unitary completions  $R_{\pm} = W \pm P_0$ . Then

$$\omega(W\beta^t(W)) = \frac{1}{4} \sum_{s,r \in \{+, -\}} \omega(R_s\beta^t(R_r)). \quad ** \quad (\text{HNM-118})$$

Four controlled-unitary products and two ancilla quadratures require eight binary settings, plus a separate mean measurement. This applies the established interferometric primitive; it assumes coherent access to the actual full both-sign evolution and the intended stationary states. The exact PVM and instrument identity do not construct those resources.

A separate V1 obstruction concerns the full local Haar operator norm. For every multiplication operator  $M_g$ , a weakly-null orthonormal sequence in a positive-measure level set of  $|g|$  gives

$\|W - M_g\| \geq \|g\|_\infty$  because  $W_F$  has finite rank. The triangle inequality also gives  $\|W - M_g\| \geq 1 - \|g\|_\infty$ , hence

$$\inf_g \|W_F - M_g\| \geq \frac{1}{2}. \quad ** \quad (\text{HNM-119})$$

The gate admits this in the full local Haar norm, not an unconstructed physical-quotient norm. It blocks this norm-transfer route to Wilson observables, not the existence of a different Wilson witness.

For V2, the actual eigenvectors defining  $W$  prove domain preservation and

$$\|[H_0, W]\| = 9\alpha/2, \quad \|[H_q, W]\| \leq \alpha(9/2 + \eta/4). \quad (\text{HNM-120})$$

Graph closure extends the finite-factor commutator, and strong integration then gives a norm-Lipschitz physical-time bound for this particular observable. For one predeclared endpoint, eighteen independent bounded-output settings,  $N$  trials per setting, and tolerance  $r$ , the exponential-moment argument and union bound give failure probability at most  $36e^{-Nr^2/2}$ . In forward's convention, full trace-norm preparation error  $s$ , complete-setting operator error  $g$ , and a common per-model time offset  $|\Delta t| \leq (\hbar/\alpha)\theta$  yield

$$N \geq \frac{2}{r^2} \log \frac{36}{\kappa}, \quad |\hat{D} - (C_q^W(T_q) - C_0^W(T_q))| \leq 8(r + s + g) + (9 + \eta/4)\theta \quad (\text{HNM-121})$$

with confidence at least  $1 - \kappa$ . Reverse permits per-shot timing jitter, uses half trace distance  $\varepsilon_p$ , and obtains instead  $8(r + 2\varepsilon_p + \varepsilon_i) + 2(9 + \eta/4)\alpha\hbar/\hbar$ ; the timing coefficients are not interchangeable.

At  $\eta = 1/2, z = 10^{-6}, q = 1 - 10^{-6}$ , forward takes  $r = s = g = \theta = 10^{-10}$  and  $N = 1.8 \cdot 10^{21}$ , giving  $3.24 \cdot 10^{22}$  preparations, a finite- $q$  true margin above  $10^{-8}$ , and confidence radius  $3.3125 \cdot 10^{-9}$  at physical time  $2 \cdot 10^{12}\hbar/\alpha$ . Reverse's different example,  $q = 1 - 10^{-8}$ , has radius  $8.4225 \cdot 10^{-10}$  and time  $2 \cdot 10^{18}\hbar/\alpha$ . Neither is an implemented or efficient experiment, an optimal resource lower bound, or confidence over a continuum of times. Sources: `research/round24/forward/v1/report.md`, `research/round24/reverse/v1/report.md`, `research/round24/skeptic/v1-review.md`, `research/round24/forward/v2/report.md`, `research/round24/reverse/v2/report.md`, `research/round24/skeptic/v2-review.md`.

### 11.3.3 \*\* Y1–Y2: spatial collars and an actual scalar limit

Starting from the eight free factors, recursively include every complete owner of every omitted face meeting the current factors. Complete selected strips must be included because reference evolution spreads an operator within its whole factor. Forward retains all omitted faces internally supported on the new factor set; reverse retains those meeting the preceding set. The conventions therefore agree on factors and links but not on their finite Hamiltonians:

| Depth | factors | links | forward faces | reverse faces |
|-------|---------|-------|---------------|---------------|
| 1     | 35      | 98    | 20            | 19            |
| 2     | 135     | 297   | 129           | 125           |

Both are unrenormalized subsets of the actual  $q$  profile, retain all Gauss actions, and still have infinite-dimensional onsite Haar spaces. Every connected word of length at most  $k$  lies in collar  $k$ ; longer retained words cancel exactly when comparing the two expansions. One full tail, not two, therefore suffices:

$$\sup_{|t| \leq T_q} \left\| \beta_q^t(W) - \beta_{q,k}^t(W) \right\| \leq D_k(15z), \quad D_k(x) = \frac{(8/3)_{k+1} x^{k+1}}{(k+1)!(1-x)^{k+4}}. \quad ** \quad (\text{HNM-122})$$

The bound includes every boundary excursion and return. Reverse independently uses the coefficientwise larger  $(3)_n/n!$  and exact rational tail

$$B_k(x) = \frac{x^{k+1}[(k+2)(k+3) - 2(k+1)(k+3)x + (k+1)(k+2)x^2]}{2(1-x)^3}. \quad (\text{HNM-123})$$

Each regional Hamiltonian has its own unique stationary ground. Comparing both states and both connected means costs  $6(d_q + d_{q,k})$ , bounded by  $12\hat{d}_q$  in forward's route. Thus the forward 98-link region retains limiting difference greater than  $z/250$ ; at  $z = 10^{-6}$  its spatial error is below  $1.101 \cdot 10^{-9}$  and depth two below  $2.568 \cdot 10^{-14}$ . Finite spatial support does not make these evaluated finite matrices. Sources: `research/round24/forward/y1/report.md`, `research/round24/reverse/y1/report.md`, `research/round24/skeptic/y1-review.md`.

For a fixed collar, its reference  $K_k = H_{0,k}/\alpha$  is a finite sum of link Casimirs plus bounded strip potentials and scalar ground shifts. Peter–Weyl projections below finite energy have finite rank; hence  $K_k$  has compact resolvent even before gauge reduction. Let  $A_k = -\sum_{f \in O_k} x_f/24$  and retain the entire degenerate energy projections in  $\bar{A}_k = \sum_E P_E A_k P_E$ . This is a bounded self-adjoint strong direct sum preserving  $D(K_k)$ ; its vacuum block vanishes.

The needed averaging argument can be proved directly. For a self-adjoint pure-point  $K$ , bounded  $A$ , and  $\ell > 0$ , put  $Z_\ell(s) = e^{isK/\ell} e^{-is(K/\ell+A)}$  and  $R_\ell(s) = \int_0^s (e^{iuK/\ell} A e^{-iuK/\ell} - \bar{A}) du$ . On a fixed energy block the nonresonant coefficient is

$$P_F R_\ell(s) P_E = \frac{\ell(e^{is(F-E)/\ell} - 1)}{i(F-E)} P_F A P_E \quad (F \neq E). \quad (\text{HNM-124})$$

Orthogonal output energies and density imply strong convergence of this primitive, uniformly on compact  $s$  intervals and compact input sets. Integration by parts in Duhamel makes it act only on the compact averaged orbit and its bounded derivative, giving strong convergence of  $Z_\ell$  and its adjoint to  $e^{-is\bar{A}}$ . If  $\Delta_E = \text{dist}(E, \text{spec } K \setminus \{E\}) > 0$ , the same proof yields the fixed-block estimate

$$\sup_{|s| \leq S} \|(Z_\ell(s) - e^{-is\bar{A}}) P_E\| \leq \frac{2\ell M}{\Delta_E} (1 + 2SM), \quad \|A\| \leq M. \quad (\text{HNM-125})$$

This is not a bound over all excited energies. Compact-resolvent doublets with vanishing splittings provide explicit counterexamples to a general operator-norm averaging inference.

In the actual collar,  $M_k = N_k/24$  and  $\|A_{q,k} - A_k\| \leq (1-q) \sum_{f \in O_k} |a_f|_1/24$ . At  $\ell = \tau_q$  and the original  $T_q$ , the exact ordered scalar is

$$e^{iEt/\hbar} \langle \Omega, W \beta_{q,k}^t(W) \Omega \rangle = \langle \psi_+, Z_{q,k} W Z_{q,k}^* \Omega \rangle. \quad (\text{HNM-126})$$

Both strong propagator limits are needed. The averaged vacuum is fixed, so for the entire energy-9/2 block  $B_k = P_{k,9/2} A_k P_{k,9/2}$  its regional limit is  $f_k(z) = \langle \psi_k, e^{-i(8z/7)B_k} \psi_k \rangle$ . The uniform spatial bound makes  $f_k$  Cauchy, and choosing  $k$  first and then  $q \uparrow 1$  proves

$$e^{iET_q/\hbar} C_q^W(T_q) \longrightarrow F_\infty(z), \quad F_\infty(z) = \lim_{k \rightarrow \infty} f_k(z), \quad |F_\infty - f_k| \leq D_k(15z). \quad ** \quad (\text{HNM-127})$$

No global  $q = 1$  generator or compact resolvent of the infinite reference is used. At this gate the actual matrices and their excited separations remained unevaluated. The forward-origin disk

$$|F_\infty(z) - 1 - iz/84| \leq D_2(15z) + \frac{1}{2}(8z/7)^2(129/24)^2 \quad (\text{HNM-128})$$

has radius below  $1.89 \cdot 10^{-11}$  at  $z = 10^{-6}$ , by spectral Taylor remainder and the known first moment  $-1/96$ . It encloses an actual scalar, not a computed trajectory. This proof exceeds the finite-spectral bounded reference hypotheses of the related norm-averaging theorem, so its pure-point strong argument is essential. Sources: `research/round24/forward/y2/report.md`, `research/round24/reverse/y2/report.md`, `research/round24/skeptic/y2-review.md`.

### 11.3.4 \*\* AA1–AA2: compute a reducing component, not the entire shell

The unit cube translated by  $O = (3, 1, 0)$  has twelve free links. On its physical energy-9/2 shell, quarter-energy arithmetic gives  $3n_{1/2} + 8n_1 + 15n_{3/2} = 18$ . Besides six half-spin links, the only solution uses one half-spin and one spin-3/2 edge and is excluded by a gauge leaf. At each cube vertex zero or two half-spin edges are required, so six occupied edges form one simple six-cycle with a unique bivalent invariant intertwiner. The sixteen six-cycles therefore form a complete orthonormal physical cube shell. Orthogonality follows from an unmatched edge-center sign.

A face can connect two such cycles only when their symmetric difference is its four-edge boundary, replacing two adjacent edges by the other two. The same Haar convolution as U1 makes every allowed matrix element  $1/4$ ; thus the compressed potential is  $-A/96$ , with **A** the exact sixteen-node flip adjacency. Enumeration and exact polynomial multiplication give

$$\begin{aligned} A(A^2 - 4I)(A^2 - 12I) &= 0, \\ \text{spec } A &= \{0^{(8)}, 2^{(3)}, (-2)^{(3)}, (2\sqrt{3})^{(1)}, (-2\sqrt{3})^{(1)}\}. \end{aligned} \quad ** \quad (\text{HNM-129})$$

Twelve vertices have degree two and four degree six. The original coherent source, reconstructed from its paths, has adjacency moments 1, 1, 4, 8, 32, 80, 320 through order six. Polynomial spectral projectors give weights  $1/3$  at zero,  $3/8$  and  $1/8$  at  $+2, -2$ , and  $1/12 \pm 1/(8\sqrt{3})$  at  $\pm 2\sqrt{3}$ . AA1 correctly remained limited because cube-shell completeness alone did not exclude resonant exterior loading.

AA2 proves that missing inclusion. A face external to the cube but sharing cube edges shares exactly one, with twenty such actual faces. Its opposite edge is free and the remaining two sides have distinct exterior reference owners. The minimum  $-3/4$  change of an occupied common edge is canceled by  $+3/4$  on the opposite free edge. A remaining free side costs  $3/4$ ; a strip side creates an endpoint-charged vector orthogonal to its gauge-invariant strip ground and costs at least  $1/8$ . Distinct ownership therefore gives a spectral support increment at least  $1/4$ , not merely a mean-energy inequality. Cube-disjoint faces excite the exterior by at least  $1/8$ . For internal faces with  $r$  occupied edges and  $m$  spin-one fusion branches, the change is

$$\Delta E = 3 - \frac{3}{2}r + 2m, \quad (\text{HNM-130})$$

whose nonzero values have magnitude at least  $1/2$ . Consequently the sixteen-state space  $\mathcal{L}$  times the unchanged exterior ground reduces every sufficiently large regional pinched operator. It need not exhaust the whole energy shell, and the instantaneous interaction still leaks. All participating nonzero frequencies from  $\mathcal{L}$  or the vacuum are at least  $1/8$  in magnitude, giving the evaluated primitive bound  $16\tau_q M_k$  and averaging error  $b_k = 16\tau_q M_k(1 + 3zM_k)$ .

These facts promote the AA1 cube scalar to the actual full canonical endpoint:

$$\begin{aligned} F_\infty(z) &= \frac{1}{3} + \frac{1}{2} \cos(2\theta) + \frac{1}{6} \cos(2\sqrt{3}\theta) + i \left[ \frac{1}{4} \sin(2\theta) + \frac{1}{4\sqrt{3}} \sin(2\sqrt{3}\theta) \right], \\ \theta &= z/84, \quad 0 < z \leq 10^{-6}, \quad |F_\infty - 1 - i\theta| \leq 2\theta^2 = \frac{z^2}{3528}. \end{aligned} \quad ** \quad (\text{HNM-131})$$

The complete second moment is four; it also gives  $|F_\infty - e^{i\theta}| \leq 3\theta^2/2$  by centering the spectral variable at its mean one. A two-state fit would lose the actual second and higher moments.

For finite  $q$ , keep the exact weighted flip matrix  $Q(q)$ , whose entries are  $q^4$  or  $q^5$  and whose symmetric row norm is at most six. Combining direct full stationary-state replacement, the collar tail, and the two averaging columns gives

$$\left| e^{iET_q/\hbar} C_q^W(T_q) - \langle \psi_+, e^{is_q Q(q)/96} \psi_+ \rangle \right| \leq 6\hat{d}_q + D_k(15z) + 32\tau_q M_k(1 + 3zM_k). \quad (\text{HNM-132})$$

A degree-eight rational scalar Taylor center adds at most  $(s_q/16)^9/9!$ ; comparison directly to the limiting formula separately adds  $s_q(1 - q^5)/16 + |s_q - 8z/7|/16$ . At  $\eta = 1/2, z = 10^{-6}, q = 1 - 10^{-12}$

and depth three with  $N_3 = 332$ , the state, spatial, averaging, and scalar errors are respectively below  $6.547 \cdot 10^{-19}$ ,  $5.455 \cdot 10^{-19}$ ,  $2.530 \cdot 10^{-34}$ , and  $1.334 \cdot 10^{-70}$ , totaling less than  $1.201 \cdot 10^{-18}$  about an exact rational center. The physical time is  $2 \cdot 10^{30} \hbar/\alpha$ . This is a mathematical certificate, not physical feasibility or clock calibration. AA1/AA2 are same-author self-review. Sources: `research/round25/solo/aa1/report.md`, `research/round25/solo/aa2/report.md`, `research/round25/advisor/aa1-gate.json`, `research/round25/advisor/aa2-gate.json`.

### 11.3.5 \*\* AD1–AD2: null and informative Wilson multipliers

AD1 tests actual elementary Wilson multiplication. Forward chooses the yz face at  $O$  and reverse the xy face there; their finite- $q$  collars and constants remain distinct. Each  $B = \text{Tr } U_f$  has norm two, reference variance one, and created vector  $\psi = B\Omega$  of energy  $3\alpha$ . Its square creates an extra spin-one state:  $B\psi = \Omega + \chi_{1,f}$ . For internal cube faces the overlap counts are  $r = 0, 1, 4$ , so (HNM-130) has no zero branch. The complete exterior charge argument excludes every external equal-energy image. Therefore  $\bar{A}_{q,k}\psi = 0 = \bar{A}_{q,k}\Omega$ , with participating nonzero frequencies bounded below by  $1/8$ . Applying the ordered scalar identity with the actual multiplier norm gives, in forward's eight-factor convention,

$$\begin{aligned} \left| e^{i3\alpha T_q/\hbar} C_q^B(T_q) - 1 \right| &\leq 24\hat{d}_q + 4D_k(15z) + 48\tau_q M_k(1 + 3zM_k), \\ \lim_{q \uparrow 1} e^{i3\alpha T_q/\hbar} C_q^B(T_q) &= 1. \end{aligned} \quad ** \quad (\text{HNM-133})$$

The right-column error costs  $2b_k$  and the remaining created-vector error  $b_k$ ; variance one does not make this multiplier norm one. Reverse's twelve-factor seed uses  $(4)_n/n!$  and its own tail  $(4)_{k+1}x^{k+1}/[(k+1)!(1-x)^{k+5}]$ , with conservative averaging cost  $4b_k$ . The null phase-removed scalar carries no slow-rate information. Nonzero variance therefore does not establish an informative probe, and AD1's scale bridge remains limited. Its selected forward finite- $q$  example has radius below  $5 \cdot 10^{-18}$  at the same impractical clock as AA2. Sources: `research/round26/forward/ad1/report.md`, `research/round26/reverse/ad1/report.md`, `research/round26/skeptic/ad1.md`.

AD2 freezes the actual original six-cycle multipliers  $B_X = \chi_X = \text{Tr}(XZ^{-1})$  and  $B_\Sigma = (\chi_X + \chi_Y)/\sqrt{2}$ . Their exact multiplication norms are two and  $2\sqrt{2}$ ; both have mean zero and variance one. Their created vectors are respectively  $\phi_X$  and  $\psi_+$ , but their fourth moments are two and  $5/2$ , versus one for the rank observable. In particular the extra squared norms of  $(B_X^2 - I)\Omega$  and  $(B_\Sigma^2 - I)\Omega$  are one and  $3/2$ . Equality of a created vector therefore transfers only the scalar mechanism proved with the actual multiplier, not the entire operator or its error budget.

The reducing component from AA2 applies to both sources. Exact polynomial spectral projectors give for the single X source weights  $2/3$  at zero,  $1/8$  at each of  $\pm 2$ , and  $1/24$  at each of  $\pm 2\sqrt{3}$ . Its moments are 1, 0, 2, 0, 16, 0, 160. Consequently, for the full stationary correlations on the original clock,

$$\begin{aligned} F_X(z) &= \frac{2}{3} + \frac{1}{4} \cos(2\theta) + \frac{1}{12} \cos(2\sqrt{3}\theta), \\ F_\Sigma(z) &= F_\infty(z) \text{ of (HNM-131), } \quad \theta = z/84, \quad 0 < z \leq 10^{-6}, \\ \theta^2 - \frac{2}{3}\theta^4 &\leq 1 - F_X(z) \leq \theta^2, \quad |F_\Sigma(z) - 1 - i\theta| \leq 2\theta^2. \end{aligned} \quad ** \quad (\text{HNM-134})$$

The positive lower quadratic witness follows from  $1 - \cos x \geq x^2/2 - x^4/24$  and the complete second/fourth moments. The coherent sum has a linear imaginary witness. A constant fitted to X's zero slope fails its curvature; a single cosine fitted to second moment two predicts fourth moment four, whereas the actual fourth moment is sixteen. A single phase fitted to the coherent mean one predicts second moment one, whereas the actual value is four. These are discriminators of named reduced descriptions, not universal model-identification claims.

For a source  $\psi_j = B_j \Omega$  of norm one and  $J_j = \|B_j\|$ , the complete finite- $q$  budget is

$$\left| e^{iET_q/\hbar} C_q^{B_j}(T_q) - \langle \psi_j, e^{is_q Q(q)/96} \psi_j \rangle \right| \leq 6J_j^2 \hat{d}_q + J_j^2 D_k(15z) + (1 + J_j)b_k. \quad (\text{HNM-135})$$

The factor  $1 + J_j$  follows by first replacing the vacuum column and then using the normalized created source on the remaining column. Use  $J_X^2 = 4$ ,  $J_\Sigma^2 = 8$ , and  $1 + 2\sqrt{2} < 4$ ; rank-norm budgets would undercount state and spatial errors. At the AA2 sample parameters, degree-eight exact rational centers have total radii below  $5 \cdot 10^{-18}$  for X and  $10^{-17}$  for the sum. Their certified displacements exceed these radii, giving actual finite- $q$  as well as limiting witnesses. If comparing to the limiting trigonometric formulas, the coefficient/time correction after (HNM-132) remains separate.

The Wilson obstruction in V1 is thus resolved by new source-specific observables, not by falsely approximating the rank operator in the forbidden norm. The remaining bridge is empirical or cross-model scale identification with frozen observables, generators, preparations, and held-out data. The elementary null probe, multiple-frequency witnesses, and physical-time budgets establish none of that matching by themselves. They also supply no global  $q = 1$  dynamics, homogeneous stability theorem, or four-dimensional continuum mass gap. Sources: `research/round26/forward/ad2/report.md`, `research/round26/reverse/ad2/report.md`, `research/round26/skeptic/ad2.md`.

## 12 What changed, and what the results can be used for

The most consequential changes are changes in what the evidence licenses. Early numerical consistency became a demand for an explicit regulator and common current/stress definition. A finite matrix gap became a full-space statement only after controlling the representation tail. An appealing local inverse became a commutator identity with a retained boundary defect. A nonzero-variance Wilson probe became an identification problem after its slow response proved null. A formal retained semigroup became an implemented approximation only after the arithmetic error and the late-time ground component were controlled. These changes are useful even when the original parent objective remains open.

### 12.1 From closed retained dynamics to complete channel loading

For a self-adjoint generator  $L$  and an orthogonal retained projection  $P$ , let

$$A = PLP, \quad B = QLP, \quad D = QLQ, \quad Q = I - P. \quad (\text{HNM-136})$$

Assume the domain and boundedness conditions used in the relevant projected problem. With  $x = Pu$  and  $y = Qu$ , the exact heat equations are  $x' = -Ax - B^*y$ ,  $y' = -Bx - Dy$ . Solving the second equation gives

$$x'(t) = -Ax(t) - B^*e^{-tD}y(0) + \int_0^t B^*e^{-(t-s)D}Bx(s) ds. \quad (\text{HNM-137})$$

This is the familiar projected-memory construction. The workbench contribution is its physical channel identification and controlled specialization, rather than invention of the formula. If  $BP_{\text{old}} = 0$ , initial leakage on an old subspace vanishes. It does not follow that  $Be^{-tA}P_{\text{old}} = 0$ :  $A$  can load a new retained channel on which  $B$  acts. AC2 exploits this delay quantitatively instead of deleting that channel.

Possible applications include certified reduced models for finite gauge systems, testing truncation designs, and preparing benchmarks for quantum-simulation algorithms. The practical advance is a bound on the full output, rather than convergence between two arbitrary finite cutoffs. A useful larger-graph application still requires a new complete basis, residual, and denominator calculation.

## 12.2 From a useful heat bound to an executable all-time result

The exact physical approximation and the finite-arithmetic approximation contribute different errors. On the declared normalized preparation class, suppose

$$\|E(\sigma)x - E_P(\sigma)x\| \leq \varepsilon_{\text{phys}}, \quad \|E_P(\sigma)x - \hat{E}_P(\sigma)x\| \leq \varepsilon_{\text{num}}, \quad \|E(\sigma)x\| \geq d_0 > 0. \quad (\text{HNM-138})$$

The triangle inequality yields

$$\frac{\|E(\sigma)x - \hat{E}_P(\sigma)x\|}{\|E(\sigma)x\|} \leq \frac{\varepsilon_{\text{phys}} + \varepsilon_{\text{num}}}{d_0}. \quad (\text{HNM-139})$$

All three quantities need proof on the same input and time class. In particular, a relative theorem may fail when the true output decays to zero, even if an absolute theorem holds.

The all-time computational design addresses a second issue: a small error in a ground-energy shift causes secular drift. If an exact ground vector has energy  $\mu$  but the program shifts by  $\hat{\mu} = \mu + \delta$ , its ground component is multiplied by  $e^{\delta\sigma}$ . A fixed nonzero  $\delta$  cannot support a uniform all-time guarantee. AF2 instead joins a certified finite-time polynomial evaluation to a certified ground projector with an exponentially bounded excited tail. The join is an algorithmic choice, not a new physical clock. Possible uses include long-heat-time state preparation, reproducible benchmark data, and testing alternative polynomial or rational approximations. None of these uses licenses real-time evolution without a separate theorem.

## 12.3 From a source correction to a nonlinear stability problem

An identity  $[J, G] = -A + R$  with  $\|R\| < \|A\|$  is useful evidence about the specified source  $A$  and generator  $G$ . The next transformation generally changes both the source and the diagonal operator. A full iteration needs, for example, bounds on a declared map

$$(G_n, A_n) \mapsto (G_{n+1}, A_{n+1}, J_n) \quad (\text{HNM-140})$$

in one complete interaction norm, along with control of scalar and diagonal updates and the summability of the corrections. This display states a future obligation, not a derived recurrence. AE2 proves finite original weighted norms and per-source contraction in its smaller coupling region; it does not prove contraction of the collective weighted interaction. The distinction prevents an unsupported homogeneous numerical mass-gap claim.

The current construction may serve as a local preconditioner, an explicit approximate homological solver, or a starting point for a future stability iteration. Such applications are proposed mathematical uses; no faster numerical algorithm or new many-body phase has been measured here.

## 12.4 From a resonant state to an informative observable

An operator  $B$  can prepare a state with nonzero variance while the selected averaged interaction annihilates that state. Then the selected slow readout is constant and cannot identify the slow parameter. AD1 supplies this control. AD2 uses actual six-cycle Wilson multiplication observables with nontrivial endpoints, retaining their own norms and higher moments. The distinction matters because a rank-one preparation surrogate and a multiplication operator may agree on  $\Omega$  but differ elsewhere.

These results can inform the selection of mathematical test observables and simulation readouts. They do not demonstrate laboratory preparation, finite-resource feasibility at the sample physical clock, or a calibration connecting the summable model to the homogeneous model. The formula depends on the parameter combinations actually present; naming a frequency is not a measurement of an independent physical scale.

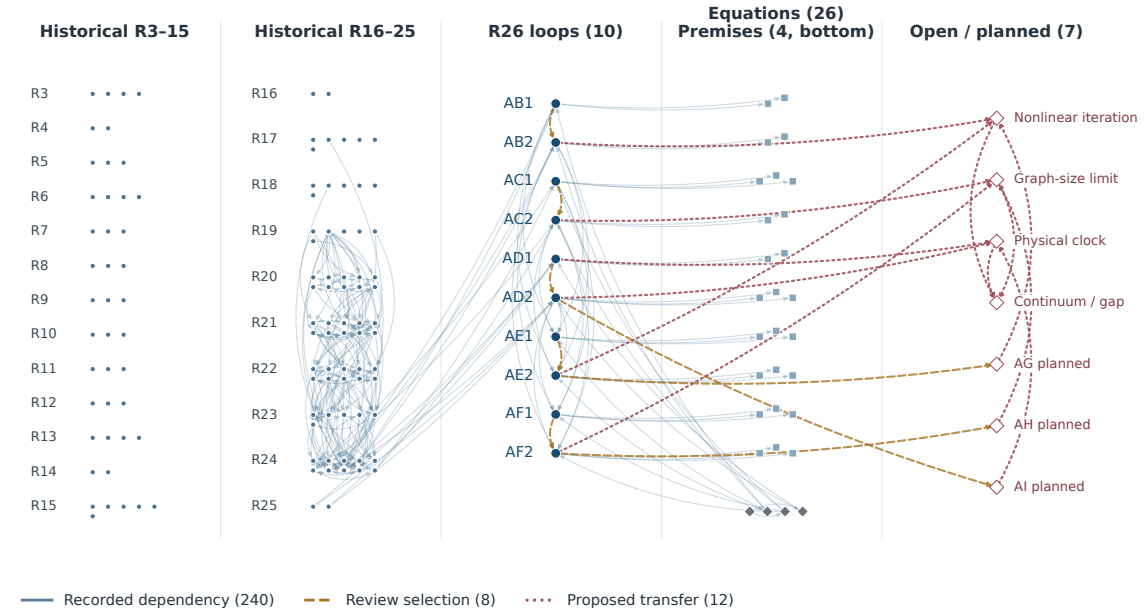
## 12.5 Evidence networks and nonphysical geometry

The released evidence network has 158 nodes and 260 directed links: 111 historical entries, 26 displayed equations, ten current loops, four premise nodes, and seven open targets. Its three edge types are source dependency, proposed transfer, and review selection. A source dependency records the declared use of a file or result; it is not necessarily a minimal mathematical dependency. Review-selection edges record chronology rather than logical implication. In particular, the drawing cannot close an edge labelled proposed transfer.

The grouped appendix and machine-readable ledgers make this network useful as a research-navigation and audit artifact. Figure 1 shows its organization; node coordinates have no physical interpretation. A new calculation can be attached to the exact obstruction it addresses, while failed or limited outcomes remain visible.

Recorded evidence relations: all 158 nodes and 260 directed edges

Schematic organization by historical round and current node kind; position and distance have no physical meaning.



Source: research/round26/network.json at commit 40960f39a3dc. Every structural node and edge is included; labels are selective. A recorded dependency is provenance, not permission to transfer a theorem between models. Planned nodes are not completed investigations.

Figure 1: The recorded evidence network, grouped for readability. Edges retain their audit meaning; geometric placement is a presentation choice. Open targets are not established conclusions.

## 13 Verification, stress tests, and interpretation of the figures

Verification has three distinct layers. First, the pinned workbench contains written model-specific derivations and source-bound historical admission records. Second, the Round26 release was replayed from a clean detached tree: 40 current producer executions and four inherited AA executions covered normal and optimized Python, 171 admission mutations were rejected in each mode, 156 generated files were compared, and seven interface suites passed. These are evidence of reproducibility and admission integrity, not a proof-assistant verification of the mathematics. Third, this manuscript adds a separately authored mathematical stress suite and parameterized calculators. Repeated execution in another interpreter mode is not counted as an additional independent derivation.

### 13.1 What the new checks challenge

The fresh calculator suite passes 194 checks; the separately authored skeptic suites pass 158 checks (90 general, 12 early, 33 middle, and 23 recent). Each corresponding normal and optimized Python run has identical result bytes. The standalone HTML formulas pass 15 additional checks. These are distinct programmed checks, not counts of independently proved theorems or measures of statistical confidence.

The independent skeptic suite derives selected identities without importing the producer implementations. It checks normalized Haar moments and their differential recurrence, tests a false moment closure, distinguishes two-trace from three-trace information, reconstructs the shared-link mixed derivative, enumerates the two-square shell minima, and recertifies the one-square continuous gap through a separate rational Sturm and full-tail argument. Further tests expose delayed omitted-channel loading, distinguish a source identity from an operator identity, count translated boundary crossings, check Wilson spectral moments and phase signs, and verify the rational preparation and combined AF2 error budgets. The audit ledger records which manuscript contributions were reviewed and which historical production runs were not rerun.

The calculator tests compare independent numerical representations where available: Bessel ratios against normalized Haar quadrature, matrix dimensions against one another as diagnostics, exact rational enclosures against the recorded coefficients, and boundary or invalid inputs against the declared domains. These tests deliberately include wrong models. For example, an approximation that drops variance can pass a selected equation and still fail the next moment identity. A retained initial column equal to zero can coexist with nonzero later leakage. A ground-energy rounding error can pass short-time tests and fail at sufficiently long times.

### 13.2 Arithmetic, analytic scope, and sampling

Exact rational arithmetic certifies the arithmetic statement it evaluates. It does not make an unevaluated transcendental quantity exact; such a quantity needs a proved enclosure. The companion distinguishes positive Taylor partial sums used as rigorous exponential bounds from high-precision floating diagnostics. Jacobi truncation comparisons are labelled diagnostics unless a separate omitted-tail and interval argument is invoked. The HTML calculators use ordinary double precision and display rounded values; they are explanatory interfaces to the formulas, not interval-proof engines.

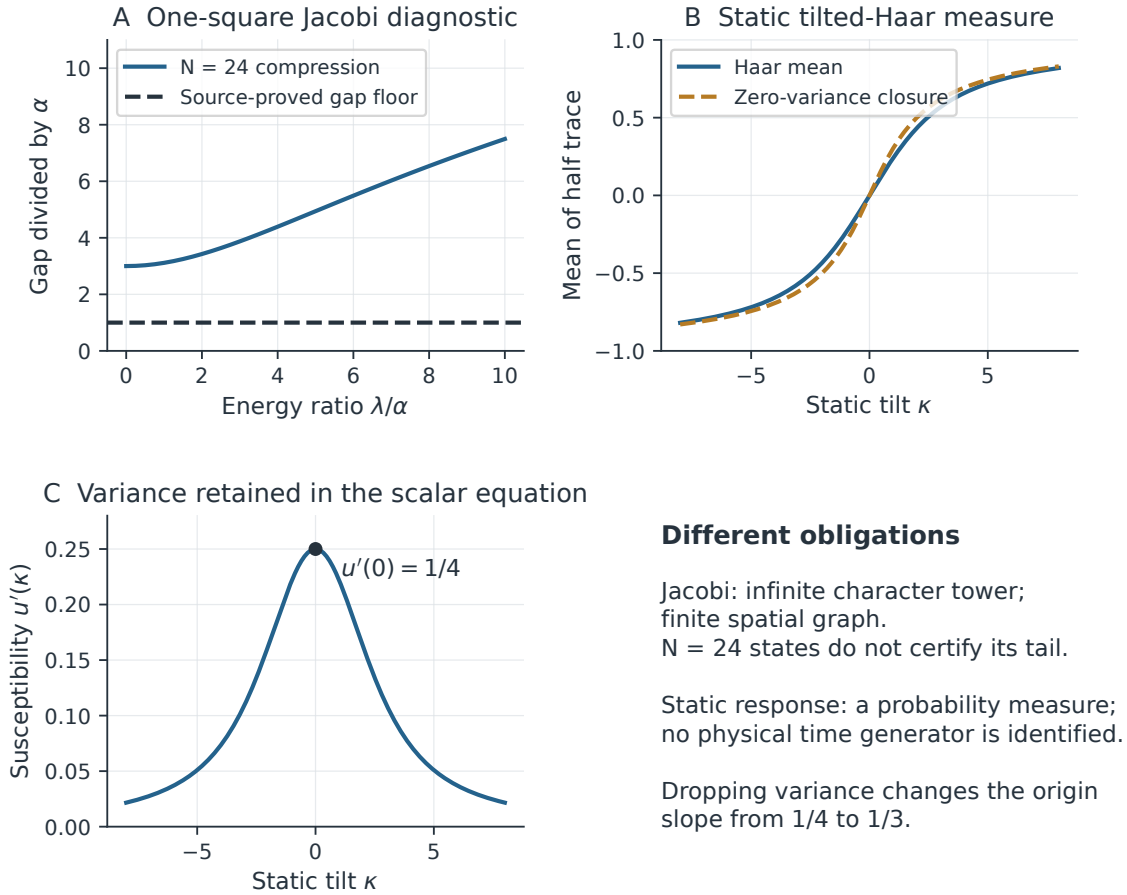
A plot samples a formula. Its visual smoothness supplies no all-time, all-coupling, or infinite-volume proof. Uniform bounds in the paper come from the stated analytic arguments: complete interval coverage with a continuity estimate, a monotone early/late envelope, a convergent majorant, or a complete spectral/channel identity. Each plot states its parameter region and whether it shows an exact function, a sufficient upper budget, or numerical diagnostic data.

### 13.3 Known limits of this audit

Neither this manuscript audit nor a finite set of stress tests can establish that the work has no limitations or that every possible error has been excluded. The complete historical numerical production archive was not regenerated for this paper. The source checkpoint, original reports, fresh calculation scripts, source hashes, and per-contribution dispositions make that boundary inspectable. No unrestricted continuum conclusion is admitted by inference from the calculations. No experimental application, literature priority, or external peer-review status is claimed.

Historical Round26 roadmap; subsequent integrated chapters record its execution.

## Distinct SU(2) reductions: spectrum and static response

**Different obligations**

Jacobi: infinite character tower;  
finite spatial graph.  
 $N = 24$  states do not certify its tail.

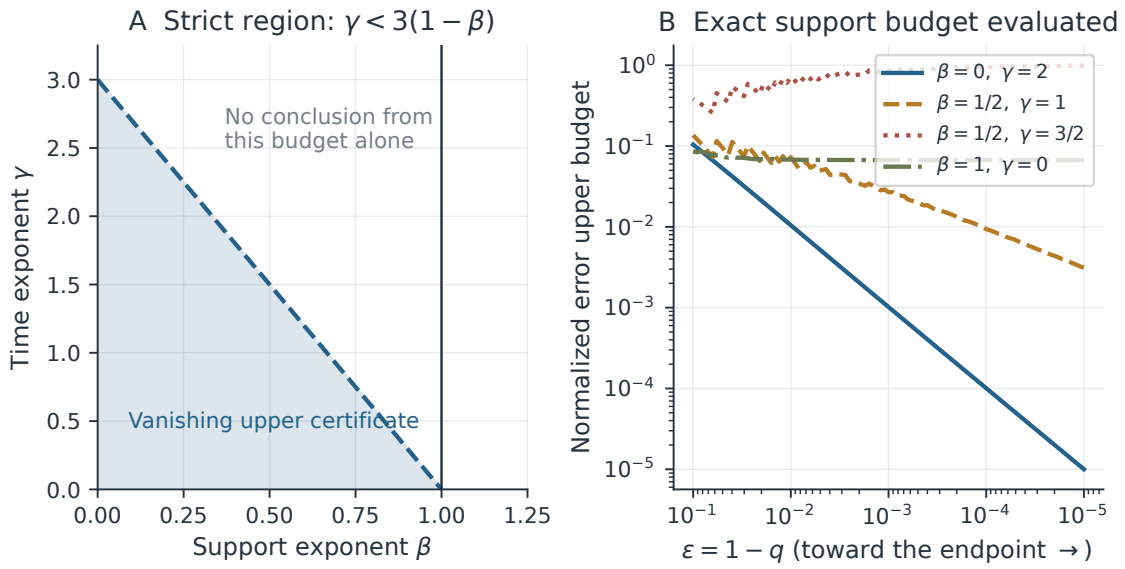
Static response: a probability measure;  
no physical time generator is identified.

Dropping variance changes the origin  
slope from  $1/4$  to  $1/3$ .

Gap floor: inherited exact Round10 certificate on  $0 \leq \lambda/\alpha \leq 10$ . Static curves: exact Bessel formula evaluated numerically. Numerical agreement and truncation refinement are diagnostics; neither proves a continuum theorem.

Figure 2: Two distinct benchmarks. The one-square Jacobi eigenvalue curves are finite-matrix numerical diagnostics; the recorded continuous full-space certificate has its own tail and interval proof. The tilted-Haar mean and variance are static quantities, not a physical trajectory. Plot data and parameters are supplied with the calculators.

### Complete support controls the available time window

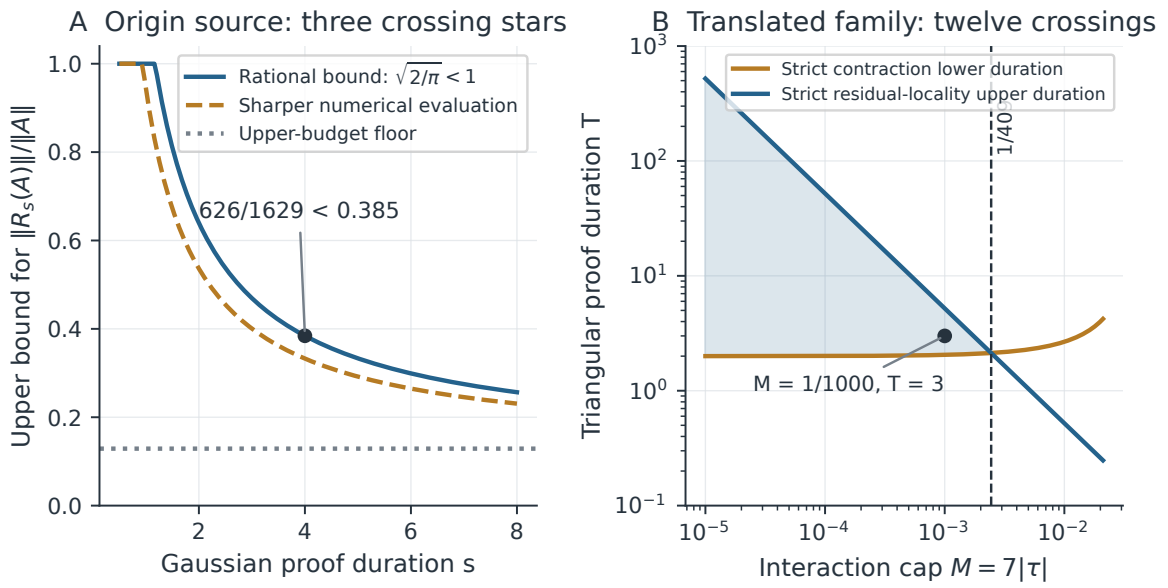


Canonical summable family,  $\eta = 1/2$ ,  $C = \ell = 1$ ;  $L = \max(1, \text{floor}(\varepsilon^{-\beta}))$ .

Boundary curves are excluded from the vanishing certificate. A positive or divergent upper bound is not a lower bound on actual correlation error. The  $q = 1$  Hamiltonian is not inserted into the summable family.

Figure 3: Sufficient support/time windows and evaluated upper budgets. The strict exponent region follows from the stated complete-support asymptotics. Values outside it indicate failure of this sufficient certificate, not proof that the actual correlation error diverges.

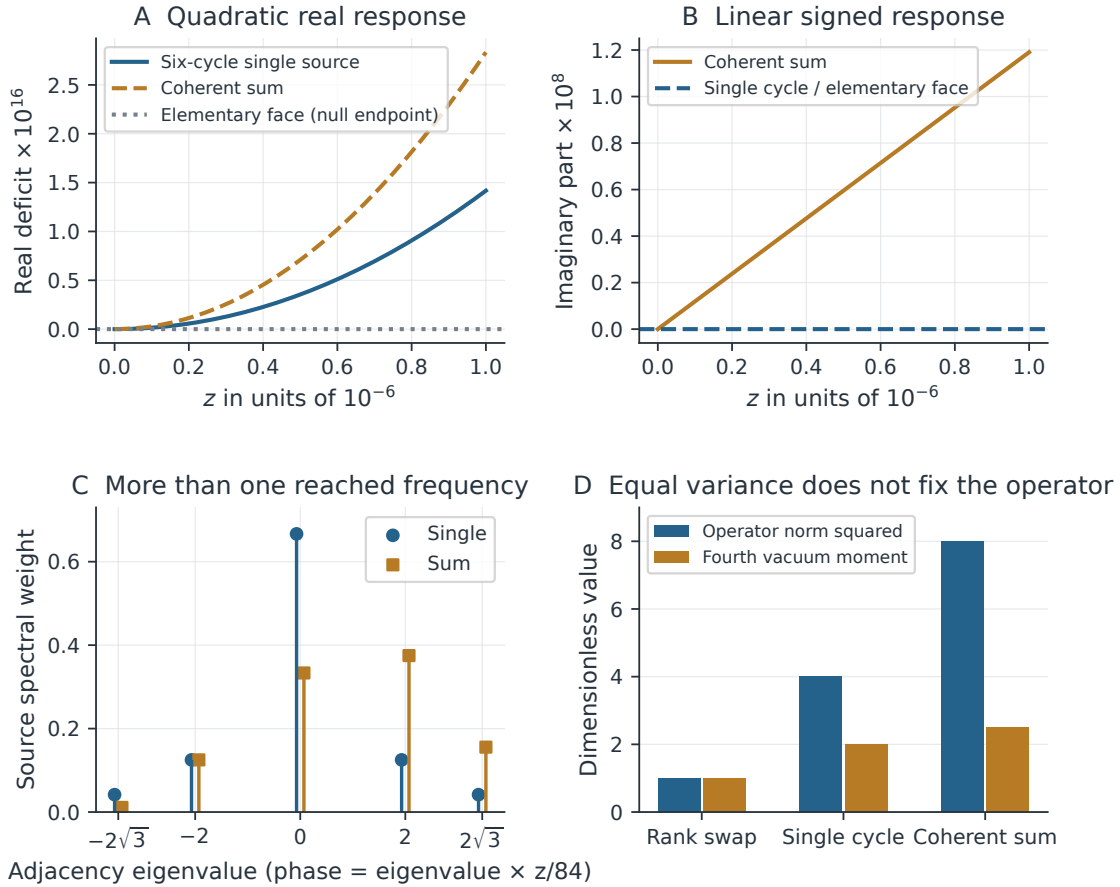
## Filtered correction: contraction and locality are separate



Panel A: AB2 original cap  $M = 35/1664$ . Panel B: AE2 requires  $2/(1-25M) < T < 1/(192M)$ . Its joint open interval exists exactly when  $409M < 1$ . The zero-frequency residual is one for both filters. These are sufficient upper certificates; neither a budget floor nor a failed locality estimate proves actual divergence.

Figure 4: Full-source Gaussian contraction and compact-filter certificate compatibility. Gaussian duration is a proof regulator. The compact construction has a smaller admitted coupling range and counts all translated crossings. The original-cap incompatibility concerns these sufficient bounds, not every possible filter or a lower bound on an actual error.

## Wilson endpoints: one clock, distinguishable probes



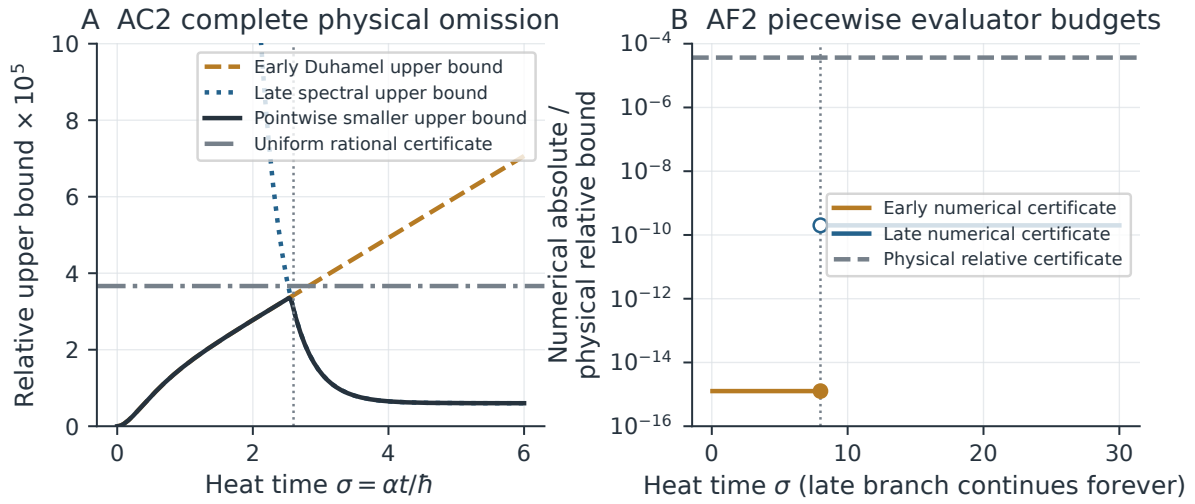
Actual multiplier endpoints,  $0 \leq z \leq 10^{-6}$ . The elementary face is a null probe at this clock.

The coherent sum shares the rank-swap endpoint scalar, but its norm squared is 8 and fourth vacuum moment is  $5/2$ .

Curves evaluate exact limiting formulas. They do not include finite- $q$  remainders, physical clock calibration or measured data.

Figure 5: Canonical endpoint functions for the specified actual Wilson observables, their reached spectral weights, and distinctions between operator norms and moments. The elementary probe supplies a null slow-response control. These are model readouts; finite- $q$  deviations and physical resource costs require their separate bounds.

## Heat approximation: physical omission and numerical error



$\lambda \leq 0.01$  for AC2;  $\lambda = 0.01$  for AF2. Inputs: normalized original P21 vectors within 0.01 of the vacuum.  
 Physical bound = 457097/12472250000; denominator  $\geq 7127/7200$ . The 293-dimensional space is an enrichment.  
 This companion recomputes scalar certificates. It does not execute the original 293-coordinate heat solver.

Figure 6: The physical AC2 early/late envelope and the separately charged AF2 numerical bound. The numerical branch changes at  $\sigma = 8$  to prevent secular center error. The denominator applies to the original normalized  $P_{21}$  preparation ball; no relative claim for arbitrary ground-null inputs is shown. The late branch extends to all later times even though the horizontal display ends.

## 14 Next planned goals and remaining proof obligations

Round26 completes AB1/AB2, AC1/AC2, AD1/AD2, AE1/AE2, and AF1/AF2. The first three goal pairs were inherited from the earlier roadmap. AE and AF were selected only after those six reviews. This paper does not execute further physics loops. The next three goals below were planned from the final evidence. Their second loops must be chosen after their first-loop reviews; two later goals remain unselected.

1. **AG: the first actual nonlinear homogeneous update.** Start within AE2's restricted coupling region. Derive the generated source, scalar term, and diagonal update from one actual transformation. Bound every translated crossing and the relevant unbounded domains in the original rooted support-cardinality norm. The decisive test is whether per-source improvement also improves the collective weighted budget. A failure would select a revised construction, not an assertion of all-stage induction.
2. **AH: larger physical graphs with complete channel control.** Freeze the next graph and build its actual Gauss-invariant retained basis. Establish all Gram, coupling, and omitted-channel identities before comparing accuracy. Compute the graph dependence of residual, ground-projection, and true-output denominator constants. A favorable graph-size trend remains weaker than a volume-uniform theorem.
3. **AI: observable and physical-scale identification.** Freeze actual Wilson observables, independent parameters, and physical observation times. Determine the parameter combinations identified by their complete readouts, using null probes and indistinguishable parameter families as controls. Add a genuinely discriminating observable or a state/domain/generator map with a bounded defect. A fitted clock that forces agreement is not an admissible substitute.

Beyond these near-term goals, a continuum implication requires an explicitly matched scale trajectory, control uniform in graph volume and cutoff, a nontrivial limiting theory, the required reconstruction/positivity properties, and a positive gap in physical units. These are separate obligations, not a consequence of the finite-graph error threshold. The present results provide controlled subproblems and counterexamples that organize this work; they do not settle it.

## Part I

# Round27: readout identification and nonlinear correction

## 15 Inherited question and research design

The unchanged Draft01 compendium records the surviving history through Round26. It distinguishes a finite Gauss-invariant  $SU(2)$  graph, a homogeneous local-stability model and a canonical summable full-link family. Its 293-coordinate heat calculation, complete-support source estimates and Wilson endpoint formulas have different hypotheses. A shared parameter name does not identify their physical clocks, generators, states or domains. Round27 therefore begins with an information question: which independent parameters do the available readouts actually identify?

The panel comprised contemporary model-agent research lenses informed by selected Newton, Tesla, Jung/Pauli, Penrose and Feynman sources. They neither impersonate those people nor claim their endorsement. The advisor chose a contract; separate forward and reverse researchers derived and implemented it without reading the other's current solution before freezing; a separate skeptic tested scope and counterexamples. The mathematical work used the paired directions where useful. The geometry of a double spiral was an organizing analogy, not a proof premise.

The first selection, AI1, analyzes exact readout fibers. Its review motivates AI2's new physical probes. Only after AI2's review is the third contract selected, addressing the homogeneous nonlinear-update obstruction. This sequencing matters: the later question must respond to an observed limitation rather than retroactively redefining the first target. Literature searches, replays, repairs and skill tests do not count as additional physics loops.

**Fixed units.** Throughout the canonical identification studies,  $\alpha > 0$  is an energy,  $\hbar > 0$  is a fixed action reference, and  $E_* > 0$  is a separately fixed physical energy scale. The dimensionless parameters satisfy  $0 < \eta, q < 1$ . Neither  $q$  nor  $1 - q$  is asserted to be a continuum lattice spacing. The homogeneous study has its own declared dimensionless coupling and support weights. A support weight is proof bookkeeping, not a new force or physical constant.

## 16 What the sources contribute

The five inventories contain 55 entries at 54 distinct URLs, consulted on 22 September 2026. They retain exact URLs, reading depth and explicit nonclaims. This is a selected survey, not exhaustive access to the works of five researchers or a complete state-of-the-art or priority search. An abstract is not a full-paper audit; a catalogue is not a read manuscript. The mathematical premises come from the frozen repository contracts and inherited reports, not from occult texts, public anecdotes or institutional sponsorship.

Newton's speculative subtle-spirit passage acknowledges inadequate experiments for a demonstrated law. His copied Hermetic text is not a modern scale-matching theorem. Tesla's patent specifies a powered mechanism; its grant establishes neither operational performance nor quantum-field implications. Vivekananda's religious interpretation records an encounter and an anticipated demonstration, not a completed Tesla derivation. The custody rumor review separates a Reddit allegation, the FBI's institutional account and a catalogue whose large scan was inaccessible. These distinctions prevent an attractive story from silently becoming a physical assumption.

Jung's selected symbolic correspondences motivate declaring observation and selection rules. The

| Lens       | Selected material and documented reading boundary                                                                                                                                                  | Question transferred to this work                                                                                |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Newton     | <i>De motu, Opticks</i> Query 31, General Scholium; selected alchemical operations and residue notes; Emerald Tablet transcription and theological imagery dictionary.                             | Does inverse reconstruction identify a unique cause? Are all residuals and support costs retained?               |
| Tesla      | Powered transmission patent; one Colorado Springs tuning entry; Vivekananda letter; public custody rumor traced to an official account and archive catalogue. The large archive scan was not read. | What are the input, load, actual observable, clock and independent calibration?                                  |
| Jung/Pauli | Publisher-hosted alchemy and synchronicity openings; correspondence context, with an inaccessible scan disclosed; two purposively sampled discussions; institutional remote-viewing history.       | Which observation rule distinguishes a pattern from an indistinguishable alternative?                            |
| Penrose    | Spin networks, first two pages of handwritten direction notes, twistor abstract; separately scoped gravitational-reduction proposal.                                                               | Do candidate states have a nondegenerate Gram matrix, and does a proposed map apply to this dimension and model? |
| Feynman    | Methodological writing; path-integral and simulation metadata/abstracts; qualitative $2 + 1$ dimensional Yang–Mills abstract.                                                                      | What measured quantity and countermodel discriminate a claim? What does the simulation actually construct?       |

Table 2: Historical sources generate operational questions. They add no Hamiltonian term or mass-gap premise. Full reading limits and links are in the expert inventories.

reviewed government history establishes that remote-viewing investigations occurred, not a mechanism or a connection to gauge dynamics. Public discussions record competing interpretations; no representative sampling, verified anomalous experience or empirical corpus experiment is claimed here. Penrose’s warning that locally admissible spin-network labels can still yield a zero-norm network reinforces an existing Gram-matrix gate. His gravitational-reduction proposal changes the physical model if imported into pure Yang–Mills. Feynman’s qualitative  $2 + 1$  dimensional paper cannot close a  $3 + 1$  dimensional quantum construction.

## 16.1 Current technical comparisons and their boundaries

The survey follows targeted primary sources through 2026. Newton’s lens inspects local plaquette-generated truncations and selected area-law theorem statements; their group, action and coupling hypotheses must be checked before transfer. Tesla’s scale-setting comparison distinguishes gradient-flow convention, finite-spacing artifacts, topology and volume. Its numerical calibration cannot supply a value of this workbench’s  $\alpha$  without a state/observable/generator map. The hardware comparison specifies measured observables and digitization errors on a two-plaquette system; government funding confirms a program, not correctness of a theorem.

The Penrose review inspects the current stochastic three-dimensional Yang–Mills–Higgs result, whose introduction retains construction and invariant-measure limitations. Its dimension, dynamics and target differ from a four-dimensional pure quantum theory. The Feynman review compares statistically bounded time evolution, Hamiltonian step scaling in  $2 + 1$  dimensional  $U(1)$  and finite-group digitization. These are useful ways to separate approximation, physical scale and continuum errors. They do not establish this workbench’s missing limits. Selected quasi-locality literature similarly motivates the distinction between strong and norm topology, while leaving our declared support-cardinality estimate to be proved.

The survey also preserves skeptical objections to current preprints. One displayed inverse map is invariant under an isotropic shift, so an unrestricted scalar cannot be recovered from it. A gauge-cooling introduction's claim about neighboring constraints is not adopted: distinct full-link vertex transformations use commuting left and right actions on a shared link. These are narrow source-screening observations, not complete refutations or literature-wide novelty findings. No recent claimed mass-gap solution was imported as an accepted premise.

## 17 HNM AI1: exact information in the declared readouts\*\*

### 17.1 Endpoint coordinates and inherited functions

Put  $\epsilon = 1 - q$ ,  $\beta = \alpha/\hbar$  and

$$\rho = \eta\epsilon^3, \quad \lambda = \frac{\beta\rho}{84}, \quad T_q(z) = \frac{z}{\beta\eta\epsilon^3}, \quad \theta = \frac{z}{84} = \lambda T_q(z). \quad (\text{HNM-141})$$

$\beta$  and  $\lambda$  have units of inverse time. The inherited endpoint theorem admits  $0 < z \leq 10^{-6}$ , with  $\alpha, \eta$  fixed and  $q \uparrow 1$ . Physical observation time then diverges. The elementary plaquette, a prescribed six-cycle  $X$ , and a coherent six-cycle pair  $S$  have respective endpoint functions

$$\begin{aligned} f_0(\theta) &= 1, \\ f_X(\theta) &= \frac{2}{3} + \frac{1}{4} \cos(2\theta) + \frac{1}{12} \cos(2\sqrt{3}\theta), \\ f_S(\theta) &= \frac{1}{3} + \frac{1}{2} \cos(2\theta) + \frac{1}{6} \cos(2\sqrt{3}\theta) + i \left[ \frac{1}{4} \sin(2\theta) + \frac{1}{4\sqrt{3}} \sin(2\sqrt{3}\theta) \right]. \end{aligned} \quad (\text{HNM-142})$$

The actual assertion is a demodulated, growing-time limit  $e^{iE_j T_q/\hbar} C_{j,q}(T_q) \rightarrow f_j(z/84)$ , with  $E_0 = 3\alpha$  and  $E_X = E_S = 9\alpha/2$ . It is not a derivative theorem at fixed physical time. Analytic extension of the finite functions in (HNM-142) does not enlarge the physical theorem's interval.

At fixed scaled coordinate  $z$ , the entire limiting readout is independent of  $\alpha, \eta, q$ . It identifies none of them. Recorded physical timestamps supply the composite rate through the protocol; that information is additional to the endpoint ordinates. At fixed physical  $t$ ,  $q \uparrow 1$  instead sends  $z = \beta\eta(1-q)^3 t$  to zero. Confusing these two limits would manufacture an inverse that the data do not provide.

### 17.2 Exact fibers and conditional identification

Define the physical-time *surrogate*, explicitly distinct from an exact finite- $q$  correlation,

$$D_j(t; \alpha, \eta, q) = f_j(\lambda t), \quad 0 < 84\lambda t \leq 10^{-6}. \quad (\text{HNM-143})$$

Substitution proves that all readouts coincide whenever

$$\alpha\eta(1-q)^3 = \text{constant}.^{**} \quad (\text{HNM-144})$$

Conversely, positivity of  $\lambda$  and strict increase of the coherent imaginary part identify this one composite rate. Indeed

$$\frac{d}{d\theta} \text{Im } f_S(\theta) = \frac{1}{2} \cos(2\theta) + \frac{1}{2} \cos(2\sqrt{3}\theta) \geq 1 - 4\theta^2 > 0 \quad (\text{HNM-145})$$

on the admitted interval. The inequality uses  $\cos x \geq 1 - x^2/2$ . The logarithmic sensitivity row is  $(1, 1, 3)$  in coordinates  $(\log \alpha, \log \eta, \log \epsilon)$ ; its rank is one. More samples of these same complete curves do not remove the two-dimensional nullspace.

For example, in fixed units the exact triples  $(\alpha/E_*, \eta, q) = (1, 1/4, 1/2)$ ,  $(2, 1/8, 1/2)$  and  $(1, 2/27, 1/4)$  all have  $\alpha\eta(1-q)^3/E_* = 1/32$ . Compensation also persists arbitrarily near  $q = 1$ . These rational countermodels disprove parameter uniqueness for (HNM-143), not equality of their actual finite- $q$  Hamiltonians.

Retain a calibrated carrier:

$$U_0(t) = e^{-3i\beta t}, \quad U_X(t) = e^{-(9/2)i\beta t} f_X(\lambda t), \quad U_S(t) = e^{-(9/2)i\beta t} f_S(\lambda t). \quad (\text{HNM-146})$$

On an open physical-time interval, complete complex surrogate curves identify  $\beta$  and  $\lambda$ . For instance  $U'_0(0) = -3i\beta$  and  $U'_S(0) = i(\lambda - 9\beta/2)$  as analytic identities of these surrogates. Their sensitivity matrix has rows  $(1, 0, 0)$  and  $(1, 1, 3)$ , rank two, leaving the  $\eta$ - $q$  fiber. Known  $q$  then gives  $\eta = 84\lambda/[\beta(1-q)^3]$ . This conclusion requires calibrated time, action and phase; it is not a measurement already implemented by the endpoint limit.

Sampling also matters. On a uniform grid  $t = n\Delta$ , the simultaneous change

$$\alpha' = \alpha + \frac{4\pi\hbar m}{3\Delta}, \quad \eta' = \eta\alpha/\alpha', \quad q' = q \quad (\text{HNM-147})$$

preserves the slow rate and gives integer- $2\pi$  shifts to both carriers, when the transformed parameters remain admissible. A bounded prior frequency range or nonaliased design is needed. Intensity-only data lose the elementary phase entirely. The elementary demodulated probe remains constant despite nonzero variance;  $X$ , however, is informative through curvature despite its zero linear coefficient.

### 17.3 Why the next loop must use actual finite-profile disks

The actual retained finite- $q$  matrix contains different face-flip weights  $q^4$  and  $q^5$ . With

$$p(q) = 2 + 5q + 5q^2 + 6q^3 + 3q^4, \quad c_S(q) = \frac{q^4(1+q)^2(1+q^2)}{32p(q)}, \quad (\text{HNM-148})$$

the coherent retained center has linear imaginary coefficient  $c_S(q)$  in  $z$ , approaching  $1/84$  but not constant in  $q$ . Thus the exact surrogate fibers need not preserve even the finite-profile center. Distinct centers do not by themselves separate full-system predictions: one must include their complete errors. Nor may a value-only bound be differentiated;  $e_N(t) = \varepsilon \sin Nt$  has uniformly small values but derivative  $\varepsilon N$  at zero. AI1 therefore closes a scoped identification question and opens a physical discrimination question.

## 18 HNM AI2: separating two actual finite-profile predictions\*\*

### 18.1 New observables require a new complete support

Use the free unit cube with origin  $(3, 1, 0)$  and the prescribed physical six-cycle  $X$ . Let  $f_4$  and  $f_5$  be the opposite  $yz$  faces with anchor sums four and five, and let  $Y_r = X \triangle f_r$  be the corresponding simple six-cycles. Define actual character multiplication operators, extended by identity,

$$B_r = (\chi_X + \chi_{Y_r})/\sqrt{2}, \quad \psi_r = B_r \Omega, \quad r = 4, 5. \quad (\text{HNM-149})$$

Each operator is bounded, real, self-adjoint and gauge-invariant. Conditional Haar integration along the common four-link path and the two independent two-link paths gives mean zero, variance one and fourth moment  $5/2$ . The exact multiplication norm is  $J = 2\sqrt{2}$ , not one: its triangle upper bound is approached on positive-measure neighborhoods of identity holonomies.

A normalized source vector therefore does not authorize replacing its multiplier by a norm-one rank operator.

Each source uses eight free links, but their union has ten. The complete-factor collar must start from that union. For depths  $k = 0, \dots, 5$ , independently reconstructed retained-face counts are

$$N_k = (2, 25, 143, 359, 695, 1171). \quad (\text{HNM-150})$$

The last collar contains 993 reference factors and 1857 complete links. Copying the earlier eight-link seed would omit part of  $B_5$ . All six cube faces occur by depth one. The inherited AA2 averaged-orbit and exterior-exclusion theorem applies to the entire sixteen-state component containing both sources; it is not an assertion that this component is the entire regional energy shell. The check reconstructs all twenty exterior touching faces and retains their loading and nonzero frequency conditions.

## 18.2 From connected words to the full scalar disk

Each complete reference factor contains at most ten links, each link meets at most four faces, and an interaction adds at most three factors. Counting every ordered connected word, repetition, factor-two commutator and time simplex yields  $(10/3)_n/n!$ . For  $x = 15z < 1$ , a conservative bound on the *whole* omitted series is

$$D_k^{(10)}(x) = \frac{(10/3)_{k+1}}{(k+1)!} \frac{x^{k+1}}{(1-x)^{k+5}}. \quad (\text{HNM-151})$$

Taylor's positive integral remainder has exponent  $k + 13/3$ ; rounding it upward to  $k + 5$  increases the bound. This accounts for paths leaving the collar and returning. It is neither a finite-spin cutoff nor a global propagator-norm statement.

The retained symmetric  $16 \times 16$  matrix  $Q(q)$  has physical face-flip entries  $q^4$  or  $q^5$  and row norm at most six. Set

$$s_q = \frac{3z(1+q)^2(1+q^2)}{p(q)}, \quad v = s_q/96, \quad h_r = \langle \psi_r, e^{ivQ(q)} \psi_r \rangle. \quad (\text{HNM-152})$$

Its exact first moments are  $\langle Q \rangle_{\psi_4} = q^4$  and  $\langle Q \rangle_{\psi_5} = q^5$ ; at  $q = 1$  both moment lists through degree six equal  $(1, 1, 4, 8, 32, 80, 320)$ . The computations evaluate rational moments through degree eight. The exponential center error is separately bounded by  $(6v)^9/9!$ .

The inherited canonical magnetic-profile sum is  $b(q) = p(q)/[24(1-q)^3(1+q)^2(1+q^2)]$ . Write  $\tau_q = \eta/[8b(q)]$ ,  $M_k = N_k/24$  and  $\hat{d}_q = \tau_q \sqrt{b(q^2)}/96/[(1-\eta)/8]$ . The complete physical-to-center radius is bounded by

$$R_{\text{phys}} = 48\hat{d}_q + 8D_k^{(10)}(15z) + 64\tau_q M_k(1 + 3zM_k), \quad R = R_{\text{phys}} + \frac{(6v)^9}{9!}. \quad (\text{HNM-153})$$

The terms respectively retain the stationary connected-state correction, full spatial tail, both source and vacuum averaging replacements, and arithmetic center error. The constants retain  $J^2 = 8$  and  $1 + J < 4$ . A rational upward square-root enclosure controls  $\hat{d}_q$ . Fractions determine the verdicts; decimal values are displays. Omitting multiplier loading, connected-state correction or the returned spatial paths would invalidate the disk.

## 18.3 A successful pairwise test and an unsuccessful ratio certificate

Freeze  $z = 10^{-6}$  and compare, for each  $u \in \{10^{-12}, 10^{-18}, 10^{-24}\}$ ,

$$H_1 : (q, \eta) = (1 - u, 1/2), \quad H_2 : (q, \eta) = (1 - 2u, 1/16). \quad (\text{HNM-154})$$

Both satisfy  $\eta(1 - q)^3 = u^3/2$ . With common independently fixed  $\alpha$ , they therefore have the same carrier and the same physical time

$$T = 2 \cdot 10^{-6} u^{-3} \hbar / \alpha. \quad (\text{HNM-155})$$

For each  $k \in \{3, 4, 5\}$  and probe  $r$ , compare rational imaginary centers. A positive margin

$$\mathfrak{m}_r = \left| \text{Im } h_{r,1}^{(8)} - \text{Im } h_{r,2}^{(8)} \right| - R_1 - R_2 > 0 \quad (\text{HNM-156})$$

proves disjoint complex disks and hence distinct actual scalar predictions. Overlap only means this sufficient test failed. It does not establish equality of the two exact models.

Only the cell  $u = 10^{-18}$ ,  $k = 5$  gives a positive margin for either probe. The conservative margins are about  $3.21452 \times 10^{-26}$  for  $B_4$  and  $4.40500 \times 10^{-26}$  for  $B_5$ . Equal additional absolute scalar-error radii must each be strictly below half the relevant margin. Its physical time is  $2 \times 10^{48} \hbar / \alpha$ . This is a conditional mathematical distinction between two named hypotheses, not feasible laboratory calibration or an inverse valid on a continuous parameter domain. Timing uncertainty cannot silently be assigned the same scalar-error allowance without a separate sensitivity bound. The retained coefficient ratio is exactly  $q^5/q^4 = q$ . For actual finite-time values  $y_r$ , however, spectral calculus first gives

$$|\text{Im } h_r - vq^r| \leq 36v^3, \quad |y_r - vq^r| \leq E_r := R_r + 36v^3. \quad (\text{HNM-157})$$

If  $vq^4 - E_4 > 0$ , interval division yields

$$\frac{vq^5 - E_5}{vq^4 + E_4} \leq \frac{y_5}{y_4} \leq \frac{vq^5 + E_5}{vq^4 - E_4}. \quad (\text{HNM-158})$$

If the denominator enclosure contains zero, no ratio inference is made. These simple ratio intervals overlap in all nine cells, including the successful scalar-discrimination cell. Their cubic sine-remainder floor is too conservative for that inverse. The result therefore resolves a pairwise scalar question while preserving the failed ratio objective. No derivative of the full-system correlation is inferred from a value-only disk.

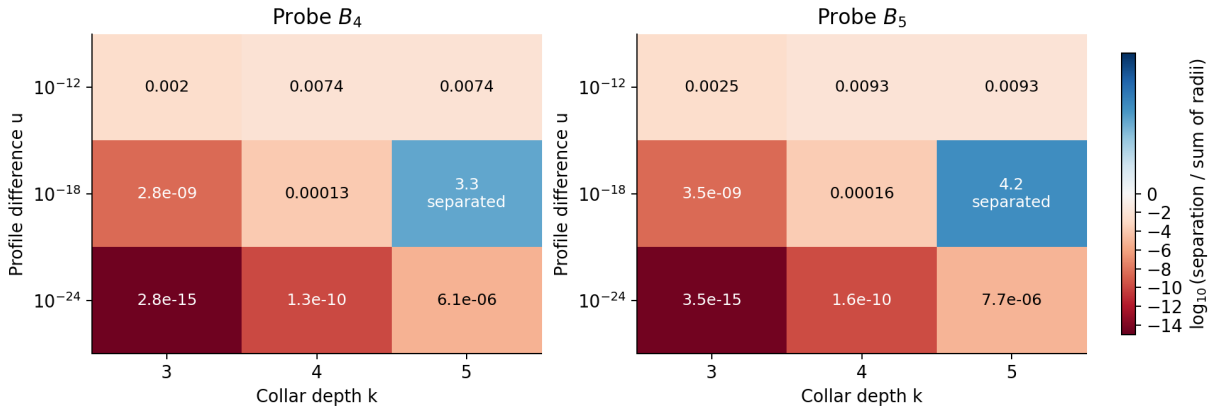


Figure 7: AI2: separation of imaginary centers divided by the sum of complete radii, computed from exact output fractions. Values above one certify disjoint disks. The only successful grid cell is  $u = 10^{-18}$ ,  $k = 5$  for both probes. Logarithmic color prevents the large failed range from hiding the threshold; the figure is a display of exact decisions, not a numerical proof.

## 19 HNM AG1: one selected-source homogeneous correction\*\*

### 19.1 The actual model and the term still outside this result

After AI2's scoped admission, the advisor selected the homogeneous-update bottleneck. Work in the fixed energy unit  $\delta = \alpha/8$  with the homogeneous reference  $G = H_0 + \sum_b D_b$  and the actual selected S1 cubic-source family  $A = \sum_b A_b$ . Each  $A_b$  is an identity-extended local mixing term on a complete four-factor star. It is a specified part of a repeated-anchor cubic contribution, not the entire cubic coefficient or transformed remainder. With  $M = 7|\tau|$ , the available source norm satisfies

$$r_* := \sup_b \|A_b\| \leq r_*^{\text{bd}} = \frac{4}{3} \left( \frac{7}{12} \right)^{3/2} |\tau|^3. \quad (\text{HNM-159})$$

Specifically, the inherited source is  $A_b = |w_b\rangle\langle\Omega_b| + \text{adjoint}$ , where

$$v_b = \phi_b \Omega_b, \quad u_b = (H_{0,Z_b}|_{Q_b})^{-1} v_b, \quad w_b = -\langle u_b, v_b \rangle u_b - \frac{\|u_b\|^2}{3} v_b. \quad (\text{HNM-160})$$

The cubic expression in (HNM-159) is an upper bound, not an evaluated equality. The frozen main interval is  $0 \leq M \leq 10^{-3}$ , with a predeclared refinement on  $0 \leq M \leq 10^{-4}$ . The auxiliary support-cardinality weight is  $\rho = 9/4$ , while the required output uses the original weight two. These are proof parameters within the same homogeneous family; the filter time  $T = 3$  is not the canonical readout clock.

Translation of identical complete blocks gives the same actual norm  $r = \|A_b\| > 0$  for every retained star when  $\tau \neq 0$ . Deleting boundary stars does not modify a retained source. If  $m_\Lambda$  is the maximum number of retained source stars containing a root, then  $1 \leq m_\Lambda \leq 4$  for a nonempty family and its declared indexed norm is exactly

$$A_a := \|\{A_b\}\|_a = m_\Lambda a^4 r, \quad \frac{A_\rho}{A_2} = \left( \frac{9}{8} \right)^4. \quad (\text{HNM-161})$$

This is a genuine input denominator even though  $r$  is not numerically evaluated. Substituting its upper bound into a denominator would be invalid. Empty families and  $\tau = 0$  have zero update and are treated without division. Physical energies multiply these dimensionless bounds by  $\delta$ , with the original positive spacing and physical reference scales fixed.

If the full previously transformed Hamiltonian contains other terms  $E$ , the correct target remains  $G + A + E$ . Controlling  $G + A$  does not control  $E$ . The exact equation below retains its transport and states the needed conditional norm premise. In particular AG1 is not a completed correction of every term in the original homogeneous Yang–Mills Hamiltonian.

### 19.2 Domain-safe conjugation and the full nonlinear remainder

For the triangular kernel  $p_T(t) = (1 - |t|/T)_+/T$  and its odd integrated primitive  $h_T(t) = \text{sgn}(t)(1 - |t|/T)^2/2$  on  $0 < |t| < T$ , let

$$\mathcal{R}_T(A_b) = \int p_T(t) e^{itG} A_b e^{-itG} dt, \quad \mathcal{J}_T(A_b) = -i \int h_T(t) e^{itG} A_b e^{-itG} dt. \quad (\text{HNM-162})$$

Their basic kernel identities are  $h'_T = \delta_0 - p_T$ ,  $\int p_T = 1$  and  $\int |h_T| = T/3$ . Weak integration by parts gives a bounded commutator  $[\mathcal{J}_T(A_b), G] = -A_b + \mathcal{R}_T(A_b)$ . The inherited adjoint-domain argument then gives graph-domain preservation. In a finite containing cuboid the sum  $S = \sum_b \mathcal{J}_T(A_b)$  is bounded and skew-adjoint, preserves  $D(G)$ , and obeys  $[S, G] = -A + R$  with  $R = \sum_b \mathcal{R}_T(A_b)$ .

On that domain the exact update is

$$e^S(G + A)e^{-S} = G + R + N,$$

$$N = \sum_{n \geq 1} \frac{\text{ad}_S^n(nA + R)}{(n+1)!} = \int_0^1 e^{s \text{ad}_S} (s[S, A] + (1-s)[S, R]) ds. \quad (\text{HNM-163})$$

This identity expands the first commutator of  $G$  only after replacing it by the bounded  $-A + R$ ; all subsequent terms involve bounded interactions. To derive the coefficient, combine  $\text{ad}_S^n A/n!$  from the bounded  $A$  conjugation with  $\text{ad}_S^n(-A + R)/(n+1)!$  from  $G$ . The result is  $\text{ad}_S^n(nA + R)/(n+1)!$ . Thus the leading nonlinear term is  $[S, A + R]/2$ . A  $G$ -only BCH formula would omit a required contribution. The resonant filter residual is retained: the Fourier multiplier  $\text{sinc}^2(\omega T/2)$  equals one at zero frequency.

### 19.3 Why the full commutator series needs weight loss

For an indexed interaction with complete connected supports, define

$$\|\Phi\|_a = \sup_x \sum_{X \ni x} a^{|X|} \|\Phi_X\|, \quad a \geq 2. \quad (\text{HNM-164})$$

Repeated terms with the same support retain their indices unless explicitly grouped. The forward complete rooted count starts at  $N_0(a) = A_a$ . Adding one interaction to a depth- $n$  connected word costs  $2Ma^3$ ; the two root locations count stars meeting the existing union and incoming stars containing the root. This gives

$$N_{n+1}(a) \leq 8Ma^3(8 + 3n)N_n(a), \quad N_n(a) \leq A_a(24Ma^3)^n(8/3)_n. \quad (\text{HNM-165})$$

All ordered repetitions, incoming overlaps and complete supports remain. Finite boundaries only delete terms. The time simplex and exact kernel moments then give, with  $x_a = 72a^3M$  and  $F_a = (1 - x_a)^{-3}$ ,

$$\|R\|_a \leq A_a \sum_{n \geq 0} \frac{2(8/3)_n x_a^n}{(n+2)!} \leq A_a F_a,$$

$$\|S\|_a \leq A_a \sum_{n \geq 0} \frac{6(8/3)_n x_a^n}{(n+3)!} \leq A_a F_a. \quad (\text{HNM-166})$$

The last inequalities bound the entire positive orbital series of exponent  $8/3$  by exponent three, using the unit absolute masses at  $T = 3$ . At the main endpoint  $x_\rho = 6561/8000 < 1$ , so these estimates hold continuously over the whole interval. The twelve interior crossings also preserve the inherited per-source residual bound  $(24M + 2/3)/(1 - M) \leq 2072/2997$ ; that alone does not prove weighted-family contraction.

For  $a > b \geq 2$ , overlap counting in a commutator has two possible root locations. After fixing one support, the overlapping second supports cost its cardinality. Nonempty overlap supplies a factor  $b^{-1} \leq 1/2$ , and  $\sup_{m \geq 1} m(b/a)^m \leq 1/[e \log(a/b)]$  absorbs that cardinality. Therefore

$$\|[\Phi, \Psi]\|_b \leq \frac{2}{e \log(a/b)} \|\Phi\|_a \|\Psi\|_a. \quad (\text{HNM-167})$$

Split the loss  $\log(\rho/2)$  equally over  $n$  commutators. Using  $n! \geq (n/e)^n$  gives

$$\frac{\|\text{ad}_S^n X\|_2}{n!} \leq \vartheta^n \|X\|_\rho, \quad \vartheta = \frac{2\|S\|_\rho}{\log(\rho/2)} \leq 18\|S\|_\rho, \quad (\text{HNM-168})$$

where  $\log(9/8) \geq 1/9$  gives the rational majorant. If a complete translated-source estimate proves a chosen upper bound  $\bar{\vartheta} < 1$ , summing every BCH term yields

$$\|N\|_2 \leq \frac{\bar{\vartheta}}{1 - \bar{\vartheta}} \left( \|A\|_\rho + \frac{1}{2} \|R\|_\rho \right). \quad (\text{HNM-169})$$

The complete rooted estimates, including repeated words, incoming overlaps and twelve possible interior crossings, are needed to justify the finite input norms. A finite-volume operator bound growing with  $\|S_\Lambda\|$  is not a substitute for a volume-independent interaction bound. Nor can the same positive weight margin be spent indefinitely without a new summable-loss induction.

For additional terms with a separately justified premise  $\|E\|_\rho \leq e_*$ ,

$$e^S(G + A + E)e^{-S} = G + R + N + e^S E e^{-S}, \quad \left\| e^S E e^{-S} \right\|_2 \leq \frac{e_*}{1 - \bar{\vartheta}}. \quad (\text{HNM-170})$$

The parameter  $e_*$  is a conditional error inventory, not a fitted physical term. This inequality does not establish that the actual omitted  $E$  satisfies it.

## 19.4 The one-step estimates and the narrower contraction

The forward rational majorant uses  $r \leq M^3/567$  and  $\bar{\vartheta} = 72\rho^4 F_\rho M^3/567$ . At  $M = 10^{-3}$  it gives  $\bar{\vartheta} < 5.5920 \times 10^{-7}$  and

$$\frac{\|N\|_2}{A_2} \leq \nu(M) := \frac{\bar{\vartheta}}{1 - \bar{\vartheta}} \left( \frac{9}{8} \right)^4 \left( 1 + \frac{F_\rho}{2} \right) < 7.7851 \times 10^{-5}. \quad (\text{HNM-171})$$

The underlying fractions, not the displayed rounding, establish the bound. Every positive factor increases with  $M$  before its pole, so endpoint inequalities cover the continuous interval. This controls the whole nonlinear remainder, while the loose main residual majorant does not establish contraction there.

For  $0 < M \leq 10^{-4}$ , split  $R$  into its exact term without a  $D_b$  interaction and the complete positive tail containing at least one  $D_b$ . On the original star the first part is  $|f_3(H_{0,Z_b})w_b\rangle\langle\Omega_b| + \text{adjoint}$ , where  $f_3(E) = \text{sinc}^2(3E/2)$ . The inherited source is orthogonal to the reference vacuum and its free local Hamiltonian has gap at least one. Thus  $|f_3(E)| \leq 4/(9E^2) \leq 4/9$  on that source. The  $n \geq 1$  part of (HNM-166) costs at most  $A_2(F_2 - 1)$ . Consequently

$$\frac{\|R + N\|_2}{A_2} \leq \frac{4}{9} + (1 - 576M)^{-3} - 1 + \nu(M) < 0.639242 < \frac{2}{3}.^{**} \quad (\text{HNM-172})$$

This contracts against the actual chosen input norm (HNM-161), not two upper budgets. The independently derived reverse count uses different positive word majorants and gives the weaker bound 0.923603 on the same interval. Post-exchange independent review verifies the stronger rooted recurrence used here. Unequal conservative constants are not conflicting evaluations of an exact observable. The narrower estimate is a predeclared refinement within AG1, not a fourth loop.

The skeptic separately retained more of the exact filter integration. With  $\mathcal{F}(x) = (4-x)/(1-x)^2$ , the positive  $n \geq 1$  tail has coefficient  $t(M) = \frac{16}{3}[\mathcal{F}(624M) - 4]$  multiplying the actual  $r_*$ . The bound follows from the exact integrated coefficient  $32(4 + 3n)x^n/[(n+1)(n+2)]$  and  $(n+1)(n+2) \geq 6$ . Since the actual input norm is at least  $16r_*$ , combining this tail with the skeptic's full nonlinear coefficient  $n_{\text{sk}}(M)$  gives

$$\frac{\|A'_{\text{new}}\|_2}{\|A\|_2} \leq \frac{4}{9} + \frac{t(M) + n_{\text{sk}}(M)}{16} < 0.605 \quad (0 < M \leq 10^{-4}). \quad (\text{HNM-173})$$

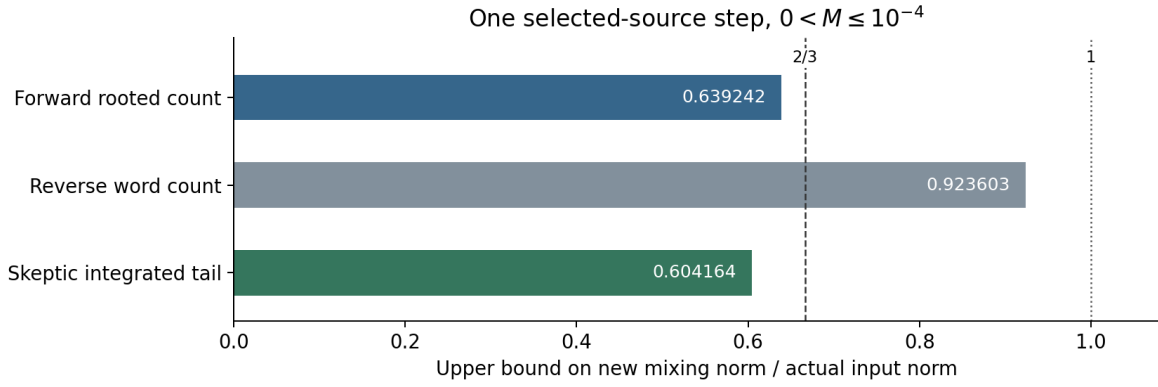


Figure 8: AG1 exact-output-backed sufficient bounds on the new selected mixing family, relative to the actual input norm. The producer bounds also cover  $R + N$  before projection. Different positive-tail estimates produce different constants; none is an exact contraction measurement. The  $2/3$  headline is supported by the reviewed forward and skeptic estimates.

This reviewer refinement, independently frozen and then cross-checked, improves a sufficient bound to about 0.604164. The conservative reviewed headline remains one-step selected-source contraction below  $2/3$ . It is neither an exact dynamical contraction factor nor an extra physics loop.

For each generated self-adjoint indexed term  $K_X$  of  $K = R + N$ , keep its full support and define  $P_X = |\Omega_X\rangle\langle\Omega_X|$ ,  $Q_X = I - P_X$  and

$$\begin{aligned} c_X &= \langle\Omega_X, K_X \Omega_X\rangle, & A'_X &= P_X K_X Q_X + Q_X K_X P_X, \\ D'_X &= Q_X (K_X - c_X I) Q_X, & K_X &= c_X I + A'_X + D'_X. \end{aligned} \quad (\text{HNM-174})$$

The costs are  $|c_X| \leq \|K_X\|$ ,  $\|A'_X\| \leq \|K_X\|$  and  $\|D'_X\| \leq 2\|K_X\|$ , hence at most four times the budget when all three are summed. The scalar keeps its declared support for a density budget and may sum extensively. It is a reference expectation, not an interacting ground energy. “Diagonal” means the reference vacuum/complement split, not all excited spectral blocks. The new mixing family alone inherits (HNM-172); the changed diagonal, scalar and conditional  $E$  cannot be discarded. The consumed weight margin, new supports and changed reference obstruct immediate iteration.

The checkers include both coupling signs, zero/empty cases, boundary multiplicities, all twelve crossings, complete positive tails, omitted-residual and wrong-BCH controls, scalar reconstruction and conditional  $E$  transport. Independent finite rational matrix fixtures test algebraic coefficients; they are diagnostics, not the physical completeness proof. The all-order support and graph-domain arguments carry the result. The full original remainder inventory, all-stage induction, homogeneous numerical spectral gap, physical matching and continuum construction remain open.

## Round27 contribution and reproduction appendix (historical checkpoint)

Historical Round27 checkpoint. References below to planned, unexecuted or next work describe the state at that checkpoint; the integrated Round29 chapter supersedes that plan.

## 20 Contributions and remaining proof obligations

| Contribution             | Earlier understanding → present change                                                                                                             | Possible application                                                                         | Limitation and next missing step                                                                                                                                         |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI1 fibers**             | Nontrivial endpoint formulas → exact classification of the combinations identified by each declared observation convention.                        | Remove redundant parameter fits; choose which calibration must be independent.               | Surrogate identities are not finite- $q$ model equivalences. Prove and use actual finite-profile error disks.                                                            |
| AI1 carrier**            | Demodulated slow signal → rank-two identification of carrier and slow rates with a residual $\eta$ - $q$ fiber and explicit sampling aliases.      | Plan phase-aware, nonaliased observation protocols.                                          | Requires calibrated physical time/action/phase. No implemented clock or derivative transfer theorem.                                                                     |
| AI2 probes and support** | One coherent face type → two actual multiplier sources with weights $q^4, q^5$ , ten-factor common seed and recomputed complete collars.           | Design observables sensitive to profile differences; prevent omitted-channel understatement. | Source-specific canonical family. The component is not the whole regional energy shell.                                                                                  |
| AI2 full-scalar test**   | Retained-center differences → disjoint complete scalar disks for one frozen grid cell. Other cells and all simple ratio tests remain insufficient. | Reject one of two named model hypotheses when the full error allowance is met.               | $T = 2 \times 10^{48} \hbar/\alpha$ and roughly $10^{-26}$ scalar margins; no feasibility claim or continuous inverse.                                                   |
| AG1 one update**         | Individual-source estimates → an entire nonlinear remainder bound and actual selected-source contraction below 2/3 on the narrower interval.       | Identify the terms, input denominator and norm costs an iteration must control.              | Restricted $G + A$ source sector and one spent weight margin. Other original terms require explicit transport; no all-stage or complete-Hamiltonian contraction follows. |

The two-star notation marks workbench specialization, derivation or implementation, not established scientific priority. These results may be useful even when a desired certificate fails: a failed scalar separation or contraction upper bound identifies a limitation of the sufficient estimate. It does not prove equality, divergence or impossibility of the underlying dynamics.

### 20.1 Why a percentage of solving Yang–Mills is not estimable

The three requested investigations have final scoped gates: the research task count is 3/3. That known denominator does not supply a percentage of a mathematical proof. The expert lenses and skeptic agree that no defensible numerical fraction of the Yang–Mills solution is identifiable from pages, loops, equations, network nodes or finite-model successes. The remaining obligations depend on each other and have no justified equal weights or known total amount of work.

The official problem asks for a nontrivial four-dimensional quantum Yang–Mills theory with the required axiomatic strength and a positive mass gap for every compact simple gauge group.

The official status remains unsolved at the recorded source check. The workbench’s outstanding obligations are:

| Obligation                        | Missing implication after Round27                                                                                     |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| Homogeneous control and matching  | Convergent changed-source/changed-diagonal induction and a proved map on states, observables, domains and generators. |
| Uniform limiting estimates        | Bounds surviving growing volume and regulator removal along a physically matched trajectory in the required topology. |
| Nontrivial continuum construction | The requested four-dimensional limiting theory and nontrivial local fields.                                           |
| Axiomatic properties              | Limiting locality, covariance, positivity/reconstruction and the required ultraviolet behavior.                       |
| Positive physical gap             | A positive finite threshold in the actual constructed limiting theory, in fixed physical units.                       |
| Gauge-group coverage              | The required compact simple groups beyond the workbench’s SU(2) emphasis.                                             |

For a dimensionless cutoff operator  $L(a)$  and energy convention  $E(a)$ , the identity  $\Delta_{\text{phys}}(a) = E(a)\Delta_L(a)$  only converts units. A positive finite-cutoff  $\Delta_L(a)$  proves no positive limiting physical gap without a justified scale, limiting construction and uniform estimate. An unknown completion percentage is not a claim of zero research value, nor an estimate of remaining effort.

The panel’s next unexecuted priority is AG2: recover the original transformed Hamiltonian at the source checkpoint, inventory every remaining term in  $E$ , prove its stronger support-norm budget, and test whether its transported mixing cost fits inside the selected-source contraction margin. Only then should a changed-diagonal step be frozen with its own domain and remaining weight hypotheses. A cancellation-aware AI3 finite-value inverse remains an alternative observable branch. Neither is executed in this three-loop continuation.

## 21 Reproduction, review and linked sources

The mathematical reports remain authoritative for their frozen assumptions, complete-support definitions and detailed controls. This manuscript summarizes them without rewriting the accepted gates or Draft01. Every displayed plot is reconstructed from exact JSON fractions; converting to floating point occurs only after the exact decision, for rendering. This is not a new physical simulation.

From the repository root, replay all three producer pairs into a fresh absolute directory outside the repository:

```
python3 -B research/round27/reproduce.py \
  --output /absolute/new/round27-replay
python3 -B research/round27/reproduce.py --optimized \
  --output /absolute/new/round27-optimized
python3 -B papers/round27-addendum/build.py
```

The first two commands verify source/snapshot bindings and byte-identical fresh outputs. Output directories must not already exist. Optimized mode checks that Python optimization does not disable the controls. They do not constitute formal verification of an operator proof or independent empirical observations. The final command builds the plot and the PDF with the installed Python scientific stack, ordinary L<sup>A</sup>T<sub>E</sub>X and Poppler; no dependency download is needed in the authoring environment. It requires all final scoped gates.

The addendum’s `manifest.json` binds its source files, final PDF and plots, together with the source gates/reports it summarizes. `pdf-qa.json` records the final PDF hash, page count,

text/layout diagnostics and the separate visual review. Temporary page rasters and TeX products are excluded from the source package. The PDF is rendered and visually checked after the final content change. Rebuilding requires repeating that visual review; successful compilation alone is insufficient.

**Repository evidence.** Within `research/round27/`, see `contracts/`, the separate `forward/` and `reverse/` reports and exact outputs, `skeptic/`, and `advisor/`. The round's `SOURCES.md` summarizes the survey; each of the five `experts/` directories contains a report and machine-readable inventory with source roles and reading limits. The earlier manuscript is preserved in `papers/draft-01/`. The public repository is [occult-kranti/yang\\_mills\\_workbench](https://github.com/occult-kranti/yang_mills_workbench).

## Selected primary links and actual reading scope

These pointers summarize the expert records; the full inventory retains all 55 entries, inaccessible endpoints and discussion links. No full copyrighted book text is redistributed. Source findings in this addendum are attributed to the recorded panel readings, not to an independent new audit of every linked work.

- [1] Newton, *Opticks*, Query 31 and General Scholium. Selected closing methodological and speculative passages. *Experiments*, Add.3973: selected dated operations and residues; [Keynes MS.28](#): English transcription, not a full Latin commentary audit.
- [2] Tesla, [US787412A](#) and [Colorado Springs entry, 7 December 1899](#). Mechanism description and one tuning table. [FBI institutional account](#): custody statement; the large catalogue scan was not read.
- [3] Jung, *Psychology and Alchemy* and *Synchronicity*. Publisher-hosted chapter openings only. [Zabriskie](#), introduction to *Atom and Archetype*: context and excerpts, not the full correspondence.
- [4] Penrose, *Angular momentum: an approach to combinatorial space-time*. Local admissibility and possible zero norm; [handwritten notes](#): first two pages visually inspected. Gravitational reduction was treated separately and at abstract depth.
- [5] Feynman, *The Qualitative Behavior of Yang-Mills Theory in 2+1 Dimensions*. Primary abstract and bibliographic scope; full text not retrieved. *Cargo Cult Science* supplies methodological context.
- [6] Jaffe–Witten, *Quantum Yang–Mills Theory*; official [Clay status page](#). Target statement and selected limit obligations; status checked 22 September 2026.
- [7] Ciavarella, Burbano and Bauer, [arXiv:2503.11888v2](#): selected local truncation and large- $N$  sections. Cao, Nissim and Sheffield, [arXiv:2505.16585v3](#): introduction and Theorem 1.2 context, not a complete proof audit.
- [8] Bonanno et al., [arXiv:2511.07355v1](#): selected scale, topology and volume sections. [Froland and Grabowska](#), [arXiv:2606.14842v1](#): selected apparatus, estimator and digitization discussion.
- [9] Chevyrev and Shen, [arXiv:2503.03060v2](#): inspected theorem and explicit remaining stochastic three-dimensional construction limits. No four-dimensional pure-theory conclusion is imported.
- [10] Kane, Hariprakash and Bauer, [arXiv:2506.16559v1](#); Negro et al., [arXiv:2606.20029v1](#); [arXiv:2609.07886v1](#). Selected continuum-error, step-scaling and digitization comparisons in their stated models.
- [11] Petrica and Popescu, [arXiv:2408.14616v2](#): local-inverse theorem and proof inspected; its finite-dimensional hypotheses are not automatically imported into this model. [Maclaren and Nicholson](#), [arXiv:1904.02826v4](#): uniqueness versus stable estimability.

- [12] Nachtergaele, Sims and Young, [arXiv:1810.02428](#): abstract and bibliographic scope checked as a quasi-locality comparison. It supplies no unverified estimate for our support norm.

## Part II

# Round28: complete remainders and physical observables

## 22 Inherited obligations and adaptive research design

Round27 established an actual one-step correction for the selected cubic-source sector  $G + A_3$ , with a narrower-coupling contraction of its selected mixing family and explicit support-weight loss. Its extra original remainder  $E$  remained conditional. The first Round28 question is therefore whether that omitted interaction can be completely reconstructed and controlled. A positive bound for an extracted reference operator must remain separate from a gap of the full Hamiltonian with its remaining mixing.

The authorized continuation has five goal pairs with two reviewed investigations per pair. The first three inherited branches concern homogeneous remainder control, cancellation-aware canonical readouts and larger physical graphs. Their six reviews led to the homogeneous representation and Wilson-nonvacuity pair; that review then led to the quantitative observable goal. Every second contract is selected after its first review, and only gated results enter this manuscript. Repairs, literature searches, replays and refinements within a frozen contract are not additional research loops.

The advisor freezes each model, observable, domain, parameter interval and desired conclusion. Separate forward and reverse researchers work before exchanging current results; a separately authored skeptic challenges the proof and exact checks. Shared inherited premises mean these are independent derivations and implementations, not independent physical observations or external peer review. Accepted gates control which chapters enter the compiled manuscript.

Physical units remain explicit. Fixed positive  $\alpha$ ,  $\hbar$  and  $E_*$  cannot be recalibrated to hide a failed matching estimate. In the homogeneous normalized Hamiltonians, physical energy is obtained using  $\delta = \alpha/8$ , with original spacing fixed. Support weights and filter durations are proof parameters. Parameters in the canonical summable family are not automatically a regulator trajectory or a homogeneous-model identification.

## 23 A bounded, attributed source survey

The initial Round28 combined ledger records **30 formal readings at 38 distinct recorded URLs**. A record is one expert-source entry or skeptic external-source entry, not one completely read book or independent study. Several lenses revisit the same paper; one entry can include abstract, PDF and publisher URLs. Three separate screening/reopening entries bring the URL union to 41. One additional unread lead would bring it to 42, but is excluded from the reviewed-source count. These distinctions are preserved in `research/round28/experts/sources.json`. The later source-only addition and its combined counts are recorded separately in Appendix 36.

The categories comprise eight historical-primary records, sixteen technical-primary records, one patent, three forum records and two official problem references. Primary provenance does not establish reliability, applicability or scientific priority. Newton explicitly deepened the historical ALCH00081 reading and selected sections of an inherited stability paper. Other new ledger entries are new recorded readings by that lens, not claims of first project discovery. This manuscript attributes the recorded findings; it does not claim a fresh audit of every external page or full proof.

| Lens       | Records | URLs | Reading character                                                                              |
|------------|---------|------|------------------------------------------------------------------------------------------------|
| Newton     | 3       | 3    | Two historical manuscripts and deeper selected stability-paper passages.                       |
| Tesla      | 5       | 5    | Autobiography, patent, forum claims and two 2026 cancellation papers.                          |
| Jung/Pauli | 9       | 9    | Historical writings at different depths, two discussions and two inference papers.             |
| Penrose    | 5       | 11   | Stability, gauge reduction, variational benchmarks, exact integration and a channel extension. |
| Feynman    | 5       | 10   | Stability, local transformations, exact algebra and two bounded current-claim inspections.     |
| Skeptic    | 3       | 3    | Official problem/status records and an older-paper bibliographic check.                        |

Table 4: Per-lens URL counts overlap and must not be summed into a unique-source count. The survey itself executes zero physics loops.

### 23.1 Historical and unconventional material: questions, not premises

Newton’s selected folios of *Of Natures obvious laws* motivate asking which terms or reservoirs have been omitted from a proposed mechanism. His [October 1666 fluxions tract](#), Proposition 7 and selected reconstruction passages, motivates testing whether an inverse returns the originating relation. Missing rendered formulas prevent a complete transcription or derivation audit. Neither source validates an alchemical mechanism or supplies a modern gauge-theory term.

Tesla’s *My Inventions*, section VI was inspected in a transcription at selected retrospective mystical and mechanistic passages. It is neither the original magazine scan nor a controlled observation. [US685957A](#), around figures 1, 3 and 4 and the claims, specifies incident radiation, storage, receiver control and output conditions. Its proposed apparatus does not establish unrestricted loading or instantaneous transmission. The associated forum exchange is a record of claims to test against the patent, not corroborating physical evidence.

Jung’s *Collected Papers on Analytical Psychology* was read in the verified 1920 second-edition reprint: selected occult-phenomena, number-dream and rumour passages are three writings at one URL, not three books. Publisher opening excerpts of the *I Ching* foreword and “Psychology and Spiritualism” do not amount to reading the full essays. The operational question is whether a fixed interpretation separates a prediction from post hoc selection, remembered matches and ignored misses. Two purposively sampled discussions do not establish prevalence, authenticate every circulating quotation or verify paranormal effects.

There is **no newly read government-primary source** in these Round28 ledgers and no newly inspected original Pauli letter. Earlier FBI, DOE and CIA material remains in Round27 at its original reading depth and is not counted again. Clay is an official mathematical-problem source, not a government program. Historical or mystical interpretations add no Hamiltonian coefficient, measured frequency, physical clock or continuum map.

### 23.2 Technical comparisons and remaining applicability tests

Three lenses revisit [Del Vecchio–Fröhlich–Pizzo, arXiv:2108.13907](#), with selected induction, domain, assumption and theorem passages recorded separately. The 2021 preprint and 2023 journal publication motivate an applicability audit. An existential stability threshold cannot be cited as an evaluated numerical threshold or a completed workbench induction. Selected local-transformation passages in [Bravyi–DiVincenzo–Loss, arXiv:1105.0675](#) and exact-algebra passages in [Ying–Li–Po, arXiv:2509.12621](#) reinforce full transformed-operator accounting. Spin

and qubit fixtures cannot replace the full  $SU(2)$  support and domain arguments.

Tesla’s selected [noise-canceling observable](#) and [spacetime mitigation](#) papers motivate retaining model bias, joint omitted events, normalization and sampling resources. They imply no automatic cancellation of this workbench’s error disks. Jung’s [relative-fit inference comparison](#) distinguishes inference under misspecification from a true-model statement. The [Hamiltonian observation-geometry comparison](#) makes sampled data a parameter pseudometric unless injectivity is proved. Deterministic interval bounds have no statistical coverage interpretation without a sampling model.

Penrose’s selected [gauge-fixed Hamiltonian](#) and [variational ground-state](#) comparisons require the full induced generator, Gram matrix and omitted channels. A variational energy is not a certified gap. [Dahlqvist’s exact integration work](#) motivates exact quotient calculations, with its surface, group and limit assumptions retained. A [quantum-channel spin-network extension](#) changes the dynamics and is not silently imported into a closed Hamiltonian model.

The supplemental Feynman review inspects selected assumptions and theorem passages of the [Faizal–Shabir four-dimensional construction claim](#); it does not audit the entire 593-page compilation. Questions about comparison direction, quantitative defect margins, time factors and tuning-map rank require checking the other supporting sections before a conclusion. The panel neither validates nor refutes the whole construction. [Nair’s lower-dimensional proof outline](#) was inspected at selected framework, equations and regularization passages. Its closed domains, regulator limits and vacuum/energy centering remain applicability obligations; it is not a four-dimensional matching theorem. The formalization lead [2603.15770](#) was unread by that lens in this pass and excluded from its reviewed count.

Version discrepancies remain explicit. Two experimental HTML headers display dates different from their version histories; no later revision is inferred. Dahlqvist’s landing abstract and version-2 theorem use different genus thresholds, so the specified theorem/version controls the theorem-scope claim. The two different math-ph identifiers in screening and bibliographic records are not merged. Root metadata checks are separate from full-proof audits. The ledger retains exact source pointers and reading limits rather than hiding these discrepancies.

**How the survey changes the tests.** For the homogeneous branch, deleting a quadratic class, overlapping ordered word, factorial coefficient, scalar correction or transported remainder must fail reconstruction. For canonical readouts, retain higher odd terms, joint/adversarial error, phase and clock uncertainty and denominators containing zero; a cancellation coefficient containing unknown  $q$  must be candidate-indexed or carry its estimation error. For graph enlargement, require exact Gram null relations, every reached and omitted channel, boundary return and an added-region observable. These are source-motivated controls for frozen goals, not new physical premises or extra loops.

The [official status](#) and [Jaffe–Witten problem statement](#) remain the target references. Nontrivial four-dimensional construction, the required axioms, compact simple groups, a positive physical mass gap and scale matching remain distinct obligations. Source counts, network edges and loop completion supply no defensible percentage of that theorem.

## 24 HNM AG2: the complete original remainder and its reference<sup>\*\*</sup>

The first admitted investigation reconstructs the original homogeneous remainder that was conditional in AG1 and transports every class through the actual AG1 correction. On  $0 \leq M \leq 10^{-3}$  the resulting indexed weight-two interaction obeys  $\|K\|_2 < 7/1000$ ; on  $0 \leq M \leq 10^{-4}$  it

obeys  $\|K\|_2 < 7/100000$ . The centered reference alone has normalized gap greater than 0.995. The full corrected Hamiltonian retains a mixing operator, so this is not its gap theorem.

## 24.1 Domain-preserving transformation and a four-part inventory

Keep the homogeneous complete 24-link coarse factors and all 21 omitted faces on each retained complete star  $Z_b = b + \{0, e_1, e_2, e_3\}$ . The onsite reference is  $H_0 = \sum_x h_x$ , with  $h_x \geq I - P_x$  and product vacuum  $\Omega$ . Write

$$\begin{aligned} v_b &= \phi_b \Omega_b, \quad u_b = (H_{0,Z_b}|_{Q_b})^{-1} v_b, \quad L_b = |u_b\rangle\langle\Omega_b| - \text{adjoint}, \\ A_{1,b} &= |v_b\rangle\langle\Omega_b| + \text{adjoint}, \quad D_b = Q_b \phi_b Q_b, \quad X = \sum_b L_b, \quad G = H_0 + D. \end{aligned} \quad (\text{HNM-175})$$

Here  $\Phi = \sum_b \phi_b$ ,  $A_1 = \sum_b A_{1,b}$ ,  $D = \sum_b D_b$ ,  $M = 7|\tau|$  and  $\|\phi_b\| \leq M$ . Exact Haar orthogonality gives  $\|v_b\|^2 = M^2/84$ . Rational upper bounds are  $\|L_b\| \leq s$  and  $\|A_{1,b}\| \leq \sigma$ , where  $s = \sigma = 4M/35$ . They are not evaluated norm equalities.

The actual vectors  $u_b, \Omega_b$  lie in the local reference domain, and  $[L_b, H_0] = -A_{1,b}$  has a bounded extension. Strong commutation with exterior energy and graph-norm control justify  $e^{\pm X} D(H_0) = D(H_0)$  in finite cuboids. Integrating the first bounded commutator before expanding the remaining bounded terms gives

$$\bar{H} = e^X (H_0 + \Phi) e^{-X} = G + R_{O1}, \quad R_{O1} = \sum_{n \geq 1} \left( \frac{\text{ad}_X^n \Phi}{n!} - \frac{\text{ad}_X^n A_1}{(n+1)!} \right). \quad (\text{HNM-176})$$

This does not postulate an operator-norm BCH series for unbounded  $H_0$ .

Pair the two terms at each ordered anchor history, retaining repetitions and the complete star union. At a same-anchor depth-two history define

$$C_b = \frac{1}{2} \text{ad}_{L_b}^2 \phi_b - \frac{1}{6} \text{ad}_{L_b}^2 A_{1,b}, \quad A_{3,b} = P_b C_b Q_b + Q_b C_b P_b. \quad (\text{HNM-177})$$

This  $A_3 = \sum_b A_{3,b}$  is precisely the earlier selected cubic source. It is distinct from  $A_1$ , the total cubic coefficient and the full remainder. The complete missing interaction is

$$\begin{aligned} E &= R_{O1} - A_3 = E_2 + E_{3,\text{same}} + E_{3,\text{other}} + E_{\geq 4}, \\ E_2 &= [X, D] + \frac{1}{2} [X, A_1], \quad E_{3,\text{same}} = \sum_b (C_b - A_{3,b}). \end{aligned} \quad (\text{HNM-178})$$

$E_{3,\text{other}}$  contains every other ordered depth-two history;  $E_{\geq 4}$  contains all depths at least three with their original factorials. The first three support maxima are seven, four and ten sites; the tail has arbitrary finite connected supports. These labels specify lowest algebraic degrees 2, 3, 3, 4, not nonvanishing of every class.

The same-anchor subtraction is an orthogonal block compression:  $C_b - A_{3,b} = P_b C_b P_b + Q_b C_b Q_b$ , of norm at most  $\|C_b\|$ . Its centered split is  $c_b I + Q_b (C_b - c_b I) Q_b$ ; using  $c_b I + Q_b C_b Q_b$  would count  $c_b Q_b$  twice. In each fixed nonempty cuboid the actual vacuum expectation has a nonzero quadratic term  $\langle \Omega, E \Omega \rangle = -a_\Lambda \tau^2 + O_\Lambda(|\tau|^3)$  with  $a_\Lambda > 0$ . Thus  $E$  cannot be a quartic-only operator-norm error in that volume. This neither computes its quadratic mixing nor proves a uniform lower coefficient.

## 24.2 Complete low-order geometry and the all-order tail

For the declared indexed norm  $\|B\|_a = \sup_x \sum_{Y \ni x} a^{|Y|} \|B_Y\|$ , let  $Q_n(a)$  count ordered relative histories with the size of their complete union as root multiplicity. The exact first polynomials

are

$$\begin{aligned} Q_0(a) &= 4a^4, & Q_1(a) &= 4a^4 + 84a^7, \\ Q_2(a) &= 4a^4 + 252a^7 + 216a^9 + 1920a^{10}. \end{aligned} \quad (\text{HNM-179})$$

An incoming anchor meeting a union  $Y$  is exactly  $y - z$  for  $y \in Y$ ,  $z \in Z_0$ . Enumerating those finite sets gives depth-two history counts  $(1, 36, 24, 192)$  at sizes  $(4, 7, 9, 10)$ . Finite boundaries delete histories; no interactions outside the original boundary are introduced.

Finite enumeration is not the infinite-tail proof. At depth  $n$  every connected union has at most  $4 + 3n$  sites. Counting both root locations, repeated anchors and incoming overlaps gives

$$N_{n+1}(a) \leq 8sa^3(8 + 3n)N_n(a), \quad N_n(a) \leq 4a^4b(24sa^3)^n(8/3)_n \quad (\text{HNM-180})$$

for a seed-norm bound  $b$ . With  $x = 24sa^3$ , the factorwise comparison  $(8/3)_n/n! \leq (n+1)(n+2)/2$  supplies the generating function  $(1-x)^{-3}$ . A complete positive majorant is

$$\begin{aligned} e_2(a) &= 2s(M + \sigma/2)Q_1(a), & u_3 &= 4s^2(M/2 + \sigma/6), \\ e_{3,\text{same}}(a) &= 4a^4u_3, & e_{3,\text{other}}(a) &= [Q_2(a) - 4a^4]u_3, \\ e_{\geq 4}(a) &= 4a^4(M + \sigma/4)[(1-x)^{-3} - 1 - 3x - 6x^2], \\ \|E\|_a &\leq e(a) := e_2(a) + e_{3,\text{same}}(a) + e_{3,\text{other}}(a) + e_{\geq 4}(a). \end{aligned} \quad (\text{HNM-181})$$

At  $a = \rho = 9/4$ ,  $x = (2187/70)M \leq 2187/70000 < 1$  on the main interval. Nonnegative power-series coefficients and monotonicity justify continuous interval bounds from exact rational endpoint evaluations. The same-anchor compression is counted once; its selected mixing norm is not added to an already complete cubic budget.

### 24.3 Transporting every original term

Let  $S$  be the admitted AG1 generator, distinct from  $X$ , with  $[S, G] = -A_3 + R_{A_3}$ . The complete corrected operator is

$$\begin{aligned} e^S \bar{H} e^{-S} &= G + K, & K &= R_{A_3} + N + e^{\text{ads}} E, \\ N &= \sum_{n \geq 1} \frac{\text{ad}_S^n(nA_3 + R_{A_3})}{(n+1)!}. \end{aligned} \quad (\text{HNM-182})$$

Graph-domain preservation applies to the actual  $S$ ; finite-volume  $E$  is bounded and need not separately regularize arbitrary domain vectors. With  $r_{\text{bd}} = M^3/567$ ,  $A_{3,\rho}^{\text{bd}} = 4\rho^4 r_{\text{bd}}$  and  $F = (1 - 72\rho^3 M)^{-3}$ , take

$$\theta = 18A_{3,\rho}^{\text{bd}}F < 1, \quad n_2 = \frac{\theta}{1-\theta}A_{3,\rho}^{\text{bd}}(1+F/2), \quad \|K\|_2 \leq k(M) := r_2 + n_2 + \frac{e(\rho)}{1-\theta}. \quad (\text{HNM-183})$$

Here  $r_2 = 64r_{\text{bd}}(1 - 576M)^{-3}$  is valid on the main interval. Only on the inherited narrow interval may it be replaced by  $64r_{\text{bd}}[4/9 + (1 - 576M)^{-3} - 1]$ . The geometric transport factor follows by allocating the  $\rho$ -to-two weight loss over every nested commutator. The entire margin is spent; the resulting source does not regain a stronger norm by renaming its weight. The reviewed conservative bounds are  $k(M) < 0.007$  on the main interval and  $k(M) < 0.00007$  on the narrow interval.

### 24.4 The centered reference bound and its exact limitation

For every generated self-adjoint  $K_Y$ , including transported  $E$ , retain its full support and split

$$\begin{aligned} c_Y &= \langle \Omega_Y, K_Y \Omega_Y \rangle, & B_Y &= P_Y K_Y Q_Y + Q_Y K_Y P_Y, \\ Z_Y &= Q_Y(K_Y - c_Y I)Q_Y, & K_Y &= c_Y I + B_Y + Z_Y. \end{aligned} \quad (\text{HNM-184})$$

The separate norm costs are one, one and two times  $\|K_Y\|$ . Every assigned support has at least four sites, including scalar terms. Double counting over sites and  $1/(|Y|2^{|Y|}) \leq 1/64$  give

$$\frac{|C_\Lambda|}{|\Lambda|} \leq \frac{k(M)}{64}, \quad \|B\|_2 \leq k(M), \quad \|Z\|_2 \leq 2k(M), \quad C_\Lambda = \sum_Y c_Y. \quad (\text{HNM-185})$$

Because  $Z_Y$  annihilates its local vacuum and  $Q_Y \leq \sum_{x \in Y} (I - P_x)$ ,

$$|\langle \psi, (D + Z)\psi \rangle| \leq \left(4M + \frac{k(M)}{8}\right) \langle \psi, H_0 \psi \rangle. \quad (\text{HNM-186})$$

The factor  $k/8 = 2 \cdot 2^{-4}k$  retains the four-site support minimum. Thus the centered reference, self-adjoint on  $D(H_0)$  in each finite cuboid, satisfies

$$G_{\text{ref}} = H_0 + D + Z \geq \left(1 - 4M - \frac{k(M)}{8}\right) H_0 \geq \left(1 - 4M - \frac{k(M)}{8}\right) (I - P_\Omega). \quad (\text{HNM-187})$$

The main interval therefore has a reference-only normalized gap greater than 0.995; physical energies multiply by  $\delta = \alpha/8$ . Adding  $C_\Lambda I$  shifts this reference ground energy. The actual corrected Hamiltonian is instead  $C_\Lambda I + G_{\text{ref}} + B$ , with  $B$  still present. No gap for it, contraction of its complete mixing, next correction or continuum theory follows from (HNM-187). The scalar is a reference expectation sum, not its interacting ground energy.

The implementations retain positive and negative coupling, zero and empty cases, exact low-order geometry, complete positive tails, wrong factorials, missing quadratic/cross histories, scalar double counting and omitted transport controls. Finite rational matrix fixtures diagnose algebra rather than simulate the full physical Hilbert space. Producer and skeptic constants differ because their sufficient positive majorants differ. The common admitted conclusions are the complete inventory, transport, stated interaction budgets and reference-only bound; scientific priority remains unverified.

## 25 HNM AG3: one complete correction and its passive comparison\*\*

The second investigation of the homogeneous goal applies a domain-preserving correction to the *complete* actual AG2 mixing family. Its bound is

$$\|R'\|_{3/2} \leq \gamma \|B\|_2, \quad \gamma = \begin{cases} 3060/3623, & 0 \leq M \leq 10^{-3}, \\ 5949/96812, & 0 \leq M \leq 10^{-4}. \end{cases} \quad (\text{HNM-188})$$

This compares different support weights. The main coefficient is weaker than the passive bound obtained by leaving  $B$  unchanged and measuring it at weight  $3/2$ . The narrow coefficient improves that universal upper ceiling, but neither interval proves improvement over the actual old lower-weight norm. The exact full conjugation, domains, retained diagonal and scalar terms are substantive parts of the accepted result.

### 25.1 A bare-reference inverse for every generated support

The actual input from AG2 is

$$H_{\text{in}} = C_\Lambda I + H_0 + D_{\text{next}} + B, \quad D_{\text{next}} = D + Z, \quad r = \|B\|_2, \quad d = \|D_{\text{next}}\|_2. \quad (\text{HNM-189})$$

Every assigned connected support  $Y$  has at least four coarse blocks, with no uniform upper size. All original and transported histories remain indexed. Coincident supports are not canceled or assigned smaller supports to alter  $r$ . With the admitted cap  $K$  from AG2,

$$r \leq K, \quad d \leq 64M + 2K, \quad |\langle \psi, D_{\text{next}} \psi \rangle| \leq (4M + K/8) \langle \psi, H_0 \psi \rangle. \quad (\text{HNM-190})$$

Here  $K = 7/1000$  or  $7/100000$  on the main or narrow interval, respectively. The actual norm  $r$  is distinct from this upper cap and from an extensive operator norm.

Since  $B_Y = P_Y K_Y Q_Y + Q_Y K_Y P_Y$ , write  $v_Y = B_Y \Omega_Y$  and define a new generator  $T$  (called  $X$  in the AG3 reports):

$$u_Y = (H_{0,Y} |_{Q_Y})^{-1} v_Y, \quad T_Y = |u_Y\rangle \langle \Omega_Y| - |\Omega_Y\rangle \langle u_Y|, \quad T = \sum_Y T_Y. \quad (\text{HNM-191})$$

The reduced local reference is at least one. Spectral calculus gives  $u_Y \in D(H_{0,Y})$ ,  $H_{0,Y} u_Y = v_Y$ , and  $\|T_Y\| \leq \|B_Y\|$ . The local products have bounded extensions and  $[T_Y, H_{0,Y}] = -B_Y$  on the domain. Extend by the exterior identity. Commuting nonnegative local and exterior energies give  $D(H_0) = D(H_{0,Y}) \cap D(H_{0,\text{ext}})$ , so the full domain is preserved. This is not regularization of an arbitrary exterior vector and is not the operator obtained with a global vacuum projector.

In each finite cuboid,

$$\sum_Y \|B_Y\| \leq |\Lambda| r / 64, \quad \sum_Y \|T_Y\| \leq |\Lambda| r / 64, \quad \|T\|_2 \leq r. \quad (\text{HNM-192})$$

Closedness of  $H_0$  and convergence of the bounded commutators give  $[T, H_0] = -B$ . The relation  $H_0 T \psi = T H_0 \psi + B \psi$  supplies the graph-norm bound; both exponentials preserve  $D(H_0)$ . No bounded infinite-volume global generator or unitary is asserted. This inverse uses  $H_0$  alone, so the changed diagonal is retained as a commutator defect.

## 25.2 The full infinite remainder with an actual denominator

First integrate the established bounded commutator on  $D(H_0)$ , then expand only bounded conjugations. The exact result is

$$e^T H_{\text{in}} e^{-T} = C_\Lambda I + H_0 + D_{\text{next}} + R', \quad R' = \sum_{n \geq 1} \left( \frac{\text{ad}_T^n D_{\text{next}}}{n!} + \frac{n \text{ad}_T^n B}{(n+1)!} \right). \quad (\text{HNM-193})$$

In particular  $[T, D_{\text{next}}] + [T, B]/2$  remains at first order; the old scalar commutes but stays in the Hamiltonian. This does not assume a norm-convergent BCH series for unbounded  $H_0$ .

For indexed interactions  $U, V$  and  $a > b \geq 3/2$ , nonempty intersection yields  $b^{|Y \cup Z|} \leq b^{|Y|} b^{|Z|} / b$ . Retaining both root placements and using  $m \exp[-m \log(a/b)] \leq 1/(e \log(a/b))$  proves

$$\|[U, V]\|_b \leq \frac{4}{b e \log(a/b)} \|U\|_a \|V\|_a. \quad (\text{HNM-194})$$

Every repeated index and complete union is included. The proof imposes no maximum support size and establishes the new output-weight range directly.

At nested order  $n$ , allocate  $\ell/n$  per bracket with  $\ell = \log(4/3)$ . All intermediate weights are at least  $3/2$ . The elementary bound  $n! \geq (n/e)^n$  gives

$$\frac{\|\text{ad}_T^n V\|_{3/2}}{n!} \leq (\bar{c} r)^n \|V\|_2, \quad \bar{c} = \frac{288}{31}. \quad (\text{HNM-195})$$

Indeed  $\ell \geq 31/108$  follows by integrating  $1/(1+t) = 1 - t + t^2 - t^3 + t^4/(1+t)$  on  $[0, 1/3]$ ; its remainder is positive. Therefore  $(8/3)/\ell \leq \bar{c}$ . With  $\theta = \bar{c}r < 1$ , the complete positive tail yields

$$\|R'\|_{3/2} \leq r \frac{\bar{c}(d+r)}{1-\bar{c}r}, \quad \text{tail after order } N \leq (d+r) \frac{\theta^{N+1}}{1-\theta}. \quad (\text{HNM-196})$$

Only the increasing coefficient uses  $r \leq K$  and  $d \leq 64M + 2K$ ; the actual  $r$  remains outside. At the main and narrow endpoints  $\theta \leq 252/3875$  and  $63/96875$ , respectively. These positive monotonicities establish continuous-interval coverage and give (HNM-188). If  $r = 0$ , each indexed  $B_Y$  vanishes,  $T = 0$  and  $R' = 0$ , without division; both coupling signs are covered by  $M = 7|\tau|$ .

### 25.3 What passive reweighting already provides

Minimum support size four gives, for the *same* indexed input family,

$$\|B\|_{3/2} \leq (3/4)^4 \|B\|_2 = \frac{81}{256} r. \quad (\text{HNM-197})$$

Exact comparison places 3060/3623 above this ceiling and 5949/96812 below it. The main active certificate consequently does not certify a benefit over doing no correction at the lower weight. The narrow active certificate is smaller than the universal passive upper ceiling, but two upper bounds do not compare the actual norms. A diagnostic family concentrated on one support of size  $m$  has passive ratio  $(3/4)^m$ , arbitrarily small as  $m$  grows. This is a counterexample to the proposed general inference, not an evaluation of the actual AG2 trajectory.

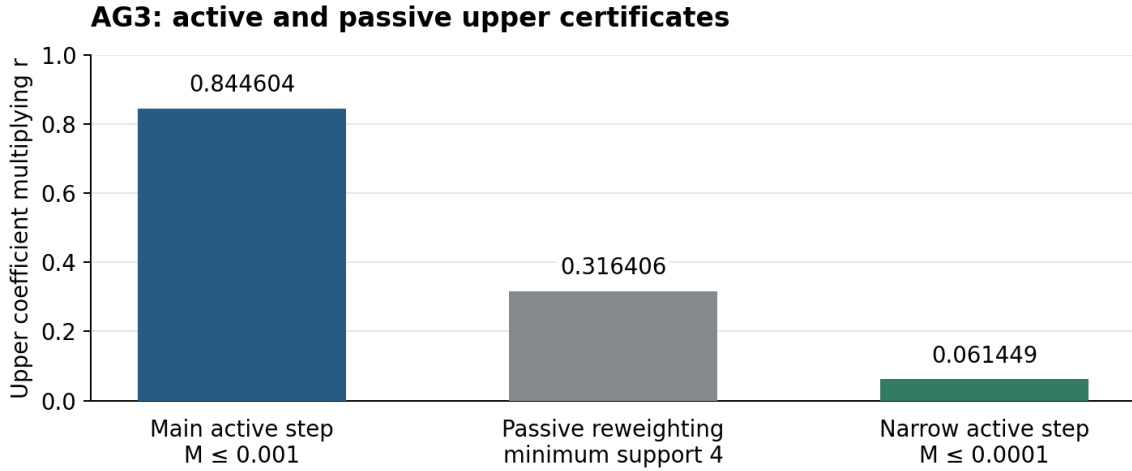


Figure 9: Sufficient coefficients multiplying the same actual  $r = \|B\|_2$ . Active bars bound the complete corrected  $\|R'\|_{3/2}$ ; the passive bar bounds the unchanged  $\|B\|_{3/2}$ . All values are exact rational certificates rendered as decimals. They do not evaluate either actual lower-weight norm, establish same-weight contraction, or measure theorem completion.

### 25.4 Retained scalars, diagonals and the reference-only gap

Split every generated self-adjoint indexed residual  $J_Y$  on its full support:

$$\begin{aligned} c'_Y &= \langle \Omega_Y, J_Y \Omega_Y \rangle, & B'_Y &= P_Y J_Y Q_Y + Q_Y J_Y P_Y, \\ Z'_Y &= Q_Y (J_Y - c'_Y I) Q_Y, & J_Y &= c'_Y I + B'_Y + Z'_Y. \end{aligned} \quad (\text{HNM-198})$$

The respective norm costs are one, one and two. Put  $W = \gamma K$  and  $C' = \sum_Y c'_Y$ . Retaining the old scalar and minimum assigned support gives

$$\begin{aligned} \|B'\|_{3/2} &\leq \gamma r \leq W, & \|Z'\|_{3/2} &\leq 2W, \\ \frac{|C_\Lambda + C'|}{|\Lambda|} &\leq \frac{K}{64} + \frac{4W}{81}, & |\langle \psi, Z'\psi \rangle| &\leq \frac{32W}{81} \langle \psi, H_0 \psi \rangle. \end{aligned} \quad (\text{HNM-199})$$

The extracted centered reference  $G' = H_0 + D_{\text{next}} + Z'$  has domain  $D(H_0)$ , annihilates the product vacuum, and satisfies

$$G' \geq \left(1 - 4M - \frac{K}{8} - \frac{32W}{81}\right) H_0. \quad (\text{HNM-200})$$

The exact endpoint coefficient is  $258975047/260856000 > 124/125$  on the main interval and  $174190074443/174261600000 > 1999/2000$  on the narrow interval. These are normalized *reference-only* gap floors. Physical reference energies multiply the unchanged  $\delta = \alpha/8$ . The full corrected operator remains  $(C_\Lambda + C')I + G' + B'$ . Its mixing is not discarded, its scalar is not identified with the true interacting ground energy, and its gap is unproved.

The reverse derivation additionally applies passive reweighting to the carried diagonal, obtaining  $\|D_{\text{next}} + Z'\|_{3/2} \leq (81/256)d + 2W$ . The forward and skeptical bound  $d + 2W$  is also valid. This difference in sufficient estimates is preserved rather than silently merged.

The bare-inverse strategy originated in the Newton post-AG2 review; the advisor's overlap refinement and proposed constant were shared before the contract. The directions independently derived and implemented the frozen target, not independently invented its premise. The skeptic's independent production preceded exchange; its additional passive interpretation and the reverse supplemental note were post-freeze reviews. Exact finite-matrix, exterior-domain, complete-support and denominator controls test the stated algebra and inference. They are not physical SU(2) cutoff simulations or observations.

Both the previous  $9/4 \rightarrow 2$  and current  $2 \rightarrow 3/2$  margins are spent. Growing-support controls reject a free return to weight two, and a generic fixed-weight commutator control rejects a universal bilinear constant for that broad class. No all-stage conclusion or repair of the earlier O2 majorant follows. A future iteration requires its own remaining weight schedule, retained-diagonal estimates and convergence theorem. This loop proves no full homogeneous gap, physical scale matching or four-dimensional continuum construction.

## 26 HNM AI3: exact retained cancellation and full physical readout limits<sup>\*\*</sup>

AI3 calculates exact odd moments and proves an all-order retained cancellation lemma. The complete physical ratio intervals nevertheless overlap in all ten frozen cells. The direct scalar distinction at  $u = 10^{-18}$ ,  $k = 5$  survives for both actual Wilson multipliers. These conclusions concern sufficient certificates in the canonical summable full-link SU(2) family, under its inherited transfer theorems. They do not prove equality of the candidates, a general identification impossibility or a homogeneous-model result.

### 26.1 Actual multipliers, complete factors and inherited transfer

Use the free unit cube based at  $(3, 1, 0)$  and the closed six-link path

$$\begin{aligned} X : (3, 1, 0), (3, 2, 0), (3, 2, 1), (4, 2, 1), \\ (4, 2, 0), (4, 1, 0), (3, 1, 0). \end{aligned}$$

The two  $yz$  faces  $f_4, f_5$  are anchored at  $(3, 1, 0)$  and  $(4, 1, 0)$ . Their symmetric differences with  $X$  give simple six-cycles  $Y_r = X \triangle f_r$  and the actual multiplication observables

$$B_r = \frac{\chi_X + \chi_{Y_r}}{\sqrt{2}}, \quad \psi_r = B_r \Omega = \frac{e_X + e_{Y_r}}{\sqrt{2}}, \quad r = 4, 5. \quad (\text{HNM-201})$$

An unmatched free-link center action gives zero means and orthogonality, so  $\|\psi_r\| = 1$ . Conditional Haar integration over exclusive paths gives  $\langle B_r^4 \rangle = 5/2$ . The exact multiplication norm is  $J = 2\sqrt{2}$ , with its supremum approached on positive-measure neighborhoods of identity holonomies. Therefore  $\|(B_r^2 - I)\Omega\|^2 = 3/2$ : a normalized source does not authorize a norm-one multiplier or rank-source replacement.

Each probe occupies eight links and their union occupies ten free factors. Independent cube enumeration reconstructs all sixteen six-cycles and the reached face-flip component at reference energy  $E = 9\alpha/2$ . This component reduces the pinched average; it is not the full regional electric shell or an invariant subspace of the unaveraged dynamics. The twenty exterior touching faces are explicitly retained. Each shares one cube edge, has a free opposite edge and two distinct external complete-factor owners. The possible  $3/4$  saving on the shared edge is canceled by the opposite free-edge cost; the side owners cost at least  $1/8$  each. With disjoint and internal nonresonant cases, this verifies applicability of the inherited participating-frequency bound  $1/8$  to both probes.

The infinite-dimensional domains, strip gaps, pinching and strong averaging theorem remain inherited premises from Y2/AA2. AD2 supplies the correctly ordered scalar identity

$$e^{iEt/\hbar} \langle \Omega, B_r \beta_q^t(B_r) \Omega \rangle = \langle \psi_r, Z_q(s) B_r Z_q(s)^* \Omega \rangle. \quad (\text{HNM-202})$$

Both source and adjoint-vacuum evolution errors are charged. No finite matrix enumeration substitutes for these full-link transfer theorems.

Starting from the common ten-factor seed, each collar adjoins every owner of every omitted face touching complete current factors. A ten-link strip is always included whole. A face is retained only when all its owners are included. The exact reconstructed counts are

| Depth $k$ | Complete factors | Complete links | Retained faces $N_k$ |
|-----------|------------------|----------------|----------------------|
| 0         | 10               | 10             | 2                    |
| 1         | 41               | 113            | 25                   |
| 2         | 149              | 329            | 143                  |
| 3         | 333              | 675            | 359                  |
| 4         | 610              | 1177           | 695                  |
| 5         | 993              | 1857           | 1171                 |
| 6         | 1497             | 2739           | 1808                 |

The depth-six extension was declared before production; it has 813 crossing faces. Complete links and owners, rather than counts alone, are exported. The old eight-link seed fails the second probe's support requirement.

At order  $n$ , connected words visit at most  $10 + 3n$  factors, each with at most forty incident faces. The interaction coefficient, commutator factor two and time simplex give  $(10/3)_n x^n / n!$ , where  $x = 10\alpha\tau|t|/\hbar$ . All full and regional words through order  $k$  agree. A complete positive tail, including repetitions and outward-return paths, is bounded at  $x \leq 15z < 1$  by

$$D_k^{(10)}(15z) = \frac{(10/3)_{k+1}}{(k+1)!} \frac{(15z)^{k+1}}{(1-15z)^{k+5}}. \quad (\text{HNM-203})$$

The integer denominator exponent conservatively rounds the positive Taylor remainder exponent  $k + 13/3$  upward. This is a spatial estimate for complete infinite-dimensional factors, not a representation cutoff.

## 26.2 Exact moments and a reflection proof at every retained order

The finite reached matrix  $Q(q)$  places weights  $q^4$  or  $q^5$  on physical face flips. Its symmetric row sums give  $\|Q\| \leq 6$  and  $\|Q'\| \leq 30$  on  $[0, 1]$ . Set

$$p(q) = 2 + 5q + 5q^2 + 6q^3 + 3q^4, \quad v = \frac{z(1+q)^2(1+q^2)}{32p(q)},$$

$$h_r = \langle \psi_r, e^{ivQ(q)} \psi_r \rangle, \quad m_{n,r} = \langle \psi_r, Q(q)^n \psi_r \rangle, \quad \Delta_n = m_{n,5} - qm_{n,4}. \quad (\text{HNM-204})$$

Exact polynomial walks, separately checked against rational matrix propagation, give

$$\begin{aligned} \Delta_1 &= 0, & \Delta_3 &= 2q^{13}(1-q^2), \\ \Delta_5 &= q^{21}(1-q^2)(12+8q+12q^2), \\ \Delta_7 &= q^{29}(1-q^2)(54+72q+164q^2+72q^3+54q^4). \end{aligned} \quad (\text{HNM-205})$$

Let  $P_r$  denote the degree-eight exponential center. The alternating imaginary signs are retained:

$$C_8 := \text{Im } P_5 - q \text{Im } P_4 = -\frac{v^3 \Delta_3}{3!} + \frac{v^5 \Delta_5}{5!} - \frac{v^7 \Delta_7}{7!}. \quad (\text{HNM-206})$$

In particular the cubic contribution is negative for  $0 < q < 1$ .

Reflection  $(x, y, z) \mapsto (7-x, y, z)$  fixes the unoriented cycle  $X$ , swaps  $Y_4, Y_5$ , and induces an involution on all sixteen cycles. Every unweighted face-flip adjacency is preserved, so its permutation  $P$  obeys  $PQ(1) = Q(1)P$ . Fundamental  $\text{SU}(2)$  characters are unchanged by reversal. Thus  $P\psi_4 = \psi_5$  and  $\Delta_n(1) = 0$  for *every*  $n$ . This is an all-order retained proof, not an extrapolation from finitely many moments. The reflection does not preserve all anchored weights at  $q < 1$  and supplies no full canonical  $q = 1$  Hamiltonian.

At fixed normalized sources, the product rule gives

$$|m'_{n,r}(q)| \leq 5n6^n, \quad |\Delta'_n(q)| \leq (10n+1)6^n, \quad |\Delta_n(q)| \leq (1-q)(10n+1)6^n. \quad (\text{HNM-207})$$

The last inequality integrates the derivative from  $q$  to one after the reflection proof. No physical error disk or parameter-dependent physical correlation is differentiated;  $v(q)$  is substituted afterward. Since  $(10n+1)/n!$  decreases for  $n \geq 9$ , for  $x = 6|v| < 1$  the whole retained tail obeys

$$|\text{Im } h_5 - q \text{Im } h_4 - C_8| \leq T_\Delta := \frac{(1-q)91x^9}{9!(1-x)}, \quad |h_r - P_r| \leq A := \frac{x^9}{9!}. \quad (\text{HNM-208})$$

Even powers are included conservatively in the first sum. Under the first hypothesis at  $u = 10^{-18}$ ,  $A$  is about  $1.334 \times 10^{-70}$  while  $T_\Delta$  is about  $1.214 \times 10^{-86}$ . The new retained arithmetic estimate does not reduce physical uncertainty.

## 26.3 Physical errors, shared denominators and valid enclosures

The unchanged canonical quantities are

$$\begin{aligned} b(q) &= \frac{p(q)}{24(1-q)^3(1+q)^2(1+q^2)}, & \tau_q &= \frac{\eta}{8b(q)}, \\ \hat{d}_q &= \frac{\tau_q \sqrt{b(q^2)/96}}{(1-\eta)/8}, & M_k &= \frac{N_k}{24}, \\ R_{\text{phys}} &= 48\hat{d}_q + 8D_k^{(10)}(15z) + 64\tau_q M_k(1+3zM_k). \end{aligned} \quad (\text{HNM-209})$$

The first term retains stationary connected-state replacement; the spatial factor is  $J^2 = 8$ ; the averaging term bounds  $(1+J)16\tau_q M_k(1+3zM_k) < 64\tau_q M_k(1+3zM_k)$  and retains both evolution

columns. Exact upward square-root enclosures differ between implementations but preserve the verdicts. Unsupplied preparation, measurement, phase and clock errors must be added separately. Write  $a_r = \text{Im } P_r$  and  $y_r = a_r + \delta_r + e_r$ , with  $|\delta_r| \leq A$  and  $|e_r| \leq R_{\text{phys}}$ . Every frozen denominator interval

$$I_4 = [a_4 - R_{\text{phys}} - A, a_4 + R_{\text{phys}} + A] \quad (\text{HNM-210})$$

has positive lower endpoint. All-corner division of the corresponding  $I_5$  by  $I_4$  gives a valid direct ratio enclosure, including signed numerator cases. A second valid construction divides  $C_8 \pm [(1+q)R_{\text{phys}} + T_\Delta]$  by  $I_4$  and adds  $q$ . Its rectangular relaxation forgets the shared  $e_4$  in numerator and denominator. Equal physical radii do not imply equal signed errors:  $e_5 = R_{\text{phys}}$  and  $e_4 = -R_{\text{phys}}$  attain the full  $(1+q)R_{\text{phys}}$  combination allowance.

The reverse derivation additionally keeps that dependence in the enlarged box

$$q + \frac{C_8 + \delta_C + e_5 - qe_4}{a_4 + \delta_4 + e_4}, \quad |\delta_C| \leq T_\Delta, \quad |\delta_4| \leq A, \quad |e_4|, |e_5| \leq R_{\text{phys}}. \quad (\text{HNM-211})$$

With positive denominator, each coordinate is linear-fractional and monotone or constant when the others are fixed, so its sixteen corners give exact extrema of this enlarged box. This is not a claim that all corners are physically attainable. Intersections of valid enclosures remain valid. Smaller width does not establish endpoint containment; no blanket containment between the direct and cancellation intervals is asserted, and no global optimality over all retained constraints is claimed. The final review finds smaller direct interval widths in all twenty hypotheses, but none of those direct intervals is wholly contained in its cancellation-centered counterpart. The forward post-freeze clarification explicitly preserves this distinction.

The measured statistic  $y_5/y_4$  contains no unknown  $q$ . A residual  $y_5 - qcy_4$  is explicitly indexed by a proposed candidate  $q_c$ ; the unknown true parameter cannot be supplied secretly as an instrument setting. The old cubic comparator, with center  $vq^r$  and additional floor  $36v^3$ , is recalculated on the same grid with complete errors and denominator checks.

## 26.4 Ten insufficient ratio cells and a surviving scalar distinction

With  $z = 10^{-6}$ , the two candidates are

$$H_1 : (q, \eta) = (1 - u, 1/2), \quad H_2 : (q, \eta) = (1 - 2u, 1/16). \quad (\text{HNM-212})$$

Both have  $\eta(1 - q)^3 = u^3/2$  and the same original physical time

$$t_u = 2 \cdot 10^{-6} u^{-3} \frac{\hbar}{\alpha} = \begin{cases} 2 \cdot 10^{30} \hbar / \alpha, & u = 10^{-12}, \\ 2 \cdot 10^{48} \hbar / \alpha, & u = 10^{-18}, \\ 2 \cdot 10^{66} \hbar / \alpha, & u = 10^{-24}. \end{cases} \quad (\text{HNM-213})$$

Positive  $\alpha$ ,  $\hbar$ ,  $\alpha/E_*$  and spacing remain fixed. No independent clock adjustment for the candidates is permitted.

For two ratio intervals  $[l_i, u_i]$ , the signed separation margin is  $\max(l_1 - u_2, l_2 - u_1)$ . Positive means certified disjointness. Every direct full-error margin below is negative; cancellation-centered, shared-corner and intersected enclosures also remain insufficient in all ten cells. Exact fractions determine signs; decimals only display scale.

| $u$        | $k$ | Direct ratio margin (approx.) | Full scalar tests $B_4, B_5$ |
|------------|-----|-------------------------------|------------------------------|
| $10^{-12}$ | 3   | $-3.811493484 \cdot 10^{-9}$  | insufficient                 |
| $10^{-12}$ | 4   | $-1.044822364 \cdot 10^{-9}$  | insufficient                 |
| $10^{-12}$ | 5   | $-1.044761496 \cdot 10^{-9}$  | insufficient                 |
| $10^{-18}$ | 3   | $-2.766731990 \cdot 10^{-9}$  | insufficient                 |
| $10^{-18}$ | 4   | $-6.086906259 \cdot 10^{-14}$ | insufficient                 |
| $10^{-18}$ | 5   | $-1.313885034 \cdot 10^{-18}$ | both disjoint                |
| $10^{-24}$ | 3   | $-2.766731990 \cdot 10^{-9}$  | insufficient                 |
| $10^{-24}$ | 4   | $-6.086901682 \cdot 10^{-14}$ | insufficient                 |
| $10^{-24}$ | 5   | $-1.268122540 \cdot 10^{-18}$ | insufficient                 |
| $10^{-24}$ | 6   | $-2.436389698 \cdot 10^{-23}$ | insufficient                 |

Replacing the separate cubic linearizations by degree-eight centers removes the material old Taylor floor, approximately  $2.0408163 \times 10^{-14}$  in the ratio comparison, and sharpens the last cells. Ordinary degree-eight arithmetic is already negligible on this grid. The further all-order  $(1 - q)$  factor improves that negligible arithmetic term; it does not cause a successful ratio discrimination or reduce the physical upper budgets. Overlap of sufficient intervals is not a lower bound on actual error, physical equality or an impossibility theorem.

Component-only recalculations diagnose these bounds and are never substituted for the full radius. At  $u = 10^{-18}$ ,  $k = 5$ , the state-only ratio margin is approximately  $-4.576149531 \times 10^{-20}$  and the spatial-only margin is  $-2.681235390 \times 10^{-19}$ . Either inherited component alone leaves this interval test insufficient. The two complete scalar radii are about  $9.011406565 \times 10^{-27}$  and  $4.761718639 \times 10^{-27}$ , combining unequal state costs with the common spatial cost  $3.774177199 \times 10^{-27}$ .

At  $u = 10^{-24}$ ,  $k = 6$ , the spatial-only ratio margin is approximately  $-2.436285122 \times 10^{-23}$ , while the state-only margin is positive, about  $9.989542385 \times 10^{-25}$ . Each spatial scalar budget is  $7.548467626 \times 10^{-32}$ ; the state budgets are about  $5.237229366 \times 10^{-36}$  and  $9.875414398 \times 10^{-37}$ . This isolates the spatial certificate as the relevant remaining bound at the predeclared depth; it does not measure the actual dominant error.

The surviving direct scalar margins at  $u = 10^{-18}$ ,  $k = 5$  are about  $3.214524214 \times 10^{-26}$  for  $B_4$  and  $4.405000405 \times 10^{-26}$  for  $B_5$ . Equal additional absolute error on each compared scalar may be strictly below half the respective exact margin. These are pairwise mathematical allowances at time  $2 \times 10^{48} \hbar / \alpha$ , not instrument sensitivity. No ratio cell has a positive margin, so no new ratio tolerance is inferred.

The common *known* reference carrier in (HNM-202) can be removed consistently. A common unknown phase generally changes an imaginary-part ratio even though it preserves complex disk distances. Independently fitted phases can also change disk distances. Thus unspecified phase, timestamp, preparation and measurement errors are not canceled, and no timing-sensitivity theorem is claimed.

The Tesla-method proposal supplied the candidate-indexed readout strategy and the advisor supplied the prospective reflection lemma. The directions and skeptic independently derived consequences of this shared context. Complete geometry, exact moment signs, finite-moment counterexamples, actual multiplier norms, omitted-vacuum-column controls, signed interval corners and phase rotations test the argument. These are mathematical controls rather than observations. No continuous inverse, realized apparatus, homogeneous matching map, nontrivial four-dimensional continuum construction or Yang–Mills mass-gap theorem follows; scientific priority remains unverified.

## 27 HNM AI4: a uniform family of complete physical ratio tests\*\*

AI3 identified a remaining spatial certificate at small  $u$ . AI4 adaptively shortens the prescribed duration to  $z = 10^{-7}$  and increases the proof collar with a dyadic schedule. Under the inherited canonical full-link transfer theorems, the two complete predicted ratio intervals have gap at least  $39u/40$  for *every real*  $0 < u \leq 10^{-24}$ . Both independent producers prove this common conclusion using different explicit upper bounds. Every state, spatial and averaging error remains; this is a conditional family of candidate-pair tests, not a continuous inverse or an apparatus.

### 27.1 The family and its unchanged physical clock

Fix  $u_0 = 10^{-24}$ ,  $z = 10^{-7}$  and let  $j \geq 0$  be the unique integer for which

$$\frac{U_j}{2} < u \leq U_j, \quad U_j = u_0 \cdot 2^{-j}, \quad k = 6 + j. \quad (\text{HNM-214})$$

An excluded lower endpoint belongs to the next band. The candidates remain (HNM-212); hence  $\eta(1 - q)^3 = u^3/2$  in both models and their common physical time is

$$t = 2zu^{-3} \frac{\hbar}{\alpha}. \quad (\text{HNM-215})$$

At the base upper endpoint it is already  $2 \times 10^{65} \hbar/\alpha$  and grows without bound. Positive  $\alpha$ ,  $\hbar$ ,  $\alpha/E_*$  and spacing remain fixed. Changing  $z$  changes the observation duration within the admitted range, not the Hamiltonian or a separately fitted candidate clock. The values  $q, \eta$  are canonical-model parameters, not lattice spacing.

The actual probes, multiplication norm  $2\sqrt{2}$ , common ten-factor seed, reached sixteen-cycle component and exterior-loading hypotheses are those of AI3. The complete physical radius remains (HNM-209), now at the shorter duration and the scheduled  $k$ . Both connected-state means, both evolution columns, complete factors, and every omitted spatial word remain. The reached averaged component is not the whole electric shell or an instantaneous invariant space. The following proof uses source-bound AA2/AD2/Y1/Y2/AI3 transfer premises; it does not replace them with matrices or revalidate the entire historical proof chain.

### 27.2 A collar envelope proved at every depth

A unit face touching a current link adds at most one to each endpoint-box coordinate. Completing a strip owner adds at most three in  $x$ , one in  $y$  and zero in  $z$ ; its actual six  $x$  links and four  $y$  links are all included. A free factor adds no completion. Induction on the complete owner rule therefore gives

$$\text{vertices}(E(F_k)) \subset [0, 4(k+1)] \times [0, 2(k+1)] \times [0, k+1] \quad \text{for every } k \geq 0. \quad (\text{HNM-216})$$

Negative-coordinate faces are absent. Every retained face lies wholly inside this box. Counting unit faces in its three orientations yields the forward envelope, with  $L = k + 1$ ,

$$N_k \leq \bar{N}_k = 24L^3 + 14L^2, \quad M_k \leq \bar{M}_k = L^2(L + 7/12). \quad (\text{HNM-217})$$

The reverse proof instead counts three orientations at every possible anchor, obtaining the valid larger bound

$$N_k \leq 3(4k+5)(2k+3)(k+2), \quad M_{6+j} \leq 435(j+1)^3. \quad (\text{HNM-218})$$

At depth six, the exact admitted face count is 1808, while the two envelopes are 8918 and 10440. Every admitted depth-zero-through-six complete link is checked against the new geometry.

No actual collar beyond depth six is enumerated; a large-depth polynomial envelope is never presented as an exact face count.

Hereafter the displayed detailed proof follows the forward constants. Put

$$M_j = (j+7)^2(j+7+7/12), \quad M_0 = 4459/12, \quad x = 15z. \quad (\text{HNM-219})$$

The full spatial tail in (HNM-203) satisfies

$$\frac{D_{k+1}}{D_k} = \frac{x}{1-x} \frac{k+13/3}{k+2}, \quad \frac{D_{6+j}}{u} \leq \frac{2^{j+1}D_{6+j}}{u_0} \leq \frac{2D_6}{u_0}, \quad (\text{HNM-220})$$

because the successive ratio of the middle majorants is at most  $2x(31/24)/(1-x) < 1$ . This proves every integer  $j$ , rather than extrapolating from sampled bands.

Each of the three positive affine factors in  $M_j$  grows by at most  $8/7$  per step. Thus  $M_{j+1} \leq gM_j$ , with  $g = (8/7)^3$ , and

$$\frac{g^2}{4} < 1, \quad \frac{g^2}{8} < 1. \quad (\text{HNM-221})$$

Consequently both  $U_j^2 M_j(1+3zM_j)$  and  $U_j^3 M_j(1+3zM_j)$  decrease. The full quadratic  $M_j^2$  term is retained. Since both candidates obey  $\tau_q \leq 3u^3/4$ , the averaging contribution is bounded by  $48u^3 M_j(1+3zM_j)$ , with its normalized and absolute maxima controlled at the base band. The reverse proof independently bounds the cubic and sixth-power sequences  $(j+1)^3/4^j \leq 2$  and  $(j+1)^6/4^j \leq 64$  by finite algebraic prefixes followed by strict ratio induction. Those prefixes are not additional physical fixtures.

### 27.3 Continuous bounds and the location of arithmetic error

For the entire family  $q \geq q_* = 99/100$ . With  $e = 1 - q$ , positive polynomial inequalities for the unchanged  $p(q), b(q)$  give

$$\frac{z}{100} \leq v \leq V := \frac{z}{64}, \quad \tau_q \leq \frac{3}{2}\eta e^3, \quad \sqrt{b(q^2)/96} \leq \frac{1}{80e^{3/2}}. \quad (\text{HNM-222})$$

For example  $p(q) - 2(1+q)^2(1+q^2) = q + q^2 + 2q^3 + q^4 \geq 0$  gives the upper  $v$  bound; rational endpoint inequalities at  $q_*$  give the others. No grid interpolation supplies these continuous claims.

The complete forward state bounds for  $H_1, H_2$  are  $(36/5)u^{3/2}$  and  $(36/25)u^{3/2}$ . An absolute mesh  $\delta = 2^{-320}$  applied to the *inner* square root adds at most

$$\frac{384\tau_q\delta}{1-\eta}, \quad \text{hence at most } 576\delta u^3 \text{ or } (1536/5)\delta u^3. \quad (\text{HNM-223})$$

The prefactor is essential. A fixed uncertainty added to the final scalar is a different error and does not gain this cubic decay. The uniform proof for arbitrary real  $u$  needs no numerical square root; the propagated allowance remains conservative, and the exact rational fixtures check its location. The reverse proof uses different analytic state bounds,  $9u^{3/2}$  and  $(9/5)u^{3/2}$ , with no numerical root needed in its theorem.

AI3's factored moments imply  $|\Delta_3| \leq 4e$ ,  $|\Delta_5| \leq 64e$ ,  $|\Delta_7| \leq 832e$ . Define exact rational constants

$$B_* := \frac{2}{3}V^3 + \frac{8}{15}V^5 + \frac{52}{315}V^7, \quad T_* := \frac{91(6V)^9}{9!(1-6V)}, \quad A_* := \frac{(6V)^9}{9!}. \quad (\text{HNM-224})$$

The complete retained combination satisfies  $|C_8| \leq eB_*$  and its remaining tail is at most  $eT_*$ . Only fixed-source retained moment polynomials enter this statement; no physical error is canceled or differentiated. The ordinary individual allowance  $A_*$  can remain constant in a positive

denominator. Replacing the vanishing numerator allowance by a constant individual Taylor error would not establish this family for arbitrarily small  $u$ .

Nonnegative retained moments and alternating center signs give

$$a_4 \geq d_{\text{poly}} := \frac{zq_*^4}{100} - 36V^3 - \frac{(6V)^7}{7!}. \quad (\text{HNM-225})$$

The positive fifth-order contribution is dropped only for this lower bound. Let

$$R_* := \frac{36}{5}u_0^{3/2} + 576\delta u_0^3 + 8D_6 + 48u_0^3M_0(1 + 3zM_0). \quad (\text{HNM-226})$$

The continuous bounds and all-band recurrences above imply  $R_h \leq R_*$  for both candidates. Exact rational evaluation proves

$$d_{\text{poly}} - A_* - R_* > d_* := z/120 > 0, \quad |y_5| < n_* := z/32. \quad (\text{HNM-227})$$

This bounds actual full scalars and the exported scalar-center denominator intervals, not just the retained matrix denominator.

## 27.4 Explicit uniform gap and the independent constants

Physical errors remain adversarially bounded:

$$|y_5 - qy_4| \leq e(B_* + T_*) + (1 + q)R_h \leq e(B_* + T_*) + 2R_h. \quad (\text{HNM-228})$$

For  $h = 1, 2$ , put  $(a_1, a_2) = (36/5, 36/25)$  and  $(b_1, b_2) = (576, 1536/5)$ , and define six exact contributions

$$\begin{aligned} c_h^{\text{state}} &= \frac{2a_h\sqrt{u_0}}{d_*}, & c_h^{\text{round}} &= \frac{2b_h\delta u_0^2}{d_*}, \\ c_h^{\text{spatial}} &= \frac{32D_6}{d_*u_0}, & c_h^{\text{average}} &= \frac{96u_0^2M_0(1 + 3zM_0)}{d_*}, \\ c_h^{\text{bias}} &= \frac{hB_*}{d_*}, & c_h^{\text{tail}} &= \frac{hT_*}{d_*}. \end{aligned} \quad (\text{HNM-229})$$

The band lower endpoint controls the spatial term; the upper endpoint controls the remaining physical terms. With  $c_h$  their sum, every real  $u$  in every band obeys

$$\frac{y_5}{y_4} \in [q_h - c_h u, q_h + c_h u], \quad \text{gap} \geq \beta u, \quad \beta := 1 - c_1 - c_2 \approx 0.9791915454711759 > 39/40. \quad (\text{HNM-230})$$

The definition makes  $\beta$  an explicit rational constant after the frozen substitutions; its full fraction is retained in the output. It is a sufficient coefficient, not an optimal physical margin or a measured signal-to-noise ratio.

The reverse producer obtains a different valid coefficient  $\Gamma \approx 0.975932912473311 > 39/40$  using its envelope and denominator  $d > z/125$ . Its compact formula is

$$\Gamma = 1 - \frac{3(B_* + T_*) + 2(P_1 + P_2)}{d}, \quad (\text{HNM-231})$$

where  $P_h$  are its complete physical-radius coefficients,  $R_h \leq uP_h$ . These include the  $9, 9/5$  state constants, the full spatial tail, and the proved growing-collar averaging bound. Its actual  $d$  is the explicit rational lower bound from its scalar-center estimate, not the relaxed number  $z/125$ . The independent skeptic obtains a third sufficient coefficient near 0.9753467151692543. These

are independently derived certificates with different conservative choices; the common public conclusion is the weaker  $39u/40$  floor.

In the forward bound the summed deficit is dominated by state, 0.020736, and spatial, approximately  $7.24545 \times 10^{-5}$ . Averaging, correctly propagated rounding, retained bias and the full retained tail are also charged. Shortening duration reduces the spatial certificate strongly while increasing the relative state cost; the new schedule succeeds only because the combined bound remains below one.

## 27.5 Vanishing scalar tolerance, exact fixtures and limits

For an additional absolute error at most  $\nu$  in each scalar, (HNM-227) gives a ratio change at most

$$L\nu, \quad L = \frac{2}{d_*}(1 + n_*/d_*) = \frac{19}{2d_*}, \quad \nu \leq d_*/2. \quad (\text{HNM-232})$$

The explicit forward choice

$$\nu(u) = \frac{d_*u}{100} = \frac{zu}{12000} \quad (\text{HNM-233})$$

charges both candidates and leaves gap at least  $(\beta - 19/100)u > u/2$ . At  $u_0$  the allowance is about  $8.33333 \times 10^{-36}$ . It decreases to zero with  $u$ : no fixed positive apparatus error is covered uniformly. The four accepted allowances have distinct denominators and remaining margins:

| Derivation             | Additional error per scalar | Remaining gap             |
|------------------------|-----------------------------|---------------------------|
| Forward                | $d_*u/100$                  | $(\beta - 19/100)u > u/2$ |
| Reverse, half margin   | $\Gamma du/24$              | $\Gamma u/2$              |
| Reverse, target margin | $(\Gamma - 1/2)du/12$       | $u/2$                     |
| Skeptic, $D = z/128$   | $39Du/1280$                 | $39u/80$                  |

Each denominator reserve is proved in its own derivation. No mixed allowance/margin theorem is asserted, and the half-margin choices need not preserve  $u/2$ . Supplied preparation, measurement or calibration uncertainty must fit the applicable total scalar allowance.

Exactly ten physical parameter fixtures are evaluated:  $j \in \{0, 1, 8, 64, 512\}$ , each at  $u = U_j$  and  $u = 3U_j/4$ . They retain exact  $q < 1$ , rational centers, complete errors and the proved face envelopes. At the two existing  $j = 0$  points only, the admitted exact  $N_6 = 1808$  is compared with the envelope; this is not an extra cell. No new collar beyond six is computed. Large-band decimal displays that round a normalized gap to one do not erase its exact nonzero deficit.

Fixture success checks implementation. The theorem's continuous and all-integer quantifiers come from the geometry induction, continuous parameter inequalities and recurrences above. Separate algebraic controls reject finite-index extrapolation, wrong endpoint assignment, incomplete strip support, a fixed collar count, dropping  $M^2$ , deleting physical error terms, a wrong candidate clock, and confusing an inner arithmetic allowance with a final scalar floor. They do not supply extra physical observations or an automatic enlarged-graph Gram or omitted-channel theorem.

The shorter-duration strategy was proposed in the Feynman-method review after AI3; the advisor proposed its dyadic extension. These shared premises are attributed, while the producers and skeptic independently establish their consequences. The measured ratio contains no unknown  $q$ ; candidate-dependent residuals and the time schedule are predictions for predeclared pairs. This is not a universal instrument setting secretly supplied an unknown  $u$ , or an inverse from fixed unrestricted data.

The common reference carrier and physical scale must be independently fixed. An unknown common phase need not preserve an imaginary-only ratio, and unsupplied timestamp or phase error requires a separate sensitivity estimate. There is no practical clock, apparatus, fixed-error

robustness, homogeneous-model matching, global  $q = 1$  Hamiltonian, continuum construction or Yang–Mills mass-gap theorem. Scientific priority remains unverified.

## 28 HNM AH1: complete physical enrichment on a larger open graph\*\*

AH1 reconstructs the open  $3 \times 2 \times 1$ -cell graph, its strict physical electric projection and every individually resolved channel in the specified enrichment. The graph has 24 vertices, 46 links and 29 faces, including seven internal faces. The initial projection has rank 30; the enriched physical space has rank 561, a positive exact Haar Gram and 2,124 nonzero retained magnetic entries. For the contracted preparation class, the exact full and retained heat evolutions, each centered at its own actual ground energy, have all-time true-output-relative error below  $11/5000 = 0.0022$ . This is an analytic certificate, not a numerical heat trajectory or a graph-size-uniform result.

### 28.1 Every original link, boundary gauge action and full domain

The vertex set is  $\{0, 1, 2, 3\} \times \{0, 1, 2\} \times \{0, 1\}$ . Links are oriented in positive coordinate directions; a face with ordered axes  $a < b$  follows the word  $+a, +b, -a, -b$ . The complete list has twelve  $xy$ , nine  $xz$  and eight  $yz$  faces. Its cycle rank  $46 - 24 + 1 = 23$  is a graph invariant, not the dimension of a linear Wilson-function basis. All boundary vertices carry the gauge action  $U_{(v,w)} \mapsto g_v U_{(v,w)} g_w^{-1}$ ; a closed face word transforms by conjugation at its base. No tree links or original electric derivatives are deleted.

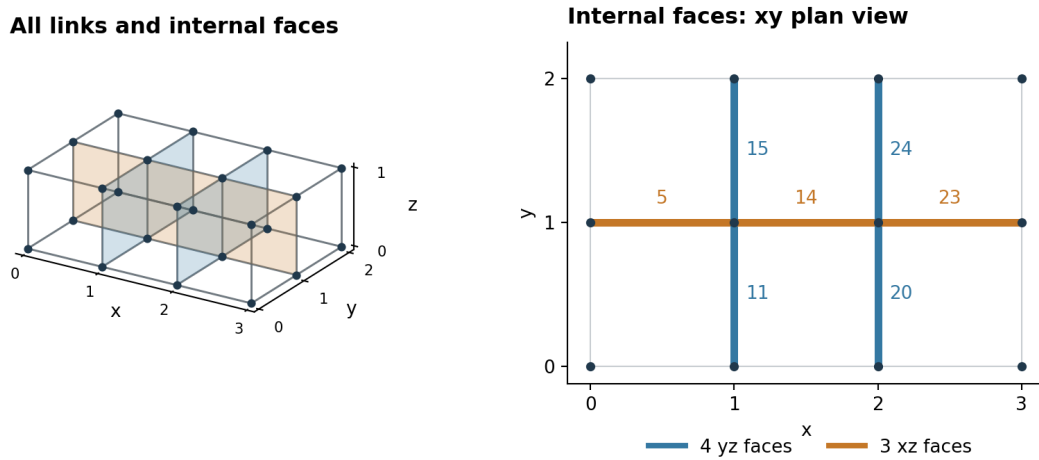


Figure 10: The actual AH1 coordinate graph and its seven internal faces. The left panel retains every one of the 46 links and 24 vertices. The right panel projects the seven vertical internal faces onto the  $xy$  plane; numbers are frozen face IDs. Colors distinguish the two plane orientations. This is graph geometry, not a proposed physical apparatus or a heat simulation.

On normalized product Haar measure, the actual model is

$$\begin{aligned} \mathcal{H}_{\text{phys}} &= L^2(\text{SU}(2)^{46})^{\text{SU}(2)^{24}}, & K &= \sum_e C_e, \\ W_p &= \frac{1}{2} \text{Tr } U_{\partial p}, & S &= \sum_{p=0}^{28} W_p, & L &= K + \lambda(29 - S). \end{aligned} \quad (\text{HNM-234})$$

Product Peter–Weyl labels give electric energies  $\sum_e j_e(j_e + 1)$  with finite multiplicities below every bound. Gauge averaging preserves each complete label block and commutes with the

Casimirs. The restricted  $K$  is self-adjoint with compact resolvent, operator domain the invariant product  $H^2$  space, form domain invariant  $H^1$ , and smooth invariant Peter–Weyl core. Since  $0 \leq 29 - S \leq 58$ , bounded perturbation gives  $D(L) = D(K)$  and the same form domain and compact resolvent. For the declared  $0 \leq \lambda \leq 1/100$ ,  $L \geq K$ . These are full representation-space statements, not consequences of a finite matrix cutoff.

Physical energy is  $H = \alpha L$ , and heat time is  $\sigma = \alpha t/\hbar$ , with  $a, E_*, \alpha/E_*, \hbar$  fixed and positive. This is a finite uniform-coupling graph model, distinct from the preceding homogeneous and canonical families. The theorem concerns Euclidean heat, not real-time evolution.

## 28.2 The strict electric cutoff and actual Haar quotient

Every nontrivial invariant spin support has no degree-one active vertex, and every active edge costs at least  $3/4$ . Below  $9/2$  at most five edges are active. A nonempty such support contains a cycle; the graph is simple, bipartite and of girth four. Two components would require at least eight edges. A five-vertex, five-edge minimum-degree-two component would be an odd five-cycle, while a bipartite support on at most four vertices has at most four edges. Hence the only possible nonempty support is a four-cycle. Complete graph enumeration identifies these exactly with the 29 elementary faces.

At bivalent vertices the invariant tensor forces a common spin around the cycle. The strict inequality  $4j(j+1) < 9/2$  selects  $j = 1/2$ , and its normalized function is  $\phi_p = 2W_p$ . Therefore

$$P_0 = \mathbf{1}_{[0,9/2)}(K), \quad \text{ran } P_0 = \text{span}\{\Omega, \phi_0, \dots, \phi_{28}\}, \quad \text{rank } P_0 = 30. \quad (\text{HNM-235})$$

The electric vacuum is unique and its full physical gap is exactly three. Six-edge fundamental loops have energy exactly  $9/2$  and are excluded from this initial interval. A total spin-degree cutoff is not identified with it.

For  $p < q$  put  $X_{pq} = W_p W_q$ . There are 96 pairs sharing one edge and 310 sharing no edge; 64 of the latter meet only at vertices. On a common edge let  $E_e$  be Haar integration, the spin-zero Casimir projection, and split  $X_s = E_e X_{pq}$ ,  $X_t = X_{pq} - X_s$ . These are the two generated branches of  $1/2 \otimes 1/2 = 0 \oplus 1$ . Haar averaging commutes with the endpoint gauge actions. With the face orientations retained, the basic conditional identity is

$$\int \chi(AU) \chi(U^{-1}B) dU = \frac{1}{2} \chi(AB). \quad (\text{HNM-236})$$

The two exterior three-link paths use disjoint variables. Their conditional Haar norms give  $\|X_s\|^2 = 1/64$ ,  $\|X_t\|^2 = 3/64$  and zero cross term. Exact quaternion-polynomial integrations check this result; no Monte Carlo or joint independence of all face holonomies is assumed.

The specified *individually resolved* enrichment consists of these actual functions:

| Function                            | Count | Squared norm | Electric energy |
|-------------------------------------|-------|--------------|-----------------|
| $\Omega$                            | 1     | 1            | 0               |
| $\phi_p = 2W_p$                     | 29    | 1            | 3               |
| $\chi_p = 4W_p^2 - 1$               | 29    | 1            | 8               |
| $d_{pq} = 4W_p W_q$ , edge-disjoint | 310   | 1            | 6               |
| $s_{pq} = 8X_s$ , shared edge       | 96    | 1            | $9/2$           |
| $t_{pq} = 8X_t$ , shared edge       | 96    | 3            | $13/2$          |

Every one of the 406 pair parity masks is nonzero, distinct from the others and unequal to a single-face mask. Independent edge-center flips prove the corresponding orthogonalities; spin-one face characters are separately distinguished by their integer representation supports. The two shared branches are orthogonal electric eigenspaces. Thus the actual Gram is diagonal with 465

ones and 96 threes, has zero kernel, and gives quotient rank 561. This is proved from the graph and Haar functions, not assumed from an older example. Smooth electric eigenvectors make the resulting orthogonal  $P_+$  reduce  $K$  and its operator and form domains. The span does not fill complete ambient electric shells, and taking only the cyclic closure of summed columns need not separate its individual degenerate channels.

### 28.3 The complete magnetic matrix and simultaneous outside loading

Assign link-center signs  $+1$  on  $x$  links,  $(-1)^x$  on  $y$  links and  $(-1)^{x+y}$  on  $z$  links. Their product is  $-1$  on every actual face. This Haar-preserving transformation commutes with  $K$  and gauge averaging and sends  $S$  to  $-S$ . Vacuum and the quadratic channels are even; the fundamental faces are odd. It proves all same-parity magnetic entries vanish. The pair-mask independence above is a separate argument.

The complete old columns are

$$\begin{aligned} S\Omega &= \frac{1}{2} \sum_p \phi_p, \\ S\phi_p &= \frac{1}{2}\Omega + \frac{1}{2}\chi_p + \frac{1}{2} \sum_{q: e(p) \cap e(q) = \emptyset} d_{pq} + \frac{1}{4} \sum_{q: |e(p) \cap e(q)|=1} (s_{pq} + t_{pq}). \end{aligned} \quad (\text{HNM-237})$$

Together with the parity zeros and metric self-adjointness  $g_i S_{ij} = g_j S_{ji}$ , these determine every retained magnetic entry. In particular a triplet-to-face coefficient is  $3/4$ , while its face-to-triplet coefficient is  $1/4$ . Blind coordinate transposition would be wrong. The exact sparse export contains 2,124 directed nonzero entries and a defining zero rule for the other 312,597 entries. The full retained generator is its sum with the listed electric diagonal and  $29\lambda I$ .

For  $A_+ = P_+ L P_+$  and  $Q_+ = I - P_+$ , define the actual full outside map

$$B = Q_+ L P_+ = -\lambda Q_+ S P_+, \quad B P_0 = 0, \quad \|B\| \leq 29\lambda. \quad (\text{HNM-238})$$

K reduction and (HNM-237) justify the first two statements. Orthogonal projection in the actual physical Gram and  $\|S\| \leq 29$  give the last bound simultaneously for every complex retained input. Every new column and unretained intertwiner is charged. Equivalently, for each basis function  $b_j$  its outside column is the full polynomial  $-\lambda[Sb_j - \sum_i b_i \langle b_i, Sb_j \rangle / g_i]$ . This specifies all columns without claiming an evaluated cubic outside Gram or exact outside norm.

The map is nonzero for  $\lambda > 0$ : the normalized spin-3/2 same-face character  $\xi_p$  has energy 15 and lies outside  $P_+$ , with

$$\langle \xi_p, B \chi_p \rangle = -\lambda/2. \quad (\text{HNM-239})$$

Other faces have the wrong center mask for this coefficient. Thus  $B P_0 = 0$  does not give autonomous retained dynamics; newly populated channels can load the exterior.

### 28.4 The full original residual and both nested ground comparisons

Put  $m = 29$  and

$$\begin{aligned} w &= \frac{\sqrt{9 + m\lambda^2} - 3}{2}, & h &= \frac{\lambda}{2(3 + w)}, & Z &= 1 + mh^2, \\ f_0 &= \frac{\Omega + h \sum_p \phi_p}{\sqrt{Z}}, & \mu_0 &= m\lambda - w. \end{aligned} \quad (\text{HNM-240})$$

These are the exact ground pair of the complete  $P_0$  star matrix. The dark face eigenvalues are  $3 + m\lambda$ . Positivity, compactness, the electric gap and min-max first establish unique full and compressed grounds with

$$0 \leq \epsilon \leq \mu_+ \leq \mu_0 \leq 29\lambda < 3, \quad E_1(L), E_1(A_+), E_1(A_0) \geq 3. \quad (\text{HNM-241})$$

Thus isolation precedes residual inference. Write their rank-one projections as  $G, G_+, G_0$ .

For  $F = S^2 - m/4$ , complete conditional Haar fourth moments and the pair masks yield

$$\begin{aligned} F &= \frac{1}{4} \sum_p \chi_p + \frac{1}{2} \sum_{\text{disjoint}} d_{pq} + \frac{1}{4} \sum_{\text{shared}} (s_{pq} + t_{pq}), \\ \|F\|^2 &= \frac{m(2m-1)}{16} = \frac{1653}{16}, \quad (L - \mu_0)f_0 = -\frac{2\lambda h}{\sqrt{Z}}F, \\ \rho_0^2 &= \frac{1653\lambda^2 h^2}{4Z}. \end{aligned} \quad (\text{HNM-242})$$

The original omitted-channel Gram has diagonal  $m/4$ , off-diagonal  $1/4$ , and bright/dark eigenvalues  $57/4$  and  $7$ . It is independently recovered from the full resolved matrix. The residual above is the actual full-Hilbert residual; keeping only the spin-one face terms would miss most of it.

For the uniform coupling cap  $\Lambda = 1/100$ , define

$$g = 3 - 29\Lambda = 271/100, \quad \bar{r}_0 = 41\Lambda^2/12, \quad \bar{p}_0 = \bar{r}_0/g. \quad (\text{HNM-243})$$

The inequalities  $h \leq \lambda/6$  and  $\sqrt{1653} < 41$  justify the first residual envelope. There are two distinct spectral arguments:

$$\|G - G_0\| \leq \bar{p}_0, \quad \|G_+ - G_0\| \leq \bar{p}_0. \quad (\text{HNM-244})$$

The first uses the full spectral measure of  $f_0$ . For the second,  $BP_0 = 0$  implies  $Lf_0 \in \text{ran } P_+$ , so the residual in the separately isolated  $A_+$  spectrum is exactly the same full residual. An internal zero residual of the enriched ground would not replace either argument.

For the exact normalized enriched ground  $f_+$ ,  $BG_0 = 0$  then gives

$$\rho_+ = \|(L - \mu_+)f_+\| = \|B(I - G_0)f_+\| \leq 29\Lambda\bar{p}_0 =: \bar{r}_+. \quad (\text{HNM-245})$$

Its full spectral measure and the nonnegative spectral polynomial  $(L - \epsilon)(L - 3)$  give

$$\|G - G_+\| \leq \bar{p}_+ := \bar{r}_+/g, \quad 0 \leq \mu_+ - \epsilon \leq \bar{d}_+ := \bar{r}_+^2/g. \quad (\text{HNM-246})$$

Smoothness supplies the second moment needed for the energy estimate. All barred quantities are upper envelopes, not evaluated actual enriched ground data. In exact fractions,

$$\bar{p}_0 = \frac{41}{325200}, \quad \bar{r}_+ = \frac{1189}{32520000}, \quad \bar{p}_+ = \frac{1189}{88129200}, \quad \bar{d}_+ = \frac{1413721}{2865961584000000}. \quad (\text{HNM-247})$$

## 28.5 True output denominator and independently centered heat

Let  $x$  be any normalized complex vector in  $\text{ran } P_0$  with  $\|x - \Omega\| \leq \eta = 1/100$ . Following the forward constants, put

$$\begin{aligned} z_0 &= 1 - 29\Lambda^2/72, \quad q_0 = (11/12)\Lambda, \quad b = q_0 + \bar{p}_0 + \eta, \\ D &= z_0 - \eta - \bar{p}_0 = \frac{193136341}{195120000} > 0. \end{aligned} \quad (\text{HNM-248})$$

The Ritz overlap and both comparisons in (HNM-244) give  $\|(I - G)x\|, \|(I - G_+)x\| \leq b$ . For the actual full heat,

$$E(\sigma) = e^{-\sigma(L-\epsilon)}, \quad E_+(\sigma) = e^{-\sigma(A_+-\mu_+)}P_+, \quad \|E(\sigma)x\| \geq \|Gx\| \geq D. \quad (\text{HNM-249})$$

The denominator is the true full output, not a Ritz norm or a regularizer. The bounds apply to complex preparations and both centered excited gaps are at least  $g$ .

Strong differentiation on smooth finite retained vectors and vectorwise integration give the exact identity

$$(E - E_+)(\sigma)x = - \int_0^\sigma E(\sigma - s)[B + (\mu_+ - \epsilon)P_+]E_+(s)x ds. \quad (\text{HNM-250})$$

The full  $E$  inside the integral retains outside evolution and return. The plus centering defect survives  $BP_0 = 0$ . No unbounded operator-norm derivative is assumed. Independent ground/excited decompositions give an increasing early bound and decreasing late bound,

$$\begin{aligned} V(\sigma) &= (\bar{r}_+ + \bar{d}_+)\sigma + \frac{29\Lambda b}{g}(1 - e^{-g\sigma}), \\ J(\sigma) &= \bar{p}_+ + (2b + \bar{p}_+)e^{-g\sigma}. \end{aligned} \quad (\text{HNM-251})$$

The extra decaying  $\bar{p}_+$  term is harmless slack. At the predeclared join  $\sigma = 2$ , use  $1 - e^{-2g} \leq 1$  and a positive rational Taylor sum proving  $e^{2g} > 225$ . Monotonicity covers  $[0, 2]$  and  $[2, \infty)$  without a time grid. The forward certificate is

$$\sup_{\sigma \geq 0} \frac{\|(E - E_+)(\sigma)x\|}{\|E(\sigma)x\|} \leq \frac{3063228992521}{1418412601938100} < \frac{11}{5000}. \quad (\text{HNM-252})$$

The skeptic independently reaches the same rational headline. The reverse proof uses the tighter valid  $q_0 = (9/10)\Lambda$  and obtains  $3037671524521/1418412601938100 < 43/20000 = 0.00215$ . These are conservative exact-semigroup bounds, not measured approximation errors; the larger graph's common certificate is weaker than the earlier graph's sharper AC2 certificate.

At zero time the two exact evolutions agree on this initial class. At zero coupling they agree on retained inputs at every time because  $P_+$  reduces  $K$ . Those exact statements supersede positive cap envelopes. Zero-extension has full-input operator norm error one at zero time, so arbitrary full-Hilbert initial accuracy is not claimed. Replacing either actual ground center by  $29\lambda$  or by an independently rounded energy can produce long-time exponential error and is outside the theorem.

Exact controls test the actual gauge words, every internal face, the strict endpoint, both shared branches, the triplet metric, duplicate-coordinate null relations, individual versus summed channels, new-input spin-3/2 leakage, negative-coupling misuse, complex preparations and center-shift sign. The reverse record preserves an initially nondiscriminating orientation fixture and its discriminating replacements; that control refinement adds no investigation. Exact generator actions are executed, but no numerical heat evaluator or numerical trajectory is supplied. Such an implementation would require an added error budget divided by the same true denominator. The outside cubic Gram, complete ambient-shell enrichment, real-time relative control, volume-uniform limit, canonical/homogeneous matching and continuum Yang–Mills mass gap remain open; scientific priority is unverified.

## 29 HNM AH2: complete delayed loading at the unchanged physical clock\*\*

AH2 sharpens the early omission estimate on the same 46-link graph and the same rank-561 physical compression. The two independent directions obtain the exact sufficient bound

$$\sup_{\sigma \geq 0} \frac{\|E(\sigma)x - E_P(\sigma)x\|}{\|E(\sigma)x\|} \leq \frac{18126701281201}{212761890290715000} \simeq 8.519712462 \times 10^{-5} < 10^{-4}. \quad (\text{HNM-253})$$

It holds for every real  $0 \leq \lambda \leq 1/100$  and every normalized complex  $x \in \text{ran } P_0$  with  $\|x - \Omega\| \leq 1/100$ . Here both semigroups use their own actual ground energies. This improves AH1's common sufficient ceiling  $11/5000$  without narrowing the preparation class, changing the graph or altering the clock. It evaluates no actual heat vector or observed approximation error.

## 29.1 Full physical domains and all three complementary sectors

Keep precisely (HNM-234): the 24 original vertex Gauss actions on all 46 original links, all 29 faces, the invariant product-Haar Hilbert space, and  $L = K + \lambda(29 - S)$ . Its operator domain is the invariant product  $H^2$  space, its form domain is invariant  $H^1$ , and smooth invariant Peter–Weyl functions form a core. Bounded real gauge-invariant multiplication preserves these domains and compact resolvent;  $0 \leq 29 - S \leq 58$  implies  $L \geq K$ . The electric vacuum remains unique with full physical gap three. Physical energy and heat time remain  $H = \alpha L$  and  $\sigma = \alpha t/\hbar$ , with all declared reference scales fixed.

Write  $P = P_+$  for the admitted enriched projection and set

$$F = P_0 - |\Omega\rangle\langle\Omega|, \quad N = P - P_0, \quad Q = I - P, \quad A = PLP, \quad B = QLP. \quad (\text{HNM-254})$$

The dimensions of  $F$  and  $N$  are 29 and 531. The full outside  $Q$  is infinite dimensional. In particular,  $NLP_0$  can be nonzero although  $QLP_0 = 0$ . Both finite projections reduce  $K$  and preserve its operator and form domains. Their ranges consist of smooth physical functions, giving the full block domain  $\text{ran } P \oplus QD(K)$  and a legitimate strong-vector Duhamel argument below.

The complete AH1 magnetic action is independently reconstructed from the actual basis, metric adjoints and all old columns: 2,124 nonzero entries and 312,597 structural zeros. The coordinate-dependent link-center transformation makes every new state even and every fundamental face odd. Hence  $NSN = 0$  on the whole new block. Its energies are 8 for the 29 spin-one faces, 6 for the 310 disjoint products,  $9/2$  for the 96 shared singlets and  $13/2$  for the 96 shared triplets; each triplet keeps metric weight three.

Let  $\epsilon, \mu$  be the actual full and retained ground energies, and  $G, G_P$  their rank-one projections. The inherited full variational ordering and separate spectral isolation give

$$0 \leq \epsilon \leq \mu \leq \mu_0 \leq 29\lambda < 3, \quad E_1(L), E_1(A) \geq 3. \quad (\text{HNM-255})$$

Consequently the complete new diagonal block, not a freely chosen damping term, is

$$D_N = N(A - \mu)N = K_N + (29\lambda - \mu)N \geq dN, \quad d = 9/2. \quad (\text{HNM-256})$$

Summing every new row in the physical metric gives

$$(NSF)^*(NSF) = 7I_{29} + \frac{1}{4}J_{29}, \quad C = NAP_0 = -\lambda NSF, \quad C\Omega = 0, \quad \|C\| = \frac{\sqrt{57}}{2}\lambda. \quad (\text{HNM-257})$$

For arbitrary complex  $v$ , the quadratic form is  $7\|v\|^2 + |\sum_p v_p|^2/4$ . Its bright eigenvalue is  $57/4$  and its 28 dark eigenvalues are seven. All 531 new rows contribute positively to the bright quadratic form; deleting any row changes the exact Gram and its unique largest endpoint. This is an all-input norm proof, not an assumption that the trajectory remains in the bright mode. The complete outside identities are still

$$B = -\lambda QSP = BN, \quad BP_0 = 0, \quad \|B\| \leq 29\lambda. \quad (\text{HNM-258})$$

The spin-3/2 same-face witness has coefficient  $-\lambda/2$  from the normalized spin-one input, so  $B \neq 0$  at positive coupling. The envelope covers all new inputs and all outside channels simultaneously; their exact outside Gram is not evaluated.

## 29.2 Stationary loading and the complete evolving face source

Put  $\Lambda = \eta = 1/100$  and retain the common AH1 spectral envelopes

$$\begin{aligned} g &= 3 - 29\Lambda = 271/100, & r_0 &= 41\Lambda^2/12, & p_0 &= r_0/g, \\ r_P &= 29\Lambda p_0, & p_P &= r_P/g, & \delta_P &= r_P^2/g. \end{aligned} \quad (\text{HNM-259})$$

The two separate comparisons  $\|G - G_0\| \leq p_0$  and  $\|G_P - G_0\| \leq p_0$  are both needed. The first follows from the original Ritz vector's full spectral residual. The second uses the same residual because  $Lf_0 \in \text{ran } P$ , with independently established retained isolation. Then  $BG_0 = 0$  bounds the enriched full residual by  $r_P$ , and the full spectral-angle and Temple estimates give

$$\|G - G_P\| \leq p_P, \quad 0 \leq \mu - \epsilon \leq \delta_P. \quad (\text{HNM-260})$$

These symbols denote envelopes, not evaluated projector distances or eigenvectors.

Use the reviewed reverse-AH1 refinement  $q_0 = 9\Lambda/10$ , with that attribution. It follows from the original Ritz parameter and  $\sqrt{29} < 27/5$ . Define

$$q = q_0 + p_0 = \frac{14839}{1626000}, \quad b = q + \eta = \frac{31099}{1626000}. \quad (\text{HNM-261})$$

The operator inequality  $\|FG_P\| \leq \|FG_0\| + \|G_P - G_0\| \leq q$  bounds the actual stationary face component. Both full and retained excited components of every allowed  $x$  have norm at most  $b$ . Therefore the complete retained evolution satisfies

$$u(\sigma) = e^{-\sigma(A-\mu)}x, \quad \|Fu(\sigma)\| \leq q + be^{-g\sigma}. \quad (\text{HNM-262})$$

This spectral decomposition already includes every return through  $N$ . The nondecaying  $q$  term cannot be discarded: at positive coupling a retained ground with no face component would, by the positive new block and the vacuum-to-face equation, have to be the vacuum, which fails the ground equation. The reverse and skeptic checks also give positive actual-model stationary-face lower witnesses.

For  $y = Nu$ , the exact coupled equation and its variation-of-constants identity are

$$\begin{aligned} y' &= -D_N y - CFu, \quad y(0) = 0, \\ y(\sigma) &= - \int_0^\sigma e^{-(\sigma-s)D_N} CFu(s) ds. \end{aligned} \quad (\text{HNM-263})$$

No term is removed from the old-sector equation, including its return term  $-C^*Nu$ . With  $\bar{C} = 19\Lambda/5 > \Lambda\sqrt{57}/2$ , equations (HNM-256)–(HNM-262) imply

$$\|Nu(\sigma)\| \leq Y(\sigma) = \bar{C} \left[ \frac{q}{d}(1 - e^{-d\sigma}) + \frac{b}{d-g}(e^{-g\sigma} - e^{-d\sigma}) \right]. \quad (\text{HNM-264})$$

Both terms are nonnegative convolutions. They account for stationary and excited loading of all 531 new states. The source is the full  $u$ , not a prescribed free face trajectory.

### 29.3 Actual centers, true denominator and a fixed all-time join

Define  $E(\sigma) = e^{-\sigma(L-\epsilon)}$  and  $E_P(\sigma) = e^{-\sigma(A-\mu)}P$ . Since  $u(s)$  stays in a finite smooth full-domain space and  $Lu(s)$  is continuous, strong vector differentiation and integration give

$$(E(\sigma) - E_P(\sigma))x = - \int_0^\sigma E(\sigma-s)[B + (\mu - \epsilon)P]u(s) ds. \quad (\text{HNM-265})$$

The plus sign of the center defect inside the integrand is essential. The full contraction  $E(\sigma-s)$  retains outside propagation and every return path. This is not an unbounded operator-norm differentiation argument.

Using  $Bu = BNu$ ,  $\|u\| \leq 1$  and  $\bar{B} = 29\Lambda$  yields the increasing early error envelope

$$\begin{aligned} V(\sigma) &= \delta_P \sigma + \bar{B}\bar{C} \left\{ \frac{q}{d} \left[ \sigma - \frac{1 - e^{-d\sigma}}{d} \right] \right. \\ &\quad \left. + \frac{b}{d-g} \left[ \frac{1 - e^{-g\sigma}}{g} - \frac{1 - e^{-d\sigma}}{d} \right] \right\}. \end{aligned} \quad (\text{HNM-266})$$

The leakage part is  $\bar{B} \int_0^\sigma Y(s) ds$ , so monotonicity and  $V(0) = 0$  follow from nonnegative kernels. Integrating the transient convolution over the whole half-line gives  $1/(gd)$ . Thus a simpler increasing majorant is

$$V(\sigma) \leq \bar{V}(\sigma) = \delta_P \sigma + \bar{B}\bar{C} \left( \frac{q\sigma}{d} + \frac{b}{gd} \right). \quad (\text{HNM-267})$$

Its positive intercept is conservative slack; the exact error at zero time is zero.

The original Ritz overlap, full projector comparison and complete complex preparation ball give the genuine output floor

$$\|E(\sigma)x\| \geq \|Gx\| \geq D = 1 - 29\Lambda^2/72 - \eta - p_0 = \frac{193136341}{195120000} > 0. \quad (\text{HNM-268})$$

This is the full output, not the retained norm or an arbitrary regularizer. Separately the full and retained spectral tails imply

$$\|(E(\sigma) - E_P(\sigma))x\| \leq J(\sigma) = p_P + (2b + p_P)e^{-g\sigma}, \quad (\text{HNM-269})$$

where the extra decaying  $p_P$  is harmless slack. At the predeclared join  $\sigma = 3$ , positive rational Taylor sums prove  $e^{3g} > 3000$ . The increasing early bound controls  $[0, 3]$  and the decreasing late bound controls  $[3, \infty)$ :

| Certificate endpoint  | Exact bound                       |
|-----------------------|-----------------------------------|
| $\bar{V}(3)$          | 18126701281201/214947118800000000 |
| $J(3)$                | $\leq 34696603/1321938000000$     |
| True output floor $D$ | 193136341/195120000               |

The early endpoint dominates. Dividing it by  $D$  proves (HNM-253). All inequalities use simultaneous cap constants and positive gaps, so they cover the full real coupling interval and all nonnegative real times. The join changes no physical evolution parameter and was not optimized. The independently frozen skeptic uses the slightly sharper rational choice  $\bar{C} = 151\Lambda/40$ , giving the separately attributed sufficient fraction

$$\frac{4501862190047}{53190472572678750} \simeq 8.463662706 \times 10^{-5} < 10^{-4}. \quad (\text{HNM-270})$$

Its stationary-loading, excited-loading and center-defect absolute contributions at the join are respectively about  $6.6605 \times 10^{-5}$ ,  $1.7170 \times 10^{-5}$  and  $1.4798 \times 10^{-9}$ . These are conservative proof budgets, not measured error components; neither certificate is claimed optimal.

## 29.4 Exact fixtures, discriminating controls and limits

The 27 prescribed records combine  $\lambda \in \{0, 1/200, 1/100\}$ ,  $\sigma \in \{0, 1, 3\}$  and three preparations. In addition to  $\Omega$ , the two nonvacuum states are

$$\sqrt{39999/40000} \Omega + \frac{1}{200} \phi_0, \quad \sqrt{39999/40000} \Omega + \frac{i}{200} \phi_0. \quad (\text{HNM-271})$$

They have exact physical norm one and vacuum distance slightly larger than their face amplitude 0.005, but below the theorem radius 0.01. Every record uses the full theorem radius. Rational Taylor and reciprocal enclosures verify scalar certificate functions in (HNM-264)–(HNM-269); no finite matrix exponential or heat vector is computed. Analytic monotonicity and spectral estimates, not these fixtures, establish continuous and all-time coverage.

Controls check every row's Gram contribution, the triplet metric, the distinction between  $N$  and  $Q$ , the nonzero outside spin-3/2 channel, stationary loading, actual centers and the fixed

clock. Feedback deletion changes the actual second derivative by  $C^*CFx$ . For either allowed nonzero face preparation its squared norm is  $(1/200)^2\lambda^4(869/16) > 0$  at positive coupling. The producers preserve their initial nondiscriminating candidates—zero coupling in one direction and vacuum-only second derivative in the other—beside the successful replacements; these are control repairs, not new investigations. A polynomial vanishing at all fixture times explicitly refutes a sampling-only all-time inference.

The forward control also proves  $\epsilon < \mu$  at positive coupling using the connected nonpositive-off-diagonal retained matrix, positive retained ground coordinates and its nonzero outside spin-3/2 coefficient. For  $x = \Omega$  this gives

$$\left. \frac{d}{d\sigma} [\|E(\sigma)x\|^2 - \|E_P(\sigma)x\|^2] \right|_{\sigma=0} = 2(\epsilon - \mu) < 0. \quad (\text{HNM-272})$$

Thus the exact retained norm can exceed the true output norm and cannot replace (HNM-268). This supplies no evaluated enriched ground or quantitative new gap. At zero time, or zero coupling for every time, the exact two heats agree on allowed inputs. Independently rounded centers cannot be exponentiated indefinitely while preserving the theorem.

Both initial freeze inventories omitted the two nested AH1 freeze snapshots. The completed records add exactly those already existing snapshots and the preserved initial draft record. Their reports, checkers, outputs and all initially bound bytes are unchanged; the source inventories and final review verify this disclosed metadata repair. AH2 inherits the reviewed AH1 construction and reverse-origin vacuum-angle refinement; each new checker parses frozen data and independently reconstructs the needed blocks without executing an older checker. No new external primary reading is claimed.

This is one analytic, fixed-graph, same-clock refinement for exact independently centered heat and the whole declared initial ball. It supplies no numerical evaluator, exact outside Gram, arbitrary-full-input or real-time relative theorem, rounded-center robustness, graph-size uniformity, model identification, thermodynamic construction or continuum Yang–Mills result. The remaining early bound is dominated by stationary loading. A smaller sufficient fraction is not an observed improvement in an actual computed trajectory.

### 30 HNM AJ1: the actual homogeneous physical GNS representation\*\*

Draft02 attribution update. See Section 1 for the primary-source comparison and its novelty limitation. HNM labels identify project records.

AJ1 proves local regularity and physical cyclic-space identification for the actual I1 positive-orthant homogeneous omitted-coupling state. The new estimate is a local reference-energy bound,

$$\text{Tr}(\rho_{\Lambda,R}H_{0,R}) \leq 2Mn_{\Lambda}(R) \leq 8M|R| = 56|\tau||R|, \quad M = 7|\tau|. \quad (\text{HNM-273})$$

Finite-region compact resolvent turns this estimate into local trace-norm convergence and normality of the represented local algebras. These properties justify continuous endpoint gauge implementation and compatible Haar averaging. The full bounded local invariant algebra generates exactly the joint fixed-vector space, which reduces the actual closed, centered I1 generator. The spectral threshold  $\alpha/16$  and symbolic small-coupling interval are inherited; no nonzero physical excitation or numerical nonzero coupling interval is constructed here.

### 30.1 The complete I1 model and the inherited limit topology

A coarse site  $b = (i, j, k) \in \mathbb{Z}_+^3$  owns every positive-direction original link whose tail lies in

$$T_b = \{(4i + r, 2j + s, k) : 0 \leq r < 4, 0 \leq s < 2\}. \quad (\text{HNM-274})$$

Euclidean division gives unique ownership. Its 24 Haar link factors contain the complete ten-link selected three-face strip and fourteen free links. Link heads can lie outside this tail-owned region and retain their original gauge actions. The tensor factors  $\mathcal{K}_b = L^2(\text{SU}(2)^{24})$  are the unreduced spaces; a tensor factorization of the Gauss-constrained space is not assumed.

The fixed selected strip coefficients obey  $|\lambda_L|, |\lambda_R| \leq \alpha/2$  and  $|\mu| \leq \alpha/8$ . With  $\delta = \alpha/8$ , let

$$h_b = \delta^{-1} \left( H_{\text{strip},b} - E_{\text{strip},b} + \alpha \sum_{e \text{ mfree at } b} C_e \right) \geq I - P_b, \quad h_b \Omega_b^0 = 0. \quad (\text{HNM-275})$$

The onsite vacuum is unique. On a finite coarse cuboid  $\Lambda$ , the actual normalized Hamiltonian is

$$\begin{aligned} \hat{H}_\Lambda &= H_{0,\Lambda} + \sum_{b+S \subset \Lambda} \phi_b, & H_{0,\Lambda} &= \sum_{b \in \Lambda} h_b, \\ S &= \{0, e_x, e_y, e_z\}, & \phi_b &= -\frac{\tau}{3} \sum_{f \in O_b} W_f, & |O_b| &= 21, & \|\phi_b\| &= M. \end{aligned} \quad (\text{HNM-276})$$

All 21 omitted faces are retained only when their *whole* four-site star is contained. Their actual two- and three-site support multiplicities are 1, 3, 1, 10, 2, 4 for  $\{0, e_x\}$ ,  $\{0, e_y\}$ ,  $\{0, e_x, e_y\}$ ,  $\{0, e_z\}$ ,  $\{0, e_x, e_z\}$  and  $\{0, e_y, e_z\}$ . The upper norm bound uses  $\|W_f\| = 1$ ; positive-Haar-measure neighborhoods of the identity attain its essential supremum. Thus a finite same-sign sum of  $N$  complete anchors has norm  $NM$ , without a bounded global homogeneous perturbation as  $N \rightarrow \infty$ .

Only the inherited range

$$|\tau| < \tau_* := \frac{\min\{c_1(S), 1/[2c_2(S)]\}}{7} \quad (\text{HNM-277})$$

is used. Its positive source constants remain unevaluated. I1 already supplies unique finite grounds  $\Omega_\Lambda$ , the specified whole-star orthant state  $\omega$ , its GNS triple  $(\pi, \mathcal{H}_\omega, \Omega)$ , and a nonnegative centered self-adjoint  $G$  with unique ground and spectrum in  $\{0\} \cup [1/2, \infty)$ . The source topology consists of convergence of each bounded local expectation and tested resolvent matrix elements:

$$\begin{aligned} \langle \Omega_\Lambda, A \Omega_\Lambda \rangle &\longrightarrow \omega(A), & A &\in \bigcup_{R \text{ mfinite}} B(\mathcal{K}_R), \\ \langle R_\Lambda(z) A \Omega_\Lambda, B \Omega_\Lambda \rangle &\longrightarrow \langle (G - z)^{-1} \pi(A) \Omega, \pi(B) \Omega \rangle, \\ R_\Lambda(z) &= (\hat{H}_\Lambda - E_\Lambda - z)^{-1}, & z &\notin \mathbb{R}. \end{aligned} \quad (\text{HNM-278})$$

Here  $A, B$  are fixed bounded local operators. This is not a same-Hilbert-space strong resolvent limit in J2's summable representation.

The orthant transfer keeps independent reference spectators at negative sites of  $\mathbb{Z}^3$  and sets outside anchor interactions to zero. Nonnegative star offsets prevent coupling to those spectators. Finite grounds factor, and finite centered resolvents on spectator-vacuum vectors equal the orthant resolvents. The local-normality argument below supplies factorization of the actual local limiting state and density of represented product excitation vectors, including for the full bounded local factor algebra. Vanishing off-sector matrix elements and the tested limit then identify a reducing orthant restriction. This is the specified boundary prescription; equality with every other thermodynamic boundary limit is not claimed.

Yarotsky's source creation core uses  $u_I \in \mathcal{H}'_I \cap D(H_{0,I})$  and  $\hat{u}_I = |u_I\rangle\langle\Omega_I^0|$ . The domain condition makes  $[H_0, \hat{u}_I] = |H_{0,I}u_I\rangle\langle\Omega_I^0|$  bounded, while only finitely many bounded star commutators meet  $I$ . Density and essential self-adjointness are inherited from the source expansion, not re-proved here. An arbitrary bounded local  $A$  need not put  $\pi(A)\Omega$  in the operator or form domain of  $G$ . The producer proofs below use the tested closed-resolvent limit to avoid that shortcut; the skeptic separately checks covariance on the domain-qualified source core and its closure.

### 30.2 A domain-valid local reset and complete incident-star counting

The 24-link electric sum has compact resolvent: below a fixed energy only finitely many Peter-Weyl labels occur, each with finite multiplicity. Bounded selected Wilson terms and the exact ground subtraction preserve its full product  $H^2$  operator domain,  $H^1$  form domain, and compact resolvent. The same holds for each finite-region sum  $H_{0,R}$ .

The reverse and skeptic additionally use the strip potential bound  $9\alpha/8$  and  $-9\alpha/8 \leq E_{\text{strip},b} \leq 0$  to obtain

$$\left\| h_b - 8 \sum_e C_e \right\| \leq 18, \quad h_b \geq 8 \sum_e C_e - 9. \quad (\text{HNM-279})$$

For  $Q_{R,T} = \mathbf{1}_{[0,T]}(H_{0,R})$ , the reverse's explicit counting bound is

$$\text{rank } Q_{R,T} \leq \left[ \sum_{k=0}^{\lfloor 4B_R/3 \rfloor} (k+1)^2 \right]^{24|R|}, \quad B_R = (T + 9|R|)/8. \quad (\text{HNM-280})$$

It uses  $k(k+2) \geq 3k$  for integer  $k = 2j \geq 0$ . This is a loose spectral-counting comparison, not equality of interacting and bare-electric spectral projections or an evaluated interacting spectrum.

Let  $\rho_\Lambda = |\Omega_\Lambda\rangle\langle\Omega_\Lambda|$  and reset only a finite region  $R \subset \Lambda$  to its reference vacuum:

$$\rho'_\Lambda = |\Omega_R^0\rangle\langle\Omega_R^0| \otimes \text{Tr}_R \rho_\Lambda, \quad \Omega_R^0 = \bigotimes_{b \in R} \Omega_b^0. \quad (\text{HNM-281})$$

The original finite ground has finite reference energy by bounded perturbation. For the reset density, bounded positive spectral truncations and the partial-trace identity preserve exterior reference energy; monotone convergence then justifies the unbounded statement. Its regional reference energy is zero. The reset therefore has finite form energy and is an admissible mixed trial density. It is unnecessary to place every vector in that density's support in the operator domain.

All relevant incident anchors are

$$I_\Lambda(R) = \{b : b + S \subset \Lambda, (b + S) \cap R \neq \emptyset\}, \quad n_\Lambda(R) = |I_\Lambda(R)|. \quad (\text{HNM-282})$$

Every anchor is  $r - s$  for some  $r \in R, s \in S$ . Thus, including incoming and outgoing stars,

$$n_\Lambda(R) \leq n_+(R) := |(R - S) \cap \mathbb{Z}_+^3| \leq 4|R|. \quad (\text{HNM-283})$$

Apply the finite variational inequality to (HNM-281). Exact scalar energies, exterior reference terms and disjoint-star expectations cancel:

$$\begin{aligned} 0 &\leq \text{Tr}(\rho'_\Lambda \hat{H}_\Lambda) - E_\Lambda \\ &= -\text{Tr}(\rho_\Lambda H_{0,R}) + \sum_{b \in I_\Lambda(R)} \text{Tr}[(\rho'_\Lambda - \rho_\Lambda)\phi_b]. \end{aligned} \quad (\text{HNM-284})$$

Each difference costs at most  $2M$  in absolute value, proving (HNM-273) for every finite region and volume. The sum on the right is nonnegative for the actual ground; positivity of each interaction difference is not assumed. The reverse retains the sharper regional constant  $C_R = 2Mn_+(R)$ , while the common forward/skeptic headline uses  $C_R \leq 8M|R|$ . No extensive global bound or numerical source threshold is substituted.

### 30.3 Local trace-norm convergence and normality of the representation

For  $T > 0$ , the finite-rank cutoffs  $Q = Q_{R,T}$  increase strongly to the identity. Spectral order gives

$$\mathrm{Tr}[\rho_{\Lambda,R}(I - Q)] \leq C_R/T =: \varepsilon_T. \quad (\text{HNM-285})$$

Finiteness of  $C_R$ , rather than its smallness, is enough. For a trace-one density  $\rho$  with  $e = \mathrm{Tr}[\rho(I - Q)]$ , the forward and skeptic use

$$\|\rho - Q\rho Q\|_1 \leq 2\sqrt{e}. \quad (\text{HNM-286})$$

Indeed  $\rho - Q\rho Q = (I - Q)\rho + Q\rho(I - Q)$  and Hilbert–Schmidt factorization bounds both trace norms by  $\sqrt{e}$ . The reverse separately proves the valid slightly looser  $2\sqrt{e} + e$  by splitting the two off-diagonal blocks and omitted positive diagonal. These different estimates lead to the same local conclusion.

For each fixed finite-rank  $Q$ , the matrix entries of  $Q\rho_{\Lambda,R}Q$  converge by the inherited bounded-local state limit. Uniform tails make the densities trace-norm precompact, and every cluster density represents that same limiting functional. Hence

$$\|\rho_{\Lambda,R} - \rho_R\|_1 \rightarrow 0, \quad \omega(A) = \mathrm{Tr}(\rho_R A), \quad \mathrm{Tr}(\rho_R H_{0,R}) \leq C_R. \quad (\text{HNM-287})$$

Equivalently the forward finite-block argument gives a limiting trace-distance upper bound  $4\sqrt{\varepsilon_T}$  before  $T \rightarrow \infty$ . This concerns every fixed finite region, including unions of supports. It supplies no volume rate or global infinite-volume trace-norm convergence. At  $\tau = 0$  the reset estimate forces the local reference vacuum.

Normality of the actual state is not yet the whole averaging premise. For bounded local  $B$  in a region  $K$ , the positive vector functional

$$A \in B(\mathcal{K}_R) \mapsto \langle \pi(B)\Omega, \pi(A)\pi(B)\Omega \rangle = \omega_{R \cup K}(B^*AB) \quad (\text{HNM-288})$$

is normal by (HNM-287) on the union region. Such cyclic vectors are dense. Their vector functionals approximate those of arbitrary vectors in norm; normal functionals form a norm-closed predual. Polarization therefore proves that  $\pi$  is normal on each original local factor. In particular, bounded local strong-\* limits and the required local weak-operator averages can be tested through this representation. No normality on a global infinite tensor-product von Neumann algebra is claimed.

The final skeptical review makes the spectator-extension density step explicit. The same reset and compactness proof applies there, with zero interaction cost for spectator sites. The normal finite-region limit is the positive-region density tensored with the pure spectator vacuum. Finite-rank spectator projections converge boundedly strongly to the identity. Normality therefore allows any bounded local operator applied to that vacuum to be approximated in GNS vector norm by finitely many spectator columns, each a bounded positive-region operator times a finite-rank spectator creation. These represented product excitation vectors are dense even though the algebraic tensor product is not asserted norm dense in  $B(\mathcal{K}_+ \otimes \mathcal{K}_-)$ . There is no use of physical cyclicity in this step.

Projection onto the spectator vacuum preserves the source’s domain-qualified creation core: spectator excitations are killed, while positive-region creations survive. It commutes with the decoupled reference and retained interaction there. Essential self-adjointness and graph closure give a reducing orthant restriction and its projected core; equivalently, off-sector tested resolvent elements vanish on the dense product vectors just justified. This completes the source-correct orthant transfer used in (HNM-278).

### 30.4 Every endpoint, continuous gauge implementation and compatible averages

An original link has the full endpoint action  $U_{(v,w)} \mapsto g_v U_{(v,w)} g_w^{-1}$ . Finite products of left/right translations are strongly continuous on the original link spaces, first on the Peter-Weyl span and then by density. They preserve every Casimir domain, selected face and omitted closed Wilson word, so they commute with each finite Hamiltonian. All external head endpoints are retained.

Uniqueness makes a finite ground transform by a continuous one-dimensional character of a finite product of  $SU(2)$ . Such a character is trivial: an  $SU(2)$  element is conjugate to its inverse, and connectedness forces the resulting value in  $\{1, -1\}$  to equal one. Thus the finite ground vectors and onsite vacua are fixed. Passing their expectations to the limit proves invariance of  $\omega$  under each local automorphism  $\beta_g$ . The prescription

$$U_\omega(g)\pi(A)\Omega = \pi(\beta_g(A))\Omega \quad (\text{HNM-289})$$

respects the GNS null ideal, preserves inner products, has inverse  $g^{-1}$  and fixes  $\Omega$ . For local  $A$ , its original orbit is bounded and strong-\* continuous within the same owned-link factor. Local normality yields

$$\| [U_\omega(g) - U_\omega(g_0)]\pi(A)\Omega \|^2 = \omega((\beta_g A - \beta_{g_0} A)^*(\beta_g A - \beta_{g_0} A)) \longrightarrow 0. \quad (\text{HNM-290})$$

Density extends this to the whole GNS space for every finite endpoint group. This is strong unitary continuity, not point-norm continuity of conjugation on every bounded local operator.

Let  $V(R)$  contain all endpoints of the  $24|R|$  owned links. In the original local factor define the weak-operator Haar average

$$\mathbb{E}_R(A) = \int_{SU(2)^{V(R)}} \beta_g(A) dg. \quad (\text{HNM-291})$$

It is a unital positive contraction, stays in  $B(\mathcal{K}_R)$  and is invariant under all local gauge transformations: outside endpoints act trivially. The averages are compatible with support enlargement. To compare it with the represented vector average, test against any  $\pi(B)\Omega$ . The normal functional  $X \mapsto \omega(B^*X)$  on the union region passes through the local weak-operator integral. Density and strong continuity then give

$$\pi(\mathbb{E}_R(A))\Omega = \int_{SU(2)^{V(R)}} U_\omega(g)\pi(A)\Omega dg. \quad (\text{HNM-292})$$

This proves the compatibility actually needed. An arbitrary GNS representation would not justify the interchange.

### 30.5 The physical null quotient and reducing closed generator

Let  $\mathcal{A}_{\text{phys}}$  be the norm closure of the *full bounded local invariant algebra*, and define

$$\begin{aligned} \mathcal{H}_{\text{cyc}} &= \overline{\pi(\mathcal{A}_{\text{phys}})\Omega}, \\ \mathcal{H}_{\text{fix}} &= \{\xi : U_\omega(g)\xi = \xi \text{ for every finite local } g\}. \end{aligned} \quad (\text{HNM-293})$$

Equation (HNM-289) gives the first inclusion. Conversely, approximate a fixed  $\xi$  by a local  $\pi(A)\Omega$ . The finite endpoint Haar projection fixes  $\xi$  and contracts the error, so (HNM-292) gives

$$\|\pi(\mathbb{E}_R(A))\Omega - \xi\| \leq \|\pi(A)\Omega - \xi\|, \quad \mathcal{H}_{\text{cyc}} = \mathcal{H}_{\text{fix}}. \quad (\text{HNM-294})$$

Each approximant uses only a finite compact endpoint product. No infinite-group Haar construction or Wilson-only density premise enters. For  $\mathcal{N}_{\text{phys}} = \{A \in \mathcal{A}_{\text{phys}} : \omega(A^*A) = 0\}$ , the

map  $[A] \mapsto \pi(A)\Omega$  is an isometry of the physical GNS quotient, with inner product  $\omega(A^*B)$ , onto a dense subspace of (HNM-294). Its completion is the desired intertwining unitary; state faithfulness is unnecessary.

Finite centered resolvents commute with the original endpoint actions and their grounds are fixed. Applying (HNM-278) to  $\beta_g(A), \beta_g(B)$  and then using density gives the actual bounded-resolvent identity

$$U_\omega(g)^*(G - z)^{-1}U_\omega(g) = (G - z)^{-1}, \quad z \notin \mathbb{R}. \quad (\text{HNM-295})$$

Both the resolvent and its adjoint preserve the joint fixed space. Hence that space reduces every spectral projection of the closed  $G$ , not merely a formal local commutator. The physical generator has the exact domains

$$\begin{aligned} G_{\text{phys}} &= G|_{\mathcal{H}_{\text{cyc}}}, \\ D(G_{\text{phys}}) &= D(G) \cap \mathcal{H}_{\text{cyc}}, \quad D(G_{\text{phys}}^{1/2}) = D(G^{1/2}) \cap \mathcal{H}_{\text{cyc}}. \end{aligned} \quad (\text{HNM-296})$$

It is a self-adjoint restriction with the same unique centered vacuum. In fixed physical units,

$$H_{\text{phys}} = \frac{\alpha}{8} G_{\text{phys}}, \quad \text{spec}(H_{\text{phys}}) \subset \{0\} \cup [\alpha/16, \infty), \quad \text{frequency generator} = H_{\text{phys}}/\hbar. \quad (\text{HNM-297})$$

The inherited form inequality restricts to the stated form domain. A vacuum-only physical space would satisfy this spectral exclusion as well, so the argument does not yet prove nonzero homogeneous Wilson variance or even a nonzero physical excitation. The threshold is inherited from I1, not a measured mass or a new numerical stability interval.

### 30.6 Three complete geometry fixtures, provenance and limits

The independently written checks enumerate original links, all endpoints, complete selected/free splits, every omitted face word and full-star retention. The declared cuboids give:

| Coarse sides | Sites | Links | Endpoints | Stars | Omitted faces |
|--------------|-------|-------|-----------|-------|---------------|
| (2, 2, 2)    | 8     | 192   | 120       | 1     | 21            |
| (3, 2, 2)    | 12    | 288   | 176       | 2     | 42            |
| (3, 3, 3)    | 27    | 648   | 342       | 8     | 168           |

The boundary singleton, adjacent pair and interior singleton have orthant incidence counts 1, 2, 4, giving reverse regional coefficients  $C_R/M = 2, 4, 8$ . In the two smaller fixtures the nominal interior singleton has no retained incident star; the all-volume count is not silently substituted for that finite-boundary value. In the largest fixture it receives three incoming stars as well as its own. Whole-star counting charges 84 group faces there, although only 49 individual faces actually touch the singleton. Individually contained omitted-face counts in the three cuboids are instead 70, 110, 336, a different boundary Hamiltonian.

Rational quaternion tests retain all head and tail actions. Finite  $Q_8$  averages verify specified fundamental first moments and closed traces, not full  $\text{SU}(2)$  Haar averaging of arbitrary operators. Entangled reset fixtures check energy cancellation; the all-volume variational and unbounded-form argument proves (HNM-273). Separate analytic countermodels show that weak-star limits of normal states can be singular when energy escapes; strong unitary continuity need not give point-norm conjugation continuity; a weak-operator average need not pass through an arbitrary representation; a bounded rank-one local operator can create a vector outside the generator domain; and a nonreducing compression cannot replace (HNM-295). Their finite diagnostics do not prove the infinite statements by sampling.

The reverse preserves a missing-head control whose first gauge element was the identity, together with its discriminating replacement. Its later administrative portability repair archives six

original files and changes only the checker's source-binding function to use owned instruction snapshots. The mathematical report and geometry bytes are unchanged; result fields outside binding metadata and all scientific checker logic are unchanged. The active freeze discloses post-exchange access and the earlier independently frozen derivation. This is an administrative repair, not a new independent proof or investigation.

The reset route was an advisor-suggested prospective method shared before the contract; the three derivations independently establish its application. They inherit the external cluster-expansion theorem and its specified core, not a fresh full proof audit. AJ1 supplies local regularity, compatible gauge averaging and the actual homogeneous physical restriction. It establishes no selected positive numerical  $\tau$ , volume convergence rate, Wilson-only algebra equality, nonzero Wilson fluctuation, boundary-state equivalence, bounded global dressing unitary, summable-model identification, continuum construction or new mass-gap solution.

### 31 HNM AJ2: a nonzero Wilson fluctuation in the homogeneous state<sup>\*\*</sup>

Draft02 attribution update. See Section 1 for the primary-source comparison and its novelty limitation. HNM labels identify project records.

AJ2 proves that one specified original Wilson multiplication observable has strictly positive variance in the actual I1/AJ1 homogeneous state. Its centered vector is a nonzero physical vector orthogonal to the vacuum, so AJ1's positive spectral sector is now shown to be nonempty. For every finite physical imaginary time,

$$0 < C(t) \leq v \exp\left(-\frac{\alpha t}{16\hbar}\right), \quad 0 \leq t < \infty, \quad v = \omega(W^2) - \omega(W)^2 > 0. \quad (\text{HNM-298})$$

The variance is not numerically evaluated. The proof uses the actual multiplier's Haar-null point levels and AJ1's normal local density, without assuming faithfulness or an interacting Haar law. The decay ceiling uses the inherited physical spectral exclusion; it identifies no mass or threshold eigenvalue.

#### 31.1 The actual face, complete factors and all endpoint actions

Keep the same positive-orthant homogeneous omitted interaction (HNM-276), fixed selected-strip reference, whole-star finite boundaries, and symbolic range (HNM-277). Zero and either sign of  $\tau$  are included; no chosen positive numerical coupling is certified. Every site still owns 24 original links, and every retained anchor carries all 21 omitted faces. AJ1 supplies the actual normal local densities, full physical bounded-local algebra, physical cyclic/fixed identification and closed centered restriction

$$H_{\text{phys}} = \frac{\alpha}{8} G_{\text{phys}}, \quad \ker H_{\text{phys}} = \mathbb{C}\Omega, \quad \text{spec}(H_{\text{phys}}) \subset \{0\} \cup [\Delta, \infty), \quad \Delta = \alpha/16. \quad (\text{HNM-299})$$

These premises refer to the actual homogeneous state, not the dyadic state or a finite AH graph. The spacing,  $\alpha$ ,  $E_*$  and  $\hbar$  retain their declared positive physical meanings.

Let  $o = (0, 0, 0)$ ,  $x = (1, 0, 0)$  and  $z = (0, 0, 1)$  denote original lattice vertices. The ordered omitted  $xz$  face is  $o \rightarrow x \rightarrow x + z \rightarrow z \rightarrow o$ , with

$$U_{\partial} = U_x(o)U_z(x)U_x(z)^{-1}U_z(o)^{-1}, \quad W = \frac{1}{2} \text{Tr } U_{\partial}. \quad (\text{HNM-300})$$

Its four distinct stored positive links have respective coarse owners  $o, o, z, o$  under the tail map  $(a_x, a_y, a_z) \mapsto (\lfloor a_x/4 \rfloor, \lfloor a_y/2 \rfloor, a_z)$ . Thus its minimal complete factor region is  $R = \{o, z\}$ ,

containing **48 original links**, twenty selected-strip links and twenty-eight free links. The 16 tails have coordinates  $0 \leq i \leq 3$ ,  $0 \leq j \leq 1$ ,  $0 \leq k \leq 1$ . Add four  $x$  heads at  $(4, j, k)$ , eight  $y$  heads at  $(i, 2, k)$  and eight  $z$  heads at  $(i, j, 2)$  to obtain all **36 endpoint vertices**.

The four loop vertices suffice to see this particular trace's cancellation, but are not the endpoint set for an arbitrary operator on the complete local factor. Under every original action  $U_{(v,w)} \mapsto g_v U_{(v,w)} g_w^{-1}$ , the ordered word telescopes to  $g_o U_{\partial} g_o^{-1}$ . Therefore  $W$  is in the full bounded local physical algebra. It is a real self-adjoint multiplier, with  $|W| \leq 1$ . The two-site observable support is distinct from the four-site Hamiltonian anchor star; no two-site Hamiltonian with that whole star retained is inferred.

### 31.2 Conditional Haar integration proves every point level is null

Work on the original unreduced local multiplication space

$$\mathcal{K}_R = L^2(X, \mu_H), \quad X = \text{SU}(2)^{48}, \quad (\text{HNM-301})$$

with normalized product Haar measure. Fix the other 47 original link variables. The reverse and skeptic integrate  $U_x(o)$ , for which  $U_{\partial} = U_x(o)A$  with fixed  $A \in \text{SU}(2)$ . Right translation preserves Haar. The forward instead integrates  $U_z(o)$ , giving  $U_{\partial} = AU_z(o)^{-1}$ ; inversion and left translation give the same conclusion. These are independent original-link coordinates, not independent plaquette holonomies. The other 44 non-face links are normalized spectators.

For a Haar unit quaternion, the half trace is its scalar coordinate  $s = \cos \theta$ . The normalized angular density  $(2/\pi) \sin^2 \theta d\theta$  on  $0 \leq \theta \leq \pi$  yields

$$d\nu_H(s) = \frac{2}{\pi} \sqrt{1-s^2} \mathbf{1}_{[-1,1]}(s) ds. \quad (\text{HNM-302})$$

Its integral is one. Every singleton has measure zero, including zero and both endpoints; outside  $[-1, 1]$  the level is empty. Conditional integration and Fubini give, for every Borel set  $B$  and every real  $c$ ,

$$\mu_H\{W \in B\} = \nu_H(B), \quad \mu_H\{W = c\} = 0. \quad (\text{HNM-303})$$

The original multiplication spectral projection is  $E_W(B) = M_{\mathbf{1}_{\{W \in B\}}}$ . It is zero exactly when  $\nu_H(B) = 0$ , hence

$$E_W(\{c\}) = 0, \quad \ker(W - cI) = \{0\} \quad (c \in \mathbb{R}). \quad (\text{HNM-304})$$

Every interior open interval has positive Haar measure. Consequently the multiplier has spectrum  $[-1, 1]$ , norm one and no point eigenprojection, including at its endpoints. These are original-operator statements. They do not say that the interacting state has the Haar probability law or assigns positive weight to every such interval.

### 31.3 Strict actual variance without faithfulness

AJ1 supplies a positive trace-class density  $\rho_R$  of trace one for the actual restriction of  $\omega$  to  $B(\mathcal{K}_R)$ . Put  $m = \omega(W) \in \mathbb{R}$ . Boundedness makes the moments finite, and

$$v = \text{Tr}[\rho_R(W - mI)^2] = \left\| (W - mI) \rho_R^{1/2} \right\|_{\text{HS}}^2. \quad (\text{HNM-305})$$

If  $v = 0$ , the range of  $\rho_R^{1/2}$  would lie in the zero kernel (HNM-304). This would force  $\rho_R = 0$ , contradicting its trace. Thus  $v > 0$  for every actual state throughout the inherited symbolic interval, without narrowing that interval by a perturbative continuity estimate.

An equivalent configuration argument writes  $\rho_R = \sum_j p_j |\psi_j\rangle\langle\psi_j|$ . Tonelli gives  $r(q) = \sum_j p_j |\psi_j(q)|^2 \geq 0$  with  $\int r d\mu_H = 1$ , and

$$v = \int_X (W(q) - m)^2 r(q) d\mu_H(q). \quad (\text{HNM-306})$$

Zero variance would concentrate all this integrable mass on a Haar-null level. This construction uses the density's spectral decomposition, not an assumed pointwise diagonal kernel. The density may be rank one or vanish on large subspaces. Neither faithfulness, global trace-class representability nor an interacting density bounded below is required. The actual Wilson distribution is absolutely continuous with respect to (HNM-302); converse absolute continuity and full support are not asserted.

Since  $W - mI$  is bounded local and invariant, its physical GNS class is nonzero:

$$\chi = (\pi(W) - mI)\Omega \in \mathcal{H}_{\text{cyc}}, \quad \langle \Omega, \chi \rangle = 0, \quad \|\chi\|^2 = v > 0. \quad (\text{HNM-307})$$

It is precisely the image of  $[W - mI]$  under AJ1's quotient isometry. The physical space therefore contains the vacuum and a nonzero orthogonal vector. No operator-domain assertion is needed for this nonvacuity statement.

### 31.4 A nonzero physical spectral measure and positive heat correlation

For the actual self-adjoint restriction (HNM-299), define

$$\nu_\chi(B) = \langle \chi, E_{H_{\text{phys}}}(B)\chi \rangle. \quad (\text{HNM-308})$$

This positive finite measure has total mass  $v$ , no atom at zero by ground uniqueness and centering, and support in  $[\Delta, \infty)$ . In particular the actual positive-energy physical sector is nonempty. At every finite real  $t \geq 0$ , bounded spectral calculus gives

$$C(t) = \langle \chi, e^{-tH_{\text{phys}}/\hbar} \chi \rangle = \int_{[\Delta, \infty)} e^{-tE/\hbar} d\nu_\chi(E). \quad (\text{HNM-309})$$

Some finite interval  $[\Delta, L]$  has positive measure, so its strictly positive integrand proves  $C(t) > 0$ . On the entire support the integrand is at most  $e^{-t\Delta/\hbar}$ , proving (HNM-298). Also  $C(0) = v$ , bounded dominated convergence gives continuity at zero, and the upper bound gives  $C(t) \rightarrow 0$  at large imaginary time. No universal lower decay estimate follows.

The actual mean is retained. The uncentered correlation would be

$$\langle \pi(W)\Omega, e^{-tH_{\text{phys}}/\hbar} \pi(W)\Omega \rangle = m^2 + C(t). \quad (\text{HNM-310})$$

There is no assumption that the interacting  $m$  vanishes. The frequency scale is  $\alpha/(16\hbar)$ ; the dimensionless normalized threshold  $1/2$  cannot be inserted as a physical frequency.

The bounded semigroup is defined on every Hilbert vector, so neither  $\chi \in D(H_{\text{phys}})$  nor  $\chi \in D(H_{\text{phys}}^{1/2})$  is used or proved. Those would require respectively finite second or first energy moments. No derivative at zero is inferred from locality. The support lower bound need not be attained, carry an atom at  $\alpha/16$ , or equal the lowest overlapping energy. Real-time phase evolution gives no magnitude-decay theorem. AJ2 identifies a nonzero positive measure, not its energies, a normalizable excited eigenvector, a sharp asymptotic rate or a measured mass.

### 31.5 Reference diagnostics, countermodels and scope

The reference Haar moments

$$\mathbb{E}_H W = 0, \quad \mathbb{E}_H W^2 = 1/4, \quad \mathbb{E}_H W^4 = 1/8 \quad (\text{HNM-311})$$

check normalization and conditional integration. They are not assigned to the interacting state. A rank-one Haar density is normal but nonfaithful and still has positive variance. Another normalized normal vector,  $2W$ , instead has variance  $1/2$ . Thus normality does not fix the moment values.

More strongly, the normal rank-one states supported in  $|W| < \varepsilon$  have positive variance at most  $\varepsilon^2$ , while the band has positive Haar measure. Their variances can approach zero. A weak-star cluster subnet can have zero variance and must be singular by (HNM-305). This disproves a positive uniform variance floor over *all normal states*; it is not a failure theorem for a possible future quantitative bound on the particular ground-state family. Nonconstancy alone also fails: a step multiplier has a positive-measure constant region and can have zero variance in a normal state supported there. Haar-null levels are the decisive premise.

Other explicit countermodels separate a vacuum-only restricted gap from a nonzero excitation, an upper heat ceiling from a lower decay law, a support endpoint from a threshold atom, and centered heat from a vacuum plateau. A positive-energy atom has constant real-time correlation modulus. Bounded self-adjoint rank-two observables can create vectors with divergent form or operator energy moments; those abstract examples do not assert divergence for the actual Wilson vector.

The exact checkers reconstruct all 48 links, 36 endpoints, coarse owners and the oriented four-link word. Rational quaternion transformations verify full head/tail covariance and charged or missing-head controls. They also check the conditional coefficient normalization and reference moments, with all attempted controls retained. These finite diagnostics do not prove the Haar-null or infinite-state claims by sampling; conditional integration, trace-class normality, the GNS quotient and the spectral theorem supply those arguments.

The advisor supplied the observable and proposed the level-set route before the three independent derivations. I1/AJ1, the immutable seven-loop draft and network are inherited premises; no new external theorem or historical reading is claimed. The original independent skeptic froze its derivation before exchange and subsequently reviewed both producers and replayed their outputs. After a reviewer handoff, another model-agent reviewer read that preserved evidence and independently audited the final admission specification and source closure. The closeout did not replace the original derivation or producer replays, add another pre-exchange proof, or execute another research loop.

This investigation proves qualitative nonvacuity and (HNM-298) in the actual specified homogeneous state. It supplies no numerical variance margin, evaluated stability radius, energy-moment result, global faithfulness, Wilson-only algebra completion, alternative boundary-state equality, physical model matching, lattice-spacing limit or continuum Yang–Mills construction. Scientific priority and a percentage of continuum-theorem completion are not established.

## 32 HNM AK1: a quantitative actual Wilson variance floor<sup>\*\*</sup>

Draft02 attribution update. See Section 1 for the primary-source comparison and its novelty limitation. HNM labels identify project records.

For the same original origin  $xz$  Wilson multiplier (HNM-300) in the actual I1/AJ1/AJ2 homogeneous state, both independent producers prove

$$\text{Var}_\omega(W) \geq \frac{119}{576} > \frac{1}{5}, \quad |\tau| < \tau_*, \quad |\tau| \leq 2^{-16}. \quad (\text{HNM-312})$$

The independently frozen skeptic proof gives the stronger, separately attributed lower bound  $131/576$  on the same regime. These are sufficient dimensionless signal bounds, not evaluated interacting variances or optimal constants. The additional numerical cap does not evaluate the

positive source constants in  $\tau_* = \min\{c_1(S), 1/[2c_2(S)]\}/7$ , or certify that any chosen positive numerical coupling lies below it. AJ2's qualitative positivity on the larger symbolic interval remains unchanged.

Keep  $|\lambda_L|, |\lambda_R| \leq \alpha/2$  and  $|\mu| \leq \alpha/8$  fixed in their inherited ranges. Both bounds are uniform over those choices, with original spacing,  $\alpha$ ,  $E_*$  and  $\hbar$  positive and fixed. The actual local density is generally mixed. The energy estimate used below concerns its normalized *reference* energy, not the physical spectral first moment of a Wilson-excited vector.

### 32.1 Complete region, incident stars and the actual energy premise

The complete unreduced region remains  $R = \{0, e_z\}$ , with 48 original links, twenty selected-strip and twenty-eight free links, and all 36 endpoint groups. Its four oriented Wilson edges retain signs  $+, +, -, -$  and owners  $0, 0, e_z, 0$ . Full endpoint transformations telescope to conjugation at the origin, so  $W$  is bounded, self-adjoint and physical, with  $\|W\| = 1$ . Neither the four loop vertices nor a tensor product of constrained physical factors replaces this region.

For the inherited  $S = \{0, e_x, e_y, e_z\}$ , each complete Hamiltonian group has all 21 omitted faces, norm  $M = 7|\tau|$ , and is retained only when  $b + S \subset \Lambda$ . An incident anchor has  $r = b + s$  for some  $r \in R, s \in S$ . Conversely every such anchor has a star meeting  $R$ . Hence

$$\mathcal{I}_+(R) = (R - S) \cap \mathbb{Z}_+^3 = \{0, e_z\}, \quad \mathcal{I}_\Lambda(R) = \{b \in \mathcal{I}_+(R) : b + S \subset \Lambda\}. \quad (\text{HNM-313})$$

The group at zero meets both sites and is counted once. The two-factor observable cover contains no whole star, although ten individual omitted faces fit within it; a two-factor boundary Hamiltonian is not substituted for the actual environment.

The independent reconstructions give the following boundary-complete counts. Charged faces are all faces in incident groups, with group multiplicity.

| Coarse cuboid | Links | Endpoints | All stars | Incident stars | Charged faces |
|---------------|-------|-----------|-----------|----------------|---------------|
| (2, 2, 2)     | 192   | 120       | 1         | 1              | 21            |
| (3, 3, 3)     | 648   | 342       | 8         | 2              | 42            |
| (4, 4, 4)     | 1536  | 736       | 27        | 2              | 42            |

For a cube with side count  $n$ , the global counts follow from  $8n^3$  original tails, three links per tail,  $14n^2$  added heads and  $(n-1)^3$  retained stars. More generally (HNM-313) proves at most two incident stars for every allowed containing cuboid; the finite table is not the argument for that universal bound.

Retain the exact strip ground scalar and all free-link energies:

$$h_b = \frac{H_{\text{strip},b} - E_{\text{strip},b} + \alpha \sum_{e \text{ free}} C_e}{\delta}, \quad \delta = \alpha/8, \quad h_b \geq I - P_b, \quad (\text{HNM-314})$$

where  $P_b$  is the unique reference-vacuum projector. The strip gap is at least  $\delta$ , and a free rotor's gap  $3\alpha/4$  equals  $6\delta$ . On the full two-factor space put  $h_R = h_0 + h_{e_z}$  and  $P = P_0 P_{e_z} = |\Omega_R^0\rangle\langle\Omega_R^0|$ , with lifted commuting projectors. The closed form inequality is

$$h_R \geq (I - P_0) + (I - P_{e_z}) = I - P + (I - P_0)(I - P_{e_z}) \geq I - P. \quad (\text{HNM-315})$$

The four joint projection sectors prove the bounded-projector identity. Closure extends it from the finite tensor form core, without a representation cutoff or an assumption that the physical spaces tensorize.

AJ1's domain-valid mixed reset replaces  $R$  in a finite ground density by  $P \otimes \text{Tr}_R \rho_\Lambda$ . Its exterior marginal is unchanged, its  $R$  reference energy is zero, and its total form energy is finite. Bounded

spectral truncations and monotone convergence justify the unbounded exterior energy cancellation. The variational principle cancels the finite ground scalar and all disjoint terms; each complete incident interaction changes expectation by at most  $2M$ . Therefore

$$\mathrm{Tr}(\rho_{\Lambda,R} h_R) \leq 2M |\mathcal{I}_\Lambda(R)| \leq 4M = 28|\tau|. \quad (\text{HNM-316})$$

This is an application of the inherited whole-group estimate, not a global small-norm perturbation argument. Pass each bounded spectral truncation of  $h_R$  through AJ1's actual local state limit, then increase the truncation. The actual normal density  $\rho_R$  obeys the same bound. By (HNM-315), its reference-vacuum deficit satisfies

$$e := 1 - \mathrm{Tr}(\rho_R P) \leq \mathrm{Tr}(\rho_R h_R) \leq 28|\tau| \leq \frac{7}{16384} =: e_*. \quad (\text{HNM-317})$$

Multiplying the reference energy by  $\delta$  converts its units; it does not make it the spectral first moment of the centered physical Wilson vector. At  $\tau = 0$ , zero deficit forces  $\rho_R = P$ . At nonzero admissible coupling it supplies an overlap bound, not purity.

### 32.2 A mixed-density trace estimate with its full normalization

For a unit vector  $\psi$ , the difference  $|\psi\rangle\langle\psi| - P$  vanishes outside the span of  $\psi$  and  $\Omega_R^0$ . In a basis formed from the reference vacuum and its orthogonal component it has trace zero and eigenvalues  $\pm\sqrt{1 - |\langle\Omega_R^0, \psi\rangle|^2}$ . The coincident and orthogonal endpoints satisfy the same formula. Thus its trace norm is twice that square root.

For an arbitrary positive trace-one density write its trace-class spectral decomposition  $\rho = \sum_j p_j |\psi_j\rangle\langle\psi_j|$ . Trace-norm convergence, the triangle inequality and weighted Cauchy–Schwarz give

$$\|\rho - P\|_1 \leq 2 \sum_j p_j \sqrt{1 - |\langle\Omega_R^0, \psi_j\rangle|^2} \quad (\text{HNM-318})$$

$$\leq 2 \sqrt{\sum_j p_j (1 - |\langle\Omega_R^0, \psi_j\rangle|^2)} = 2\sqrt{1 - \mathrm{Tr}(\rho P)}. \quad (\text{HNM-319})$$

Partial sums and monotone convergence justify the countable case. This is an upper bound for mixed densities; imposing the pure-projector equality would be false. The symbol  $T = \|\rho_R - P\|_1$  denotes the *full* trace norm, not its half-distance convention. Exact arithmetic in (HNM-317) gives

$$T \leq 2\sqrt{e}, \quad T^2 \leq \frac{7}{4096} < \frac{1}{576}, \quad T \leq \frac{1}{24}, \quad (\text{HNM-320})$$

since  $7 \cdot 576 = 4032 < 4096$ . No floating square root or evaluated source constant enters this cap.

### 32.3 Actual reference moments, unknown mean and attributed refinement

The reference vector can contain entanglement within each selected strip, but every original  $z$  link is reference-free. Condition on all link variables except  $u = U_z(0)$ . The Wilson holonomy is  $Au^{-1}$  with  $A$  fixed in this conditional integral. Inversion and left translation preserve normalized Haar, and the remaining squared reference wavefunction is independent of  $u$  and integrates to one. For a uniform unit quaternion, sign symmetry gives zero mean of its scalar coordinate, and its four equal second moments sum to one. Consequently

$$\mathrm{Tr}(PW) = 0, \quad \mathrm{Tr}(PW^2) = \frac{1}{4}. \quad (\text{HNM-321})$$

These are moments of the actual selected reference vector, uniform over its allowed coefficients. They are not an interacting conditional Haar law or an assumption of independent plaquette variables.

Let  $m = \omega(W)$  be the actual unknown real mean. Trace duality and  $\|W\| = \|W^2\| = 1$  give the common producer estimates

$$|m| \leq T, \quad \omega(W^2) \geq \frac{1}{4} - T, \quad v := \text{Var}_\omega(W) \geq \frac{1}{4} - T - T^2. \quad (\text{HNM-322})$$

The polynomial  $T + T^2$  is increasing on  $T \geq 0$ , since its difference at  $y \geq x$  is  $(y-x)(1+y+x) \geq 0$ . Thus

$$v \geq \frac{1}{4} - \frac{1}{24} - \frac{1}{576} = \frac{119}{576} = \frac{1}{5} + \frac{19}{2880}. \quad (\text{HNM-323})$$

The squared mean is essential. Before this rational relaxation, the common route gives the lower expression  $1/4 - 2\sqrt{28|\tau|} - 112|\tau|$ .

The independent skeptic had already frozen a sharper second-moment argument before exchange. Since  $\text{Tr}(\rho_R - P) = 0$  and  $0 \leq W^2 \leq I$ ,

$$\left| \text{Tr}[(\rho_R - P)W^2] \right| = \left| \text{Tr}[(\rho_R - P)(W^2 - I/2)] \right| \leq \frac{T}{2}. \quad (\text{HNM-324})$$

The signed first-moment estimate still costs  $T$ . Therefore this separately attributed proof gives

$$v \geq \frac{1}{4} - \frac{T}{2} - T^2 \geq \frac{131}{576} = \frac{1}{5} + \frac{79}{2880}. \quad (\text{HNM-325})$$

The improvement is valid without another physical premise, but is not assigned retroactively to the two producers. Neither bound is sharpness or an actual variance measurement. At zero coupling the actual variance is exactly  $1/4$ .

### 32.4 Discriminating controls and admitted limits

The exact pure-projector fixtures reject removing the square root or the factor two from (HNM-319); mixed matrices reject imposing its pure-state equality on  $\rho_R$ . For example the reverse density  $(\begin{smallmatrix} 17 & 6 \\ 6 & 8 \end{smallmatrix})/25$  has determinant  $4/25$ , deficit  $8/25$  and trace distance  $4/5$ . Both defective upper bounds fail. The skeptic's signed-observable and positive-effect controls distinguish  $T$  from  $T/2$ ; the effect saving cannot be copied to the mean of  $W$ .

On the same original configuration space,  $\sqrt{1+W}\Omega_R^0$  is a normalized alternative normal vector. Conditional reference moments give mean  $1/4$ , second moment  $1/4$  and variance  $3/16 < 1/5$ . It exposes a discarded squared mean or an unjustified interacting Haar substitution. Its nonzero mean and (HNM-319) require reference-vacuum deficit at least  $1/64 > e_*$ , so the added actual energy premise excludes it. It is not an interacting ground-state calculation.

Likewise, normalize  $\mathbf{1}_{\{|W|<1/4\}}\Omega_R^0$  by the square root of its reference probability  $p$ . Conditional Haar gives  $p > 0$ , mean zero and positive variance at most  $1/16$ . The density  $(2/\pi)\sqrt{1-w^2}$  and  $\pi > 2$  give  $p \leq 1/2$ , hence deficit and reference energy at least  $1/2$ . These normal states violate (HNM-317); they show why normality alone gives no uniform floor. The possibly infinite energy of a discontinuous indicator is not claimed finite.

At the origin, counting only outgoing anchors or only individually touching faces is blind: all incident anchors lie in  $R$  and all their group faces touch it. The retained incidence-only region  $R' = \{(1,1,1), (1,1,2)\}$  resolves this. Its seven orthant anchors yield four and seven retained groups in cubes of sides three and four, versus one and two outgoing anchors. The inherited group budget charges 84 and 147 faces, versus 49 and 82 individually touching faces. These diagnostics reject substituting a different count into AJ1's whole-group estimate. They do not rule out a

separately derived estimate for a finer decomposition into face interactions, and establish no translated-observable variance theorem.

Scalar and scale controls retain  $E_{\text{strip},b}$  and  $\delta = \alpha/8$ . Full noncommuting original-link gauge transformations preserve  $W$  while missing-head actions fail. Hypothetical positive source constants with  $\tau_* < 2^{-16}$  reject treating the cap as an evaluated stability radius; those controls do not assign the actual source constants.

The advisor suggested the overlap route before production. Separate forward, reverse and independent skeptical derivations froze before exchange; the stronger effect estimate was in that independent freeze. This skeptic's new AK1 derivation is distinct from its earlier AJ2 reviewer-handoff closeout. The post-review read both full reports and checkers and replayed both complete output sets under normal and optimized Python with byte equality. Administrative input-path and inventory-selector histories, blind incidence controls and first successful scientific outputs remain preserved. They add no research loop or independent observation.

The controlling AJ1/AJ2 proofs were read by all three workers. Forward and skeptic read the complete current-manuscript text; reverse recorded focused reading. The skeptic's text reading is not visual PDF review. The I1 stability theorem remains inherited, and no new external primary retrieval or scientific-priority comparison is claimed. Finite rational and geometry diagnostics audit this proof without replacing its analytic infinite-volume premises.

AK1 proves the variance floors for the fixed original observable and specified conditional homogeneous state. It proves no physical excited-vector form or operator domain, first energy moment, commutator sum rule, finite spectral window, lower heat bound, numerical mass or AK2 theorem. Other translates, boundary states and models require their own estimates. No physical matching, continuum Yang–Mills construction, historical validation or defensible percentage of the Millennium problem follows.

### 33 HNM AK2: physical Wilson form energy, a finite window and lower heat<sup>\*\*</sup>

**Current extension: HNM-C-AO2.** The historical limitation below remains the scope of this original result. The later, separately gated statement in Section 47 records the extension: For the same actual I1/AJ1/AK2 orthant physical state,  $\chi$  belongs to  $D(H_{\text{phys}})$ , its second spectral moment is at most  $\alpha^2(36+28|\tau|) < 37\alpha^2$ , and its first moment equals  $\alpha \omega(1-W^2)$ . Uniform first-energy tails bounded by  $B2/R$  close the former AK2 moment-loss limitation. The proof uses the actual centered spectral measures, compact spectral tests and bounded-local expectations. Its own model and state assumptions remain required.

Draft02 attribution update. See Section 1 for the primary-source comparison and its novelty limitation. HNM labels identify project records.

For the same actual homogeneous state and original origin  $xz$  Wilson observable, AK2 proves

$$\chi \in D(H_{\text{phys}}^{1/2}), \quad \mu_1 := \int E d\nu(E) \leq \alpha, \quad \nu([\alpha/16, 10\alpha]) \geq \frac{1}{10}, \quad (\text{HNM-326})$$

and, for every finite physical imaginary time,

$$C(t) = \langle \chi, e^{-tH_{\text{phys}}/\hbar} \chi \rangle \geq \frac{1}{5} e^{-5\alpha t/\hbar}, \quad 0 \leq t < \infty. \quad (\text{HNM-327})$$

Here  $\chi = (\pi(W) - \omega(W)I)\Omega$  and  $\nu$  is its actual physical energy spectral measure. The regime remains *both*  $|\tau| < \tau_*$  with unevaluated source constants and  $|\tau| \leq 2^{-16}$ , with the same fixed

selected coefficients, positive scales and whole-star positive-orthant limit. The numerical cap alone certifies no chosen positive admissible coupling.

The proof uses only the common AK1 consequence  $v = \|\chi\|^2 \geq 1/5$ . Its producer floor 119/576 and independent-skeptic refinement 131/576 retain their attribution and are not needed to optimize this target. AJ1/AJ2 supply

$$\nu(\mathbb{R}) = v, \quad \nu(\{0\}) = 0, \quad \text{supp } \nu \subset [\alpha/16, \infty), \quad C(t) \leq v e^{-\alpha t/(16h)}. \quad (\text{HNM-328})$$

AK1's local reference-density energy is not a physical first moment of  $\chi$ . The new calculation supplies that missing premise from the actual original-link kinetic operator. It proves a form-domain statement and a first-moment upper bound, without asserting  $\chi \in D(H_{\text{phys}})$ , a second moment, a threshold eigenatom or a measured mass.

### 33.1 Finite physical operator, complete support and multiplier domains

Retain the full  $R = \{0, e_z\}$ , all 48 original links and all 36 endpoint actions from AJ2/AK1. The four distinct signed Wilson links have owners  $0, 0, e_z, 0$ ; all other 44 local links remain present even though their derivatives of  $W$  vanish. Each finite volume contains every owned link, three selected faces per block and every complete retained 21-face star. For a coarse cube of side  $n$ , the original counts are  $24n^3$  links,  $8n^3 + 14n^2$  endpoints,  $3n^3$  selected faces and  $21(n-1)^3$  retained omitted faces. The original tail ownership and complete-star rule establish these formulas at every volume; the  $n = 2, 3, 4$  diagnostics are finite checks of that geometry.

On the finite product Haar space, choose the standard half-Pauli left fields

$$X_{e,a} f(U_e) = \left. \frac{d}{dt} f(e^{tT_a} U_e) \right|_{t=0}, \quad T_a = -i\sigma_a/2, \quad C_e = - \sum_{a=1}^3 X_{e,a}^2. \quad (\text{HNM-329})$$

Thus a fundamental matrix coefficient has Casimir eigenvalue  $3/4$ . Reversing the overall Lie-algebra basis convention, as in the forward derivation, leaves this Casimir and the squared-gradient identity unchanged; the signs within each inverse-link derivative still matter.

Write the raw physical operator and reference scalar as

$$H_{\Lambda}^{\text{raw}} = \alpha \sum_{e \text{ owned}} C_e - \sum_{f \text{ selected}} \lambda_f W_f - \frac{\alpha\tau}{24} \sum_{b+S \subset \Lambda} \sum_{f \in O_b} W_f, \quad (\text{HNM-330})$$

$$B_{\Lambda} = \sum_{b \in \Lambda} E_{\text{strip},b}, \quad \delta \hat{H}_{\Lambda} = H_{\Lambda}^{\text{raw}} - B_{\Lambda}, \quad \delta = \alpha/8, \quad |O_b| = 21. \quad (\text{HNM-331})$$

The selected  $\lambda_f$  include both endpoint and bridge coefficients. If  $\hat{E}_{\Lambda}$  is the actual normalized ground energy, then

$$E_{\Lambda}^{\text{raw}} = B_{\Lambda} + \delta \hat{E}_{\Lambda}, \quad K_{\Lambda} = \delta(\hat{H}_{\Lambda} - \hat{E}_{\Lambda}) = H_{\Lambda}^{\text{raw}} - E_{\Lambda}^{\text{raw}} \geq 0. \quad (\text{HNM-332})$$

Both reference and true ground scalars are retained before cancellation. For the actual normalized ground  $\Omega_{\Lambda}$ , put

$$m_{\Lambda} = \langle \Omega_{\Lambda}, W \Omega_{\Lambda} \rangle, \quad \chi_{\Lambda} = (W - m_{\Lambda}) \Omega_{\Lambda}, \quad K_{\Lambda} \Omega_{\Lambda} = 0. \quad (\text{HNM-333})$$

The finite sum of link Casimirs is an elliptic product Laplacian with operator domain  $H^2$  and form domain  $H^1$  on the compact group product. Finite Peter–Weyl sums form a graph core. Every selected and omitted potential is smooth real bounded multiplication at each fixed volume, so bounded perturbation preserves these domains. A uniform bound on the entire extensive potential is not assumed. In particular the finite ground is in  $H^2$ .

Smoothness, rather than bounded locality alone, supplies the multiplier regularity. Both  $W$  and  $W^2$  have bounded derivatives through order two. On the finite core, for  $F = W$  or  $W^2$  and  $Q_\Lambda = \sum_e C_e$ ,

$$Q_\Lambda(F\psi) = FQ_\Lambda\psi + (Q_\Lambda F)\psi - 2 \sum_{e,a} (X_{e,a}F)X_{e,a}\psi. \quad (\text{HNM-334})$$

The first-derivative norm satisfies  $\sum_{e,a} \|X_{e,a}\psi\|^2 = \langle \psi, Q_\Lambda\psi \rangle$  and is bounded by the graph norm using  $x \leq 1 + x^2$ . A graph-approximating sequence and closedness therefore prove  $FD(Q_\Lambda) \subset D(Q_\Lambda)$ . The weak product rule similarly gives  $H^1$  preservation. The same statements hold for  $W - m_\Lambda$  and its square. These are the required finite-volume domains; no infinite-volume operator-domain conclusion is imported from them.

### 33.2 Four original derivatives and the exact finite form identity

On the common finite core, Haar integration by parts and the product rule give

$$[C_e, W]\psi = (C_e W)\psi - 2 \sum_a (X_{e,a}W)X_{e,a}\psi, \quad [W, [C_e, W]] = 2M_{\sum_a (X_{e,a}W)^2}. \quad (\text{HNM-335})$$

The first commutator is a first-order operator, not an asserted bounded  $L^2$  operator. The second has the displayed bounded multiplication extension; its domain identity extends to  $H^2$  by the preceding argument. All selected and retained omitted magnetic terms commute with  $W$  as multiplication operators, including overlapping plaquettes. Both scalars commute as well. Their cancellation does not replace the actual interacting ground by Haar.

For a positive original link occurrence, varying  $U$  by  $e^{tT_a}U$  differentiates it as  $T_a U$ . For an inverse occurrence,

$$(e^{tT_a}U)^{-1} = U^{-1}e^{-tT_a}, \quad \left. \frac{d}{dt}(e^{tT_a}U)^{-1} \right|_0 = -U^{-1}T_a. \quad (\text{HNM-336})$$

Both occurrences have second derivative  $-U^{\pm 1}/4$ . Fixed left/right factors and inversion are isometries of the bi-invariant metric. If the full holonomy is the unit quaternion  $(w, \mathbf{q})$ , then  $W = w$  and  $|\mathbf{q}|^2 = 1 - w^2$ . The three half-Pauli fields give scalar-coordinate derivatives equal to components of  $\mathbf{q}/2$ , up to an orthogonal adjoint rotation and the required inverse sign. Thus *each* of the four distinct original links gives

$$\sum_a (X_{e,a}W)^2 = \frac{1 - W^2}{4}, \quad C_e W = \frac{3}{4}W. \quad (\text{HNM-337})$$

Every other link derivative is zero. Retaining all four original kinetic terms yields

$$\Gamma(W) := \sum_{e,a} (X_{e,a}W)^2 = 1 - W^2, \quad [W, [K_\Lambda, W]] = 2\alpha M_{1-W^2}. \quad (\text{HNM-338})$$

The identity does not substitute a reduced one-plaquette kinetic operator.

The previously proved domains make  $WK_\Lambda W$ ,  $W^2 K_\Lambda$  and  $K_\Lambda W^2$  well-defined at the ground vector. In  $2WK_\Lambda W - W^2 K_\Lambda - K_\Lambda W^2$ , the last two ground expectations vanish. Centering  $W$  leaves its form energy unchanged because  $K_\Lambda \Omega_\Lambda = 0$ . Therefore, for the actual finite spectral measure  $\nu_\Lambda$ ,

$$\mu_{1,\Lambda} = \langle \chi_\Lambda, K_\Lambda \chi_\Lambda \rangle = \frac{1}{2} \langle \Omega_\Lambda, [W, [K_\Lambda, W]] \Omega_\Lambda \rangle \quad (\text{HNM-339})$$

$$= \alpha \omega_\Lambda (1 - W^2) \leq \alpha. \quad (\text{HNM-340})$$

The forward proof independently obtains this by subtracting the weak ground equation tested against  $f^2\Omega_\Lambda$  from the form at  $f\Omega_\Lambda$ ,  $f = W - m_\Lambda$ . For each real derivative,

$$|X(f\Omega_\Lambda)|^2 - \operatorname{Re}(\overline{X\Omega_\Lambda} X(f^2\Omega_\Lambda)) = (Xf)^2|\Omega_\Lambda|^2. \quad (\text{HNM-341})$$

The potential and scalar terms cancel, and  $H^1$  continuity extends the identity from the smooth core. Both routes give the same physical coefficient and factor one-half. The bound is uniform in volume and the declared coefficients, despite their effect on the unevaluated actual expectation.

### 33.3 Tested resolvents, scalar measure convergence and the limiting form

The source-qualified I1/AJ1 theorem gives centered finite-resolvent matrix-element convergence tested on fixed bounded-local vectors  $A\Omega_\Lambda, B\Omega_\Lambda$ . Its domain-qualified creation core is retained separately. The theorem does not assert common-concrete-space strong resolvent convergence or place every bounded-local vector in  $D(G)$ . Fixed physical energy is restored through

$$(K_\Lambda - z)^{-1} = \delta^{-1}(\hat{H}_\Lambda - \hat{E}_\Lambda - z/\delta)^{-1}, \quad \operatorname{Im} z \neq 0. \quad (\text{HNM-342})$$

Local state convergence gives  $m_\Lambda \rightarrow m = \omega(W)$  and  $v_\Lambda = \omega_\Lambda(W^2) - m_\Lambda^2 \rightarrow v$ , with  $v_\Lambda \leq 1$ . First use the source theorem on the fixed  $A = W - mI$ . Replacing its finite cyclic vector by  $\chi_\Lambda$  changes it by  $(m_\Lambda - m)\Omega_\Lambda$ . The resolvent norm  $1/|\operatorname{Im} z|$  and both vector norms at most two bound the difference of quadratic resolvent tests by

$$\frac{4|m_\Lambda - m|}{|\operatorname{Im} z|} \rightarrow 0. \quad (\text{HNM-343})$$

This justifies the moving ground-state mean. The resulting nonreal transforms converge to those of the actual spectral measure  $\nu$  of  $(H_{\text{phys}}, \chi)$ ; AJ1's reducing physical identification preserves this measure.

The linear span of nonreal resolvent functions is dense in  $C_0(\mathbb{R})$ . Conjugation stays in the span, products at distinct poles reduce by the resolvent identity, and repeated poles are uniform limits of difference quotients with poles away from the real line. Its uniform closure is therefore an algebra that separates points and vanishes nowhere. The locally compact Stone–Weierstrass theorem gives the claimed density. The uniform measure-mass bound extends convergence to every  $C_0$  test. The reverse proof equivalently works on  $[0, \infty)$ .

For  $L > 0$  and  $E \geq 0$  use the continuous compactly supported functions

$$F_L(E) = E \min\{1, \max(0, 2 - E/L)\}, \quad F_L(E) = 0 \quad (E < 0). \quad (\text{HNM-344})$$

They satisfy  $0 \leq F_L \leq E$  on the positive half-line and increase pointwise to  $E$ . The measure convergence and (HNM-340) give  $\int F_L d\nu \leq \alpha$  for every  $L$ , and monotone convergence yields

$$\mu_1 = \int E d\nu(E) \leq \alpha, \quad \|H_{\text{phys}}^{1/2}\chi\|^2 = \mu_1 < \infty. \quad (\text{HNM-345})$$

This is the spectral characterization of the actual form domain. The independent skeptic instead uses  $\min(E, L) = L - (L - |E|)_+$  on nonnegative energies: mass convergence and the  $C_0$  test give the same monotone argument.

Equation (HNM-340) is an *equality at finite volume*. Its bounded-local right side converges, but the limiting spectral first moment is only bounded above by this method. Uniform integrability has not been established. Neither equality of first moments, a second moment nor  $\chi \in D(H_{\text{phys}})$  is inferred. Scalar measure convergence replaces no missing common-Hilbert-space operator theorem.

The independently frozen skeptic derivation also retains the more specific upper bound  $\mu_1 \leq \alpha[1 - \omega(W^2)]$ , by passing the bounded-local right side through the same argument. It remains an inequality with an unevaluated interacting expectation. The common ceiling  $\alpha$  suffices for the frozen window and heat targets below.

### 33.4 A bounded energy interval and all finite imaginary times

With  $\text{supp } \nu \subset [\alpha/16, \infty)$ ,  $v \geq 1/5$  and (HNM-345), the elementary tail inequality gives

$$\nu((10\alpha, \infty)) \leq \frac{\mu_1}{10\alpha} \leq \frac{1}{10}, \quad \nu([\alpha/16, 10\alpha]) \geq v - \frac{1}{10} \geq \frac{1}{10}. \quad (\text{HNM-346})$$

The removed tail is open and the window's upper endpoint is included. Positive interval mass does not identify an eigenatom, the lowest overlapping energy or a sharp threshold.

Normalize the positive measure by  $v$  to obtain a probability measure with finite mean  $\bar{E} = \mu_1/v \leq 5\alpha$ . For  $s = t/\hbar \geq 0$ , the convex tangent inequality

$$e^{-sE} \geq e^{-s\bar{E}}[1 - s(E - \bar{E})] \quad (\text{HNM-347})$$

is integrable because the first moment is finite. Integrating proves Jensen's inequality here without a small-time expansion:

$$C(t) \geq ve^{-t\mu_1/(\hbar v)} \geq ve^{-5\alpha t/\hbar} \geq \frac{1}{5}e^{-5\alpha t/\hbar}. \quad (\text{HNM-348})$$

It includes  $t = 0$ , where  $C(0) = v$ . Together with (HNM-328), this gives two physical imaginary-time envelopes for every finite time. The variance, spectral mass and  $C(t)$  are dimensionless;  $\alpha$ ,  $\mu_1$  and the window endpoints have energy units, and  $\alpha t/\hbar$  is dimensionless. No real-time magnitude-decay or exact asymptotic mass is claimed.

### 33.5 Controls, preserved failures and the stopping boundary

At the explicitly labeled corner where *all* selected and omitted couplings vanish, the Haar vacuum gives  $\|W\Omega\|^2 = 1/4$ , excitation energy  $3\alpha$  and first moment  $3\alpha/4$ . This checks the four-link normalization, without assigning those values to an arbitrary interacting state. Doubling the half-Pauli generator quadruples the gradient and Casimir budget. Omitting the double-commutator factor one-half gives the wrong moment. Scalar and units fixtures separately detect a missing actual ground subtraction,  $\delta$  conversion or division by  $\hbar$ ; simultaneous raw-operator and ground shifts cancel.

Exact noncommuting quaternion derivatives, supplemented by signed group-displacement or Cayley-path tests, verify all four original occurrences and their inverse derivatives. Squaring an inverse derivative hides its wrong overall sign, so that blind gradient test is preserved beside signed first-derivative controls. The independent skeptic's first identity-link fixture has  $\Gamma = 0$  and is also blind to a kinetic rescaling; the later noncommuting fixture discriminates. Full endpoint gauge transformations preserve the actual word, while missing-head actions fail.

Both producers give abstract bounded self-adjoint creating operators with vectors of divergent form energy. The skeptic additionally uses actual free original-link characters, with an auxiliary observable distinct from  $W$ . On the full 48-link region, let  $\phi_n$  be the spin- $n/2$  character of the same four-link holonomy and let  $\Omega_R^0$  be its free constant vacuum. Conditional Haar gives orthonormal  $\phi_n$ , with four-link energy  $\alpha n(n+2)$ . The gauge-invariant vector

$$f_R = \sqrt{3} \sum_{k \geq 1} 2^{-k} \phi_{2k} \quad (\text{HNM-349})$$

has norm one and is orthogonal to  $\Omega_R^0$ . Its first  $N$  terms have squared norm  $1 - 4^{-N}$  and form energy divided by  $\alpha$  equal to  $3N + 6(1 - 2^{-N})$ , which diverges. The bounded local physical self-adjoint operator

$$A_R = |f_R\rangle\langle\Omega_R^0| + |\Omega_R^0\rangle\langle f_R| \quad \text{on } \mathcal{K}_R, \quad A = A_R \otimes I_{R^c} \quad (\text{HNM-350})$$

has norm one and creates the divergent-energy vector in the free product representation. It is extended by the exterior *identity*, not a global vacuum projector. The example rejects automatic form regularity of bounded local observables; it does not undermine the separately proved smooth-Wilson argument.

The three independent moment-loss diagnostics are abstract positive spectral measures. For example the reverse uses, in units of  $\alpha$ ,

$$\eta_n = (1 - 1/n)\delta_{1/16} + (1/n)\delta_{1/16+n/2}. \quad (\text{HNM-351})$$

Masses are one, bounded-test differences from  $\delta_{1/16}$  are at most  $2\|f\|_\infty/n$ , and every first moment is  $9/16$ , while the limiting first moment is  $1/16$ . This proves that bounded first moments and mass/resolvent convergence do not force moment equality. The forward and skeptic retain their different valid measures and exact prefixes.

Noncentered observables with nonzero means retain vacuum mass and heat plateaus. By contrast, abstract mass  $1/5$  at  $20\alpha$  has the inherited lower support and variance floor but neither mass in the target window nor the proposed lower heat curve; its first moment exceeds  $\alpha$ . The skeptic uses  $11\alpha$  for the same inference control and separately checks inclusion of  $10\alpha$  in the closed window. These are alternative measures, not observed spectral atoms of the actual state. Hypothetical source constants below the numerical cap retain the strict symbolic coupling requirement.

The forward first invocation stopped at a missing closing parenthesis before executing science. The original checker and SyntaxError are preserved, followed by corrected normal and optimized output agreement. Blind controls and replacements remain attributed; none is erased or counted as another investigation. Frozen sources and independent implementations support reproducibility, while the finite domain proof, original-link algebra, source-qualified measure limit and monotone convergence establish the result. The original I1 cluster-expansion theorem remains inherited; no full reproof, new primary retrieval or scientific-priority claim is made.

The final post-review also preserves one unused stale label in a reverse centering diagnostic: its metadata says `abstract_gap=1`, while the implemented positive level is 4 in units of  $\alpha$ . The correctly implemented centered mass is  $1/16$ , first moment divided by  $\alpha$  is  $1/4$ , and centered and raw heat values are  $1/256$  and  $65/256$ . The reverse author's Newton-method comparison acknowledges the stale label; it is not reinterpreted as an intentional lower-bound certificate. No accepted target depends on it, and no frozen field is rewritten. That comparison shares authorship with reverse and is not another independent producer.

The final skeptic authored its AK2 derivation before exchange, then read both complete producer reports and checkers and reproduced each complete output set under normal and optimized Python with byte equality. Its separately bound final specification audit found no blocking issue. The local character diagnostic in (HNM-350) is the additive post-review clarification of its own independently frozen control. Source reading remains differentiated: the skeptic's current manuscript coverage was differential text coverage against its previously read draft, while both producers recorded focused current readings. These activities are not visual PDF review or fresh audits of the full external cluster proof.

This is investigation ten and the authorized scientific stop boundary. The fixed-lattice, fixed-observable result supplies (HNM-326)–(HNM-327) in the same conditional homogeneous model. It supplies no numerical stability radius, infinite first-moment equality, second moment, infinite operator-domain assertion, eigenatom, exact mass, other-boundary identification, physical model matching or continuum Yang–Mills theorem. Further questions remain plans rather than extra scientific executions.

Historical Round28 checkpoint. References below to planned, unexecuted or next work describe the state at that checkpoint; the integrated Round29 chapter supersedes that plan.

## 34 Completed investigation count and remaining obligations

All ten requested investigations are admitted, completing five adaptive goal pairs. This is 100% of the requested investigation count, including limited outcomes. It is not a percentage of the Yang–Mills existence-and-mass-gap proof: equations, loops, numerical error percentages and network size provide no calibrated denominator for that theorem.

The final assessment identifies the strongest structural advance as a controlled, nonvacuous physical sector of the declared fixed-spacing  $SU(2)$  homogeneous omitted interaction on its selected-strip reference. AJ1 supplies the normal local representation and reducing physical generator; AJ2 supplies a nonzero original Wilson fluctuation; AK1/AK2 add conditional variance, physical form-energy, finite-window mass and imaginary-time control. These are improvements within the specified whole-star orthant state. The inherited  $\alpha/16$  threshold is not a new continuum gap, and the unnormalized Wilson spectral mass is not a particle count or an evaluated mass.

The final retrospective assessment is one contemporary model-agent review applying the documented historical-method questions. It is not five new independent reviewers, participation by historical figures, external human peer review or another scientific investigation. The independent productions share inherited premises and model authorship. The Newton AK2 comparison is by the reverse producer; the separate skeptical production and later clarifications retain their recorded attribution.

The advisor's 22 September 2026 [official problem-status check](#) records the problem as unsolved. Selected sections 3–5 and 6.5 of the [Jaffe–Witten description](#) were checked for scope, not as an exhaustive current research survey. The required nontrivial four-dimensional theory for any compact simple gauge group, suitable axioms, required short-distance behavior and strictly positive finite physical mass gap remain open here. Confinement and an isolated one-particle state are discussed extensions, not additional mandatory clauses of the stated Millennium problem. Numerical lattice stability constants, a particular correction iteration and exact Wilson moment equality are possible technical milestones, likewise not compulsory clauses in themselves.

For this workbench's proposed route, full generator, observable, state, coupling, spacing and clock matching remains unresolved. So does identification of specified bulk and boundary limits. A boundary-local orthant witness cannot silently become a bulk observable, and the finite, canonical, summable and homogeneous models cannot be interchanged. The finite first-moment equality has not been proved to survive the limit; no actual strict moment loss is established.

The final roadmap ranks the following *unselected* questions. Its ordering concerns missing implications and bounded test value, not success probabilities or equal units of proof work.

| Goal | Proposed direction                             | First unresolved premise                                                                                                                                               |
|------|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AL   | Full model and physical-scale bridge           | A complete coefficient, observable and clock dictionary with one admissible scaling path; compare the existing bare-coupling obstruction before claiming a new bridge. |
| AM   | Numerical homogeneous stability interval       | Evaluate every constant required for the complete original model, retaining an insufficient verdict if a necessary constant or infinite estimate remains open.         |
| AN   | Specified bulk-boundary comparison             | Compare deep-bulk orthant translates with one declared centered-box route using a bound that vanishes with boundary distance.                                          |
| AO   | Same-Wilson moment identification              | Prove sufficient uniform integrability or form convergence before transferring the finite identity as equality; assume no new energy-tail or second-moment theorem.    |
| AP   | Canonical inference at a prior-selected design | Separately test one design fixed from coarse prior information and decisions across independently varying candidate parameters.                                        |

Every second loop remains unselected until its first review. A certified numerical evaluator for the declared 561-state AH enrichment is a lower-priority computational alternative; no such heat vector was produced by this cycle's assessment. The detailed models, dependency IDs and proposed first tests are recorded in `research/round28/advisor/roadmap.json`. Investigation ten is the stopping boundary. This closeout executes no further research target.

The assessment, roadmap and official scope check are separately source-bound. They do not alter the 33-record exploratory reading ledger or claim new primary research by the retrospective reviewer. Scientific priority remains unverified; the two-star marker identifies contributions in this workbench.

## Round28 contribution and reproduction appendix (historical checkpoint)

### 35 Admitted scope and contribution ledger

| Loop | Admitted contribution                                                                                                                                                                            | Possible application                                                                                                                          | Remaining limitation                                                                                                                                                                                       |
|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AG2  | Complete original remainder and transport; output weight-two budget below 0.007, and centered-reference gap above 0.995 on the main interval.                                                    | Inventory the missing original interaction and supply explicit scalar, mixing and reference-diagonal budgets for the next review.             | The remaining mixing is retained. No gap of the full corrected Hamiltonian, complete-source contraction or next iteration is proved.                                                                       |
| AG3  | Complete one-step correction of actual AG2 mixing, with coefficients 3060/3623 and 5949/96812 from weight 2 to 3/2; full scalar and reference-diagonal updates.                                  | Control one complete original-source correction and compare it with passive reweighting before choosing a further method.                     | Main coefficient is weaker than passive reweighting. Narrow coefficient beats only its universal upper ceiling. No actual lower-weight improvement, same-weight contraction, iteration or full gap.        |
| AI3  | Exact odd moments and an all-order retained reflection-tail bound. All ten complete ratio certificates remain insufficient; both direct scalar probes still separate at $u=1e-18$ , $k=5$ .      | Remove the material cubic arithmetic floor and identify the state or spatial upper budget that a later physical readout method must improve.  | Finite-grid gain comes primarily from degree-eight centers. No physical error cancellation, continuous inverse, unknown-phase tolerance, homogeneous matching or mass-gap result.                          |
| AI4  | Complete physical ratio intervals separate by at least $39u/40$ for every real $0 < u \leq 1e-24$ under the dyadic collar and shorter-duration schedule.                                         | A uniform family of conditional candidate-pair tests with proved all-depth resource envelopes and attributed scalar-error allowances.         | Clock grows as the inverse cube of $u$ ; all scalar allowances vanish with $u$ . No fixed-error apparatus, continuous inverse, automatic enlarged-graph theorem, homogeneous matching or continuum result. |
| AH1  | Complete new 46-link graph; strict rank-30 cutoff and rank-561 physical enrichment, exact Gram/magnetic action, and all-time relative heat error below 0.0022 for the declared preparation ball. | A complete finite-graph generator and conservative heat certificate that retains every new-input outside channel and both ground comparisons. | Exact own-ground heat only; no numerical trajectory, whole-shell completion, evaluated outside Gram, real-time theorem, graph-size uniformity or continuum result.                                         |
| AH2  | Complete 531-channel delayed loading with feedback gives a proved all-time true-relative exact heat error below 0.0001 on the unchanged graph and preparation ball.                              | Improve the same-clock physical omission certificate by integrating stationary and excited loading before applying the full outside bound.    | Sufficient bound for exact separate ground centers; no numerical heat vector, evaluated outside Gram, real-time theorem, graph-size uniformity, model matching or continuum result.                        |

| Loop | Admitted contribution                                                                                                                                                                                                               | Possible application                                                                                                                                                       | Remaining limitation                                                                                                                                                                                                                |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AJ1  | Actual homogeneous I1 local energy bound, local trace-norm limits and normal representations; compatible Haar averaging, physical cyclic/fixed equality and closed reducing generator.                                              | Identify the physical GNS restriction of the specified homogeneous orthant state with justified gauge continuity, local averages and operator/form domains.                | Symbolic coupling interval and inherited $\alpha/16$ threshold only; no nonzero physical excitation or Wilson variance, numerical interval, boundary equivalence, model matching or continuum result.                               |
| AJ2  | The actual fixed xz Wilson observable has positive variance in the homogeneous state; its nonzero physical spectral measure gives strictly positive finite-time heat correlation and the inherited $\alpha/16$ upper decay ceiling. | Close the physical-sector nonvacuity omission using Haar-null multiplier levels, local normality, the physical GNS quotient and bounded spectral calculus.                 | Symbolic coupling interval only; no quantitative variance floor, energy moment, threshold atom, measured mass, lower decay law, real-time magnitude decay, boundary identification or continuum result.                             |
| AK1  | Actual origin Wilson variance at least $119/576$ from both producers; the independent skeptic improves the lower certificate to $131/576$ on the same conditional subregime.                                                        | Turn actual local reference energy into a mixed-state vacuum-overlap bound and a quantitative physical signal margin, retaining the unknown squared mean.                  | Requires both symbolic stability and the additional coupling cap $1/65536$ ; no evaluated stability radius, actual variance, physical excited-vector energy moment/domain, spectral window, translated theorem or continuum result. |
| AK2  | Actual Wilson physical form domain and first moment at most $\alpha$ ; spectral mass at least $1/10$ in $[\alpha/16, 10\alpha]$ , and lower heat envelope $(1/5) \exp[-5\alpha t/\hbar]$ at every finite nonnegative time.          | Transfer the finite original-link energy identity through tested resolvents and nonnegative cutoffs, then combine the upper moment with the admitted variance and support. | Same conditional fixed-lattice regime only; no limiting moment equality, infinite operator domain, second moment, spectral atom, evaluated mass, real-time decay, model transfer or continuum result.                               |

## 36 Additive source deepening after the first two goal pairs

The initial survey remains **30 formal reading records at 38 distinct URLs**. The later source-only supplement adds three attributed records: further selected passages of the already known Del Vecchio–Fröhlich–Pizzo paper, an ETH institutional publication record, and a targeted topology reading of the already known Yarotsky paper. These are not three new independent studies. Together the current ledgers contain **33 formal reading records at 41 distinct recorded URLs**, plus five screening or reopen records; the URL union including those screens is 46. The intermediate two-record supplement had totals 32/39 and a screened URL union of 44; its creation checkpoint remains preserved. The one unread lead stays outside the reviewed count. No government-primary document was newly read in this supplement, and it adds zero research investigations.

The technical text is [arXiv:2108.13907v1](#), a 54-page preprint. The [ETH institutional record](#) verifies publication in *Journal of Functional Analysis* 284(1), 109734 (2023), [doi:10.1016/j.jfa.2022.109734](#). Final journal text was not compared with v1. The recorded technical passages are Sections 1.1–1.2 and Remark 1.1; the Section 2 opening; definitions (2.29)–(2.32); Lemma 2.8’s statement and initial proof; selected Section 2.4 gap estimates through Corollary 2.12; the Section 3 induction introduction and selected Theorem 3.1 inequalities; Theorem 3.3 and its closing proof; Lemma 4.1’s statement; and Lemma 4.3 with the adjoining inverse-series justification. The long induction and appendices were not audited end to end.

The Feynman-method deepening initially compared an application to the original finite-range homogeneous Hamiltonian  $H_0 + \Phi$  with organizing AG3’s generated interaction inside the cited paper’s evolving-reference induction. Its later archive correction identifies an omitted established result: the current Round21 I1 admission, revision 2, already supplies the complete 24-link/21-face dictionary, qualitative original-model stability and the specified positive-orthant full-link-observable GNS construction. With source constants left symbolic,

$$\tau_* = \frac{\min\{c_1(S), 1/[2c_2(S)]\}}{7} > 0, \quad 0 < |\tau| < \tau_* \implies \Delta \geq \alpha/16. \quad (\text{HNM-352})$$

I1 certifies no selected nonzero numerical  $\tau$ . Its homogeneous *omitted* coupling on the selected-strip reference is not a uniform coefficient on every plaquette, a summable-model representation, or a continuum construction. The specified orthant GNS conclusion is not independence of every boundary sequence.

Consequently another theorem application giving only unspecified-small-coupling existence would duplicate the admitted qualitative outcome. A distinct new target must supply a numerical interval, complete generated-interaction iteration, or an additional representation or boundary statement beyond I1. The archive comparison uses the current repository admission and source dictionary; it neither rereads nor reconstructs the external Yarotsky proof, and adds no primary reading record or investigation. The unchanged deeper-source note and this additive correction are separately bound in `research/round28/experts/source-supplement.json`.

A different published theorem would need its own hypothesis matching while reusing I1’s established site, star, domain and normalization dictionary. Infinite onsite dimension is permitted by the Del Vecchio paper’s stated assumptions; its initial range condition is not confined to the expository nearest-neighbor presentation. Theorem-specific boundary and gauge-observable restrictions must remain explicit, and a constrained physical space cannot silently become a tensor product. Re-entry with AG3’s generated interaction additionally requires a proved passage from arbitrary indexed supports to rectangular energy-weighted data, changed local gaps, retained scalars and diagonals, and the complete induction. These are inferred transfer obligations, not defects asserted in either external theorem or missing work already discharged by I1.

The third supplemental reading, record R28-FGNS1, inspects Yarotsky’s 17-page [math-ph/0411042v1 PDF](#) and corresponding [HTML](#). Its depth is the definitions and Theorems 1–3 on

printed pages 2–4, creation operators on pages 6–7, selected expansion and state-limit passages on pages 8–9, and the ending generator/core argument on pages 9–10. References on page 17 were bibliographic pointers only. The long proof and cited predecessor papers were not independently reconstructed; an older I1 download hash is not claimed as a hash of these newly inspected browser bytes.

The source states convergence of finite-ground expectations for each observable in the full local algebra  $\bigcup_F B(\mathcal{H}_F)$ , and resolvent matrix-element convergence tested on local excitation vectors. Its generator discussion uses local bounded commutators and creation vectors with  $u_I \in \mathcal{H}'_I \cap D(H_{I,0})$ , followed by density and essential self-adjointness statements. These are not trace-norm convergence of local density matrices, strong resolvent convergence in the earlier summable representation, or a declaration that every bounded-local cyclic vector lies in the generator domain. No explicit local-normality or gauge-average compatibility claim was located in the inspected passages; this limited reading does not prove that such properties fail or cannot follow from the full source theory.

For a later homogeneous physical-GNS identification, the source note therefore distinguishes pointwise state limits and a generator core from the regularity needed to pass local weak-operator Haar averages through the actual representation. It calls for local normality or another explicit continuity-and-averaging argument, continuous gauge implementers on all endpoint groups, the full bounded local invariant algebra, and a justified reducing spectral restriction. I1 already supplies the specified homogeneous full-algebra state and generator; J2's cyclic/invariant identification belongs to its dyadic summable representation and does not automatically transfer. These are prospective transfer obligations only. The reading constructs no vacuum-reset bound, local tightness estimate, gauge-cyclic equality or new gap.

The AG3 passive comparison and spent  $9/4 \rightarrow 2 \rightarrow 3/2$  margins remain unchanged. AG2/AG3's complete-remainder and one-step results do not evaluate I1's source constants; their numerical extracted-reference gaps are not new numerical full-Hamiltonian gap theorems. The supplement supplies no new estimate, fixture, threshold, theorem application or continuum premise. It informed later planning without selecting a fourth or fifth goal at the time of the review. The source and reading-depth records, including the correction, are included in the manuscript source manifest.

The later admitted AJ1 investigation, Section 30, now supplies the previously prospective local-energy and normality argument, compatible finite-endpoint averaging, physical cyclic/fixed equality and closed reducing restriction for the actual I1 state. Its post-review clarification uses local normality and finite spectator columns to justify represented product-vector density; it does not assume operator-norm density of the algebraic tensor product. This mathematical admission is distinct from the earlier source-only note. AJ1's primary retrieval records and restricted proof-reading claims are bound through its gate; they do not silently change the 33-record survey ledger. AJ2, Section 31, then proves a nonzero homogeneous Wilson fluctuation using this normality and the actual multiplier's Haar-null levels. The coupling interval remains symbolic, and neither investigation evaluates a mass or the source stability constants.

## 37 Reproduction and artifact integrity

The frozen contracts, producer reports, exact outputs and skeptical gates in `research/round28/` govern each result. Scientific replay is separate from building this manuscript. From the repository root the addendum is built with

```
python3 -B papers/round28-addendum/build.py
```

All ten investigations have source-bound advisor gates and a reviewed chapter here. The final build requires those ten gates and verifies their complete bindings. The explicitly marked intermediate-document mode `--draft` remains available; it includes only admitted chapters and writes solely to its temporary build directory. Neither mode promotes a report to acceptance. Figures are computed from admitted exact evidence; floating-point conversion is only for rendering unless a figure explicitly states a different numerical scope.

The build uses ordinary  $\text{\LaTeX}$ , Poppler and the installed Python scientific stack. It installs no dependencies. Final publication requires rendering and visually reviewing every page after the last meaningful change. A clean compiler log, text-extraction check or targeted intermediate review does not replace that all-page inspection. The final artifact manifest binds the editable sources, figures, PDF and review record; a separate source manifest binds the admitted reports, outputs and gates. Hash agreement establishes integrity, not mathematical truth or independent experimental confirmation. Draft01 and the Round27 artifact bytes remain unchanged.

**Open target.** The admitted homogeneous representation and Wilson results, canonical readout certificates, finite-graph calculations and reference-only bounds retain their separate models and conclusions. Their combination does not supply the missing continuum construction, axioms or physical matching. The scope ledger states each investigation's result, possible application and remaining limitation; later results do not silently rewrite earlier admissions.

## Part III

# Round29: source comparison and next admitted investigations

## 38 Round29: models, sources and the ten-investigation contract

The inherited Round28 roadmap selected the full physical-scale dictionary (AL), numerical homogeneous stability (AM), and a specified bulk-boundary comparison (AN). Each second investigation responds to its first skeptical review. The fourth and fifth goal pairs are selected only after those six admissions. This continuation preserves fixed positive  $E_\star$  and  $\hbar$ , complete original interactions, the actual physical gauge completion, and the distinction between sufficient membership and an actual spectral gap. Two directions and a separate skeptic share inherited premises; their derivations and code are not independent physical observations or external human peer review.

The new literature pass records 13 selected primary sources in `research/round29/experts/sources.json`. The comparison to Gauvin [1] changes the novelty boundary described in Section 1. Earlier historical and unconventional source inventories remain available in the integrated addenda. No rumor, patent or institutional program is imported as a new Hamiltonian premise. The panel uses those historical methods as questions about inverse identification, full-system loading, representation, selection and actual observables.

Each following chapter carries its admitted forward derivation, common gate result, explicit limitations, and links to the separate reverse and skeptical records. An insufficient numerical radius or a failed bridge is retained as a limited completed investigation. A project label is not a missing proof step.

## 39 Round29 primary sources and reading boundaries

This table records the selected source review used by the advisor. A screened source is not an admitted theorem transfer, and the list is not an exhaustive state-of-the-art or priority search.

| Primary source                                                                                                                                                                     | Actual reading depth and use                                                                                                                                                                                                     | Transfer limitation                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Pinched Multi Affine Geometry and Confinement: Describing the Yang-Mills Mass Gap</a><br>Shoshauna Gauvin<br>arXiv:2503.15539v3                                        | Full named sections and associated equations; selected relevant pages, not every result<br>Closest checked overlap with I1/AJ/AK and AL: regulated gauge-sector gap, Wilson survival, spectral moments and conditional transfer. | SU(3) coefficients and hypotheses do not identify I1 SU(2). No continuum construction or boundary independence is supplied. General mechanisms cannot be renamed as new Hruday discoveries. |
| <a href="#">Supplemental Material for Pinched Multi Affine Geometry and Confinement: Describing the Yang-Mills Mass Gap</a><br>Shoshauna Gauvin<br>ancillary to arXiv:2503.15539v3 | A.6–A.11 read in full from downloaded PDF text; screenshots obtained for S4–S8; no claim to read all 67 pages<br>AM template and attribution comparison: support-dependent majorant, full shifted equation and cutoff removal.   | Pure-electric Haar reference and three-block supports differ from I1 selected-strip vacua and four-site stars. Numerical constants cannot transfer without a new proof.                     |
| <a href="#">Quantum Yang–Mills Theory</a><br>Arthur Jaffe, Edward Witten<br>official 14-page PDF at supplied URL                                                                   | Sections 3–5 and 6.1, 6.5 read; selected later caveat<br>Defines continuum existence, physical gap, axioms, gauge groups and short-distance identification.                                                                      | A fixed-lattice gap or compactness subsequence alone does not satisfy the stated problem.                                                                                                   |

| Primary source                                                                                                                                                                                                                                     | Actual reading depth and use                                                                                                                                                                                                                                                                                                              | Transfer limitation                                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Quasi-particles in weak perturbations of non-interacting quantum lattice systems</a><br>D. A. Yarotsky<br>arXiv:math-ph/0411042v1                                                                                                      | Definitions, Theorems 1–3 and Section 2 proof outline inspected<br>Rechecks I1 stability and resolvent premises.                                                                                                                                                                                                                          | Range-dependent constants remain unevaluated in inspected statements; the article refers to preceding proofs. It supplies no convenient numerical tau. |
| <a href="#">Local stability of ground states in locally gapped and weakly interacting quantum spin systems</a><br>Joscha Henheik, Stefan Teufel, Tom Wessel<br>Letters in Mathematical Physics 112, article 9 (2022); companion arXiv:2106.13780v3 | Definitions 1,2,6; Theorems 3,7; Lemma 8 and proofs read; unbounded-operator appendix caveat checked<br>Boundary comparison after a complete I1 dictionary.                                                                                                                                                                               | Requires its own symbolic smallness constant and common bulk ground-state hypotheses. Boundary deletion conventions require proof, not a name match.   |
| <a href="#">A stochastic analysis approach to lattice Yang–Mills at strong coupling</a><br>Hao Shen, Rongchan Zhu, Xi-angchan Zhu<br>arXiv:2204.12737v1; Communications in Mathematical Physics 400 (2023), 805–851                                | Abstract and publication metadata only<br>Separates Euclidean strong-coupling correlation/mixing results from the I1 Hamiltonian problem.                                                                                                                                                                                                 | Langevin clock and Euclidean measure cannot replace the physical Hamiltonian without an identifying map; full proof not audited.                       |
| <a href="#">U(N) lattice Yang-Mills in the 't Hooft regime</a><br>Ron Nissim<br>arXiv:2510.22788v1                                                                                                                                                 | Abstract and version history only<br>Current example combining a random-environment decomposition and cluster expansion when a direct curvature route fails.                                                                                                                                                                              | Different group and regime; no theorem or constants imported into I1.                                                                                  |
| <a href="#">Deconfinement For SO(3) Lattice Yang-Mills at Strong Coupling</a><br>Ron Nissim<br>arXiv:2605.16162v1                                                                                                                                  | Abstract and version history only<br>Checks that a lattice spectral question and Wilson confinement criterion are distinct targets.                                                                                                                                                                                                       | SO(3) center structure differs from SU(2); full proof not audited and no result transferred.                                                           |
| <a href="#">On Charge Conjugation, Correlations, Elitzur's Theorem and the Mass Gap Problem in Lattice SU(N) Yang-Mills Models in d=4 Dimensions</a><br>Paulo A. Faria da Veiga, Michael O'Carroll<br>arXiv:2509.03513v2                           | Abstract, Theorem 1 and concluding discussion inspected; full proofs not audited<br>Gauge-invariant observable and group-sector check for Wilson spectral claims.                                                                                                                                                                         | Its $N!=2$ charge-conjugation split does not carry over to SU(2); a local Haar-density term is not a spectral-gap proof.                               |
| <a href="#">Response theory for locally gapped systems</a><br>Joscha Henheik, Tom Wessel<br>arXiv:2410.10809v1                                                                                                                                     | Abstract, Section 1.1 limitation and Section 1.2 definitions inspected<br>Possible later local-response work after AN, with a new domain audit.                                                                                                                                                                                           | The inspected presentation restricts to finite-dimensional spaces and bounded operators; direct untruncated-rotor application is not admitted.         |
| <a href="#">Locality in Quantum Lattice Systems</a><br>Tom Wessel<br>University of Tübingen dissertation, 430 PDF pages                                                                                                                            | Title, abstract, contents, publication list and Section 3.8 selected passages only<br>Recent locality map and pointer to published HTW theorem.                                                                                                                                                                                           | No claim to have read all 430 pages. Thermal extensions have additional hypotheses and supply no automatic I1 quantitative radius.                     |
| <a href="#">On the dynamics of lattice systems with unbounded on-site terms in the Hamiltonian</a><br>Bruno Nachtergaele, Robert Sims<br>arXiv:1410.8174v1                                                                                         | Section3 model/F-norm setup and Theorem3.1 statement pp7–8; Theorem4.1 and proof pp11–13; selected continuity preliminaries. Not every preliminary result audited.<br>Actual infinite-dimensional onsite dynamics and norm thermodynamic approximation on bounded local operators, uniform compact physical times after clock conversion. | Does not establish the chosen state or full-algebra norm time continuity; AQ requires its own local-normal state/GNS argument.                         |

| Primary source                                                                                                                                                                                                                                                                     | Actual reading depth and use                                                                                                                                                                                                                                                                                                                      | Transfer limitation                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">Quasi-Locality Bounds for Quantum Lattice Systems. Part I. Lieb-Robinson Bounds, Quasi-Local Maps, and Spectral Flow Automorphisms</a><br>Bruno Nachtergaele, Robert Sims, Amanda Young<br>arXiv:1810.02428v2;<br>J.Math.Phys.60,061101(2019),<br>DOI10.1063/1.5095769 | Section3.2 setup and Theorem3.3 statement; Theorem3.4 and3.5 statements/proofs and Corollary3.6, printed pp24–28. Section7.1 setup and Theorem7.6 applicability warning screened. Not all106 pages.<br>Modern framework for AQ local dynamics with unbounded onsite Hamiltonians, bounded whole-star interactions, explicit topology distinction. | Theorem7.6 spin/bounded-generator gap continuity is not a direct rotor gap-transfer theorem. Full-B local norm convergence is not full-B norm time continuity. |

**Current admitted count: 10/10.** Publication requires the completed ten-gate source build.

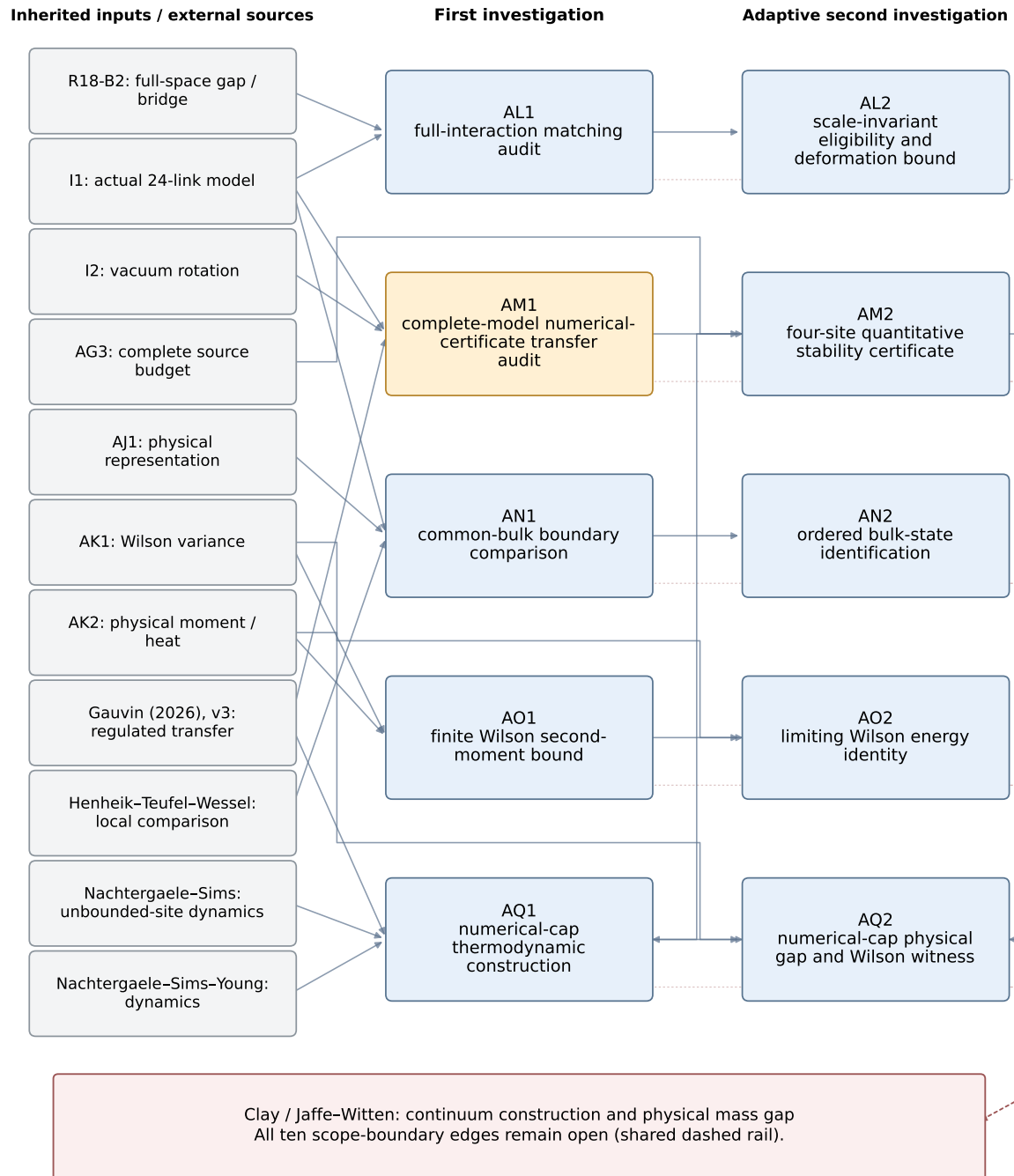


Figure 11: Current Round29 dependency projection. Solid arrows retain the canonical graph's scoped input relations; a dependency does not transfer every parent conclusion. Amber marks the limited AM1 outcome. External sources retain their authorship. The ten continuum scope-boundary edges are grouped on the labelled dashed rail: no branch reaches a continuum construction. The full graph retains equation nodes, earlier ancestors, future goals and source links; the companion projection record binds its exact source hash.

research/round29/network.json  
figures/round29-dependencies.json

## 40 HNM AL1: full-interaction matching audit\*\*

**Statement alias HNM-P-AL1.** This identifier denotes the scoped mathematical claim admitted by this chapter's gate. It adds no assumption and establishes no literature-priority claim.

**Admission: accepted within scope.** The complete uniform SU(2) bulk dictionary gives  $r = 4/g^4$ ,  $\tau = 96/g^4$  and local norm  $672/g^4$ . The actual selected bridge requires  $g^4 \geq 32$ ; the frozen  $a_n = a_0/n$ ,  $g_n^2 = 1/n$  path violates this sufficient condition for every  $n \geq 1$ . Finite whole-star and all-contained-plaquette boundaries differ.

The typeset body below preserves the source-bound forward derivation. Its prospective language is historical; this boxed gate records the final reviewed verdict. The separately authored reverse and skeptical records remain linked. The common admission and attributed refinements are determined by the gate, not by a renamed equation.

Human project author: **Hruday N M (BUNZEEY)**. HNM labels identify this project's derivation; priority is unverified. The candidate was supplied by the advisor. This forward derivation and executable were completed without reading current reverse or skeptic results.

**Result: the frozen candidate has an exact bulk coefficient dictionary but the specified weak-bare-coupling path is outside the actual I1 sufficient regime.** This is failure of a sufficient certificate, not absence of a mass gap. The bridge restriction and whole-star boundary must be retained.

### 40.1 Complete dictionary

Use the untruncated Haar link Hilbert spaces with the original Casimir and all endpoint gauge actions. With the primary convention of Bauer et al. [12], arXiv:2307.11829, p.6, Eqs.55–56, and  $W_p = Tr U_p/2$ , the raw SU(2) magnetic term is  $\lambda \sum_p (1 - W_p)$ , where

$$\alpha = \frac{g^2}{2a}, \quad \lambda = \frac{2}{g^2 a}, \quad r = \frac{\lambda}{\alpha} = \frac{4}{g^4}. \quad (\text{HNM-AL1.1})$$

The identity link and Wilson map preserves Hilbert space, every link Casimir, all endpoint actions, and all bounded local physical observables. On a finite fixed link/face inventory, bounded magnetic multiplication leaves the Casimir-sum operator domain unchanged; equal coefficients give equal operators on that domain. The additive magnetic scalar is  $\lambda N_p$ ; it is not the interacting ground energy.

The I1 cell owns all three outgoing links from each of its eight  $(\mathbf{r}, \mathbf{s}, \mathbf{k})$  tail vertices,  $r = 0, \dots, 3, s = 0, 1$ : 24 links, partitioned into ten selected-strip and fourteen free reference links. Each cell anchors 8 xy, 8 xz and 8 yz faces. Selected xy classes are  $(r, s) = (0, 0), (1, 0), (2, 0)$ , giving two endpoints and one bridge. The remaining five xy, eight xz and eight yz classes are the 21 omitted faces. Unique Euclidean division of coordinates proves completeness at every translated cell; the finite checker audits these phase classes.

For a face with base p in directions i,j, its exact owner support is  $\{\pi(p), \pi(p + e_i), \pi(p + e_j)\}$ . Its fourth vertex is not another link tail. The union of all omitted owner supports is  $S = \{0, e_x, e_y, e_z\}$ . The uniform bulk coefficient assignment matches I1 only when

$$\lambda_L = \lambda_R = \mu = \nu = \lambda, \quad \tau = \frac{24\lambda}{\alpha} = \frac{96}{g^4}, \quad \epsilon = 7|\tau| = \frac{672}{g^4}. \quad (\text{HNM-AL1.2})$$

The exact I1 inequalities therefore read

$$r \leq \frac{1}{2} \quad (\text{ends}), \quad r \leq \frac{1}{8} \quad (\text{bridge}), \quad \frac{672}{g^4} < c_1(S), \quad \frac{672c_2(S)}{g^4} < \frac{1}{2}. \quad (\text{HNM-AL1.3})$$

Here  $c_1, c_2$  remain the positive, unevaluated source constants. Equivalently the selected bridge requires  $g^4 \geq 32$ , and the omitted interaction requires  $g^4 > 672/c_1$  and  $g^4 > 1344c_2$ . No convenient numerical value is assigned to either source constant. The inherited R18-B2 bound  $g^4 \geq 32/3$  concerned a different finite-graph coefficient box. The stronger bridge condition here and the omitted-coefficient dictionary are the incremental audit; the general weak-coupling obstruction is inherited in kind.

## 40.2 Path and fixed clock

For the prospectively frozen sequence  $a_n = a_0/n, g_n^2 = 1/n$ ,

$$\alpha_n = \frac{1}{2a_0}, \quad \lambda_n = \frac{2n^2}{a_0}, \quad r_n = 4n^2, \quad \tau_n = 96n^2, \quad \epsilon_n = 672n^2. \quad (\text{HNM-AL1.4})$$

Thus  $\alpha/E_\star$  is fixed and positive, but even the endpoint condition fails for every positive integer  $n$ ; the bridge condition fails more strongly. For any fixed positive finite  $c_1, c_2$  both omitted inequalities eventually fail. This is an explicit algebraic test path, not a claim that this is the physically correct renormalization trajectory. More generally every path with  $g \rightarrow 0$  eventually violates the bridge condition.

Let  $\tilde{H}$  denote the finite I1 normalized Hamiltonian, including the selected-strip ground subtraction, and let  $\tilde{E}$  be its actual ground energy. On a matched finite face inventory,

$$H_{\text{KS}} = (\alpha/8)\tilde{H} + \sum_b E_{\text{strip},b} + \lambda N_p.$$

Consequently its centered operator is exactly  $(\alpha/8)(\tilde{H} - \tilde{E})$ ; the frequency generator is this operator divided by the same fixed  $\hbar$ . The raw scalar and selected reference scalar cancel under actual centering. Replacing the frequency by normalized  $G/\hbar$  changes the clock unless  $\alpha/8 = 1$  in already fixed units. Positive  $E_\star$  is a physical reference, never a regulator chosen as zero.

## 40.3 Boundary and model limits

The preceding equality uses the same retained inventory. I1 retains an omitted anchor group only if its entire star lies in the coarse volume. For  $B = \{0, e_z\}$ , no whole star is retained, although ten omitted faces anchored at zero have their actual supports in B. Thus I1's boundary prescription is not automatically the ordinary all-contained-plaquette boundary. The checker derives this exact mismatch. Deep bulk coefficient agreement supplies no boundary-state identification, which remains AN's obligation.

An interior local site belongs to four star anchors  $x, x - e_x, x - e_y, x - e_z$ . Dropping incoming stars loses three interaction groups. A summable profile is not the uniform omitted interaction: over  $N$  complete anchors the same-sign norm budget is  $7N|\tau|$ , not bounded independently of  $N$ . Neither substitution repairs (HNM-AL1.3).

## 40.4 Evidence and interpretation

`check.py` independently constructs face phases/supports and computes all scale identities using exact rational arithmetic. Its controls reject omitting the bridge,  $\tau = r$ , lost incoming stars, a

wrong electric coefficient, normalized-clock replacement, zero reference energy and summable-profile substitution. Finite arithmetic checks audit the all-n argument; they are not physical simulations or observations.

Newton's analysis/synthesis method here is the explicit observable/domain dictionary followed by the all-n consequence. Tesla's whole-device method is the retention of omitted plaquettes, incoming stars and the physical clock. Neither historical analogy supplies a modern theorem premise.

Primary reading: Bauer et al. PDF introduction pp.1–2 and Hamiltonian definitions p.6 were freshly checked on 2026-09-22; their truncation numerics are not used. I1 and R19-A1 are inherited full reports. The local stability constants, matched continuum construction and continuum mass gap remain unresolved.

Reproduce from the repository root:

```
python -B research/round29/forward/a11/check.py \
  --output /absolute/new/output
```

## Final admitted limits and evidence

- This is failure of a specified sufficient certificate, not a no-gap or continuum-impossibility theorem.
- The positive source constants  $c_1(S)$ ,  $c_2(S)$  remain unevaluated.
- Bulk interaction equality does not identify different boundary-state limits; common energy and actual ground centering must be retained.
- General weak-coupling obstruction is inherited from R18-B2; the full I1 dictionary, bridge cap and boundary audit are the incremental result. Scientific priority unverified.

research/round29/advisor/a11-gate.json

research/round29/reverse/a11/report.md

research/round29/skeptic/a11.md

*Sequencing note.* Prospective statements in this frozen derivation describe the moment of its admission. The final Round29 roadmap below governs subsequent work.

## 41 HNM AL2: scale-invariant eligibility and deformation bound\*\*

**Statement alias HNM-P-AL2.** This identifier denotes the scoped mathematical claim admitted by this chapter's gate. It adds no assumption and establishes no literature-priority claim.

**Admission: accepted within scope.** Common positive energy multiplication and scalar shifts preserve all sufficient-membership ratios. With electric  $\alpha$  fixed, a nonnegative common magnetic multiplier  $s$  belongs to the inherited coefficient domain exactly when  $s \leq g^4/32$ ,  $s < c_1 * g^4/672$  and  $s < g^4/(1344 * c_2)$ . Therefore along  $g \rightarrow 0$  any such  $s$  tends to zero.

The typeset body below preserves the source-bound forward derivation. Its prospective language is historical; this boxed gate records the final reviewed verdict. The separately authored reverse and skeptical records remain linked. The common admission and attributed refinements are determined by the gate, not by a renamed equation.

Human project author: **Hruday N M (BUNZEEY)**. This is the independent forward derivation under the frozen AL2 contract, before reading the current reverse or skeptic results. The label denotes a project result, with scientific priority unverified.

**A common positive energy rescaling or additive scalar does not repair the AL1 coupling-ratio obstruction.** Changing the magnetic coefficient independently can enter the sufficient region, but is an action deformation. The following is an if-and-only-if statement about inherited sufficient membership, not a criterion for an actual spectral gap.

Let  $g, a, E_*, \hbar$  be strictly positive, let  $\alpha = g^2/(2a)$ ,  $\lambda = 2/(g^2a)$ , and let  $c_1, c_2$  denote the positive unevaluated I1 source constants. For  $k > 0$  and real  $C$ , the operator  $H' = kH + CI$  has  $\alpha' = k\alpha$ ,  $\lambda' = k\lambda$ ,  $\delta' = k\delta$ , so

$$\frac{\lambda'}{\alpha'} = \frac{\lambda}{\alpha}, \quad \tau' = \frac{24\lambda'}{\alpha'} = \tau, \quad \epsilon' = 7|\tau'| = \epsilon. \quad (\text{HNM-AL2.1})$$

The scalar cancels under actual ground subtraction:  $E'_0 = kE_0 + C$  and  $H' - E'_0 = k(H - E_0)$ . Thus a scalar alone does not change gaps. Positive common multiplication does scale physical gaps, even though it preserves eligibility ratios. At fixed  $\hbar$  and the same physical time it changes evolution; equivalent time parameters obey  $t' = t/k$ . At fixed  $E_*$ ,  $\alpha'/E_* = k\alpha/E_*$ , so this is not secretly the same fixed physical energy calibration. A mere units conversion would convert the reference unit as well and would still preserve every eligibility ratio.

Now hold  $\alpha$  fixed and take  $\lambda_s = s\lambda$ , uniformly on both selected and omitted original plaquettes. For  $s > 0$ , exact AL1 membership is equivalent to

$$0 < s \leq \frac{g^4}{32}, \quad s < \frac{c_1 g^4}{672}, \quad s < \frac{g^4}{1344 c_2}. \quad (\text{HNM-AL2.2})$$

If the word suppression is used, impose additionally  $s \leq 1$ ; that linguistic restriction is separate from the theorem inequalities. Necessity follows from the selected bridge ratio  $4s/g^4 \leq 1/8$ , normalized omitted norm  $672s/g^4 < c_1$ , and the positive half-gap condition  $672c_2s/g^4 < 1/2$ . Sufficiency follows by reversing these steps: the endpoint condition is weaker than the bridge condition, all complete support/domain/boundary premises remain those of AL1, and the two strict source inequalities are exactly retained. Equality is permitted only at the selected bridge cap, not at either omitted cap.

In particular every eligible  $s(g)$  along  $g \rightarrow 0$  obeys  $0 < s(g) \leq g^4/32$ , hence tends to zero. On AL1's exact sequence,  $s_n \leq 1/(32n^2)$ . The needed change is therefore not a uniformly small

relative correction to the magnetic coefficient: its retained fraction vanishes. Writing an HNM label or auxiliary parameter does not produce physical equivalence. This does not exclude a continuum construction through a different proof regime.

At  $s = 0$ , the formal inequalities hold, but the nonzero omitted interaction requested for I1 is absent. The model becomes the all-free electric reference (selected terms also vanish), with its inherited tensor-product vacuum and gap; it is a degenerate control, not a successful nonzero homogeneous bridge. Negative  $s$  is outside the declared nonnegative uniform branch. A zero or negative  $k$  is not an allowed positive common energy rescaling;  $k = 0$  destroys the reference gap and  $k < 0$  reverses lower-boundedness. Zero physical normalization is rejected.

The exact checker verifies the two formulations of membership over rational diagnostic constants and coefficients, including isolated strict-threshold equality controls. Those rational constants are **test parameters**, never evaluated Yarotsky constants. It separately tests scalar centering and physical-clock mismatch, the omitted bridge, and the misleading identification of magnetic-only suppression with a units conversion. No SU(2) truncation or numerical mass calculation is performed.

The incremental result beyond AL1 is the invariance statement and the required deformation fraction, rather than a new mass-gap theorem. Newton's inverse-map discipline identifies what scale changes cannot alter; Tesla's energy/clock accounting keeps the physical generator fixed. The next missing premise is still an actual full-model quantitative stability interval or a different matched continuum strategy.

Reproduce from the repository root:

```
python -B research/round29/forward/al2/check.py \
    --output /absolute/new/output
```

## Final admitted limits and evidence

- The iff concerns the inherited sufficient domain, not necessary conditions for an actual gap.
- The  $s = 0$  case is a separate decoupled electric control, not a nonzero magnetic bridge.
- Common energy multiplication changes physical gaps and requires reciprocal time rescaling;  $E_*$  and  $\hbar$  remain fixed.
- The c1,c2 constants are unevaluated. Independent magnetic suppression changes the action and does not prove physical equivalence or a continuum limit. Scientific priority unverified.

research/round29/advisor/al2-gate.json

research/round29/reverse/al2/report.md

research/round29/skeptic/al2.md

*Sequencing note.* Prospective statements in this frozen derivation describe the moment of its admission. The final Round29 roadmap below governs subsequent work.

## 42 HNM AM1: complete-model numerical-certificate transfer audit\*\*

**Current extension: HNM-C-AM2.** The historical limitation below remains the scope of this original result. The later, separately gated statement in Section 43 records the extension: For every nonempty finite I1 complete-factor volume with the actual selected-strip reference and full whole-star omitted interaction, both signs of  $|\tau| \leq 1/100000000$  admit a unique ground and a physical gap at least  $\alpha/16$ . The proof derives the four-site creation majorant  $G(t)=16 \exp(8t)(1+10t)$ , full scalar fixed point, centered excited-sector exclusion and removal of onsite representation cutoffs. The forward full-Hilbert-space strengthening is separately proved and reviewed. Its own model and state assumptions remain required.

**Admission: limited.** The source-specific audit constructs a nonzero physical Wilson excitation within one complete 24-link factor and proves the grouped site budget  $J = 28|\tau|$ . Gauvin’s four-active-fine-link argument, Haar vacuum and at-most-three-site support do not transfer unchanged to the actual selected-strip reference and four-site grouping. The numerical I1 radius remains insufficient in this investigation.

The typeset body below preserves the source-bound forward derivation. Its prospective language is historical; this boxed gate records the final reviewed verdict. The separately authored reverse and skeptical records remain linked. The common admission and attributed refinements are determined by the gate, not by a renamed equation.

Human project author: **Hruday N M (BUNZEEY)**. Independent forward report under the frozen AM1 contract. Current reverse and skeptic work were not read. **Verdict: the inspected explicit SU(3) proof template cannot be copied into I1 with its constants; the numerical I1 radius remains insufficient in this loop.** The new result is the exact failed dictionary, not another claim that unspecified constants were just discovered.

### 42.1 What the older source still leaves unevaluated

I1 uses  $h_b \geq I - P_b$ , a complete 24-link factor,  $S = \{0, e_x, e_y, e_z\}$ , and  $\|\phi_b\| = 7|\tau|$ . Theorem1 of Yarotsky [17], math-ph/0411042v1 introduces range-dependent  $c_1, c_2$ . Section2 uses variable constants; its Lemma1 requires a sufficiently small interaction to obtain weighted ground-expansion coefficients. The required fixed-point radius/contraction constant is not evaluated. Equation14 then bounds a dressed commutator with a further unspecified factor; the resolvent argument of Eq15 needs that factor times the interaction norm below one. The I2 report already isolated these stops. This fresh source read does not convert them into numbers or a new obstruction.

### 42.2 A precise template comparison

Fresh reading of Gauvin [1], arXiv:2503.15539v3, companion supplement A.6–A.11, identifies a different reference: three outgoing **fine links** at each fine vertex, constant Haar vacuum and pure-electric onsite operator. Lemma2.14 separately counts at least four nontrivially labelled fine links from graph girth and at least one active link for each excited outgoing block. It does **not** say every physical excitation occupies four blocks. Lemma2.15 uses an interaction support of at most three fine blocks, a per-site interaction sum, and physical-sector electric inverse estimates. The numerical SU(3) result follows only with that dictionary and its complete fixed-point, spectral-exclusion and cutoff-removal argument.

I1 instead uses the selected-strip ground within each 24-link coarse factor, and its omitted interactions are grouped into four-site stars. Its onsite theorem provides normalized gap at least one; it does not provide the template’s stronger inverse bound with a four-link electric floor.

Its vacuum is generally not the constant Haar vector on selected links. Replacing source fine sites with coarse labels therefore changes the hypotheses. Individual omitted faces have at most three coarse owners, but this is not the contracted complete four-site grouping; changing the decomposition would require a new norm and source dictionary.

The source's dimensionless SU(3) constants also depend on its Casimir, trace range and support count. In the present normalization the SU(2) fundamental Casimir is 3/4, while the corresponding SU(3) value is 4/3; SU(2)'s normalized real trace has range  $[-1, 1]$ , not the SU(3) fundamental real-trace range. Copying 1/648 into I1 is unsupported. The source is treated as a proposed proof template, not a verified continuum solution.

### 42.3 An actual one-coarse-factor physical excitation

Fix a selected xy plaquette p entirely inside the strip of factor b. Let  $\Omega_b$  be the actual normalized onsite reference ground, and set

$$m_p = \langle \Omega_b, W_p \Omega_b \rangle, \quad v_b = (W_p - m_p) \Omega_b, \quad v = v_b \otimes \bigotimes_{c \neq b} \Omega_c. \quad (\text{HNM-AM1.1})$$

The finite compact rotor ground is smooth and strictly positive: the positive heat kernel of the connected compact product together with bounded real potential gives a positivity-improving semigroup and its unique positive ground. The selected strip ground times the free Haar factors has this property. Since  $W_p$  is continuous and nonconstant, its variance in the full-support measure  $|\Omega_b|^2 dm$  is strictly positive. Thus  $v_b \neq 0$  and  $v_b$  is perpendicular to  $\Omega_b$ . Gauge transformations at all original endpoints fix the ground and the closed Wilson multiplication, so  $v$  is physical. Its smoothness also gives the onsite operator domain.

With actual coarse reference projectors  $P_c = |\Omega_c \rangle \langle \Omega_c|$ , the vector satisfies  $Q_b v = v$  and  $Q_c v = 0$  for every  $c \neq b$ . Exactly **one coarse factor** is excited. The selected square still has four distinct fine vertices and four links; there is no contradiction. This rejects an imported assertion that physicality forces four excited coarse factors. It does not prove the falsity of any proposed numerical lower energy bound: the energy of this vector has not been bounded above by that number.

At zero selected coefficients, a useful exact subcase is  $\Omega_b = 1$ ,  $m_p = 0$ ,  $\|v_b\|^2 = 1/4$ : condition on three plaquette links and the remaining holonomy is Haar SU(2), whose scalar coordinate has second moment one quarter. This subcase audits the variance mechanism. The positivity argument, rather than substituting Haar moments for the dressed state, proves the nonzero-coefficient statement.

### 42.4 Complete incidence and norm conversion

There are three selected and 21 omitted anchored face classes. Each omitted group has norm  $M = 7|\tau|$ . A bulk coarse site  $x$  belongs to exactly the anchors  $x, x - e_x, x - e_y, x - e_z$ . Thus the source-style indexed site sum for this actual grouping is

$$J_{\text{HNM}} = \sup_x \sum_{b: x \in b+S} \|\phi_b\| = 4M = 28|\tau|. \quad (\text{HNM-AM1.2})$$

Boundary clipping can only decrease it. The 84 face occurrences in these four grouped stars are retained in the indexed budget even though only 49 individual faces actually touch an interior factor. Contracting supports after grouping would change the declared norm. Counting only the outgoing anchor incorrectly divides  $J$  by four.

Scalar and retained diagonal terms are independent obstacles to a shortcut: an interaction splitting must retain  $cI$ , the centered complement term and the mixing term. Removing a scalar changes raw ground energy while leaving centered gaps; removing the complement diagonal generally changes gaps. The exact diagnostic matrix in the checker separates these effects. It is not an  $SU(2)$  truncation.

## 42.5 Outcome and next missing premise

The executable reconstructs the one-factor square and complete star incidence, checks the exact Haar subcase and unit conventions, and runs the wrong-transfer controls. No unknown source constant is replaced by a convenient rational. The template's mathematical proof is not disproved; its direct **transfer** has missing premises. A later loop could prospectively rederive the creation estimates for I1's actual four-site support and available onsite inverse, including every stage of its fixed-point and spectral argument. That work has not been executed here.

Source reading: Yarotsky Theorem1 and Section2 Lemma1/Eqs14–15; Gauvin supplementary A.6–A.11, downloaded from [https://arxiv.org/src/2503.15539v3/anc/PinchedMultiAffineGeometry\\_Supplement.pdf](https://arxiv.org/src/2503.15539v3/anc/PinchedMultiAffineGeometry_Supplement.pdf). Historical methods enter through complete mechanism and inverse-dictionary checks, not as mathematical authority.

Reproduce from the repository root:

```
python -B research/round29/forward/am1/check.py \
  --output /absolute/new/output
```

## Final admitted limits and evidence

- The one-factor witness rejects an incorrect coarse-factor inference, not the external source's actual fine-link statement or its numerical energy bound.
- The qualitative I1 theorem and I2 missing-constant ledger remain inherited; no convenient  $c_1$  or  $c_2$  has been inserted.
- This audit does not prove that explicit constants cannot be derived, that the actual gap fails, or that the external paper is false.
- Scientific priority is unverified and no continuum construction follows.

research/round29/advisor/am1-gate.json

research/round29/reverse/am1/report.md

research/round29/skeptic/am1.md

*Sequencing note.* Prospective statements in this frozen derivation describe the moment of its admission. The final Round29 roadmap below governs subsequent work.

## 43 HNM AM2: four-site quantitative stability certificate\*\*

**Statement alias HNM-T-AM2.** This identifier denotes the scoped mathematical claim admitted by this chapter's gate. It adds no assumption and establishes no literature-priority claim.

**Current extension: HNM-C-AQ1.** The historical limitation below remains the scope of this original result. The later, separately gated statement in Section 48 records the extension: At both signs of  $|\tau| \leq 1/100000000$ , actual centered full-Z3 whole-star boxes have a subsequence defining a compatible locally normal gauge-invariant stationary state on the norm closure of full bounded local algebras. The actual reference reset bound is  $56|\tau||R|$ ; compact-resolvent cutoffs prove local trace-norm precompactness. The matched unbounded-onsite dynamics theorem and separate local-normality argument supply strongly continuous GNS evolution with a nonnegative self-adjoint physical energy generator. The forward physical cyclic restriction is reducing; full joint-fixed completion and its positive gap remain for AQ2. Its own model and state assumptions remain required.

**Current extension: HNM-C-AQ2.** The historical limitation below remains the scope of this original result. The later, separately gated statement in Section 49 records the extension: For the actual AQ1 centered full-Z3 subsequential state at both signs of  $|\tau| \leq 1/100000000$ , the complete original-endpoint gauge-fixed space equals the invariant-local cyclic completion and reduces the physical energy generator. Centered Fourier tests including zero give  $H_{\text{phys}} \geq (\alpha/16)(I-P_{\Omega})$  and a simple vacuum in this representation. The full-GNS strengthening explicitly uses AM2's separately reviewed full-Hilbert finite gap. For the original xz Wilson in the same state, the actual 48-link/36-endpoint cover has seven incident stars, local reference energy  $\leq 98|\tau|$  and variance  $\geq 61999/250000 > 1/5$ . The reverse proof separately sharpens that floor to  $3099951/12500000$ . Its own model and state assumptions remain required.

**Admission: accepted within scope.** For every nonempty finite  $I1$  complete-factor volume with the actual selected-strip reference and full whole-star omitted interaction, both signs of  $|\tau| \leq 1/100000000$  admit a unique ground and a physical gap at least  $\alpha/16$ . The proof derives the four-site creation majorant  $G(t) = 16 \exp(8t)(1+10t)$ , full scalar fixed point, centered excited-sector exclusion and removal of onsite representation cutoffs. The forward full-Hilbert-space strengthening is separately proved and reviewed.

The typeset body below preserves the source-bound forward derivation. Its prospective language is historical; this boxed gate records the final reviewed verdict. The separately authored reverse and skeptical records remain linked. The common admission and attributed refinements are determined by the gate, not by a renamed equation.

Human project author: **Hruday N M (BUNZEEY)**. The advisor/panel supplied the proposed route and constants. This independent forward derivation checks the entire implication before reading current reverse or skeptic results. Creation expansions are established methods; scientific priority of this application is unverified.

**Proposed admission:** for every finite  $I1$  complete-factor volume, with its actual selected reference, whole-star omitted interactions and both signs of  $|\tau| \leq 10^{-8}$ , the full untruncated Hamiltonian has a unique ground and centered gap at least  $\alpha/16$ . Its physical restriction has the same ground and this lower gap. Every nonempty complete-factor volume has nonzero physical excited vectors; the empty volume has none and its gap exclusion is vacuous. This is an alternative explicit finite-volume theorem, not an evaluation of Yarotsky's named constants or a continuum result.

### 43.1 Actual model and available inverse

Write the normalized operator on a finite set  $\Lambda$  of complete 24-link factors as

$$H = H_0 + V, \quad H_0 = \sum_{x \in \Lambda} h_x, \quad h_x \Omega_x = 0, \quad h_x \geq Q_x = I - |\Omega_x\rangle\langle\Omega_x|, \quad V = \sum_{b: b+S \subset \Lambda} \phi_b. \quad (\text{HNM-AM2.1})$$

All 21 omitted faces per retained anchor are present,  $S = \{0, e_x, e_y, e_z\}$ , and  $\|\phi_b\| = 7|\tau|$ . The indexed per-site sum is

$$J := \max_u \sum_{X \ni u} \|V_X\| \leq 28|\tau| \leq J_0 = \frac{7}{25000000}. \quad (\text{HNM-AM2.2})$$

Here each retained  $X$  is its complete four-site star, including incoming anchors. The interaction is bounded in each finite volume, though not uniformly bounded as a global extensive operator. All onsite domains and scalar selected ground subtractions are inherited from  $I_1$ . No interaction term is deleted.

For a nonempty excitation set  $M$ , put  $P_M = (\otimes_{x \in M} Q_x)(\otimes_{y \notin M} P_y)$  and identify its range with the excited factor tensor product. On it  $H_M = \sum_{x \in M} h_x \geq |M|$ . This is a full-Hilbert-space inequality and needs no extra physical four-link floor. For real  $|z| < 1/2$ ,

$$\|H_M^{-1}\| \leq |M|^{-1}, \quad \|(H_M - z)^{-1}\| \leq 2|M|^{-1}. \quad (\text{HNM-AM2.3})$$

First impose finite-rank full spectral cutoffs of each  $h_x$  containing its vacuum. The proof below is finite dimensional and constants do not depend on these ranks. Cutoff removal is proved in Section6.

### 43.2 Creation algebra and complete anchored estimate

For a vector  $c_I \in \otimes_{x \in I} Q_x \mathcal{H}_x$ , define the identity-extended creation  $\hat{c}_I = |c_I\rangle\langle\Omega_I| \otimes I_{I^c}$ . Two creations commute: disjoint supports commute by tensor factorization, and overlapping supports have product zero in both orders. The latter follows because the vacuum bra annihilates an excited ket at any common factor, first for simple tensors and then by finite-dimensional linearity. Set

$$C = \sum_{I \neq \emptyset} \hat{c}_I, \quad \|c\|_a = \max_u \sum_{I \ni u} \|c_I\|.$$

This is a norm on the finite-dimensional direct sum of coefficient spaces. For collections  $c_1, \dots, c_k$ , define

$$(L_k(c_1, \dots, c_k))_M = H_M^{-1} P_M \text{ad}_{C_1} \cdots \text{ad}_{C_k}(V) \Omega_0, \quad \Omega_0 = \bigotimes_x \Omega_x. \quad (\text{HNM-AM2.4})$$

We derive, rather than copy, the proposed estimate

$$\|L_k(c_1, \dots, c_k)\|_a \leq J L_k^{\text{num}} \prod_j \|c_j\|_a, \quad L_k^{\text{num}} = 16 \cdot 8^k (1 + 5k/4). \quad (\text{HNM-AM2.5})$$

Fix an indexed interaction  $V_X$ ,  $|X| \leq p = 4$ , and creation supports  $I_j$ . Every  $I_j$  must meet  $X$ : commuting adjoint actions allow a disjoint creation to be moved innermost, where it commutes with  $V_X$ . Write  $N = \bigcup_j I_j$ . In any nonzero term, outside  $X$  a singly touched factor is excited and a multiply touched factor annihilates the term. Therefore  $N \setminus X \subset M \subset N \cup X$ . There are at most  $2^p = 16$  possible output support sets  $M$ , independent of Hilbert dimensions, and  $|I_j| \leq |M| + p$ . The commutator expansion has  $2^k$  products; each projected product norm is at most  $\|V_X\| \prod_j \|c_{j, I_j}\|$ .

For a fixed root  $u$  in  $M$ , first count terms with  $u$  in  $X$ . Use inverse bound one, sum  $X$  containing  $u$  at cost  $J$ , and for every collection use  $\sum_{I: I \cap X \neq \emptyset} \|c_I\| \leq p \|c\|_a$ . This contributes  $2^p (2p)^k J \prod_j \|c_j\|_a$ . Otherwise  $u$  lies in at least one  $I_l$ ; overcount all  $k$  choices of  $l$ . Since  $M$  is nonempty,  $1/|M| \leq (p+1)/|I_l|$ .

Sum interactions meeting  $I_l$  at cost  $J|I_l|$ , canceling that denominator; sum the remaining collections meeting  $X$  at cost  $p(k-1)$ , and sum  $I_l$  containing  $u$  with its anchored norm. This contributes  $2^p (2p)^k k(p+1)/p$  times the same  $J$  product. Adding the two root placements and setting  $p = 4$  gives (HNM-AM2.5). All sums are finite; repeated interaction supports and repeated creation histories are counted, not canceled to improve the bound.

The nested commutator is zero for  $k > 2p = 8$ . In each expanded product, one side of  $V_X$  would contain at least five creations all meeting the four-site  $X$ . Two share a site, so that product is zero. An exact four-qubit algebra fixture gives  $\text{ad}_C^8(V)\Omega = 8!|1111\rangle$  and  $\text{ad}_C^9(V) = 0$ ; thus importing a support-three termination at order six would be wrong. This fixture audits the algebra, not a rotor truncation or a numerical spectrum.

### 43.3 Fixed point, scalar energy and all orders

Because creators commute,  $[C, H_0] = -\sum_I \widehat{H_I c_I}$  and its next creation commutator vanishes. Thus

$$e^C(H_0 + sV)e^{-C} = H_0 - \sum_I \widehat{H_I c_I} + se^C V e^{-C}. \quad (\text{HNM-AM2.6})$$

All exponential products are finite polynomials:  $C$  is nilpotent since nonzero creator products have disjoint nonempty supports. For real  $|s| \leq 1$ ,  $\psi(s) = e^{-C(s)}\Omega_0$  is an eigenvector precisely when

$$c = s \sum_{k=0}^8 \frac{L_k(c, \dots, c)}{k!}, \quad E(s) = s \langle \Omega_0, e^{C(s)} V e^{-C(s)} \Omega_0 \rangle. \quad (\text{HNM-AM2.7})$$

The vacuum coefficient of  $\psi$  is one, so it never vanishes. The scalar equation is retained even when the original reference vacuum mean of  $V$  is zero. All diagonal pieces of  $V$  remain inside every commutator and the scalar; this is not an offdiagonal-only model.

The positive all-order majorant and derivative are

$$G(t) = \sum_{k \geq 0} \frac{L_k^{\text{num}} t^k}{k!} = 16e^{8t}(1 + 10t), \quad G'(t) = 16e^{8t}(18 + 80t). \quad (\text{HNM-AM2.8})$$

For  $R = 1/64$ ,  $\exp(1/8) < 8/7$ , by comparison of its positive power series with the geometric series (strict already at second order). Hence

$$\begin{aligned} G(R) &< 148/7, & G'(R) &< 352, \\ J_0 G(R) &< 148/25000000 < R, \\ J_0 G'(R) &< 77/781250, & 2J_0 G'(R) &< 77/390625 < 1. \end{aligned} \quad (\text{HNM-AM2.9})$$

The multilinear bound proves the ball self-map. Telescoping each  $k$ -linear term between two coefficient collections in the radius- $R$  ball yields the Lipschitz bound  $JG'(R)$ , with all  $k$  possible replacements counted. Banach's theorem supplies a unique coefficient solution there. Subtracting the fixed-point equations and absorbing the common Lipschitz factor gives continuity in  $s$ . Therefore  $E(s)$  is a continuous real eigenvalue branch. At  $J = 0$ ,  $c = 0$  and  $E = 0$  exactly.

### 43.4 Excited-sector exclusion: construction alone is not enough

Suppose a second eigenvector  $\phi$  of  $H_0 + sV$  has eigenvalue  $E(s) + z$ , real  $|z| < 1/2$ , and is independent of  $\psi$ . Decompose

$$e^{C(s)}\phi = b_0\Omega_0 + \sum_{M \neq \emptyset} b_M, \text{ and set } B = \sum_M \widehat{b_M}.$$

Then  $\phi = (b_0 + B)\psi$  because creations commute, and  $b$  is nonzero. Subtract the eigenvector equations, conjugate by  $e^C$ , and project each nonempty sector. Creation commutation cancels the creation term in (HNM-AM2.6), giving

$$b_M = s(H_M - z)^{-1}P_M[B, e^{C(s)}V e^{-C(s)}]\Omega_0. \quad (\text{HNM-AM2.10})$$

The scalar equation may additionally constrain  $b_0$ , but is unnecessary for excluding  $b! = 0$ . Expanding the remaining commutator keeps all terms through total creation order eight. The shifted inverse multiplies (HNM-AM2.5) by at most two. Treating  $b$  as one input and summing the other orders gives

$$\|b\|_a \leq 2JG'(R)\|b\|_a < \|b\|_a, \text{ a contradiction.}$$

Thus the branch is simple and no other eigenvalue lies within one half. At  $s = 0$  it is the unique lowest eigenvalue; continuous finite-dimensional eigenvalues cannot cross this isolated branch. It remains the ground for every real  $|s| \leq 1$ , with full-space gap at least one half. The argument includes  $\tau$  of either sign and equality at the contracted  $\tau$  cap. It does not interpret rejection beyond the chosen cap as actual gap failure.

### 43.5 Gauge and physical sector

Every onsite selected-strip/free-link operator commutes with the original gauge actions at all owned-link endpoints, including heads outside the tail volume. Its unique positive ground is fixed by these actions. Hence  $P_x, Q_x$ , every spectral cutoff and  $H_M$  commute with gauge actions. The fixed-point map preserves invariant collections; uniqueness gives invariant  $C$  and  $\psi$ . Equivalently the unique full ground is fixed under the finite product of endpoint  $SU(2)$  groups. The physical subspace reduces the full Hamiltonian, so the full-space gap restricts to it with the same vacuum.

For a nonempty complete-factor volume, choose a selected square in any factor. The full finite-volume ground is smooth and strictly positive on the connected compact product of link groups, including both signs of its bounded real potential. The gauge-invariant vector  $(W_p - \langle W_p \rangle)\Omega_H$  is nonzero, is in the operator domain and is orthogonal to the ground. This proves a genuine nonzero physical excited sector. The empty volume is one dimensional and is treated separately.

### 43.6 Untruncated compact-domain passage

Each actual  $h_x$  is an elliptic Casimir sum plus bounded smooth selected potential and a scalar shift on the compact manifold  $SU(2)^{24}$ . It is self-adjoint on the Casimir domain and has compact resolvent. Choose increasing full spectral projections  $Q_{x,N}$  containing its ground and tending strongly to the identity. They commute with gauge actions. The product cutoff  $Q_N$  commutes with  $H_0$ , and compressed interactions retain support and do not increase  $J$ . The preceding estimates are uniform in  $N$ .

Finite tensor products of onsite eigenvectors form a form core for  $H_0$ : spectral tails of  $\|u\|^2 + \langle u, H_0 u \rangle$  tend to zero. Since  $V$  is bounded in each fixed finite volume, the same union is a form core for  $H_0 + sV$ , after a harmless scalar lower-bound shift. Compact resolvent persists. By min-max, the first two cutoff eigenvalues converge downward to the first two full eigenvalues: form approximation to their finite-dimensional eigenspaces proves the upper limit, while the variational

principle gives the lower limit. The uniform cutoff inequality therefore gives  $E_1 - E_0 \geq 1/2$  for the untruncated operator. The same core argument applies on the physical subspace because  $Q_N$  commutes with gauge actions; its nonzero sector was established separately.

No infinite-volume bounded global creator, similarity transform or unitary is asserted. The normalized physical energy is restored using the original scalar relation: after actual ground subtraction,

$$\Delta_{\text{physical}} \geq \frac{\alpha}{8} \frac{1}{2} = \frac{\alpha}{16}, \quad \Delta/E_* \geq \frac{\alpha}{16E_*}, \quad \text{frequency threshold } \alpha/(16\hbar). \quad (\text{HNM-AM2.11})$$

### 43.7 Evidence and scope

The exact checker verifies the anchored combinatorial constants, complete four-site termination example, certified exponential enclosure, radius inequalities, shifted-inverse bound, zero/equality cases, scalar/diagonal controls and the I1 J conversion. These audit the analytic proof; passing a polynomial or a finite cutoff alone is not this theorem. The admitted result, if independent review confirms the argument, is a conservative uniform **finite-volume** radius for the complete original I1 family. It does not evaluate Yarotsky's c1,c2, select an infinite-volume state, prove AN's boundary comparison or repair AL's weak-coupling obstruction. Subsequent infinite-volume or continuum claims need their own complete arguments.

Reproduce from the repository root:

```
python -B research/round29/forward/am2/check.py \
--output /absolute/new/output
```

### Final admitted limits and evidence

- This is a deliberately conservative finite-volume theorem uniform over the declared volumes, not an infinite-volume state-identification theorem.
- The proof does not evaluate Yarotsky's named c1/c2 or HTW's boundary constants; inherited infinite-state claims retain their own restrictions.
- The empty volume has only the vacuum and spectral exclusion is vacuous; nonzero physical excitations are asserted for nonempty complete-factor volumes.
- Fixed positive physical scales, actual onsite reference and complete interaction grouping are essential. No weak-bare-coupling continuum bridge or continuum mass-gap construction follows.
- The candidate constants were shared panel proposals and the creation method is established mathematics; scientific priority remains unverified.

research/round29/advisor/am2-gate.json

research/round29/reverse/am2/report.md

research/round29/skeptic/am2.md

*Sequencing note.* Prospective statements in this frozen derivation describe the moment of its admission. The final Round29 roadmap below governs subsequent work.

## 44 HNM AN1: common-bulk boundary comparison\*\*

**Statement alias HNM-P-AN1.** This identifier denotes the scoped mathematical claim admitted by this chapter's gate. It adds no assumption and establishes no literature-priority claim.

**Current extension: HNM-C-AN2.** The historical limitation below remains the scope of this original result. The later, separately gated statement in Section 45 records the extension: With the original orthant smallness and independent HTW condition retained, deep-bulk translates of the actual orthant state converge in local state norm to a compatible locally normal centered-box state on every fixed finite complete-factor region  $F$ . The uniform enclosure is  $\min(2, \exp(C1|F| - C2(n-r-1)))$  for  $F$  contained in  $[-r, r]^3$ . Independent outer cutoffs, all-pairs centered Cauchy convergence, compatibility, full gauge invariance and coarse translation invariance are established before identifying the ordered limits. Its own model and state assumptions remain required.

**Admission: accepted within scope.** Under the separately retained HTW smallness condition, the translated actual orthant box  $[-n, n]^3$  and centered box  $[-2n, 2n]^3$  obey the uniform local bound  $\min(2, \exp(2C1 - C2(n-2)))$  on the unit ball of  $B(H_Y)$ ,  $Y = \{0, e_z\}$ ,  $n \geq 3$ . The proof retains the actual whole-star boundary, all 48 links and 36 endpoints, and the domain-preserving bounded-commutator ground condition.

The typeset body below preserves the source-bound forward derivation. Its prospective language is historical; this boxed gate records the final reviewed verdict. The separately authored reverse and skeptical records remain linked. The common admission and attributed refinements are determined by the gate, not by a renamed equation.

Human project author: **Hruday N M (BUNZEEY)**. Independent forward execution under the frozen contract, without current reverse or skeptic results. The theorem below applies an existing locality theorem to this exact boundary dictionary; scientific priority is unverified.

**Finite comparison:** write  $A_n = [-n, n]^3$ , the coarse translate of  $[0, 2n]^3$ , and  $B_n = [-2n, 2n]^3$ . For the contracted identical coarse-homogeneous selected coefficient triple and omitted tau, the actual whole-star ground states obey, for  $Y = \{0, e_z\}$  and  $n \geq 3$ ,

$$\sup_{\|A\| \leq 1, A \in B(H_Y)} |\rho_{A_n}(A) - \rho_{B_n}(A)| \leq \min\{2, \exp(2C_1 - C_2(n-2))\}. \quad (\text{HNM-AN1.1})$$

This uses the **additional** source smallness  $7|\tau| \leq c_{\text{HTW}}(1, 1)$ , with positive unevaluated source constants. They are not AM2's numerical radius or I1's  $c1, c2$ . Infinite-state identification is not this loop's result.

### 44.1 Common Hilbert space and canonical interior

Extend  $\rho_{A_n}$  to  $B_n$  by the actual onsite vacuum product outside  $A_n$ ; call the resulting normal state rho1. It is the ground state of

$$H_1 = H_{A_n} + \sum_{x \in B_n \setminus A_n} h_x.$$

Let rho2 be the actual whole-star ground of  $H_2 = H_{B_n}$ . Both are trace-one densities on the **same** finite tensor Hilbert space. The original compact elliptic rotor ground exists and is unique/positive for the bounded real finite interaction; this assertion does not require guessing a theorem constant.

Set  $\Lambda_* = A_n$  and define the source canonical restriction

$$H_* = \sum_{x \in A_n} h_x + \sum_{x: \text{dist}_1(x, A_n^c) > 1} \phi_x. \quad (\text{HNM-AN1.2})$$

The retained source anchors are exactly  $[-n+1, n-1]^3$ . Each source  $\phi_x$  is supported in the radius-one l1 ball about  $x$  because the whole star  $S$  is contained in that ball, even though  $S$  has l1 diameter two. Its norm is  $M = 7|\tau|$ , and the onsite normalized gap is at least one. This matches HTW Definition1 with  $R = 1$  and  $g = 1$ ;  $R = 0$  would discard actual neighboring factors.

The source bulk from Definition6 is the twice-eroded set

$$D_n = A_n^c = \{x : \text{dist}_1(x, A_n^c) > 2\} = [-n+2, n-2]^3. \quad (\text{HNM-AN1.3})$$

Actual whole-star anchors of  $A_n$  are  $[-n, n-1]^3$ , not the canonical source anchors. Their extra groups have some base coordinate  $-n$ , and every point of their star has that coordinate at most  $-n+1$ ; therefore their support misses  $D_n$ . For  $H_2$ , every anchor not in the canonical set has a coordinate  $\leq -n$  or  $\geq n$ . Its star has that coordinate  $\leq -n+1$  or  $\geq n$ , again outside  $D_n$ . Onsite terms outside  $A_n$  also miss  $D_n$ . Thus each  $H_i - H_*$  is supported outside  $D_n$ , including all missing/extra incoming stars. Equality of boundary Hamiltonians has not been assumed.

## 44.2 Unbounded-domain ground condition

It remains to prove the source's ground-in-the-bulk hypothesis, rather than equate states from a common gap. A source test operator  $T$  is bounded and supported in  $D_n$ , preserves  $D(H\{0, D_n\})$ , and has bounded commutator with that local onsite sum. For a finite sum of nonnegative commuting tensor onsite operators,

$$D(H_{0,B_n}) = D(H_{0,D_n}) \cap D(H_{0,B_n \setminus D_n}).$$

Identity extension of  $T$  commutes with the exterior spectral projections and preserves the latter domain. It consequently preserves the full domain. Every finite interaction is bounded, so each  $H_i$  has that same onsite domain. Every interaction in  $H_i - H_*$  acts in the disjoint tensor factor outside  $D_n$ , and commutes with  $T$ ; the exterior onsite terms commute on the stated domain as well. Hence

$$[H_i, T] = [H_*, T] \quad \text{as bounded extensions,} \quad i = 1, 2. \quad (\text{HNM-AN1.4})$$

Their genuine ground vectors  $\Psi_i$  therefore give

$$\text{Tr}(\rho_i T^* [H_*, T]) = \langle T \Psi_i, (H_i - E_i) T \Psi_i \rangle \geq 0.$$

This is exactly HTW Definition6's ground-in-the-common-bulk condition. It does not assume every bounded local  $T$  preserves an unbounded generator domain. The code retains a rank-one counterexample to that false universal domain claim. The output observable  $A$  in (HNM-AN1.1), in contrast, may be any bounded operator in the local factor, as allowed by the source theorem after the ground condition has been established.

## 44.3 The theorem and exact distance

Henheik–Teufel–Wessel [18], arXiv:2106.13780v3, Definitions1,2,6, Theorem7 and Lemma8, permit infinite-dimensional onsite factors and a bound on two normal states ground in a common weakly interacting bulk. The required interaction threshold  $c_{\text{HTW}}$  and decay constants  $C_1, C_2$  are existential. Their Theorem7, attributed there to Yarotsky, applies to  $\rho_1, \rho_2$  and  $H_*$  and gives  $|\text{Tr}((\rho_1 - \rho_2)A)| \leq e^{C_1|Y| - C_2 \text{dist}(Y, \mathbb{Z}^3 \setminus D_n)} \|A\|$ .

For  $Y = \{0, e_z\}$ ,  $|Y| = 2$  and the exact l1 distance to the complement of  $D_n$  is  $n-2$  for  $n \geq 3$  (the  $e_z$  point is nearest the upper  $z$  boundary). This proves (HNM-AN1.1). The bound is uniform over the full norm-one bounded local algebra, hence is also the trace-norm bound on the two  $Y$ -marginals and applies to its gauge-invariant subalgebra.

Y consists of **48 actual links and 36 distinct original endpoints**. Gauge actions at all those endpoints are preserved by the coarse translation  $(4n, 2n, n)$  in fine coordinates and by both boundary prescriptions. Coarse translation invariance is essential: the same selected triple is used in every factor. Mere membership in coefficient ranges would not identify the translated interaction.

#### 44.4 Controls and limits

The checker reconstructs both boundary inventories and all star supports for  $n = 3, 4, 5$ , computes the eroded set and exact distance, and verifies that every boundary-difference group misses  $D_n$ . It reconstructs four incoming/outgoing anchors of an interior singleton and seven for Y; keeping only the two anchors in Y loses five groups. It also checks the 48-link/36-endpoint completion.

A near-boundary region with a point at  $(-n, 0, 0)$  has distance zero to the complement of  $D_n$  for all  $n$ ; this comparison gives no vanishing enclosure there. A changed interaction at a bulk anchor generally fails (HNM-AN1.4); an exact matrix commutator demonstrates that support overlap cannot be dropped. Changing a translated selected coefficient also destroys the common model. The additional source smallness is retained symbolically; it cannot be deleted or replaced by the explicit AM2 gap radius without a new theorem.

The numerical fixtures are geometry and domain countercontrols, not simulated expectation values. We have proved a conditional finite comparison, and have not yet identified the actual inherited orthant state with any centered-box state. That requires a separate limit argument under the full simultaneous premises, including I1's  $|\tau| < \tau_*$  whenever that inherited state is invoked.

Reproduce from the repository root:

```
python -B research/round29/forward/an1/check.py \
  --output /absolute/new/output
```

#### Final admitted limits and evidence

- This loop proves a finite comparison only; diagonal finite agreement does not identify infinite-volume states.
- The selected coefficient triple must be fixed and coarse-translation invariant. A changed bulk interaction is outside this comparison.
- The additional HTW threshold  $7|\tau| \leq c_{\text{HTW}}(1, 1)$  and decay constants remain unevaluated; AM2 does not supply them.
- Invoking the actual inherited orthant state also requires its original  $|\tau| < \tau_*$  restriction. Near-boundary controls show failure of this decay certificate, not actual unequal expectations.
- This is a model-specific application of HTW/Yarotsky, not a continuum construction or an established priority claim.

```
research/round29/advisor/an1-gate.json
```

```
research/round29/reverse/an1/report.md
```

```
research/round29/skeptic/an1.md
```

*Sequencing note.* Prospective statements in this frozen derivation describe the moment of its admission. The final Round29 roadmap below governs subsequent work.

## 45 HNM AN2: ordered bulk-state identification\*\*

**Statement alias HNM-T-AN2.** This identifier denotes the scoped mathematical claim admitted by this chapter's gate. It adds no assumption and establishes no literature-priority claim.

**Admission: accepted within scope.** With the original orthant smallness and independent HTW condition retained, deep-bulk translates of the actual orthant state converge in local state norm to a compatible locally normal centered-box state on every fixed finite complete-factor region  $F$ . The uniform enclosure is  $\min(2, \exp(C_1|F| - C_2(n-r-1)))$  for  $F$  contained in  $[-r, r]^3$ . Independent outer cutoffs, all-pairs centered Cauchy convergence, compatibility, full gauge invariance and coarse translation invariance are established before identifying the ordered limits.

The typeset body below preserves the source-bound forward derivation. Its prospective language is historical; this boxed gate records the final reviewed verdict. The separately authored reverse and skeptical records remain linked. The common admission and attributed refinements are determined by the gate, not by a renamed equation.

Human project author: **Hruday N M (BUNZEEY)**. Independent forward execution under the frozen AN2 contract. Current reverse and skeptic results were not read. This is a source-theorem application and project boundary dictionary, with scientific priority unverified.

**Result:** under the inherited  $|\tau| < \tau_*$  and additional  $7|\tau| \leq c_{\text{HTW}}(1, 1)$ , deep-bulk translates of the actual  $\Pi$  orthant state converge locally in state norm to the compatible normal state obtained from centered whole-star boxes. This identifies these specified local-state routes. It is not a generator, dynamics or continuum identification.

### 45.1 Independent outer cutoffs

Fix a nonempty finite complete-factor region  $F$  in  $[-r, r]^3$  and  $n \geq r + 2$ . Let  $b_n = n(1, 1, 1)$ , and compare the translated orthant cutoff

$O_{n,L} = [-n, L - n]^3$ , for every integer  $L \geq 2n$ ,

with the centered cutoff  $C_M = [-M, M]^3$ , for every integer  $M \geq n$ . Both contain the same inner cube  $A_n = [-n, n]^3$ . Extend each finite normal ground state by the onsite vacuum product to their common finite union (or a containing rectangle). These are grounds of their respective actual Hamiltonians plus decoupled exterior onsite terms.

Use exactly AN1's canonical reference on  $A_n$ , retaining star anchors  $[-n + 1, n - 1]^3$ , with twice-eroded bulk  $D_n = [-n + 2, n - 2]^3$ . Every difference term between either extended Hamiltonian and this canonical Hamiltonian is supported outside  $D_n$ . For an excluded anchor, some coordinate is  $\leq -n$  or  $\geq n$ ; its star cannot reach  $D_n$ . Exterior onsite terms are disjoint too. The tensor-domain proof of AN1 therefore gives equality of bounded local commutators and the source ground-in-the-common-bulk condition for **every** allowed  $L, M$ .

HTW Theorem7 now gives

$$\|\rho_{O_{n,L},F} - \rho_{C_M,F}\|_1 \leq \varepsilon_n(F) := \min\{2, \exp(C_1|F| - C_2(n - r - 1))\}. \quad (\text{HNM-AN2.1})$$

Indeed the exact distance from  $F$  to the complement of  $D_n$  is at least  $n-r-1$ . The norm is the trace norm, equal to the supremum over norm-one bounded operators on this same  $F$ . The constants do not depend on  $L, M, n$  or local Hilbert dimensions. The added smallness and positive source constants are unevaluated; AM2 did not supply them.

## 45.2 Centered marginals are Cauchy, not just a chosen subsequence

For any  $M, M' \geq n$ , repeat the same common-bulk argument directly for  $C_M$  and  $C_{M'}$ . It gives  $\|\rho_{C_M, F} - \rho_{C_{M'}, F}\|_1 \leq \varepsilon_n(F)$ .

For fixed  $F$ ,  $\varepsilon_n$  tends to zero because  $C_2 > 0$ . This is the Cauchy criterion for **all sufficiently large pairs**, rather than convergence along a diagonal. Completeness of the trace-class space gives a positive trace-one limit  $\rho_F$ . Positivity and trace are preserved by trace-norm convergence.

For  $F$  subset  $F'$ , finite marginals satisfy  $\text{Tr}_{F' \setminus F} \rho_{C_M, F'} = \rho_{C_M, F}$ . Partial trace is trace-norm contractive on trace-class operators, so the limits have the same compatibility. Define  $\omega_{\mathbb{Z}}(A) = \text{Tr}(\rho_F A)$  for a bounded observable in  $F$ . Compatibility makes this independent of its chosen containing factor. It is positive, normalized and bounded by the operator norm on the local algebra; it uniquely extends to its norm closure. Each local restriction is normal by construction. One marginal alone would not define this state; the compatibility step is essential.

## 45.3 Ordered identification with the actual inherited orthant state

For **fixed**  $n$  and any fixed  $M \geq n$ , take  $L$  to infinity in (HNM-AN2.1). Before translation this is the inherited I1 orthant exhaustion  $[0, L]^3$ , tested on the fixed observable translated into  $F + b_n$ . I1's local-state convergence applies because  $n$  is fixed during this passage. Coarse homogeneity identifies the translated finite state exactly; all endpoint gauge actions move with the original links. The result is, for every bounded  $A$  in  $F$ ,

$$|\omega_+(T_{b_n} A) - \rho_{C_M, F}(A)| \leq \varepsilon_n(F) \|A\|.$$

Now take  $M$  to infinity using Section2. Finally let  $n$  tend to infinity:

$$\sup_{\|A\| \leq 1, A \in B(\mathcal{H}_F)} |\omega_+(T_{b_n} A) - \omega_{\mathbb{Z}}(A)| \leq \varepsilon_n(F) \longrightarrow 0. \quad (\text{HNM-AN2.2})$$

The order is  $L \rightarrow \infty$  at fixed  $n$ , then the independent centered limit, then  $n \rightarrow \infty$ . Uniformity in the outer cutoffs supplies this result; bare pointwise thermodynamic convergence on fixed observables would not justify a growing translation. The rate is a symbolic enclosure rather than a numerically evaluated convergence rate.

## 45.4 Gauge and coarse translations

Every finite ground is fixed by all its original endpoint gauge actions. On a bounded local factor those actions are the original left/right link actions, including outgoing heads outside the tail set, and preserve that factor. Local trace-norm convergence passes gauge invariance to  $\omega_{\mathbb{Z}}$ . No gauge-factor tensor decomposition or Wilson-only observable completion is substituted.

For a fixed coarse vector  $k$ , homogeneity gives

$$\rho_{C_M}(T_k A) = \rho_{C_M - k}(A).$$

When  $M \geq n + \|k\| \rightarrow \infty$ , both  $C_M$  and  $C_M - k$  contain  $A_n$ . Applying the same common-bulk comparison and then passing  $M \rightarrow \infty$  bounds  $|\omega_{\mathbb{Z}}(T_k A) - \omega_{\mathbb{Z}}(A)|$  by  $\varepsilon_n(F)$ . Let  $n \rightarrow \infty$  to obtain equality. Thus  $\omega_{\mathbb{Z}}$  is coarse-translation invariant. This needs the same selected triple at every cell, not merely arbitrary coefficients inside common ranges. The fine-coordinate translation associated with coarse  $k$  is  $(4k_x, 2k_y, k_z)$ .

The result does not assert that a fixed-origin orthant observable already has its bulk expectation. Near-boundary translates whose distance from the missing coordinate half-spaces stays bounded are outside the proved vanishing-distance hypothesis. Nor does local-state equality by itself identify closed generators or their domains, all possible boundary limits, physical energy calibration,

or a four-dimensional field theory.

## 45.5 Exact controls

The checker varies  $L$  and  $M$  independently, reconstructs all retained stars and the common interior, and verifies the  $n$ - $r$ -1 distance and containment. It checks a translated centered box as required in the invariance proof. An abstract triangular array  $a_{n,L} = 1_{L \geq n^2}$  has  $\lim_n a_{(n,2n)}=0$  but  $\lim_n \lim_L a_{(n,L)}=1$ ; all its diagonal points satisfy  $L \geq 2n$ . This discriminates the illegal diagonal substitution. The control is not an actual ground-state example.

An exact two-bit density fixture verifies marginal compatibility and rejects an inconsistent proposed singleton marginal. Near-boundary support has zero source-bulk distance; altered selected coefficients and absent HTW smallness fail the declared dictionary. These are geometric/logical controls, not computed  $SU(2)$  expectation values. The prior source-domain proof remains required for the actual states.

Reproduce from the repository root:

```
python -B research/round29/forward/an2/check.py \
  --output /absolute/new/output
```

## Final admitted limits and evidence

- This identifies the two specified local-state routes, not their closed generators or dynamics, all possible boundary limits, or a fixed-origin boundary expectation with a bulk expectation.
- Both  $|\tau| < \tau_*$  and  $7|\tau| \leq c_{\text{HTW}}(1,1)$  remain; source thresholds and decay constants are positive but unevaluated.
- The repeated selected coefficient triple and coarse translation convention are essential. Fine translation invariance is not asserted.
- This is an application of HTW/Yarotsky under a complete boundary and domain dictionary, not a new generic theorem of boundary independence.
- No physical continuum construction, mass calibration, or established literature priority follows.

research/round29/advisor/an2-gate.json

research/round29/reverse/an2/report.md

research/round29/skeptic/an2.md

*Sequencing note.* Prospective statements in this frozen derivation describe the moment of its admission. The final Round29 roadmap below governs subsequent work.

## 46 HNM AO1: finite Wilson second-moment bound\*\*

**Statement alias HNM-P-AO1.** This identifier denotes the scoped mathematical claim admitted by this chapter's gate. It adds no assumption and establishes no literature-priority claim.

**Current extension: HNM-C-AO2.** The historical limitation below remains the scope of this original result. The later, separately gated statement in Section 47 records the extension: For the same actual  $\Pi/\text{AJ1}/\text{AK2}$  orthant physical state,  $\chi$  belongs to  $D(\text{H}_{\text{phys}})$ , its second spectral moment is at most  $\alpha^2(36+28|\tau|) < 37\alpha^2$ , and its first moment equals  $\alpha\omega(1-W^2)$ . Uniform first-energy tails bounded by  $B_2/R$  close the former AK2 moment-loss limitation. The proof uses the actual centered spectral measures, compact spectral tests and bounded-local expectations. Its own model and state assumptions remain required.

**Admission: accepted within scope.** For the same actual origin xz Wilson and every finite whole-star orthant volume containing its complete region, the centered physical second moment obeys  $\mu_2 \leq \alpha^2(36+28|\tau|) \leq \alpha^2(36+7/16384) < 37\alpha^2$ . The proof derives the kinetic bound from the actual selected-strip reference, retains the exact ground scalar before its legitimate upper-bound relaxation, and proves the finite domain-valid electric commutator including its derivative cross term.

The typeset body below preserves the source-bound forward derivation. Its prospective language is historical; this boxed gate records the final reviewed verdict. The separately authored reverse and skeptical records remain linked. The common admission and attributed refinements are determined by the gate, not by a renamed equation.

Human project author: **Hruday N M (BUNZEEY)**. Independent forward derivation of the shared prospective commutator route, without reading current reverse or skeptic results. The method uses standard product rules; scientific priority is unverified.

**Finite-volume result:** the original origin xz Wilson in every contracted actual whole-star orthant ground state satisfies

$$\int E^2 d\nu_{\Lambda}(E) = \|K_{\Lambda}\chi_{\Lambda}\|^2 \leq \alpha^2(36+28|\tau|) \leq \alpha^2(36+7/16384) < 37\alpha^2. \quad (\text{HNM-AO1.1})$$

Here  $\chi_{\Lambda} = (W - \omega_{\Lambda}(W))\Omega_{\Lambda}$  and  $K_{\Lambda} = H_{\Lambda}^{\text{raw}} - E_{\Lambda}^{\text{raw}}$  is the actual centered physical operator. The inherited  $|\tau| < \tau_*$  and  $|\tau| \leq 2^{-16}$  remain in force. Reference-ground energy is not an excited-vector spectral moment.

### 46.1 Exact local kinetic reconstruction

The complete cover is  $R = \{0, e_z\}$ : 48 links and 36 endpoint actions. Its actual orthant incident anchor union is  $(R - S) \cap \mathbb{Z}_+^3 = \{0, e_z\}$ , counting the star at zero once. AK1 therefore gives  $\omega_{\Lambda}(h_R) \leq 28|\tau|$ .

Let  $T_R = \alpha \sum_{e \text{ owned by } R} C_e$ ,  $V_R^{\text{sel}} = - \sum_{f \text{ in } R}^{\text{selected}} \lambda_f W_f$ , and  $B_R = \sum_{b \in R} E_{\text{strip}, b}$ . Keeping every selected ground scalar,

$$\delta h_R = T_R + V_R^{\text{sel}} - B_R, \quad T_R = \delta h_R - V_R^{\text{sel}} + B_R, \quad \delta = \alpha/8. \quad (\text{HNM-AO1.2})$$

The constant Haar trial on each strip has zero electric and selected-Wilson mean, so  $E_{\text{strip}, b} \leq 0$ . Each strip has selected absolute budget at most  $\alpha/2 + \alpha/8 + \alpha/2 = 9\alpha/8$ . Thus

$$\omega_\Lambda(T_R) \leq \frac{\alpha}{8}28|\tau| + \frac{9\alpha}{4} = \alpha(9/4 + (7/2)|\tau|). \quad (\text{HNM-AO1.3})$$

Both coupling signs and every allowed signed selected coefficient are covered. Omitting  $B_R$  from the exact identity is wrong; discarding its nonpositive contribution after writing an upper inequality is legitimate. Omitting  $V_R^{\text{sel}}$  from the reconstruction is also wrong. At  $\tau = 0$  a nontrivial selected ground has zero reference energy while its electric expectation need not vanish.

## 46.2 Domains and four original links

At finite volume the Casimir sum is elliptic on compact  $SU(2)^L$ , and the full magnetic potential is a bounded smooth real multiplier. Its operator domain is the product  $H^2$  domain. The ground is smooth.  $W$  has bounded derivatives through order two, so multiplication preserves this domain. Hence  $\chi_\Lambda$  lies in  $D(K_\Lambda)$ , and its second spectral moment equals the squared generator norm. No infinite-volume domain is assumed.

All selected and omitted magnetic multipliers commute with  $W$ . For  $C_e = -\sum_a X_{ea}^2$ ,

$$[C_e, W]u = (C_e W)u - 2\sum_a (X_{ea} W)X_{ea}u.$$

The original four stored links are distinct. Each contributes  $C_e W = 3W/4$ , with inverse occurrences retaining their derivative side and sign. Consequently

$$K_\Lambda \chi_\Lambda = 3\alpha W \Omega_\Lambda - 2\alpha Z_\Lambda, \quad Z_\Lambda = \sum_{e \in p, a} (X_{ea} W)X_{ea} \Omega_\Lambda. \quad (\text{HNM-AO1.4})$$

AK2's half-Pauli identity gives  $\sum_{e \in p, a} (X_{ea} W)^2 = 1 - W^2 \leq 1$ . Pointwise Cauchy-Schwarz followed by integration yields

$$\|Z_\Lambda\|^2 \leq \int (1 - W^2) \sum_{e \in p, a} |X_{ea} \Omega_\Lambda|^2 \leq \omega_\Lambda(T_R)/\alpha. \quad (\text{HNM-AO1.5})$$

The derivative cross term is retained: the actual selected reference is generally not constant. Using  $\|W\Omega\| \leq 1$  and  $\|u + v\|^2 \leq 2\|u\|^2 + 2\|v\|^2$ ,

$$\|K_\Lambda \chi_\Lambda\|^2 \leq 18\alpha^2 + 8\alpha\omega_\Lambda(T_R) \leq \alpha^2(36 + 28|\tau|). \quad (\text{HNM-AO1.6})$$

This proves the claimed bound in physical energy-squared units. A frequency second moment would divide by  $\hbar^2$ . The fixed energy reference and clock are unchanged.

## 46.3 Controls and finite scope

At the fully free corner, where all selected coefficients and  $\tau$  vanish, the Haar ground is constant, variance is  $1/4$ , and the Wilson energy is  $3\alpha$ . Its exact second moment is  $9\alpha^2/4$  despite zero reference-ground energy. This rejects equating those two energies.

The checker reconstructs origin versus interior incidence (two versus seven stars), the complete link cover, and exact coefficients. A smooth one-rotor diagnostic  $\Omega(q) = 1 + q_0/3$ ,  $W = q_0$  at  $q_0 = 3/5$  has a nonzero gradient cross term; dropping it changes the commutator. It is an algebra control, not an I1 ground. A finite matrix expectation fixture tests the negative ground scalar in the exact reconstruction and its safe removal only for an upper inequality. Doubling Lie generators quadruples squared gradients and is rejected. Measures with bounded first moments but escaping high-energy tails have growing second moments, so first-moment boundedness cannot replace this proof.

The regional constant is specific to the original orthant cover; interior or summable-model substitution needs a new dictionary. No limit is taken. Uniform integrability, limiting moment equality and the actual infinite-volume operator domain remain for adaptive AO2.

Reproduce from the repository root:

```
python -B research/round29/forward/ao1/check.py \
  --output /absolute/new/output
```

## Final admitted limits and evidence

- This is a finite-volume uniform moment theorem only; infinite operator-domain membership and first-moment equality require a separate limit argument.
- The actual orthant state and both inherited conditions  $|\tau| < \tau_*$  and  $|\tau| \leq 2^{-16}$  are retained; this does not evaluate  $\tau_*$ .
- The two-star origin incidence cannot be substituted for an interior region or a different summable model.
- Bounded second moments do not imply equality of limiting second moments. Standard commutator methods and the shared prospective panel route are acknowledged; scientific priority remains unverified.
- Fixed positive physical scales and original half-Pauli normalization are essential. No continuum or experimental claim follows.

research/round29/advisor/ao1-gate.json

research/round29/reverse/ao1/report.md

research/round29/skeptic/ao1.md

*Sequencing note.* Prospective statements in this frozen derivation describe the moment of its admission. The final Round29 roadmap below governs subsequent work.

## 47 HNM AO2: limiting Wilson energy identity\*\*

**Statement alias HNM-P-AO2.** This identifier denotes the scoped mathematical claim admitted by this chapter's gate. It adds no assumption and establishes no literature-priority claim.

**Admission: accepted within scope.** For the same actual I1/AJ1/AK2 orthant physical state,  $\chi$  belongs to  $D(H_{\text{phys}})$ , its second spectral moment is at most  $\alpha^2(36 + 28|\tau|) < 37\alpha^2$ , and its first moment equals  $\alpha \omega(1 - W^2)$ . Uniform first-energy tails bounded by B2/R close the former AK2 moment-loss limitation. The proof uses the actual centered spectral measures, compact spectral tests and bounded-local expectations.

The typeset body below preserves the source-bound forward derivation. Its prospective language is historical; this boxed gate records the final reviewed verdict. The separately authored reverse and skeptical records remain linked. The common admission and attributed refinements are determined by the gate, not by a renamed equation.

Human project author: **Hruday N M (BUNZEEY)**. Independent forward execution, before reading current reverse or skeptic results. The spectral-measure method is standard; this is a source-qualified application to the actual AK2 state. Scientific priority is unverified.

**The former first-moment-loss limitation is closed for the contracted state:**

$$\chi \in D(H_{\text{phys}}), \quad \|H_{\text{phys}}\chi\|^2 \leq B_2 := \alpha^2(36 + 28|\tau|) < 37\alpha^2, \quad \langle \chi, H_{\text{phys}}\chi \rangle = \alpha \omega(1 - W^2). \quad (\text{HNM-AO2.1})$$

This is exactly the inherited I1/AJ1/AK2 homogeneous whole-star orthant state, original 48-link/36-endpoint Wilson cover, actual physical ground subtraction and fixed energy clock. Both  $|\tau| < \tau_*$  and  $|\tau| \leq 2^{-16}$  remain. It is not an identity for a newly constructed AQ state without its own dictionary.

### 47.1 The actual limiting measures

Use the positive finite measures  $\nu_\Lambda$  of  $\chi_\Lambda = (W - m_\Lambda)\Omega_\Lambda$  for the centered physical  $K_\Lambda$ , and the actual spectral measure  $\nu$  of  $\chi = (\pi(W) - m)\Omega$  for the reducing self-adjoint  $H_{\text{phys}}$ . AK2 already proved  $C_0$  spectral-test convergence from its tested resolvent limit, with moving finite means handled explicitly. Their masses converge to the actual Wilson variance. These are not measures on an assumed common concrete finite/infinite Hilbert space.

AO1 gives  $\int E^2 d\nu_\Lambda \leq B_2$  uniformly. The finite exact identity is  $\mu_{1,\Lambda} = \alpha\omega_\Lambda(1 - W^2)$ . Since  $1 - W^2$  is bounded local, these first moments converge numerically to  $s = \alpha\omega(1 - W^2)$ . The remaining task is to show that this number equals the limiting spectral first moment; bounded-local convergence alone had not done that in AK2.

### 47.2 Second moment transfer and uniform energy tails

For a physical energy cutoff  $L > 0$ , choose the continuous function  $\theta(u)$  equal to one for  $u \leq 1$ ,  $2-u$  for  $1 < u < 2$ , and zero for  $u \geq 2$ . On  $E \geq 0$  set  $F_L(E) = E\theta(E/L)$  and  $Q_L(E) = E^2\theta(E/L)$ , extending both by zero on negative  $E$ . These are nonnegative continuous compact tests; as  $L$  increases, they increase to  $E$  and  $E^2$  on the positive half-line.

The admitted  $C_0$  convergence gives  $\int Q_L d\nu = \lim_\Lambda \int Q_L d\nu_\Lambda \leq B_2$ . Monotone convergence therefore proves

$$\int E^2 d\nu(E) \leq B_2. \quad (\text{HNM-AO2.2})$$

For every finite measure and for  $\nu$  itself,

$$\int_{E>L} E d\nu_\Lambda(E) \leq B_2/L, \quad \int_{E>L} E d\nu(E) \leq B_2/L. \quad (\text{HNM-AO2.3})$$

This follows from  $E \leq E^2/L$  above  $L$  and is uniform integrability of the first moment. The bound has energy units because  $B_2$  has energy-squared units. The analogous probability tail is at most  $B_2/L^2$ . Neither statement is uniform integrability of the **second** moment.

### 47.3 Equality and operator domain

Because  $F_L$  agrees with  $E$  below  $L$  and satisfies  $0 \leq F_L \leq E$ ,

$$0 \leq \mu_{1,\Lambda} - \int F_L d\nu_\Lambda \leq B_2/L.$$

At fixed  $L$  take the volume limit. The left first term converges to  $s$ , and the compact test converges to  $\int F_L d\nu$ , so

$$0 \leq s - \int F_L d\nu \leq B_2/L.$$

Then take  $L$  to infinity. Monotone convergence proves  $\int E d\nu = s$ , establishing the equality in (HNM-AO2.1). No formal differentiation at  $t = 0$  or unjustified interchange of an unbounded spectral test is used.

For a self-adjoint operator, membership of  $\chi$  in  $D(H_{\text{phys}})$  is equivalent to finiteness of its spectral second moment. Equation(HNM-AO2.2) therefore supplies the **actual** operator-domain statement and  $\|H_{\text{phys}}\chi\|^2 \leq B_2$ . Its physical restriction and original ground centering are inherited intact. We have not proved convergence of finite second moments to that number, nor a common-Hilbert-space strong convergence of the vectors  $K_\Lambda \chi_\Lambda$ .

### 47.4 Countercontrols and scope

The old AK2 measures  $(1 - 1/(2n))\delta_{1/4} + (1/(2n))\delta_n$  have bounded first moments but second moments growing at least  $n/2$ . They violate the new hypothesis and cannot support a counterexample to this argument.

Conversely,  $\eta_n = (1 - 1/n^2)\delta_1 + (1/n^2)\delta_n$  converges on bounded tests to  $\delta_1$ , has uniformly bounded second moments  $2 - 1/n^2$ , and first moments tending to one. Its second moments tend to two, whereas the limiting second moment is one. The escaping second-moment tail equals one. This proves that our new upper bound does not license equality of second moments.

The exact checker computes both families, their rational resolvent tests, tail inequalities, and energy-unit and finite-centering controls. It retains the original symbolic source restriction. Removing finite mean or ground subtraction changes the spectral measure, while an alpha/delta mismatch changes the physical first and second moments by different powers. These abstract controls are not additional data about the actual  $SU(2)$  state.

The result closes the specified AK2 energy-identity and operator-domain obligations. It does not supply a continuum construction, identify another infinite-volume state, or determine the actual first positive eigenenergy.

Reproduce from the repository root:

```
python -B research/round29/forward/ao2/check.py \
--output /absolute/new/output
```

Pre-admission repair: the first frozen checker gave identical labels to two distinct cutoff fixture positions when  $n = 2$ , since  $1 = n/2$ . Root admission correctly rejected the duplicate executed-check IDs. The original complete report, checker, inputs, outputs and freeze are preserved under `attempts/duplicate-check-ids-001/original/`. The repaired checker adds the cutoff position to each label, preserves all 67 executed checks and their mathematics, and is freshly rerun in normal and optimized modes. This is an evidence-label repair, not a ninth investigation or a changed theorem.

## Final admitted limits and evidence

- Both inherited conditions  $|\tau| < \tau_*$  and  $|\tau| \leq 2^{-16}$  remain. This does not evaluate  $\tau_*$  or identify a later numerical-cap state.
- The second moment has an upper bound; equality or convergence of finite second moments is not proved, as an explicit escaping-second-moment control shows.
- No common-Hilbert-space strong-resolvent or excited-vector convergence is asserted. Reverse cutoff formulas are on  $E \geq 0$  and extended by zero to negative energies for  $C0(\mathbb{R})$  tests, as clarified in review.
- The methods are established spectral and uniform-integrability arguments applied to the actual scoped Wilson state; scientific priority remains unverified.
- No lowest excited energy, experimental signal, weak-coupling bridge or continuum Yang–Mills construction follows.

`research/round29/advisor/ao2-gate.json`

`research/round29/reverse/ao2/report.md`

`research/round29/skeptic/ao2.md`

*Sequencing note.* Prospective statements in this frozen derivation describe the moment of its admission. The final Round29 roadmap below governs subsequent work.

## 48 HNM AQ1: numerical-cap thermodynamic construction\*\*

**Statement alias HNM-T-AQ1.** This identifier denotes the scoped mathematical claim admitted by this chapter's gate. It adds no assumption and establishes no literature-priority claim.

**Current extension: HNM-C-AQ2.** The historical limitation below remains the scope of this original result. The later, separately gated statement in Section 49 records the extension: For the actual AQ1 centered full-Z3 subsequential state at both signs of  $|\tau| \leq 1/100000000$ , the complete original-endpoint gauge-fixed space equals the invariant-local cyclic completion and reduces the physical energy generator. Centered Fourier tests including zero give  $H_{\text{phys}} \geq (\alpha/16)(I-P_{\Omega})$  and a simple vacuum in this representation. The full-GNS strengthening explicitly uses AM2's separately reviewed full-Hilbert finite gap. For the original xz Wilson in the same state, the actual 48-link/36-endpoint cover has seven incident stars, local reference energy  $\leq 98|\tau|$  and variance  $\geq 61999/250000 > 1/5$ . The reverse proof separately sharpens that floor to  $3099951/12500000$ . Its own model and state assumptions remain required.

**Admission: accepted within scope.** At both signs of  $|\tau| \leq 1/100000000$ , actual centered full-Z3 whole-star boxes have a subsequence defining a compatible locally normal gauge-invariant stationary state on the norm closure of full bounded local algebras. The actual reference reset bound is  $56|\tau||R|$ ; compact-resolvent cutoffs prove local trace-norm precompactness. The matched unbounded-onsite dynamics theorem and separate local-normality argument supply strongly continuous GNS evolution with a nonnegative self-adjoint physical energy generator. The forward physical cyclic restriction is reducing; full joint-fixed completion and its positive gap remain for AQ2.

The typeset body below preserves the source-bound forward derivation. Its prospective language is historical; this boxed gate records the final reviewed verdict. The separately authored reverse and skeptical records remain linked. The common admission and attributed refinements are determined by the gate, not by a renamed equation.

Human project author: **Hruday N M (BUNZEEY)**. Independent forward execution of the frozen AQ1 contract, without current reverse or skeptic work. The compactness/dynamics/GNS/Fourier strategy has direct prior overlap with Gauvin [1], arXiv:2503.15539v3 Supplement A.10. The dynamics theorem is Nachtergaele–Sims arXiv:1410.8174v1 Section3/Theorem4.1 (also cross-checked by the source panel against NSY). This is a model-specific SU(2) application, not invention of that strategy; scientific priority is unverified.

**Result:** at either sign of  $|\tau| \leq 10^{-8}$ , centered complete-factor whole-star boxes have a subsequence defining a locally normal stationary state on the full coarse  $Z^3$  algebra. Its GNS evolution is strongly continuous, with a nonnegative self-adjoint physical energy generator and an invariant physical cyclic restriction. No  $\tau_*$  or HTW smallness is used. Whole-sequence convergence, state uniqueness, translation invariance and a physical spectral gap are not claimed in this loop.

### 48.1 Actual full-lattice factor model

For any integer fine coordinates, set  $\pi(x, y, z) = (\lfloor x/4 \rfloor, \lfloor y/2 \rfloor, z)$ . Euclidean division, including negative x,y, gives unique residues  $r = 0, \dots, 3, s = 0, 1$ . Each coarse factor owns the three positive links at its eight tail vertices, giving 24 untruncated SU(2) Haar factors. Its ten selected-strip links and fourteen free links are unchanged. The same selected coefficient triple, within the inherited ranges, repeats at every coarse site. All original endpoint gauge actions are retained, including heads outside tail blocks.

For  $\Lambda_N = [-N, N]^3$ , the normalized finite operator is

$$\hat{H}_N = \sum_{b \in \Lambda_N} h_b + \sum_{b+S \subset \Lambda_N} \phi_b,$$

where  $h_b \geq I - P_b$ ,  $P_b$  is the actual selected-reference vacuum projection,  $S = \{0, e_x, e_y, e_z\}$ , and  $\|\phi_b\| = M = 7|\tau|$ . The raw physical operator differs from  $\delta \hat{H}_N$  by the selected ground scalar, with  $\delta = \alpha/8$ . Actual centered physical energy is  $K_N = \delta(\hat{H}_N - \hat{E}_N)$ . AM2 supplies the finite full-Hilbert construction at the numerical cap, but a finite gap by itself is not the state construction below.

## 48.2 Reset energy and trace-norm compactness

Let  $F$  be any fixed finite complete-factor region contained in  $\Lambda_N$ . Reset the ground density on  $F$  to  $P_F = \bigotimes_{b \in F} P_b$ , keeping its exterior marginal. The mixed trial state has finite reference form energy; nonnegative spectral truncations justify cancellation of unchanged unbounded exterior energy. Every incident retained star changes its expectation by at most  $2M$ . Its anchor lies in  $F$ - $S$ , whose size is at most  $4|F|$ . The ground variational principle therefore gives

$$\text{Tr}(\rho_{N,F} h_F) \leq 2MN_{F,N} \leq 8M|F| = 56|\tau||F| =: C_F, \quad h_F = \sum_{b \in F} h_b. \quad (\text{HNM-AQ1.1})$$

All incoming stars are included. This is actual selected-reference energy, not an assumed Haar/pure-electric energy bound. At  $\tau = 0$  it forces the product reference locally.

Each  $h_b$  is a compact-manifold elliptic Casimir sum plus bounded selected potential and its actual ground shift, hence has compact resolvent. So does the finite tensor sum  $h_F$ . Its spectral projection  $Q_{F,L} = \mathbf{1}_{[0,L]}(h_F)$  is finite rank. For  $L > 0$ ,

$$\text{Tr} \rho_{N,F}(I - Q_{F,L}) \leq C_F/L, \quad \|\rho_{N,F} - Q_{F,L} \rho_{N,F} Q_{F,L}\|_1 \leq 2\sqrt{C_F/L}. \quad (\text{HNM-AQ1.2})$$

The second inequality follows by purifying the density, bounding the projected rank-one difference by  $2\sqrt{\text{tail}}$ , and contracting under partial trace. For fixed  $L$  the compressed positive trace-at-most-one matrices form a compact finite-dimensional set. Equation(HNM-AQ1.2) makes the full family trace-norm precompact.

Diagonal extraction over the countable nested coarse cubes produces a subsequence  $N_k$  whose densities converge in trace norm on every finite  $F$ . Limits are positive and trace one. Partial-trace contractivity passes compatibility. Thus  $\omega_{\text{num}}(A) = \text{Tr}(\rho_F A)$  defines a normalized positive functional on the full local algebra and extends by norm continuity to its  $C^*$ -closure. Every local restriction is normal. No compactness of an extensive global density, Haar replacement, or uniqueness of the selected subsequence is asserted.

## 48.3 Exact placement in the unbounded-onsite dynamics theorem

Use the  $l_1$  metric on  $\mathbb{Z}^3$  and  $F(r) = (1+r)^{-4}$ . The shell at integer  $r \geq 1$  contains  $4r^2 + 2$  sites. Since this is at most  $6(r+1)^2$ , and  $\sum_{r \geq 1} (r+1)^{-2} \leq 1$ ,

$$\|F\| \leq 7.$$

In the convolution sum, either  $d(x, z) \geq d(x, y)/2$  or  $d(z, y) \geq d(x, y)/2$ . On that half, the corresponding  $F$  factor is at most  $16F(d(x, y))$ ; summing both halves gives convolution constant  $C \leq 32\|F\| \leq 224$ .

Whole stars have  $l_1$  diameter two. Each site belongs to four anchor stars. Therefore the actual normalized interaction satisfies

$$J \leq 4M = 28|\tau|, \quad \|\Phi\|_F \leq 3^4 J = 81J \leq 2268|\tau|. \quad (\text{HNM-AQ1.3})$$

This is a finite interaction norm, not the extensive total norm. The countable lattice, separable onsite Hilbert spaces, self-adjoint unbounded  $h_b$ , and bounded self-adjoint finite-range  $\Phi(X)$  now satisfy Nachtergaele–Sims Section3 and Theorem4.1. Its native restriction  $\sum_{X \subset \Lambda} \Phi(X)$  is exactly the retained whole-star prescription here. The theorem gives, for each bounded local  $A$ , the norm limit of its finite evolutions, uniform on compact **normalized** time intervals. It extends to a group of isometric star automorphisms of the quasi-local algebra.

Restore physical time by  $u = \delta t / \hbar$ :

$$\mathcal{T}_t(A) = \lim_N e^{iu\hat{H}_N} A e^{-iu\hat{H}_N} = \lim_N e^{itK_N/\hbar} A e^{-itK_N/\hbar}. \quad (\text{HNM-AQ1.4})$$

All raw/reference ground scalars cancel in this conjugation. A physical compact time window corresponds to a fixed compact  $u$  window. The theorem does **not** assert norm continuity in time for every operator of the full local  $B(H)$  algebra. Its proof handles strongly continuous interaction-picture operators; no unjustified norm differentiability is added here.

#### 48.4 Stationarity and separate GNS continuity

Local density convergence extends along the same subsequence to every fixed quasi-local observable by norm approximation. Finite ground invariance and the norm volume limit imply

$$\omega_{\text{num}}(\mathcal{T}_t(A)) = \omega_{\text{num}}(A).$$

Indeed replace  $T_t(A)$  by a fixed finite-region approximation, pass the local state limit, then use the uniform norm errors to compare with each finite-volume invariant expectation. The construction yields well-defined unitaries

$$U_t \pi(A) \Omega = \pi(\mathcal{T}_t(A)) \Omega, \text{ with } U_t \Omega = \Omega.$$

Strong time continuity requires a separate argument. For a fixed finite containing region  $D$ , its finite unitary evolution is strong-\* continuous on every bounded  $A$ . The normal density  $\rho_D$  makes  $\omega_{\text{num}}(A^* \mathcal{T}_t^D(A))$  continuous by bounded dominated convergence in its trace-class decomposition. Uniform-on-compact-time norm volume approximation then makes  $\omega_{\text{num}}(A^* \mathcal{T}_t(A))$  continuous. Consequently

$$\|(U_t - I)\pi(A)\Omega\|^2 = 2\omega_{\text{num}}(A^*A) - 2\text{Re } \omega_{\text{num}}(A^* \mathcal{T}_t(A)) \longrightarrow 0.$$

Local cyclic vectors are dense; unitarity extends this to the full GNS Hilbert space. Stone's theorem gives a self-adjoint physical energy  $H_{\text{num}}$  such that  $U_t = \exp(itH_{\text{num}}/\hbar)$  and  $H_{\text{num}} \Omega = 0$ . There is no implicit reset of the physical clock.

#### 48.5 Nonnegative spectral support

For fixed bounded local  $A$ , the finite correlation

$$c_{N,A}(t) = \omega_N(A^* \mathcal{T}_t^N(A)) = \langle A \Omega_N, e^{itK_N/\hbar} A \Omega_N \rangle$$

has a positive spectral measure supported on  $[0, \infty)$ . Along the selected subsequence, local state and norm dynamical convergence give its limit  $c_A(t) = \langle \pi(A)\Omega, e^{itH_{\text{num}}/\hbar} \pi(A)\Omega \rangle$ . All correlations are bounded by  $\|A\|^2$ .

Take any nonnegative smooth compactly supported energy test  $f$  inside  $(-\infty, 0)$ . Its  $\hbar$ -scaled Fourier transform is integrable. Fourier inversion and dominated convergence pass  $\int f d\nu_{N,A} = 0$  to the actual Stone spectral measure  $\nu_A$ . Such tests exhaust the negative half-line. Therefore the negative spectral projection annihilates every local cyclic vector and, by density, is zero. Thus  $H_{\text{num}} \geq 0$ . No finite positive gap is passed in this loop.

Finite ground densities are gauge invariant under every original finite endpoint group, so  $\omega_{\text{num}}$  is gauge invariant. Each finite evolution takes a bounded local gauge-invariant operator to another

such operator, and its norm limit lies in their norm-closed algebra  $A_{\text{phys}}$ . Both signs of time preserve  $A_{\text{phys}}$ ; hence the physical cyclic space  $\overline{\pi(\mathcal{A}_{\text{phys}})\Omega}$  reduces  $U_t$  and  $H_{\text{num}}$ . Its restriction is a nonnegative self-adjoint physical generator. Equality with the full joint fixed-vector sector and a nonzero excited physical signal remain for AQ2.

## 48.6 Controls and limitations

The checker audits negative-coordinate ownership, actual endpoint counts, complete incoming-star incidence, exact shell/convolution estimates and physical units. The densities  $|e_n\rangle\langle e_n|$  show how local trace mass can escape without an energy-tightness bound. Inconsistent finite marginals are rejected.

On  $\ell_2(\mathbb{N})$ ,  $U(t)e_j = e^{ijt}e_j$  is strongly continuous, while  $Ae_j = e_{2j}$  obeys  $\|U(\pi/n)AU(\pi/n)^* - A\| = 2$  for every  $n$ , witnessed by  $e_n$ . This preserves the distinction between norm volume convergence and full-B norm time continuity.

The prior strategy credit is explicit. No  $SU(3)$  coefficient, old  $\tau_*$  interval, HTW constant, prior orthant-state identity, whole-sequence limit, translation invariance or boundary uniqueness is imported. The state is a constructed full-lattice subsequential state at a fully numerical coupling cap. Subsequent physical-gap claims require the next reviewed loop.

Reproduce from the repository root:

```
python -B research/round29/forward/aq1/check.py \
    --output /absolute/new/output
```

## Final admitted limits and evidence

- This is existence of a subsequential full-lattice state, not uniqueness of all states, whole-sequence convergence, translation invariance or all-boundary independence.
- No old symbolic  $\tau_*$  or HTW restriction is used, but no identity with the earlier conditional orthant or bulk state is thereby established.
- Norm convergence of each local evolution as volume grows is distinct from norm continuity in time on all  $B(H)$  local operators; only the separately proved GNS strong continuity is asserted.
- The compactness/dynamics/GNS/Fourier strategy is already explicit in Gauvin Supplement A.10 and uses Nachtergaele–Sims/NSY. The contribution is the reviewed  $SU(2)$  selected-strip dictionary and numerical-cap application, not invention of the generic method.
- No positive physical spectral gap, Wilson operator-domain identity, calibrated mass or continuum construction is admitted by this loop.

research/round29/advisor/aq1-gate.json

research/round29/reverse/aq1/report.md

research/round29/skeptic/aq1.md

*Sequencing note.* Prospective statements in this frozen derivation describe the moment of its admission. The final Round29 roadmap below governs subsequent work.

## 49 HNM AQ2: numerical-cap physical gap and Wilson witness\*\*

**Statement alias HNM-T-AQ2.** This identifier denotes the scoped mathematical claim admitted by this chapter's gate. It adds no assumption and establishes no literature-priority claim.

**Draft03 current extension: HNM-C-AT1.** The statements and future goals below retain their Round29 admission date. Round30 separately proves the original Wilson operator domain and energy-moment certificate in the actual AQ state; see Section 54. This fresh seven-star proof has second-moment ceiling  $\alpha^2(36 + 98|\tau|)$  and does not identify AQ with AO or replace the earlier orthant coefficient. Historical statements that the AQ moment theorem is missing are superseded only by that new scoped gate.

**Admission: accepted within scope.** For the actual AQ1 centered full-Z3 subsequential state at both signs of  $|\tau| \leq 1/100000000$ , the complete original-endpoint gauge-fixed space equals the invariant-local cyclic completion and reduces the physical energy generator. Centered Fourier tests including zero give  $H_{\text{phys}} \geq (\alpha/16)(I - P_\Omega)$  and a simple vacuum in this representation. The full-GNS strengthening explicitly uses AM2's separately reviewed full-Hilbert finite gap. For the original xz Wilson in the same state, the actual 48-link/36-endpoint cover has seven incident stars, local reference energy  $\leq 98|\tau|$  and variance  $\geq 61999/250000 > 1/5$ . The reverse proof separately sharpens that floor to  $3099951/12500000$ .

The typeset body below preserves the source-bound forward derivation. Its prospective language is historical; this boxed gate records the final reviewed verdict. The separately authored reverse and skeptical records remain linked. The common admission and attributed refinements are determined by the gate, not by a renamed equation.

Human project author: **Hruday N M (BUNZEEY)**. This is the independent forward execution of the frozen tenth contract; no current reverse or skeptic solution was read. The compact-group averaging and Fourier gap-passage strategy is explicitly present in Gauvin [1], arXiv:2503.15539v3 Supplement A.10, pp.S7–S8. We apply that established strategy with the actual SU(2) selected-reference dictionary and AM2 constants. Scientific priority is unverified.

**Result:** for the actual AQ1 subsequential state at either sign of  $|\tau| \leq 10^{-8}$ , the full original-endpoint gauge-fixed space equals the physical local cyclic completion and reduces the physical generator. Its vacuum is simple and its spectral gap is at least  $\alpha/16$ . Using the separately reviewed **full-Hilbert** AM2 strengthening gives this gap and vacuum simplicity also on the full AQ1 GNS representation. The original xz Wilson has variance at least  $61999/250000 > 1/5$  in this same state. There is no identification with the earlier orthant state and no moment or operator-domain assertion for this Wilson vector.

### 49.1 The actual state, algebra and compact endpoint group

Keep exactly AQ1's chosen subsequence of centered boxes  $\Lambda_N = [-N, N]^3$ , actual whole-star Hamiltonians, complete 24-link factors and locally normal limit  $\omega_{\text{num}}$ . Set  $\delta = \alpha/8$ , and let its GNS data be  $(\pi, \mathcal{H}_{\text{num}}, \Omega)$ . AQ1 constructs the strongly continuous time group  $U_t = \exp(itH_{\text{num}}/\hbar)$  with  $H_{\text{num}} \geq 0$  and  $H_{\text{num}}\Omega = 0$ . Every physical scale including  $\alpha/E_*$ , spacing and  $\hbar$  is fixed.

Let  $V$  be the countable set of **all original fine-lattice vertices**, and  $G = \prod_{v \in V} \text{SU}(2)$ , with its compact product topology and product normalized Haar measure. For every original positively stored link  $(x, j)$ ,

$$U_{x,j} \mapsto g_x U_{x,j} g_{x+e_j}^{-1}.$$

This is the original tail-and-head action. A bounded operator  $A$  in a finite complete-factor algebra depends under conjugation on only the finite endpoint set  $V(F)$ . The representation on

its finite link Hilbert space is strongly continuous, so  $g \mapsto \beta_g(A)$  is bounded and strong-star continuous in those endpoint coordinates. This does not imply operator-norm continuity for arbitrary  $A$  in  $B(H_F)$ .

Each such action agrees locally with a gauge transformation supported on finitely many vertices. AQ1 state invariance consequently extends to all  $G$ . Define

$$V_g \pi(A) \Omega = \pi(\beta_g(A)) \Omega. \quad (\text{HNM-AQ2.1})$$

State invariance makes this well-defined and isometric; the inverse action makes it unitary. For local  $A$ , normality of the local density gives continuity of  $\omega_{\text{num}}(A^* \beta_g(A))$  under bounded strong-star convergence. The norm-square identity used in AQ1 then proves strong continuity of  $V_g$  on local cyclic vectors, and unitarity extends it by density to the full GNS space. This also proves continuity for the compact product topology, since a local vector needs only finitely many endpoint coordinates. No physical tensor factorization is assumed.

## 49.2 Haar projection and complete physical density

Strong vector integration defines  $P_G \xi = \int_G V_g \xi dg$ . Strong continuity makes the orbit of each vector compact and separable; its bounded vector integral exists. Haar invariance and inversion give  $P_G^2 = P_G = P_G^*$ , with range the joint fixed-vector space. Finitely supported gauge transformations are dense in  $G$ , so their joint fixed space is the same range.

For local  $A$  define instead the **ultraweak** operator integral

$$\mathcal{E}_F(A) = \int_{SU(2)^{V(F)}} \beta_g(A) dg \in B(H_F). \quad (\text{HNM-AQ2.2})$$

It exists by pairing with trace-class functionals, is contractive and positive, and is invariant under every endpoint action. We do not use a norm-Bochner integral of the full-B operator orbit. To identify its GNS vector, pair with any local cyclic vector  $\pi(B)\Omega$ . Both operators lie in a larger finite complete-factor region  $D$ . The functional  $X \mapsto \omega_{\text{num}}(B^* X)$  is normal on that region, since its density is trace class and  $B$  is bounded. It therefore commutes with this ultraweak integral. The scalar equality, followed by density, proves

$$P_G \pi(A) \Omega = \pi(\mathcal{E}_F(A)) \Omega. \quad (\text{HNM-AQ2.3})$$

Every local invariant cyclic vector is fixed. Conversely apply  $P_G$  to a sequence of local cyclic vectors approaching any fixed vector and use (HNM-AQ2.3). Hence

$$\mathcal{H}_{\text{phys}} = \overline{\pi(\mathcal{A}_{\text{loc}}^G) \Omega} = \{\xi : V_g \xi = \xi \text{ for all } g \in G\}. \quad (\text{HNM-AQ2.4})$$

This is the full bounded local invariant completion, not only Wilson polynomials or a tensor product of blockwise physical spaces. Shared original vertices act on incoming and outgoing links across factor boundaries.

Every finite Hamiltonian is invariant under its complete original endpoint group. Its evolution commutes with  $\beta_g$ ; the AQ1 norm limit retains that identity. Thus  $V_g$  commutes with  $U_t$ , as does  $P_G$ . The physical space reduces  $H_{\text{num}}$ :  $P_G D(H_{\text{num}}) \subset D(H_{\text{num}})$  and  $H_{\text{num}} P_G = P_G H_{\text{num}}$  there, by the spectral theorem. Its restriction is self-adjoint and nonnegative.

## 49.3 Centered spectral measures, including zero

Write the actual raw finite Hamiltonian as  $H_{\text{raw\_N}}$ , its ground energy as  $E_{\text{raw\_N}}$ , and

$$K_N = H_N^{\text{raw}} - E_N^{\text{raw}} = \delta(\hat{H}_N - \hat{E}_N).$$

No ground scalar is omitted. Put  $m_{\text{gap}} = \delta/2 = \alpha/16$ . For any bounded local **gauge-invariant**  $A$ , its finite centered vector

$$\chi_{N,A} = (A - \omega_N(A))\Omega_N$$

is physical, orthogonal to the unique finite vacuum, and has spectral measure supported on  $[m_{\text{gap}}, \infty)$  by AM2's finite physical theorem. For arbitrary bounded local  $A$  the identical support statement uses the **separately reviewed full-Hilbert AM2 strengthening**; it does not follow merely from a physical finite gap.

For either allowed class, let  $a_N = \omega_N(A)$  and  $a = \omega_{\text{num}}(A)$ . Local trace-norm convergence gives  $a_N \rightarrow a$  and

$$\nu_{N,A}(\mathbb{R}) = \omega_N(A^*A) - |a_N|^2 \longrightarrow \omega_{\text{num}}(A^*A) - |a|^2. \quad (\text{HNM-AQ2.5})$$

Complex means require the absolute square. The centered correlations are exactly

$$c_{N,A}(t) = \omega_N(A^* \mathcal{T}_t^N(A)) - |a_N|^2 \longrightarrow c_A(t) = \omega_{\text{num}}(A^* \mathcal{T}_t(A)) - |a|^2. \quad (\text{HNM-AQ2.6})$$

The convergence follows along the actual AQ1 subsequence by a fixed finite-region dynamical approximation and local trace-norm convergence, uniformly on compact times. The limiting function is the actual Stone spectral correlation of  $\chi_A = (\pi(A) - a)\Omega$ ; it is continuous at zero. All correlations are bounded in magnitude by  $\|A\|^2$ . Thus mass and Fourier transforms converge to the same actual spectral measure, without any assumed unbounded moment convergence.

For a nonnegative  $f \in C_c^\infty((-\infty, m_{\text{gap}}))$ , use the physical Fourier pair

$$f(E) = \int_{\mathbb{R}} \hat{f}_h(t) e^{itE/\hbar} dt,$$

whose inverse-transform coefficient is integrable. Finite support exclusion gives  $\int f d\nu_{N,A} = 0$ . Dominated convergence in (HNM-AQ2.6) gives  $\int f d\nu_A = 0$ . Such functions exhaust the entire interval below  $m_{\text{gap}}$ , **including neighborhoods of  $E = 0$** . Consequently  $\nu_A((-\infty, m_{\text{gap}})) = 0$ . Testing only  $(0, m_{\text{gap}})$  would leave a spurious centered zero atom undetected.

By (HNM-AQ2.4), the centered local invariant vectors are dense in  $\mathcal{H}_{\text{phys}} \cap \Omega^\perp$ . Therefore the physical spectral projection below  $m_{\text{gap}}$  vanishes on that orthogonal complement. The physical restriction obeys

$$H_{\text{num,phys}} \geq \frac{\alpha}{16}(I - |\Omega\rangle\langle\Omega|) \quad \text{as a quadratic-form inequality.} \quad (\text{HNM-AQ2.7})$$

The vacuum is simple there. Applying the same argument to **all** local  $A$  with the reviewed full-Hilbert finite gap makes the centered vectors dense in the whole  $\Omega^\perp$  and gives (HNM-AQ2.7) on the full GNS space as well. Equivalently, in either space  $\mathbf{1}_{[0, \alpha/16)}(H) = |\Omega\rangle\langle\Omega|$ . The energy threshold is  $\alpha/16$ ; the corresponding frequency is  $\alpha/(16 \hbar)$ .

#### 49.4 Original Wilson cover and seven-star reset

Take the original xz face at the fine origin:

$$W = \frac{1}{2} \text{Tr}(U_{0,x} U_{e_x,z} U_{e_z,x}^{-1} U_{0,z}^{-1}).$$

The four distinct stored links belong exactly to the complete factors  $R = \{0, e_z\}$  under  $\pi(x, y, z) = (\lfloor x/4 \rfloor, \lfloor y/2 \rfloor, z)$ . The complete cover contains **48 links and 36 original endpoints**. Each factor has 22 endpoints; their intersection has eight. These shared endpoints forbid replacing the joint gauge action by two unrelated blockwise physical actions.

The full-lattice incident anchors are exactly

$$R - S = \{0, -e_x, -e_y, -e_z, e_z, e_z - e_x, e_z - e_y\}.$$

There are seven, all retained in every centered box with  $N \geq 2$ . The orthant's two-anchor count is not applicable. Resetting the ground density on  $R$  to the **actual selected-reference product**  $P_R = P_0 \otimes P_{e_z}$  changes each incident star expectation by at most  $2\|\phi_b\| = 14|\tau|$ . The finite ground variational principle, with unchanged exterior reference energy, gives

$$\omega_N(h_R) \leq 98|\tau|. \quad (\text{HNM-AQ2.8})$$

The reset trial need not be physical: AM2 identifies the full ground. It has finite reference energy; bounded spectral truncation justifies cancellation of the exterior unbounded terms as in AQ1. Passing bounded positive spectral truncations of  $h_R$  through local trace-norm convergence and taking their increasing supremum yields the same inequality in the actual AQ1 state.

Since  $h_R \geq I - P_R$ , set  $\varepsilon = 1 - \text{Tr}(\rho_R P_R) \leq 98|\tau|$ . The pure-reference trace distance satisfies

$$\|\rho_R - P_R\|_1 \leq 2\sqrt{\varepsilon} \leq \frac{1}{500} \quad (|\tau| \leq 10^{-8}). \quad (\text{HNM-AQ2.9})$$

For completeness, decompose  $\rho_R$  into pure states. The trace distance from each pure state to  $P_R$  is  $2\sqrt{1 - \text{overlap}}$ ; convexity and concavity of the square root yield the displayed bound. The numerical inequality uses  $98/100000000 < 1/1000000$ .

## 49.5 Reference integration and interacting variance

Every original  $z$  link is one of the fourteen free links in its reference factor. Conditional on all other original links, one  $z$  link of this face remains Haar in  $P_R$ . Left/right multiplication and inversion preserve its normalized Haar law, so the face product is Haar under this **reference** integration. For  $\text{SU}(2)$ , write a Haar matrix as  $q_0 I + i\mathbf{q} \cdot \boldsymbol{\sigma}$  on the unit three-sphere. Symmetry gives mean  $q_0$  zero and second moment  $1/4$ . Thus

$$\text{Tr}(P_R W) = 0, \quad \text{Tr}(P_R W^2) = 1/4.$$

This does not assume that either the selected strip vacuum or the interacting local density is a product Haar state. Boundedness  $\|W\| \leq 1$ ,  $\|W^2\| \leq 1$  and (HNM-AQ2.9) give

$$|\omega_{\text{num}}(W)| \leq 1/500, \quad \omega_{\text{num}}(W^2) \geq 1/4 - 1/500, \quad (\text{HNM-AQ2.U10})$$

$$\text{Var}_{\omega_{\text{num}}}(W) \geq \frac{1}{4} - \frac{1}{500} - \frac{1}{250000} = \frac{61999}{250000} > \frac{1}{5}. \quad (\text{HNM-AQ2.10})$$

The vector  $(\pi(W) - \omega_{\text{num}}(W))\Omega$  is therefore nonzero, physical and orthogonal to the vacuum in the **same** AQ1 representation. Its bounded-vector spectral measure has positive total mass and is supported on  $[\alpha/16, \infty)$ . No operator-domain or first/second-moment identity is asserted here.

## 49.6 Scope, controls and stop

The executable checker independently reconstructs the complete link/endpoints, shared gauge vertices, full seven-anchor incidence and rational reset/variance constants. Actual  $\text{SU}(2)$  matrix fixtures distinguish correct tail/head actions from missing heads. A joint-invariant singlet fixture rejects an independent blockwise fixed-sector substitution. Centered measures with a zero atom reject tests only in the open positive interval; complex and moving-mean controls retain the true finite centering and ground shift. A bounded local density fixture shows why reference Haar moments cannot be assigned directly to the interacting density. Two distinct gapped representations of a direct-sum algebra reject equating a simple vacuum in one representation

with uniqueness of all states. These finite diagnostics support the analytic arguments; they do not prove infinite completeness by enumeration.

The Newton-inspired check reconstructs the original graph and limiting premises before synthesizing the conclusion. The Tesla-inspired mechanism check retains the endpoint channels, source/state distinction and fixed physical clock. Neither historical figure is asserted to endorse this result.

Vacuum simplicity here is a property of the chosen GNS representation. It does not establish uniqueness of every thermodynamic ground state, whole-sequence convergence, translation invariance, boundary independence, or identity with the older symbolic-smallness state. No continuum limit or Clay mass-gap solution follows. This is investigation ten: after review, scientific production stops; further goals are advice only.

Reproduce from the repository root:

```
python -B research/round29/forward/aq2/check.py \
  --output /absolute/new/output
```

## Final admitted limits and evidence

- This is the chosen AQ1 representation at fixed lattice spacing and fixed physical scales. Vacuum simplicity there does not establish uniqueness of all thermodynamic ground states, whole-sequence convergence, translation invariance or all-boundary independence.
- The numerical-cap state has not been identified with the older conditional orthant or bulk state. No AO2 Wilson operator-domain or moment identity is transferred to it.
- Local gauge averaging is ultraweak on full bounded operator algebras and strong on GNS vectors; full-B norm continuity and physical tensor factorization are not assumed.
- The general gauge-averaging and Fourier gap-transfer strategy is already explicit in Gauvin Supplement A.10 and relies on the AQ1 source-matched dynamics. The reviewed contribution is the specific SU(2) dictionary, constants and same-state witness; scientific priority is unverified.
- A lower spectral gap and positive Wilson variance do not evaluate a particle mass, spectral atom, dispersion relation or experimental signal. AL's weak-bare-coupling certificate obstruction and the continuum Yang-Mills construction remain unresolved.

research/round29/advisor/aq2-gate.json

research/round29/reverse/aq2/report.md

research/round29/skeptic/aq2.md

## Separately admitted reverse refinement

The common synthesis uses the forward floor  $61999/250000 > 1/5$ . The same final gate explicitly admits the reverse derivation's sharper rational floor, under the same chosen state and coupling conditions:

$$\text{Var}_{\omega_{\text{num}}}(W) \geq \frac{3099951}{12500000} > \frac{1}{5}. \quad (\text{HNM-AQ2.R1})$$

This refinement is attributed to `research/round29/reverse/aq2/report.md` and its bound checks. It is not substituted into the frozen forward report and does not identify another state or establish literature priority.

*Sequencing note.* Prospective statements in this frozen derivation describe the moment of its admission. The final Round29 roadmap below governs subsequent work.

## 50 An exact arithmetic companion for HNM-C-AM2

The manuscript includes `calculators/hnm_round29.py`, a Python standard-library companion that evaluates the admitted majorant budgets using exact rational arithmetic. It preserves the physical conversions by  $\alpha$ ,  $\hbar$  and fixed  $E_*$ . It checks scalar inequalities only: it does not examine an input Hamiltonian, establish the onsite hypotheses, select an infinite-volume state or measure an actual gap.

At the inclusive cap  $|\tau| = 10^{-8}$ , both signs give

$$J \frac{148}{7} = \frac{37}{6250000}, \quad 2J \, 352 = \frac{77}{390625} < 1, \quad J = 28|\tau|. \quad (\text{HNM-353})$$

The first budget is less than the coefficient-ball radius  $1/64$ . For  $\alpha = 2$  in the declared energy unit, the conditional physical lower bound is  $1/8$  in that unit. Outside the admitted cap the companion prints an explicit out-of-scope status even when these displayed scalar inequalities pass; it does not promote that arithmetic to a reviewed extension of the theorem. Decimal outputs are for display; the rational strings are the authoritative calculations.

From the repository root:

```
python3 -B papers/draft-02/calculators/hnm_round29.py \
  --tau=-1/1000000000 --alpha 2 --hbar 1 \
  --energy-reference 1
```

This is an editorial/reproduction companion, not another physics investigation. The original producer and skeptic programs retain their separate source-bound proof checks and controls.

## 51 Round29 closeout and the next five goals

The five selected goal pairs are complete: AL1–AL2, AM1–AM2, AN1–AN2, AO1–AO2 and AQ1–AQ2. These are ten reviewed investigations, not ten independent discoveries. AM1’s insufficient source-template transfer remains visible; AM2 supplies a separately proved alternative certificate. The last two pairs were chosen only after the first six reviews, and each second investigation responded to its preceding review. Scientific production stops at this checkpoint. Subsequent builds, evidence replays and presentation repairs do not count as research loops.

**Draft03 current extension: HNM-C-AT1.** The statements and future goals below retain their Round29 admission date. Round30 separately proves the original Wilson operator domain and energy-moment certificate in the actual AQ state; see Section 54. This fresh seven-star proof has second-moment ceiling  $\alpha^2(36 + 98|\tau|)$  and does not identify AQ with AO or replace the earlier orthant coefficient. Historical statements that the AQ moment theorem is missing are superseded only by that new scoped gate.

### 51.1 What the completed cycle changes

| Pair | Admitted change                                                                                                                       | Surviving boundary                                                                                                         |
|------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| AL   | Exact complete-model scale dictionary and common-scale invariants; the declared weak-bare-coupling path leaves the sufficient regime. | Magnetic-only suppression changes the action. A failed sufficient certificate does not prove that the physical gap closes. |

| Pair | Admitted change                                                                                                                                                                                                                                 | Surviving boundary                                                                                                                                                                              |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AM   | A full four-site creation proof gives a unique finite-volume ground and gap at least $\alpha/16$ at both signs of $ \tau  \leq 10^{-8}$ , with representation cutoffs removed.                                                                  | The named external stability constants are not numerically evaluated. Finite-volume data alone do not construct a thermodynamic state; AQ is the separate extension.                            |
| AN   | Specified deep-bulk translates of the old orthant state converge in local state norm to the compatible centered-box state, with independent outer cutoffs.                                                                                      | The original symbolic radius and independent HTW smallness remain; this is state identification under those conditions, not generator identification or all-boundary uniqueness.                |
| AO   | The actual old orthant Wilson vector has finite second spectral moment, belongs to the physical operator domain and has exact first energy $\alpha\omega(1 - W^2)$ .                                                                            | Both inherited restrictions remain. Equality of second moments is unproved; this result is not automatically a statement about the new AQ state.                                                |
| AQ   | At the numerical cap, a full-lattice locally normal subsequential state has complete original-endpoint physical sector, gap at least $\alpha/16$ , simple vacuum in its representation and original Wilson variance $\geq 61999/250000 > 1/5$ . | No uniqueness of all thermodynamic states, whole-sequence limit, translation invariance, old-state identity, Wilson moment/domain theorem, particle mass or continuum construction is inferred. |

The stronger full-GNS AQ2 conclusion explicitly uses AM2's separately reviewed full-Hilbert finite-volume gap. The reverse AQ2 variance refinement is separately attributed in that chapter; the table uses the common forward floor. Gauvin's Supplement A.10 and the matched unbounded-on-site dynamics sources retain credit for the general limiting strategy. HNM labels locate this project's checked model application and constants, without settling scientific priority.

The two state branches must remain distinct. AO refers to the inherited conditional orthant state with  $|\tau| < \tau_*$  and  $|\tau| \leq 2^{-16}$ . AN's bulk identification further retains the HTW hypothesis. AQ uses the fully numerical cap  $|\tau| \leq 10^{-8}$  and a selected centered full-lattice subsequence; it has not been identified with those older states. A simple vacuum in the chosen AQ representation is not a theorem that every thermodynamic limit equals it. The new numerical interval also does not repair AL's weak-bare-coupling obstruction. Fixed positive  $\alpha$ ,  $\hbar$ , lattice spacing and energy reference remain part of each physical claim.

## 51.2 Prospective roadmap: planned and unexecuted

The following five goals are the advisor's actual post-ten selection in `research/round29/advisor/roadmap.json`. Their order is qualitative scientific dependency and utility, not a novelty score or a percentage of the Clay problem. The first AT deliverable is especially bounded, even though state identification and the weak-coupling strategy have higher scientific priority. Each first contract must be frozen prospectively; each second investigation is selected only after independent directions and skeptical review. None of these future goals is executed in this release.

### 51.2.1 1. HNM roadmap AR: Identify the numerical-cap bulk state

**Target.** Derive an explicit spatial boundary estimate for the AQ model and use it to test whole-sequence convergence and uniqueness of the declared full-lattice limit.

**Missing premise.** AQ proves subsequential existence and a simple vacuum within its GNS representation; it does not give a quantitative comparison of different thermodynamic limits.

**Prospective first test.** Freeze two centered/bulk exhaustions of the same repeated selected-strip Hamiltonian. Derive or reject one explicit norm-one local-observable boundary estimate at a declared numerical interval, including all complete factors and original endpoints.

**Adaptive second test.** After the first skeptical review, either use the actual bound to prove the specified all-pairs local Cauchy/state-identification result, or isolate the missing estimate with a discriminating countercontrol.

**Exact model.** Full coarse Z3, actual complete 24-link factors, fixed repeated selected triple, whole-star omission and physical scales; AQ state at  $|\tau| \leq 10^{-8}$ . Any comparison to AN or old orthant states retains their additional common hypothesis range and deep-bulk observation map.

**Reason for priority.** Closes the main identification gap exposed by the newly completed numerical-cap construction. Rank reflects scientific dependency, not predicted ease.

**Limits.** Do not identify a fixed-origin orthant boundary state with the full-lattice bulk state. AM2 does not evaluate HTW constants. A simple GNS vacuum is not uniqueness of all states.

### 51.2.2 2. HNM roadmap AS: Test a different weak-coupling reference strategy

**Target.** Select one source-grounded alternative reference or multiscale candidate and test whether its complete dimensionless interaction dictionary can follow a weak-bare-coupling trajectory.

**Missing premise.** The current selected-strip sufficient regime has  $\tau = 96/g^4$  and cannot approach  $g=0$ . Common energy/time rescaling leaves this obstruction unchanged.

**Prospective first test.** Freeze one alternate reference, full original Hamiltonian, boundary prescription and physical-clock map; compute the actual normalized local interaction along a specified weak-coupling path. Preserve an insufficient outcome if no eligible route is found.

**Adaptive second test.** Only if the first dictionary passes review, derive or simulate one bounded stability/scale-matching test with explicit error; otherwise select a concrete obstruction or model-distinction test.

**Exact model.** A newly declared candidate must map to the actual uniform  $SU(2)$  Hamiltonian or explicitly state its deformation; no automatic transfer from AQ.

**Reason for priority.** Addresses the most consequential unresolved route toward the continuum target, while recognizing that it is substantially harder than a routine corollary.

**Limits.** Independent magnetic suppression changes the action and is not merely a change of units. No continuum construction follows from one finite-scale test.

### 51.2.3 3. HNM roadmap AT: Certify spectral response in the numerical-cap state

**Target.** Build a same-AQ-state Wilson spectral-response certificate, beginning with operator-domain and energy-moment control and then testing one bounded spectral-weight or decay question.

**Missing premise.** AQ2 supplies a gap and nonzero fluctuation but no Wilson domain or moment theorem in that state. AO2 belongs to a different state and hypothesis range.

**Prospective first test.** Reconstruct the Wilson electric commutator and complete seven-star regional energy budget directly in AQ finite boxes; pass actual compact spectral tests to that same AQ subsequential state and state exactly which moment bounds/equalities survive.

**Adaptive second test.** Choose one nontrivial, certified spectral-weight/window or response

conclusion from the reviewed first-loop output. If the premises cannot support it, record the obstruction instead of inventing a particle pole.

**Exact model.** The actual AQ1/AQ2 numerical-cap representation, original xz Wilson, 48-link/36-endpoint region, seven incident stars and fixed physical energy/time.

**Reason for priority.** Most bounded early subgoal for deliverability. The domain/moment repair belongs to a larger spectral-response question and is not inflated into several discoveries.

**Limits.** AO is a proof pattern, not an automatic state transfer. A uniform second-moment bound does not imply second-moment equality. A lower gap and moments do not determine an isolated mass, atom or dispersion law.

#### 51.2.4 4. HNM roadmap AU: Evaluate the certified 561-state heat output

**Target.** Turn the inherited exact-semigroup error theorem into an exported, reproducible numerical vector/evaluator with a rigorous full-output error budget.

**Missing premise.** AH1/AH2 provide structural and exact-semigroup estimates; the final certified 561-coordinate output has not been evaluated.

**Prospective first test.** Freeze graph, representation span, input, Haar Gram metric, separate ground centers and a requested tolerance. Implement one certified heat-action evaluation with interval or rigorous residual/tail control.

**Adaptive second test.** After reviewing the actual output and error denominator, extend to one declared time range/input or diagnose the failing resource bound.

**Exact model.** The inherited finite physical graph and AH1/AH2 561-coordinate generated span; complete outside-channel and truncation controls remain mandatory.

**Reason for priority.** Converts an existing theorem into a usable reproducible calculator without confusing finite numerical evidence with a continuum proof.

**Limits.** All-time true-relative error requires the actual full-output denominator. An exact matrix-exponential symbol is not an evaluated numerical artifact.

#### 51.2.5 5. HNM roadmap AP: Test inference with a design fixed from the prior

**Target.** Test canonical-model parameter inference under one schedule chosen before observing candidate outcomes, with independently varying nuisance parameters and complete physical errors.

**Missing premise.** Prior results identify conditional candidate separations; they do not yet validate one prior-fixed design over independently varying candidates.

**Prospective first test.** Freeze a compact prior, observable/clock dictionary, one observation schedule and acceptance margin, then compute candidate intervals with independent nuisance variations and all inherited error terms.

**Adaptive second test.** Choose the second investigation from the first review: certify a supported separation region, or derive an unresolved/inseparable counterfamily and revise the future design rule.

**Exact model.** The separate canonical summable model and physical readout maps of AI3/AI4; do not import the homogeneous AQ state or its gap.

**Reason for priority.** Preserves the previously proposed AP question under its original identity and keeps useful inference work distinct from homogeneous-gap evidence.

**Limits.** An unresolved result is allowed and must remain visible. A design optimized after seeing candidate outcomes does not answer the prior-fixed question.

The current priorities preserve the deferred AP identity and the finite-graph heat evaluation as separate questions. Improving a conservative constant is not automatically an extra research pair. All later conclusions still require their own prospective contracts, source dictionary and actual reviewed evidence.

## Part IV

# Round30: same-state spectral certificates and their limits

## 52 Round30: three adaptive investigations

Human author: **Hruday N M (BUNZEEY)**. This cycle contains exactly three reviewed investigations. AT1 was selected from the inherited roadmap; the later contracts were selected only after the preceding evidence and skeptical feedback. Source research, replay, editorial integration and publication do not count as extra investigations.

The current conclusions concern the original Wilson observable in the numerical-cap AQ state. HNM names identify this specific contribution record. Earlier eponyms, authors, physical symbols and frozen evidence remain intact. The three contributions and their separate equation aliases are not three proofs of continuum Yang–Mills.

## 53 Round30 source comparison and reading limits

The following records describe selected source reading, not an exhaustive state-of-the-art survey. Historical documents, patents, government custody and subreddit discussions may motivate controls, but none establishes a new Hamiltonian term. An unread lead remains a lead. Standard methods and their authors retain attribution. The modern comparison uses moment duality of Bertsimas–Popescu [27] and the spectral-kernel formulation of Abbott and collaborators [28]. These precede this application. The positive pointwise certificates in the following derivations do not claim a new generic moment method. The magnetic-basis trial-state route of Spriggs and collaborators [29] remains prospective; finite numerical energy improvements do not prove a lower gap.

| Source / version                                                                                                                                                             | Actual reading and relevance                                                                                                                                                                                                                                          | Limits                                                                                  |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <a href="#">Pinched Multi Affine Geometry and Confinement: Describing the Yang–Mills Mass Gap</a><br>v3, 2026-09-14; v1 2025-03-06;<br>v2 2026-09-11                         | Selected statements and adjacent explanation; not entire paper<br>Abstract and version history; II.3 Proposition 2.4; II.4 Theorem 2.5 and Corollary 2.6; Proposition 2.7<br>Closest prior comparison for AQ construction and a proposed spectral-response follow-up. | No wholesale validation of all manuscript claims; no scientific-priority determination. |
| <a href="#">Pinched Multi Affine Geometry: supplementary proofs</a><br>v3 ancillary supplement, 2026-09-14                                                                   | Targeted proof passages, printed S6-S8; no whole-supplement audit<br>A.9 excitation-energy passage; A.10 local compactness, dynamics, GNS continuity and spectral passage; A.11 proof of Corollary 2.6<br>Prevents renaming the generic strategy as an HNM invention. | AQ group, reference factors, seven-star region and constants require their own proofs.  |
| <a href="#">Quantum Yang–Mills Theory: the Clay problem statement</a><br>Official Clay-hosted problem statement; URL directory<br>2022/06 is not treated as publication date | Targeted pages 5-6 of PDF<br>Section 3 Quantum Fields; Section 4 The Problem; Beginning of Section 5 Comments<br>Scope gate for all fixed-spacing workbench claims.                                                                                                   | The workbench has not constructed the required continuum fields.                        |

| Source / version                                                                                                                                               | Actual reading and relevance                                                                                                                                                                                                                                                                                                                               | Limits                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| <a href="#">On the dynamics of lattice systems with unbounded on-site terms in the Hamiltonian</a><br>v1, 2014-10-29                                           | Targeted theorem and assumptions<br>Section 3 countable metric lattice and summability/convolution hypotheses; Theorem 4.1 and beginning of its proof<br>Actual AQ dynamics ancestry; limit in volume and continuity in time are distinct.                                                                                                                 | No new reproof of the external theorem; actual matching is inherited from AQ1.                      |
| <a href="#">Quasi-Locality Bounds for Quantum Lattice Systems, Part I</a><br>v2, 2019-03-01; J. Math. Phys. 60, 061101 (2019)                                  | Metadata and abstract only in this pass<br>Abstract and version history<br>Further theorem route; not a replacement for the explicitly checked NS theorem.                                                                                                                                                                                                 | Specific NSY theorem transfer not checked anew.                                                     |
| <a href="#">Angular momentum: an approach to combinatorial space-time</a><br>1971, Quantum Theory and Beyond, original pp.151-180; hosted transcription        | Selected opening passages; later combinatorial rules not audited<br>PDF pages 1-6: opening motivation, spin labels, model caveat, relational-angle proposal<br>Inspires representation and inner-product checks, not a mass-gap premise.                                                                                                                   | No transfer of emergent geometry into the workbench model; no endorsement.                          |
| <a href="#">Cargo Cult Science</a><br>1974 Caltech commencement address                                                                                        | Full short web text inspected; this ledger phrases only the named passages<br>Opening mystical-experience discussion; Integrity and disconfirming details paragraphs; Publishing unfavorable outcomes; Baseline replication discussion<br>Discriminating controls and source-qualified novelty; personal experiences do not establish physical mechanisms. | Historical anecdotes were not independently investigated; no physical conclusion relies on them.    |
| <a href="#">Simulating physics with computers: bibliographic lead</a><br>1982 article metadata, Int. J. Theor. Phys. 21, 467-488                               | Metadata only; linked resolver produced retrieval error<br>Simulating physics with computers bibliographic entry<br>Lead retained without pretending full-text reading.                                                                                                                                                                                    | Article text unread in this pass.                                                                   |
| <a href="#">Accurate ground states of SU(2) lattice gauge theory in 2+1D and 3+1D</a><br>v1, 2025-09-15                                                        | Selected model/ansatz sections and code-availability statement<br>Abstract; II.1 Hamiltonian and representation; II.2 Variational model; V Code and data availability<br>Concrete future AS reference strategy preserving the actual coupling dictionary.                                                                                                  | Code not installed or run; variational values do not certify a lower spectral gap.                  |
| <a href="#">Untruncated SU(2) theory with dynamical fermions: selected author preprint</a><br>v1, 2026-07-28                                                   | Selected technical model passages, not numerical validation<br>Abstract; Model paragraph and Eqs.(1)-(2); Variational wavefunction description<br>Model-deformation warning and current simulation lead.                                                                                                                                                   | Different dimension and matter content from AQ; numerical conclusions not independently reproduced. |
| <a href="#">Finite-temperature spectral functions in the lattice Schwinger model: selected author paper</a><br>v2, 2025-12-09; Phys. Rev. D 112, 114505 (2025) | Selected response definitions and numerical-scope passages<br>Lattice Schwinger model setup; Spectral functions subsection Eqs.(10)-(13); Finite-lattice peak/width qualification paragraphs<br>Operational spectral-response design and resolution warning.                                                                                               | Not the AQ model; no spectral pole or continuum inference transfers automatically.                  |

| Source / version                                                                                                                                                                                                 | Actual reading and relevance                                                                                                                                                                                                                                                                                                                                                                                                          | Limits                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <a href="#">Optimal Inequalities in Probability Theory: A Convex Optimization Approach</a><br>SIAM J. Optim. 15(3),780-804 (2005), DOI 10.1137/S1052623401399903; paper states electronic publication 2005-04-08 | Selected formulation and proof passages; no whole-paper reproduction<br>Section 2, dual problem, upper-moment sign rule, Theorems 2.1-2.2; Section 3, Proposition 3.1 positivity certificates<br>Closest classical method for prospective Loop2; no generic novelty claim.                                                                                                                                                            | No claim of strong duality or optimality for workbench certificates.                                      |
| <a href="#">The Causal Bootstrap: Bounding Smeared Spectral Functions from Non-Perturbative Euclidean Data</a><br>v1, 2026-05-19                                                                                 | Selected method and interpretation passages; no numerical reproduction<br>III.1 dual exact-data certificates; IV.1-IV.3 rational and piecewise polynomial kernels; VII.4 distinction between extremal spectra and true spectrum; VIII kernel and positivity caveats<br>Direct current prior overlap for proposed response bounds; exact arithmetic recommended for this small case.                                                   | No imported stochastic covariance or all-results validation; AQ inputs are deterministic analytic bounds. |
| <a href="#">Keynes MS.28: Hermes / Tabula Smaragdina</a><br>Indiana University normalized transcription                                                                                                          | Metadata and English ff.2r-2v read; Latin commentary visible but not fully translated; manuscript images not collated<br>ff.2r-2v: above/below correspondence and separation/recombination; catalogue distinction between English translation, Latin transcription and commentary<br>Ask for an explicit state/generator/scale map before identifying theories.                                                                       | No complete Latin commentary translation or manuscript collation                                          |
| <a href="#">Add.3973: Experiments</a><br>Indiana University normalized transcription                                                                                                                             | Metadata; ff.1r-2r and ff.5r-6r read; no complete notebook claim<br>December 1678 residues and mixtures; January 1679/80 uncertain dry weights, cracked vessel and escaped material<br>Keep actual residual, omitted-channel and state-replacement costs in AT/AU certificates.                                                                                                                                                       | Modern chemical identities and reproducibility not investigated                                           |
| <a href="#">US787412A: Art of transmitting electrical energy through the natural mediums</a><br>Patent text mirrored by Google Patents                                                                           | Description of terrestrial standing waves, three resonance requirements and receiving apparatus read; no physical performance validation<br>Diameter/quarter-wave condition and approximate six-per-second lowest rate; Duration requirement; Tuned receiver and location-dependent zero response at nodes<br>Specify the full generator, source, receiver and null geometry; do not equate a measured peak with a unique eigenvalue. | OCR not collated against original patent image; no performance audit                                      |
| <a href="#">Letter to E. T. Sturdy, 13 February 1896</a><br>Complete Works V, Epistles First Series LVII in this transcription                                                                                   | Letter text read, with attention to Tesla paragraph and cosmology discussion<br>Tesla's reported interest in Prana, Akasha and Kalpas; Expected later mathematical demonstration; Vivekananda's cosmological interpretation<br>Separate promised unification from a supplied mathematical map.                                                                                                                                        | No original manuscript facsimile collated; demonstration not located                                      |

| Source / version                                                                                                                                   | Actual reading and relevance                                                                                                                                                                                                                                                                                                                                                                                                                     | Limits                                                                                                         |
|----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <a href="#">Collected Papers on Analytical Psychology</a><br>Authorized translation, second edition reprinted; Project Gutenberg ebook 48225       | Title/edition, author's second-edition preface and Chapter I Automatisms pp.50–54 read; later synchronicity works not read here<br>1917 preface distinction between subject's numerical associations and author's explanation; Automatisms: experimenter cueing, table responses, and meaningless outputs preceding interpreted ones<br>Freeze tests and selection rules; retain null outputs and distinguish observed data from interpretation. | No modern clinical evaluation; no claim to survey all Jung writings                                            |
| <a href="#">The Pauli Letter Collection</a><br>CERN collection guide CERN-ARCH-PLC                                                                 | Collection scope, archival history, finding aids and allied materials read<br>CERN collection includes philosophical, psychological and epistemological correspondence; Pauli–Jung correspondence is held in ETH's Pauliana collection<br>Trace claims to the correct document and correspondent before interpreting them.                                                                                                                       | Letter 1417 from Pauli to Fierz, 1952-06-03, is an unread lead; its CDS page returned an anti-bot interstitial |
| <a href="#">Nikola Tesla, with his equipment for producing high-frequency alternating currents</a><br>Public Reddit thread as retrieved 2026-09-22 | Opening post and first displayed replies; no representative forum sample<br>Post's Tesla/Solfeggio/528 Hz/consciousness connection; Author's link to an original musical composition<br>Reject the jump from an emotionally salient or resonant output to its alleged physical cause without a model and control.                                                                                                                                | Attributed quotation not authenticated; physical or physiological claim not tested                             |
| <a href="#">Nikola Tesla Part 03 (Final)</a><br>Official FBI release catalogue and linked 64-page PDF                                              | Catalogue and PDF metadata only; text extraction empty; requested page screenshots produced text-only placeholders and were not visually inspected<br>Catalogue title and download description<br>Distinguish documentary custody from experimental confirmation.                                                                                                                                                                                | Underlying pages remain unread; no claim regarding death rays, energy technology or hidden equations           |
| <a href="#">NIST DLMF: Quadrature, trapezoidal rules</a><br>Online reference as read 2026-09-22; no release-version claim                          | Selected rule definitions and error formulas; no whole-chapter review<br>Section 3.5(i); Equations 3.5.1–3.5.3 and adjacent definitions<br>Standard trapezoidal rule and convex-error sign; the integrated derivative estimate is derived explicitly in the workbench application.                                                                                                                                                               | No novel general quadrature method or optimality claim. Actual AQ Euclidean samples have not been computed.    |

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research/round30/experts/historical/sources.json

## 54 Hrunday AT1: reviewed admission

**Verdict: accepted within scope.** In the actual numerical-cap AQ state at either sign of  $|\tau| \leq 10^{-8}$ , the centered original Wilson vector satisfies  $\chi \in D(H_{\text{phys}})$ . Its second-energy moment is at most  $\alpha^2(36 + 98|\tau|) < 37\alpha^2$ , and its first-energy moment equals  $\alpha\omega_{\text{num}}(1 - W^2)$ .

### Limits and independent review

The producer reports were frozen separately before exchange. Their common ancestry, contract and inherited methods remain disclosed. The skeptic reviewed both proofs and executed controls; this does not constitute external human peer review.

- No equality of finite-to-infinite second moments
- No AQ/AO/AN state identification
- No particle pole, exact mass or dispersion
- No uniqueness of all thermodynamic states or whole-sequence convergence
- No continuum Yang-Mills construction
- Scientific priority unverified; established methods retain attribution

research/round30/advisor/at1-gate.json

research/round30/skeptic/at1.md

### Substantive directional differences

- Both directions recover the same coefficient. The reverse checker separately tests inverse-link derivative signs through  $U U^{-1}$ ; agreement uses shared inherited premises and proof patterns.

## 55 Hruday AT1: same-state Wilson energy certificate (forward derivation)

**Source-bound forward derivation.** This is the complete typeset mathematical report of one producer. Prospective claims describe its original freeze; the shared reviewed result and permitted refinements are stated in the admission section. The two agents share inherited premises and model ancestry. This is neither an independent physical observation nor external human peer review.

Human project author: **Hruday N M (BUNZEEY)**. AI-assisted independent forward derivation, before reading any current AT1 reverse or skeptic solution. HNM labels below are traceable project aliases; established identities and external authors retain their names. Scientific priority is unverified. This is a model-specific extension, with no new axiom.

**Result.** In the actual AQ1 chosen subsequential centered-full-Z3 state at either sign of  $|\tau| \leq 10^{-8}$ , let  $H = H_{\text{num}}, \text{phys}$ ,  $\chi = (\pi(W) - \omega_{\text{num}}(W))\Omega$ , where  $W$  is the original xz plaquette. Then

$$\chi \in D(H), \quad \|H\chi\|^2 \leq B_2 := \alpha^2(36 + 98|\tau|) < 37\alpha^2, \quad \langle \chi, H\chi \rangle = \alpha\omega_{\text{num}}(1 - W^2). \quad (\text{HNM-AT1-F01})$$

This closes these obligations in AQ's own state without identifying it with AO's orthant state. Second-moment equality is not proved. The actual AQ gap and variance remain inherited hypotheses, not recomputed discoveries.

### 55.1 Model, source, scales and proof dependencies

Use exactly the AQ1 subsequence of whole-star centered boxes  $\Lambda_N = [-N, N]^3$  and the same fixed repeated selected coefficient triple. Every factor owns all 24 positive links whose tail is  $(4b_x + r, 2b_y + s, b_z)$ ,  $0 \leq r < 4, 0 \leq s < 2$ . Both tail and head gauge actions remain. Write

$$H_N^{\text{raw}} = \alpha \sum_{e \text{ owned}} C_e + V_{\text{sel}, N} - \frac{\alpha\tau}{24} \sum_{b+S \subset \Lambda_N} \sum_{f \in O_b} W_f, \quad S = \{0, e_x, e_y, e_z\}, \quad |O_b| = 21. \quad (\text{HNM-AT1-F02})$$

$C_e = -\sum_a X_{ea}^2$ , with generators  $i\sigma_a/2$ . The selected potential is  $-\sum \lambda_f W_f$ , with per-factor absolute coefficient budget at most  $9\alpha/8$ . With the actual selected-reference ground scalar  $B_N = \sum_b E_{\text{strip}, b}$  and  $\delta = \alpha/8$ ,

$$\delta \hat{H}_N = H_N^{\text{raw}} - B_N, \quad E_N^{\text{raw}} = B_N + \delta \hat{E}_N, \quad K_N = H_N^{\text{raw}} - E_N^{\text{raw}} = \delta(\hat{H}_N - \hat{E}_N). \quad (\text{HNM-AT1-F03})$$

The normalized interaction per anchor obeys  $\|\phi_b\| \leq 7|\tau|$ . All these terms, including selected bridges and both ground scalars, are retained. Alpha, hbar, lattice spacing and the positive comparison  $E_{\text{star}}$  are fixed. Tau is an interaction coefficient. L below is a proof energy cutoff, not a modified Hamiltonian; no new free physical variable is introduced.

AQ1 supplies local trace-norm convergence along its chosen subsequence, norm convergence of bounded-local dynamics uniformly on compact physical-time intervals, and the actual strongly continuous GNS time group. AQ2 identifies the full physical reducing sector and gives  $H \geq \alpha/16$  on its vacuum orthogonal complement. No AO state limit or old symbolic smallness interval is imported. AO1/AO2 are disclosed shared proof patterns only.

The closest inspected primary comparison is Gauvin [1], arXiv:2503.15539v3 Supplement A.9-A.11 (saved pp.S6-S8): compactness, centered Fourier transfer, local Wilson first-energy control

and spectral-envelope methods already occur there for another model. Nachtergaele-Sims arXiv:1410.8174 is the dynamics theorem already matched in AQ1. Here the specific contribution is the complete  $SU(2)$  seven-star commutator estimate and its first-moment/domain conclusion in the actual AQ state. This does not claim invention of spectral uniform integrability or establish priority. Live abstract/version pages were rechecked; reading depth is recorded in inputs/reading.json.

Newton analysis/synthesis guides reconstructing the necessary local energy bound and deriving the limiting statement; Tesla energy accounting guides retaining every gradient channel and scalar. These are modern methodological uses, not claims that those historical people supplied or reviewed this proof. Historical/esoteric interpretations add no premise.

## 55.2 Exact region and seven-star reset

The original observable is

$$W = \frac{1}{2} \text{Tr}[U_{0,x} U_{e_x,z} U_{e_z,x}^{-1} U_{0,z}^{-1}]. \quad (\text{HNM-AT1-F04})$$

Its four stored links have owners  $0, 0, e_z, 0$ . Its complete-factor cover is  $R = \{0, e_z\}$ , containing 48 distinct links. A single factor has eight tails and outgoing slabs of sizes 2,4,8, hence 22 distinct endpoints. For the two stacked factors the tail block has 16 vertices and outgoing slabs of sizes 4,8,8, hence 36 endpoints. These are the original shared endpoint actions; blockwise physical tensor factors are not substituted.

Every incident anchor lies in the exact set

$$R - S = \{0, -e_x, -e_y, -e_z, e_z, e_z - e_x, e_z - e_y\}. \quad (\text{HNM-AT1-F05})$$

All seven whole stars are retained once  $N \geq 2$ . In the positive orthant only  $0, e_z$  survive, giving the discriminating two-star control. The bulk proof must use seven.

Reset the finite ground density on  $R$  to the actual selected-reference rank-one product  $P_R$ , preserving the exterior reduced density. The reset has zero reference energy on  $R$  and leaves exterior reference expectations unchanged. Each incident retained interaction changes by at most  $2\|\phi_b\| \leq 14|\tau|$ . Positivity and the finite full-ground variational principle give

$$\omega_N(h_R) \leq 98|\tau|, \quad h_R = h_0 + h_{e_z}. \quad (\text{HNM-AT1-F06})$$

The mixed trial need not be gauge invariant because the AQ/AM full ground is the minimum used. All reference terms have finite expectation in the finite ground; replacing only finitely many factors by reference ground vectors keeps finite form energy. Nonnegative spectral truncations justify cancellation of unchanged exterior expectations. This reproduces the regional argument in the actual centered Hamiltonian; it is not a limit-state identification.

Define  $T_R = \alpha \sum_{e \text{ owned by } R} C_e$ ,  $V_R^{\text{sel}} = - \sum_{f \text{ selected}} \lambda_f W_f$ , and  $B_R = \sum_{\text{bin } R} E_{\text{strip},b}$ . Exactly,

$$T_R = \delta h_R - V_R^{\text{sel}} + B_R. \quad (\text{HNM-AT1-F07})$$

The Haar trial has zero electric and selected Wilson means, so each  $E_{\text{strip},b} \leq 0$ . The selected absolute budget on  $R$  is at most  $9\alpha/4$ . Consequently

$$\omega_N(T_R) \leq \alpha \left( \frac{9}{4} + \frac{49}{4} |\tau| \right). \quad (\text{HNM-AT1-F08})$$

Discarding the nonpositive  $B_R$  only after this exact reconstruction is valid. Setting it to zero in the identity is not. The selected-reference ground is generally not Haar: for one nonzero selected plaquette, the trial  $1 + tW$  has energy numerator  $3\alpha t^2/4 - \lambda t/2$ , negative at  $t = \lambda/(3\alpha)$ . At  $\tau=0$  reference energy is zero but actual electric energy need not be zero.

### 55.3 Finite domains, inverse links and the full commutator

For each finite box, its manifold is the compact product  $SU(2)^L$ . The elliptic Casimir sum has Sobolev operator domain  $H^2$  and form domain  $H^1$ . The real smooth bounded magnetic multiplication is a bounded perturbation, preserving these domains. Finite ground vectors belong to  $H^2$  (indeed elliptic regularity makes them smooth). Smooth multipliers  $W, W^2$  and their centered versions preserve  $H^{2/H^1}$  by the weak product rule and compactness. Thus  $\chi_N = (W - m_N)\Omega_N$  belongs to  $D(K_N)$ , with  $m_N = \omega_N(W)$ . This is a finite-volume domain result, not an assumed infinite-volume one.

All magnetic multipliers and ground scalars commute with  $W$ . For any smooth  $u$ ,

$$[C_e, W]u = (C_e W)u - 2 \sum_a (X_{ea} W) X_{ea} u. \quad (\text{HNM-AT1-F09})$$

Each of the four stored link occurrences contributes  $C_e W = 3W/4$ . To check both orientation and normalization, write its  $SU(2)$  holonomy as a quaternion  $(w, v)$ . Half-Pauli left multiplication gives scalar derivative  $-v_a/2$ , so its summed squared derivatives are  $(1-w^2)/4$ . For a positive occurrence  $Q=LUR$ , the basis is conjugated by  $L$ . For an inverse occurrence  $Q = LU^{-1}R$ , differentiating  $(\exp(tT_a)U)^{-1}$  gives  $-U^{-1}T_a$ ; on  $Q$  this is right multiplication by  $-Ad_{R^{-1}}T_a$ . Adjoint rotations preserve the summed squares, and the inverse sign/side remain in the actual derivative. Also  $T_a^2 = -I/4$  on either occurrence. Hence on all four distinct original links,

$$\sum_{e,a} (X_{ea} W)^2 = 1 - W^2, \quad \sum_e C_e W = 3W. \quad (\text{HNM-AT1-F10})$$

No other link derivative contributes. Define  $Z_N = \sum_{e \in p,a} (X_e a W) X_e a \Omega_N$ . Ground centering and (F09) give the  $L^2$  identity

$$K_N \chi_N = 3\alpha W \Omega_N - 2\alpha Z_N. \quad (\text{HNM-AT1-F11})$$

The derivative cross term is generally nonzero. Pointwise Cauchy-Schwarz and positivity of the four-link electric form inside the complete  $R$  sum yield

$$\|Z_N\|^2 \leq \int (1 - W^2) \sum_{e \in p,a} |X_{ea} \Omega_N|^2 \leq \omega_N(T_R)/\alpha. \quad (\text{HNM-AT1-F12})$$

Using  $\|W \Omega_N\| \leq 1$  and  $\|u + v\|^2 \leq 2\|u\|^2 + 2\|v\|^2$  now gives

$$\int E^2 d\nu_N(E) = \|K_N \chi_N\|^2 \leq 18\alpha^2 + 8\alpha\omega_N(T_R) \leq \alpha^2(36 + 98|\tau|) = B_2. \quad (\text{HNM-AT1-F13})$$

At the exact cap,  $B_2/\alpha^2 = 1800000049/50000000 < 37$ . The estimate is volume uniform along the actual AQ boxes for both  $\tau$  signs and all allowed selected coefficients. Its units are energy squared; the analogous frequency moment is divided by  $\hbar^2$ .

### 55.4 Finite first-energy identity

Let  $f_N = W - m_N$ . Subtract the weak ground equation tested on  $f_N^2 \Omega_N$  from the quadratic form evaluated on  $f_N \Omega_N$ . The valid  $H^1$  product identities give, for every  $X$ ,

$$|X(f_N \Omega_N)|^2 - \operatorname{Re}(\operatorname{conj}(X \Omega_N) X(f_N^2 \Omega_N)) = (X f_N)^2 |\Omega_N|^2.$$

All potential and scalar terms cancel because they commute with  $f_N$ . Thus, in the actual ground rather than a Haar substitute,

$$\int E d\nu_N(E) = \langle \chi_N, K_N \chi_N \rangle = \alpha \omega_N (1 - W^2). \quad (\text{HNM-AT1-F14})$$

Equivalently the double commutator has bounded extension  $2\alpha(1-W^2)$  and its ground expectation is twice this energy. The factor one half is essential. At the entirely free corner the exact variance is  $1/4$ , Wilson excitation energy  $3\alpha$ , first moment  $3\alpha/4$  and second moment  $9\alpha^2/4$ . That corner is a normalization fixture only.

### 55.5 Actual AQ spectral-test convergence

AQ provides local trace-norm state convergence and actual dynamics  $T_t = \lim_N \exp(it K_N / \hbar) \exp(-it K_N / \hbar)$  in norm on local observables uniformly on compact  $t$  intervals. Its GNS implementation is strongly continuous, with the actual physical self-adjoint generator  $H$ . Consequently, along the chosen AQ subsequence,

$$c_N(t) = \omega_N(W T_t^N(W)) - m_N^2 \longrightarrow c(t) = \omega_{num}(W T_t(W)) - m^2 = \langle \chi, e^{itH/\hbar} \chi \rangle. \quad (\text{HNM-AT1-F15})$$

Here  $m_N \rightarrow m$  and the masses  $v_N = \omega_N(W^2) - m_N^2 \rightarrow v$ . To justify the dynamical expectations, first approximate both volume evolutions by one fixed finite-region evolution; its state error is bounded by local trace distance times  $\|W\|^2$ , uniformly in time. Then remove the approximation using AQ's norm volume estimate. No common finite/infinite Hilbert-space embedding is asserted. The centered correlations are bounded in magnitude by  $v_N \leq 1$ .

For  $f \in C_c^\infty(\mathbb{R})$ , choose the physical Fourier pair  $f(E) = \int \hat{f}_h(t) e^{itE/\hbar} dt$ . Its transform is integrable. The bound above and dominated convergence imply  $\int f d\nu_N \rightarrow \int f d\nu$ , where  $\nu$  is the actual  $H$  spectral measure of  $\chi$ . Smooth compact functions are uniformly dense in  $C_0(\mathbb{R})$ ; all masses are at most one. A three-term uniform approximation therefore extends convergence to every  $C_0$  test. This explicitly builds the compact-test convergence from AQ correlations rather than importing AO tested resolvents. The inherited AQ gap supports these centered measures on  $[\alpha/16, \infty)$ ; only nonnegativity is needed for the moment argument.

Uncentering adds  $m_N^2 \delta_0$  and changes the total mass and zero-atom statement, although it leaves positive moments unchanged. The computation therefore cannot silently substitute uncentered measures.

### 55.6 The two limiting passages and domain conclusion

For  $L > 0$  let  $\theta(u) = 1$  for  $u \leq 1$ ,  $2-u$  for  $1 < u < 2$ , and  $0$  for  $u \geq 2$ . On  $E \geq 0$  define  $F_L(E) = E\theta(E/L)$  and  $Q_L(E) = E^2\theta(E/L)$ , extending by zero on negative  $E$ . They are continuous compact tests and increase pointwise to  $E$  and  $E^2$  as  $L$  increases. At fixed  $L$ , compact-test convergence and (F13) give  $\int Q_L d\nu \leq B_2$ . Monotone convergence yields

$$\int E^2 d\nu(E) \leq B_2. \quad (\text{HNM-AT1-F16})$$

For  $\nu_N$  and  $\nu$ , the explicit tails obey

$$\int_{E>L} E d\nu_N \leq B_2/L, \quad \nu_N((L, \infty)) \leq B_2/L^2, \quad (\text{HNM-AT1-F17})$$

and identically for  $\nu$ . The first inequality is uniform integrability of the first moment. It is not uniform integrability of the second.

The local bounded expectation in (F14) tends to  $s = \alpha\omega_{\text{num}}(1 - W^2)$ . At each fixed  $L$ ,

$$0 \leq \int E d\nu_N - \int F_L d\nu_N \leq B_2/L.$$

Take the AQ volume subsequence limit first, obtaining  $0 \leq s - \int F_L d\nu \leq B_2/L$ ; then send  $L$  to infinity. This proves

$$\int E d\nu(E) = \alpha\omega_{\text{num}}(1 - W^2). \quad (\text{HNM-AT1-F18})$$

For the actual self-adjoint  $H$ , the spectral criterion  $\int E^2 d\nu < \infty$  is equivalent to  $\chi$  in  $D(H)$ , and its value equals  $\|H\chi\|^2$ . Thus (F16) and (F18) prove (F01). No differentiation of the limiting correlation at zero or assumed strong convergence of finite unbounded vectors is used.

## 55.7 Falsifying controls, limitations and reproduction

The independent stdlib checker uses exact rationals and explicit sets. It reconstructs the 48 links, 36 endpoints, four original Wilson links, and seven incoming stars versus two orthant stars. Quaternion checks use positive and inverse occurrences with nontrivial stored-link holonomies. The  $Q(W^2) = 8W^2 - 2$  diagnostic gives  $[Q, W]W = 5W^2 - 2$ , distinguishing the missing-gradient surrogate  $3W^2$ .

The spectral family  $(1 - 1/n)\delta_1 + (1/n)\delta_n$  converges on bounded tests to  $\delta_1$ , while its first moments tend to two and its second moments grow; it falsifies first-moment boundedness as a transfer argument. In contrast  $(1 - 1/n^2)\delta_1 + (1/n^2)\delta_n$  has uniformly bounded second moments tending to two, but limiting second moment one. It confirms that the proved second-moment inequality cannot be promoted to equality without stronger tails. These abstract measure controls are not actual  $SU(2)$  measures. A two-state exact fixture demonstrates that the same gap/variance bounds do not identify a state; it licenses no AO-to-AQ transfer.

Run `python-Bresearch/round30/forward/at1/check.py--output/absolute/fresh/directory`. The normal and optimized interpreter results agree. Executed fixture checks support algebra and catch wrong models; the domain and limiting proof above supplies the infinite-volume theorem. Source snapshots and SHA-256 inventories bind the report, checker, actual inputs and outputs.

No second-moment equality, AO/AQ state identification, continuum solution, particle pole, exact mass, dispersion, translation invariance, whole-sequence convergence or uniqueness of all thermodynamic states is claimed. The same-state numerical-cap result is at fixed lattice spacing and scales. Next investigation remains for advisor selection after independent skeptical review.

`research/round30/forward/at1/report.md`

## 56 Hrudy AT1: same-state Wilson energy certificate (reverse derivation)

**Source-bound reverse derivation.** This is the complete typeset mathematical report of one producer. Prospective claims describe its original freeze; the shared reviewed result and permitted refinements are stated in the admission section. The two agents share inherited premises and model ancestry. This is neither an independent physical observation nor external human peer review.

Human author: **Hrudy N M (BUNZEEY)**. AI-assisted mathematical reconstruction and exact controls. HNM labels are project aliases, not priority claims; scientific priority remains unverified. The current forward and skeptic AT1 solutions were not read. The contract, inherited reports, gates and primary-source excerpt are copied in `inputs/`. Newton's backward-analysis method and Tesla's complete-system energy accounting motivate the workflow; neither historical authority nor mystical material supplies a mathematical premise.

### 56.1 Reverse target and necessary bridge

The target is the actual AQ1/AQ2 centered-full-Z3 subsequential state, not the older AO orthant state. To prove a local Wilson vector belongs to the domain of its actual energy generator, the spectral theorem requires a finite second energy moment in that same representation. Compact spectral-test convergence plus a uniform finite-volume second-moment bound suffice. To obtain an exact first-moment identity, the same bound supplies the missing uniformly integrable first-energy tails. A mere first-moment ceiling would be insufficient. This identifies the finite regional kinetic estimate, full commutator and same-state Fourier convergence as the necessary bridge; each is established below.

Keep exactly AQ's chosen subsequence of centered whole-star boxes, fixed repeated selected coefficient triple, and complete original 24-link SU(2) factors. Fix positive physical energy  $\alpha$ , action  $\hbar$ , spacing  $a$  and comparison energy  $E_{\text{star}}$ ;  $\delta = \alpha/8$ . Both signs of  $\tau$  with  $|\tau| \leq 10^{-8}$  are allowed. The spectral cutoff  $L > 0$  used below has energy units and is a proof parameter, not an interaction deformation. No symbolic smallness threshold from AO is assumed.

AQ1 provides locally normal compatible limits and the physical-time stationary dynamics. AQ2 gives its reducing physical sector, gap  $\alpha/16$  and nonzero original Wilson fluctuation. Write  $H = H_{\text{phys}}$ ,  $U_t = \exp(itH/\hbar)$ ,  $H_{\Omega} = 0$ , and

$$W = \frac{1}{2} \text{Tr}[U_x(0)U_z(e_x)U_x(e_z)^{-1}U_z(0)^{-1}], \quad \chi = (\pi(W) - \omega(W))\Omega, \quad \nu(B) = \langle \chi, \mathbf{1}_B(H)\chi \rangle. \quad (\text{HNM-AT1-R01})$$

The corresponding finite quantities are  $K_N = H_N^{\text{raw} - E_N^{\text{raw}}}$ ,  $m_N = \omega_N(W)$ ,  $\chi_N = (W - m_N)\Omega_N$  and  $\nu_N$ . Their measures have support in  $[\alpha/16, \infty)$  and mass at most one. This inherited gap is not needed for the moment proof except to preserve the exact physical state and positive-energy setting.

### 56.2 Complete geometry and seven-star reset

The coarse factor  $b = (i, j, k)$  owns every positive-direction original link with tail  $(4i + r, 2j + s, k)$ ,  $0 \leq r < 4$ ,  $0 \leq s < 2$ . The four stored Wilson links have owners  $0, 0, e_z, 0$ ; inverse occurrences retain the original stored link and its original endpoints. The complete region  $R = \{0, e_z\}$  therefore contains 48 links. Its 16 tail vertices plus outgoing boundary slabs of sizes 4, 8, 8 give 36 endpoint actions. All remain present, including heads outside the tail box.

A whole-star support is  $b+S$ ,  $S=\{0, e_x, e_y, e_z\}$ . The incident anchors are precisely  $R-S$ :

$$\{0, -e_x, -e_y, -e_z, e_z, e_z - e_x, e_z - e_y\}. \quad (\text{HNM-AT1-R02})$$

There are seven, as set enumeration and the shared anchor 0 show. Near a finite boundary there can be fewer retained stars, never more. For sufficiently large centered boxes all seven are retained. An origin region in the positive orthant instead meets only anchors  $0, e_z$ ; its two-star budget cannot be used here.

Reset the reduced state on  $R$  to the product of the actual selected-reference vacua while preserving its exterior marginal. Outside onsite energies are unchanged; bounded spectral truncation and monotone convergence justify this for unbounded energies. The actual finite-volume full ground energy, admitted in  $AQ$ 's premises, permits this full-Hilbert mixed trial. All terms disjoint from  $R$  agree. Each incident star has normalized operator norm  $\leq 7|\tau|$  and its expectation changes by at most twice that norm. Hence

$$\omega_N(h_R) \leq 2 \cdot 7 \cdot 7|\tau| = 98|\tau|. \quad (\text{HNM-AT1-R03})$$

This is a bound for the selected reference energy. It is not yet the pure kinetic energy or the Wilson excitation energy.

### 56.3 Reconstruct kinetic energy with all scalar terms

For a complete factor  $b$ , let  $Q_b = \sum_{e \in b} C_e$ ,  $C_e = \sum_a X_{ea}^2$  with half-Pauli generators  $i \sigma_a/2$ . The exact dictionary is

$$\delta h_b = \alpha Q_b - \sum_{p \in b, \text{selected}} \lambda_p W_p - E_{\text{strip}, b}, \quad \sum_{p \in b, \text{selected}} |\lambda_p| \leq 9\alpha/8. \quad (\text{HNM-AT1-R04})$$

The ground scalar is that of the actual selected strip, not zero by convention. The constant Haar trial has zero kinetic expectation and zero selected Wilson means, so  $E_{\text{strip}, b} \leq 0$ . Rearranging (R04), summing the two factors and only then dropping its nonpositive scalar contribution gives

$$\omega_N(Q_R) \leq \frac{1}{8}\omega_N(h_R) + \frac{9}{4} \leq \frac{9}{4} + \frac{49}{4}|\tau|. \quad (\text{HNM-AT1-R05})$$

All 48 regional links enter this estimate. Positivity permits the kinetic form on just the four Wilson links to be bounded by  $Q_R$ . The selected potential cannot generally be discarded: at an allowed nonzero one-face coefficient  $\lambda$ , the Haar-based trial  $1+tW$  has energy numerator  $3\alpha t^2/4 - \lambda t/2$ . Choosing  $t = \lambda/(3\alpha)$  makes it negative. Thus even at  $\tau=0$  the true selected ground generally differs from Haar and its raw ground scalar is negative.

### 56.4 Finite domains, derivatives and first-moment identity

At fixed  $N$  the full compact manifold is  $SU(2)$  to the number of owned original links. Its elliptic  $Q_N$  has operator domain  $H^2$  and form domain  $H^1$ . Every selected and omitted Wilson potential is bounded, real and smooth. Therefore  $H_N^{\text{raw}} = \alpha Q_N + V_N$  is self-adjoint on  $H^2$ , with form domain  $H^1$  and compact resolvent. The actual ground vector belongs to  $H^2$ . Multiplication by  $W$ ,  $W^2$ ,  $W \cdot m_N$  and  $(W \cdot m_N)^2$  preserves  $H^2$  and  $H^1$ : use their bounded derivatives through order two and the weak Leibniz rule, initially on smooth functions and then by completion. No volume-uniform global potential norm is needed for these fixed-volume domain statements.

The exact finite physical subtraction is

$$\delta \hat{H}_N = H_N^{raw} - \sum_b E_{strip,b}, \quad E_N^{raw} = \sum_b E_{strip,b} + \delta \hat{E}_N, \quad K_N = \delta(\hat{H}_N - \hat{E}_N). \quad (\text{HNM-AT1-R06})$$

Both scalars are retained before cancellation. Selected and omitted magnetic multipliers and these scalars all commute with  $W$ .

Write the Wilson holonomy as a unit quaternion  $(w,v)$ . For one positive link, left variation by a half-Pauli generator gives scalar derivatives equal to the components of  $-v/2$  after an orthogonal adjoint rotation. For an inverse stored link,  $d(U(t)^{-1})/dt = -U^{-1} T_a$  when  $U(t) = \exp(t T_a) U$ ; the sign and right-multiplication placement are retained. Left/right multiplication, inversion and adjoint rotation preserve the quadratic sum. Since  $T_a^2 = -I/4$ , each of the four distinct link occurrences satisfies  $C_e W = 3W/4$  and contributes  $(1-W^2)/4$  to the gradient square. Thus

$$Q_N W = 3W, \quad \Gamma(W) := \sum_{e,a} (X_{ea} W)^2 = 1 - W^2. \quad (\text{HNM-AT1-R07})$$

Now put  $f = W - m_N$ . Expand the form on  $f \Omega_N$  and subtract the real part of the ground equation tested against  $f^2 \Omega_N$ . For each  $X$  the kinetic integrand difference is

$$|X(f \Omega_N)|^2 - \Re\{\overline{X \Omega_N} X(f^2 \Omega_N)\} = (Xf)^2 |\Omega_N|^2. \quad (\text{HNM-AT1-R08})$$

All scalar and multiplication terms cancel. The established domains justify the ground test. Consequently, without differentiating a limiting correlation,

$$\mu_{1,N} = \langle \chi_N, K_N \chi_N \rangle = \alpha \omega_N (\Gamma(W)) = \alpha \omega_N (1 - W^2). \quad (\text{HNM-AT1-R09})$$

Equivalently one half of the ground expectation of  $[W, [K_N, W]]$  gives the same identity; the double commutator is  $2\alpha$  times multiplication by  $\Gamma(W)$ . This one-half factor is essential.

## 56.5 Complete commutator and finite second moment

The finite domain statements allow the actual L2 identity

$$K_N \chi_N = \alpha [Q_N, W] \Omega_N = \alpha (3W \Omega_N - 2S_N), \quad S_N = \sum_{e,a} (X_{ea} W)(X_{ea} \Omega_N). \quad (\text{HNM-AT1-R10})$$

The first commutator is not claimed to be bounded. Pointwise Cauchy-Schwarz and (R07) show  $|S_N|^2 \leq (1-W^2) \sum_{\text{four links}, a} |X_{ea} \Omega_N|^2$ . Integrating bounds its L2 norm squared by  $\omega_N(Q_R)$ . Using  $(x+y)^2 \leq 2x^2 + 2y^2$  and  $|W| \leq 1$  yields

$$\mu_{2,N} = \|K_N \chi_N\|^2 \leq \alpha^2 [18 + 8\omega_N(Q_R)] \leq B_2 := \alpha^2 (36 + 98|\tau|) < 37\alpha^2. \quad (\text{HNM-AT1-R11})$$

At the maximal cap  $B_2/\alpha^2 = 1800000049/50000000$ . The gradient cross term cannot be deleted: applied to the test vector  $W$ ,  $Q_N(W^2) = 8W^2 - 2$ , hence  $[Q_N, W]W = 5W^2 - 2$  rather than  $3W^2$ . The bounds cover every finite centered whole-star box containing  $R$  and every allowed selected triple, with either sign of  $\tau$ . They retain the actual reference potential and energy clock.

## 56.6 Compact spectral tests in AQ's actual representation

No common finite/infinite Hilbert space is assumed. AQ1's dynamics converge in norm on every bounded local observable, uniformly on compact physical time intervals. Its chosen states converge in trace norm on every finite region. Approximate the evolved  $W$  by the evolution inside one fixed large region; the norm error is uniform on a fixed time interval, and trace-norm state convergence controls the bounded local expectation. Thus

$$\omega_N(W\alpha_t^N(W)) \longrightarrow \omega(W\alpha_t(W)). \quad (\text{HNM-AT1-R12})$$

uniformly on compact times. Local convergence gives  $m_N \rightarrow m$ . Therefore the actual centered correlations

$$C_N(t) = \omega_N(W\alpha_t^N(W)) - m_N^2 = \int e^{itE/\hbar} d\nu_N(E) \longrightarrow C(t) = \int e^{itE/\hbar} d\nu(E). \quad (\text{HNM-AT1-R13})$$

Their masses are variances and are  $\leq 1$ , so the correlations are uniformly bounded. For  $f$  in  $C_c^\infty(\mathbb{R})$ , use physical-energy Fourier inversion  $f(E) = \int k_f(t) \exp(itE/\hbar) dt$ , with  $k_f$  integrable. Bounded correlations and dominated convergence give  $\int f d\nu_N \rightarrow \int f d\nu$ . Smooth compact tests are uniformly dense in  $C_0(\mathbb{R})$ ; the common mass bound extends this convergence to every  $C_0$  test. This is a passage of scalar spectral measures for the actual Stone generator, not a formal differentiation or a strong-resolvent statement across different spaces. AQ2's original-endpoint gauge proof places  $\chi$  in the reducing physical sector.

## 56.7 Second-moment inequality, domain and exact first moment

For  $E \geq 0$  set  $\theta_L(E) = 1$  on  $[0, L]$ ,  $2 - E/L$  on  $[L, 2L]$ , and zero above  $2L$ . Let  $f_L(E) = E\theta_L(E)$  and  $g_L(E) = E^2\theta_L(E)$ , extending both by zero to  $E < 0$ . These are continuous compactly supported functions, nonnegative, and increase pointwise to  $E$  and  $E^2$  as  $L$  increases. Compact-test convergence and (R11) imply  $\int g_L d\nu \leq B^2$ . Monotone convergence gives

$$\int E^2 d\nu(E) \leq B^2, \quad \chi \in D(H), \quad \|H\chi\|^2 = \int E^2 d\nu(E). \quad (\text{HNM-AT1-R14})$$

The equality here is the spectral-domain characterization for the limiting generator. It is not equality with the limiting sequence of finite second moments.

For all finite volumes and for the limiting measure,

$$\int_{E > L} E d\nu_N(E) \leq B^2/L, \quad \int_{E > L} E^2 d\nu(E) \leq B^2/L. \quad (\text{HNM-AT1-R15})$$

This follows pointwise from  $E \leq E^2/L$  on that tail and supplies the required uniform integrability. Moreover  $0 \leq E - f_L(E) \leq E - 1$  ( $E > L$ ). The bounded-local right side of (R09) tends to  $\alpha\omega(1 - W^2)$ . For each  $L$ , take  $N$  to infinity in

$$0 \leq \alpha\omega_N(1 - W^2) - \int f_L d\nu_N \leq B^2/L. \quad (\text{HNM-AT1-R16})$$

Then let  $L$  increase to infinity using monotone convergence. The exact same-state energy certificate is

$$\langle \chi, H\chi \rangle = \int E d\nu(E) = \alpha\omega(1 - W^2), \quad \|H\chi\|^2 \leq \alpha^2(36 + 98|\tau|) < 37\alpha^2. \quad (\text{HNM-AT1-R17})$$

The left inner product is justified by the operator domain already proved. No  $\hbar$  belongs in an energy moment;  $H/\hbar$  would instead give frequency moments. Division by  $E_{\text{star}}$  expresses all energies on the declared common physical comparison scale without changing the model.

## 56.8 Executed counter-inferences and scope

The exact standard-library checker reconstructs all link/endpoints and incident stars, evaluates noncommuting rational-quaternion derivative fixtures including inverse signs, and checks scalar, tail and normalization controls. Its named controls execute genuinely different values:

- Two-star substitution gives coefficient 28 instead of the required seven-star 98; centered geometry rejects it.
- A nonzero selected-face trial has negative raw energy, rejecting a pure-electric Haar reference with zero scalar.
- The computed  $Q(W^2)$  and commutator on  $W$  reject omission of the gradient cross term. The inverse-sign derivative square is insensitive to a global sign, so an additional derivative of  $U U^{-1}$  rejects that sign error.
- Free-reference first and second moments scale as  $\alpha$  and  $\alpha^2$ . Doubling the generators quadruples energy and multiplies the second moment by sixteen. Ground shifts, the one-half double-commutator factor and  $\hbar$  frequency conversion are separately exercised.
- A two-state matrix example has an uncentered zero-energy contribution that disappears under the actual mean subtraction; positive moments alone cannot detect that change in mass.
- The abstract family  $(1-1/n)\delta_g + (1/n)\delta_{(g+n/2)}$ ,  $g=1/16$ , has first moment  $9/16$ , but second moment  $g^2 + g + n/4$ . Its bounded-test limit is  $\delta_g$ . It rejects first-moment equality from bounded first moments alone.
- The abstract family  $(1-1/n^2)\delta_g + (1/n^2)\delta_{(g+n)}$  has second moments  $g^2 + 2g/n + 1$ , uniformly bounded; its limiting measure  $\delta_g$  has second moment  $g^2$ . This rejects finite-to-infinite second-moment equality under the proved assumptions.
- Two explicit normalized ground states for two positive gapped matrix Hamiltonians both satisfy a variance floor above  $1/5$  yet give different Wilson means. Common gap/variance properties therefore do not identify states. These fixtures are logical controls, not claimed AQ or AO states.

No second-moment convergence, identification of AQ with AO/AN, whole-sequence or all-ground-state uniqueness, particle pole, exact mass, dispersion or continuum Yang–Mills claim follows. No new axiom is introduced.

The nearest checked primary source is Gauvin [1], arXiv:2503.15539v3, Supplement A.9–A.10: its ground-state Wilson energy identity and compactness/dynamics/Fourier construction are directly relevant established proof patterns. The copied selected-page excerpt was read. Its  $SU(3)$ , pure-electric link dictionary and numerical constants are not imported. The new work here is a same-state  $SU(2)$  application retaining AQ’s selected onsite potential, full 48-link geometry and seven-star numerical second-moment coefficient. AQ’s dynamics remain source-matched to Nachtergaele–Sims, arXiv:1410.8174v1. The current abstract/version pages were checked; no broad literature-priority claim is made.

Reproduce with `python-Bresearch/round30/reverse/at1/check.py--output/absolute/fresh-directory`. The checker requires a fresh absolute directory, copied-source hashes and explicit exceptions, so optimized Python cannot disable gates. Normal and optimized output bytes are compared. The manifest binds deterministic report/code/input bytes; freeze.json binds every owned scientific source and output. This ends reverse AT1 production. No next loop is selected or executed here.

`research/round30/reverse/at1/report.md`

## 57 Hrunday AT2: reviewed admission

**Verdict: accepted within scope.** For the same state, the unnormalized spectral mass satisfies  $\nu([\alpha/16, 8\alpha]) \geq 307992/1984375$ . On the physical vacuum-orthogonal sector let  $R = \langle \chi, H_{\text{phys}}^{-1} \chi \rangle$ . Its dimensionless value has lower bound  $3843876001/47000000000$  and upper bound  $28462237549/7503125000$ : these are bounds on  $\alpha R$ , not an evaluated response or a unique spectral measure.

### Limits and independent review

The producer reports were frozen separately before exchange. Their common ancestry, contract and inherited methods remain disclosed. The skeptic reviewed both proofs and executed controls; this does not constitute external human peer review.

- The window is unnormalized spectral mass, not probability or a hard cutoff
- No evaluated actual AQ inverse response or optimality
- No particle pole, exact mass or identified spectral type
- Abstract equal-moment fixtures are not actual lattice realizations
- No static susceptibility or deformed thermodynamic-state differentiability
- No continuum result; scientific priority unverified

`research/round30/advisor/at2-gate.json`

`research/round30/skeptic/at2.md`

### Substantive directional differences

- Both directions derive the same rational endpoints and atomic/atomless controls.
- The frozen tangent lower certificate is valid but the standard Cauchy baseline is slightly stronger; the combined interval reports the stronger value.
- Forward susceptibility control checks a contract exclusion; reverse and skeptic give substantive logical controls.

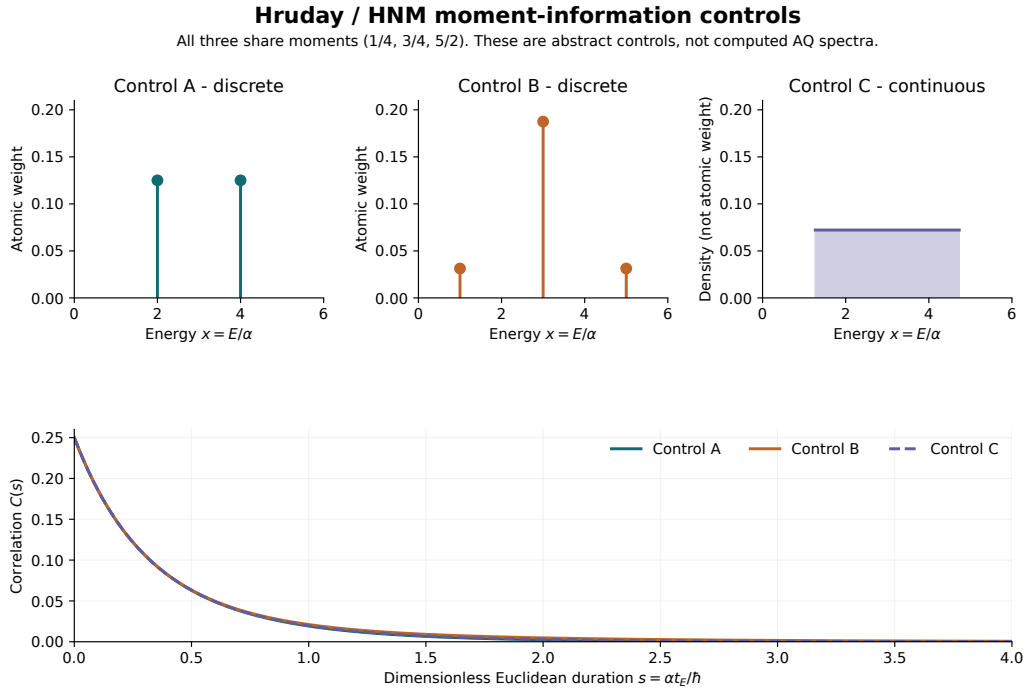


Figure 12: Three positive spectral measures share the same zeroth, first and second moments but differ in atomic structure and Euclidean correlations. A and B are the exact discrete controls; C is the uniform density on  $[3-\sqrt{3}, 3+\sqrt{3}]$  with total mass  $1/4$ . Density and atomic-weight axes are distinguished. These are information-limit controls, not spectra computed for the AQ lattice Hamiltonian. Floating-point renderings of exact analytic formulas; the proof and rational inverse-moment enclosures are in AT2.

## 58 Hrudy AT2: spectral-window and inverse-energy certificate (forward derivation)

**Source-bound forward derivation.** This is the complete typeset mathematical report of one producer. Prospective claims describe its original freeze; the shared reviewed result and permitted refinements are stated in the admission section. The two agents share inherited premises and model ancestry. This is neither an independent physical observation nor external human peer review.

Human project author: **Hrudy N M (BUNZEEY)**. AI-assisted forward derivation independently executed before reading any current AT2 reverse or skeptic solution. HNM tags are project aliases, not claims to have invented moment duality, Jensen/Cauchy inequalities or reciprocal polynomial identities. Model-specific application; scientific priority is unverified.

**Result in the actual AQ state.** The original centered Wilson spectral measure satisfies

$$\nu([\alpha/16, 8\alpha]) \geq \frac{307992}{1984375}, \quad \frac{3843876001}{47000000000\alpha} \leq \langle \chi, H_{phys}^{-1} \chi \rangle \leq \frac{28462237549}{7503125000\alpha}. \quad (\text{HNM-AT2-F01})$$

The interval is a conservative certificate, not an optimal reconstruction. The lower inverse endpoint in (F01) uses the standard Cauchy baseline, which slightly improves the frozen  $c=3$  tangent. None of these bounds identifies a particle pole or makes an infinite-volume susceptibility assertion.

### 58.1 Exact shared state, normalization and uncertainty

Take exactly the AQ numerical-cap state and physical reducing generator admitted in AQ2 and AT1, with the original centered Wilson vector  $\chi$ . The inherited gap is  $H_{phys} \geq \alpha/16$  on the vacuum-orthogonal sector, and  $\chi$  belongs to that sector. Define the dimensionless pushforward  $\eta(B) = \nu(\alpha B)$ ,  $x = E/\alpha$ ,  $a = 1/16$ . With  $w = \omega_{num}(W)$ ,  $q = \omega_{num}(W^2)$ , put

$$\begin{aligned} s &= \int d\eta = q - w^2, \quad u = \int x d\eta = 1 - q, \\ v &= \int x^2 d\eta \leq B = 36 + 98|\tau|, \quad \text{supp } \eta \subseteq [a, \infty). \end{aligned} \quad (\text{HNM-AT2-F02})$$

These are actual AQ moments. The second moment is bounded, not evaluated. AQ2's trace-norm estimate on the actual 48-link cover gives  $|w| \leq 1/500$ ,  $|q - 1/4| \leq 1/500$ . Write  $z = w^2$ ; the joint declared domain is

$$\begin{aligned} q_- := 31/125 \leq q \leq 63/250 =: q_+, \quad 0 \leq z \leq 1/250000 =: z_+, \\ s = q - z \geq 61999/250000, \quad 187/250 \leq u \leq 94/125. \end{aligned} \quad (\text{HNM-AT2-F03})$$

Actual  $q$  and  $w$  are not replaced by Haar values. In particular  $s$  and  $u$  share  $q$  and must be propagated jointly. The rectangular  $q, z$  enclosure may include more points than the actual Hamiltonian realizes; bounds valid throughout it are conservative for the actual state. It is deterministic analytic uncertainty, not a sampling covariance.

The frozen  $L=8$ ,  $c=3$  and  $b=49$  are dimensionless certificate choices; they change neither state nor Hamiltonian. Positive  $\alpha, \hbar, E_{\text{star}}$  and lattice spacing remain fixed. No AO/AQ state identification or continuum limit is used.

## 58.2 Full-half-line window minorant

For any  $L > a$  and  $x \geq a$ , the affine function  $p_L(x) = (L - x)/(L - a)$  obeys

$$\frac{L - x}{L - a} \leq \mathbf{1}_{[a, L]}(x). \quad (\text{HNM-AT2-F04})$$

For  $a \leq x \leq L$  its value lies between zero and one. For  $x > L$  it is negative while the indicator is zero. At  $x = L$  the minorant equals zero and the closed-window indicator equals one. Thus a possible atom at  $L$  is retained and never erroneously subtracted. Integrating this pointwise inequality is justified because  $\eta$  has finite first moment. For  $L = 8$ ,

$$\eta([a, 8]) \geq \frac{8s - u}{8 - a} = \frac{9q - 8z - 1}{127/16} \geq \frac{307992}{1984375}. \quad (\text{HNM-AT2-F05})$$

The last expression increases with  $q$  and decreases with  $z$ , so its minimum over (F03) is exactly at  $q = q_-$ ,  $z = z_+$ . This certificate uses the mass, first moment and support bound only; it does not use  $v$  or  $B$ . It proves at least this much unnormalized spectral weight in the window. It is not a hard upper spectral cutoff, a probability of one, or an assertion that the true spectrum ends at  $8\alpha$ .

## 58.3 Meaning and units of the inverse-energy form

The physical vacuum-orthogonal subspace reduces the self-adjoint  $H_{\text{phys}}$  and its spectrum there lies in  $[\alpha/16, \infty)$ . The spectral reciprocal on that subspace is a bounded positive operator with norm at most  $16/\alpha$ . Since  $\chi$  is centered, its quadratic form is well-defined:

$$R := \langle \chi, (H_{\text{phys}}|_{\Omega^\perp})^{-1} \chi \rangle = \int E^{-1} d\nu(E) = \alpha^{-1} I, \quad I := \int x^{-1} d\eta(x). \quad (\text{HNM-AT2-F06})$$

One may extend the inverse by zero on the vacuum, but not call it an inverse of the full operator at zero.  $R$  has inverse-energy units.  $\hbar$  is absent from an energy inverse; a frequency inverse would carry a different factor. This is a mathematically defined inverse-energy quadratic form. Identifying  $2R$  with a derivative of a thermodynamic ground-state expectation would require a separately specified deformation, existence and differentiability of its state branch, and an interchange of thermodynamic/differentiation limits. None is assumed here, so no static-susceptibility assertion is made.

## 58.4 Frozen tangent lower certificate

For any  $c > 0$  and  $x > 0$ ,

$$\frac{1}{x} - \left( \frac{2}{c} - \frac{x}{c^2} \right) = \frac{(x - c)^2}{c^2 x} \geq 0. \quad (\text{HNM-AT2-F07})$$

The denominator is strictly positive throughout the support. This is an algebraic identity and a whole-half-line proof, not a positivity grid. With  $c = 3$ , integration and joint uncertainty propagation give

$$I \geq \frac{2}{3}s - \frac{1}{9}u = \frac{7q - 6z - 1}{9} \geq \frac{91997}{1125000}. \quad (\text{HNM-AT2-F08})$$

Again the minimum is at  $q_-$ ,  $z_+$ . No second-moment value is used. The tangent may be negative for large  $x$ ; that does not spoil a lower bound.

### 58.5 Frozen quadratic upper certificate and correct moment sign

For  $a > 0$ ,  $b \geq a$  and  $x \geq a$ , define the quadratic numerator  $N_b(x) = x^2 - (2b + a)x + b^2 + 2ab$ . Direct polynomial expansion gives

$$\frac{N_b(x)}{ab^2} - \frac{1}{x} = \frac{(x - a)(x - b)^2}{ab^2x} \geq 0. \quad (\text{HNM-AT2-F09})$$

The denominator is strictly positive, the first factor is nonnegative, and the last factor is a square. This proves the upper bound at every continuous  $x \geq a$ , including  $x = a, b$  and the infinite tail. Integrating and using the **positive**  $x^2$  coefficient gives

$$I \leq \frac{v - (2b + a)u + (b^2 + 2ab)s}{ab^2} \leq \frac{B - (2b + a)(1 - q) + (b^2 + 2ab)(q - z)}{ab^2}. \quad (\text{HNM-AT2-F10})$$

The replacement  $v \leq B$  has the required direction only because  $1/(ab^2) > 0$ . It is not an equality claim. For  $b = 49$ , the numerator increases with  $B$  and  $q$  and decreases with  $z$ . Hence use  $B_{\text{cap}} = 1800000049/50000000$ ,  $q = q_+$ ,  $z = 0$ , obtaining

$$I \leq \frac{28462237549}{7503125000}. \quad (\text{HNM-AT2-F11})$$

The declared  $q, w$  relation remains intact during this calculation. These coefficients were frozen before execution; there is no tuning on outputs or claim that  $b = 49$  is optimal.

### 58.6 Baseline comparisons

Cauchy-Schwarz applied to  $\sqrt{x}$  and  $1/\sqrt{x}$  in  $L^2(\eta)$ , and the support reciprocal bound, give

$$\frac{s^2}{u} \leq I \leq \frac{s}{a}. \quad (\text{HNM-AT2-F12})$$

Both integrals needed for Cauchy are finite;  $u > 0$  follows explicitly from (F03). The function  $(q - z)^2/(1 - q)$  decreases with  $z$  and increases with  $q$  throughout the declared domain: its  $q$  derivative is  $2(q - z)/(1 - q) + (q - z)^2/(1 - q)^2 > 0$ . Therefore the uniform Cauchy baseline is

$$I \geq \frac{(61999/250000)^2}{94/125} = \frac{3843876001}{47000000000} > \frac{91997}{1125000}, \quad I \leq 16q_+ = \frac{504}{125}. \quad (\text{HNM-AT2-F13})$$

The frozen quadratic upper bound is strictly smaller than  $504/125$ . Combining the stronger of the two valid lower bounds with the stronger upper bound produces (F01). All numbers are exact rationals. No strong-duality theorem, extremizer existence or global optimality is asserted.

### 58.7 Equal moments do not identify an energy atom

The prescribed abstract controls are

$$\eta_A = \frac{\delta_2 + \delta_4}{8}, \quad \eta_B = \frac{\delta_1}{32} + \frac{3\delta_3}{16} + \frac{\delta_5}{32}, \quad \eta_C = \frac{1}{4} \text{Uniform}[3 - \sqrt{3}, 3 + \sqrt{3}]. \quad (\text{HNM-AT2-F14})$$

For A and B, direct finite sums give  $(s, u, v) = (1/4, 3/4, 5/2)$ . For C, write  $x=3+y$  with  $y$  uniformly distributed on  $[-\sqrt{3}, \sqrt{3}]$ . Symmetry gives  $E[y]=0$ , and elementary integration gives  $E[y^2]=3/3=1$ . Multiplying the resulting moments 1,3,10 by  $1/4$  gives the same triple. The support of C lies strictly above one because  $\sqrt{3}<2$ , hence all three supports are above  $a$ . All obey the moment relations at  $q=1/4, w=0$  and satisfy  $v \leq B$ . A and B have distinct atom sets; C has a bounded interval density and no atoms.

Their inverse moments are nevertheless different:

$$I_A = \frac{3}{32}, \quad I_B = \frac{1}{10}, \quad I_C = \frac{1}{8\sqrt{3}} \log \frac{3+\sqrt{3}}{3-\sqrt{3}} = \frac{1}{12} \sum_{k=0}^{\infty} \frac{1}{3^k(2k+1)}. \quad (\text{HNM-AT2-F15})$$

The integral formula follows by integrating  $1/x$  against the uniform density  $1/(8\sqrt{3})$ . The series follows from integrating the geometric series for  $1/(1-t^2)$  from zero to  $1/\sqrt{3}$ , where it converges absolutely and uniformly on that closed interval. There is no uncontrolled logarithm evaluation. The first three terms are  $17/180$ . For  $k \geq 3$ ,  $1/(2k+1) \leq 1/7$  and the geometric sum of  $3^{-k}$  is  $1/18$ . Therefore

$$\frac{3}{32} < \frac{17}{180} < I_C \leq \frac{17}{180} + \frac{1}{1512} = \frac{719}{7560} < \frac{1}{10}. \quad (\text{HNM-AT2-F16})$$

Strictness on the lower side follows from the positive remaining series. Thus even exact knowledge of all three contracted moments, stronger than AT1's second-moment ceiling, does not determine spectral atom structure or inverse-energy response. Each fixture is realized abstractly by multiplication by  $x$  on  $L^2(\text{eta})$ , with source vector 1; adding an orthogonal one-dimensional zero-energy vacuum realizes a centered nonnegative spectral problem. This is not a realization by the AQ lattice Hamiltonian. The controls establish insufficiency of the moment information, not actual thermodynamic-state nonuniqueness or multiple AQ phases.

## 58.8 Controls, source ancestry and scope

The exact checker expands the rational identities as polynomial coefficient arrays, checks endpoint propagation with rational arithmetic, and integrates the finite and uniform moment fixtures symbolically. Whole-half-line validity comes from the factorization and sign proofs, not the diagnostic sample evaluations. A polynomial  $(x - 3/2)^2 - 1/16$  is positive at all integer sample nodes but negative at  $3/2$ , exposing grid-only positivity. A negative-quadratic kernel control shows why replacing  $v$  by  $B$  in an upper bound with a negative coefficient is invalid. Further controls preserve an atom at  $x=8$ , distinguish a second-moment ceiling from equality, reject substituting  $q=1/4, w=0$  for the allowed interacting unknowns, and verify the  $\alpha^{-1}$  scaling.

Bertsimas and Popescu, *SIAM J. Optim.* 15 (2005), Sections 2-3, supply classical ancestry for polynomial moment certificates and the upper-moment coefficient-sign rule. Abbott, Fields, Jay, Oare and Saccardi, arXiv:2605.20509v1, III.1, IV.1-IV.3 and VII.4, provide current spectral-kernel certificate context and explicitly distinguish extremal feasible spectra from the true spectrum. Selected passages and the panel's source record were consulted, not the entirety of either research program. The contribution here is this particular AQ dictionary, evaluated enclosure and exact controls; generic methods are not Hruday inventions.

Run `python-Bresearch/round30/forward/at2/check.py--output/absolute/new/directory`. Normal and optimized results must match. The checker and report use only `stdlib/exact` arithmetic. No physical susceptibility, pole, exact mass, sharpness, state identification or continuum Yang-Mills result is claimed. Loop 3 awaits advisor selection after the separate skeptical review.

`research/round30/forward/at2/report.md`

## 59 Hrudy AT2: spectral-window and inverse-energy certificate (reverse derivation)

**Source-bound reverse derivation.** This is the complete typeset mathematical report of one producer. Prospective claims describe its original freeze; the shared reviewed result and permitted refinements are stated in the admission section. The two agents share inherited premises and model ancestry. This is neither an independent physical observation nor external human peer review.

Human author: **Hrudy N M (BUNZEEY)**. AI-assisted independent reverse reconstruction; scientific priority unverified. HNM labels are traceable project aliases. No current forward or skeptic AT2 result was read. The shared prospective certificate choices and inherited admitted premises are disclosed and copied in `inputs/`.

### 59.1 Reverse reconstruction from the requested observable

The two target functionals are the mass of a fixed finite energy interval and the inverse-energy quadratic form. They are integrals against the same positive spectral measure. A complete pointwise minorant/majorant on its entire support therefore suffices. Only a second-moment upper bound is available, so the quadratic coefficient's sign must permit replacing that moment by its ceiling. This reverse argument determines the proof obligations before numerical evaluation. It does not reconstruct a unique spectrum from moments.

Use exactly AQ's chosen state and original centered Wilson vector, now with AT1's operator-domain and first-moment identity. Let  $\eta$  be the pushforward of its positive spectral measure under  $x=E/\alpha$ . Define

$$\begin{aligned} a &= \frac{1}{16}, \quad w = \omega_{\text{num}}(W), \\ q &= \omega_{\text{num}}(W^2), \quad s = \int d\eta = q - w^2, \\ u &= \int x d\eta = 1 - q, \quad v = \int x^2 d\eta \leq B = 36 + 98|\tau|. \end{aligned} \tag{HNM-AT2-R01}$$

The support is contained in  $[a, \infty)$ . AQ2 and  $|\tau| \leq 10^{-8}$  provide the convenient weak rational bounds

$$q_- := \frac{31}{125} \leq q \leq q_+ := \frac{63}{250}, \quad 0 \leq z := w^2 \leq z_+ := \frac{1}{250000}, \quad B \leq B_* := \frac{1800000049}{50000000}. \tag{HNM-AT2-R02}$$

These follow because  $2\sqrt{98|\tau|} < 1/500$ . The  $q, w$  uncertainty is deterministic analytic uncertainty, not sampled statistical error. We retain the joint relations  $s=q-z$  and  $u=1-q$ ; independent optimization of unrelated moment intervals is not claimed. The rectangular  $q, z$  enclosure can contain points not realized by the actual model. Monotone endpoint substitution below gives valid uniform bounds, without asserting sharpness over its actual realizable states.

The frozen choices  $L=8, c=3, b=49$  are dimensionless certificate parameters. They are not Hamiltonian couplings, regulators, fitted energies or predictions of pole locations.  $\alpha, \hbar, \text{spacing}$  and  $E_{\text{star}}$  remain fixed positive.

### 59.2 A closed finite spectral window

For  $x \geq a$  the affine function  $p_L(x) = (L-x)/(L-a)$  is at most one on  $[a, L]$ , and is negative above  $L$ . Since  $L > a$ , this proves on the whole half-line

$$\frac{L-x}{L-a} \leq \mathbf{1}_{[a,L]}(x), \quad \eta([a, L]) \geq \frac{Ls-u}{L-a} = \frac{(L+1)q - Lz - 1}{L-a}. \quad (\text{HNM-AT2-R03})$$

At  $x=L$  the minorant is zero but the indicator is one. A possible endpoint atom is therefore retained rather than silently discarded. Finite first moment makes the affine function integrable even though it becomes negative without bound at large  $x$ .

The last expression increases with  $q$  and decreases with  $z$ . Insert  $q_-$ ,  $z_+$  and  $L=8$  to obtain the exact Hruday window certificate

$$\boxed{\nu([\alpha/16, 8\alpha]) = \eta([1/16, 8]) \geq \frac{307992}{1984375} > \frac{31}{200}.} \quad (\text{HNM-AT2-R04})$$

The floor is approximately 0.15520857 of the unnormalized Wilson spectral weight. It is a mass guarantee, not a statement that all weight lies below  $8\alpha$ . This bound uses the gap and first moment, not the second-moment ceiling.

### 59.3 Inverse on the actual physical subspace and units

AQ2 makes the physical vacuum-orthogonal subspace reducing, and  $H$  restricted to it obeys  $H \geq \alpha a$ . Functional calculus thus defines a bounded positive inverse on the whole subspace, with norm at most  $1/(\alpha a)$ , and also a bounded inverse square root. Since  $\chi$  is centered, it lies in that subspace. Therefore

$$R := \langle \chi, H^{-1} \chi \rangle = \|H^{-1/2} \chi\|^2 = \frac{1}{\alpha} \int_a^\infty \frac{1}{x} d\eta(x), \quad 0 \leq R \leq \frac{s}{\alpha a}. \quad (\text{HNM-AT2-R05})$$

The inverse is not defined by inverting the vacuum eigenvalue zero. It is the reduced inverse (or equivalently the bounded pseudoinverse set to zero on the vacuum).  $R$  has units inverse energy;  $\alpha R$  is dimensionless. The inverse of the frequency generator  $H/\hbar$  instead gives  $\hbar R$ . No differentiability of a deformed infinite-volume ground state, or static susceptibility, follows from this functional calculus alone.

### 59.4 Entire-half-line reciprocal certificates

For every  $x > 0$ , the tangent at  $c > 0$  has the exact positive residual

$$\ell_c(x) = \frac{2}{c} - \frac{x}{c^2}, \quad \frac{1}{x} - \ell_c(x) = \frac{(x-c)^2}{c^2 x} \geq 0. \quad (\text{HNM-AT2-R06})$$

For every  $x \geq a > 0$  and  $b \geq a$  the proposed quadratic satisfies

$$P_b(x) = \frac{x^2 - (2b+a)x + b^2 + 2ab}{ab^2}, \quad P_b(x) - \frac{1}{x} = \frac{(x-a)(x-b)^2}{ab^2 x} \geq 0. \quad (\text{HNM-AT2-R07})$$

The denominators are strictly positive throughout the support. These factorizations prove global half-line domination, including arbitrarily large energies; a numerical grid is unnecessary. Integrating positive inequalities is weak duality. No strong duality or optimality theorem is used.

The coefficient of  $x^2$  in  $P_b$  is  $+1/(ab^2) > 0$ . Consequently  $v \leq B$  may be substituted in the upper-bound direction. A negative coefficient would reverse that substitution inequality. The tangent has zero quadratic coefficient and uses no second-moment information. With  $c=3, b=49$ , let  $A=2b+a$  and  $D=b^2+2ab$ . Then

$$\frac{7q - 6z - 1}{9} \leq \alpha R \leq \frac{B - A + (A + D)q - Dz}{ab^2}. \quad (\text{HNM-AT2-R08})$$

Here  $A, D, A+D$  and  $ab^2$  are positive. The lower expression is smallest at  $q_-, z_+$ ; the upper expression is largest in this enclosure at  $B_*, q_+, z=0$ . Exact rational arithmetic gives

$$\boxed{\frac{91997}{1125000} \leq \alpha R \leq \frac{28462237549}{7503125000}}. \quad (\text{HNM-AT2-R09})$$

These are approximately 0.08177511 and 3.79338443. The second moment remains an upper bound throughout; no value  $v=B$  is assigned to the actual AQ measure.

## 59.5 Preserve and compare the elementary baselines

Cauchy–Schwarz applied to  $\sqrt{x}$  and  $1/\sqrt{x}$  in  $L2(\eta)$  yields  $s^2 \leq u \int x^{-1} d\eta$ . Both functions are square integrable:  $u$  is finite by AT1, and inverse integrability follows from  $a > 0$ . Also  $u = 1 - q > 1 - q_- > 0$ . The support bound remains as in (R05):

$$\frac{s^2}{u} \leq \alpha R \leq \frac{s}{a}. \quad (\text{HNM-AT2-R10})$$

The lower expression  $(q-z)^2/(1-q)$  decreases with  $z$ , because  $q-z > 0$ . Its  $q$  derivative is  $(q-z)(2-q-z)/(1-q)^2 > 0$  throughout the declared box. Thus its uniform floor is attained at  $q_-, z_+$  within that box. The upper support ceiling is at  $q_+, z=0$ . Therefore

$$\frac{3843876001}{47000000000} \leq \alpha R \leq \frac{504}{125}. \quad (\text{HNM-AT2-R11})$$

The Cauchy floor is slightly stronger than the frozen  $c=3$  tangent, while the polynomial upper bound is stronger than the support ceiling. The certificate package consequently permits the combined interval

$$\boxed{\frac{3843876001}{47000000000\alpha} \leq R \leq \frac{28462237549}{7503125000\alpha}}. \quad (\text{HNM-AT2-R12})$$

Its lower endpoint is approximately  $0.08178460/\alpha$ . The tangent remains recorded as a independently checked frozen certificate, with its honest baseline comparison. No parameter was retuned to hide this comparison, and no globally optimal interval is claimed.

## 59.6 Equal moments do not specify spectral type or inverse energy

Consider the following abstract dimensionless positive measures, solely as information-content controls:

$$\eta_A = \frac{\delta_2 + \delta_4}{8}, \quad \eta_B = \frac{\delta_1}{32} + \frac{3\delta_3}{16} + \frac{\delta_5}{32}, \quad \eta_C = \frac{1}{4} \text{Uniform}[3 - \sqrt{3}, 3 + \sqrt{3}]. \quad (\text{HNM-AT2-R13})$$

Their supports lie above  $a$ :  $3\sqrt{3} > 1 > a$  because  $\sqrt{3} < 2$ .  $A$  and  $B$  have different finite atom sets.  $C$  has a bounded Lebesgue density on its interval and no atoms. Direct sums give the first three moments of  $A$  and  $B$ . For  $C$ , write  $X=3+Y$  with  $Y$  uniform on  $[-\sqrt{3}, \sqrt{3}]$ . Symmetry gives  $E Y=0$  and  $E Y^2=(\sqrt{3})^2/3=1$ . Hence all three have

$$\left(\int d\eta, \int x d\eta, \int x^2 d\eta\right) = \left(\frac{1}{4}, \frac{3}{4}, \frac{5}{2}\right). \quad (\text{HNM-AT2-R14})$$

These match the contracted relation at  $q=1/4, w=0$  and lie strictly below  $B$  even at  $\tau=0$ . They show insufficiency even if the second moment were supplied exactly, more information than the actual AT1 ceiling.

Their inverse moments are

$$r_A = \frac{3}{32}, \quad r_B = \frac{1}{10}, \quad r_C = \frac{1}{8\sqrt{3}} \log \frac{3+\sqrt{3}}{3-\sqrt{3}} = \frac{1}{12} \sum_{k=0}^{\infty} \frac{3^{-k}}{2k+1}. \quad (\text{HNM-AT2-R15})$$

For the series identity, expand  $1/(3+Y)$  in its uniformly convergent geometric series on  $|Y| \leq \sqrt{3} < 3$  and integrate: odd powers vanish and the even uniform moments are  $3^k/(2k+1)$ . The exact remainder after terms  $k=0, \dots, N-1$  is positive and bounded by

$$0 < r_C - \frac{1}{12} \sum_{k=0}^{N-1} \frac{3^{-k}}{2k+1} \leq \frac{3^{-N}}{8(2N+1)}. \quad (\text{HNM-AT2-R16})$$

At  $N=8$  the rational enclosure is

$$\frac{3}{32} < \frac{1872586}{19702683} < r_C \leq \frac{254674699}{2679564888} < \frac{1}{10}. \quad (\text{HNM-AT2-R17})$$

Thus identical zeroth, first and second moments permit different inverse responses, and an atomic or atomless measure. Each fixture has an abstract self-adjoint multiplication realization on  $L_2(\eta)$ , source vector 1, and an added vacuum summand. This construction does not realize the actual AQ Hamiltonian. It proves neither multiple AQ thermodynamic states nor any claim about which spectral type AQ actually possesses. In particular moments alone do not identify a particle pole, an exact mass, or  $b=49$  as an energy atom.

## 59.7 Exact controls and provenance

The standard-library checker verifies polynomial coefficient identities, all rational endpoints, monotonicity coefficient signs and all equal-moment arithmetic. Its executed controls include:

- A negative quadratic coefficient with  $v < B$  reverses the putative upper-bound substitution. In contrast the actual  $P_b$  coefficient is positive, and the strict substitution slack is computed on  $\eta_A$ .
- The polynomial  $(x-3/2)^2 - 1/16$  is positive at the frozen sampled points  $a, 1, 2, 3, 8, 49, 100$ , but negative at  $x=3/2$ . This rejects a sampled-grid certificate. The actual certificates use exact factorizations instead.
- A permitted joint moment point  $q=q_-, w=1/500$  has  $s$  and  $u$  different from the Haar values. A one-atom abstract measure of mass  $s$  at  $u/s$  realizes those moments with  $v=u^2/s < B$ . It invalidates an assignment of Haar values to unknown interacting moments; it does not claim every Haar-substituted numerical bound happens to fail.
- The endpoint atom  $\delta_L$  has window mass one while  $p_L(L)=0$ . A separate measure with positive high-energy weight above  $L$  verifies that a mass floor is not a hard spectral cutoff.
- $A/B/C$  have exactly equal moments and different inverse responses, including atomless  $C$ . The  $v$  ceiling is strictly larger than their exact second moments.
- Rescaling  $\alpha$  changes  $R$  inversely, and replacing energy by frequency introduces  $\hbar$ . Dimensionless inverse integrals cannot be presented as an energy inverse unchanged.

- In a two-state diagnostic  $H=\text{diag}(0,3)$ ,  $W=\sigma_x/2$ ,  $R=1/12$ . Perturbations  $H\text{-f}W$  and  $H\text{-}2\text{f}W$  have different derivatives of the  $W$  expectation,  $2R$  and  $4R$ , despite identical undeformed  $R$ . This demonstrates the missing source-calibration requirement. Even after fixing a source, existence and interchange of derivatives for an infinite-volume deformed AQ state remain unproved; this finite fixture does not establish that differentiability.

Bertsimas and Popescu, *SIAM J. Optim.* 15 (2005), Sections 2–3, provide the moment-duality and polynomial-positivity ancestry, including the upper-moment coefficient-sign condition. Abbott et al., arXiv:2605.20509v1, Sections III.1, IV.1–IV.3 and VII.4, give the current spectral-kernel certificate context and caution about interpreting extremal feasible measures as the true spectrum. Those primary passages were checked with the source panel and direct reading. Their general methods retain attribution. The Hruday/HNM contribution is this model-specific AQ application, exact physical interval and discriminating countercontrols; priority is unverified.

Reproduce with `python-Bresearch/round30/reverse/at2/check.py--output/absolute/fresh-directory`. Every source, report, checker and output is hash-bound; normal and optimized Python output bytes are compared. Finite controls audit algebra and counter-inferences; the full-half-line proofs above establish certificate validity. No global optimization, actual susceptibility, unique spectrum or continuum conclusion is asserted. AT2 ends here; Loop 3 remains unselected pending its review.

`research/round30/reverse/at2/report.md`

## 60 Hrunday AT3: reviewed admission

**Verdict: accepted within scope.** The fixed 4097-node protocol conditionally encloses  $I = \alpha R$  when supplied certified centered AQ samples with absolute error at most  $10^{-6}$ . Exact rational data give interval width below  $1/500$ ; arithmetic bins contribute their explicitly bounded trapezoid width. Both executed abstract A/B benchmarks pass, and their full admissible-error output envelopes are disjoint. No actual AQ correlation samples or response value are supplied.

### Limits and independent review

The producer reports were frozen separately before exchange. Their common ancestry, contract and inherited methods remain disclosed. The skeptic reviewed both proofs and executed controls; this does not constitute external human peer review.

- No actual AQ correlation samples or inverse response evaluated
- Forward permits arbitrary arithmetic bins and reports width verdict; reverse admits bins  $\leq 1e - 12$  and guarantees the width target
- Input validation does not certify centering, model/state provenance or scientific error bounds
- A/B and the separately disclosed damaging tail control are abstract measures, not AQ simulations
- No susceptibility, identified pole, optimal design, state uniqueness or continuum construction
- Scientific priority unverified; established methods retain attribution
- Exactly three investigations; no fourth scientific execution

research/round30/advisor/at3-gate.json

research/round30/skeptic/at3.md

### Substantive directional differences

- Forward uses degree39/40 and reverse degree33/32 alternating exponential brackets; both enclose every node using outward rational arithmetic.
- Forward accepts arbitrary arithmetic bins and reports a width verdict; reverse requires bins no wider than  $10^{-12}$  to guarantee the target.
- The reverse omitted-tail diagnostic internally evaluates a third abstract curve; this is required countercontrol work, not an additional investigation or AQ simulation.

### The two reusable input domains

The forward `evaluator.py` accepts exact rational sample values or enclosing pairs. Wider arithmetic bins still yield a conditional enclosing interval, but can fail its width verdict. The reverse `check.py` evaluator requires enclosing pairs, arithmetic width at most  $10^{-12}$ , and its additional mass/error envelope checks; its CSV route also validates the complete time grid. These are different admitted API domains, not interchangeable promises.

For both APIs, callers must establish centering, the actual AQ state/model provenance and the deterministic sample-error premise. Syntax checks cannot certify those physical conditions. The executed CSV data are synthetic A/B controls, and no actual AQ correlation samples or response value are supplied.

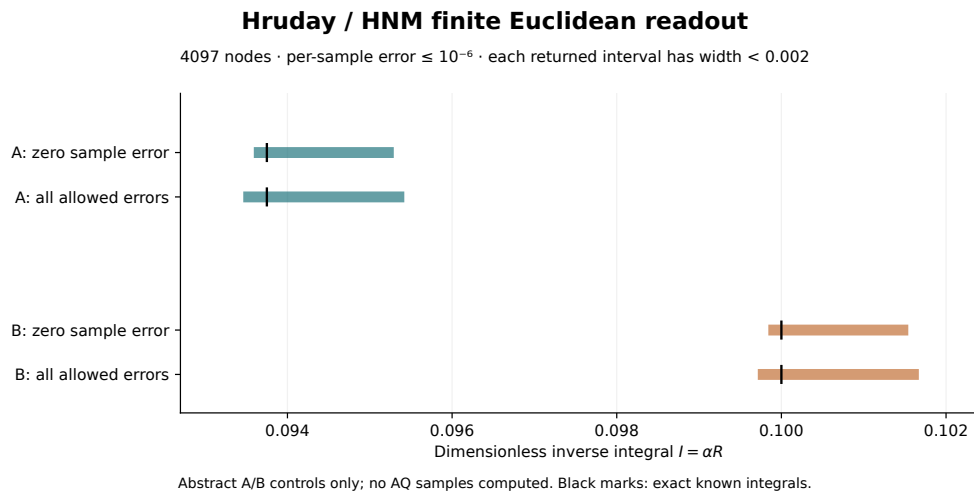


Figure 13: Certified inverse-integral intervals for the two abstract equal-moment controls A and B. The zero-error rows still include the protocol error allowances. The all-errors rows are envelopes of possible returned intervals: they include both the observed-sum shift and the sample allowance inside each certificate. Each individual returned interval has width below 0.002; its full varying-data envelope can be wider. The envelopes remain separated by more than 0.004293. Black marks show the known exact values  $3/32$  and  $1/10$ . These are synthetic controls, not AQ measurements or computed AQ response values. Floating-point rendering of frozen exact rational endpoints; the source output supplies rigorous enclosures.

## 61 Hrudy AT3: finite Euclidean readout certificate (forward derivation)

**Source-bound forward derivation.** This is the complete typeset mathematical report of one producer. Prospective claims describe its original freeze; the shared reviewed result and permitted refinements are stated in the admission section. The two agents share inherited premises and model ancestry. This is neither an independent physical observation nor external human peer review.

Human project author: **Hrudy N M (BUNZEEY)**. AI-assisted independent forward execution, without reading current AT3 reverse or skeptic solutions. HNM tags are project aliases for this application of established quadrature and spectral methods. Scientific priority is unverified.

**Outcome:** the frozen design certifies a conditional AQ inverse-form readout with interval width below  $1/500$  for exact rational reported data meeting its error contract. Its two executed abstract fixtures pass, including all-positive/all-negative sample errors; their worst-case output envelopes are disjoint. **No actual AQ correlation samples or actual AQ inverse value have been produced.**

### 61.1 Actual-state conditional theorem and physical clock

Use exactly AT1/AT2's actual AQ centered Wilson spectral measure  $\eta$  in dimensionless energy  $x=E/\alpha$ . Its inherited hypotheses are support  $x \geq a=1/16$ , mass  $m \leq M = 63/250$ , first moment  $u \leq U = 94/125$ , and finite second moment  $v \leq B = 36 + 98|\tau|$ . To avoid confusing Euclidean duration with mass, denote mass by  $m$  in this report and duration by  $s$ . Define

$$C(s) = \int e^{-sx} d\eta(x), \quad I = \int_0^\infty C(s) ds = \int x^{-1} d\eta(x) = \alpha R. \quad (\text{HNM-AT3-F01})$$

Positivity permits Tonelli, and  $x \geq a > 0$  gives  $I \leq M/a < \infty$ . Here  $R$  is precisely AT2's reduced inverse-energy quadratic form, not a thermodynamic susceptibility. In physical Euclidean time,

$$s = \alpha t_E / \hbar, \quad C_{phys}(t_E) = C(\alpha t_E / \hbar), \quad R = \hbar^{-1} \int_0^\infty C_{phys}(t_E) dt_E. \quad (\text{HNM-AT3-F02})$$

The frozen design is

$$T = 128, \quad h = 1/32, \quad N = 4096, \quad Nh = T, \quad \epsilon = 10^{-6}, \quad N + 1 = 4097. \quad (\text{HNM-AT3-F03})$$

Thus the physical endpoint is  $128\hbar/\alpha$  and the physical sample spacing is  $\hbar/(32\alpha)$ . Sample errors are deterministic absolute errors on dimensionless  $C(s)$ , not Gaussian noise, independent random errors or confidence intervals. This protocol supplies no claim that obtaining those data is experimentally or computationally feasible.

### 61.2 Differentiability and integrated curvature from actual moments

Since  $m, u, v$  are finite, dominated convergence on each compact nonnegative duration interval yields right derivatives at zero and ordinary derivatives thereafter:

$$C'(s) = - \int x e^{-sx} d\eta(x), \quad C''(s) = \int x^2 e^{-sx} d\eta(x) \geq 0. \quad (\text{HNM-AT3-F04})$$

Continuity of  $C''$  at zero follows from domination by the integrable  $x^2$ . There is no third-moment assumption. More usefully than a pointwise supremum, Tonelli gives

$$\int_0^T C''(s) ds = \int x(1 - e^{-Tx}) d\eta(x) \leq u \leq U. \quad (\text{HNM-AT3-F05})$$

Only nonnegative integrands are interchanged. The second moment justifies  $C^2$  regularity; the final quadrature constant uses the smaller integrated first moment  $U$ .

### 61.3 Exact Peano identity and quadrature sign

On cell  $[jh, (j+1)h]$ , take  $k_j(s) = (s - jh)((j+1)h - s)/2$ . It vanishes at both endpoints, with derivatives  $h/2$  and  $-h/2$  there and second derivative  $-1$ . Two integrations by parts yield

$$\frac{h}{2}[C(jh) + C((j+1)h)] - \int_{jh}^{(j+1)h} C(s) ds = \int_{jh}^{(j+1)h} k_j(s) C''(s) ds. \quad (\text{HNM-AT3-F06})$$

The kernel satisfies  $0 \leq k_j \leq h^2/8$  throughout the continuous cell. Summing and using (F05), with  $\text{Trap}(C) = h(C(0)/2 + \sum_{j=1}^{N-1} C(jh) + C(T)/2)$ , proves

$$0 \leq \text{Trap}(C) - \int_0^T C(s) ds \leq Q := h^2 U/8 = 47/512000. \quad (\text{HNM-AT3-F07})$$

The trapezoid rule overestimates the finite integral because  $C$  is convex. Treating  $Q$  as an amount to add to a lower endpoint gives the wrong sign. No observed mesh convergence or finite positivity grid supplies this theorem.

### 61.4 Infinite-time tail and deterministic sample budget

Support above  $a$  gives

$$0 \leq \int_T^\infty C(s) ds = \int e^{-Tx} x^{-1} d\eta(x) \leq D := \frac{M}{a} e^{-aT} = \frac{504}{125} e^{-8}. \quad (\text{HNM-AT3-F08})$$

For reported real samples  $y_j$  satisfying  $|y_j - C(jh)| \leq \epsilon$ , all trapezoid weights are positive and  $\sum hN = T$ . Therefore

$$|\text{Trap}(y) - \text{Trap}(C)| \leq T\epsilon = 2/15625 =: J. \quad (\text{HNM-AT3-F09})$$

The vectors with every error  $+\epsilon$  and every error  $-\epsilon$  attain  $+J$  and  $-J$ . A root- $N$  allowance would be invalid for this deterministic model. Combining (F07)-(F09) yields the conditional interval

$$I \in [\text{Trap}(y) - Q - J, \text{Trap}(y) + D + J]. \quad (\text{HNM-AT3-F10})$$

For arithmetic enclosures  $y_j \in [l_j, r_j]$ , use the corresponding positive-weight lower and upper sums  $P_-, P_+$ . With a proved rational upper  $D_+$  for  $D$ ,

$$I \in [P_- - Q - J, P_+ + D_+ + J], \quad \text{width} = (P_+ - P_-) + Q + D_+ + 2J. \quad (\text{HNM-AT3-F11})$$

The arithmetic width is separate from the physical/sample error budget. The API never treats unknown exponential roundoff as free. For exact rational  $y_j$  the arithmetic width is zero. The

certified error calculation gives  $Q + D_+ + 2J < 170039/100000000 < 1/500$ . For the executed enclosures their extra width is below  $10^{-24}$ , so the frozen width target also passes.

## 61.5 Directed rational exponential arithmetic and complete node semantics

The arithmetic scale  $G = 10^{30}$  was fixed in evaluator.py before generating any fixture outcome. To enclose  $\exp(-y)$  for rational  $y \geq 0$ , choose  $k$  so  $r = y/2^k \leq 1/2$ . The alternating exponential Taylor terms decrease in absolute magnitude. Consequently the odd partial sum at degree39 and even partial sum at degree40 give

$$\sum_{n=0}^{39} \frac{(-r)^n}{n!} \leq e^{-r} \leq \sum_{n=0}^{40} \frac{(-r)^n}{n!}. \quad (\text{HNM-AT3-F12})$$

Both sums are exact Fractions. Since  $r \leq 1/2$ , the odd lower sum is at least  $1-r \geq 1/2$  by grouping consecutive positive/negative terms. Round the lower sum down to the  $G$  grid and the upper sum up. Repeatedly square the nonnegative interval  $k$  times, rounding every lower product down and every upper product up. Monotonicity of squaring on nonnegative numbers preserves enclosure. This produces rational bounds for the needed  $\exp(-8)$  tail and each fixture ratio  $\exp(-h x)$ . No floating exponential, library transcendental value or uncontrolled arbitrary precision number is used in a claimed bound.

For a fixture atom of energy  $x$ , let  $[q_-, q_+]$  enclose  $\exp(-hx)$ , and start  $[p_0^-, p_0^+] = [1, 1]$ . At every step compute

$$p_{j+1}^- = \lfloor G p_j^- q_- \rfloor / G, \quad p_{j+1}^+ = \lceil G p_j^+ q_+ \rceil / G. \quad (\text{HNM-AT3-F13})$$

Induction using positive multiplication proves  $p_j^- \leq \exp(-jh x) \leq p_j^+$  for every  $j=0, \dots, 4096$ . Positive weighted summation over atoms encloses each  $C(jh)$ . **Every one of the 4097 nodes is generated and included with its actual trapezoid weight.** No geometric closed-sum substitution is used. The saved CSVs contain exactly those node enclosures, explicitly marked as synthetic in the result metadata. The total arithmetic trapezoid widths are computed exactly and both are below  $10^{-24}$ .

## 61.6 Executed equal-moment benchmarks and all-error separation

Only AT2's abstract atomic fixtures are evaluated:

$$\eta_A = (\delta_2 + \delta_4)/8, \quad \eta_B = \delta_1/32 + 3\delta_3/16 + \delta_5/32, \quad I_A = 3/32, \quad I_B = 1/10. \quad (\text{HNM-AT3-F14})$$

For each fixture and each sign  $\sigma = -1, 0, +1$ , shift every exact node enclosure by  $\sigma \epsilon$ . These enclose the real observations  $C(jh) + \sigma \epsilon$ , so the extreme errors are exactly the contracted epsilon, not epsilon plus an unrecorded midpoint error. Feed all 4097 shifted enclosures through the same reusable evaluator.

The following decimal intervals are widened outward to nine decimal places; exact rational results are saved:

| Fixture and sample error | Certified I interval       |
|--------------------------|----------------------------|
| A, every error -epsilon  | [0.093463226, 0.095163609] |
| A, zero error            | [0.093591226, 0.095291609] |
| A, every error +epsilon  | [0.093719226, 0.095419609] |

| Fixture and sample error | Certified I interval       |
|--------------------------|----------------------------|
| B, every error -epsilon  | [0.099713226, 0.101413609] |
| B, zero error            | [0.099841226, 0.101541609] |
| B, every error +epsilon  | [0.099969226, 0.101669609] |

Each interval contains its known exact inverse integral and has width less than 0.001701, passing the requested 0.002 target. For every possible error vector, not merely the two tested constant vectors, positive weights place its trapezoid result between those extremal sums. Hence compare **A-plus's upper output endpoint against B-minus's lower output endpoint**. This includes both the shift J in the observed sum and the already-present allowance J inside the reported interval. The independently executed rational comparison gives

$$\text{Lower}(B, -\epsilon) - \text{Upper}(A, +\epsilon) > 4293/1000000 > 0. \quad (\text{HNM-AT3-F15})$$

Thus the whole permitted output envelopes are disjoint. Merely comparing the two noiseless intervals would not prove worst-error separation. The result discriminates these abstract controls using this finite protocol. It does not identify the true AQ spectral type, response, particle mass or pole.

## 61.7 Reusable deterministic API and rejected inputs

`evaluator.py` exposes `evaluate(samples, *, T='128', h='1/32', N=4096, epsilon='1/1000000')`. Supply a list/tuple of exactly 4097 entries. Each entry is an exact rational string, integer or Fraction, or a two-element lower/upper enclosure of those types. Decimal strings denote exact rationals. Floats, NaN, infinities, reversed enclosures, wrong sample count, changed design, and changed error contract are rejected. The output contains rational lower/upper endpoints, width, trapezoid arithmetic width, all three physical allowances, and the target-width verdict.

The caller must independently establish that each enclosure contains its actual reported sample and that the actual reported sample is within epsilon of the relevant AQ correlation. The API cannot verify that physical premise from numbers alone; it labels its output conditional and never claims to have supplied AQ samples. Wider arithmetic enclosures remain mathematically valid but can fail the target-width verdict. Divide a returned I interval by positive alpha to obtain R with inverse-energy units.

Reproduce all declared tests with `python-Bresearch/round30/forward/at3/check.py--output/absolute/new/directory`. Normal and optimized Python semantic outputs and manifests must match. Saved fixture CSVs are optional inspection data, not measurement records.

## 61.8 Damaging controls, ancestry and final stop

The checker tests the quadrature sign against exactly enclosed finite fixture integrals; deleting the tail fails by an analytic finite-integral/curvature bound for a slow abstract atom at the gap. No samples from that control are generated. That atom satisfies the gap/mass/first-moment hypotheses used by the quadrature theorem, and is not called an AQ realization. Root-N error scaling is rejected by the exact constant-sign vector. Adding a zero-energy atom creates a constant correlation and divergent full inverse integral, exposing missing centering and violation of the gap support assumption. Energy/time rescaling, float inputs, malformed designs, and absent source provenance are distinguished explicitly. Coarsening h to  $1/2$  gives  $Q > 1/500$ ; truncating at  $T=16$  gives a tail bound  $> 1/500$ . These failed alternatives are controls, not retuning or additional investigations.

NIST DLMF §3.5(i), Eqs.3.5.1-3.5.3, supplies the standard trapezoidal rule and sign. The Peano-kernel integrated-moment argument is derived fully above for this model's moment information. Abbott et al., arXiv:2605.20509v1, supplies current context for spectral bounds from finite Euclidean data; no new generic quadrature, bootstrap or Laplace-inversion method is claimed. This selected source comparison does not establish scientific priority.

This is investigation3 of3. Production stops after its review. Only repair, validation, integration, prospective planning and publication follow. No actual AQ sample acquisition, susceptibility proof, spectral-pole inference, continuum construction or fourth scientific investigation is included.

`research/round30/forward/at3/report.md`

## 62 Hrudy AT3: finite Euclidean readout certificate (reverse derivation)

**Source-bound reverse derivation.** This is the complete typeset mathematical report of one producer. Prospective claims describe its original freeze; the shared reviewed result and permitted refinements are stated in the admission section. The two agents share inherited premises and model ancestry. This is neither an independent physical observation nor external human peer review.

Human author: **Hrudy N M (BUNZEEY)**. AI-assisted independent reverse reconstruction; scientific priority unverified. HNM identifiers are project aliases, not claims to invent quadrature. The contract and shared prospective Peano-kernel suggestion were fixed before fixture production. No current forward or skeptic AT3 results were read. This is the third and final authorized investigation.

### 62.1 Reverse target: a finite readout of a specified spectral functional

AT2 produced a broad interval for an inverse-energy form and demonstrated that moments do not identify a spectrum. This loop asks whether a finite, explicitly accurate Euclidean sample set could certify that same scalar. Reconstructing backward from the desired integral requires four controlled errors: quadrature on a finite interval, the unobserved infinite-time tail, deterministic sample errors, and arithmetic enclosures. The actual AQ premises establish a conditional theorem. The executed inputs below are only AT2's declared abstract A/B controls.

Let  $\eta$  be AQ's centered Wilson spectral measure in  $x=E/\alpha$ , supported on  $x \geq a=1/16$ , with mass  $\leq s_{up}=63/250$ , first moment  $\leq u_{up}=94/125$ , and second moment  $\leq B=36+98|\tau|$ . Use the dimensionless Euclidean time  $s=\alpha t_E/\hbar$ ; it is distinct from the mass notation used in AT2. Define

$$C(s) = \int_a^\infty e^{-sx} d\eta(x), \quad I = \int_0^\infty C(s) ds = \int_a^\infty \frac{1}{x} d\eta(x) = \alpha R. \quad (\text{HNM-AT3-R01})$$

Tonelli applies to the nonnegative integrand, and the support bound gives  $I \leq s_{up}/a < \infty$ . Thus the desired scalar is dimensionless  $I$ ; physical  $R=I/\alpha$  has inverse-energy units. An integral in physical time would instead equal  $\hbar R$ . No static susceptibility is identified.

The frozen sample design is

$$\begin{aligned} T &= 128, \quad h = 1/32, \quad N = 4096, \\ s_j &= jh \quad (0 \leq j \leq N), \quad \epsilon = 10^{-6}, \quad \text{target width} = 1/500. \end{aligned} \quad (\text{HNM-AT3-R02})$$

There are 4097 nodes. Physically these correspond to  $t_{E,j}=(\hbar/\alpha)jh$  and total duration  $128\hbar/\alpha$ . The error premise is  $|d_j - C(s_j)| \leq \epsilon$  separately at every node. It is a deterministic absolute-error contract; neither independence nor a probability distribution is assumed. Feasibility or cost of producing actual AQ data at this precision is not established.

### 62.2 Differentiability, integrated curvature and the exact Peano kernel

The finite second moment permits differentiating twice under the measure integral for every  $s \geq 0$ , including right derivatives and continuity at zero. The majorants  $x$  and  $x^2$  are integrable; dominated convergence also gives continuity of the derivatives. Consequently  $C$  is  $C^2$  on every closed finite nonnegative time interval, with

$$C'(s) = - \int x e^{-sx} d\eta(x), \quad C''(s) = \int x^2 e^{-sx} d\eta(x) \geq 0. \quad (\text{HNM-AT3-R03})$$

Tonelli supplies a stronger integrated budget than a supremum estimate:

$$\int_0^T C''(s) ds = \int x(1 - e^{-Tx}) d\eta(x) \leq \int x d\eta(x) \leq u_{up}. \quad (\text{HNM-AT3-R04})$$

The second moment supplies regularity at the first node; the first moment supplies this global quadrature budget.

On a cell  $[jh, (j+1)h]$ , put  $r=s-jh$  and  $k_j(s)=r(h-r)/2$ . Twice integrating by parts, with  $k=0$  at both endpoints,  $k'(jh)=h/2$ ,  $k'((j+1)h)=-h/2$  and  $k''=-1$ , gives

$$\frac{h}{2}\{C(jh) + C((j+1)h)\} - \int_{jh}^{(j+1)h} C(s) ds = \int_{jh}^{(j+1)h} k_j(s) C''(s) ds, \quad 0 \leq k_j(s) \leq h^2/8. \quad (\text{HNM-AT3-R05})$$

This proves the excess sign as well as the bound. Summing all cells defines the usual composite trapezoid and yields

$$Q_h(C) = h \left[ \frac{1}{2} C(0) + \sum_{j=1}^{N-1} C(jh) + \frac{1}{2} C(T) \right], \quad (\text{HNM-AT3-R06})$$

$$0 \leq Q_h(C) - \int_0^T C(s) ds \leq q_{err} := \frac{h^2 u_{up}}{8} = \frac{47}{512000}.$$

The proof holds on the full time interval and retains all  $N$  cells. It is not an inference from observed grid convergence, nor  $N$  times a sampled second derivative.

### 62.3 Tail, deterministic errors and interval arithmetic

For all  $s \geq T$  the gap gives  $C(s) \leq s_{up} \exp(-as)$ . Therefore

$$0 \leq \int_T^\infty C(s) ds \leq t_{err} := \frac{s_{up}}{a} e^{-aT} = \frac{504}{125} e^{-8}. \quad (\text{HNM-AT3-R07})$$

Every trapezoid weight is positive and the weights sum to  $Nh=T$ . If  $|d_j - C(s_j)| \leq \epsilon$ , the triangle inequality gives

$$|Q_h(d) - Q_h(C)| \leq T\epsilon =: n_{err} = \frac{16}{125000}. \quad (\text{HNM-AT3-R08})$$

The two constant-sign error vectors attain the positive and negative extremes. A root- $N$  estimate is unjustified for this premise.

The implementation permits a rational interval  $[d_j^-, d_j^+]$  enclosing each observed real  $d_j$ ; these are arithmetic representation intervals, separate from the sample error. Positive weights produce  $Q_h(d)$  in  $[Q^-, Q^+]$ . A rigorous upper exponential enclosure  $t_{err}^+$  then gives the conditional certificate

$$I \in [Q^- - q_{err} - n_{err}, Q^+ + t_{err}^+ + n_{err}]. \quad (\text{HNM-AT3-R09})$$

Its width is  $Q^+ - Q^- + q_{err} + t_{err}^+ + 2n_{err}$ . Exact rational arithmetic supplies the endpoints, so no later floating-point rounding is hidden in them.



contains its exact I and satisfies the target width. The following decimal displays are for readability; the JSON contains exact rational endpoints.

| Input case                                | Lower endpoint | Upper endpoint | Exact benchmark |
|-------------------------------------------|----------------|----------------|-----------------|
| A, zero error                             | 0.09359122636  | 0.09529160856  | 0.09375         |
| B, zero error                             | 0.09984122636  | 0.10154160856  | 0.1             |
| Envelope of all<br>admissible A<br>errors | 0.09346322636  | 0.09541960856  | 0.09375         |
| Envelope of all<br>admissible B<br>errors | 0.09971322636  | 0.10166960856  | 0.1             |

The envelopes include two distinct noise effects. When  $d$  varies, its observed trapezoid itself shifts by at most  $T$  epsilon; each returned certificate also has its own allowance  $T$  epsilon. Let  $[Q\_A^-, Q\_A^+]$  and  $[Q\_B^-, Q\_B^+]$  enclose the true noise-free trapezoids. The largest possible A upper endpoint and smallest possible B lower endpoint are bounded by

$$U_A = Q_A^+ + t_{err}^+ + 2T\epsilon, \quad L_B = Q_B^- - q_{err} - 2T\epsilon, \quad L_B - U_A > \frac{4}{1000}. \quad (\text{HNM-AT3-R14})$$

The exact computed margin is approximately 0.00429361781. The checker independently executes A with all-positive observations and B with all-negative observations, each passed through the full output-interval evaluator. Positive weights prove these are the damaging extremes among all allowed error vectors, even when A and B errors are unrelated. This establishes discrimination of these two abstract controls by the frozen protocol. It does not identify the AQ spectrum, its poles or its actual inverse-energy value.

## 62.6 Reusable input contract

`check.py` provides deterministic functions `evaluate(samples, *, T_value=128, h_value=Fraction(1, 32), N_value=4096, epsilon=Fraction(1, 1000000))` and `evaluate_csv(path)`. Each sample is a lower/upper pair of exact rational numbers. Allowed numeric values are Python `Fraction`, integers, exact decimal strings or rational strings; binary floats are rejected. The CSV has exactly this header:

```
index,s,value_lower,value_upper
```

It must have 4097 data rows, consecutive indices 0 through 4096, and exact times  $j/32$ . Both sample-bin endpoints may be equal. Sample values may be slightly negative under the allowed observation error near the tail; the evaluator does not clip them and bias the sum. Intervals wholly incompatible with the mass/error envelope are rejected. Inverted intervals, excessive arithmetic width, wrong sample count, time grid, design, error level, nonfinite text and malformed rational fields are rejected.

For actual future observations, the caller must independently certify centering, AQ model/state provenance and the epsilon absolute-error premise. Parsing data and producing an interval cannot establish these scientific conditions. Even the output flag stating `actual_aq_samples_computed: false` remains false in external-input mode: this program does not compute those samples itself.

Run the benchmark and its exact controls with:

```
python-Bresearch/round30/reverse/at3/check.py--output/absolute/fresh-directory
```

Evaluate a CSV meeting the declared design with:

```
python-Bresearch/round30/reverse/at3/check.py--csv/absolute/input.csv--output/absolute/fresh-directory
```

The program writes exact rational certificates and a file manifest. Division of its I endpoints by a specified alpha gives physical R endpoints; no implicit choice of alpha or hbar is made.

## 62.7 Damaging controls and limitations

The executed controls reject the following inferences:

- **Wrong trapezoid sign.** For fixture A, the rigorous lower bound for its trapezoid exceeds the exact  $I_A$ . Reversing the convex-quadrature correction would exclude the truth.
- **Omitted tail.** A further diagnostic with mass  $1/10$  at  $x=1/16$  and  $3/20$  at  $x=119/24$  preserves total mass  $1/4$  and first moment  $3/4$ . Its no-tail upper endpoint lies strictly below its exact inverse integral, even including the sample allowance. It is a damaging abstract kernel control, not another AQ computation.
- **Root-N deterministic noise.** All-positive errors shift the sum by exactly  $T$  epsilon. The proposed root-N replacement  $T$  epsilon/64 is smaller by 64 and fails this explicit allowed error vector.
- **Missing centering.** Adding a positive zero-energy atom leaves a nondecaying constant in C. Its time integral diverges and it violates the positive-gap tail estimate; inversion on the vacuum-orthogonal subspace cannot be replaced by integrating that uncentered correlation.
- **Wrong time/energy conversion.** An exact  $\alpha=4, \hbar=3$  fixture distinguishes physical time from dimensionless duration and I from R.
- **Insufficient settings.** With  $T=16$  the allowed tail budget alone exceeds the target; with  $h=1/4$  the quadrature budget alone exceeds it. These outcomes concern the uniform certificate and do not prove actual reconstruction failure for every measure.
- **Uncontrolled floating exponentials and malformed input.** The API rejects these rather than treating a point float or invalid grid as a rigorous enclosure. Source snapshots, report, code, CSVs and output bytes are frozen and replayed normally and with optimized Python.

NIST DLMF Section 3.5(i), equations 3.5.1–3.5.3, is the primary source for standard trapezoidal quadrature and its sign. The positive kernel and integrated-moment application are derived explicitly above. Abbott et al., arXiv:2605.20509v1, supplies established spectral-functional certification from finite Euclidean information. Neither general technique is attributed as a new Hruday invention. No actual AQ correlation data, AQ response value, physical susceptibility, optimal sampling design, spectral pole, experimental apparatus or continuum theory has been produced. After this freeze only review, repairs, integration and publication remain; no fourth scientific investigation is started.

```
research/round30/reverse/at3/report.md
```

## 63 Round30 closeout and next unexecuted goals

All three investigations in this cycle have recorded reviews. The complete evidence and current next-step plan are the source-bound machine-readable records below. A selected next goal is not an executed fourth investigation.

research/round30/advisor/findings.json

research/round30/advisor/roadmap.json

### Exact certificate calculator

The standalone calculator below evaluates the admitted AT2 window and inverse-energy bounds with rational arithmetic. It requires positive energy coefficient  $\alpha$  and  $|\tau| \leq 10^{-8}$ ; those validations do not enlarge the theorem range. It returns a bound on the actual unknown response, not a sampled AQ spectrum or an identified susceptibility.

```
python3 calculators/hnm_round30.py --alpha 1 --tau 1/100000000
```

The command is relative to the unpacked manuscript directory. The source-bound repository original is `research/round30/calculators/spectral_certificate.py`.

**AR: Identify the numerical-cap bulk state** Derive an explicit spatial boundary estimate for the AQ model and use it to test whole-sequence convergence and uniqueness of the declared full-lattice limit.

**Status.** `planned_not_executed`. AQ proves subsequential existence and a simple vacuum within its GNS representation; it does not give a quantitative comparison of different thermodynamic limits.

**First prospective test.** Freeze two centered/bulk exhaustions of the same repeated selected-strip Hamiltonian. Derive or reject one explicit norm-one local-observable boundary estimate at a declared numerical interval, including all complete factors and original endpoints.

**Adaptive follow-up rule.** After the first skeptical review, either use the actual bound to prove the specified all-pairs local Cauchy/state-identification result, or isolate the missing estimate with a discriminating countercontrol.

**Exact model boundary.** Full coarse Z3, actual complete 24-link factors, fixed repeated selected triple, whole-star omission and physical scales; AQ state at  $|\tau| \leq 10^{-8}$ . Any comparison to AN or old orthant states retains their additional common hypothesis range and deep-bulk observation map.

**Reason for rank.** Closes the main identification gap exposed by the newly completed numerical-cap construction. Rank reflects scientific dependency, not predicted ease.

**AS: Test a different weak-coupling reference strategy** Select one source-grounded alternative reference or multiscale candidate and test whether its complete dimensionless interaction dictionary can follow a weak-bare-coupling trajectory.

**Status.** `planned_not_executed`. The current selected-strip sufficient regime has  $\tau = 96/g^4$  and cannot approach  $g = 0$ . Common energy/time rescaling leaves this obstruction unchanged.

**First prospective test.** Freeze one alternate reference, full original Hamiltonian, boundary prescription and physical-clock map; compute the actual normalized local interaction along a specified weak-coupling path. Preserve an insufficient outcome if no eligible route is found.

**Adaptive follow-up rule.** Only if the first dictionary passes review, derive or simulate one bounded stability/scale-matching test with explicit error; otherwise select a concrete obstruction or model-distinction test.

**Exact model boundary.** A newly declared candidate must map to the actual uniform  $SU(2)$  Hamiltonian or explicitly state its deformation; no automatic transfer from AQ.

**Reason for rank.** Addresses the most consequential unresolved route toward the continuum target, while recognizing that it is substantially harder than a routine corollary.

**AT4: Certify actual AQ Euclidean data** Produce one rigorously enclosed centered Euclidean correlation sample in the chosen AQ state, or isolate the missing estimate that prevents the required absolute accuracy.

**Status.** `planned_not_executed`. AT3 supplies a conditional finite-data readout but no actual AQ samples. Finite-volume/boundary/state, representation-cutoff, solver and arithmetic errors have not been bounded together for such a sample.

**First prospective test.** Freeze one nonzero dimensionless time, physical clock, exact chosen AQ state and Wilson centering, finite graph/cutoff and absolute tolerance before computing. Bound all errors and retain an insufficient verdict if any required estimate is missing.

**Adaptive follow-up rule.** After a new first-loop review, decide whether to extend to the frozen4097-node AT3 schedule, improve a necessary bound, or stop with a quantified obstruction.

**Exact model boundary.** Same chosen AQ numerical-cap state, centered original Wilson, complete original factors, fixed physical energy and clock. Abstract A/B controls and separate AU/AP models cannot supply AQ observations.

**Reason for rank.** Most bounded next response deliverable, conditional on a genuine sample-production certificate; difficulty and physical significance differ from the finite readout arithmetic.

**AU: Evaluate the certified 561-state heat output** Turn the inherited exact-semigroup error theorem into an exported, reproducible numerical vector/evaluator with a rigorous full-output error budget.

**Status.** `planned_not_executed`. AH1/AH2 provide structural and exact-semigroup estimates; the final certified 561-coordinate output has not been evaluated.

**First prospective test.** Freeze graph, representation span, input, Haar Gram metric, separate ground centers and a requested tolerance. Implement one certified heat-action evaluation with interval or rigorous residual/tail control.

**Adaptive follow-up rule.** After reviewing the actual output and error denominator, extend to one declared time range/input or diagnose the failing resource bound.

**Exact model boundary.** The inherited finite physical graph and AH1/AH2 561-coordinate generated span; complete outside-channel and truncation controls remain mandatory.

**Reason for rank.** Converts an existing theorem into a usable reproducible calculator without confusing finite numerical evidence with a continuum proof.

**AP: Test inference with a design fixed from the prior** Test canonical-model parameter inference under one schedule chosen before observing candidate outcomes, with independently varying nuisance parameters and complete physical errors.

**Status.** `planned_not_executed`. Prior results identify conditional candidate separations; they do not yet validate one prior-fixed design over independently varying candidates.

**First prospective test.** Freeze a compact prior, observable/clock dictionary, one observation

schedule and acceptance margin, then compute candidate intervals with independent nuisance variations and all inherited error terms.

**Adaptive follow-up rule.** Choose the second investigation from the first review: certify a supported separation region, or derive an unresolved/inseparable counterfamily and revise the future design rule.

**Exact model boundary.** The separate canonical summable model and physical readout maps of AI3/AI4; do not import the homogeneous AQ state or its gap.

**Reason for rank.** Preserves the previously proposed AP question under its original identity and keeps useful inference work distinct from homogeneous-gap evidence.

## Attribution and reproducibility

Hruday N M (BUNZEEY) is the human project author. AI systems assisted source retrieval, derivation, implementation and review. The result is a source-qualified research record, not an externally peer-reviewed paper or arXiv acceptance. General moment methods and operator constructions retain their original authors. No new axioms are asserted.

`research/round30/reproduce.py`

`papers/draft-03/build.py`

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## A Current priority-ranked results and their limits

This qualitative ranking emphasizes dependencies and usable conclusions for the current route. It is neither a scientific-priority judgment nor a percentage of continuum Yang–Mills solved. The old gate limitations are preserved in the full catalog; a separately gated current extension is shown beside the earlier result. Recency and equation counts are not ranking criteria.

| Rank | Hruday record                                                          | Why useful                                                                                                                                                                                                                                                                                           | Scope / surviving limit                                                                                                                                                                                                                                                                                                                                                                     |
|------|------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | <b>HNM-C-AQ2</b> Hruday numerical-cap physical gap and Wilson witness  | The numerical finite-volume certificate reaches the complete physical sector of a locally normal fixed-lattice GNS representation, with a nonzero original Wilson witness. This is a model-specific application of an established transfer strategy, without uniqueness of all thermodynamic states. | Chosen GNS representation only; no all-state uniqueness, boundary identification, particle pole or continuum construction. AT1 separately supplies same-state domain/moment control.                                                                                                                                                                                                        |
| 2    | <b>HNM-C-AM2</b> Hruday four-site quantitative stability certificate   | A complete alternative creation proof supplies an explicit finite-volume coupling cap for the actual untruncated model, including excitation exclusion and removal of representation cutoffs. It does not identify an infinite-volume state.                                                         | Fixed-spacing complete-factor model; no weak-bare-coupling bridge or evaluated general boundary constant.                                                                                                                                                                                                                                                                                   |
| 3    | <b>HNM-C-AT1</b> Hruday same-state Wilson energy certificate           | Closes the domain and moment gap for the actual numerical-cap AQ state, enabling the next two spectral certificates without identifying distinct states.                                                                                                                                             | No equality of finite-to-infinite second moments No AQ/AO/AN state identification No particle pole, exact mass or dispersion No uniqueness of all thermodynamic states or whole-sequence convergence No continuum Yang–Mills construction Scientific priority unverified; established methods retain attribution                                                                            |
| 4    | <b>HNM-C-AT2</b> Hruday spectral-window and inverse-energy certificate | Turns actual-state moment information into a nonzero finite-window mass guarantee and rigorous inverse-energy bounds while exposing non-identification.                                                                                                                                              | The window is unnormalized spectral mass, not probability or a hard cutoff No evaluated actual AQ inverse response or optimality No particle pole, exact mass or identified spectral type Abstract equal-moment fixtures are not actual lattice realizations No static susceptibility or deformed thermodynamic-state differentiability No continuum result; scientific priority unverified |

| Rank | Hrunday record                                                                           | Why useful                                                                                                                                                                                                    | Scope / surviving limit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------|------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5    | <b>HNM-C-AT3</b> Hrunday finite Euclidean readout certificate                            | Supplies a conditional finite-data protocol after moment ambiguity; abstract benchmark success remains separate from producing actual AQ samples.                                                             | No actual AQ correlation samples or inverse response evaluated Forward permits arbitrary arithmetic bins and reports width verdict; reverse admits bins $\leq 1e - 12$ and guarantees the width target Input validation does not certify centering, model/state provenance or scientific error bounds A/B and the separately disclosed damaging tail control are abstract measures, not AQ simulations No susceptibility, identified pole, optimal design, state uniqueness or continuum construction Scientific priority unverified; established methods retain attribution Exactly three investigations; no fourth scientific execution |
| 6    | <b>HNM-C-RECENT25</b> Hrunday full-system endpoint witness from connected-word tails     | A full-model counterexample marks the critical growing-time boundary; useful for rejecting an unjustified interchange of limits.                                                                              | Existential probe and growing clock; separate canonical model and same-author historical review.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| 7    | <b>HNM-C-AL1</b> Hrunday full-interaction matching audit                                 | Prevents the existing small-interaction theorem from being mistaken for a weak-coupling continuum strategy.                                                                                                   | Obstruction to the sufficient theorem, not a no-gap theorem.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 8    | <b>HNM-C-AO2</b> Hrunday limiting Wilson energy identity                                 | The actual inherited Wilson vector gains an operator-domain statement and exact first-energy identity, closing a documented AK2 loss-of-moment limitation while retaining both original smallness conditions. | Original orthant state and smallness range; no AO/AQ state equality.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 9    | <b>HNM-C-EARLY10</b> Hrunday one-square full-space spectral enclosure and coupling cover | Full representation-space spectral enclosure distinguishes a certified infinite tail from a finite matrix diagonalization.                                                                                    | Finite graph; continuum matching remains open.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 10   | <b>HNM-C-AH2</b> Hrunday delayed-loading heat certificate                                | A complete 561-state finite-graph enrichment improves the all-time omitted-channel heat certificate on a fixed preparation class.                                                                             | Finite generated span; not an evaluated 561-coordinate heat vector or continuum certificate.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

## B Complete HNM naming and equation concordance

The HNM-C IDs below map every contribution in the original 114-row catalog and the two integrated addenda. The equation keys are stable source aliases; printed equation numbers are locators in this edition. No alias changes a model, variable, proof premise, external eponym or literature-priority status.

### B.1 Contribution aliases

| HNM ID              | Legacy ID     | Hruday project name                                                                                                                                                  | Classification                         |
|---------------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| HNM-C-EARLY01       | early01       | Hruday matched finite Maxwell–Dirac closure and continuation                                                                                                         | application of established mathematics |
| HNM-C-EARLY02       | early02       | Hruday complete causal tangent and stationary second variation                                                                                                       | scoped derivation                      |
| HNM-C-EARLY03       | early03       | Hruday finite-window coefficient certificate and cofinal matching obstruction                                                                                        | scoped obstruction or negative control |
| HNM-C-EARLY04       | early04       | Hruday anisotropic gravitational and regulated Dirac interface                                                                                                       | scoped derivation                      |
| HNM-C-EARLY05       | early05       | Hruday action-derived scalar coupling and reference-subtraction taxonomy                                                                                             | scoped derivation                      |
| HNM-C-EARLY06       | early06       | Hruday positive spectral-measure diagnostics and conditional gap bridges                                                                                             | application of established mathematics |
| HNM-C-EARLY07       | early07       | Hruday finite Haar hierarchy characterization and Wilson identities                                                                                                  | scoped derivation                      |
| HNM-C-EARLY08       | early08       | Hruday exact central-convolution spectrum and continuum-time rotor limit                                                                                             | scoped computation or certificate      |
| HNM-C-EARLY09       | early09       | Hruday product-L1 operator-norm counterexample and transfer stability                                                                                                | scoped obstruction or negative control |
| HNM-C-EARLY10       | early10       | Hruday one-square full-space spectral enclosure and coupling cover                                                                                                   | application of established mathematics |
| HNM-C-EARLY11       | early11       | Hruday three-trace quotient, induced operator and complete electric shells                                                                                           | scoped derivation                      |
| HNM-C-EARLY12       | early12       | Hruday interacting two-square certificates and global heat comparison                                                                                                | scoped computation or certificate      |
| HNM-C-EARLY13       | early13       | Hruday volume, measure, clock and nontriviality obstructions                                                                                                         | scoped obstruction or negative control |
| HNM-C-EARLY14       | early14       | Hruday factorial representation error and certified finite numerical state                                                                                           | scoped computation or certificate      |
| HNM-C-EARLY15       | early15       | Hruday compact hierarchy fluctuation and formula audits                                                                                                              | scoped derivation                      |
| HNM-C-EARLY16       | early16       | Hruday finite obligation planner and historical admission repairs                                                                                                    | scoped derivation                      |
| HNM-C-EARLY17       | early17       | Hruday inherited context and logical gravity controls                                                                                                                | scoped obstruction or negative control |
| HNM-C-R13-LOCALITY  | R13-locality  | Hruday local finite-time volume limit                                                                                                                                | scoped derivation                      |
| HNM-C-R20-G1        | R20-g1        | Hruday literal-box real-time dynamics converges in operator norm on each local observable at fixed physical time, with a strongly continuous Hilbert implementation. | scoped derivation                      |
| HNM-C-R13-STABILITY | R13-stability | Hruday qualitative homogeneous stability                                                                                                                             | scoped derivation                      |
| HNM-C-R17-A1        | R17-a1        | Hruday exact local projector and incidence bounds; physical counterexample to the full pure relative-form shortcut. Dense uniform threshold remains open.            | scoped obstruction or negative control |

| HNM ID             | Legacy ID    | Hruday project name                                                                                                                                                                                   | Classification                         |
|--------------------|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| HNM-C-R21-I1       | R21-i1       | Hruday complete 24-link blocking gives 21 omitted anchor faces and exact normalized local interaction norm $7 \tau $ ; qualitative fixed-spacing stability applies with unevaluated source constants. | scoped derivation                      |
| HNM-C-R21-I2       | R21-i2       | Hruday exact one-star vacuum rotation cancels first-order vacuum mixing and retains a bounded quadratic remainder; a bare pure relative bound for the original interaction fails.                     | scoped derivation                      |
| HNM-C-R22-O1       | R22-o1       | Hruday complete simultaneous homogeneous rotation has explicit weighted remainder with preserved unbounded operator domain.                                                                           | scoped derivation                      |
| HNM-C-R13-MOMENTS  | R13-moments  | Hruday compact moment certificates and uniqueness                                                                                                                                                     | scoped computation or certificate      |
| HNM-C-R13-RESPONSE | R13-response | Hruday exact scalar response                                                                                                                                                                          | scoped derivation                      |
| HNM-C-R14-LOOP1    | R14-loop1    | Hruday exact conditional Haar reduction with removable zero field.                                                                                                                                    | scoped derivation                      |
| HNM-C-R14-LOOP2    | R14-loop2    | Hruday 23 exact finite-measure covariance enclosures independently replayed using characters.                                                                                                         | scoped derivation                      |
| HNM-C-R15-A1       | R15-A1       | Hruday all8 positive exact midpoints, all8 coarse transported lower bounds insufficient.                                                                                                              | scoped obstruction or negative control |
| HNM-C-R15-A2       | R15-A2       | Hruday refined failing cell radii rather than increasing already sufficient point precision.                                                                                                          | scoped derivation                      |
| HNM-C-R15-B1       | R15-B1       | Hruday closed cube has six-way dependence while any five face class observables factorize; use actual character gluing and distinct face derivatives in B2.                                           | scoped derivation                      |
| HNM-C-R15-B2       | R15-B2       | Hruday character dimension and separate face derivatives retained; coarse degree failures remain explicit.                                                                                            | scoped derivation                      |
| HNM-C-R16-LOOP1    | R16-loop1    | Hruday exact finite two-cube Haar/fusion identities and separate conditional physical energy-scale audit.                                                                                             | scoped derivation                      |
| HNM-C-R16-LOOP2    | R16-loop2    | Hruday positive normalized all-eleven observable difference for outer $1/8$ and shared $1/8$ versus 0, exact width $\leq 1e-12$ .                                                                     | scoped derivation                      |
| HNM-C-R15-C1       | R15-C1       | Hruday cube lower bound is zero at common $\lambda=\alpha/2$ ; accepted B1 trial moments permit a variational ground-energy upper bound.                                                              | scoped derivation                      |
| HNM-C-R15-C2       | R15-C2       | Hruday the independent E1 lower bound and trial E0 upper bound repair the C1 zero-margin fixture; no trial gap was substituted for the full gap.                                                      | scoped derivation                      |
| HNM-C-R17-B1       | R17-b1       | Hruday improved full physical gap lower bound on the dense eleven-face two-cube graph, positive for common $ \lambda /\alpha < 12/43$ ; finite volume only.                                           | scoped derivation                      |
| HNM-C-R17-B2       | R17-b2       | Hruday exact positive full physical gap lower $\alpha \cdot 1388/284767457$ at the dense finite coupling $\lambda/\alpha=12/43$ , with a proved open variational-amplitude interval.                  | scoped derivation                      |

| HNM ID       | Legacy ID | Hruday project name                                                                                                                                                                                                                                                                                                                                                                                                                                    | Classification                         |
|--------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| HNM-C-R18-B1 | R18-b1    | Hruday complete physical complement threshold $9\alpha/2$ and exact QVP Gram on the dense two-cube graph; no finite trial spectrum substituted for the entire omitted space.                                                                                                                                                                                                                                                                           | scoped derivation                      |
| HNM-C-R18-B2 | R18-b2    | Hruday full untruncated physical two-cube signed coefficient-box gap lower bound, separate endpoint ground trial, and bare-coupling domain obstruction. No volume-uniform or continuum claim.                                                                                                                                                                                                                                                          | scoped obstruction or negative control |
| HNM-C-R19-B1 | R19-b1    | Hruday goal B Loop 1 only: complete strict-cutoff physical projector and exact full cross Gram on the actual finite two-cube $SU(2)$ graph. No continuous coefficient box, dense limit, continuum Yang-Mills or mass-gap claim is accepted here.                                                                                                                                                                                                       | scoped derivation                      |
| HNM-C-R19-B2 | R19-b2    | Hruday goal B Loop 2 only: finite two-cube continuous signed coefficient-box gap certificate using the accepted B1 48-channel projector, full compressed PVP ledger and exact ordered $W^*W$ cross Gram. No dense limit, continuum Yang-Mills, Clay mass gap or physical-scale matching claim is accepted.                                                                                                                                             | scoped computation or certificate      |
| HNM-C-R17-A2 | R17-a2    | Hruday volume-uniform physical gap for the pairwise link-disjoint magnetic-support family with a common positive energy scale; dense overlapping family remains open.                                                                                                                                                                                                                                                                                  | scoped derivation                      |
| HNM-C-R18-A1 | R18-a1    | Hruday finite dressed three-square bridge; physical gap at least $\alpha/8$ throughout the declared primary family; no homogeneous volume extension.                                                                                                                                                                                                                                                                                                   | scoped derivation                      |
| HNM-C-R18-A2 | R18-a2    | Hruday all finite volumes of the specified spatially decaying inhomogeneous family have physical gap at least $7\alpha_{\min}/64$ ; boundary reclassification and homogeneous dense/thermodynamic limits remain explicit.                                                                                                                                                                                                                              | scoped derivation                      |
| HNM-C-R19-A1 | R19-a1    | Hruday for every reachable selected clipped component $L$ , $M$ , $R$ , $LM$ , $MR$ or $LMR$ under the frozen coefficient bounds $ \lambda_L /\alpha \leq 1/2$ , $ \mu_M /\alpha \leq 1/8$ and $ \lambda_R /\alpha \leq 1/2$ , the full-link local gap lower estimate is at least $\alpha/8$ . Positive full-link uniqueness and gauge invariance permit the same lower estimate after Gauss-sector restriction for the finite local component result. | scoped derivation                      |
| HNM-C-R19-A2 | R19-a2    | Hruday goal A Loop 2 only: direct incomplete product-representation construction for a summable omitted-face perturbation of the accepted A1 complete-strip/free reference. This is not finite clipped-restriction convergence, homogeneous dense stability, continuum Yang-Mills or a mass-gap theorem.                                                                                                                                               | scoped derivation                      |
| HNM-C-R20-D1 | R20-d1    | Hruday finite-factor completion gives norm-resolvent convergence with explicit dyadic tail on the exact A2 exterior-reference lift.                                                                                                                                                                                                                                                                                                                    | scoped derivation                      |

| HNM ID       | Legacy ID | Hruday project name                                                                                                                                                                                                                                                                                                                                                       | Classification    |
|--------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| HNM-C-R20-D2 | R20-d2    | Hruday ground energy, rank-one projections and bounded observables converge with explicit dyadic errors for D1 lifts.                                                                                                                                                                                                                                                     | scoped derivation |
| HNM-C-R20-E1 | R20-e1    | Hruday literal aligned boxes $N=4m+3$ converge after explicit reference-energy subtraction and exact exterior-reference lift.                                                                                                                                                                                                                                             | scoped derivation |
| HNM-C-R20-E2 | R20-e2    | Hruday all upper-boundary phases of literal growing rectangular boxes converge to the A2 ground state on bounded local observables.                                                                                                                                                                                                                                       | scoped derivation |
| HNM-C-R17-C1 | R17-c1    | Hruday complete four-adjoint Haar tensor and realized four-cube conditional boundary; no bulk or spectral claim.                                                                                                                                                                                                                                                          | scoped derivation |
| HNM-C-R17-C2 | R17-c2    | Hruday same-action conditional four-face expectations with complete rational numerator and partition errors; action-only observable closure rejected, full bulk and spectrum open.                                                                                                                                                                                        | scoped derivation |
| HNM-C-R18-C1 | R18-c1    | Hruday complete joint-Gram sufficiency for the declared central S3 observable/action class, exact rational admissibility including singular cases, and independently matched degree0–6 diagnostics. No minimality, finite-t tail, surrounding measure or mass-gap conclusion.                                                                                             | scoped derivation |
| HNM-C-R18-C2 | R18-c2    | Hruday full normalized two-link conditional six-face integral at the declared signed fixtures, exact tails, actual surrounding Gram/measure connection and preserved omission/symmetry controls. Thirty-one links remain fixed; no physical spectrum or full bulk/continuum claim.                                                                                        | scoped derivation |
| HNM-C-R19-C1 | R19-c1    | Hruday goal C Loop 1 only: exact finite static U-V-W chain integral on the actual four-cube graph with all other links fixed. This accepts a static mathematical checkpoint and common-V failed-independence discriminator; it does not accept physical energy/time-scale matching, dense convergence, continuum Yang-Mills, a spectral gap theorem or the Clay mass gap. | scoped derivation |
| HNM-C-R19-C2 | R19-c2    | Hruday static common-V integral only; physical energy/time matching remains open. No Hamiltonian spectral gap or continuum claim.                                                                                                                                                                                                                                         | scoped derivation |
| HNM-C-R20-F1 | R20-f1    | Hruday even among local reversible diffusions the C1 static law does not identify an energy/time coefficient: $H_{\text{kappa},c}=c A_{\text{kappa}}$ .                                                                                                                                                                                                                   | scoped derivation |
| HNM-C-R20-F2 | R20-f2    | Hruday exact shared-middle gradient closure, action range $[-5,7]$ and conditional diffusion gap $c/6$ on $ \text{kappa}  \leq 1/8$ .                                                                                                                                                                                                                                     | scoped derivation |
| HNM-C-R21-K1 | R21-k1    | Hruday a positive state-dependent mobility preserves the static density but changes reversible dynamics; two chosen Haar correlation slopes identify $c$ and $\zeta$ within this conditional family.                                                                                                                                                                      | scoped derivation |
| HNM-C-R21-K2 | R21-k2    | Hruday a tunable observable gives a uniform two-rate inverse across the declared static interval, while a positive blind cubic proves that two slopes do not identify arbitrary mobility.                                                                                                                                                                                 | scoped derivation |

| HNM ID       | Legacy ID | Hruday project name                                                                                                                                                                       | Classification                    |
|--------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------|
| HNM-C-R21-L1 | R21-l1    | Hruday a third observable detects the previously hidden cubic mobility and gives an exact three-parameter inverse in the declared conditional family.                                     | scoped derivation                 |
| HNM-C-R21-L2 | R21-l2    | Hruday any finite list of initial correlation slopes leaves strictly positive mobility freedom; a second derivative detects the explicit hidden fifth-degree direction.                   | scoped derivation                 |
| HNM-C-R20-G2 | R20-g2    | Hruday the E2 limiting local state GNS on the full factor algebra is the A2 representation with canonical generator H-e and its inherited gap.                                            | scoped derivation                 |
| HNM-C-R21-J1 | R21-j1    | Hruday an actual omitted Wilson loop has a nonzero gauge-invariant fluctuation in the perturbed dyadic vacuum, with a uniform rational variance floor.                                    | scoped derivation                 |
| HNM-C-R21-J2 | R21-j2    | Hruday physical vacuum GNS space equals the gauge-invariant subspace of this representation, with the restricted shifted energy generator and a nonzero imaginary-time Wilson correlator. | scoped derivation                 |
| HNM-C-R20-H1 | R20-h1    | Hruday the continuous omitted-face profile has an exact budget and a rigorously delimited sufficient gap certificate at fixed physical scale.                                             | scoped computation or certificate |
| HNM-C-R20-H2 | R20-h2    | Hruday every strict uniform global perturbation-budget ray returns the actual ground projection to the selected-strip reference, with an exact vacuum variance and a quantitative rate.   | scoped derivation                 |
| HNM-C-R21-M1 | R21-m1    | Hruday the fixed-budget summable perturbation converges strongly to zero, giving strong-resolvent and compact-time strong dynamics convergence to the selected-strip reference.           | scoped derivation                 |
| HNM-C-R21-M2 | R21-m2    | Hruday connected physical correlations converge on explicitly growing time windows in the canonical summable profile, with a nonzero actual physical Wilson fluctuation.                  | scoped derivation                 |
| HNM-C-R22-N1 | R22-n1    | Hruday stationary Heisenberg comparison proves the sharper local connected-correlation bound in the canonical summable model.                                                             | scoped derivation                 |
| HNM-C-R22-N2 | R22-n2    | Hruday complete tail-block cubes have an exact weighted incident-face budget, yielding a joint support/time convergence certificate.                                                      | scoped computation or certificate |
| HNM-C-R22-O2 | R22-o2    | Hruday the bare-reference elimination has explicit all-support bounds, but its retained-diagonal loss majorants cannot certify an infinite iteration at nonzero coupling.                 | scoped derivation                 |
| HNM-C-R22-R1 | R22-r1    | Hruday the actual initial diagonal has a uniform local inverse, while its local homological generator leaves an exact nonzero SU2 crossing-boundary defect.                               | scoped derivation                 |
| HNM-C-R22-R2 | R22-r2    | Hruday the actual initial boundary-complete vacuum response converges with explicit spatial and graph tails; its source-sector and specified volume-weight limits are exact.              | scoped derivation                 |

| HNM ID         | Legacy ID | Hruday project name                                                                                                                                                           | Classification                         |
|----------------|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|
| HNM-C-R22-P1   | R22-p1    | Hruday the prescribed section is nonclosable and its fitted clock fails the reserved correlation.                                                                             | scoped derivation                      |
| HNM-C-R22-P2   | R22-p2    | Hruday a genuine Haar reduction exactly transports the finite electric dynamics; the declared conditional family fails the reserved data.                                     | scoped derivation                      |
| HNM-C-R22-Q1   | R22-q1    | Hruday the full magnetic action has correlated positive conditional variance, proving selected-image leakage and a quantitative nonautonomous dynamic defect.                 | scoped derivation                      |
| HNM-C-R22-Q2   | R22-q2    | Hruday the actual magnetic projection has exact memory/resolvent identities, seven-energy fixed-channel leading matrices and explicit cubic full-Hilbert operator remainders. | scoped derivation                      |
| HNM-C-RECENT01 | recent01  | Hruday nonzero indexed cubic source                                                                                                                                           | scoped derivation                      |
| HNM-C-RECENT02 | recent02  | Hruday interior inverse and complete connected crossing defect                                                                                                                | scoped derivation                      |
| HNM-C-RECENT03 | recent03  | Hruday compact-time graph-domain commutator with retained sinc residual                                                                                                       | scoped derivation                      |
| HNM-C-RECENT04 | recent04  | Hruday connected weighted bounds and infinite initial-operator identification                                                                                                 | scoped derivation                      |
| HNM-C-RECENT05 | recent05  | Hruday the short-time residual cannot be made small at the admitted cap                                                                                                       | scoped obstruction or negative control |
| HNM-C-RECENT06 | recent06  | Hruday iterated filter converges on the vacuum column but retains resonant blocks                                                                                             | scoped derivation                      |
| HNM-C-RECENT07 | recent07  | Hruday the resonant block and spectral windows are boundary controlled                                                                                                        | scoped derivation                      |
| HNM-C-RECENT08 | recent08  | Hruday gaussian boundary identity gives strict operator residual contraction                                                                                                  | scoped derivation                      |
| HNM-C-RECENT09 | recent09  | Hruday gaussian collar locality and graph-domain truncation error                                                                                                             | scoped derivation                      |
| HNM-C-RECENT10 | recent10  | Hruday compact residual contraction and original weighted finiteness                                                                                                          | scoped derivation                      |
| HNM-C-RECENT11 | recent11  | Hruday full-space leading-memory return model and unshifted heat error                                                                                                        | scoped derivation                      |
| HNM-C-RECENT12 | recent12  | Hruday separately centered full-selected heat remains uniformly close                                                                                                         | scoped derivation                      |
| HNM-C-RECENT13 | recent13  | Hruday electric-energy cutoff and isolated-ground truncation theorem                                                                                                          | scoped derivation                      |
| HNM-C-RECENT14 | recent14  | Hruday representation-degree path truncation and the assembled 21-state seed                                                                                                  | scoped computation or certificate      |
| HNM-C-RECENT15 | recent15  | Hruday complete residual of the 21-state ground trial                                                                                                                         | scoped computation or certificate      |
| HNM-C-RECENT16 | recent16  | Hruday two-sided ground enclosure and evaluated late heat                                                                                                                     | scoped computation or certificate      |
| HNM-C-RECENT17 | recent17  | Hruday uniform relative heat accuracy fails on ground-null inputs                                                                                                             | scoped obstruction or negative control |
| HNM-C-RECENT18 | recent18  | Hruday exact omitted Gram matrix and all-time relative repair for prepared inputs                                                                                             | scoped computation or certificate      |
| HNM-C-RECENT19 | recent19  | Hruday the 293-state physical channel enlargement                                                                                                                             | scoped computation or certificate      |

| HNM ID                | Legacy ID       | Hrunday project name                                                              | Classification                         |
|-----------------------|-----------------|-----------------------------------------------------------------------------------|----------------------------------------|
| HNM-C-RECENT20        | recent20        | Hrunday improved ground certificate does not yet prove an improved heat bound     | scoped obstruction or negative control |
| HNM-C-RECENT21        | recent21        | Hrunday delayed channel loading closes the improved all-time relative bound       | scoped computation or certificate      |
| HNM-C-RECENT22        | recent22        | Hrunday actual 293-coordinate heat at one declared time                           | scoped computation or certificate      |
| HNM-C-RECENT23        | recent23        | Hrunday piecewise all-time computed heat and the fixed numerical scope            | scoped computation or certificate      |
| HNM-C-RECENT24        | recent24        | Hrunday canonical rank observable has an actual eight-link first-order resonance  | scoped derivation                      |
| HNM-C-RECENT25        | recent25        | Hrunday full-system endpoint witness from connected-word tails                    | scoped obstruction or negative control |
| HNM-C-RECENT26        | recent26        | Hrunday finite-outcome and coherent readout of the complex correlation            | scoped derivation                      |
| HNM-C-RECENT27        | recent27        | Hrunday full-Haar operator-norm obstruction to multiplying-observable replacement | scoped obstruction or negative control |
| HNM-C-RECENT28        | recent28        | Hrunday clock and finite-sample error budget for the coherent protocol            | scoped computation or certificate      |
| HNM-C-RECENT29        | recent29        | Hrunday complete-factor spatial collars with factorial connected tails            | scoped computation or certificate      |
| HNM-C-RECENT30        | recent30        | Hrunday actual scalar endpoint limit from strong block averaging                  | scoped derivation                      |
| HNM-C-RECENT31        | recent31        | Hrunday exact sixteen-cycle compression and spectral measure                      | scoped computation or certificate      |
| HNM-C-RECENT32        | recent32        | Hrunday exterior selection proves the reached component reduces                   | scoped derivation                      |
| HNM-C-RECENT33        | recent33        | Hrunday finite-q approach and numerical endpoint enclosure                        | scoped computation or certificate      |
| HNM-C-RECENT34        | recent34        | Hrunday nonzero-variance elementary Wilson probe is endpoint-null                 | scoped obstruction or negative control |
| HNM-C-RECENT35        | recent35        | Hrunday informative single-cycle and sum Wilson endpoints                         | scoped derivation                      |
| HNM-C-R27-AI1-FIBERS  | R27-AI1-FIBERS  | Hrunday exact readout fibers                                                      | scoped derivation                      |
| HNM-C-R27-AI1-CARRIER | R27-AI1-CARRIER | Hrunday carrier and slow-rate identification                                      | scoped derivation                      |
| HNM-C-R27-AI2-SUPPORT | R27-AI2-SUPPORT | Hrunday complete physical multiplier support                                      | scoped computation or certificate      |
| HNM-C-R27-AI2-DISKS   | R27-AI2-DISKS   | Hrunday complete scalar prediction-disk separation                                | scoped computation or certificate      |
| HNM-C-R27-AG1         | R27-AG1         | Hrunday one selected-source homogeneous correction                                | application of established mathematics |
| HNM-C-AG2             | AG2             | Hrunday complete original-remainder transport                                     | scoped computation or certificate      |
| HNM-C-AG3             | AG3             | Hrunday one complete-source homogeneous correction                                | scoped computation or certificate      |
| HNM-C-AI3             | AI3             | Hrunday odd-moment and reflection-tail certificate                                | scoped computation or certificate      |

| HNM ID    | Legacy ID | Hrunday project name                                                      | Classification                         |
|-----------|-----------|---------------------------------------------------------------------------|----------------------------------------|
| HNM-C-AI4 | AI4       | Hrunday uniform conditional physical ratio separation                     | scoped computation or certificate      |
| HNM-C-AH1 | AH1       | Hrunday 561-state physical graph enrichment                               | scoped computation or certificate      |
| HNM-C-AH2 | AH2       | Hrunday delayed-loading heat certificate                                  | scoped computation or certificate      |
| HNM-C-AJ1 | AJ1       | Hrunday locally normal physical representation                            | application of established mathematics |
| HNM-C-AJ2 | AJ2       | Hrunday nonzero original Wilson fluctuation                               | application of established mathematics |
| HNM-C-AK1 | AK1       | Hrunday conditional Wilson variance floor                                 | application of established mathematics |
| HNM-C-AK2 | AK2       | Hrunday wilson form-energy, spectral-window and heat-envelope certificate | application of established mathematics |
| HNM-C-AL1 | AL1       | Hrunday full-interaction matching audit                                   | scoped obstruction or negative control |
| HNM-C-AL2 | AL2       | Hrunday scale-invariant eligibility and deformation bound                 | scoped obstruction or negative control |
| HNM-C-AM1 | AM1       | Hrunday complete-model numerical-certificate transfer audit               | scoped obstruction or negative control |
| HNM-C-AM2 | AM2       | Hrunday four-site quantitative stability certificate                      | scoped derivation                      |
| HNM-C-AN1 | AN1       | Hrunday common-bulk boundary comparison                                   | application of established mathematics |
| HNM-C-AN2 | AN2       | Hrunday ordered bulk-state identification                                 | application of established mathematics |
| HNM-C-AO1 | AO1       | Hrunday finite Wilson second-moment bound                                 | scoped derivation                      |
| HNM-C-AO2 | AO2       | Hrunday limiting Wilson energy identity                                   | application of established mathematics |
| HNM-C-AQ1 | AQ1       | Hrunday numerical-cap thermodynamic construction                          | application of established mathematics |
| HNM-C-AQ2 | AQ2       | Hrunday numerical-cap physical gap and Wilson witness                     | application of established mathematics |

## B.2 Scoped statement aliases

These semantic names index admitted claims, not newly invented axioms. Each linked chapter supplies the complete hypotheses, proof record, source attribution and final limitations. The theorem/proposition naming does not imply external peer review.

| Statement alias | Type / chapter                         | Named scoped statement                                   |
|-----------------|----------------------------------------|----------------------------------------------------------|
| HNM-P-AL1       | scoped proposition; Section 40         | Hruday full-interaction matching audit                   |
| HNM-P-AL2       | scoped proposition; Section 41         | Hruday scale-invariant eligibility and deformation bound |
| HNM-T-AM2       | scoped theorem; Section 43             | Hruday four-site quantitative stability certificate      |
| HNM-P-AN1       | source-theorem application; Section 44 | Hruday common-bulk boundary comparison                   |
| HNM-T-AN2       | source-theorem application; Section 45 | Hruday ordered bulk-state identification                 |
| HNM-P-AO1       | scoped proposition; Section 46         | Hruday finite Wilson second-moment bound                 |
| HNM-P-AO2       | scoped proposition; Section 47         | Hruday limiting Wilson energy identity                   |
| HNM-T-AQ1       | source-theorem application; Section 48 | Hruday numerical-cap thermodynamic construction          |
| HNM-T-AQ2       | source-theorem application; Section 49 | Hruday numerical-cap physical gap and Wilson witness     |

### B.3 Named quantity aliases

| HNM quantity               | Retained notation / units          | Meaning and limit                                                                                                              |
|----------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| HNM-Q-ENERGY-REFERENCE     | $E_*$ ; energy                     | Hruday registry: fixed positive physical energy reference. Reference, not a fitted constant.                                   |
| HNM-Q-KINETIC-SCALE        | $\alpha$ ; energy                  | Hruday registry: declared kinetic energy scale. Model-local scale; no continuum identification supplied.                       |
| HNM-Q-ACTION               | $\hbar$ ; action                   | Hruday registry: fixed action reference. Established physical constant, not a project invention.                               |
| HNM-Q-HOMOGENEOUS-UNIT     | $\delta = \alpha/8$ ; energy       | Hruday registry: homogeneous conversion unit. Inherited normalization, not a new fundamental constant.                         |
| HNM-Q-HOMOGENEOUS-COUPLING | $\tau$ ; dimensionless             | Hruday registry: homogeneous omitted-interaction coupling. Admissible symbolic interval also required.                         |
| HNM-Q-STABILITY-RADIUS     | $\tau_*$ ; dimensionless           | Hruday registry: unevaluated homogeneous stability radius. Positive theorem parameter; not yet a certified numerical interval. |
| HNM-Q-CANONICAL-PROFILE    | $q$ ; dimensionless                | Hruday registry: canonical summable profile parameter. Not identified with lattice spacing.                                    |
| HNM-Q-CANONICAL-STRENGTH   | $\eta$ ; dimensionless             | Hruday registry: canonical coupling strength. Not automatically a homogeneous coupling.                                        |
| HNM-Q-PROFILE-DEFECT       | $\epsilon = 1 - q$ ; dimensionless | Hruday registry: canonical profile deficit. Separate from a continuum regulator.                                               |

| HNМ quantity               | Retained notation / units                             | Meaning and limit                                                                                                                                                  |
|----------------------------|-------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HNМ-Q-SLOW-COMPOSITE       | $\rho = \eta(1 - q)^3$ ; dimensionless                | Hruday registry: canonical slow composite. Readout information is model and protocol dependent.                                                                    |
| HNМ-Q-CARRIER-RATE         | $\beta = \alpha/\hbar$ ; inverse time                 | Hruday registry: carrier scale. Requires calibrated physical time and action.                                                                                      |
| HNМ-Q-SLOW-RATE            | $\lambda = \beta\rho/84$ ; inverse time               | Hruday registry: surrogate slow rate. This lambda is distinct from couplings reused in other chapters.                                                             |
| HNМ-Q-WILSON-WITNESS       | $\chi = (\pi(W) - \omega(W)I)\Omega$ ; Hilbert vector | Hruday registry: centered physical Wilson vector. Specified origin plaquette and conditional orthant state.                                                        |
| HNМ-Q-WILSON-VARIANCE      | $v = \ \chi\ ^2$ ; dimensionless                      | Hruday registry: Wilson fluctuation mass. Variance, not number of particles.                                                                                       |
| HNМ-Q-WILSON-MEASURE       | $\nu$ ; dimensionless mass                            | Hruday registry: Wilson spectral measure. Positive actual state-dependent measure, not an evaluated spectrum.                                                      |
| HNМ-Q-FIRST-MOMENT         | $\mu_1 = \int E d\nu(E)$ ; energy                     | Hruday registry: Wilson first energy moment. Upper bound passes to limit; equality needs additional control.                                                       |
| HNМ-Q-HEAT-READOUT         | $C(t)$ ; dimensionless                                | Hruday registry: physical imaginary-time Wilson correlation. Not a real-time signal or measured mass.                                                              |
| HNМ-Q-HEAT-CLOCK           | $\sigma = \alpha t/\hbar$ ; dimensionless             | Hruday registry: finite-graph heat coordinate. Each finite and full heat uses its correct ground center.                                                           |
| HNМ-Q-SUPPORT-WEIGHT       | $a$ ; dimensionless                                   | Hruday registry: rooted support-cardinality proof weight. Symbol $a$ elsewhere denotes spacing; meaning is local to model.                                         |
| HNМ-Q-MATCHING-RATIO       | $r = \lambda/\alpha = 4/g^4$ ; dimensionless          | Hruday registry: full uniform magnetic/electric matching ratio. AL1 convention; no new field and no proof of a continuum trajectory.                               |
| HNМ-Q-COMMON-SCALE         | $k > 0$ ; dimensionless                               | Hruday registry: common positive energy multiplier. AL2: preserves eligibility ratios but changes physical gaps at fixed units.                                    |
| HNМ-Q-MAGNETIC-DEFORMATION | $s \geq 0$ ; dimensionless                            | Hruday registry: independent magnetic multiplier. AL2: changing it with electric scale fixed changes the action; it is not a units conversion.                     |
| HNМ-Q-FOUR-SITE-MAJORANT   | $G(t) = 16e^{8t}(1 + 10t)$ ; dimensionless            | Hruday registry: four-site creation majorant. AM2 proof bound, not a physical generator or invented law.                                                           |
| HNМ-Q-CREATION-RADIUS      | $R = 1/64$ ; dimensionless                            | Hruday registry: creation coefficient-ball radius. AM2 proof parameter; no optimality claim.                                                                       |
| HNМ-Q-GROUPED-SITE-BUDGET  | $J \leq 28 \tau $ ; dimensionless normalized norm     | Hruday registry: complete indexed interaction budget. Includes all four incident whole-star anchors; not a global extensive operator norm.                         |
| HNМ-Q-FINITE-VOLUME-CAP    | $ \tau  \leq 10^{-8}$ ; dimensionless                 | Hruday registry: explicit finite-volume coupling cap. AM2 full untruncated finite-volume theorem only; no numerical infinite-state interval follows automatically. |

| HNM quantity                     | Retained notation / units                                               | Meaning and limit                                                                                                                                                                                 |
|----------------------------------|-------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HNM-Q-CANDIDATE-SCALE-PATH       | $(a_n, g_n^2) = (a_0/n, 1/n)$ ; length and dimensionless coupling       | Hruday registry: prospectively frozen weak-bare-coupling test path. AL1 algebraic candidate only; not a derived physical renormalization trajectory.                                              |
| HNM-Q-HOMOGENEOUS-LOCAL-NORM     | $\epsilon = 7 \tau $ ; dimensionless                                    | Hruday registry: complete one-anchor normalized omitted norm. I1/AL context; distinct from the canonical profile deficit also denoted epsilon.                                                    |
| HNM-Q-SOURCE-CONSTANTS           | $c_1(S), c_2(S)$ ; dimensionless                                        | Hruday registry: inherited range-dependent stability constants. Yarotsky source parameters keep their external attribution and remain unevaluated by AM2.                                         |
| HNM-Q-ONE-FACTOR-WITNESS         | $v_b = (W_p - m_p)\Omega_b$ ; Hilbert vector                            | Hruday registry: actual one-coarse-factor physical witness. AM1: no implication of four excited coarse factors and no measured excitation energy.                                                 |
| HNM-Q-FINITE-CREATOR             | $C = \sum_{I \neq \emptyset} \hat{c}_I$ ; dimensionless                 | Hruday registry: finite-volume nilpotent creation operator. AM2 cutoff proof construction; no bounded infinite-volume similarity operator is asserted.                                            |
| HNM-Q-SPECTRAL-SHIFT             | $ z  < 1/2$ ; normalized energy                                         | Hruday registry: centered normalized excitation shift. AM2 exclusion window; distinct from the canonical scaled-time z.                                                                           |
| HNM-Q-INTERPOLATION-PARAMETER    | $ s  \leq 1$ ; dimensionless                                            | Hruday registry: finite-model proof interpolation parameter. AM2 interpolation, distinct from AL2 magnetic suppression despite its reused symbol.                                                 |
| HNM-Q-COMMON-BULK-REGION         | $D_n = [-n + 2, n - 2]^3$ ; coarse lattice coordinates                  | Hruday registry: twice-eroded comparison bulk. AN1 exact boundary dictionary; physical lattice spacing is unchanged.                                                                              |
| HNM-Q-BOUNDARY-COMPARISON-STATES | $\rho_{A_n}, \rho_{B_n}$ ; normalized states                            | Hruday registry: matched finite-box normal ground states. AN1 compares actual whole-star models with the same coarse-translation-invariant selected coefficients.                                 |
| HNM-Q-HTW-SMALLNESS              | $7 \tau  \leq c_{\text{HTW}}(1, 1)$ ; dimensionless                     | Hruday registry: independent locality smallness condition. HTW source condition remains unevaluated and is not supplied by AM2.                                                                   |
| HNM-Q-HTW-DECAY                  | $\min\{2, e^{2C_1 - C_2(n-2)}\}$ ; dimensionless norm bound             | Hruday registry: conditional local comparison enclosure. AN1: source decay constants remain symbolic; this is not a simulated expectation or an assertion about near-boundary equality.           |
| HNM-Q-INDEPENDENT-CUTOFFS        | $O_{n,L} = [-n, L - n]^3, C_M = [-M, M]^3$ ; coarse lattice coordinates | Hruday registry: independent orthant and centered cutoff boxes. AN2 requires every $L \geq 2n$ and $M \geq n$ before the ordered state limits; a diagonal sequence alone does not suffice.        |
| HNM-Q-BULK-LOCAL-STATE           | $\omega_{\mathbb{Z}}(A) = \text{Tr}(\rho_F A)$ ; normalized state       | Hruday registry: compatible locally normal centered-box state. AN2 identifies this specified state under both inherited I1 and independent HTW smallness; equality of generators does not follow. |

| HNM quantity                    | Retained notation / units                                                              | Meaning and limit                                                                                                                                                                                                                          |
|---------------------------------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HNM-Q-BULK-STATE-ENCLOSURE      | $\varepsilon_n(F)$<br>$\min\{2, e^{C_1 F -C_2(n-r-1)}\};$<br>dimensionless trace norm  | =<br>Hruday registry: ordered local-state identification enclosure. AN2 on a fixed finite complete-factor region F inside $[-r,r]^3$ ; C1 and C2 remain unevaluated source constants.                                                      |
| HNM-Q-INHERITED-STATE-CAP       | $ \tau  < \tau_*$ ; dimensionless                                                      | Hruday registry: inherited orthant-state sufficient smallness. I1 source-dependent threshold remains distinct from the explicit AM2 finite-volume cap and the additional HTW restriction.                                                  |
| HNM-Q-WILSON-SECOND-MOMENT      | $\mu_{2,\Lambda} = \int E^2 d\nu_\Lambda(E)$ ; energy squared                          | Hruday registry: actual finite physical Wilson second moment. AO1 proves the uniform bound $\alpha^2 \times (36+28 \tau )$ for the original origin cover under the inherited state conditions.                                             |
| HNM-Q-REGIONAL-KINETIC-ENERGY   | $T_R = \alpha \sum_{e \text{ owned by } R} C_e$ ; energy                               | Hruday registry: actual regional electric energy. AO1 reconstructs it from the selected reference, retaining the exact ground scalar before an upper-bound relaxation.                                                                     |
| HNM-Q-WILSON-DERIVATIVE-TERM    | $Z_\Lambda$<br>$\sum_{e \in p,a} (X_{ea} W) X_{ea} \Omega_\Lambda$ ;<br>Hilbert vector | =<br>Hruday registry: Wilson electric derivative cross term. AO1 requires the original half-Pauli normalization; the interacting selected reference is not replaced by a constant Haar vector.                                             |
| HNM-Q-LIMITING-MOMENT-BUDGET    | $B_2 = \alpha^2(36 + 28 \tau )$ ; energy squared                                       | Hruday registry: inherited-state physical second-moment budget. AO2 bounds the limiting second moment; it does not identify it with a limit of finite second moments.                                                                      |
| HNM-Q-UNIFORM-FIRST-ENERGY-TAIL | $\int_{E>L} E d\nu_\Lambda(E) \leq B_2/L$ ; energy                                     | Hruday registry: uniform first-energy tail enclosure. AO2 uses $L > 0$ as a physical spectral cutoff and retains the inherited orthant-state restrictions.                                                                                 |
| HNM-Q-LIMITING-FIRST-ENERGY     | $\langle \chi, H_{\text{phys}} \chi \rangle = \alpha \omega(1 - W^2)$ ; energy         | Hruday registry: actual inherited-state Wilson first-energy identity. AO2 establishes equality for the same I1/AJ1/AK2 state; another thermodynamic construction needs its own state dictionary.                                           |
| HNM-Q-NUMERICAL-CAP-STATE       | $\omega_{\text{num}}$ ; normalized state                                               | Hruday registry: locally normal numerical-cap subsequential state. AQ1 constructs a full coarse-lattice state without the old symbolic thresholds; uniqueness and identification with the earlier orthant/bulk states are not established. |
| HNM-Q-LOCAL-RESET-ENERGY        | $C_F = 56 \tau  F $ ; normalized reference energy                                      | Hruday registry: actual reference reset-energy budget. AQ1 includes all incoming whole-star terms; this is not a Haar or pure-electric replacement.                                                                                        |
| HNM-Q-COMPACT-RESOLVENT-CUTOFF  | $Q_{F,L} = \mathbf{1}_{[0,L]}(h_F)$ ; projection                                       | Hruday registry: finite-rank local reference-energy cutoff. AQ1 combines the actual compact resolvent with a trace-tail bound; L is normalized reference energy, distinct from AO2 physical energy.                                        |

| HNM quantity                      | Retained notation / units                                                   | Meaning and limit                                                                                                                                                                                                             |
|-----------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HNM-Q-DYNAMICS-DECAY-KERNEL       | $F(r) = (1 + r)^{-4}$ ; dimensionless                                       | Hruday registry: admissible lattice decay kernel. The Nachtergaele–Sims source framework retains its authorship; this choice only supplies the model dictionary.                                                              |
| HNM-Q-DYNAMICS-INTERACTION-NORM   | $\ \Phi\ _F \leq 2268 \tau $ ; dimensionless normalized norm                | Hruday registry: complete whole-star interaction norm bound. AQ1 norm volume convergence is not norm continuity in time on the full local bounded-operator algebra.                                                           |
| HNM-Q-NUMERICAL-CAP-GENERATOR     | $H_{\text{num}} \geq 0$ ; energy                                            | Hruday registry: strongly continuous numerical-cap GNS energy generator. AQ1 proves nonnegativity and a reducing physical cyclic restriction; a positive physical gap requires the separate AQ2 gate.                         |
| HNM-Q-ORIGINAL-ENDPOINT-GROUP     | $G = \prod_{v \in V} \text{SU}(2)$ ; compact group                          | Hruday registry: complete original-endpoint gauge group. AQ2 retains every fine-lattice tail and head; the compact product and Haar construction are established mathematics.                                                 |
| HNM-Q-FULL-PHYSICAL-PROJECTION    | $P_G = \int_G V_g dg$ ; orthogonal projection                               | Hruday registry: complete physical fixed-sector projection. AQ2 uses strong vector integration and the actual locally normal GNS state, without a physical tensor-factor decomposition.                                       |
| HNM-Q-LOCAL-GAUGE-AVERAGE         | $\mathcal{E}_F(A) = \int \beta_g(A) dg$ ; observable                        | Hruday registry: bounded local endpoint-gauge average. AQ2 operator averaging is ultraweak, not an assumed norm-Bochner integral on all bounded operators.                                                                    |
| HNM-Q-NUMERICAL-PHYSICAL-GAP      | $m_{\text{gap}} = \alpha/16$ ; energy                                       | Hruday registry: numerical-cap physical spectral-gap lower bound. AQ2 proves it in the chosen AQ1 representation, with simple vacuum there; this is not uniqueness of all thermodynamic states or a particle-mass evaluation. |
| HNM-Q-SEVEN-STAR-RESET            | $\omega_{\text{num}}(h_R) \leq 98 \tau $ ; normalized reference energy      | Hruday registry: full-lattice seven-star reference reset bound. AQ2 applies to the same original two-factor Wilson cover; the orthant two-star count cannot be substituted.                                                   |
| HNM-Q-NUMERICAL-STATE-TRACE-BOUND | $\ \rho_R - P_R\ _1 \leq 1/500$ ; dimensionless trace norm                  | Hruday registry: same-state reference trace-distance enclosure. AQ2 compares the interacting marginal to its actual selected-reference product at $ \tau  \leq 10^{-8}$ ; it does not call the interacting state Haar.        |
| HNM-Q-NUMERICAL-WILSON-VARIANCE   | $\text{Var}_{\omega_{\text{num}}}(W) \geq 61999/250000$ ; dimensionless     | Hruday registry: common numerical-cap Wilson variance floor. AQ2 makes the physical excitation nonzero in the same state, without importing the older AO2 operator domain or moment identity.                                 |
| HNM-Q-REVERSE-WILSON-VARIANCE     | $\text{Var}_{\omega_{\text{num}}}(W) \geq 3099951/12500000$ ; dimensionless | Hruday registry: separately admitted reverse variance refinement. AQ2 reverse proof and gate support this stronger floor; common synthesis retains the forward floor and explicit attribution.                                |

## B.4 Labelled equation aliases

| Stable equation alias           | Equation in this edition |
|---------------------------------|--------------------------|
| HNM-E-EARLY-BLOCH               | (HNM-1)                  |
| HNM-E-EARLY-MATCHED             | (HNM-2)                  |
| HNM-E-EARLY-WORK                | (HNM-3)                  |
| HNM-E-EARLY-EXISTENCE           | (HNM-4)                  |
| HNM-E-EARLY-WEAKRESPONSE        | (HNM-5)                  |
| HNM-E-EARLY-TANGENT             | (HNM-6)                  |
| HNM-E-EARLY-SECONDVARIATION     | (HNM-7)                  |
| HNM-E-EARLY-WINDOW              | (HNM-8)                  |
| HNM-E-EARLY-RATIONALZ           | (HNM-9)                  |
| HNM-E-EARLY-COFINAL             | (HNM-10)                 |
| HNM-E-EARLY-CONSTRAINT          | (HNM-11)                 |
| HNM-E-EARLY-PRESSURES           | (HNM-12)                 |
| HNM-E-EARLY-LOGICALCONTROLS     | (HNM-13)                 |
| HNM-E-EARLY-SCALAR              | (HNM-14)                 |
| HNM-E-EARLY-EFFECTIVE           | (HNM-15)                 |
| HNM-E-EARLY-DENSEBRIDGE         | (HNM-16)                 |
| HNM-E-EARLY-WARD                | (HNM-17)                 |
| HNM-E-EARLY-CONVOLUTION         | (HNM-18)                 |
| HNM-E-EARLY-NORMCOUNTER         | (HNM-19)                 |
| HNM-E-EARLY-JACOBI              | (HNM-20)                 |
| HNM-E-EARLY-ONETAIL             | (HNM-21)                 |
| HNM-E-EARLY-QUOTIENT            | (HNM-22)                 |
| HNM-E-EARLY-OPERATOR            | (HNM-23)                 |
| HNM-E-EARLY-SHELL               | (HNM-24)                 |
| HNM-E-EARLY-TWOTAIL             | (HNM-25)                 |
| HNM-E-EARLY-MOMENTS             | (HNM-26)                 |
| HNM-E-EARLY-GLOBALGAP           | (HNM-27)                 |
| HNM-E-EARLY-GIBBS               | (HNM-28)                 |
| HNM-E-EARLY-FACTORIAL           | (HNM-29)                 |
| HNM-E-EARLY-MIDPOINT            | (HNM-30)                 |
| HNM-E-EARLY-CLOSURE             | (HNM-31)                 |
| HNM-E-EQ-MIDDLE-MOMENTS         | (HNM-32)                 |
| HNM-E-EQ-MIDDLE-RESPONSE        | (HNM-33)                 |
| HNM-E-EQ-MIDDLE-CUBE            | (HNM-34)                 |
| HNM-E-EQ-MIDDLE-TWO-CUBE        | (HNM-35)                 |
| HNM-E-EQ-MIDDLE-GRAM            | (HNM-36)                 |
| HNM-E-EQ-MIDDLE-STATIC-SIGN     | (HNM-37)                 |
| HNM-E-EQ-MIDDLE-ARROW           | (HNM-38)                 |
| HNM-E-EQ-MIDDLE-48GAP           | (HNM-39)                 |
| HNM-E-EQ-MIDDLE-DYADIC          | (HNM-40)                 |
| HNM-E-EQ-MIDDLE-SUMMABLE-GAP    | (HNM-41)                 |
| HNM-E-EQ-MIDDLE-GNS             | (HNM-42)                 |
| HNM-E-EQ-MIDDLE-LR              | (HNM-43)                 |
| HNM-E-EQ-MIDDLE-PROFILE         | (HNM-44)                 |
| HNM-E-EQ-MIDDLE-VARIANCE        | (HNM-45)                 |
| HNM-E-EQ-MIDDLE-STATIONARY      | (HNM-46)                 |
| HNM-E-EQ-MIDDLE-MOVING          | (HNM-47)                 |
| HNM-E-EQ-MIDDLE-RATES           | (HNM-48)                 |
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| HNM-E-R29-AN1-04            | (HNM-AN1.4)              |
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| HNM-E-R29-AN2-02            | (HNM-AN2.2)              |
| HNM-E-R29-AO1-01            | (HNM-AO1.1)              |
| HNM-E-R29-AO1-02            | (HNM-AO1.2)              |
| HNM-E-R29-AO1-03            | (HNM-AO1.3)              |
| HNM-E-R29-AO1-04            | (HNM-AO1.4)              |
| HNM-E-R29-AO1-05            | (HNM-AO1.5)              |
| HNM-E-R29-AO1-06            | (HNM-AO1.6)              |
| HNM-E-R29-AO2-01            | (HNM-AO2.1)              |
| HNM-E-R29-AO2-02            | (HNM-AO2.2)              |
| HNM-E-R29-AO2-03            | (HNM-AO2.3)              |
| HNM-E-R29-AQ1-01            | (HNM-AQ1.1)              |
| HNM-E-R29-AQ1-02            | (HNM-AQ1.2)              |
| HNM-E-R29-AQ1-03            | (HNM-AQ1.3)              |
| HNM-E-R29-AQ1-04            | (HNM-AQ1.4)              |
| HNM-E-R29-AQ2-01            | (HNM-AQ2.1)              |
| HNM-E-R29-AQ2-02            | (HNM-AQ2.2)              |
| HNM-E-R29-AQ2-03            | (HNM-AQ2.3)              |
| HNM-E-R29-AQ2-04            | (HNM-AQ2.4)              |
| HNM-E-R29-AQ2-05            | (HNM-AQ2.5)              |
| HNM-E-R29-AQ2-06            | (HNM-AQ2.6)              |
| HNM-E-R29-AQ2-07            | (HNM-AQ2.7)              |
| HNM-E-R29-AQ2-08            | (HNM-AQ2.8)              |
| HNM-E-R29-AQ2-09            | (HNM-AQ2.9)              |
| HNM-E-R29-AQ2-10            | (HNM-AQ2.U10)            |
| HNM-E-R29-AQ2-11            | (HNM-AQ2.10)             |
| HNM-E-R29-AQ2-REV-01        | (HNM-AQ2.R1)             |
| HNM-E-R29-CALCULATOR-BUDGET | (HNM-353)                |

## B.5 Round30 contribution and statement aliases

The following three records extend the inherited catalog. Aliases for both directions preserve provenance and do not double the number of contributions. No new axioms are asserted.

**HNМ-C-AT1: Hruday same-state Wilson energy certificate** In the actual numerical-cap AQ state at either sign of  $|\tau| \leq 10^{-8}$ , the centered original Wilson vector satisfies  $\chi \in D(H_{\text{phys}})$ . Its second-energy moment is at most  $\alpha^2(36 + 98|\tau|) < 37\alpha^2$ , and its first-energy moment equals  $\alpha\omega_{\text{num}}(1 - W^2)$ . **Scoped statement alias: HNМ-T-AT1.** The admission and its limits are in Section 54. **Status and scope.** accepted\_within\_scope. No equality of finite-to-infinite second moments No AQ/AO/AN state identification No particle pole, exact mass or dispersion No uniqueness of all thermodynamic states or whole-sequence convergence No continuum Yang-Mills construction Scientific priority unverified; established methods retain attribution

**HNМ-C-AT2: Hruday spectral-window and inverse-energy certificate** For the same state, the unnormalized spectral mass satisfies  $\nu([\alpha/16, 8\alpha]) \geq 307992/1984375$ . On the physical vacuum-orthogonal sector let  $R = \langle \chi, H_{\text{phys}}^{-1} \chi \rangle$ . Its dimensionless value has lower bound  $3843876001/47000000000$  and upper bound  $28462237549/7503125000$ : these are bounds on  $\alpha R$ , not an evaluated response or a unique spectral measure. **Scoped statement alias: HNМ-T-AT2.** The admission and its limits are in Section 57. **Status and scope.** accepted\_within\_scope. The window is unnormalized spectral mass, not probability or a hard cutoff No evaluated actual AQ inverse response or optimality No particle pole, exact mass or identified spectral type Abstract equal-moment fixtures are not actual lattice realizations No static susceptibility or deformed thermodynamic-state differentiability No continuum result; scientific priority unverified

**HNМ-C-AT3: Hruday finite Euclidean readout certificate** The fixed 4097-node protocol conditionally encloses  $I = \alpha R$  when supplied certified centered AQ samples with absolute error at most  $10^{-6}$ . Exact rational data give interval width below  $1/500$ ; arithmetic bins contribute their explicitly bounded trapezoid width. Both executed abstract A/B benchmarks pass, and their full admissible-error output envelopes are disjoint. No actual AQ correlation samples or response value are supplied. **Scoped statement alias: HNМ-T-AT3.** The admission and its limits are in Section 60. **Status and scope.** accepted\_within\_scope. No actual AQ correlation samples or inverse response evaluated Forward permits arbitrary arithmetic bins and reports width verdict; reverse admits bins  $\leq 1e - 12$  and guarantees the width target Input validation does not certify centering, model/state provenance or scientific error bounds A/B and the separately disclosed damaging tail control are abstract measures, not AQ simulations No susceptibility, identified pole, optimal design, state uniqueness or continuum construction Scientific priority unverified; established methods retain attribution Exactly three investigations; no fourth scientific execution

## B.6 Round30 equation concordance

| Stable equation alias | Printed locator             | Frozen derivation                                      |
|-----------------------|-----------------------------|--------------------------------------------------------|
| HNМ-E-R30-AT1-F-01    | <a href="#">HNМ-AT1-F01</a> | <a href="#">research/round30/forward/at1/report.md</a> |
| HNМ-E-R30-AT1-F-02    | <a href="#">HNМ-AT1-F02</a> | <a href="#">research/round30/forward/at1/report.md</a> |
| HNМ-E-R30-AT1-F-03    | <a href="#">HNМ-AT1-F03</a> | <a href="#">research/round30/forward/at1/report.md</a> |
| HNМ-E-R30-AT1-F-04    | <a href="#">HNМ-AT1-F04</a> | <a href="#">research/round30/forward/at1/report.md</a> |
| HNМ-E-R30-AT1-F-05    | <a href="#">HNМ-AT1-F05</a> | <a href="#">research/round30/forward/at1/report.md</a> |
| HNМ-E-R30-AT1-F-06    | <a href="#">HNМ-AT1-F06</a> | <a href="#">research/round30/forward/at1/report.md</a> |

| Stable equation alias | Printed locator             | Frozen derivation                      |
|-----------------------|-----------------------------|----------------------------------------|
| HNМ-E-R30-AT1-F-07    | <a href="#">HNМ-AT1-F07</a> | research/round30/forward/at1/report.md |
| HNМ-E-R30-AT1-F-08    | <a href="#">HNМ-AT1-F08</a> | research/round30/forward/at1/report.md |
| HNМ-E-R30-AT1-F-09    | <a href="#">HNМ-AT1-F09</a> | research/round30/forward/at1/report.md |
| HNМ-E-R30-AT1-F-10    | <a href="#">HNМ-AT1-F10</a> | research/round30/forward/at1/report.md |
| HNМ-E-R30-AT1-F-11    | <a href="#">HNМ-AT1-F11</a> | research/round30/forward/at1/report.md |
| HNМ-E-R30-AT1-F-12    | <a href="#">HNМ-AT1-F12</a> | research/round30/forward/at1/report.md |
| HNМ-E-R30-AT1-F-13    | <a href="#">HNМ-AT1-F13</a> | research/round30/forward/at1/report.md |
| HNМ-E-R30-AT1-F-14    | <a href="#">HNМ-AT1-F14</a> | research/round30/forward/at1/report.md |
| HNМ-E-R30-AT1-F-15    | <a href="#">HNМ-AT1-F15</a> | research/round30/forward/at1/report.md |
| HNМ-E-R30-AT1-F-16    | <a href="#">HNМ-AT1-F16</a> | research/round30/forward/at1/report.md |
| HNМ-E-R30-AT1-F-17    | <a href="#">HNМ-AT1-F17</a> | research/round30/forward/at1/report.md |
| HNМ-E-R30-AT1-F-18    | <a href="#">HNМ-AT1-F18</a> | research/round30/forward/at1/report.md |
| HNМ-E-R30-AT1-R-01    | <a href="#">HNМ-AT1-R01</a> | research/round30/reverse/at1/report.md |
| HNМ-E-R30-AT1-R-02    | <a href="#">HNМ-AT1-R02</a> | research/round30/reverse/at1/report.md |
| HNМ-E-R30-AT1-R-03    | <a href="#">HNМ-AT1-R03</a> | research/round30/reverse/at1/report.md |
| HNМ-E-R30-AT1-R-04    | <a href="#">HNМ-AT1-R04</a> | research/round30/reverse/at1/report.md |
| HNМ-E-R30-AT1-R-05    | <a href="#">HNМ-AT1-R05</a> | research/round30/reverse/at1/report.md |
| HNМ-E-R30-AT1-R-06    | <a href="#">HNМ-AT1-R06</a> | research/round30/reverse/at1/report.md |
| HNМ-E-R30-AT1-R-07    | <a href="#">HNМ-AT1-R07</a> | research/round30/reverse/at1/report.md |
| HNМ-E-R30-AT1-R-08    | <a href="#">HNМ-AT1-R08</a> | research/round30/reverse/at1/report.md |
| HNМ-E-R30-AT1-R-09    | <a href="#">HNМ-AT1-R09</a> | research/round30/reverse/at1/report.md |
| HNМ-E-R30-AT1-R-10    | <a href="#">HNМ-AT1-R10</a> | research/round30/reverse/at1/report.md |
| HNМ-E-R30-AT1-R-11    | <a href="#">HNМ-AT1-R11</a> | research/round30/reverse/at1/report.md |
| HNМ-E-R30-AT1-R-12    | <a href="#">HNМ-AT1-R12</a> | research/round30/reverse/at1/report.md |
| HNМ-E-R30-AT1-R-13    | <a href="#">HNМ-AT1-R13</a> | research/round30/reverse/at1/report.md |
| HNМ-E-R30-AT1-R-14    | <a href="#">HNМ-AT1-R14</a> | research/round30/reverse/at1/report.md |
| HNМ-E-R30-AT1-R-15    | <a href="#">HNМ-AT1-R15</a> | research/round30/reverse/at1/report.md |
| HNМ-E-R30-AT1-R-16    | <a href="#">HNМ-AT1-R16</a> | research/round30/reverse/at1/report.md |
| HNМ-E-R30-AT1-R-17    | <a href="#">HNМ-AT1-R17</a> | research/round30/reverse/at1/report.md |
| HNМ-E-R30-AT2-F-01    | <a href="#">HNМ-AT2-F01</a> | research/round30/forward/at2/report.md |
| HNМ-E-R30-AT2-F-02    | <a href="#">HNМ-AT2-F02</a> | research/round30/forward/at2/report.md |
| HNМ-E-R30-AT2-F-03    | <a href="#">HNМ-AT2-F03</a> | research/round30/forward/at2/report.md |
| HNМ-E-R30-AT2-F-04    | <a href="#">HNМ-AT2-F04</a> | research/round30/forward/at2/report.md |
| HNМ-E-R30-AT2-F-05    | <a href="#">HNМ-AT2-F05</a> | research/round30/forward/at2/report.md |
| HNМ-E-R30-AT2-F-06    | <a href="#">HNМ-AT2-F06</a> | research/round30/forward/at2/report.md |
| HNМ-E-R30-AT2-F-07    | <a href="#">HNМ-AT2-F07</a> | research/round30/forward/at2/report.md |
| HNМ-E-R30-AT2-F-08    | <a href="#">HNМ-AT2-F08</a> | research/round30/forward/at2/report.md |
| HNМ-E-R30-AT2-F-09    | <a href="#">HNМ-AT2-F09</a> | research/round30/forward/at2/report.md |
| HNМ-E-R30-AT2-F-10    | <a href="#">HNМ-AT2-F10</a> | research/round30/forward/at2/report.md |
| HNМ-E-R30-AT2-F-11    | <a href="#">HNМ-AT2-F11</a> | research/round30/forward/at2/report.md |
| HNМ-E-R30-AT2-F-12    | <a href="#">HNМ-AT2-F12</a> | research/round30/forward/at2/report.md |

| Stable equation alias | Printed locator             | Frozen derivation                      |
|-----------------------|-----------------------------|----------------------------------------|
| HNМ-E-R30-AT2-F-13    | <a href="#">HNМ-AT2-F13</a> | research/round30/forward/at2/report.md |
| HNМ-E-R30-AT2-F-14    | <a href="#">HNМ-AT2-F14</a> | research/round30/forward/at2/report.md |
| HNМ-E-R30-AT2-F-15    | <a href="#">HNМ-AT2-F15</a> | research/round30/forward/at2/report.md |
| HNМ-E-R30-AT2-F-16    | <a href="#">HNМ-AT2-F16</a> | research/round30/forward/at2/report.md |
| HNМ-E-R30-AT2-R-01    | <a href="#">HNМ-AT2-R01</a> | research/round30/reverse/at2/report.md |
| HNМ-E-R30-AT2-R-02    | <a href="#">HNМ-AT2-R02</a> | research/round30/reverse/at2/report.md |
| HNМ-E-R30-AT2-R-03    | <a href="#">HNМ-AT2-R03</a> | research/round30/reverse/at2/report.md |
| HNМ-E-R30-AT2-R-04    | <a href="#">HNМ-AT2-R04</a> | research/round30/reverse/at2/report.md |
| HNМ-E-R30-AT2-R-05    | <a href="#">HNМ-AT2-R05</a> | research/round30/reverse/at2/report.md |
| HNМ-E-R30-AT2-R-06    | <a href="#">HNМ-AT2-R06</a> | research/round30/reverse/at2/report.md |
| HNМ-E-R30-AT2-R-07    | <a href="#">HNМ-AT2-R07</a> | research/round30/reverse/at2/report.md |
| HNМ-E-R30-AT2-R-08    | <a href="#">HNМ-AT2-R08</a> | research/round30/reverse/at2/report.md |
| HNМ-E-R30-AT2-R-09    | <a href="#">HNМ-AT2-R09</a> | research/round30/reverse/at2/report.md |
| HNМ-E-R30-AT2-R-10    | <a href="#">HNМ-AT2-R10</a> | research/round30/reverse/at2/report.md |
| HNМ-E-R30-AT2-R-11    | <a href="#">HNМ-AT2-R11</a> | research/round30/reverse/at2/report.md |
| HNМ-E-R30-AT2-R-12    | <a href="#">HNМ-AT2-R12</a> | research/round30/reverse/at2/report.md |
| HNМ-E-R30-AT2-R-13    | <a href="#">HNМ-AT2-R13</a> | research/round30/reverse/at2/report.md |
| HNМ-E-R30-AT2-R-14    | <a href="#">HNМ-AT2-R14</a> | research/round30/reverse/at2/report.md |
| HNМ-E-R30-AT2-R-15    | <a href="#">HNМ-AT2-R15</a> | research/round30/reverse/at2/report.md |
| HNМ-E-R30-AT2-R-16    | <a href="#">HNМ-AT2-R16</a> | research/round30/reverse/at2/report.md |
| HNМ-E-R30-AT2-R-17    | <a href="#">HNМ-AT2-R17</a> | research/round30/reverse/at2/report.md |
| HNМ-E-R30-AT3-F-01    | <a href="#">HNМ-AT3-F01</a> | research/round30/forward/at3/report.md |
| HNМ-E-R30-AT3-F-02    | <a href="#">HNМ-AT3-F02</a> | research/round30/forward/at3/report.md |
| HNМ-E-R30-AT3-F-03    | <a href="#">HNМ-AT3-F03</a> | research/round30/forward/at3/report.md |
| HNМ-E-R30-AT3-F-04    | <a href="#">HNМ-AT3-F04</a> | research/round30/forward/at3/report.md |
| HNМ-E-R30-AT3-F-05    | <a href="#">HNМ-AT3-F05</a> | research/round30/forward/at3/report.md |
| HNМ-E-R30-AT3-F-06    | <a href="#">HNМ-AT3-F06</a> | research/round30/forward/at3/report.md |
| HNМ-E-R30-AT3-F-07    | <a href="#">HNМ-AT3-F07</a> | research/round30/forward/at3/report.md |
| HNМ-E-R30-AT3-F-08    | <a href="#">HNМ-AT3-F08</a> | research/round30/forward/at3/report.md |
| HNМ-E-R30-AT3-F-09    | <a href="#">HNМ-AT3-F09</a> | research/round30/forward/at3/report.md |
| HNМ-E-R30-AT3-F-10    | <a href="#">HNМ-AT3-F10</a> | research/round30/forward/at3/report.md |
| HNМ-E-R30-AT3-F-11    | <a href="#">HNМ-AT3-F11</a> | research/round30/forward/at3/report.md |
| HNМ-E-R30-AT3-F-12    | <a href="#">HNМ-AT3-F12</a> | research/round30/forward/at3/report.md |
| HNМ-E-R30-AT3-F-13    | <a href="#">HNМ-AT3-F13</a> | research/round30/forward/at3/report.md |
| HNМ-E-R30-AT3-F-14    | <a href="#">HNМ-AT3-F14</a> | research/round30/forward/at3/report.md |
| HNМ-E-R30-AT3-F-15    | <a href="#">HNМ-AT3-F15</a> | research/round30/forward/at3/report.md |
| HNМ-E-R30-AT3-R-01    | <a href="#">HNМ-AT3-R01</a> | research/round30/reverse/at3/report.md |
| HNМ-E-R30-AT3-R-02    | <a href="#">HNМ-AT3-R02</a> | research/round30/reverse/at3/report.md |
| HNМ-E-R30-AT3-R-03    | <a href="#">HNМ-AT3-R03</a> | research/round30/reverse/at3/report.md |
| HNМ-E-R30-AT3-R-04    | <a href="#">HNМ-AT3-R04</a> | research/round30/reverse/at3/report.md |
| HNМ-E-R30-AT3-R-05    | <a href="#">HNМ-AT3-R05</a> | research/round30/reverse/at3/report.md |

| Stable equation alias | Printed locator             | Frozen derivation                      |
|-----------------------|-----------------------------|----------------------------------------|
| HNМ-E-R30-AT3-R-06    | <a href="#">HNМ-AT3-R06</a> | research/round30/reverse/at3/report.md |
| HNМ-E-R30-AT3-R-07    | <a href="#">HNМ-AT3-R07</a> | research/round30/reverse/at3/report.md |
| HNМ-E-R30-AT3-R-08    | <a href="#">HNМ-AT3-R08</a> | research/round30/reverse/at3/report.md |
| HNМ-E-R30-AT3-R-09    | <a href="#">HNМ-AT3-R09</a> | research/round30/reverse/at3/report.md |
| HNМ-E-R30-AT3-R-10    | <a href="#">HNМ-AT3-R10</a> | research/round30/reverse/at3/report.md |
| HNМ-E-R30-AT3-R-11    | <a href="#">HNМ-AT3-R11</a> | research/round30/reverse/at3/report.md |
| HNМ-E-R30-AT3-R-12    | <a href="#">HNМ-AT3-R12</a> | research/round30/reverse/at3/report.md |
| HNМ-E-R30-AT3-R-13    | <a href="#">HNМ-AT3-R13</a> | research/round30/reverse/at3/report.md |
| HNМ-E-R30-AT3-R-14    | <a href="#">HNМ-AT3-R14</a> | research/round30/reverse/at3/report.md |

## B.7 Round30 quantity aliases

**HNМ-Q-AT1-AQ-SECOND-MOMENT** Hruday AQ Wilson second-moment ceiling:  $B_2 = \alpha^2(36 + 98|\tau|)$ . energy squared. Ceiling, not an equality or second-moment convergence theorem.

**HNМ-Q-AT1-AQ-FIRST-MOMENT** Hruday AQ Wilson first-energy identity:  $\mu_1 = \alpha\omega_{\text{num}}(1 - W^2)$ . energy. Same chosen numerical-cap state; no identification with AO.

**HNМ-Q-R30-SPECTRAL-MASS** Hruday centered Wilson spectral mass registry:  $s = q - w^2$ . dimensionless. q and w are actual state expectations; not reference values.

**HNМ-Q-R30-INVERSE-ENERGY** Hruday inverse-energy Wilson functional:  $R = \int E^{-1} d\nu(E)$ . inverse energy. A spectral functional; a perturbed-state susceptibility requires additional differentiability.

**HNМ-Q-AT2-WINDOW-CHOICE** Hruday window-certificate design choice:  $L = 8$ . dimensionless energy cutoff  $x=E/\alpha$ . Proof choice;  $L/\alpha$  is the physical upper endpoint. Neither an eigenvalue nor a new physical constant.

**HNМ-Q-AT2-TANGENT-CHOICE** Hruday reciprocal-minorant tangent choice:  $c = 3$ . dimensionless energy coordinate. Chosen certificate parameter; no optimizer or physical-pole interpretation.

**HNМ-Q-AT2-MAJORANT-CHOICE** Hruday reciprocal-majorant factor choice:  $b = 49$ . dimensionless energy coordinate. Chosen polynomial factor parameter; no optimizer or physical-pole interpretation.

**HNМ-Q-AT3-EUCLIDEAN-HORIZON** Hruday Euclidean readout horizon choice:  $T = 128$ . dimensionless Euclidean coordinate conjugate to  $x=E/\alpha$ . Protocol design parameter; physical Euclidean duration is  $\hbar T/\alpha$ . No real-time observation or measured AQ data is supplied.

**HNM-Q-AT3-EUCLIDEAN-STEP** Hrudy Euclidean readout grid step:  $h = 1/32$ . dimensionless Euclidean coordinate. Protocol design parameter; not Planck constant, lattice spacing or a new dynamics clock.

**HNM-Q-AT3-INTERVAL-COUNT** Hrudy Euclidean readout interval count:  $N = 4096$ . dimensionless integer count.  $N$  grid intervals and  $N+1$  sample locations; a design count rather than degrees of freedom in the AQ Hamiltonian.

**HNM-Q-AT3-SAMPLE-ALLOWANCE** Hrudy Euclidean sample-error allowance:  $\varepsilon = 10^{-6}$ . dimensionless correlation amplitude. Deterministic per-sample absolute error assumption, not measured uncertainty, a confidence interval or a claimed capability to produce AQ samples.

**HNM-Q-AT3-QUADRATURE-ALLOWANCE** Hrudy integrated-curvature quadrature allowance:  $Q = h^2 u_{\text{up}}/8 = 47/512000$ . dimensionless inverse-energy functional  $I=\alpha R$ . Certified one-sided quadrature allowance, not an observed integration error or a new quadrature method.

**HNM-Q-AT3-TAIL-ALLOWANCE** Hrudy omitted-Euclidean-tail allowance:  $D = (504/125)e^{-8}$ . dimensionless inverse-energy functional  $I=\alpha R$ . Uniform upper bound for the unobserved tail; rational arithmetic must enclose the exponential outward.

**HNM-Q-AT3-NOISE-ALLOWANCE** Hrudy deterministic readout-error allowance:  $J = T\varepsilon = 2/15625$ . dimensionless inverse-energy functional  $I=\alpha R$ . Worst-case signed weighted error; not a root-sample-count stochastic uncertainty.

**HNM-Q-AT3-ARITHMETIC-GRID** Hrudy directed-rational arithmetic grid choice:  $G = 10^{30}$  (forward),  $D = 10^{30}$  (reverse). dimensionless integer denominator. Implementation design, distinct from the forward tail allowance also denoted  $D$ ; the source-local symbols remain unchanged.

**HNM-Q-AT3-REVERSE-BIN-ALLOWANCE** Hrudy reverse-API arithmetic-bin width allowance:  $10^{-12}$ . dimensionless correlation amplitude. Reverse API input restriction. Forward API permits wider bins and returns the resulting interval-width verdict; the two domains are not asserted equal.

## C Contribution catalog, applications, and remaining obligations

The double star marks workbench derivation, computation, repair, or specialization; it does not assert literature priority. Related early studies are consolidated; later loop increments remain separately indexed and grouped by subject. Every row points to a derivation or source and states a possible application, the change from earlier understanding, and a remaining obligation. Application means a supported mathematical use or proposed use within the stated model, not a measured technological benefit. The separate skeptical ledger records the level of fresh verification. Additional claim details and source paths are retained in `audit/contribution-catalog.json`.

Table 17: Organized contributions and their limits.

| Contribution and derivation                                                                                                                                                                                                                                                                        | Application and change                                                                                                                                                                                                                                                                                                                                         | Limitation and next obligation                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Origins and finite gauge benchmarks</b>                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                |
| <b>HNM-C-EARLY01</b> (legacy early01)**<br>Matched finite Maxwell–Dirac closure and continuation<br>Eqs. (HNM-1), (HNM-2), (HNM-3), (HNM-4), (HNM-5).<br><i>Scope</i> : model-specific derivation and implementation using established finite ODE theory<br>evidence/qeg-research/round3/theory.md | <i>Use</i> : Finite homogeneous backreaction and a global finite-dimensional existence theorem under $Z \geq z^* > 0$ .<br><i>Change</i> : A source-off electric reversal became a reproducible finite-model result; energy conservation alone does not determine whether matching is physically correct.                                                      | Static B, flat geometry, finite positive quadrature, finite Landau range; adjacent refinements are not rigorous trajectory enclosures. The 1e-18 advisor benchmark remains failed.<br><i>Next</i> : Common covariant current/stress renormalization, regulator convergence and quantum fluctuation control.                    |
| <b>HNM-C-EARLY02</b> (legacy early02)**<br>Complete causal tangent and stationary second variation<br>Eqs. (HNM-6), (HNM-7).<br><i>Scope</i> : model-specific derivative, positive quadratic form and explicit falsifier<br>evidence/qeg-research/round4/response_contract.md                      | <i>Use</i> : Causal finite-source response; stationary electric/transformed tangent control.<br><i>Change</i> : The signed first variation is not a norm; even a conserved positive second variation does not control raw potential/spin tangents.                                                                                                             | Fixed grid and matching; stationary pure-vacuum base and norm-preserving tangents $h_i \cdot \eta_i = 0$ ; no time-uniform driven stability theorem.<br><i>Next</i> : Construct and test an appropriate coercive response energy for a declared driven base solution.                                                          |
| <b>HNM-C-EARLY03</b> (legacy early03)**<br>Finite-window coefficient certificate and cofinal matching obstruction<br>Eqs. (HNM-8), (HNM-9), (HNM-10).<br><i>Scope</i> : exact integral derivation, rational certificate and scoped obstruction<br>evidence/qeg-research/round6/extensions.md       | <i>Use</i> : $Z > 3/4$ for fixed $b=10, K=20$ , $\alpha_{em} < 1/137$ , every finite Landau cutoff and positive constant-exact quadrature; obstruction to unchanged matching in joint cutoff removal.<br><i>Change</i> : The continuous integral maximizer does not transfer to a quadrature rule, and a positive finite coefficient is not regulator-uniform. | Continuous-window max theorem differs from discrete certificate; tail bound controls only C; bounded-addition obstruction requires uniform field-domain bounds and fixed/bounded potential.<br><i>Next</i> : A different controlled matching/renormalization path and full dynamical error estimates.                          |
| <b>HNM-C-EARLY04</b> (legacy early04)**<br>Anisotropic gravitational and regulated Dirac interface<br>Eqs. (HNM-11), (HNM-12).<br><i>Scope</i> : model-specific consistency derivations and implementation controls<br>evidence/qeg-research/round3/gravity.md                                     | <i>Use</i> : Bianchi-I constraint residual, shear diagnostics, regulated mode current/stress and counterterm Ward checks.<br><i>Change</i> : A fixed-FLRW occupation calculation is not Einstein–Maxwell–Dirac closure. Energy conservation leaves a pressure nullspace; the correct off-constraint coefficient here is $-\theta$ .                            | Bare finite comoving mode quantities; Ohmic and scalar fixtures have their own classical matter; no renormalized quantum pressure or self-consistent strong-curvature solution.<br><i>Next</i> : Common covariant subtractions and a specified suitable state, then compatible constraints and metric/current/stress response. |
| <b>HNM-C-EARLY05</b> (legacy early05)**<br>Action-derived scalar coupling and reference-subtraction taxonomy<br>Eqs. (HNM-14).<br><i>Scope</i> : reduced-model derivation and separate classical implementation<br>evidence/qeg-research/round7/research_report.md                                 | <i>Use</i> : Finite scalar-Maxwell–Dirac continuation and independently conserved classical scalar-gravity fixtures.<br><i>Change</i> : A dynamical scalar, a renormalization scale, a subtraction reference and a numerical cutoff are different objects.                                                                                                     | New scalar matter changes the theory; the classical gravity model and quantum finite model have no shared renormalized stress.<br><i>Next</i> : Prove any proposed identifying/decoupling map; retain scalar exchange and stress in new models.                                                                                |

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| Contribution and derivation                                                                                                                                                                                                                                                           | Application and change                                                                                                                                                                                                                                                                                                      | Limitation and next obligation                                                                                                                                                                                                                                       |
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| <b>HNM-C-EARLY06</b> (legacy early06)**<br>Positive spectral-measure diagnostics and conditional gap bridges<br>Eqs. (HNM-15), (HNM-16).<br><i>Scope:</i> known spectral theorems rederived; workbench counter-benchmark implementation<br><b>research/round8/spectral-proof.md</b>   | <i>Use:</i> False-plateau, hidden-channel and vanishing finite-gap controls; conditional common-gap survival and dense-channel decay theorem.<br><i>Change:</i> Effective mass approaches the lowest overlapping threshold from above and cannot certify a lower bound from a short plateau.                                | No reconstructed continuum QFT or regulator-uniform physical rate; periodic massless plot excludes its zero mode.<br><i>Next:</i> Construct the physical limit and a common all-time rate on a dense centered observable family.                                     |
| <b>HNM-C-EARLY07</b> (legacy early07)**<br>Finite Haar hierarchy characterization and Wilson identities<br>Eqs. (HNM-17).<br><i>Scope:</i> elementary finite-measure theorem rederived and authentic lattice implementation<br><b>research/round9/advisor/advisor.md</b>              | <i>Use:</i> Full compact connected-group Gibbs-law characterization; SU(2) staple residual and non-Gibbs atomic falsifier.<br><i>Change:</i> One Ward identity, especially the identically zero beta=0 residual, cannot identify the probability law.                                                                       | Finite graph, smooth action, all generators/insertions for converse; independent-link staple premise; beta=2.2 autocorrelation insufficient.<br><i>Next:</i> Controlled physical transfer/observable estimates and adequately sampled finite-volume data.            |
| <b>HNM-C-EARLY08</b> (legacy early08)**<br>Exact central-convolution spectrum and continuum-time rotor limit<br>Eqs. (HNM-18).<br><i>Scope:</i> known harmonic analysis rederived and numerically implemented<br><b>research/round9/transfer/PROOF.md</b>                             | <i>Use:</i> All-spin Bessel spectrum, exact gap, negative-beta positivity falsifier and strong rotor semigroup limit.<br><i>Change:</i> Pointwise positive kernels need not be positive operators, and a rotor spectrum is not the locally gauge-projected interacting transfer spectrum.                                   | Auxiliary SU(2) operator; fixed-spin asymptotics; continuum time only, no continuum space; strong convergence, not norm convergence.<br><i>Next:</i> Actual many-link gauge transfer construction, interactions and matched physical clock.                          |
| <b>HNM-C-EARLY09</b> (legacy early09)**<br>Product-L1 operator-norm counterexample and transfer stability<br>Eqs. (HNM-19).<br><i>Scope:</i> exact family of counterexamples and standard conditional norm lemma<br><b>research/round9/advisor/advisor.md</b>                         | <i>Use:</i> Positive symmetric Markov family with product L1 tending to zero and fixed operator discrepancy; common-vacuum gap and finite-time comparison.<br><i>Change:</i> Kernel product-L1 does not bound essential row/column norms; $o(a)$ , not merely $o(1)$ , is sufficient for telescoped fixed-time convergence. | Local objection to a displayed estimate, not a verdict on every claim of a cited preprint; shared vacuum and common Hilbert space required.<br><i>Next:</i> Uniform Schur or Hilbert-Schmidt bounds on the actual model and a justified vacuum/space identification. |
| <b>HNM-C-EARLY10</b> (legacy early10)**<br>One-square full-space spectral enclosure and coupling cover<br>Eqs. (HNM-20), (HNM-21).<br><i>Scope:</i> certified implementation and explicit error contract for known one-square reduction<br><b>research/round10/advisor/advisor.md</b> | <i>Use:</i> Full-representation gap $\geq 0.999999$ alpha on $\lambda/\alpha$ in $[0,10]$ ; correct radial and character normalizations.<br><i>Change:</i> Finite graph does not mean finite-dimensional Hilbert space; sampled coupling points do not cover an interval.                                                   | $\alpha > 0$ , $\lambda \geq 0$ for displayed tail; finite graph; all-coupling monotonicity unproved; $\alpha > 0$ counterexample is an unmatched path.<br><i>Next:</i> More plaquettes with shared-link Gauss constraints and coupled tails.                        |

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| Contribution and derivation                                                                                                                                                                                                                                                                         | Application and change                                                                                                                                                                                                                                                                                                                         | Limitation and next obligation                                                                                                                                                                                                                                                                |
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| <b>HNM-C-EARLY11</b> (legacy early11)**<br>Three-trace quotient, induced operator and complete electric shells Eqs. (HNM-22), (HNM-23), (HNM-24), (HNM-26).<br><i>Scope:</i> graph-specific derivation and exact polynomial implementation<br>research/round10/skeptic/coordinate-note.md           | <i>Use:</i> Complete physical quotient measure/domain, shared-link kinetic operator, spin-network spectrum and exact degree tail threshold.<br><i>Change:</i> Two separate traces miss a physical variable; two independent rotors reproduce some low tests but fail Kz and K(xy).                                                             | Two adjacent open squares, seven links, all six vertex Gauss constraints; no arbitrary boundary condition on quotient singularities.<br><i>Next:</i> Extend exact induced operators/degree bounds to larger connected graphs.                                                                 |
| <b>HNM-C-EARLY12</b> (legacy early12)**<br>Interacting two-square certificates and global heat comparison Eqs. (HNM-25), (HNM-27).<br><i>Scope:</i> graph-specific exact boundary algebra, gap certificate implementation and explicit conservative estimate<br>research/round11/advisor/advisor.md | <i>Use:</i> Coupling-square floor $>0.96$ alpha; specific full gap interval at (1,2,3,1); explicit positive bound for every finite nonnegative pair.<br><i>Change:</i> A monomial form matrix is not an ordinary Euclidean Hamiltonian, and positive pointwise gaps need quantified tails and coverage.                                        | $\alpha > 0$ , nonnegative magnetic coefficients, $k=0,1$ when $\mu_1 < R < \tau$ ; finite graph only; global heat bound deteriorates.<br><i>Next:</i> Connected-volume control retaining locality; sharper actual boundary norm and matched observables.                                     |
| <b>HNM-C-EARLY13</b> (legacy early13)**<br>Volume, measure, clock and non-triviality obstructions Eqs. (HNM-28).<br><i>Scope:</i> exact scoped counterbenchmarks and standard conditional limit criteria<br>research/round12/volume/volume-bridge.md                                                | <i>Use:</i> Free graph gap 3 alpha; explicit extensive bound; exact tensor counterbenchmark; Wilson Gibbs/ground-state inequivalence; conditional true-ground-state curvature route.<br><i>Change:</i> Decay of a lower certificate need not mean a vanishing actual gap; stochastic, Euclidean and physical Hamiltonian gaps are distinct.    | Tensor copies are disconnected; no uniform true-ground-state curvature bound; no physical continuum construction or vacuum convergence.<br><i>Next:</i> Actual connected Hamiltonian estimates, embeddings, physical scale, strongly continuous nontrivial limit and full spectral exclusion. |
| <b>HNM-C-EARLY14</b> (legacy early14)**<br>Factorial representation error and certified finite numerical state Eqs. (HNM-29), (HNM-30).<br><i>Scope:</i> finite-graph theorem and exact rational computation; priority unverified<br>research/round12/advisor/advisor.md                            | <i>Use:</i> Full-Hilbert-space error for finite-support preparation and fixed action; rigorous residual/midpoint/Taylor layers; exact two-step state certificate.<br><i>Change:</i> Historical work/norm/refinement diagnostics did not certify omitted representations; R12 supplies a new theorem and a separate computed-state certificate. | Fixed finite graph and action horizon, signed drivers require absolute values, phase must match; numerical state need not be normalized; energy unbounded.<br><i>Next:</i> Sharper trajectory boundary integrals and preparation certificates; connected graph and physical-time scaling.     |
| <b>HNM-C-EARLY15</b> (legacy early15)**<br>Compact hierarchy fluctuation and formula audits Eqs. (HNM-31).<br><i>Scope:</i> exact explicit closure counterbenchmark and version-specific algebra checks<br>research/round12/closure/closure-audit.md                                                | <i>Use:</i> Exact recurrence, nonzero variance lower bound, residual $\kappa \cdot v$ and rank/counting/phase/prefactor tests.<br><i>Change:</i> Adding a variance symbol exposes another obligation; it does not close an infinite hierarchy.                                                                                                 | One-plaquette Euclidean measure differs from Hamiltonian dynamics; tested closure is point concentration; external display checks apply to 2608.05415v1.<br><i>Next:</i> Quantified hierarchy truncation or an independently constructed measure with controlled moments.                     |

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| Contribution and derivation                                                                                                                                                                                                                                                                             | Application and change                                                                                                                                                                                                                                                                                                           | Limitation and next obligation                                                                                                                                                                                                                                                                                                                                                |
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| <b>HNM-C-EARLY16</b> (legacy early16)**<br>Finite obligation planner and historical admission repairs<br><i>Scope:</i> workflow implementation, not a mathematical proof kernel<br><a href="#">evidence/qeg-research/round4/README.md</a>                                                               | <i>Use:</i> Finite forward/backward rule search and replay; source binding and retained defect regressions.<br><i>Change:</i> Successful script exit, exact binary input rationalization and counts of passing checks do not establish the physical theorem or transcendental rounding.                                          | Finite reviewed rule libraries; R8 symbolic checker unavailable; no comprehensive fresh execution in this manuscript.<br><i>Next:</i> Independent theorem-premise and evidence admission review with explicit complete dependencies.                                                                                                                                          |
| <b>HNM-C-EARLY17</b> (legacy early17)**<br>Inherited context and logical gravity controls<br>Eqs. (HNM-13).<br><i>Scope:</i> surviving context; workbench counterexamples, with missing earlier packages<br><a href="#">evidence/qeg-research/round3/chapters/01_master_action.md</a>                   | <i>Use:</i> Common-functional target, rotating-mode connection, energy/noise counterexample, spurious truncated concurrence and horizon integrability screen.<br><i>Change:</i> The same means can have different noise; nonlinear witnesses at an omitted order can be artifacts; fixed geometry modes are not gravity closure. | No independently identifiable R1/R2 contracts or source packages. Imported JSON has provenance and numerical cases but is not a new replay. Contextual conjectures and black-hole/mixing endpoints remain open.<br><i>Next:</i> Recover missing source packages if available; otherwise keep provenance gap and separately formulate the next physical initial-value problem. |
| <b>Static, spectral, and local-limit development / dynamics</b>                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                               |
| <b>HNM-C-R13-LOCALITY</b> (legacy R13-locality)**<br>Local finite-time volume limit<br>Eqs. (HNM-43).<br><i>Scope:</i> historical study — scope below<br><a href="#">research/round13/locality/locality.md</a>                                                                                          | <i>Use:</i> Local finite-time boundary comparison<br><i>Change:</i> Removed the extensive global-volume factor for local dynamics                                                                                                                                                                                                | Fixed spacing and common envelope; no vacuum comparison or continuum-uniform velocity<br><i>Next:</i> Sharper physical scaling estimate and an appropriate observable algebra                                                                                                                                                                                                 |
| <b>HNM-C-R20-G1</b> (legacy R20-g1)**<br>Literal-box real-time dynamics converges in operator norm on each local observable at fixed physical time, with a strongly continuous Hilbert implementation.<br>Section 8.<br><i>Scope:</i> accepted<br><a href="#">research/round20/forward/g1/report.md</a> | <i>Use:</i> All-phase fixed-time literal-box dynamics limit<br><i>Change:</i> Bounded strong Duhamel avoids false operator-norm differentiation                                                                                                                                                                                  | Full bounded local algebra need not have point-norm time continuity<br><i>Next:</i> Limiting state's GNS generator                                                                                                                                                                                                                                                            |
| <b>Static, spectral, and local-limit development / homogeneous</b>                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                               |
| <b>HNM-C-R13-STABILITY</b> (legacy R13-stability)**<br>Qualitative homogeneous stability<br>Section 10.1.<br><i>Scope:</i> historical study — scope below<br><a href="#">research/round13/advisor/weak-coupling-stability.md</a>                                                                        | <i>Use:</i> Apply a known product-vacuum theorem to full rotor links<br><i>Change:</i> Matched infinite-dimensional sites, boundaries and gauge-invariant uniqueness                                                                                                                                                             | Numerical smallness constants unevaluated; weak bare coupling exits this regime<br><i>Next:</i> Extract a matched explicit local stability threshold                                                                                                                                                                                                                          |

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| Contribution and derivation                                                                                                                                                                                                                                                                                                        | Application and change                                                                                                                                             | Limitation and next obligation                                                                                                                     |
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| <b>HNM-C-R17-A1</b> (legacy R17-a1)**<br>Exact local projector and incidence bounds; physical counterexample to the full pure relative-form shortcut. Dense uniform threshold remains open.<br>Section 10.1.<br><i>Scope</i> : accepted<br><b>research/round17/forward/a1/report.md</b>                                            | <i>Use</i> : Diagnose diagonal control and vacuum mixing<br><i>Change</i> : Actual physical trial disproves pure relative bound for full W                         | Does not refute valid additive bounds or dressing<br><i>Next</i> : Restricted link-disjoint support theorem                                        |
| <b>HNM-C-R21-I1</b> (legacy R21-i1)**<br>Complete 24-link blocking gives 21 omitted anchor faces and exact normalized local interaction norm $7 \tau $ ; qualitative fixed-spacing stability applies with unevaluated source constants.<br>Section 10.1.<br><i>Scope</i> : limited<br><b>research/round21/forward/i1/report.md</b> | <i>Use</i> : Exact homogeneous theorem dictionary<br><i>Change</i> : Complete ownership, all crossing supports and boundary-aware padding                          | Yarotsky constants remain unevaluated; no declared numerical $\tau$<br><i>Next</i> : Construct explicit local rotations and audit global iteration |
| <b>HNM-C-R21-I2</b> (legacy R21-i2)**<br>Exact one-star vacuum rotation cancels first-order vacuum mixing and retains a bounded quadratic remainder; a bare pure relative bound for the original interaction fails.<br>Section 10.1.<br><i>Scope</i> : limited<br><b>research/round21/forward/i2/report.md</b>                     | <i>Use</i> : Local domain-safe vacuum dressing<br><i>Change</i> : Exact cancellation sign and full quadratic remainder                                             | One-star result; overlapping generated terms not yet controlled<br><i>Next</i> : Simultaneous finite-volume rotation                               |
| <b>HNM-C-R22-O1</b> (legacy R22-o1)**<br>Complete simultaneous homogeneous rotation has explicit weighted remainder with preserved unbounded operator domain.<br>Eqs. (HNM-52).<br><i>Scope</i> : limited<br><b>research/round22/reverse/o1/report.md</b>                                                                          | <i>Use</i> : One simultaneous homogeneous rotation<br><i>Change</i> : All ordered connected supports, root counts and unbounded domain retained                    | One-step radius is not gap threshold or iteration closure<br><i>Next</i> : Generated-support recurrence including diagonal part                    |
| <b>Static, spectral, and local-limit development / moments</b>                                                                                                                                                                                                                                                                     |                                                                                                                                                                    |                                                                                                                                                    |
| <b>HNM-C-R13-MOMENTS</b> (legacy R13-moments)**<br>Compact moment certificates and uniqueness<br>Eqs. (HNM-32).<br><i>Scope</i> : historical study — scope below<br><b>research/round13/moments/README.md</b>                                                                                                                      | <i>Use</i> : Exact scalar Haar moment enclosures<br><i>Change</i> : Retained variance and all recurrence/support constraints; proved complete-hierarchy uniqueness | No hierarchy rate; code only levels 1–6 and slack must vanish<br><i>Next</i> : Connected shared-link hierarchy with complete identities            |

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| Contribution and derivation                                                                                                                                                                                                                                    | Application and change                                                                                                                                                       | Limitation and next obligation                                                                                                                             |
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| <b>HNM-C-R13-RESPONSE</b><br>(legacy R13-response)**<br>Exact scalar response<br>Eqs. (HNM-33).<br><i>Scope:</i> historical study — scope below<br><b>research/round13/response/response.md</b>                                                                | <i>Use:</i> Coupling-response analysis and residual error transport<br><i>Change:</i> Separated regular Haar branch, singular branches and changed-prior countermodel        | Static response; sampled floating error is not a global enclosure<br><i>Next:</i> Certify a full residual envelope between numerical nodes                 |
| <b>HNM-C-R14-LOOP1</b> (legacy R14-loop1)**<br>Exact conditional Haar reduction with removable zero field.<br>Section 6.1.<br><i>Scope:</i> accepted within declared finite-model scope<br><b>research/round14/forward/loop1.md</b>                            | <i>Use:</i> Controlled two-holonomy interaction and local covariance sign<br><i>Change:</i> Exact conditional field and removable zero-field reduction replace factorization | Small-eta bound insufficient at selected central fixture; floating comparisons diagnostic<br><i>Next:</i> Exact rational same-action covariance enclosures |
| <b>HNM-C-R14-LOOP2</b> (legacy R14-loop2)**<br>23 exact finite-measure covariance enclosures independently replayed using characters.<br>Section 6.1.<br><i>Scope:</i> accepted within declared finite-model scope<br><b>research/round14/forward/loop2.md</b> | <i>Use:</i> Normalized rational covariance certification<br><i>Change:</i> 23 exact enclosures certify selected central target including full tails                          | Static finite action; cap48 distinct from uncapped convergence theorem<br><i>Next:</i> Continuous target-interval coverage                                 |
| <b>HNM-C-R15-A1</b> (legacy R15-A1)**<br>All8 positive exact midpoints, all8 coarse transported lower bounds insufficient.<br>Section 6.1.<br><i>Scope:</i> accepted correctly diagnosed insufficiency<br><b>research/round15/forward/A1/derivation.md</b>     | <i>Use:</i> Transport exact covariance point intervals<br><i>Change:</i> Eight positive midpoint results correctly fail all eight coarse cells                               | Positive point values do not prove complete interval positivity<br><i>Next:</i> Bisect failing cells using unchanged point accuracy                        |
| <b>HNM-C-R15-A2</b> (legacy R15-A2)**<br>Refined failing cell radii rather than increasing already sufficient point precision.<br>Section 6.1.<br><i>Scope:</i> accepted exact complete positive cover<br><b>research/round15/forward/A2/result.md</b>         | <i>Use:</i> Complete positive cover of eta in $[1/8, 1/4]$<br><i>Change:</i> Refined geometry rather than unnecessarily increasing point precision                           | Finite $SU(2)^2$ action only; cap failures retained<br><i>Next:</i> Higher-dimensional or larger-graph certified covers                                    |

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| Contribution and derivation                                                                                                                                                                                                                                                                                                               | Application and change                                                                                                                                   | Limitation and next obligation                                                                                                   |
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| <b>Static, spectral, and local-limit development / gluing</b>                                                                                                                                                                                                                                                                             |                                                                                                                                                          |                                                                                                                                  |
| <b>HNM-C-R15-B1</b> (legacy R15-B1)**<br>Closed cube has six-way dependence while any five face class observables factorize; use actual character gluing and distinct face derivatives in B2.<br>Eqs. (HNM-34).<br><i>Scope:</i> accepted exact finite cube graph and Haar identities<br><b>research/round15/forward/B1/derivation.md</b> | <i>Use:</i> Graph-faithful closed-cube Haar contraction<br><i>Change:</i> Proved six-way dependence alongside any-five class-function factorization      | Does not assert six independent holonomies or Hamiltonian gap<br><i>Next:</i> Full normalized Wilson-weighted cube integral      |
| <b>HNM-C-R15-B2</b> (legacy R15-B2)**<br>Character dimension and separate face derivatives retained; coarse degree failures remain explicit.<br>Section 6.2.<br><i>Scope:</i> accepted exact finite cube integral and rejected factorization<br><b>research/round15/forward/B2/derivation.md</b>                                          | <i>Use:</i> Finite closed-cube Wilson expectation<br><i>Change:</i> Retained dimension factors and distinct-face derivatives with full Taylor tails      | Product expectation is not a connected six-face cumulant; coarse degrees fail<br><i>Next:</i> Shared-face higher-incidence graph |
| <b>HNM-C-R16-LOOP1</b> (legacy R16-loop1)**<br>Exact finite two-cube Haar/fusion identities and separate conditional physical energy-scale audit.<br>Eqs. (HNM-35).<br><i>Scope:</i> accepted<br><b>research/round16/forward/loop1/derivation.md</b>                                                                                      | <i>Use:</i> Shared-face Haar fusion on 12-vertex, 20-link, 11-face graph<br><i>Change:</i> Replaced invalid single-surface label rule by two-disk fusion | Static graph; full 11-face and outer 10-face actions differ<br><i>Next:</i> Same observable comparison with shared action on/off |
| <b>HNM-C-R16-LOOP2</b> (legacy R16-loop2)**<br>Positive normalized all-eleven observable difference for outer1/8 and shared1/8 versus 0, exact width $\leq 1e-12$ .<br>Section 6.2.<br><i>Scope:</i> accepted<br><b>research/round16/forward/loop2/derivation.md</b>                                                                      | <i>Use:</i> Quantitative shared-face action comparison<br><i>Change:</i> Retained nonzero omitted-action baseline and both normalization errors          | Finite static target; no general monotonicity theorem<br><i>Next:</i> Additional recoupling and surrounding-link dependence      |

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| Contribution and derivation                                                                                                                                                                                                                                                                                           | Application and change                                                                                                  | Limitation and next obligation                                                                                                              |
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| <b>Static, spectral, and local-limit development / finite-gap</b>                                                                                                                                                                                                                                                     |                                                                                                                         |                                                                                                                                             |
| <b>HNM-C-R15-C1</b> (legacy R15-C1)**<br>Cube lower bound is zero at common $\lambda=\alpha/2$ ; accepted B1 trial moments permit a variational ground-energy upper bound.<br>Section 7.1.<br>Scope: accepted finite-graph side theorem and retained original source gap<br>research/round15/forward/C1/derivation.md | Use: Full untruncated finite-graph gap bound<br>Change: Distinguished physical girth gap from full-link $3\alpha/4$ gap | Cube bound zero at $\lambda=\alpha/2$ ; global budget grows with volume<br>Next: Variational E0 upper bound with independent E1 lower bound |
| <b>HNM-C-R15-C2</b> (legacy R15-C2)**<br>The independent E1 lower bound and trial E0 upper bound repair the C1 zero-margin fixture; no trial gap was substituted for the full gap.<br>Eqs. (HNM-38).<br>Scope: accepted finite physical cube variational improvement<br>research/round15/forward/C2/derivation.md     | Use: Repair finite cube zero margin<br>Change: Combined one-sided full E1 and trial E0 bounds correctly                 | Radical interval encloses lower-bound formula, not actual gap<br>Next: Dense two-cube and complete complement improvements                  |
| <b>HNM-C-R17-B1</b> (legacy R17-b1)**<br>Improved full physical gap lower bound on the dense eleven-face two-cube graph, positive for common $ \lambda /\alpha < 12/43$ ; finite volume only.<br>Eqs. (HNM-38).<br>Scope: accepted<br>research/round17/forward/b1/report.md                                           | Use: Dense finite two-cube spectral guarantee<br>Change: Rebuilt every repeated-index trial entry on actual graph       | Fixed graph; zero margin at $12/43$ is certificate boundary<br>Next: Add actual shared-face adjoint trial channel                           |
| <b>HNM-C-R17-B2</b> (legacy R17-b2)**<br>Exact positive full physical gap lower $\alpha*1388/284767457$ at the dense finite coupling $\lambda/\alpha=12/43$ , with a proved open variational-amplitude interval.<br>Section 7.1.<br>Scope: accepted<br>research/round17/forward/b2/report.md                          | Use: Positive full gap at previous endpoint<br>Change: Actual adjoint trial lowers E0 below independent E1 bound        | Amplitude midpoint maximizes numerator, not quotient; no volume theorem<br>Next: Complete low-energy complement                             |

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| Contribution and derivation                                                                                                                                                                                                                                                                                                                                                                                                              | Application and change                                                                                                                                      | Limitation and next obligation                                                                                                                     |
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| <b>HNM-C-R18-B1</b> (legacy R18-b1)**<br>Complete physical complement threshold $9\alpha/2$ and exact QVP Gram on the dense two-cube graph; no finite trial spectrum substituted for the entire omitted space.<br>Section 7.1.<br><i>Scope:</i> accepted<br><b>research/round18/forward/b1/report.md</b>                                                                                                                                 | <i>Use:</i> Complete physical complement and exact leakage Gram<br><i>Change:</i> Classified all states below threshold and retained projection subtraction | Independent Haar faces agree at fourth order; Gaussian fourth moment is false control<br><i>Next:</i> Codimension-one full excited-energy estimate |
| <b>HNM-C-R18-B2</b> (legacy R18-b2)**<br>Full untruncated physical two-cube signed coefficient-box gap lower bound, separate endpoint ground trial, and bare-coupling domain obstruction. No volume-uniform or continuum claim.<br>Section 7.1.<br><i>Scope:</i> accepted<br><b>research/round18/forward/b2/report.md</b>                                                                                                                | <i>Use:</i> Full untruncated dense two-cube gap<br><i>Change:</i> Converted exact complement/Gram into continuous signed-box theorem                        | Fixed graph; standard bare-coupling membership requires $g^4 \geq 32/3$<br><i>Next:</i> Larger complete projector and better cross envelope        |
| <b>HNM-C-R19-B1</b> (legacy R19-b1)**<br>Goal B Loop 1 only: complete strict-cutoff physical projector and exact full cross Gram on the actual finite two-cube $SU(2)$ graph. No continuous coefficient box, dense limit, continuum Yang-Mills or mass-gap claim is accepted here.<br>Section 7.1.<br><i>Scope:</i> accepted<br><b>research/round19/forward/b1/report.md</b>                                                             | <i>Use:</i> Exact physical channel and cross-Gram inventory<br><i>Change:</i> Includes all 36 six-cycles and threshold intertwiner multiplicities           | Strict cutoff is full spectral classification, not chosen variational list<br><i>Next:</i> Continuous signed-box full E1 estimate                  |
| <b>HNM-C-R19-B2</b> (legacy R19-b2)**<br>Goal B Loop 2 only: finite two-cube continuous signed coefficient-box gap certificate using the accepted B1 48-channel projector, full compressed PVP ledger and exact ordered $W^*W$ cross Gram. No dense limit, continuum Yang-Mills, Clay mass gap or physical-scale matching claim is accepted.<br>Eqs. (HNM-39).<br><i>Scope:</i> accepted<br><b>research/round19/forward/b2/report.md</b> | <i>Use:</i> Wider finite signed coefficient box<br><i>Change:</i> Improved complement $6\alpha$ and exact 47-state row/cross envelopes                      | $r = 1/2$ fails only this sufficient envelope<br><i>Next:</i> Sharper fixed-graph envelopes or separate volume method                              |

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| Contribution and derivation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Application and change                                                                                                                                    | Limitation and next obligation                                                                                                                     |
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| <b>Static, spectral, and local-limit development / strips</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                           |                                                                                                                                                    |
| <b>HNM-C-R17-A2</b> (legacy R17-a2)**<br>Volume-uniform physical gap for the pairwise link-disjoint magnetic-support family with a common positive energy scale; dense overlapping family remains open.<br>Section 7.2.<br>Scope: accepted<br>research/round17/forward/a2/report.md                                                                                                                                                                                                                                                                  | <i>Use:</i> Volume-uniform sparse magnetic family<br><i>Change:</i> Full-link factorization plus gauge-invariant unique ground                            | Pairwise link-disjoint support; dense overlaps excluded<br><i>Next:</i> Internally overlapping dressed strip                                       |
| <b>HNM-C-R18-A1</b> (legacy R18-a1)**<br>Finite dressed three-square bridge; physical gap at least $\alpha/8$ throughout the declared primary family; no homogeneous volume extension.<br>Section 7.2.<br>Scope: accepted<br>research/round18/forward/a1/report.md                                                                                                                                                                                                                                                                                   | <i>Use:</i> Bridge two dressed sparse blocks<br><i>Change:</i> Free bridge link proves zero reference mean and one-norm improvement                       | Finite cluster; exact dressed wavefunction unknown and unnecessary<br><i>Next:</i> Repeat clusters with complete remainder ledger                  |
| <b>HNM-C-R18-A2</b> (legacy R18-a2)**<br>All finite volumes of the specified spatially decaying inhomogeneous family have physical gap at least $7\alpha_{\min}/64$ ; boundary reclassification and homogeneous dense/thermodynamic limits remain explicit.<br>Section 7.2.<br>Scope: accepted<br>research/round18/forward/a2/report.md                                                                                                                                                                                                              | <i>Use:</i> All-finite-volume inhomogeneous full-support family<br><i>Change:</i> Repeated link-disjoint overlapping strips plus summable remaining faces | Boundary rule reclassifies coefficients; not yet literal restrictions<br><i>Next:</i> Fixed infinite assignment and clipped-component theorem      |
| <b>HNM-C-R19-A1</b> (legacy R19-a1)**<br>For every reachable selected clipped component L, M, R, LM, MR or LMR under the frozen coefficient bounds $ \lambda_L /\alpha \leq 1/2$ , $ \mu_M /\alpha \leq 1/8$ and $ \lambda_R /\alpha \leq 1/2$ , the full-link local gap lower estimate is at least $\alpha/8$ . Positive full-link uniqueness and gauge invariance permit the same lower estimate after Gauss-sector restriction for the finite local component result.<br>Section 7.2.<br>Scope: accepted<br>research/round19/forward/a1/report.md | <i>Use:</i> Literal finite restrictions of one infinite assignment<br><i>Change:</i> Removes Round18 coefficient reclassification                         | Finite gap statement alone does not construct thermodynamic representation<br><i>Next:</i> Direct product representation and summable perturbation |

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| Contribution and derivation                                                                                                                                                                                                                                                                                                                                                                                                                      | Application and change                                                                                                                          | Limitation and next obligation                                                                                                                         |
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| <b>HNM-C-R19-A2</b> (legacy R19-a2)**<br>Goal A Loop 2 only: direct incomplete product-representation construction for a summable omitted-face perturbation of the accepted A1 complete-strip/free reference. This is not finite clipped-restriction convergence, homogeneous dense stability, continuum Yang-Mills or a mass-gap theorem.<br>Eqs. (HNM-40), (HNM-41).<br><i>Scope:</i> accepted<br><b>research/round19/forward/a2/report.md</b> | <i>Use:</i> Explicit infinite-volume summable Hamiltonian<br><i>Change:</i> Closed product-reference form and rank-one spectral isolation       | Initially no finite clipped-limit convergence; fixed representation and spacing<br><i>Next:</i> Quantitative finite-factor and literal-box convergence |
| <b>HNM-C-R20-D1</b> (legacy R20-d1)**<br>Finite-factor completion gives norm-resolvent convergence with explicit dyadic tail on the exact A2 exterior-reference lift.<br>Section 7.2.<br><i>Scope:</i> accepted<br><b>research/round20/forward/d1/report.md</b>                                                                                                                                                                                  | <i>Use:</i> Quantitative finite-factor completion<br><i>Change:</i> Correct exterior-reference lift preserves infinite generator                | Finite factors remain infinite-dimensional; no literal arbitrary-phase theorem yet<br><i>Next:</i> Ground energies, projections and states             |
| <b>HNM-C-R20-D2</b> (legacy R20-d2)**<br>Ground energy, rank-one projections and bounded observables converge with explicit dyadic errors for D1 lifts.<br>Section 7.2.<br><i>Scope:</i> accepted<br><b>research/round20/forward/d2/report.md</b>                                                                                                                                                                                                | <i>Use:</i> Ground-state convergence with explicit dyadic errors<br><i>Change:</i> Zero trial mean and rank-one residual give sharp denominator | Requires $\beta < \delta$ and correct embedding<br><i>Next:</i> Literal geometric boxes and energy origins                                             |
| <b>HNM-C-R20-E1</b> (legacy R20-e1)**<br>Literal aligned boxes $N=4m+3$ converge after explicit reference-energy subtraction and exact exterior-reference lift.<br>Section 7.2.<br><i>Scope:</i> accepted<br><b>research/round20/forward/e1/report.md</b>                                                                                                                                                                                        | <i>Use:</i> Aligned literal-box convergence<br><i>Change:</i> Retained exact strip energy subtraction and orientation-specific tail             | Unshifted absolute energy/resolvent convergence not proved<br><i>Next:</i> All boundary phases in local-state topology                                 |
| <b>HNM-C-R20-E2</b> (legacy R20-e2)**<br>All upper-boundary phases of literal growing rectangular boxes converge to the A2 ground state on bounded local observables.<br>Section 7.2.<br><i>Scope:</i> accepted<br><b>research/round20/forward/e2/report.md</b>                                                                                                                                                                                  | <i>Use:</i> All-phase rectangular local-state limit<br><i>Change:</i> Observable support included in common complete central factors            | No global vector or arbitrary-phase norm-resolvent conclusion<br><i>Next:</i> Local dynamics and GNS identification                                    |

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| Contribution and derivation                                                                                                                                                                                                                                                                                                                                                                             | Application and change                                                                                                                          | Limitation and next obligation                                                                                                              |
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| <b>Static, spectral, and local-limit development / conditional</b>                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                 |                                                                                                                                             |
| <b>HNM-C-R17-C1</b> (legacy R17-c1)**<br>Complete four-adjoint Haar tensor and realized four-cube conditional boundary; no bulk or spectral claim.<br>Section 6.3.<br><i>Scope:</i> accepted<br>research/round17/forward/c1/report.md                                                                                                                                                                   | <i>Use:</i> Complete central four-adjoint Haar tensor<br><i>Change:</i> Three invariant channels and negative inverse-Gram entries retained     | One central link with 32 others fixed<br><i>Next:</i> Nonzero-action joint observable with actual boundary                                  |
| <b>HNM-C-R17-C2</b> (legacy R17-c2)**<br>Same-action conditional four-face expectations with complete rational numerator and partition errors; action-only observable closure rejected, full bulk and spectrum open.<br>Section 6.3.<br><i>Scope:</i> accepted<br>research/round17/forward/c2/report.md                                                                                                 | <i>Use:</i> Reject action-only observable closure<br><i>Change:</i> Equal partition and action coexist with different joint observables         | Conditional central result; surrounding marginal not integrated<br><i>Next:</i> Sufficient complete action-observable coordinates           |
| <b>HNM-C-R18-C1</b> (legacy R18-c1)**<br>Complete joint-Gram sufficiency for the declared central S3 observable/action class, exact rational admissibility including singular cases, and independently matched degree0–6 diagnostics. No minimality, finite-tail, surrounding measure or mass-gap conclusion.<br>Eqs. (HNM-36).<br><i>Scope:</i> accepted<br>research/round18/forward/c1/report.md      | <i>Use:</i> Coordinate sufficiency for central integral<br><i>Change:</i> Handles rank-deficient, reflected and zero-action data exactly        | Sufficiency is not minimality or surrounding measure closure<br><i>Next:</i> Integrate actual surrounding Haar link                         |
| <b>HNM-C-R18-C2</b> (legacy R18-c2)**<br>Full normalized two-link conditional six-face integral at the declared signed fixtures, exact tails, actual surrounding Gram/measure connection and preserved omission/symmetry controls. Thirty-one links remain fixed; no physical spectrum or full bulk/continuum claim.<br>Section 6.3.<br><i>Scope:</i> accepted<br>research/round18/forward/c2/report.md | <i>Use:</i> Two-link normalized six-face conditional integral<br><i>Change:</i> Restored all surrounding-only weights and partition reweighting | 31 links fixed; consistent $V \rightarrow V_{\text{dagger}}$ is a valid integrated exception<br><i>Next:</i> Three-link shared-middle chain |

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| Contribution and derivation                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Application and change                                                                                                                                     | Limitation and next obligation                                                                                                                      |
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| <b>HNM-C-R19-C1</b> (legacy R19-c1)**<br>Goal C Loop 1 only: exact finite static U-V-W chain integral on the actual four-cube graph with all other links fixed. This accepts a static mathematical checkpoint and common-V failed-independence discriminator; it does not accept physical energy/time-scale matching, dense convergence, continuum Yang-Mills, a spectral gap theorem or the Clay mass gap.<br>Section 6.3.<br><i>Scope:</i> accepted<br><b>research/round19/forward/c1/report.md</b> | <i>Use:</i> Actual three-link conditional chain integral<br><i>Change:</i> Shared-middle dependence retained and freezeW regression checked                | Thirty surrounding links remain fixed; static scale unmatched<br><i>Next:</i> Continuous signed-kappa result                                        |
| <b>HNM-C-R19-C2</b> (legacy R19-c2)**<br>Static common-V integral only; physical energy/time matching remains open. No Hamiltonian spectral gap or continuum claim. Eqs. (HNM-37).<br><i>Scope:</i> accepted<br><b>research/round19/forward/c2/report.md</b>                                                                                                                                                                                                                                          | <i>Use:</i> Certified continuous positivity and sign asymmetry<br><i>Change:</i> Converted exact coefficients into normalized all-interval inequalities    | Degree8 boundary failures do not prove nonpositivity; static model only<br><i>Next:</i> More surrounding variables or independent physical matching |
| <b>Static, spectral, and local-limit development / clock</b>                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                            |                                                                                                                                                     |
| <b>HNM-C-R20-F1</b> (legacy R20-f1)**<br>Even among local reversible diffusions the C1 static law does not identify an energy/time coefficient: $H_{\kappa,c} = c A_{\kappa}$ .<br>Section 9.1.<br><i>Scope:</i> accepted<br><b>research/round20/forward/f1/report.md</b>                                                                                                                                                                                                                             | <i>Use:</i> Demonstrate static clock non-identifiability<br><i>Change:</i> Same static density supports independently scaled reversible dynamics           | Added conditional generator, not physical lattice derivation<br><i>Next:</i> Full derivative closure and independently observed slope               |
| <b>HNM-C-R20-F2</b> (legacy R20-f2)**<br>Exact shared-middle gradient closure, action range[-5,7] and conditional diffusion gap $c/6$ on $ \kappa  \leq 1/8$ .<br>Section 9.1.<br><i>Scope:</i> accepted<br><b>research/round20/forward/f2/report.md</b>                                                                                                                                                                                                                                              | <i>Use:</i> Exact chosen-generator transform and conditional clock inverse<br><i>Change:</i> Retained actual common-V derivative and improved action range | No physical slope measured; $c$ unmatched to $\alpha$<br><i>Next:</i> Mobility deformations and identifying observables                             |

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| Contribution and derivation                                                                                                                                                                                                                                                                                                           | Application and change                                                                                                                                       | Limitation and next obligation                                                                                                  |
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| <b>HNM-C-R21-K1</b> (legacy R21-k1)**<br>A positive state-dependent mobility preserves the static density but changes reversible dynamics; two chosen Haar correlation slopes identify $c$ and $zeta$ within this conditional family.<br>Section 9.1.<br><i>Scope:</i> accepted<br><code>research/round21/forward/k1/report.md</code> | <i>Use:</i> Two-rate affine-mobility identification<br><i>Change:</i> Static law independent of $c$ , $zeta$ ; gradmobility drift retained                   | Haar inverse only known $\kappa = 0$ ; synthetic rates only<br><i>Next:</i> Uniform nonzero- $\kappa$ rank and conditioning     |
| <b>HNM-C-R21-K2</b> (legacy R21-k2)**<br>A tunable observable gives a uniform two-rate inverse across the declared static interval, while a positive blind cubic proves that two slopes do not identify arbitrary mobility.<br>Section 9.1.<br><i>Scope:</i> accepted<br><code>research/round21/forward/k2/report.md</code>           | <i>Use:</i> Robust rank certificate in declared affine family<br><i>Change:</i> Monotone double-integral determinant and blind cubic counter-model           | Finite-family injectivity does not identify arbitrary mobility<br><i>Next:</i> Third rate detects hidden cubic                  |
| <b>HNM-C-R21-L1</b> (legacy R21-l1)**<br>A third observable detects the previously hidden cubic mobility and gives an exact three-parameter inverse in the declared conditional family.<br>Eqs. (HNM-48).<br><i>Scope:</i> accepted<br><code>research/round21/forward/l1/report.md</code>                                             | <i>Use:</i> Three-parameter conditional inverse<br><i>Change:</i> Third observable exposes $p_3$ invisible to old pair                                       | Inverse at known $\kappa = 0$ ; polynomial itself classical<br><i>Next:</i> Finite-data nonidentifiability beyond chosen family |
| <b>HNM-C-R21-L2</b> (legacy R21-l2)**<br>Any finite list of initial correlation slopes leaves strictly positive mobility freedom; a second derivative detects the explicit hidden fifth-degree direction.<br>Section 9.1.<br><i>Scope:</i> accepted<br><code>research/round21/forward/l2/report.md</code>                             | <i>Use:</i> Exact finite-slope limitation and held-out curvature<br><i>Change:</i> Explicit positive $p_5$ mobility keeps $p_3$ slopes but changes generator | Does not show complete-time-curve ambiguity<br><i>Next:</i> Specify finite basis and reserve dynamical data                     |

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| Contribution and derivation                                                                                                                                                                                                                                                                                      | Application and change                                                                                                                                           | Limitation and next obligation                                                                                                                |
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| <b>Static, spectral, and local-limit development / gns</b>                                                                                                                                                                                                                                                       |                                                                                                                                                                  |                                                                                                                                               |
| <b>HNM-C-R20-G2</b> (legacy R20-g2)**<br>The E2 limiting local state GNS on the full factor algebra is the A2 representation with canonical generator H-e and its inherited gap.<br>Section 7.3.<br><i>Scope:</i> accepted<br>research/round20/forward/g2/report.md                                              | <i>Use:</i> Identify physical energy origin of full-algebra limiting state<br><i>Change:</i> Irreducibility and cyclicity proved on dense finite-excitation core | Gauss-only algebra not yet identified; $\beta < \delta$ restored<br><i>Next:</i> Nonvacuous invariant local algebra and physical GNS          |
| <b>HNM-C-R21-J1</b> (legacy R21-j1)**<br>An actual omitted Wilson loop has a nonzero gauge-invariant fluctuation in the perturbed dyadic vacuum, with a uniform rational variance floor.<br>Section 7.3.<br><i>Scope:</i> accepted<br>research/round21/forward/j1/report.md                                      | <i>Use:</i> Nonzero physical Wilson fluctuation<br><i>Change:</i> Transfers actual reference variance to perturbed dyadic ground                                 | Invariant cyclic space only inclusion at this stage<br><i>Next:</i> Prove equality with invariant Hilbert subspace                            |
| <b>HNM-C-R21-J2</b> (legacy R21-j2)**<br>Physical vacuum GNS space equals the gauge-invariant subspace of this representation, with the restricted shifted energy generator and a nonzero imaginary-time Wilson correlator.<br>Eqs. (HNM-42).<br><i>Scope:</i> accepted<br>research/round21/forward/j2/report.md | <i>Use:</i> Physical GNS space and imaginary-time spectrum<br><i>Change:</i> Finite gauge averaging proves reverse cyclic inclusion                              | No real-time magnitude decay or measured glueball mass<br><i>Next:</i> Different homogeneous/continuum representation requires separate proof |
| <b>Static, spectral, and local-limit development / profile</b>                                                                                                                                                                                                                                                   |                                                                                                                                                                  |                                                                                                                                               |
| <b>HNM-C-R20-H1</b> (legacy R20-h1)**<br>The continuous omitted-face profile has an exact budget and a rigorously delimited sufficient gap certificate at fixed physical scale.<br>Eqs. (HNM-44).<br><i>Scope:</i> accepted<br>research/round20/forward/h1/report.md                                             | <i>Use:</i> Continuous spatial decay budget<br><i>Change:</i> Exact class sums locate sufficient budget boundary                                                 | qcrit is not physical transition; q1 infinite sum diverges<br><i>Next:</i> Constant-budget rays and actual ground-state behavior              |

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| Contribution and derivation                                                                                                                                                                                                                                                                                           | Application and change                                                                                                                                | Limitation and next obligation                                                                                                       |
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| <b>HNM-C-R20-H2</b> (legacy R20-h2)**<br>Every strict uniform global perturbation-budget ray returns the actual ground projection to the selected-strip reference, with an exact vacuum variance and a quantitative rate.<br>Eqs. (HNM-45).<br><i>Scope:</i> accepted<br><b>research/round20/forward/h2/report.md</b> | <i>Use:</i> Ground-state return along fixed-budget rays<br><i>Change:</i> Exact two-free-link variance beats generic overlap bound                    | Endpoint retains selected strips; fixed omitted coefficients vanish<br><i>Next:</i> Strong operator and correlation limits           |
| <b>HNM-C-R21-M1</b> (legacy R21-m1)**<br>The fixed-budget summable perturbation converges strongly to zero, giving strong-resolvent and compact-time strong dynamics convergence to the selected-strip reference.<br>Section 8.1.<br><i>Scope:</i> accepted<br><b>research/round21/forward/m1/report.md</b>           | <i>Use:</i> Strong-resolvent and compact-time strong dynamics limit<br><i>Change:</i> Complete-factor commutator extends vacuum residual to dense set | Norm resolvent/dynamics status unresolved; endpoint is selected strip<br><i>Next:</i> Connected correlations on growing time windows |
| <b>Static, spectral, and local-limit development / windows</b>                                                                                                                                                                                                                                                        |                                                                                                                                                       |                                                                                                                                      |
| <b>HNM-C-R21-M2</b> (legacy R21-m2)**<br>Connected physical correlations converge on explicitly growing time windows in the canonical summable profile, with a nonzero actual physical Wilson fluctuation.<br>Section 8.2.<br><i>Scope:</i> accepted<br><b>research/round21/forward/m2/report.md</b>                  | <i>Use:</i> Connected physical correlations with $\gamma_3 < 3/2$<br><i>Change:</i> All centering, phases and factor supports retained                | Endpoint failure is failure of this upper certificate<br><i>Next:</i> Use stationary Heisenberg comparison                           |
| <b>HNM-C-R22-N1</b> (legacy R22-n1)**<br>Stationary Heisenberg comparison proves the sharper local connected-correlation bound in the canonical summable model.<br>Eqs. (HNM-46).<br><i>Scope:</i> accepted<br><b>research/round22/reverse/n1/report.md</b>                                                           | <i>Use:</i> Improved fixed-support growing-time certificate<br><i>Change:</i> Stationarity removes unnecessary time-times-vacuum-residual term        | Actual $\gamma_3$ behavior unresolved; no real-time decay assertion<br><i>Next:</i> Moving complete-factor support budgets           |
| <b>HNM-C-R22-N2</b> (legacy R22-n2)**<br>Complete tail-block cubes have an exact weighted incident-face budget, yielding a joint support/time convergence certificate.<br>Eqs. (HNM-47).<br><i>Scope:</i> accepted<br><b>research/round22/reverse/n2/report.md</b>                                                    | <i>Use:</i> Joint support and time design certificate<br><i>Change:</i> Exact incident budget including outgoing links and origin boundary            | Necessity only for nonnegative certificate; translated cubes differ<br><i>Next:</i> Actual endpoint or stronger dynamics estimate    |

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| Contribution and derivation                                                                                                                                                                                                                                                                                | Application and change                                                                                                                                       | Limitation and next obligation                                                                                                                                                        |
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| <b>Static, spectral, and local-limit development / inverse</b>                                                                                                                                                                                                                                             |                                                                                                                                                              |                                                                                                                                                                                       |
| <b>HNM-C-R22-O2</b> (legacy R22-o2)**<br>The bare-reference elimination has explicit all-support bounds, but its retained-diagonal loss majorants cannot certify an infinite iteration at nonzero coupling.<br>Eqs. (HNM-53).<br><i>Scope:</i> limited<br><b>research/round22/reverse/o2/report.md</b>     | <i>Use:</i> Precise failed iteration certificate<br><i>Change:</i> Retained diagonal term forces eventual failure of stated loss majorant                    | Not actual algorithm failure; regrouped1/64 differs indexed(9N+16)/(64(3N+1))<br><i>Next:</i> Structure-sensitive interacting inverse and full boundary terms                         |
| <b>HNM-C-R22-R1</b> (legacy R22-r1)**<br>The actual initial diagonal has a uniform local inverse, while its local homological generator leaves an exact nonzero SU2 crossing-boundary defect.<br>Eqs. (HNM-54).<br><i>Scope:</i> accepted<br><b>research/round22/reverse/r1/report.md</b>                  | <i>Use:</i> Initial interacting local inverse and boundary audit<br><i>Change:</i> Absorbs interior diagonal while proving actual SU2 crossing defect        | Diagnostic source not identified as generated residual; initial D only<br><i>Next:</i> Boundary-complete inverse response with domain control                                         |
| <b>HNM-C-R22-R2</b> (legacy R22-r2)**<br>The actual initial boundary-complete vacuum response converges with explicit spatial and graph tails; its source-sector and specified volume-weight limits are exact.<br>Eqs. (HNM-55).<br><i>Scope:</i> accepted<br><b>research/round22/reverse/r2/report.md</b> | <i>Use:</i> Initial boundary-complete response with spatial/graph tails<br><i>Change:</i> Proves actual response regularity and exact source-sector mismatch | Global vacuum cancellation not local identity extension; cardinality upper sum fails<br><i>Next:</i> Full-source cancellation plus useful weighted decomposition and later-stage gaps |
| <b>Static, spectral, and local-limit development / tree</b>                                                                                                                                                                                                                                                |                                                                                                                                                              |                                                                                                                                                                                       |
| <b>HNM-C-R22-P1</b> (legacy R22-p1)**<br>The prescribed section is nonclosable and its fitted clock fails the reserved correlation.<br>Section 9.2.<br><i>Scope:</i> accepted<br><b>research/round22/reverse/p1/report.md</b>                                                                              | <i>Use:</i> Falsify proposed physical section and clock<br><i>Change:</i> Nonclosable map and held-out full-time discrepancy on actual graph                 | Rejects this section only; other reductions remain possible<br><i>Next:</i> Genuine Haar maximal-tree isometry                                                                        |
| <b>HNM-C-R22-P2</b> (legacy R22-p2)**<br>A genuine Haar reduction exactly transports the finite electric dynamics; the declared conditional family fails the reserved data.<br>Eqs. (HNM-49).<br><i>Scope:</i> accepted<br><b>research/round22/reverse/p2/report.md</b>                                    | <i>Use:</i> Exact pure-electric selected reduction<br><i>Change:</i> Unitary16chord map, Haar adjoint, full tree/mixed derivatives and domains               | Different observable completion from P1; conditional family still fails<br><i>Next:</i> Restore all magnetic faces and quantify leakage                                               |

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| Contribution and derivation                                                                                                                                                                                                                                                                                 | Application and change                                                                                                                                                                                                                                                        | Limitation and next obligation                                                                                                                                                                                                                                                                 |
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| <b>Static, spectral, and local-limit development / memory</b>                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                |
| <b>HNM-C-R22-Q1</b> (legacy R22-q1)**<br>The full magnetic action has correlated positive conditional variance, proving selected-image leakage and a quantitative nonautonomous dynamic defect.<br>Eqs. (HNM-50).<br><i>Scope:</i> accepted<br><b>research/round22/reverse/q1/report.md</b>                 | <i>Use:</i> Exact actual-model magnetic leakage<br><i>Change:</i> Three correlated faces rule out autonomous selected compression                                                                                                                                             | Haar is not interacting ground; no operator-norm Taylor promotion<br><i>Next:</i> Exact projected memory and complementary spectrum                                                                                                                                                            |
| <b>HNM-C-R22-Q2</b> (legacy R22-q2)**<br>The actual magnetic projection has exact memory/resolvent identities, seven-energy fixed-channel leading matrices and explicit cubic full-Hilbert operator remainders.<br>Eqs. (HNM-51).<br><i>Scope:</i> accepted<br><b>research/round22/reverse/q2/report.md</b> | <i>Use:</i> Projected dynamics with explicit kernel and self-energy errors<br><i>Change:</i> Full return factor, offdiagonal weight1/4 and complete leading spectrum                                                                                                          | No autonomous2-channel closure; absolute not late-time relative accuracy<br><i>Next:</i> Controlled interacting projected approximation with original clock                                                                                                                                    |
| <b>Full-source, observable, and computed-heat results</b>                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                |
| <b>HNM-C-RECENT01</b> (legacy recent01)**<br>Nonzero indexed cubic source<br>Eqs. (HNM-56), (HNM-57).<br><i>Scope:</i> model-specific derivation, obstruction, or certified computation<br><b>research/round23/forward/s1/report.md</b>                                                                     | <i>Use:</i> Provides the actual order- $\tau^3$ off-diagonal source for the later filter and loading tests.<br><i>Change:</i> The cubic term cannot be dismissed as algebraically zero; it is an indexed term rather than the full grouped cubic remainder.                   | Initial retained diagonal and one specified indexed term only; $\tau=0$ is the trivial zero control. No all-sector inverse follows.<br><i>Next:</i> Control its complete crossing defect and the residual in the declared interaction norm.                                                    |
| <b>HNM-C-RECENT02</b> (legacy recent02)**<br>Interior inverse and complete connected crossing defect<br>Eqs. (HNM-58), (HNM-59).<br><i>Scope:</i> model-specific derivation, obstruction, or certified computation<br><b>research/round23/forward/s1/report.md</b>                                          | <i>Use:</i> One-step full-source inverse-with-defect and weighted connected-family certificate.<br><i>Change:</i> The local vacuum inverse is valid only after its boundary defect is included; reverse improves 192 Mell $e^{\wedge}(7\mu)$ to 168 Mell $e^{\wedge}(7\mu)$ . | The nonzero value of the actual crossing defect and an actual nonzero SU(2) resonance are not evaluated. No nonlinear induction or all-order summability.<br><i>Next:</i> Remove or contract the full defect without changing the action or replacing exterior identity by a vacuum projector. |
| <b>HNM-C-RECENT03</b> (legacy recent03)**<br>Compact-time graph-domain commutator with retained sinc residual<br>Eqs. (HNM-60), (HNM-61).<br><i>Scope:</i> model-specific derivation, obstruction, or certified computation<br><b>research/round23/forward/s2/report.md</b>                                 | <i>Use:</i> Bounded regulated inverse on the actual initial graph domain.<br><i>Change:</i> The commutator equation has a residual, including its entire zero-frequency block.                                                                                                | $0 < \Theta \leq 1/16$ at the historical coupling cap. $\Theta$ is a dimensionless proof parameter, not a calibrated physical time. No residual contraction is claimed here.<br><i>Next:</i> Quantify the residual on this particular cubic source, then preserve locality while reducing it.  |

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| Contribution and derivation                                                                                                                                                                                                                                                                                  | Application and change                                                                                                                                                                                                                                                                                                           | Limitation and next obligation                                                                                                                                                                                                                                                |
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| <b>HNM-C-RECENT04</b> (legacy recent04)**<br>Connected weighted bounds and infinite initial-operator identification<br>Eqs. (HNM-62), (HNM-63), (HNM-64), (HNM-65).<br><i>Scope</i> : model-specific derivation, obstruction, or certified computation<br><code>research/round23/forward/s2/report.md</code> | <i>Use</i> : Complete weight-2 short-time interaction bounds and a qualified infinite-volume initial-stage construction.<br><i>Change</i> : Finite-volume formulas acquire an actual infinite initial-generator interpretation in the forward proof; the extensive translated sum is still not a bounded Hilbert-space operator. | Forward-only infinite identification; no global norm unitary limit, later-diagonal iteration or all-time positive-weight bound.<br><i>Next</i> : Establish the topology and locality required for a later renormalized diagonal, with its own domain control.                 |
| <b>HNM-C-RECENT05</b> (legacy recent05)**<br>The short-time residual cannot be made small at the admitted cap<br>Eqs. (HNM-66), (HNM-67).<br><i>Scope</i> : model-specific derivation, obstruction, or certified computation<br><code>research/round24/forward/w1/report.md</code>                           | <i>Use</i> : Discriminating lower certificate against deleting the short-filter residual.<br><i>Change</i> : Smaller proof time preserves most of the source; this is an actual-source failure certificate rather than a generic toy resonance.                                                                                  | The statement concerns the original cubic source and admitted short-filter range. It does not establish a nonzero zero-frequency atom or forbid longer or different filters.<br><i>Next</i> : Use a filter with a controlled long-time or smooth-kernel mechanism.            |
| <b>HNM-C-RECENT06</b> (legacy recent06)**<br>Iterated filter converges on the vacuum column but retains resonant blocks<br>Eqs. (HNM-68), (HNM-69), (HNM-70).<br><i>Scope</i> : model-specific derivation, obstruction, or certified computation<br><code>research/round24/forward/w2/report.md</code>       | <i>Use</i> : Certified column inverse and a precise surviving-block target for all-sector work.<br><i>Change</i> : Column convergence and strong operator convergence are different results; neither compact resolvent nor pointwise spectral damping gives an operator-norm inverse.                                            | Old weighted estimates close only for $n \leq 3$ ; no nonzero actual resonance block is proved and no all-order weighted inverse follows.<br><i>Next</i> : Bound $D_{\text{G A}}$ directly and find a filter that contracts the complete source in a usable interaction norm. |
| <b>HNM-C-RECENT07</b> (legacy recent07)**<br>The resonant block and spectral windows are boundary controlled<br>Eqs. (HNM-71), (HNM-72).<br><i>Scope</i> : model-specific derivation, obstruction, or certified computation<br><code>research/round26/forward/ab1/report.md</code>                           | <i>Use</i> : Quantifies possible resonance from the complete connected boundary defect.<br><i>Change</i> : A full source can have a much smaller resonant block than its short-filter residual; this supplies an upper bound, not a nonzero resonance proof.                                                                     | The historical gate remains limited because actual crossing resonance values and all-sector inversion are unresolved.<br><i>Next</i> : Compute an actual resonance block or contract the entire source without assuming that the block vanishes.                              |
| <b>HNM-C-RECENT08</b> (legacy recent08)**<br>Gaussian boundary identity gives strict operator residual contraction<br>Eqs. (HNM-73).<br><i>Scope</i> : model-specific derivation, obstruction, or certified computation<br><code>research/round26/forward/ab2/report.md</code>                               | <i>Use</i> : Strict full operator residual contraction for the original cubic source at the old coupling cap.<br><i>Change</i> : Removes the W1 short-time obstruction through a different proof filter while keeping the same Hamiltonian.                                                                                      | Operator contraction does not itself give the original weighted interaction norm or contraction of later transformed sources.<br><i>Next</i> : Localize the Gaussian map quantitatively and obtain a compatible weighted contraction theorem.                                 |

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| Contribution and derivation                                                                                                                                                                                                                                                          | Application and change                                                                                                                                                                                                                                                                                                      | Limitation and next obligation                                                                                                                                                                                                                                                              |
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| <b>HNM-C-RECENT09</b> (legacy recent09)**<br>Gaussian collar locality and graph-domain truncation error Eqs. (HNM-74), (HNM-75).<br><i>Scope:</i> model-specific derivation, obstruction, or certified computation<br>research/round26/forward/ae1/report.md                         | <i>Use:</i> Explicit finite-collar approximations in operator norm and, by the reverse extension, the graph domain.<br><i>Change:</i> Quasi-locality is established in radius/diameter; this does not close an exponential penalty in support cardinality.                                                                  | The displayed upper budget $2^{\sim R}$ cannot absorb $\exp(\mu O(R^3))$ ; failure of that upper estimate is not a proof that the weighted norm diverges.<br><i>Next:</i> Change the combinatorial summation or kernel to establish finiteness in the original weighted norm.               |
| <b>HNM-C-RECENT10</b> (legacy recent10)**<br>Compact residual contraction and original weighted finiteness Eqs. (HNM-76), (HNM-77), (HNM-78).<br><i>Scope:</i> model-specific derivation, obstruction, or certified computation<br>research/round26/forward/ae2/report.md            | <i>Use:</i> First simultaneous original-source operator contraction and original weight-2 finiteness under a stricter coupling cap.<br><i>Change:</i> The old cap cannot satisfy the present contraction and series-radius inequalities; shrinking the declared parameter range resolves this method-level incompatibility. | Weighted finiteness is not weighted contraction, and the theorem is only for the original cubic source. No nonlinear homogeneous gap proof.<br><i>Next:</i> Prove contraction in the actual weighted interaction norm and propagate it through the nonlinear update.                        |
| <b>HNM-C-RECENT11</b> (legacy recent11)**<br>Full-space leading-memory return model and unshifted heat error Eqs. (HNM-79), (HNM-80), (HNM-81), (HNM-82).<br><i>Scope:</i> model-specific derivation, obstruction, or certified computation<br>research/round23/forward/t1/report.md | <i>Use:</i> Unique full-space Volterra approximation with fixed-time cubic and uniform absolute quadratic coupling errors.<br><i>Change:</i> The two-channel second derivative has a nonzero omitted defect; direct selected propagation is not the full magnetic heat.                                                     | Unshifted absolute error only at T1; fixed finite graph, constant coupling, Euclidean heat. No relative, real-time, graph-size or continuum theorem.<br><i>Next:</i> Center each full-space evolution at its own minimum and certify a finite implementation.                               |
| <b>HNM-C-RECENT12</b> (legacy recent12)**<br>Separately centered full-selected heat remains uniformly close Eqs. (HNM-83).<br><i>Scope:</i> model-specific derivation, obstruction, or certified computation<br>research/round23/forward/t2/report.md                                | <i>Use:</i> Time-uniform absolute comparison of two individually centered full-space heat models.<br><i>Change:</i> Correct energy centering is part of the observable approximation, not a harmless plotting convention.                                                                                                   | T2 uses correlated same-author derivations and skeptical self-review. No relative theorem for arbitrary inputs and no finite executable truncation at this stage.<br><i>Next:</i> Control a concrete finite physical basis and specify input preparation before claiming relative accuracy. |
| <b>HNM-C-RECENT13</b> (legacy recent13)**<br>Electric-energy cutoff and isolated-ground truncation theorem Eqs. (HNM-84), (HNM-85), (HNM-86).<br><i>Scope:</i> model-specific derivation, obstruction, or certified computation<br>research/round24/forward/x1/report.md             | <i>Use:</i> Explicit delayed uniform truncation and numerical perturbation criteria.<br><i>Change:</i> A mathematically adequate cutoff can be enormous and unassembled; the theorem alone is not an executed solver.                                                                                                       | The $R=1024$ matrix is not assembled. No claim of an arbitrary retained-block theorem from external real-time truncation literature.<br><i>Next:</i> Assemble a smaller physical basis and exploit its actual residual.                                                                     |

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| Contribution and derivation                                                                                                                                                                                                                                                                         | Application and change                                                                                                                                                                                                                                                                                 | Limitation and next obligation                                                                                                                                                                                                               |
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| <b>HNM-C-RECENT14</b> (legacy recent14)**<br>Representation-degree path truncation and the assembled 21-state seed<br>Eqs. (HNM-87), (HNM-88).<br><i>Scope</i> : model-specific derivation, obstruction, or certified computation<br>research/round24/forward/x1/report.md                          | <i>Use</i> : A second truncation mechanism plus the explicit 21-dimensional Ritz model.<br><i>Change</i> : Large theoretical cutoffs and the actually assembled small basis are different deliverables; the latter has nonzero omitted coupling.                                                       | X1 remains limited; its assembled 21-state space does not yet have the later residual and all-time relative certificates.<br><i>Next</i> : Compute all omitted channels and their norm, rather than infer accuracy from matrix size.         |
| <b>HNM-C-RECENT15</b> (legacy recent15)**<br>Complete residual of the 21-state ground trial<br>Eqs. (HNM-89), (HNM-90).<br><i>Scope</i> : model-specific derivation, obstruction, or certified computation<br>research/round24/forward/x2/report.md                                                 | <i>Use</i> : Exact model-specific residual norm and spectral distribution for ground and heat certification.<br><i>Change</i> : The residual is a physical channel calculation, not a norm estimate based solely on discarded basis count.                                                             | The electric spectral weights concern this residual vector; they are not the full interacting spectrum.<br><i>Next</i> : Use both its norm and energy moment to bound the ground shift and late heat.                                        |
| <b>HNM-C-RECENT16</b> (legacy recent16)**<br>Two-sided ground enclosure and evaluated late heat<br>Eqs. (HNM-91), (HNM-92).<br><i>Scope</i> : model-specific derivation, obstruction, or certified computation<br>research/round24/forward/x2/report.md                                             | <i>Use</i> : Certified late-time absolute full-space heat and evaluated output in the 21-state seed.<br><i>Change</i> : A two-sided energy interval proves strict Ritz bias, enabling the later relative-error falsifier.                                                                              | Absolute late-time theorem; arbitrary nearly ground-null input can still have unbounded relative error.<br><i>Next</i> : Declare a preparation class and bound the true output denominator.                                                  |
| <b>HNM-C-RECENT17</b> (legacy recent17)**<br>Uniform relative heat accuracy fails on ground-null inputs<br>Eqs. (HNM-93).<br><i>Scope</i> : model-specific derivation, obstruction, or certified computation<br>research/round24/forward/z1/report.md                                               | <i>Use</i> : Discriminating falsifier against extending an absolute semigroup estimate to arbitrary-input relative accuracy.<br><i>Change</i> : Ground overlap and denominator convention must be part of the theorem; exact agreement at $\lambda=0$ or $\sigma=0$ does not repair late-time failure. | Constructed bad inputs are not the later normalized vacuum-neighborhood preparation class.<br><i>Next</i> : Prove relative accuracy on an explicit input class whose true output has a certified floor.                                      |
| <b>HNM-C-RECENT18</b> (legacy recent18)**<br>Exact omitted Gram matrix and all-time relative repair for prepared inputs<br>Eqs. (HNM-94), (HNM-95), (HNM-96), (HNM-97).<br><i>Scope</i> : model-specific derivation, obstruction, or certified computation<br>research/round24/forward/z2/report.md | <i>Use</i> : Full-time same-input true-output-relative error with stated preparation hypotheses.<br><i>Change</i> : The obstruction is repaired by an input contract and exact dynamic loading, not by redefining away preparation bias.                                                               | Different forward/reverse preparation sets must not be conflated. No arbitrary-ground-null-input relative claim.<br><i>Next</i> : Include the complete omitted image as new physical channels and check whether the error actually improves. |

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| Contribution and derivation                                                                                                                                                                                                                                                                                 | Application and change                                                                                                                                                                                                                             | Limitation and next obligation                                                                                                                                                                                                                                                                                                                                                                                                                     |
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| <b>HNM-C-RECENT19</b> (legacy recent19)**<br>The 293-state physical channel enlargement<br>Eqs. (HNM-98), (HNM-99).<br><i>Scope:</i> model-specific derivation, obstruction, or certified computation<br>research/round26/forward/ac1/report.md                                                             | <i>Use:</i> An actually assembled sparse gauge-invariant 293-dimensional Galerkin matrix.<br><i>Change:</i> Exact zero leakage on the old subspace is a first-step property; it is not invariance under the enlarged retained evolution.           | The 293-state space is not a complete electric-energy shell. Channel count alone does not prove the desired heat improvement.<br><i>Next:</i> Bound delayed occupation of the new channels before applying their remaining omitted coupling.                                                                                                                                                                                                       |
| <b>HNM-C-RECENT20</b> (legacy recent20)**<br>Improved ground certificate does not yet prove an improved heat bound<br>Eqs. (HNM-100).<br><i>Scope:</i> model-specific derivation, obstruction, or certified computation<br>research/round26/forward/ac1/report.md                                           | <i>Use:</i> Ground-state improvement and a documented failed coarse all-time error target.<br><i>Change:</i> The historical AC1 gate correctly remains limited despite successful matrix assembly and a tighter ground certificate.                | No implication from static ground improvement to the requested dynamic relative improvement.<br><i>Next:</i> Use the initially empty new channels and their actual loading equation.                                                                                                                                                                                                                                                               |
| <b>HNM-C-RECENT21</b> (legacy recent21)**<br>Delayed channel loading closes the improved all-time relative bound<br>Eqs. (HNM-101), (HNM-102), (HNM-103), (HNM-104), (HNM-105).<br><i>Scope:</i> model-specific derivation, obstruction, or certified computation<br>research/round26/forward/ac2/report.md | <i>Use:</i> Strict improvement over the Z2 physical all-time preparation-class certificate.<br><i>Change:</i> The necessary improvement comes from the time ordering of physical channels, not the dimension 293 itself.                           | The input remains in the original P21 class. No all-input relative bound, real-time theorem, graph-size uniformity or executed solver for every coupling.<br><i>Next:</i> Execute the finite heat action with rigorous ground centering and numerical output error.                                                                                                                                                                                |
| <b>HNM-C-RECENT22</b> (legacy recent22)**<br>Actual 293-coordinate heat at one declared time<br>Eqs. (HNM-106).<br><i>Scope:</i> model-specific derivation, obstruction, or certified computation<br>research/round26/forward/af1/report.md                                                                 | <i>Use:</i> Executed 293-coordinate output with Haar metric, polynomial, centering and export errors accounted for.<br><i>Change:</i> The heat theorem becomes a finite evaluated artifact at the fixed parameter-time-input contract.             | AF1 executes one coupling, time and vacuum input only; it is not the all-time evaluator.<br><i>Next:</i> Prevent approximate centering from causing secular error and cover the full declared preparation class.                                                                                                                                                                                                                                   |
| <b>HNM-C-RECENT23</b> (legacy recent23)**<br>Piecewise all-time computed heat and the fixed numerical scope<br>Eqs. (HNM-107), (HNM-108), (HNM-109).<br><i>Scope:</i> model-specific derivation, obstruction, or certified computation<br>research/round26/forward/af2/report.md                            | <i>Use:</i> Complete executed all-time finite-graph heat approximation on the declared preparation class.<br><i>Change:</i> The early/late switch makes the error budget uniform while retaining a finite computation at requested rational times. | Fixed lambda=0.01 only. Reverse CLI accepts real rational 293-vectors, with 11e-30 rounding; complex operator bounds and forward complex demonstrations use the separate 15e-40 allowance, not an arbitrary-complex CLI. A tiny switch discontinuity is allowed within budget.<br><i>Next:</i> Validate graph-size or observable dependence and an explicit physical scale-identification contract; do not infer continuum or calibration results. |

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| Contribution and derivation                                                                                                                                                                                                                                                                                  | Application and change                                                                                                                                                                                                                                                                            | Limitation and next obligation                                                                                                                                                                                                                                                                         |
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| <b>HNM-C-RECENT24</b> (legacy recent24)**<br>Canonical rank observable has an actual eight-link first-order resonance<br>Eqs. (HNM-110), (HNM-111), (HNM-112), (HNM-113).<br><i>Scope</i> : model-specific derivation, obstruction, or certified computation<br><b>research/round23/forward/u1/report.md</b> | <i>Use</i> : A concrete nonzero secular coefficient for the canonical summable family.<br><i>Change</i> : This supplies a physical-model resonance mechanism rather than a two-level analogy, while leaving higher bulk terms unresolved at U1.                                                   | Rank-two bounded operator, not Wilson multiplication. U1 is limited and has only same-author self-review. No exact endpoint limit yet.<br><i>Next</i> : Bound all connected higher orders of the full model uniformly on the stated endpoint window.                                                   |
| <b>HNM-C-RECENT25</b> (legacy recent25)**<br>Full-system endpoint witness from connected-word tails<br>Eqs. (HNM-114), (HNM-115), (HNM-116).<br><i>Scope</i> : model-specific derivation, obstruction, or certified computation<br><b>research/round23/forward/u2/report.md</b>                              | <i>Use</i> : A full-model counterexample to extending the earlier subcritical window to the critical cubic time exponent for every bounded local probe.<br><i>Change</i> : The gamma=3 endpoint obstruction is established in the frozen canonical model; it is not merely a formal secular term. | Existence of a suitable probe, not a universal statement for all observables. No exact trajectory at U2, Wilson measurement or physical scale calibration. Same-author self-review.<br><i>Next</i> : Provide an operational readout and prove an actual scalar limit using controlled spatial collars. |
| <b>HNM-C-RECENT26</b> (legacy recent26)**<br>Finite-outcome and coherent readout of the complex correlation<br>Eqs. (HNM-117), (HNM-118).<br><i>Scope</i> : model-specific derivation, obstruction, or certified computation<br><b>research/round24/forward/v1/report.md</b>                                 | <i>Use</i> : An ideal coherent measurement contract with explicit operator approximation tolerance.<br><i>Change</i> : Finite outcome count does not establish a finite laboratory implementation; complex order must be measured coherently.                                                     | Ideal bounded operators and controlled unitaries are assumed, not physically calibrated or implemented.<br><i>Next</i> : Include clock, state, gate and statistical errors in the same correlation budget.                                                                                             |
| <b>HNM-C-RECENT27</b> (legacy recent27)**<br>Full-Haar operator-norm obstruction to multiplying-observable replacement<br>Eqs. (HNM-119).<br><i>Scope</i> : model-specific derivation, obstruction, or certified computation<br><b>research/round24/forward/v1/report.md</b>                                 | <i>Use</i> : Rules out an unjustified identification of the original rank observable with Wilson multiplication.<br><i>Change</i> : Same vacuum-created vector or a formal compressed agreement does not imply that the complete observables are equal.                                           | This full-space norm obstruction does not forbid separately analyzing a Wilson multiplier or a declared restricted norm.<br><i>Next</i> : Construct and certify a Wilson observable on its own operator and state contract.                                                                            |
| <b>HNM-C-RECENT28</b> (legacy recent28)**<br>Clock and finite-sample error budget for the coherent protocol<br>Eqs. (HNM-120), (HNM-121).<br><i>Scope</i> : model-specific derivation, obstruction, or certified computation<br><b>research/round24/forward/v2/report.md</b>                                 | <i>Use</i> : A complete ideal-protocol error and shot-count certificate at explicit finite q.<br><i>Change</i> : Operational tolerances are mathematically quantified and can be extremely demanding even when the ideal signal is nonzero.                                                       | Sufficient theoretical budget, not evidence of experimental attainability. State, gate and timing conventions differ between directions and are not interchangeable.<br><i>Next</i> : Find feasible apparatus-specific approximations with a justified physical scale calibration.                     |

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| Contribution and derivation                                                                                                                                                                                                                                                                     | Application and change                                                                                                                                                                                                                                                      | Limitation and next obligation                                                                                                                                                                                                                                                                                        |
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| <b>HNM-C-RECENT29</b> (legacy recent29)**<br>Complete-factor spatial collars with factorial connected tails<br>Eqs. (HNM-122), (HNM-123).<br><i>Scope:</i> model-specific derivation, obstruction, or certified computation<br>research/round24/forward/y1/report.md                            | <i>Use:</i> Finite spatial reduction with explicit error that preserves the endpoint signal.<br><i>Change:</i> A controlled physical collar replaces an arbitrary geometric crop; its small tail does not by itself justify averaging the regional spectrum.                | Collars and counts depend on the stated forward/reverse construction. No regional matrix or infinite limiting generator is computed at this stage.<br><i>Next:</i> Establish strong averaging on fixed energy blocks and then pass the scalar limit through the collar bound.                                         |
| <b>HNM-C-RECENT30</b> (legacy recent30)**<br>Actual scalar endpoint limit from strong block averaging<br>Eqs. (HNM-124), (HNM-125), (HNM-126), (HNM-127), (HNM-128).<br><i>Scope:</i> model-specific derivation, obstruction, or certified computation<br>research/round24/forward/y2/report.md | <i>Use:</i> An actual limiting endpoint function with quantitative small-window location.<br><i>Change:</i> U2 nonconvergence is strengthened to existence of the relevant scalar limit; the whole energy shell and the q=1 global generator are not thereby constructed.   | Fixed small window, scalar limit and strong block topology. Compact resolvent is not a global operator-norm averaging theorem.<br><i>Next:</i> Compute a reducing reached component or else retain the full regional energy block.                                                                                    |
| <b>HNM-C-RECENT31</b> (legacy recent31)**<br>Exact sixteen-cycle compression and spectral measure<br>Eqs. (HNM-129).<br><i>Scope:</i> model-specific derivation, obstruction, or certified computation<br>research/round25/solo/aa1/report.md                                                   | <i>Use:</i> An exact finite spectral candidate for the endpoint function.<br><i>Change:</i> An autonomous two-state approximation is replaced by the correct sixteen-state compression, while exterior closure remains an explicit missing hypothesis at AA1.               | AA1 gate is limited. The cube compression alone does not transfer to the surrounding or full model. Same-author serial self-review, not independent review.<br><i>Next:</i> Prove that exterior faces cannot return resonantly into the shell and identify a reducing reached component.                              |
| <b>HNM-C-RECENT32</b> (legacy recent32)**<br>Exterior selection proves the reached component reduces<br>Eqs. (HNM-130), (HNM-131).<br><i>Scope:</i> model-specific derivation, obstruction, or certified computation<br>research/round25/solo/aa2/report.md                                     | <i>Use:</i> Transfers the exact spectral measure to the actual canonical endpoint scalar on the admitted window.<br><i>Change:</i> AA1 compression becomes a full-model result because exterior resonant escape is excluded, not because the entire energy shell is finite. | Only the reached component of the averaged generator is reducing; instantaneous evolution can leave it. No large-z stationary-limit claim follows from the analytic formula. Same-author review.<br><i>Next:</i> Attach finite-q and arithmetic errors to the full correlation and analyze simpler multiplier probes. |
| <b>HNM-C-RECENT33</b> (legacy recent33)**<br>Finite-q approach and numerical endpoint enclosure<br>Eqs. (HNM-132).<br><i>Scope:</i> model-specific derivation, obstruction, or certified computation<br>research/round25/solo/aa2/report.md                                                     | <i>Use:</i> An explicit finite-q scalar enclosure for the original rank observable.<br><i>Change:</i> The endpoint spectral formula is accompanied by an actual full-model finite-q error budget.                                                                           | Numerical point is extremely close to q=1 and has an enormous uncalibrated physical duration. No experimental feasibility or larger-z theorem.<br><i>Next:</i> Check the same full finite-q transfer for actual Wilson multipliers with their own operator norms.                                                     |

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| Contribution and derivation                                                                                                                                                                                                                            | Application and change                                                                                                                                                                                                                                                                                     | Limitation and next obligation                                                                                                                                                                                                                                                                                           |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>HNM-C-RECENT34</b> (legacy recent34)**<br>Nonzero-variance elementary Wilson probe is endpoint-null Eqs. (HNM-133).<br><i>Scope</i> : model-specific derivation, obstruction, or certified computation<br>research/round26/forward/ad1/report.md    | <i>Use</i> : A physical-model null control against inferring a slow coupling from a constant phase-removed scalar.<br><i>Change</i> : Nonzero variance and an apparently clean signal are insufficient for scale identification; this probe has no endpoint sensitivity to that coupling.                  | AD1 remains limited as a rate-bridge target. A constant endpoint does not rule out subleading finite-q dependence or sensitivity of another observable.<br><i>Next</i> : Choose a Wilson multiplier whose created vector reaches a nontrivial averaged block and prove the full scalar transfer.                         |
| <b>HNM-C-RECENT35</b> (legacy recent35)**<br>Informative single-cycle and sum Wilson endpoints Eqs. (HNM-134), (HNM-135).<br><i>Scope</i> : model-specific derivation, obstruction, or certified computation<br>research/round26/forward/ad2/report.md | <i>Use</i> : Actual Wilson-multiplication probes with quadratic or linear small-window endpoint sensitivity.<br><i>Change</i> : The rank observable is no longer silently identified with multiplication; different physical observables can share an endpoint spectral component yet differ as operators. | Fixed canonical summable model and $0 < z \leq 1e-6$ . The coefficient is not an empirical coupling calibration, and no large-z limit or homogeneous/continuum transfer is proved.<br><i>Next</i> : Declare the observable, clock, physical scales and comparison norm in an actual parameter-identification experiment. |

## C.1 Round30 additions to the complete catalog

The inherited rows above preserve their original gates. The current same-state AT1 extension is separately reviewed and is not a retroactive change to AO or AQ. These three additional records complete the current catalog; their equation aliases do not add further discoveries.

**HNM-C-AT1: Hrudyay same-state Wilson energy certificate** In the actual numerical-cap AQ state at either sign of  $|\tau| \leq 10^{-8}$ , the centered original Wilson vector satisfies  $\chi \in D(H_{\text{phys}})$ . Its second-energy moment is at most  $\alpha^2(36 + 98|\tau|) < 37\alpha^2$ , and its first-energy moment equals  $\alpha\omega_{\text{num}}(1 - W^2)$ . **Application.** Makes subsequent spectral and inverse-energy statements meaningful in the numerical-cap state itself. **Surviving limitations.** No equality of finite-to-infinite second moments No AQ/AO/AN state identification No particle pole, exact mass or dispersion No uniqueness of all thermodynamic states or whole-sequence convergence No continuum Yang-Mills construction Scientific priority unverified; established methods retain attribution **Evidence.** Section 54; research/round30/advisor/at1-gate.json

**HNM-C-AT2: Hrudyay spectral-window and inverse-energy certificate** For the same state, the unnormalized spectral mass satisfies  $\nu([\alpha/16, 8\alpha]) \geq 307992/1984375$ . On the physical vacuum-orthogonal sector let  $R = \langle \chi, H_{\text{phys}}^{-1} \chi \rangle$ . Its dimensionless value has lower bound  $3843876001/47000000000$  and upper bound  $28462237549/7503125000$ : these are bounds on  $\alpha R$ , not an evaluated response or a unique spectral measure. **Application.** Guarantees nonzero weight in one finite energy window and bounds the inverse-energy quadratic form. Shows that these moments do not identify a particle pole or the actual AQ spectrum. **Surviving limitations.** The window is unnormalized spectral mass, not probability or a hard cutoff No evaluated actual AQ inverse response or optimality No particle pole, exact mass or identified spectral type Abstract equal-moment fixtures are not actual lattice realizations No static susceptibility or deformed thermodynamic-state differentiability No continuum result; scientific priority unverified **Evidence.** Section 57; research/round30/advisor/at2-gate.json

**HNM-C-AT3: Hruday finite Euclidean readout certificate** The fixed 4097-node protocol conditionally encloses  $I = \alpha R$  when supplied certified centered AQ samples with absolute error at most  $10^{-6}$ . Exact rational data give interval width below  $1/500$ ; arithmetic bins contribute their explicitly bounded trapezoid width. Both executed abstract A/B benchmarks pass, and their full admissible-error output envelopes are disjoint. No actual AQ correlation samples or response value are supplied. **Application.** Evaluate externally supplied certified centered samples through a precise rational API. Distinguish the specified equal-moment abstract controls without claiming that they are the actual AQ spectrum. **Surviving limitations.** No actual AQ correlation samples or inverse response evaluated Forward permits arbitrary arithmetic bins and reports width verdict; reverse admits bins  $\leq 1e - 12$  and guarantees the width target Input validation does not certify centering, model/state provenance or scientific error bounds A/B and the separately disclosed damaging tail control are abstract measures, not AQ simulations No susceptibility, identified pole, optimal design, state uniqueness or continuum construction Scientific priority unverified; established methods retain attribution Exactly three investigations; no fourth scientific execution **Evidence.** Section 60; `research/round30/advisor/at3-gate.json`

## D Complete historical record

This table preserves all 121 recorded study or loop entries. It is an index of the history, not a list of 121 new discoveries. Historical verdicts retain their original scope. The contribution and skeptic ledgers identify the fresh manuscript checks; inclusion here does not claim that every old production simulation was rerun. Complete source paths and SHA-256 bindings are in `audit/history-ledger.json`.

Table 18: All recorded entries, grouped by round.

| Record           | Historical entry                             | Verdict and first source                                                              |
|------------------|----------------------------------------------|---------------------------------------------------------------------------------------|
| R3<br>PRODUCTION | <b>Source-off electric reversal</b>          | historical study — scope below<br><code>evidence/qeg-research/round3/README.md</code> |
| R3<br>REFINEMENT | <b>Regulator and node refinements</b>        | historical study — scope below<br><code>evidence/qeg-research/round3/README.md</code> |
| R3<br>CONTROLS   | <b>Matching and independent formulations</b> | historical study — scope below<br><code>evidence/qeg-research/round3/README.md</code> |
| R3<br>BENCHMARK  | <b>Retained precision failure</b>            | historical study — scope below<br><code>evidence/qeg-research/round3/README.md</code> |
| R4<br>RESPONSE   | <b>Source-amplitude tangent</b>              | historical study — scope below<br><code>evidence/qeg-research/round4/README.md</code> |
| R4<br>AUDIT      | <b>Source and report reconciliation</b>      | historical study — scope below<br><code>evidence/qeg-research/round4/README.md</code> |
| R5<br>THEOREM    | <b>Finite-model existence and response</b>   | historical study — scope below<br><code>evidence/qeg-research/round5/README.md</code> |
| R5<br>PLANNER    | <b>Two-front proof planning</b>              | historical study — scope below<br><code>evidence/qeg-research/round5/README.md</code> |

Continued on next page

| Record                   | Historical entry                                        | Verdict and first source                                                                 |
|--------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------|
| R5<br>GRAVITY            | <b>Bianchi-I constraint bridge</b>                      | historical study — scope below<br><a href="#">evidence/qeg-research/round5/README.md</a> |
| R6<br>COEFFICIENT        | <b>Exact finite-window coefficient</b>                  | historical study — scope below<br><a href="#">evidence/qeg-research/round6/README.md</a> |
| R6<br>COFINAL            | <b>Cofinal cutoff obstruction</b>                       | historical study — scope below<br><a href="#">evidence/qeg-research/round6/README.md</a> |
| R6<br>TANGENT            | <b>Stationary second variation</b>                      | historical study — scope below<br><a href="#">evidence/qeg-research/round6/README.md</a> |
| R6<br>REPAIR             | <b>Core and coefficient audit</b>                       | historical study — scope below<br><a href="#">evidence/qeg-research/round6/README.md</a> |
| R7<br>FINITE-<br>SCALAR  | <b>Scalar-coupled finite model</b>                      | historical study — scope below<br><a href="#">evidence/qeg-research/round7/README.md</a> |
| R7<br>GRAVITY-<br>SCALAR | <b>Reduced gravitational simulation</b>                 | historical study — scope below<br><a href="#">evidence/qeg-research/round7/README.md</a> |
| R7<br>SUBTRACTION        | <b>Reference-subtraction and bounded-addition tests</b> | historical study — scope below<br><a href="#">evidence/qeg-research/round7/README.md</a> |
| R8<br>SPECTRAL           | <b>Spectral and finite-volume counterexamples</b>       | historical study — scope below<br><a href="#">research/round8/README.md</a>              |
| R8<br>DELAYED            | <b>Delayed scalar-response study</b>                    | historical study — scope below<br><a href="#">research/round8/README.md</a>              |
| R8<br>PROOF              | <b>Conditional proof replay</b>                         | historical study — scope below<br><a href="#">research/round8/README.md</a>              |
| R9<br>HAAR               | <b>Nonabelian identity hierarchy</b>                    | historical study — scope below<br><a href="#">research/round9/README.md</a>              |
| R9<br>WILSON             | <b>Four-dimensional Wilson lattice runs</b>             | historical study — scope below<br><a href="#">research/round9/README.md</a>              |
| R9<br>TRANSFER           | <b>Central-convolution spectrum</b>                     | historical study — scope below<br><a href="#">research/round9/README.md</a>              |
| R10<br>GAP               | <b>Certified full one-plaquette gap</b>                 | historical study — scope below<br><a href="#">research/round10/README.md</a>             |
| R10<br>DRIVE             | <b>Driven one-square diagnostics</b>                    | historical study — scope below<br><a href="#">research/round10/README.md</a>             |
| R10<br>QUOTIENT          | <b>Three-trace reconstruction</b>                       | historical study — scope below<br><a href="#">research/round10/README.md</a>             |
| R11<br>OPERATOR          | <b>Physical two-plaquette operator</b>                  | historical study — scope below<br><a href="#">research/round11/README.md</a>             |
| R11<br>GAP               | <b>Continuous two-plaquette gap certificate</b>         | historical study — scope below<br><a href="#">research/round11/README.md</a>             |
| R11<br>DYNAMICS          | <b>Driven checks and falsifiers</b>                     | historical study — scope below<br><a href="#">research/round11/README.md</a>             |

Continued on next page

| Record              | Historical entry                                                                                                                                            | Verdict and first source                                                                                                           |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| R12<br>DYNAMIC-TAIL | <b>Rigorous representation-error theorem</b>                                                                                                                | historical study — scope below<br><a href="#">research/round12/README.md</a>                                                       |
| R12<br>VOLUME       | <b>Growing-graph bound and counter-benchmark</b>                                                                                                            | historical study — scope below<br><a href="#">research/round12/README.md</a>                                                       |
| R12<br>CLOSURE      | <b>Compact hierarchy audit</b>                                                                                                                              | historical study — scope below<br><a href="#">research/round12/README.md</a>                                                       |
| R13<br>LOCALITY     | <b>Local finite-time volume limit</b>                                                                                                                       | historical study — scope below<br><a href="#">research/round13/README.md</a>                                                       |
| R13<br>STABILITY    | <b>Qualitative homogeneous stability</b>                                                                                                                    | historical study — scope below<br><a href="#">research/round13/README.md</a>                                                       |
| R13<br>MOMENTS      | <b>Compact moment certificates and uniqueness</b>                                                                                                           | historical study — scope below<br><a href="#">research/round13/README.md</a>                                                       |
| R13<br>RESPONSE     | <b>Exact scalar response</b>                                                                                                                                | historical study — scope below<br><a href="#">research/round13/README.md</a>                                                       |
| R14<br>LOOP1        | <b>Exact conditional Haar reduction with removable zero field.</b>                                                                                          | accepted within declared finite-model scope<br><a href="#">research/round14/advisor/loop1_gate.json</a>                            |
| R14<br>LOOP2        | <b>23 exact finite-measure covariance enclosures independently replayed using characters.</b>                                                               | accepted within declared finite-model scope<br><a href="#">research/round14/advisor/loop2_gate.json</a>                            |
| R15<br>A1           | <b>All8 positive exact midpoints, all8 coarse transported lower bounds insufficient.</b>                                                                    | accepted correctly diagnosed insufficiency<br><a href="#">research/round15/advisor/loop-gates/A1.json</a>                          |
| R15<br>A2           | <b>Refined failing cell radii rather than increasing already sufficient point precision.</b>                                                                | accepted exact complete positive cover<br><a href="#">research/round15/advisor/loop-gates/A2.json</a>                              |
| R15<br>B1           | <b>Closed cube has six-way dependence while any five face class observables factorize; use actual character gluing and distinct face derivatives in B2.</b> | accepted exact finite cube graph and Haar identities<br><a href="#">research/round15/advisor/loop-gates/B1.json</a>                |
| R15<br>B2           | <b>Character dimension and separate face derivatives retained; coarse degree failures remain explicit.</b>                                                  | accepted exact finite cube integral and rejected factorization<br><a href="#">research/round15/advisor/loop-gates/B2.json</a>      |
| R15<br>C1           | <b>Cube lower bound is zero at common <math>\lambda=\alpha/2</math>; accepted B1 trial moments permit a variational ground-energy upper bound.</b>          | accepted finite-graph side theorem and retained original source gap<br><a href="#">research/round15/advisor/loop-gates/C1.json</a> |
| R15<br>C2           | <b>The independent E1 lower bound and trial E0 upper bound repair the C1 zero-margin fixture; no trial gap was substituted for the full gap.</b>            | accepted finite physical cube variational improvement<br><a href="#">research/round15/advisor/loop-gates/C2.json</a>               |
| R16<br>LOOP1        | <b>Exact finite two-cube Haar/fusion identities and separate conditional physical energy-scale audit.</b>                                                   | accepted<br><a href="#">research/round16/advisor/loop1_gate.json</a>                                                               |
| R16<br>LOOP2        | <b>Positive normalized all-eleven observable difference for outer1/8 and shared1/8 versus0, exact width<math>\leq 1e-12</math>.</b>                         | accepted<br><a href="#">research/round16/advisor/loop2_gate.json</a>                                                               |

Continued on next page

| Record    | Historical entry                                                                                                                                                                                                                                                      | Verdict and first source                          |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| R17<br>A1 | Exact local projector and incidence bounds; physical counterexample to the full pure relative-form shortcut. Dense uniform threshold remains open.                                                                                                                    | accepted<br>research/round17/advisor/a1-gate.json |
| R17<br>A2 | Volume-uniform physical gap for the pairwise link-disjoint magnetic-support family with a common positive energy scale; dense overlapping family remains open.                                                                                                        | accepted<br>research/round17/advisor/a2-gate.json |
| R17<br>B1 | Improved full physical gap lower bound on the dense eleven-face two-cube graph, positive for common $ \lambda /\alpha < 12/43$ ; finite volume only.                                                                                                                  | accepted<br>research/round17/advisor/b1-gate.json |
| R17<br>B2 | Exact positive full physical gap lower $\alpha \cdot 1388/284767457$ at the dense finite coupling $\lambda/\alpha = 12/43$ , with a proved open variational-amplitude interval.                                                                                       | accepted<br>research/round17/advisor/b2-gate.json |
| R17<br>C1 | Complete four-adjoint Haar tensor and realized four-cube conditional boundary; no bulk or spectral claim.                                                                                                                                                             | accepted<br>research/round17/advisor/c1-gate.json |
| R17<br>C2 | Same-action conditional four-face expectations with complete rational numerator and partition errors; action-only observable closure rejected, full bulk and spectrum open.                                                                                           | accepted<br>research/round17/advisor/c2-gate.json |
| R18<br>A1 | Finite dressed three-square bridge; physical gap at least $\alpha/8$ throughout the declared primary family; no homogeneous volume extension.                                                                                                                         | accepted<br>research/round18/advisor/a1-gate.json |
| R18<br>A2 | All finite volumes of the specified spatially decaying inhomogeneous family have physical gap at least $7\alpha_{\min}/64$ ; boundary reclassification and homogeneous dense/thermodynamic limits remain explicit.                                                    | accepted<br>research/round18/advisor/a2-gate.json |
| R18<br>B1 | Complete physical complement threshold $9\alpha/2$ and exact QVP Gram on the dense two-cube graph; no finite trial spectrum substituted for the entire omitted space.                                                                                                 | accepted<br>research/round18/advisor/b1-gate.json |
| R18<br>B2 | Full untruncated physical two-cube signed coefficient-box gap lower bound, separate endpoint ground trial, and bare-coupling domain obstruction. No volume-uniform or continuum claim.                                                                                | accepted<br>research/round18/advisor/b2-gate.json |
| R18<br>C1 | Complete joint-Gram sufficiency for the declared central S3 observable/action class, exact rational admissibility including singular cases, and independently matched degree 0–6 diagnostics. No minimality, finite-tail, surrounding measure or mass-gap conclusion. | accepted<br>research/round18/advisor/c1-gate.json |

Continued on next page

| Record    | Historical entry                                                                                                                                                                                                                                                                                                                                                                                                                | Verdict and first source                          |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| R18<br>C2 | Full normalized two-link conditional six-face integral at the declared signed fixtures, exact tails, actual surrounding Gram/measure connection and preserved omission/symmetry controls. Thirty-one links remain fixed; no physical spectrum or full bulk/continuum claim.                                                                                                                                                     | accepted<br>research/round18/advisor/c2-gate.json |
| R19<br>A1 | For every reachable selected clipped component L, M, R, LM, MR or LMR under the frozen coefficient bounds $ \lambda_L /\alpha \leq 1/2$ , $ \mu_M /\alpha \leq 1/8$ and $ \lambda_R /\alpha \leq 1/2$ , the full-link local gap lower estimate is at least $\alpha/8$ . Positive full-link uniqueness and gauge invariance permit the same lower estimate after Gauss-sector restriction for the finite local component result. | accepted<br>research/round19/advisor/a1-gate.json |
| R19<br>A2 | Goal A Loop 2 only: direct incomplete product-representation construction for a summable omitted-face perturbation of the accepted A1 complete-strip/free reference. This is not finite clipped-restriction convergence, homogeneous dense stability, continuum Yang-Mills or a mass-gap theorem.                                                                                                                               | accepted<br>research/round19/advisor/a2-gate.json |
| R19<br>B1 | Goal B Loop 1 only: complete strict-cutoff physical projector and exact full cross Gram on the actual finite two-cube SU(2) graph. No continuous coefficient box, dense limit, continuum Yang-Mills or mass-gap claim is accepted here.                                                                                                                                                                                         | accepted<br>research/round19/advisor/b1-gate.json |
| R19<br>B2 | Goal B Loop 2 only: finite two-cube continuous signed coefficient-box gap certificate using the accepted B1 48-channel projector, full compressed PVP ledger and exact ordered $W^*W$ cross Gram. No dense limit, continuum Yang-Mills, Clay mass gap or physical-scale matching claim is accepted.                                                                                                                             | accepted<br>research/round19/advisor/b2-gate.json |
| R19<br>C1 | Goal C Loop 1 only: exact finite static U-V-W chain integral on the actual four-cube graph with all other links fixed. This accepts a static mathematical checkpoint and common-V failed-independence discriminator; it does not accept physical energy/time-scale matching, dense convergence, continuum Yang-Mills, a spectral gap theorem or the Clay mass gap.                                                              | accepted<br>research/round19/advisor/c1-gate.json |
| R19<br>C2 | Static common-V integral only; physical energy/time matching remains open. No Hamiltonian spectral gap or continuum claim.                                                                                                                                                                                                                                                                                                      | accepted<br>research/round19/advisor/c2-gate.json |

Continued on next page

| Record    | Historical entry                                                                                                                                                                               | Verdict and first source                          |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| R20<br>D1 | Finite-factor completion gives norm-resolvent convergence with explicit dyadic tail on the exact A2 exterior-reference lift.                                                                   | accepted<br>research/round20/advisor/d1-gate.json |
| R20<br>D2 | Ground energy, rank-one projections and bounded observables converge with explicit dyadic errors for D1 lifts.                                                                                 | accepted<br>research/round20/advisor/d2-gate.json |
| R20<br>E1 | Literal aligned boxes $N=4m+3$ converge after explicit reference-energy subtraction and exact exterior-reference lift.                                                                         | accepted<br>research/round20/advisor/e1-gate.json |
| R20<br>E2 | All upper-boundary phases of literal growing rectangular boxes converge to the A2 ground state on bounded local observables.                                                                   | accepted<br>research/round20/advisor/e2-gate.json |
| R20<br>F1 | Even among local reversible diffusions the C1 static law does not identify an energy/time coefficient: $H\_kappa, c = A\_kappa$ .                                                              | accepted<br>research/round20/advisor/f1-gate.json |
| R20<br>F2 | Exact shared-middle gradient closure, action range $[-5, 7]$ and conditional diffusion gap $c/6$ on $ kappa  \leq 1/8$ .                                                                       | accepted<br>research/round20/advisor/f2-gate.json |
| R20<br>G1 | Literal-box real-time dynamics converges in operator norm on each local observable at fixed physical time, with a strongly continuous Hilbert implementation.                                  | accepted<br>research/round20/advisor/g1-gate.json |
| R20<br>G2 | The E2 limiting local state GNS on the full factor algebra is the A2 representation with canonical generator $H-e$ and its inherited gap.                                                      | accepted<br>research/round20/advisor/g2-gate.json |
| R20<br>H1 | The continuous omitted-face profile has an exact budget and a rigorously delimited sufficient gap certificate at fixed physical scale.                                                         | accepted<br>research/round20/advisor/h1-gate.json |
| R20<br>H2 | Every strict uniform global perturbation-budget ray returns the actual ground projection to the selected-strip reference, with an exact vacuum variance and a quantitative rate.               | accepted<br>research/round20/advisor/h2-gate.json |
| R21<br>I1 | Complete 24-link blocking gives 21 omitted anchor faces and exact normalized local interaction norm $7 \tau $ ; qualitative fixed-spacing stability applies with unevaluated source constants. | limited<br>research/round21/advisor/i1-gate.json  |
| R21<br>I2 | Exact one-star vacuum rotation cancels first-order vacuum mixing and retains a bounded quadratic remainder; a bare pure relative bound for the original interaction fails.                     | limited<br>research/round21/advisor/i2-gate.json  |

Continued on next page

| Record    | Historical entry                                                                                                                                                                                                           | Verdict and first source                          |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| R21<br>J1 | <b>An actual omitted Wilson loop has a nonzero gauge-invariant fluctuation in the perturbed dyadic vacuum, with a uniform rational variance floor.</b>                                                                     | accepted<br>research/round21/advisor/j1-gate.json |
| R21<br>J2 | <b>Physical vacuum GNS space equals the gauge-invariant subspace of this representation, with the restricted shifted energy generator and a nonzero imaginary-time Wilson correlator.</b>                                  | accepted<br>research/round21/advisor/j2-gate.json |
| R21<br>K1 | <b>A positive state-dependent mobility preserves the static density but changes reversible dynamics; two chosen Haar correlation slopes identify <math>c</math> and <math>\zeta</math> within this conditional family.</b> | accepted<br>research/round21/advisor/k1-gate.json |
| R21<br>K2 | <b>A tunable observable gives a uniform two-rate inverse across the declared static interval, while a positive blind cubic proves that two slopes do not identify arbitrary mobility.</b>                                  | accepted<br>research/round21/advisor/k2-gate.json |
| R21<br>L1 | <b>A third observable detects the previously hidden cubic mobility and gives an exact three-parameter inverse in the declared conditional family.</b>                                                                      | accepted<br>research/round21/advisor/l1-gate.json |
| R21<br>L2 | <b>Any finite list of initial correlation slopes leaves strictly positive mobility freedom; a second derivative detects the explicit hidden fifth-degree direction.</b>                                                    | accepted<br>research/round21/advisor/l2-gate.json |
| R21<br>M1 | <b>The fixed-budget summable perturbation converges strongly to zero, giving strong-resolvent and compact-time strong dynamics convergence to the selected-strip reference.</b>                                            | accepted<br>research/round21/advisor/m1-gate.json |
| R21<br>M2 | <b>Connected physical correlations converge on explicitly growing time windows in the canonical summable profile, with a nonzero actual physical Wilson fluctuation.</b>                                                   | accepted<br>research/round21/advisor/m2-gate.json |
| R22<br>N1 | <b>Stationary Heisenberg comparison proves the sharper local connected-correlation bound in the canonical summable model.</b>                                                                                              | accepted<br>research/round22/advisor/n1-gate.json |
| R22<br>N2 | <b>Complete tail-block cubes have an exact weighted incident-face budget, yielding a joint support/time convergence certificate.</b>                                                                                       | accepted<br>research/round22/advisor/n2-gate.json |
| R22<br>O1 | <b>Complete simultaneous homogeneous rotation has explicit weighted remainder with preserved unbounded operator domain.</b>                                                                                                | limited<br>research/round22/advisor/o1-gate.json  |

Continued on next page

| Record    | Historical entry                                                                                                                                                         | Verdict and first source                          |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| R22<br>O2 | The bare-reference elimination has explicit all-support bounds, but its retained-diagonal loss majorants cannot certify an infinite iteration at nonzero coupling.       | limited<br>research/round22/advisor/o2-gate.json  |
| R22<br>P1 | The prescribed section is nonclosable and its fitted clock fails the reserved correlation.                                                                               | accepted<br>research/round22/advisor/p1-gate.json |
| R22<br>P2 | A genuine Haar reduction exactly transports the finite electric dynamics; the declared conditional family fails the reserved data.                                       | accepted<br>research/round22/advisor/p2-gate.json |
| R22<br>Q1 | The full magnetic action has correlated positive conditional variance, proving selected-image leakage and a quantitative nonautonomous dynamic defect.                   | accepted<br>research/round22/advisor/q1-gate.json |
| R22<br>Q2 | The actual magnetic projection has exact memory/resolvent identities, seven-energy fixed-channel leading matrices and explicit cubic full-Hilbert operator remainders.   | accepted<br>research/round22/advisor/q2-gate.json |
| R22<br>R1 | The actual initial diagonal has a uniform local inverse, while its local homological generator leaves an exact nonzero SU2 crossing-boundary defect.                     | accepted<br>research/round22/advisor/r1-gate.json |
| R22<br>R2 | The actual initial boundary-complete vacuum response converges with explicit spatial and graph tails; its source-sector and specified volume-weight limits are exact.    | accepted<br>research/round22/advisor/r2-gate.json |
| R23<br>S1 | The genuine cubic indexed O1 source is nonzero; its initial interior inverse preserves the graph domain and leaves an explicitly bounded connected crossing interaction. | limited<br>research/round23/advisor/s1-gate.json  |
| R23<br>S2 | A compact-time regulated full-source commutator map preserves the actual graph domain and has complete connected weight2 bounds; its sinc residual is retained exactly.  | accepted<br>research/round23/advisor/s2-gate.json |
| R23<br>T1 | The full selected magnetic evolution is approximated by the unique leading-memory return equation with cubic fixed-time and uniform absolute quadratic coupling errors.  | accepted<br>research/round23/advisor/t1-gate.json |
| R23<br>T2 | Separately ground-centered full-selected heat evolutions differ by at most $60\lambda^2$ uniformly for all $t \geq 0$ on the frozen finite physical graph.               | accepted<br>research/round23/advisor/t2-gate.json |
| R23<br>U1 | Actual eight-link gauge-invariant resonance and first-order endpoint coefficient $C_{\eta/84}$ ; higher-order endpoint behavior unresolved in U1.                        | limited<br>research/round23/advisor/u1-gate.json  |

Continued on next page

| Record     | Historical entry                                                                                                                                                                                                     | Verdict and first source                                        |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| R23<br>U2  | <b>Actual full-system endpoint nonconvergence for the frozen bounded local gauge-invariant autocorrelation: <math>\liminf \text{error} &gt; C \eta/168</math> for fixed <math>0 &lt; C \eta \leq 10^{-6}</math>.</b> | accepted<br>research/round23/advisor/u2-gate.json               |
| R24<br>V1  | <b>Three outcomes with a uniform error budget</b>                                                                                                                                                                    | accepted_within_scope<br>research/round24/advisor/v1-gate.json  |
| R24<br>V2  | <b>Finite confidence, with explicit resource costs</b>                                                                                                                                                               | accepted_within_scope<br>research/round24/advisor/v2-gate.json  |
| R24<br>W1  | <b>The actual short-filter residual stays cubic</b>                                                                                                                                                                  | accepted_within_scope<br>research/round24/advisor/w1-gate.json  |
| R24<br>W2  | <b>What repeated filtering actually converges to</b>                                                                                                                                                                 | limited<br>research/round24/advisor/w2-gate.json                |
| R24<br>X1  | <b>An actual 21-state gauge sector, with its leakage</b>                                                                                                                                                             | limited<br>research/round24/advisor/x1-gate.json                |
| R24<br>X2  | <b>A small computation certifies the full graph</b>                                                                                                                                                                  | accepted_within_scope<br>research/round24/advisor/x2-gate.json  |
| R24<br>Y1  | <b>A fixed spatial collar retains the endpoint signal</b>                                                                                                                                                            | accepted_within_scope<br>research/round24/advisor/y1-gate.json  |
| R24<br>Y2  | <b>The endpoint correlation has an actual scalar limit</b>                                                                                                                                                           | accepted_within_scope<br>research/round24/advisor/y2-gate.json  |
| R24<br>Z1  | <b>An actual obstruction to uniform relative accuracy</b>                                                                                                                                                            | accepted_within_scope<br>research/round24/advisor/z1-gate.json  |
| R24<br>Z2  | <b>Useful relative accuracy for specified inputs, at every heat time</b>                                                                                                                                             | accepted_within_scope<br>research/round24/advisor/z2-gate.json  |
| R25<br>AA1 | <b>Complete physical 16-state cube shell, compressed matrix and U spectral measure</b>                                                                                                                               | limited<br>research/round25/advisor/aa1-gate.json               |
| R25<br>AA2 | <b>Proved 16-state observable-reachable reducing component; exact canonical scalar endpoint and evaluated finite-q error</b>                                                                                         | accepted_within_scope<br>research/round25/advisor/aa2-gate.json |
| R26<br>AB1 | <b>AB1 / Locate the resonant source in the boundary defect</b>                                                                                                                                                       | limited<br>research/round26/advisor/ab1-gate.json               |
| R26<br>AB2 | <b>AB2 / Contract the full source at finite spectral resolution</b>                                                                                                                                                  | proved-in-model<br>research/round26/advisor/ab2-gate.json       |

Continued on next page

| Record     | Historical entry                                                                  | Verdict and first source                                  |
|------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------|
| R26<br>AC1 | <b>AC1 / Retain every old residual channel in 293 physical states</b>             | limited<br>research/round26/advisor/ac1-gate.json         |
| R26<br>AC2 | <b>AC2 / Control delayed loading and sharpen all-time heat accuracy</b>           | proved-in-model<br>research/round26/advisor/ac2-gate.json |
| R26<br>AD1 | <b>AD1 / Identify an elementary Wilson observable with a null slow endpoint</b>   | limited<br>research/round26/advisor/ad1-gate.json         |
| R26<br>AD2 | <b>AD2 / Certify nontrivial endpoints for actual six-cycle Wilson observables</b> | proved-in-model<br>research/round26/advisor/ad2-gate.json |
| R26<br>AE1 | <b>AE1 / Localize the Gaussian correction with all-time control</b>               | proved-in-model<br>research/round26/advisor/ae1-gate.json |
| R26<br>AE2 | <b>AE2 / Combine source contraction with the original support-size norm</b>       | proved-in-model<br>research/round26/advisor/ae2-gate.json |
| R26<br>AF1 | <b>AF1 / Compute and certify the actual 293-coordinate heat vector</b>            | proved-in-model<br>research/round26/advisor/af1-gate.json |
| R26<br>AF2 | <b>AF2 / Certify heat evaluation at every time on the original input class</b>    | proved-in-model<br>research/round26/advisor/af2-gate.json |

## E Reproduction and manuscript build

### E.1 Carried historical companions

The commands and dependency versions in this subsection document the carried Draft01 companions. They do not certify the current PDF or the three Round30 investigations. Current replay and build instructions follow separately below.

The accompanying source bundle contains this manuscript's  $\text{\LaTeX}$  files, bibliography, vector figures, calculator code, machine-readable historical and contribution ledgers, and independent audit scripts. The standalone `calculators.html` runs locally without a server. Use a fresh Python environment for the companion requirements; plot rendering dependencies are listed separately by the calculator guide. The companion was exercised with Python 3.12.14, mpmath 1.3.0, NumPy 2.3.5, and Matplotlib 3.10.8. Its default working precision is 70 decimal digits. Exact rational outputs use Python's standard-library fraction arithmetic; a 70-digit diagnostic is not thereby an interval enclosure.

Representative commands, executed from the unpacked manuscript directory, are:

```
python3 -m pip install -r calculators/requirements.txt
python3 calculators/ym_calculators.py jacobi --ratio 2 --dimension 24
python3 calculators/ym_calculators.py static --kappa 1
python3 calculators/ym_calculators.py filter --kind gaussian \
  --M 35/1664 --duration 4
python3 calculators/ym_calculators.py wilson --z 1/1000000
python3 calculators/ym_calculators.py heat-certificates
```

```
python3 calculators/test_calculators.py --output /tmp/ym-calculator-tests.json
python3 audit/skeptic_checks.py --output /tmp/ym-skeptic-checks
python3 audit/skeptic_early_checks.py --output /tmp/ym-early-checks
python3 audit/skeptic_middle_checks.py --output /tmp/ym-middle-checks
python3 audit/skeptic_recent_checks.py --output /tmp/ym-recent-checks
node audit/test_web_calculators.cjs
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The skeptic commands require fresh output directories; use new names on a second run. The heat companion computes certificate budgets; it is not a replacement for the original 293-coordinate solver. To obtain a certified vector, clone the workbench, check out the exact commit in Section 2, and use a fresh absolute output directory:

```
python3 -B research/round26/reverse/af2/check.py \
  --sigma 7/2 --output /absolute/new/heat-at-seven-halves
```

The optional `--input-vector PATH` takes a JSON list of 293 real rational strings and checks exact normalization, original  $P_{21}$  support, and the radius-0.01 condition. The reverse command interface does not accept arbitrary complex coordinates. Forward exports include complex demonstrations under a separate export-error allowance. The operator estimate is broader than the implemented input interface; those scopes must remain distinct.

The workbench's original historical guides remain authoritative for their production commands and dependencies. Some early programs write outputs in place, so run them only in an isolated checkout. Never remove a bound source hash or reinterpret a missing test as a proved premise to make a replay pass. The new source bundle adds no research loop and changes no frozen historical record.

## E.2 Current source-bound replay and manuscript

The current Round30 replay is `research/round30/reproduce.py`. Its machine-readable report binds the three reviewed investigations, inherited evidence and retained limitations. The paired producers and the independent model-agent skeptic are separate computational roles, not external human peer reviewers.

From the repository root, the current editorial commands are:

```
python3 -B research/round30/editorial/integrate.py
python3 -B papers/draft-03/build.py
```

The first requires all three completed reviews and the resulting roadmap. The second checks the source hashes and compiles the manuscript with a fixed build clock. Every new build resets visual-review status. The top-level `papers/draft-03/pdf-qa.json` identifies the current PDF, its actual source inventory, resolved references, geometry checks, and the pages inspected as images. The checksum inventory is `papers/draft-03/artifact-manifest.json`. Earlier copied audit reports retain historical scope.